

eBook

A COMPLETE BOOK OF

DATA INTERPRETATION

& ANALYSIS

Useful for Banking & Insurance examinations
like IBPS, SBI, RBI, LIC, UIIC & Others

2000+
Questions

**SECOND
EDITION**

- **2000 + Questions Based on the Latest Pattern**
- **All types of DI: Table, Pie, Bar, Line, Caselets, Radar, Arithmetic DI & Missing DI**
- **Essential for Both Prelims and Mains Exams**
- **Previous Year Questions asked in**
IBPS PO 2018, SBI PO 2018, RBI PO 2018 & Other Banking Exams

Introduction to Data Interpretation

Data: A series of observations, measurements or facts associated with any event (Physical, Social or Economic). Data can be in the form of figures or statements.

Data interpretation: Act of organizing and interpreting data to get meaningful information.

In Data interpretation, a large volume of data is organized and is represented into a compact and precise form which is easier to interpret than the raw data. Students are required to draw conclusions and inferences from a comprehensive data presented numerically in these organized forms by means of a table or a graphical image (Graphs, Pie-Chart etc.). It tests speed as well as understanding, analytical and decision making capabilities of the students.

Basic tools to solve Data Interpretation:

- Calculation
- Percentage
- Ratio
- Average

1. **Calculation:** Below are some essential tools which help in faster calculations. Students must learn these by heart as much as they can:
 - (i) Tables
 - (ii) Squares and Cubes
 - (iii) Square roots & cube root
 - (iv) Reciprocal Values
2. **Percentage:** Percentage means every hundred. It is a ratio with base of 100. Percentage calculation is the most important aspect in the representation as well as in the interpretation of the data. Students must know about various basic properties of percentage and tricks involved in the faster calculation of percentages.

Some important formula:

$$\text{Percentage Increase} = \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}} \times 100$$

$$\text{Percentage Decrease} = \frac{\text{Initial Value} - \text{Final Value}}{\text{Initial Value}} \times 100$$

$$\text{Quantity I is how much percent of Quantity II} = \frac{\text{Quantity I}}{\text{Quantity II}} \times 100$$

$$\text{Quantity I is how much percent more than Quantity II} = \frac{\text{Quantity I} - \text{Quantity II}}{\text{Quantity II}} \times 100$$

$$\text{Quantity I is how much percent less than Quantity II} = \frac{\text{Quantity II} - \text{Quantity I}}{\text{Quantity II}} \times 100$$

Students must learn by heart the fractional values of some important frequently used percentages:

$$5\% = \frac{1}{20}$$

$$40\% = \frac{2}{5}$$

$$75\% = \frac{3}{4}$$

$$125\% = \frac{5}{4}$$

$$10\% = \frac{1}{10}$$

$$50\% = \frac{1}{2}$$

$$80\% = \frac{4}{5}$$

$$150\% = \frac{3}{2}$$

$$20\% = \frac{1}{5}$$

$$60\% = \frac{3}{5}$$

$$120\% = \frac{6}{5}$$

Fractional values of some important frequently used fractional percentages:

$$33\frac{1}{3}\% = \frac{1}{3}$$

$$14\frac{2}{7}\% = \frac{1}{7}$$

$$12\frac{1}{2}\% = \frac{1}{8}$$

$$9\frac{1}{11}\% = \frac{1}{11}$$

$$16\frac{2}{3}\% = \frac{1}{6}$$

$$7\frac{1}{7}\% = \frac{1}{14}$$

$$6\frac{1}{4}\% = \frac{1}{16}$$

$$8\frac{1}{3}\% = \frac{1}{12}$$

$$11\frac{1}{9}\% = \frac{1}{9}$$

$$6\frac{2}{3}\% = \frac{1}{15}$$

Students can also learn some other percentages based on the above tables:

For example:

$$\frac{1}{20} = 5\%$$

$$\therefore 15\% = 3 \times 5\% = 3 \times \frac{1}{20} = \frac{3}{20} \Rightarrow \frac{1}{3} = 33\frac{1}{3}\%$$

$$\therefore 66\frac{1}{2}\% = 2 \times 33\frac{1}{3}\% = 2 \times \frac{1}{3} = \frac{2}{3} \Rightarrow \frac{1}{8} = 12\frac{1}{2}\%$$

$$\therefore 37\frac{1}{2}\% = 3 \times 12\frac{1}{2}\% = 3 \times \frac{1}{8} = \frac{3}{8}$$

Note: If the percentage value is increased by 100%, then the equivalent fraction value will also be increased by 1.

For example:

$$25\% = \frac{1}{4}$$

$$\therefore 125\% = 1 + \frac{1}{4} = \frac{5}{4} \Rightarrow 33\frac{1}{3}\% = \frac{1}{3}$$

$$\therefore 133\frac{1}{3}\% = 1 + \frac{1}{3} = \frac{4}{3} \Rightarrow 8\frac{1}{3}\% = \frac{1}{12}$$

$$\therefore 108\frac{1}{3}\% = 1 + \frac{1}{12} = \frac{13}{12}$$

Must know: If any quantity doubles itself then it is 200% of its previous value.

If any quantity triples itself then it is 300% of its previous value.

If any quantity becomes 5 times of itself then it is 500% of its previous value.

But

If any quantity doubles itself then it is increased by 100%.

If any quantity triples itself then it is increased by 200%.

If any quantity becomes 5 times of itself then it is increased by 400%.

Note: Always break the single percentage into easier percentages wherever possible.

For example:

$$65\% = 50\% + 10\% + 5\% \Rightarrow 45\% = 50\% - 5\%$$

$$95\% = 100\% - 5\% \Rightarrow 87\frac{1}{2}\% = 100\% - 12\frac{1}{2}\%$$

$$43\frac{1}{3}\% = 50\% - 6\frac{2}{3}\%$$

3. **Ratio:** It is defined as the reduced form of values of quantities to lowest integers for the purpose of comparison between the values of quantities. It is the result of value of one quantity divided by another. Ratios can be expressed as fractions, decimals or even as percentages. It is necessary that the two figures compared should have the same characteristics and should be expressed either in same unit or in comparable units. For the calculation of ratios, students must learn tables, divisibility of numbers and simplification of expressions etc.

$$\text{Ratio} = \frac{\text{Quantity I}}{\text{Quantity II}}$$

4. **Average:** It is defined as the central value of values of all the quantities taken into consideration. It is the result of sum of values of all the quantities divided by the number of quantities. Average is always between the highest and the lowest values among the values of all the quantities. It is necessary that the quantities taken in consideration should have the same characteristics and should be expressed either in same unit or in comparable units. For the calculation of averages, students must learn the various properties related to average.

$$\text{Average} = \frac{\text{Sum of values of all quantities}}{\text{Number of quantities}}$$

Important Points to Remember:

1. **Read the question carefully:** The first and the most important step in solving any Data Interpretation question is to read the question carefully. You should read all the data that comes with the graphs or table in the question. Many a times, the data given above/below the graph (additional instructions) turn out to be more important than most of the numbers in the graphs.
2. **Analyze the data carefully:** The next step is to analyze the given graph/data carefully. Do not try to see the questions first and find out the answers accordingly. You will waste your time following that method. Try to understand the graph. Look at the type of data given in each graph, chart, table or pie chart. Look carefully at the labels.
3. **Don't worry about too much data:** Try to understand the question. Sometimes, the question contains lots of data that is unrelated and is not required for answering the questions. When you look at the question you may get discouraged by the lengthy tables or by the amount of information given above/below the graphs. But, if you try to understand what the data is about and then look at the question, you may find that you only have to use part of the data. Hence, it is important that you do not get disheartened by the size of the data and skip the question without looking closely at it.
4. **Learn to skim through data:** Some graphs have a lot of data associated with them and not all of which is required to solve the questions. Skimming through the data and avoiding mess is an important part of the process. So, just focus on what is required in the question, rather than on all the data at one time.
5. **Avoid unnecessary calculations:** We have a habit from our school days to solve questions in a step by step method. This is a very good habit for school exams but a really bad habit when it comes to the competitive exams. There are many unnecessary calculations that we do while attempting the questions which cost us a precious few seconds per question. Sometimes, there are many steps that can be skipped but we still do it as we are trained to solve in a step by step method. Learn to skip those steps.
6. **Learn to approximate:** You do not need to calculate the exact answer for every question. Many a times, the options given are far enough from each other to give you enough room for approximation. So, instead of finding the accurate answer, try to find an approximate answer. This will give you the correct answer more often than not. If however, the options are close, you will still be able to eliminate 1 or 2 options easily.
7. **Pay close attention to the units used:** Sometimes, the questions may use a different unit for the question and another unit for the data. If you do not pay close attention to the unit, you may be ended up choosing the wrong answer. Always convert the units into the ones which are asked in the question.
8. **Skip questions that need too much calculation:** Some questions ask too much from you. They require lots of calculation in order to be solved. These questions are known as the speed-breakers. Such questions are best left alone, at least in the first round of attempt. Once you have finished solving all the easy questions and still have time left for the section, you should attempt these questions. If you try such questions, you will lose your precious time on them and may not be able to attempt some simple questions that may follow.
9. **Don't assume anything:** Sometimes there are questions which need to find out some data which cannot be calculated even with the help of the data given in the graph and the question. Always be alert enough to see whether the data given is enough to answer the question or not and do not go forward with answering the questions based on assumptions. Sometimes, 'cannot be determined' can also be the correct answer.

Classification: Data interpretation is broadly classified as follow:

- Table
- Line Graph
- Bar Graph
- Pie Chart
- Radar Graph
- Mixed Graph
- Caselet

Table (Tabulation): It is the most fundamental and the most versatile way of representing data and an easier format to comprehend. Data is arranged in columns and rows in a table in either alphabetic or chronologic order (as A, B, C or month wise, year wise). Either the columns or the rows will represent different values to describe the variables. Other different kind of data representation formats like bar graph, line graph, pie chart etc., originate from the table. In other words, representing the data in a tabular format is the first step in forming other types of data representation formats.

Example:

Directions (1-8): Following table shows the percentage of population below poverty line out of the total population of a particular state for six states and the ratios of male and female below and above poverty line in these states:

State	Percentage population below poverty line	Proportion of male and female	
		Below poverty line M : F	Above poverty line M : F
A	16	5 : 3	4 : 3
B	10	3 : 7	5 : 4
C	22	6 : 5	7 : 6
D	28	3 : 4	5 : 7
E	12	1 : 3	6 : 5
F	20	2 : 3	3 : 5

Types of question asked:

- Find the population of males above poverty line in state C if the total population of the state is 60 lakh.
Sol; Percentage of population above poverty line in state C = $100 - 22 = 78\%$
 Percentage of males above poverty line in state C out of total population = $\frac{7}{13} \times 78 = 42\%$
 Population of males above poverty line in state C = 42% of 60 lakh = 25.2 lakh
- Find the difference of population of males below poverty line and females above poverty line in state A if the total population of the state is 35 lakh.
Sol; Percentage of males below poverty line in state A out of total population = $\frac{5}{8} \times 16 = 10\%$
 Percentage of population above poverty line in state A = $100 - 16 = 84\%$
 Percentage of females above poverty line in state A out of total population = $\frac{3}{7} \times 84 = 36\%$
 Required Difference = $(36\% - 10\%)$ of 35 lakh = 26% of 35 lakh = 9.1 lakh
- If the population of males above poverty line in state D is 13.5 lakh then find the total population of the state.
Sol. Percentage of population above poverty line in state D = $100 - 28 = 72\%$
 Percentage of males above poverty line in state D out of total population = $\frac{5}{12} \times 72 = 30\%$
 Population of males above poverty line in state D = 30% of total population = 13.5 lakh
 Total population = $\frac{100}{30} \times 13.5 = 45$ lakh
- Find the ratio of population of females below poverty line and males above poverty line in state F.
Sol; Percentage of females below poverty line in state F out of total population = $\frac{3}{5} \times 20 = 12\%$
 Percentage of population above poverty line in state F = $100 - 20 = 80\%$
 Percentage of males above poverty line in state F out of total population = $\frac{3}{8} \times 80 = 30\%$
 Required Ratio = 12% of total population : 30% of total population = 2 : 5
- Find the ratio of population of males below poverty line in state C and females above poverty line in state D if the ratio of total populations of state C and D is 7 : 4.
Sol; Let the total populations of state C and D be 7x and 4x respectively.
 Percentage of males below poverty line in state C out of total population = $\frac{6}{11} \times 22 = 12\%$
 Percentage of population above poverty line in state D = $100 - 28 = 72\%$
 Percentage of females above poverty line in state D out of total population = $\frac{7}{12} \times 72 = 42\%$
 Required Ratio = 12% of total population of state C : 42% of total population of state D
 = 12% of 7x : 42% of 4x = 1 : 2
- Find the difference of population of males below poverty line in state C and females above poverty line in state D if the ratio of total populations of state C and D is 7 : 4.
Sol; The difference of populations cannot be determined because only the ratio of the populations of states is given, not the actual populations.

7. Population of females below poverty line in state B is how much percent of males above poverty line in state F if the ratio of total populations of state B and F is 6 : 7?

Sol; Let the total populations of state B and F be $6x$ and $7x$ respectively.

Percentage of females below poverty line in state B out of total population = $\frac{7}{10} \times 10 = 7\%$

Percentage of population above poverty line in state F = $100 - 20 = 80\%$

Percentage of males above poverty line in state F out of total population = $\frac{3}{8} \times 80 = 30\%$

Required Percentage = $\frac{7\% \text{ of } 6x}{30\% \text{ of } 7x} \times 100 = 20\%$

8. For which state, the population of females below poverty line is the maximum?

Sol. The state with maximum female below poverty line population cannot be determined because only the percentages and ratios for below poverty line populations are given, not the actual populations of each state.

Line Graph: It is a type of graph in which the variable does not change according to any law but changes abruptly (broken off suddenly). It indicates the variation of one parameter with respect to another (X-axis, Y-axis). It determines trends and rate of change over the time. We can easily see data movement in case of line graph.

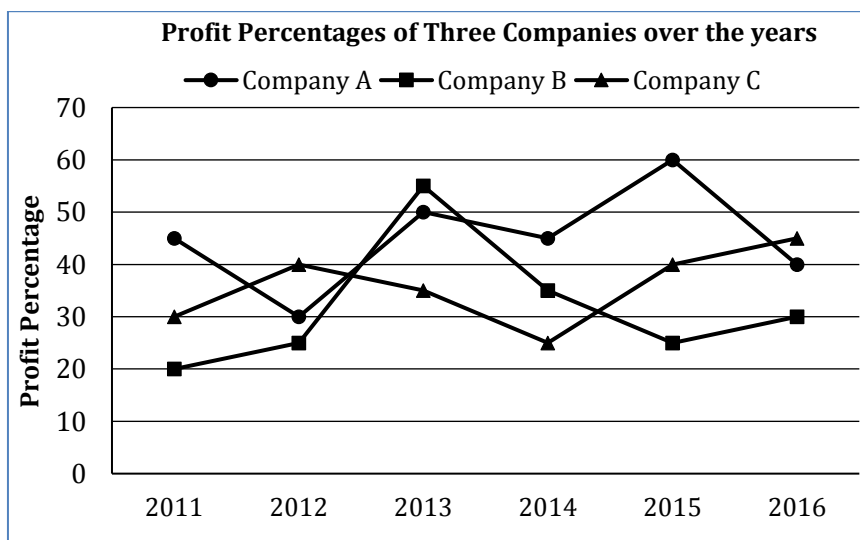
This graph can be classified into following categories.

- (i) Simple line graph
- (ii) Multiple lines graph

- (i) **Simple line graph:** It is also known as single dependent variable graph. It is plotted against the independent factor. The former is plotted on Y-axis while the latter is plotted on the X-axis.



- (ii) **Multiple lines graph:** In this graph more than one dependent variable is plotted against the independent variable. The Y-axis is common to all the variables.

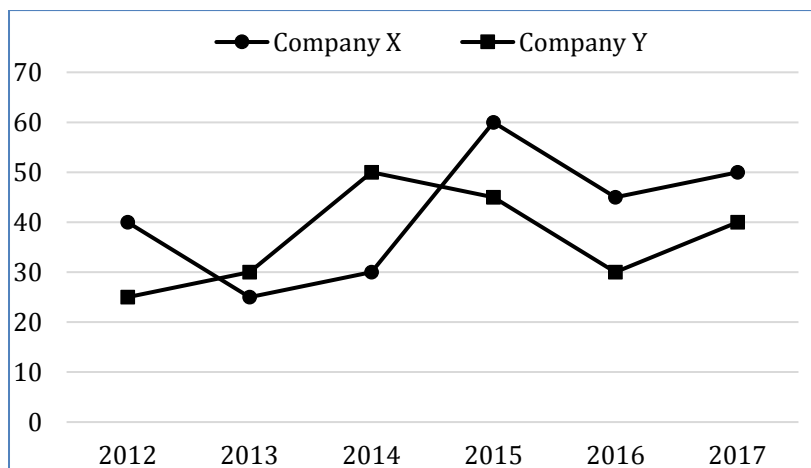


Example:

Directions (1-7): study the following graph to answer the given questions.

$$\text{Profit \%} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$

Percent profit earned by two companies over the given years:

**Types of question asked:**

1. If the expenditure of Company X in 2015 was equal to the expenditure of Company Y in that year, then what was the ratio of their respective incomes?

Sol; Let the expenditure of Company X and Y in 2015 be Rs.x

$$\begin{aligned} \text{Ratio of Income of Company X and Y in 2015} &= (100\% + 60\%) \text{ of } x : (100\% + 45\%) \text{ of } x \\ &= 160\% \text{ of } x : 145\% \text{ of } x = 32 : 29 \end{aligned}$$

2. For Company Y, the income in 2012 was equal to the expenditure in 2014. What was the ratio of its respective incomes in these two years?

Sol; Let the income in 2012 and expenditure in 2014 of Company Y be Rs.x

$$\text{Income of Company Y in 2014} = (100\% + 50\%) \text{ of } x = 150\% \text{ of } x = \text{Rs.}1.5x$$

$$\text{Ratio of Income of Company Y in 2012 and in 2014} = x : 1.5x = 2 : 3$$

3. In 2017, the income of Company Y was Rs.35 crore. What was the expenditure of the company in that year?

Sol; Expenditure of Company Y in 2017 = $\frac{35}{100 + 40} \times 100 = \frac{35}{140} \times 100 = \text{Rs.}25 \text{ crore}$

4. In 2016, the income of Company Y was Rs.52 crore and the expenditures for both the companies were same in that year. What was the average of incomes of both the companies in that year?

Sol; Expenditure of Company Y in 2016 = $\frac{52}{100 + 30} \times 100 = \frac{52}{130} \times 100 = \text{Rs.}40 \text{ crore}$

$$\text{Expenditure of Company X in 2016} = \text{Rs.}40 \text{ crore}$$

$$\text{Income of Company X in 2016} = (100\% + 45\%) \text{ of } 40 = \text{Rs.}58 \text{ crore}$$

$$\text{Average of incomes of both the companies in 2016} = \frac{58 + 52}{2} = \frac{110}{2} = \text{Rs.}55 \text{ crore}$$

5. For which year, the actual profit amount for Company X is the maximum?

Sol; Actual profit amount cannot be determined because only the profit percentages are given but the actual amounts of expenditure or income for the company are not given.

6. For which year, the difference in the actual profit amount for both the companies is the maximum?

Sol; Difference in the actual profit amounts for the companies cannot be determined because only the profit percentages are given but the actual amounts of expenditure or income for the companies are not given.

7. If the ratio of expenditures of Company X in 2013 and 2017 is 4 : 3 respectively, then the income of Company X in 2013 is how much percent more/less than the income of Company X in 2017?

Sol; Let the expenditures of Company X in 2013 and 2017 be Rs.4x and Rs.3x respectively.

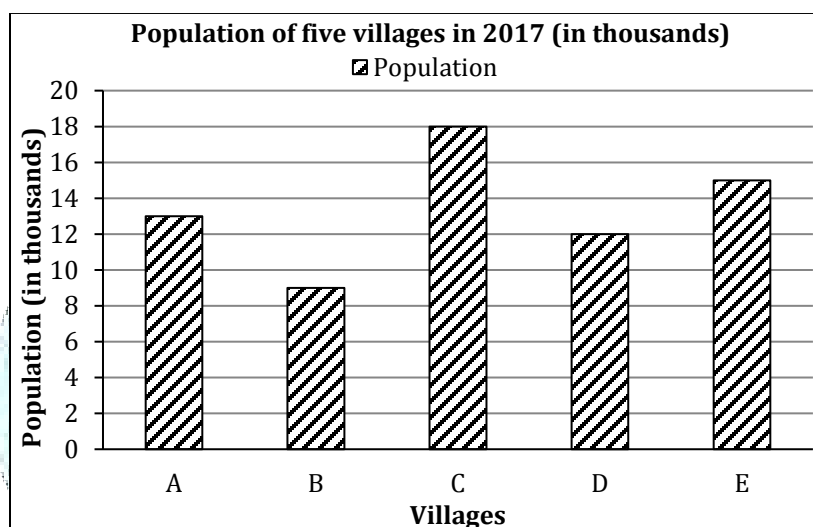
Income of Company X in 2013 = (100% + 25%) of 4x = Rs.5x

Income of Company X in 2017 = (100% + 50%) of 3x = Rs.4.5x

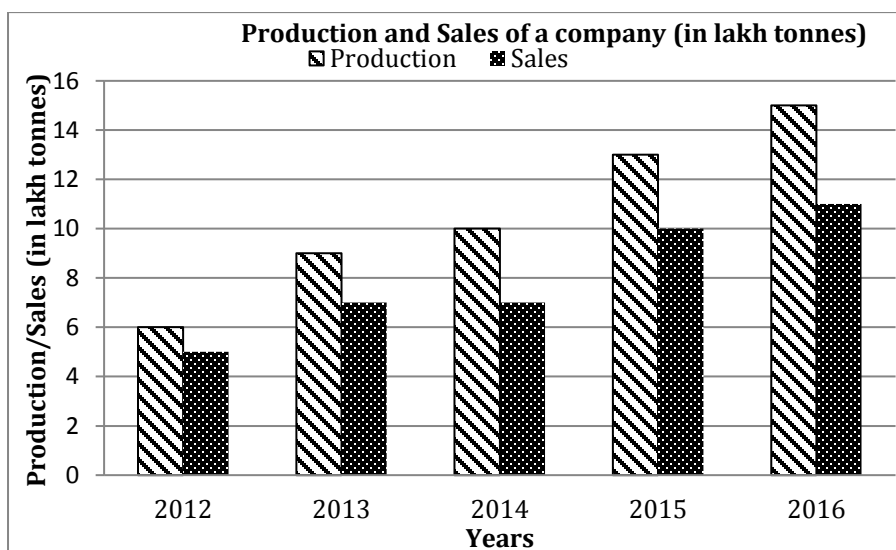
$$\text{Required Percentage} = \frac{5x - 4.5x}{4.5x} \times 100 = \frac{0.5x}{4.5x} \times 100 = 11\frac{1}{9}\% \text{ more}$$

Bar Graph: Bar Graph is the most commonly used method of representing data among the graphs. It is drawing the form of rectangular bars of uniform width with equal spaces between them where the length of the bars is proportional to the values they represent. It can be drawn either horizontally or vertically. Effective representation of Bar graph is mainly classified into the followings categories:

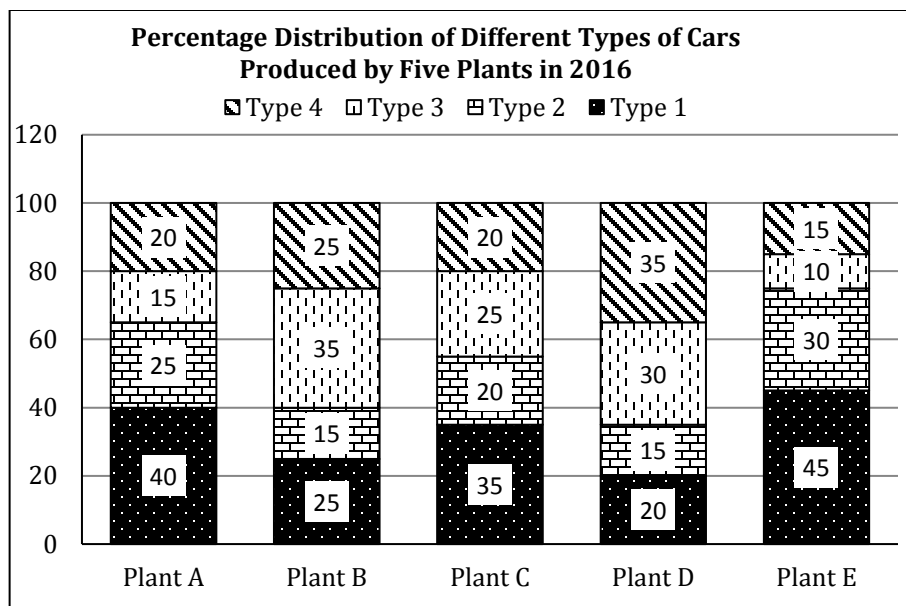
- (i) Simple bar group
 - (ii) Multiple Bar graph
 - (iii) Sub-dividend Bar graph or cumulative Bar graph
- (i) **Simple Bar graph:** It represents only one variable with equal width but of varying heights in proportion to the values of the variable.



- (ii) **Multiple Bar group:** In this graph, two or more bar graphs are constructed adjoining one another in a single graph, to represent either different multiple variables or different components of a single variable.

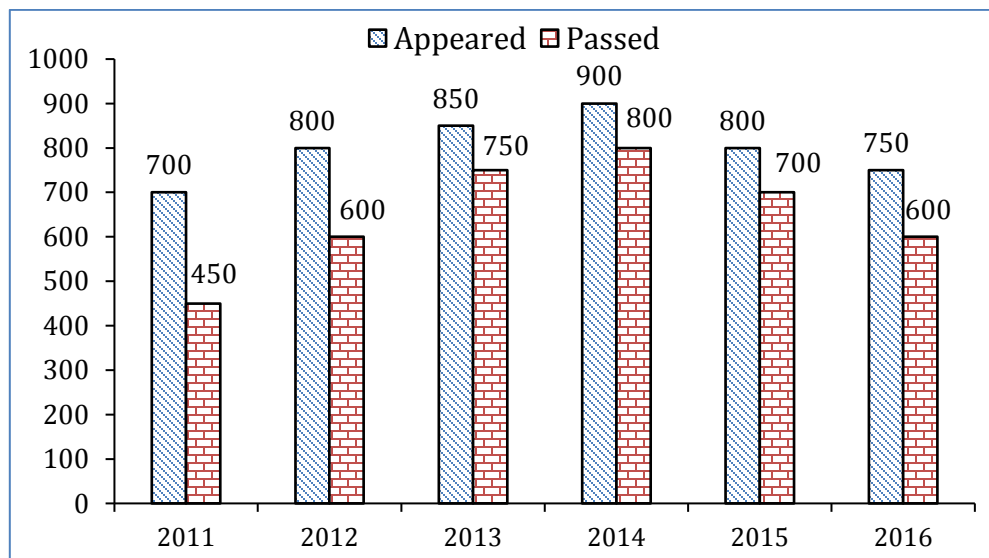


- (iii) **Sub-dividend or cumulative Bar graph:** In this graph, total value as well as individual component values of a variable are pictorially represented as a single bar. The variable is to be divided into various components. It is drawn proportionally in length to the total and divided in the ratios of their components.



Example:

Directions (1-6): Study the following bar graph carefully and answer the questions given below:
Number of students appeared and passed in an exam over the years:



Types of question asked:

1. What was the average number of candidates appeared in the exam over the years?

Sol; Required Average = $\frac{700 + 800 + 850 + 900 + 800 + 750}{6} = \frac{4800}{6} = 800$

2. What is the ratio of number of students who did not pass the exam in 2011 to that in 2016?

Sol; Required Ratio = $(700 - 450) : (750 - 600) = 250 : 150 = 5 : 3$

3. For which year, the percentage of students who passed the exam is the maximum?

Sol; Percentage of passed students:

$$\text{In 2011} = \frac{450}{700} \times 100 = 64\frac{2}{7}\%$$

$$\text{In 2012} = \frac{600}{800} \times 100 = 75\%$$

$$\text{In 2013} = \frac{750}{850} \times 100 = 88\frac{4}{17}\%$$

$$\text{In 2014} = \frac{800}{900} \times 100 = 88\frac{8}{9}\%$$

$$\text{In 2015} = \frac{700}{800} \times 100 = 87\frac{1}{2}\%$$

$$\text{In 2016} = \frac{600}{750} \times 100 = 80\%$$

Hence, the percentage of passed students is the maximum for the year 2014.

4. For which year, the percentage increase/decrease in the number of passed students from the previous year is the minimum?

Sol; Percentage increase/decrease in the number of passed students:

$$\text{In 2012} = \frac{150}{450} \times 100 = 33\frac{1}{3}\%$$

$$\text{In 2013} = \frac{150}{600} \times 100 = 25\%$$

$$\text{In 2014} = \frac{50}{750} \times 100 = 6\frac{2}{3}\%$$

$$\text{In 2015} = \frac{100}{800} \times 100 = 12\frac{1}{2}\%$$

$$\text{In 2016} = \frac{100}{700} \times 100 = 14\frac{2}{7}\%$$

Hence, the percentage increase/decrease in the number of passed students is the minimum for the year 2014.

5. What is the total number of students who did not pass the exam over the years?

Sol; Total number of students who did not pass the exam

$$= 250 + 200 + 100 + 100 + 100 + 150 = 900$$

6. The total number of students who passed the exam from 2011 to 2013 is how much percent more/less than the total number of students who passed the exam from 2014 to 2016?

Sol; Number of students passed from 2011 to 2013 = $450 + 600 + 750 = 1800$

Number of students passed from 2014 to 2016 = $800 + 700 + 600 = 2100$

$$\text{Required Percentage} = \frac{2100 - 1800}{2100} \times 100 = 14\frac{2}{7}\% \text{ less}$$

Pie Chart: It is circular representation of data where the data is represented as a part of a circle. The total quantity is distributed over a total angle of 360° . The circle represents the total value (360° or 100%) and the different parts or sectors represent certain proportions (degree or percentage value) of the total. The value of each component is in proportion to the circular area (or central angle) representing the component. It may be classified in the following categories:

- (i) Simple Pie chart
- (ii) Multiple Pie Charts

Note: The sector of circle is divided mainly into two ways:

(a) In degrees: In this representation, the given data is distributed over a total angle of 360° . Each part makes a certain angle called central angle.

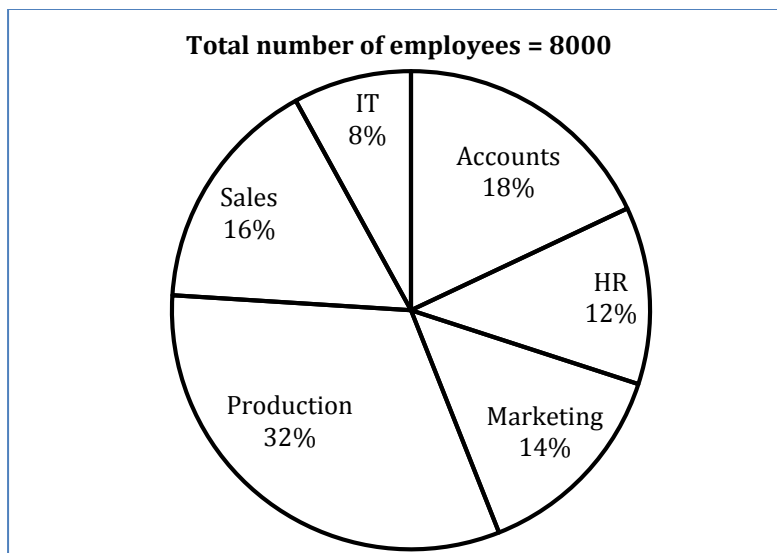
$$\therefore \text{Central angle of a sector} = (\text{Value of sector}) / (\text{Total value}) \times 360^\circ$$

(b) In percentage – In this representation, the given data is distributed over a total of 100%. For solution we take base 100.

$$\therefore \text{Percentage value of a sector} = (\text{Value of sector}) / (\text{Total value}) \times 100$$

Example:

Directions (1-8): Study the following Pie Chart carefully and answer the questions given below:
Percentage Distribution of Employees in Different Departments of a Company:



Types of question asked:

1. What is the total number of employees in Accounts and Marketing departments together?

Sol; Number of employees in Accounts and Marketing together = $(18\% + 14\%)$ of total employees
 $= 32\%$ of 8000 = 2560

2. What is the difference between the number of employees in Production and Sales departments?

Sol; Difference between number of employees in Production and Sales = $(32\% - 16\%)$ of total employees = 16% of 8000 = 1280

3. What is the ratio of number of employees in IT and HR department together to the number of Sales and Marketing departments together?

Sol; Required ratio = $(8\% + 12\%)$ of total employees : $(16\% + 14\%)$ of total employees = 2 : 3

4. The number of employees in HR department is how much percent more/less than the number of employees in Production department?

Sol; Required percentage = $\frac{32\% \text{ of total employees} - 12\% \text{ of total employees}}{32\% \text{ of total employees}} \times 100 = \frac{20}{32} \times 100 = 62.5\%$ less

5. What is the average of total number of employees in Accounts, HR, Marketing and Production departments together?

Sol; Total number of employees in Accounts, HR, Marketing and Production together
 $= (18\% + 12\% + 14\% + 32\%)$ of total employees = 76% of total employees
 Required average = $\frac{76\% \text{ of total employees}}{4} = 19\%$ of total employees = 19% of 8000 = 1520

6. If number of employees in Sales department is increased by 25% in the next year, then what is number of employees in the department in the next year?

Sol; Number of employees in Sales department in next year
 $= (100\% + 25\%)$ of number of employees in Sales department this year
 $= 125\%$ of 16% of 8000 = 20% of 8000 = 1600

7. If numbers of employees in IT and Marketing departments are increased by 60% and 20% respectively in the next year, then what is the ratio of numbers of employees in these departments in the next year?

Sol; Required Ratio = 160% of 8% of total employees : 120% of 14% of total employees
 $= 16 : 21$

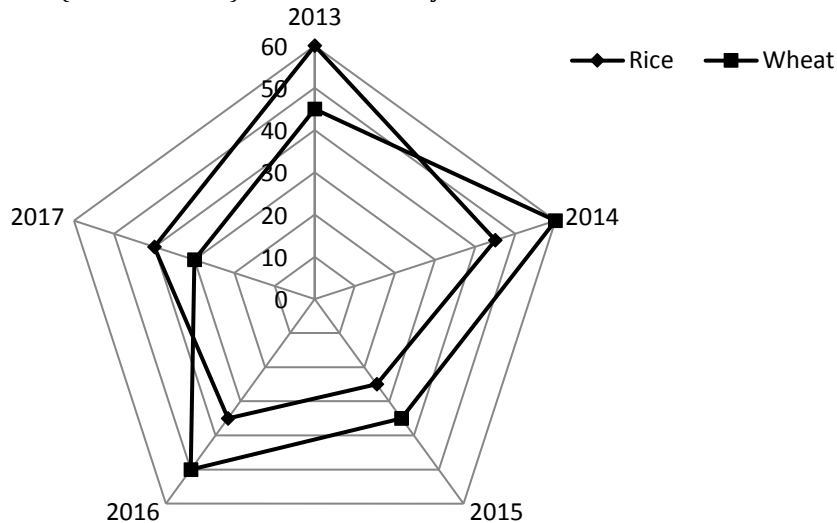
8. If the ratio of numbers of male and female employees in Accounts department is 5 : 4 and 60% of the employees in the HR department are females, then what is the total number of male employees in these departments together?

Sol; Number of male employees in Accounts department = $\frac{5}{9} \times 18\%$ of 8000 = 10% of 8000 = 800
 Number of male employees in HR department = $(100\% - 60\%)$ of 12% of 8000
 $= 40\%$ of 12% of 8000 = 4.8% of 8000 = 384
 Total Number = 800 + 384 = 1184

Radar Graph: In this graph, the values of variables are represented with respect to a central point. The values are represented in proportion with the distances from this central point. This graph can be seen as a circular line graph. This graph is also known as spider or web graph.

Example:**Directions (1-5):** Study the following radar graph carefully and answer the questions that follow:

The production of rice and wheat (in lakh tonnes) in a state in five years:

**Types of question asked:**

1. What is the average of total wheat production in the state over the years?

Sol; Required Average = $\frac{45 + 60 + 35 + 50 + 30}{5} = \frac{220}{5} = 44$ lakh tonnes

2. What is the ratio of productions of wheat and rice from year 2013 to 2016?

Sol; Required Ratio = $(45 + 60 + 35 + 50) : (60 + 45 + 25 + 35) = 190 : 165 = 38 : 33$

3. Production of rice in 2016 is how much percent more/less than the production of wheat in the same year?

Sol; Required Percentage = $\frac{50 - 35}{50} \times 100 = \frac{3}{10} \times 100 = 30\%$ less

4. For which year, the difference between the production of rice and wheat is the minimum?

Sol; Difference between the production of rice and wheat:

For 2013 = 15 lakh tonnes

For 2014 = 15 lakh tonnes

For 2015 = 10 lakh tonnes

For 2016 = 15 lakh tonnes

For 2017 = 10 lakh tonnes

Hence, the difference is the minimum for the years 2015 and 2017.

5. For which year, the percentage increase/decrease in the production of rice from the previous year is the maximum?

Sol; Percentage increase/decrease in the production of rice:

For 2014 = 25% decrease

For 2015 = $44\frac{4}{9}\%$ decrease

For 2016 = 40% increase

For 2017 = $14\frac{2}{7}\%$ increase

Hence, the percentage increase/decrease is the maximum for the year 2015.

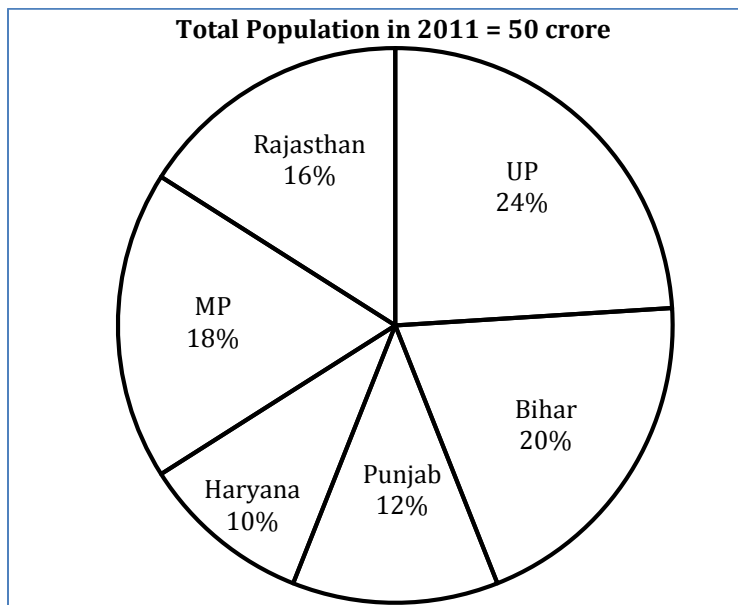
Mixed Graph: It is not based on a single graph but on a combination of two or more graphs. These graphs may or may not represent similar variables. If the variables represented by these graphs are not similar, then the relationship between these variables is mentioned in the question along with the other data which is useful in solving the questions.

It can mainly be classified into the following categories:

- (i) Table and Bar Graph
- (ii) Table and Line Graph
- (iii) Table and Pie Chart
- (iv) Bar Graph and Line Graph
- (v) Pie Chart and Bar Graph
- (vi) Pie Chart and Line Graph

Example:**Directions (1-6):** Study the pie chart and table carefully and answer the following questions:

Percentage Distribution of population of 6 states out of total population:



Ratio of male to female population and literate to illiterate population in these states:

State	Sex M : F	Literacy Literate : Illiterate
UP	7 : 5	3 : 5
Bihar	3 : 2	2 : 3
Punjab	5 : 7	3 : 1
Haryana	7 : 3	3 : 2
MP	4 : 5	1 : 2
Rajasthan	5 : 3	1 : 3

Types of question asked:

1. What is the total population of illiterate people in Punjab and Haryana together?

Sol; Total population of illiterate people in Punjab and Haryana together

$$= \frac{1}{4} \text{ of population of Punjab} + \frac{2}{5} \text{ of population of Haryana}$$

$$= \frac{1}{4} \text{ of } 12\% \text{ of } 50 \text{ crore} + \frac{2}{5} \text{ of } 10\% \text{ of } 50 \text{ crore}$$

$$= 3\% \text{ of } 50 \text{ crore} + 4\% \text{ of } 50 \text{ crore}$$

$$= 7\% \text{ of } 50 \text{ crore} = 3.5 \text{ crore}$$

2. Total literate population in MP is how much percent of the total illiterate population in UP?

Sol; Required percentage = $\frac{\text{Total literate population in MP}}{\text{Total illiterate population in UP}} \times 100$

$$= \frac{\frac{1}{3} \text{ of } 18\% \text{ of total population}}{\frac{5}{8} \text{ of } 24\% \text{ of total population}} \times 100 = \frac{6}{15} \times 100 = 40\%$$

3. If the total population of Rajasthan and Bihar is increased by 20% and 25% respectively in comparison to the previous year, then what will be ratio of male population of these two states if the ratio of male to female population remains the same as previous year?

Sol; Required Ratio = $120\% \text{ of } \frac{5}{8} \text{ of } 16\% : 125\% \text{ of } \frac{3}{5} \text{ of } 20\% = 12\% : 15\% = 4 : 5$

4. What is the population of literate females in Bihar?

Sol; Population of literate females in Bihar cannot be determined because ratio of literate to illiterate population is given for total population of the state, not for male or female population.

5. If 50% of male population is literate in MP, then what percent of female population in MP is literate?

Sol; Percentage of literate population in MP = $\frac{1}{3}$ of 18% of total population = 6% of total population

Percentage of male population in MP = $\frac{4}{9}$ of 18% of total population = 8% of total population

Percentage of female population in MP = 18% of total population - 8% of total population
= 10% of total population

Percentage of literate male population in MP = 50% of 8% of total population
= 4% of total population

Percentage of literate female population in MP = 6% of total population - 4% of total population
= 2% of total population

Required Percentage = $\frac{\text{literate female population in MP}}{\text{Total female population in MP}} \times 100 = \frac{2\% \text{ of total population}}{10\% \text{ of total population}} \times 100 = 20\%$

6. What is the average male population of Bihar, Haryana and MP?

Sol; Total male population of Bihar, Haryana and MP

= $(\frac{3}{5} \text{ of } 20\% + \frac{7}{10} \text{ of } 10\% + \frac{4}{9} \text{ of } 18\%) \text{ of total population}$

= $(12\% + 7\% + 8\%) \text{ of total population}$

= 27% of total population

Required Average = 9% of total population = 9% of 50 crore = 4.5 crore

Caselet: It is a comprehensive type question where the information is given in the form of paragraphs or multiple sentences which provide the details of all the parameters involved including their inter-relationships. The information can be converted into either tabular form or Venn-Diagram to solve the questions. In the recent patterns, caselets related to the various quantitative aptitude topics are also seen where a situation is described with in the form of a paragraph with data and conditions. We have to use the data and solve the questions according to the given conditions.

Example:

Directions (1-8): Study the following information carefully and answer the questions that follow:

There are four departments in a company – Production, Marketing, Sales and HR. 40% of the total employees in the company works in Production department and 60% of the total employees working the Production department are males. Half of the rest of the employees works in Marketing department and the ratio of male and female employees in the department is 3 : 7. The number of employees working in the Sales department is one-fifth of the total employees. Number of females working in Sales department is 16 less than number of females working in production department. Number of males working in HR department is 40% of the number of males working in Sales department. Total number of the employees working the company is 1600.

Sol; The information given above can be converted a table as follows:

Departments	Total Employees		Males		Females	
	Percentage	Number	Percentage	Number	Percentage	Number
Production	40%	640	24%	384	16%	256
Marketing	30%	480	9%	144	21%	336
Sales	20%	320	5%	80	15%	240
HR	10%	160	2%	32	8%	128

Note: Percentages are given out of total number of employees in the company.

Types of question asked:

1. What is the percentage of female employees working in the company?

Sol; Percentage of female employees in the company = $16\% + 21\% + 15\% + 8\% = 60\%$

2. What is the total number of female employees working in Marketing, Sales and HR departments?

Sol; Number of female employees working in Marketing, Sales and HR departments
= $336 + 240 + 128 = 704$

3. What is the average of number of male employees working in all the departments together?

Sol; Number of male employees working in all the departments = 40% of total employees
Required Average = 10% of total employees = 160

4. What is the ratio of number of male employees working in Production department to the number of female employees working in Marketing department?

Sol; Required Ratio = 24% of total employees : 21% of total employees = 8 : 7

5. The number of male employees working in Sales department is how much percent more/less than the number of female employees working in HR department?

Sol; Required Percentage = $\frac{8\% \text{ of total employees} - 5\% \text{ of total employees}}{8\% \text{ of total employees}} \times 100 = \frac{3}{8} \times 100 = 37.5\% \text{ less}$

6. In which department, the number of female employees working is the maximum?
Sol; Marketing department
7. 40% of the female employees working in Sales department are postgraduates, then what is number of female employees working in Sales department who are not postgraduates?
Sol; Number of female employees working in Sales department who are not postgraduates
= (100% - 40%) of number of female employees working in Sales department
= 60% of 240 = 144
8. 40% of the female employees working in Sales department are postgraduates, then what is number of male employees working in Sales department who are not postgraduates?
Sol; The number of male employees working in Sales department, who are not postgraduates, cannot be determined because the information about the educational qualifications of male employees in the department is not given.



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Table

Tables are one of the most versatile methods of systematic representation of quantitative data where the data is represented through horizontal rows and vertical columns. In fact, the data that can be represented on any type of graph/chart can also be represented on a table, but the reverse is not always true. Also, the amount of data that can be represented on a table is much higher than that can be represented on any other graph/chart. But, tables are a little harder to interpret when the number of variables represented is higher, due to their less visual impact.

This chapter contains:

- Solved Examples
- Previous Year Questions
- Practice Set Level I
- Practice Set Level II

SOLVED EXAMPLES

Directions (1-5): Given below is the table which shows the total numbers of voters in 5 different villages, % of voters who did not vote and percentage of female out of total persons who have voted.

Villages	Total voters	% of voters who do not vote	% of females out of total persons who have voted
A	20000	12%	40%
B	18000	13%	60%
C	16000	20%	45%
D	13500	10%	30%
E	22500	8%	40%

1. Total people who have not voted from village A and C together is what % of total males who have voted from these village?

(a) $17\frac{2}{5}\%$ (b) $31\frac{9}{11}\%$ (c) $20\frac{2}{5}\%$ (d) $13\frac{14}{17}\%$ (e) $8\frac{2}{11}\%$

Sol. (b); Total persons who have not voted from A and C together = $12 \times 200 + 20 \times 160 = 5600$

$$\text{Total males who have voted from these villages} = \frac{88}{100} \times 20,000 \times \frac{60}{100} + \frac{80}{100} \times 16000 \times \frac{55}{100} \\ = 10560 + 7040 = 17600$$

$$\text{Required percentage} = \frac{5600}{17600} \times 100 = 31\frac{9}{11}\%$$

2. What is the ratio of males who have voted from village D and E together to the females who have voted from village D and E together?

(a) 63 : 43 (b) 93 : 53 (c) 73 : 33 (d) 82 : 23 (e) 53 : 33

Sol. (b); Required ratio = $\frac{\frac{90}{100} \times 13500 \times \frac{70}{100} + \frac{92}{100} \times 22500 \times \frac{60}{100}}{\frac{90}{100} \times 13500 \times \frac{30}{100} + \frac{92}{100} \times 22500 \times \frac{40}{100}}$

$$= \frac{8505 + 12420}{3645 + 8280} = \frac{20925}{11925} = \frac{93}{53}$$

3. What is the difference between number of male from village A who have voted and female from village D who have voted?

(a) 6020 (b) 4200 (c) 6300 (d) 5585 (e) 6915

Sol. (e); Required difference = $\left(\frac{88}{100} \times 20000 \times \frac{60}{100} - \frac{90}{100} \times 13500 \times \frac{30}{100}\right)$

$$= 10560 - 3645 = 6915$$

4. If ratio of male to female who do not vote is 5 : 3 from village C then what is the ratio of total male voters and total female voters from village C ?

(a) 113 : 87 (b) 96 : 53 (c) 83 : 43 (d) 104 : 101 (e) 117 : 97

Sol. (a); Total males who voted = $\frac{80}{100} \times 16000 \times \frac{55}{100} = 7040$

$$\text{Total females who voted} = \frac{80}{100} \times 16000 \times \frac{45}{100} = 5760$$

$$\text{Total male voters} = 7040 + \frac{20}{100} \times 16000 \times \frac{5}{8} = 9040$$

$$\text{Total female voters} = 5760 + \left(\frac{20}{100} \times 16000 \times \frac{3}{8}\right) = 6960$$

$$\text{Required ratio} = \frac{9040}{6960} = \frac{113}{87}$$

5. What is the average number of voters who have not voted from village A, C and D together?

(a) $1526\frac{1}{3}$ (b) $1820\frac{2}{3}$ (c) $2316\frac{2}{3}$ (d) $1015\frac{3}{5}$ (e) $2112\frac{2}{3}$

Sol. (c); Required average = $\frac{12 \times 200 + 20 \times 160 + 10 \times 135}{3}$

$$= \frac{2400 + 3200 + 1350}{3} = 2316\frac{2}{3}$$

Directions (6-10): Study the table carefully to answer the questions that follow:

Total number of students = 32500

The percentage distribution of students with respect to total number of students studying in various universities and within each university percentage of boys and girls is given.

Universities	Percentage of students	Percentage of boys	Percentage of girls
A	12	55	45
B	15	60	40
C	8	30	70
D	28	75	25
E	17	20	80
F	20	64	36

6. What is the ratio of total number of boys in University B and D together to the total number of girls in the same universities together?

(a) 25 : 19 (b) 35 : 21 (c) 20 : 7 (d) 30 : 13 (e) 5 : 3

Sol. (d); Total number of Boys in B = $\frac{15}{100} \times 32500 \times \frac{60}{100} = 2925$

Total number of Boys in D = $\frac{28}{100} \times 32500 \times \frac{75}{100} = 6825$

Total number of girls in B = $\frac{15}{100} \times 32500 \times \frac{40}{100} = 1950$

Total number of girls in D = $\frac{28}{100} \times 32500 \times \frac{25}{100} = 2275$

Req. Ratio = $(2925 + 6825) : (1950 + 2275) = 9750 : 4225 = 30 : 13$

7. The total number of students in the University A is what per cent of the total number of students in University F?

(a) 55% (b) 67% (c) 60% (d) 48% (e) 58%

Sol. (c); Req. % = $\frac{12 \times 32500}{20 \times 32500} \times 100 = 60\%$

8. What is the total number of boys from university A, C and E together?

(a) 6030 (b) 5030 (c) 7030 (d) 4030 (e) 3030

Sol. (d); Total number of boys from university A, C and E together

= $\frac{32500}{100 \times 100} [12 \times 55 + 8 \times 30 + 17 \times 20] = 4030$

9. The girls in the University B are what per cent of the boys in the University F?

(a) 39.88% (b) 46.875% (c) 49.23% (d) 42.63% (e) 51.02%

Sol. (b); Req. % = $\frac{15 \times 40}{20 \times 64} \times 100 = 46.875$

10. What is the ratio of the number of boys in the University C to the number of boys in the University D?

(a) 9 : 39 (b) 3 : 8 (c) 17 : 35 (d) 13 : 32 (e) 4 : 35

Sol. (e); Ratio = $\frac{8 \times 30}{28 \times 75} = 4 : 35$

Directions (11-15): Study the table and answer the given questions.

Number of bangles sold by 6 stores during 5 months

Month	Stores				
	P	Q	R	S	T
May	154	129	87	89	165
June	121	120	87	89	172
July	145	88	105	133	104
August	169	102	130	114	129
September	128	177	94	220	131

11. Out of the total number of bangles sold by store R in June, July and August together, 35% were made of gold. What was the total number of gold bangles sold by store R in June, July and August together? (approx. value)

(a) 127 (b) 139 (c) 121 (d) 145 (e) 113

Sol. (e); Total number of gold bangles sold by store R in June, July and August together = 35% of $(87 + 105 + 130)$

= 35% of 322 = $\frac{35}{100} \times 322 = 112.7$

≈ 113

12. What is the difference between the total number of bangles sold by store S in June and July together and the total number of bangles sold by store P in the same months together?

(a) 72 (b) 58 (c) 44 (d) 64 (e) 62

Sol. (c); Req'd difference = $(121 + 145) - (89 + 133) = 266 - 222 = 44$

13. What is the average number of bangles sold by stores Q, R and T in May?

(a) 131 (b) 117 (c) 127 (d) 135 (e) 123

Sol. (c); Req'd average = $\frac{129+87+165}{3} = \frac{381}{3} = 127$

14. What is the ratio of the total number of bangles sold by stores S and T together in August to the total number of bangles sold by the same stores together in September?

(a) 9 : 13 (b) 7 : 9 (c) 11 : 13 (d) 9 : 11 (e) 7 : 11

Sol. (a); Req'd ratio = $\frac{(S+T)_{\text{August}}}{(S+T)_{\text{September}}} = \frac{114+129}{220+131} = \frac{243}{351} = \frac{9}{13} = 9 : 13$

15. The number of bangles sold by store Q increased by what per cent from June to September?

(a) 42.5 (b) 45 (c) 40 (d) 47.5 (e) 44.5

Sol. (d); Req'd% increase = $\frac{177-120}{120} \times 100 = \frac{57}{120} \times 100 = \frac{57 \times 5}{6} = 47.5\%$

Directions (16-20): The table given below shows the profit earned by six friends in five different years. Study the table carefully to answer the question that follow.

Friends	2011	2012	2013	2014	2015
Rajat	8925	9310	7250	8200	6050
Sunny	9100	8172	7520	9100	8000
Harshit	6550	8500	6880	8000	10,200
Ritu	9170	7550	7250	5010	6520
Neetu	10520	7000	6580	6810	8050
Rajiv	6150	9005	8172	9015	8670

16. Harshit and Neetu earned their profit in 2012 by investing their money in a business for 8 months and 10 months respectively. Find the amount invested by Neetu if total profit was distributed in capital ratio and investment of Harshit was Rs. 20,400.

(a) Rs. 13,340 (b) Rs. 13,440 (c) Rs. 13,000 (d) Rs. 14,340 (e) Rs. 14,480

Sol. (b); Ratio of profit of Harshit & Neetu = $8500 : 7000 = 17 : 14$

Let Neetu made an investment of x

$$\therefore \frac{20400 \times 8}{10x} = \frac{17}{14} \quad \text{Or, } x = \text{Rs. } 13,440$$

17. Find the amount which when lend on C.I. at 20% interest being compounded annually for 3 years, gives total interest equal to profit earned by Sunny in 2014.

(a) Rs. 12550 (b) Rs. 15200 (c) Rs. 12500 (d) Rs. 12700 (e) Rs. 13000

Sol. (c); Profit earned by Sunny in 2014 = Rs. 9100

$$\therefore 9100 + P = P \left(1 + \frac{20}{100}\right)^3 \Rightarrow 9100 + P = 1.728P$$

or, $P = \text{Rs. } 12500$

18. Find the ratio of average profit earned by Rajat in 2012 and 2013 together to average profit earned by Ritu in 2011, 2014 and 2015 together.

(a) 5 : 6 (b) 6 : 7 (c) 7 : 5 (d) 6 : 5 (e) 7 : 6

Sol. (d); Required ratio = $\frac{\frac{1}{2} \times (9310 + 7250)}{\frac{1}{3} \times (9170 + 5010 + 6520)} = \frac{8280}{6900} = \frac{6}{5}$

19. Maximum profit of Rajiv is approximately what percent more than the 2nd minimum profit of Neetu in the given years?

(a) 30% (b) 55% (c) 42% (d) 28% (e) 32%

Sol. (e); Required percentage = $\frac{9015 - 6810}{6810} \times 100 \approx 32\%$

20. In which year Rajat earned a profit 5% more than $\frac{5}{6}$ th of Harshit's profit in year 2015?

(a) 2014 (b) 2011 (c) 2015 (d) 2013 (e) 2012

Sol. (b); $\frac{5}{6}$ th of Harshit's profit in 2015 = 8500

$$\therefore \frac{105}{100} \text{ of } 8500 = 8925 \text{ which is equal to Rajat's profit in year 2011.}$$

Directions (21-25): The following table shows the monthly income and various expenditures of six friends in absolute value or in percentage (in terms of total income). Some values are missing which you are expected to calculate if required.

Friends	Salary (in Rs.)	Incentive (in Rs.)	Expenditure (in Rs.) on			
			Travel	Parties	Accommodation	Marketing
Babu	46000	-	-	5480	10%	15%
Gaurav	-	7200	7640	8500	6200	-
Arunoday	-	6300	12%	8%	-	12%
Mohit	44000	-	-	7560	9%	8400
Kamal	40000	-	5%	-	4200	5620
Mohan	-	5700	4200	8%	-	6860

Note:

1. Incentive amounts to 15% of salary and all friends save 40% of their total income (salary + incentive)
2. there is no expenditure other than the given expenditures.

Solutions (21-25):

Friends	Salary (in Rs.)	Incentive (in Rs.)	Expenditure (in Rs.)				Saving (in Rs.)
			Travel	Parties	Accom.	Marketing	
Babu	46000	6900	13035	5480	5290	7935	21160
Gaurav	48000	7200	7640	8500	6200	10780	22080
Arunoday	42000	6300	5796	3864	13524	5796	19320
Mohit	44000	6600	9846	7560	4554	8400	20240
Kamal	40000	6000	2300	15480	4200	5620	18400
Mohan	38000	5700	4200	3496	11664	6860	17480

21. Find the total amount (in Rs) expended by all friends together on travelling?

- (a) 42817 (b) 42871 (c) 41817 (d) 41781 (e) None of these

Sol. (a); Expenditure on travelling = $13035 + 7640 + 5796 + 9846 + 2300 + 4200 = 42817$ Rs

22. Find the difference in the amount spent by Gaurav on parties and Marketing together and that of Arunoday on Accommodation?

- (a) 5656 (b) 5776 (c) 5756 (d) 5576 (e) None of these

Sol. (c); Required difference = $8500 + 10780 - 13524 = \text{Rs } 5756$

23. What amount is saved by all friends together?

- (a) Rs 126880 (b) Rs 118680 (c) Rs 118860 (d) Rs 181680 (e) None of these

Sol. (b); Total saving = $21160 + 22080 + 19320 + 20240 + 18400 + 17480 = \text{Rs } 118680$

24. Total annual income of Mohit is by what amount less than that of Babu?

- (a) Rs 2300 (b) Rs 23600 (c) Rs 27600 (d) Rs 2700 (e) None of these

Sol. (c); Required difference = $(52900 - 50600) \times 12 = \text{Rs } 27600$

25. Expenditure by Babu on Travelling constitutes what percent (Approx.) of salary of Mohan?

- (a) 30% (b) 38% (c) 32% (d) 34% (e) 40%

Sol. (d); Required percentage = $\frac{13035}{38000} \times 100 \approx 34\%$

Directions (26-30): Study the table and answer the given questions.

Data related to Human Resource Dept. of a multinational company (X) which has 145 offices across 8 countries in year 2016.

Countries	Offices	Total Employees	Respective Ratio of male & female employees	% of post graduate
A	16	2568	5 : 7	75
B	18	2880	11 : 5	65
C	14	2310	10 : 11	40
D	22	3575	3 : 2	60
E	13	2054	7 : 6	50
F	17	2788	20 : 21	75
G	24	3720	8 : 7	55
H	21	3360	8 : 6	80

26. The number of male post graduate employees in country H is 1800. If number of female post graduates increase by 50% in the next year, what percent of female employees in that particular country are post graduate in year 2017? (Given that total female remain same in next year)

(a) 76.8% (b) 74% (c) 92.5% (d) 90% (e) 80%

Sol. (c); Number of post-Graduate in country H = $3360 \times \frac{4}{5} = 2688$

$$\text{Female post-graduate} = 2688 - 1800 = 888$$

$$\text{Total Female employee} = 3360 \times \frac{6}{14} = 1440$$

$$\text{Female graduate next year} = 888 \times \frac{3}{2} = 1332$$

$$\% \text{ of female graduate} = \frac{1332}{1440} \times 100 = 92.5\%$$

27. What is the average of post graduate employees of country A, B and H together ?

(a) 2262 (b) 2153 (c) 2162 (d) 2272 (e) 2252

Sol. (c); Required average = $\frac{2568 \times 0.75 + 2880 \times 0.65 + 3360 \times 0.80}{3} = \frac{6486}{3} = 2162$

28. What is the ratio between total number of male employees in countries B and H together and total number of post graduate employees in same countries?

(a) 76 : 65 (b) 86:85 (c) 75 : 76 (d) 65 : 76 (e) 12 : 33

Sol. (d); $\frac{\frac{11}{16} \times 2880 + \frac{8}{14} \times 3360}{2880 \times \frac{65}{100} + 3360 \times \frac{4}{5}} = \frac{1980 + 1920}{1872 + 2688} = \frac{3900}{4560} = 65 : 76$

29. What is the difference between average number of post graduate employees in countries A, B and D together and average number of post graduate employees in countries F, G and H together ?

(a) 294 (b) 282 (c) 284 (d) 280 (e) 200

Sol. (a); A $\Rightarrow 2568 \times \frac{3}{4} = 1926$

$$F \Rightarrow 2788 \times \frac{3}{4} = 2091$$

$$B \Rightarrow 2880 \times \frac{65}{100} = 1872$$

$$G \Rightarrow 3720 \times \frac{55}{100} = 2046$$

$$D \Rightarrow 3575 \times \frac{3}{5} = 2145$$

$$H \Rightarrow 3360 \times \frac{4}{5} = 2688$$

$$\text{Total post-graduate in A, B and D} = 5943$$

$$\text{Total post-graduate in F, G and H} = 6825$$

$$\text{Difference} = 6825 - 5943 = 882$$

$$\text{Required average} = \frac{882}{3} = 294$$

30. If the number of male employee which are not graduate in country C is 786. Then non-graduate male employees is what percent of non-graduate female employees ?

(a) 131% (b) 125% (c) 143% (d) 113% (e) 137%

Sol. (a); Non-graduate employees in country C = $2310 \times \frac{60}{100} = 1386$

$$\therefore \text{Non-graduate female employees in country C} = 1386 - 786 = 600$$

$$\therefore \text{percentage required} = \frac{786}{600} \times 100 = 131\%$$

Direction (31-35): Refer the following table and answer the questions based on it.

Total number of books and ratio of Indian and Foreign author books out of total books in six public libraries in a city in year 2016:

Libraries	Total No. of Books	Indian : Foreign
A	56250	7 : 8
B	48750	4 : 9
C	49500	11 : 7
D	31500	9 : 5
E	38250	4 : 13
F	52250	10 : 9

31. What is the average number of Indian author books in library A, C and F together?

(a) 28400 (b) 21000 (c) 26300 (d) 28000 (e) 24000

Sol. (d); Number of Indian author books in given libraries = $26250 + 30250 + 27500 = 84000$

$$\text{Required Average} = \frac{84000}{3} = 28000$$

32. What will be the difference between number of Indian author books in libraries A and B in year 2017, if the total number of books in library A is decreased by 20% and that in library B is increased by 10% from the previous year while the ratio of Indian and foreign author books in both the libraries remains the same?

(a) 4000 (b) 4500 (c) 3500 (d) 3000 (e) 2500

Sol. (b); Number of Indian author books in library A in 2017 = $26250 - 20\% \text{ of } 26250 = 21000$
 Number of Indian author books in library B in 2017 = $15000 + 10\% \text{ of } 15000 = 16500$
 Required difference = $21000 - 16500 = 4500$

33. The number of foreign author books in library C is how much percent less than the number of Indian author books in library F?

(a) 40% (b) 35% (c) 30% (d) 25% (e) 20%

Sol. (c); Number of foreign author books in library C = 19250
 Number of Indian author books in library F = 27500
 Required percentage = $\frac{27500 - 19250}{27500} \times 100 = 30\%$

34. What is the ratio of number of Indian author book in libraries C and D together to the number of foreign author books in libraries E and F together?

(a) 101 : 108 (b) 108 : 101 (c) 103 : 105 (d) 105 : 103 (e) 101 : 109

Sol. (a); Number of Indian author book in libraries C and D = $30250 + 20250 = 50500$
 Number of foreign author books in libraries E and F = $29250 + 24750 = 54000$
 Required Ratio = $\frac{50500}{54000} = \frac{101}{108} = 101 : 108$

35. For which library, the difference between Indian and foreign author books is the maximum.

(a) B (b) E (c) B and E (d) C and E (e) C

Sol. (b); Difference between number of Indian and foreign author books:

For Library A = 3750

For Library B = 18750

For Library C = 11000

For Library D = 9000

For Library E = 20250

For Library F = 2750

Hence, maximum difference is for the Library E.

Directions (36-40): The table given below shows production of five types of Cars (in thousands) by a company in years 2011 to 2015. Study the table of answer the given question.

<u>Cars</u> <u>Years</u>	A	B	C	D	E	Total
2011	20	15	25	12	18	90
2012	18	20	10	12	20	80
2013	—	10	—	20	—	75
2014	12	23	15	18	10	78
2015	22	18	20	25	15	100

Note: Few values are missing in table, candidate is expected to calculate the missing values if it is required to answer the given questions.

36. If Car A and Car D manufactured in 2012 increases by $11\frac{1}{9}\%$ and $8\frac{1}{3}\%$ respectively. Then what will be percentage change in total car manufactured in 2012.

(a) 2.75% (b) 3.75% (c) 4.75% (d) 3% (e) 3.25%

Sol. (b); Increase in production of Car A = $18,000 \times \frac{1}{9} = 2,000$
 Increase in production of Car D = $12,000 \times \frac{1}{12} = 1,000$
 Total increase = $2,000 + 1,000 = 3,000$ cars
 Desired percentage = $\frac{3,000}{80,000} \times 100 = 3.75\%$

37. What will be the average of car A, B and C manufactured in 2013, If the ratio of number cars A, C and E manufactured in 2013 is 2 : 3 : 4 respectively.

(a) 12,666.67 (b) 14,167.67 (c) 12,167.67 (d) 11,666.67 (e) 16,167.67

Sol. (d); Number of cars A, C and E manufactured in 2013 = $75,000 - 20,000 - 10,000 = 45,000$

$$A : C : E = 2x : 3x : 4x$$

$$9x = 45,000$$

$$x = 5,000$$

Number of cars manufactured in 2013

$$A = 2 \times 5,000 = 10,000$$

$$C = 3 \times 5,000 = 15,000$$

$$B = 10,000$$

$$\text{Average} = \frac{10,000 + 15,000 + 10,000}{3} = 11,666.67$$

38. Car A and B manufactured together in 2011 is how much % less or more than Cars A and B manufactured together in 2015.

(a) 12.5% less (b) 12.5% more (c) 14.5% less (d) 10% less (e) 12% more

- Sol. (a); Car produced by (A + B) in 2011 = 20,000 + 15,000 = 35,000

Car produced by (A + B) in 2015 = 22,000 + 18,000 = 40,000

$$\text{percentage change} = \frac{40,000 - 35,000}{40,000} \times 100$$

$$= \frac{5,000}{40,000} \times 100 = 12.5\% \text{ less}$$

39. Among 5 cars, which car is the 2nd highest manufactured car in the duration of five year If the ratio of number cars A, C and E manufactured in 2013 is 4 : 3 : 2 ?

(a) C (b) A (c) B (d) E (e) D

- Sol. (e); Number of cars A, C and E manufactured in 2013 = 75,000 - 30,000 = 45,000

$$A : C : E = 4x : 3x : 2x$$

$$\text{Total} = 9x = 45,000$$

$$x = 5,000$$

$$A = 4 \times 5,000 = 20,000$$

$$C = 3 \times 5,000 = 15,000$$

$$E = 2 \times 5,000 = 10,000$$

$$\text{Total Cars manufacture A} = 92,000$$

$$\text{Total Cars manufacture B} = 86,000$$

$$\text{Total Cars manufacture C} = 85,000$$

$$\text{Total Cars manufacture D} = 87,000$$

$$\text{Total Cars manufacture E} = 73,000$$

$$2^{\text{nd}} \text{ highest D} = 87,000$$

40. Total cars manufactured in 2013 is how much percent more than the number of cars B and D manufactured together in 2013.

(a) 160% (b) 120% (c) 150% (d) 130% (e) 140 %

- Sol. (c); Total cars in 2013 = 75,000

Total cars B & D in 2013 = 30,000

$$\text{percentage change} = \frac{75,000 - 30,000}{30,000} \times 100 = 150\%$$

Directions (41-45): The following table shows the percentage of marks obtained by five students in six different subjects. Answer the following questions based on this table.

Students	Maths (200)	Biology (75)	Hindi (100)	English (100)	Physics (75)	Chemistry (50)
Ajay	67%	42%	49%	72%	84%	32%
Kavita	58%	84%	77%	79%	68%	66%
Neeraj	63%	78%	56%	89%	62%	74%
Rohit	72%	60%	46%	92%	72%	56%
Mamta	66%	54%	68%	63%	54%	44%

41. What is the half of average marks scored by all students in Physics?

(a) 25.5 (b) 26 (c) 26.5 (d) 28.5 (e) 30

- Sol. (a); Total Marks obtained by all students in Physics = $84 \times \frac{3}{4} + 68 \times \frac{3}{4} + 62 \times \frac{3}{4} + 72 \times \frac{3}{4} + 54 \times \frac{3}{4} = \frac{3}{4} [340] = 255$

$$\text{Average Marks} = \frac{255}{5} = 51$$

Required Answer = 25.5

42. Marks scored by Ajay in Maths and Hindi is what percent of maximum marks of Maths and Hindi together ?
 (a) 68% (b) 42% (c) 61% (d) 71% (e) 53%

Sol. (c); Required percentage = $\frac{67 \times 2 + 49}{200 + 100} \times 100 = \frac{183}{300} \times 100 = 61\%$

43. What is 25% of the difference between the total marks obtained by Neeraj in Maths & Biology together and that obtained by Rohit in Hindi and English together?

(a) 11.12 (b) 11.625 (c) 12.225 (d) 12 (e) 13

Sol. (b); Neeraj in (M + B) = 126 + 58.50 = 184.5

Rohit in (H + E) = 46 + 92 = 138

25% of Difference = $\frac{46.5}{4} = 11.625$

44. What is the ratio between maximum marks and overall marks obtained by Mamta and Kavita together?

(a) 115 : 157 (b) 120 : 115 (c) 120 : 155 (d) 120 : 157 (e) 157 : 120

Sol. (d); Maximum Marks = 600

Kavita's in Marks = 116 + 63 + 77 + 79 + 51 + 33 = 419

Mamata's Marks = 132 + 40.5 + 68 + 63 + 40.5 + 22 = 366

Required Ratio = $\frac{600}{366 + 419} = \frac{600}{785} = \frac{120}{157}$

45. Marks scored by Rohit in Chemistry is approximately how many times of the marks scored by him in Physics?

(a) 2 (b) 0.5 (c) 1.5 (d) 3 (e) 1

Sol. (b); $\frac{28}{54} = 0.5$ times (approximately)

Directions (46-50): The following table shows the total number of students appeared in an entrance exam from six different schools in different years, and the ratio of passed to failed students among them. Answer the given questions based on this table.

Note: Total appeared in any year = Total pass + Total fail.

School	2010		2011		2012	
	Total Appeared	Pass : Fail	Total Appeared	Pass : Fail	Total Appeared	Pass : Fail
A	646	11 : 8	754	7 : 6	672	3 : 5
B	847	4 : 7	845	8 : 5	952	9 : 8
C	810	8 : 7	792	7 : 4	637	4 : 3
D	876	7 : 5	828	11 : 7	988	7 : 12
E	870	3 : 2	726	7 : 4	715	8 : 5
F	986	17 : 12	867	12 : 5	924	8 : 13

46. What is the difference between total number of passed students from school D in the year 2010 and three fourth of failed students from school B in the year 2012.

(a) 165 (b) 176 (c) 175 (d) 180 (e) 111

Sol. (c); School D, Passed in 2010, $876 \times \frac{7}{12} = 511$

School B, year 2012, failed = $952 \times \frac{8}{17} = 448$

Required value = $511 - \frac{3}{4} \times 448 = 175$

47. What is the total number of failed students from school A and D in all three years together?

(a) 1036 (b) 1311 (c) 2351 (d) 2446 (e) 1600

Sol. (c); A : Failed = $646 \times \frac{8}{19} + 754 \times \frac{6}{13} + 672 \times \frac{5}{8} = 1040$

D : Failed = $876 \times \frac{5}{12} + 828 \times \frac{7}{18} + 988 \times \frac{12}{19} = 1311$

Total failed = 2351

48. What is the difference between the number of passed students from A, B and D together in 2011 and the number of failed students from A, C and F together in 2012?

(a) 167 (b) 177 (c) 217 (d) 157 (e) 180

Sol. (a); A : 2011 : Passed = $754 \times \frac{7}{13} = 406$

B : 2011 : Passed = $845 \times \frac{8}{13} = 520$

$$D : 2011 : \text{Passed} = 828 \times \frac{11}{18} = 506$$

Total pass = 1432

Failed in 2012

$$A = 672 \times \frac{5}{8} = 420$$

$$C = 637 \times \frac{3}{7} = 273$$

$$F = 924 \times \frac{13}{21} = 572$$

Total failed = 1265

$$\text{Required Difference} = 1432 - 1265 = 167$$

49. By what percent (approx) the failed students from school D are more or less than the passed students from E in all three years together ?

(a) 10% more (b) 15% less (c) 12% more (d) 5% less (e) 8% less

Sol. (e); Failed; D :

$$2010 : 876 \times \frac{5}{12} = 365$$

$$2011 : 828 \times \frac{7}{18} = 322$$

$$2012 : 988 \times \frac{12}{19} = 624$$

Total failed = 1311

Passed; E:

$$2010 : 870 \times \frac{3}{5} = 522$$

$$2011 : 726 \times \frac{7}{11} = 462$$

$$2012 : 715 \times \frac{8}{13} = 440$$

Total passed = 1424

$$\text{Required percentage} = \frac{113}{1424} \times 100 \approx 8\% \text{ less}$$

50. The total number of passed students from school E in 2010 is approximately what percent of the total number of failed students from school C in 2012?

(a) 191.2% (b) 190% (c) 188.4% (d) 178% (e) 185%

Sol. (a); Required % = $\frac{522}{273} \times 100 = 191.2\%$

Directions (51-55): Given below is the table showing the data related to sales of chips packets of different brands by a shopkeeper in 6 months. Some of the values are missing which you are expected to calculate, if necessary and answer the following questions.

Months→ Brands↓	July	August	September	October	November	December	Total
Lay's	—	6531	6690	1820	3334	4545	28540
Ruffles	5890	—	6050	2018	—	4382	24100
Pringles	4320	4690	—	1990	5835	7830	28700
Utz	6130	3500	3862	3965	3413	—	29850
Kettle	—	—	—	—	4931	7997	29082
Popchips	1928	3032	3164	3640	8882	—	25772
Total	29520	24800	27735	—	28875	38860	

51. Find the difference between total number of Lay's Chips sold in July and August together and that of Lay's and Ruffles together in November.

(a) 6737 (b) 6337 (c) 3667 (d) 6037 (e) None of these

Sol. (b); Total Lay's in July and August together = 28,540 - 6,690 - 1,820 - 3,334 - 4,545 = 12151

Lay's and Ruffles in November = 3334 + 2480 = 5814

∴ Required difference = 6,337

52. Find the total number of sold packets of Kettle chips in July, August and September together.

(a) 13133 (b) 23333 (c) 11333 (d) 13333 (e) None of these

Sol. (d); Kettle chips in July, August and September = 5632 + 3767 + 3934 = 13,333

53. Pringles Chips sold in September are what percent of Lay's chips sold in July (approximately)?
 (a) 72% (b) 75% (c) 65% (d) 70% (e) 80%

Sol. (a); Required percentage = $\frac{4035}{5620} \times 100 \approx 72\%$

54. Utz chips packets sold in August are by what percent more or less than kettle chips sold in September (approximately)?
 (a) 13% (b) 11% (c) 15% (d) 21% (e) 8%

Sol. (b); Required percentage = $\frac{3934-3500}{3934} \times 100 \approx 11\%$

55. What is the ratio between Pringles chips sold in July and Lay's chips sold in December?
 (a) 9 : 11 (b) 101 : 96 (c) 96 : 101 (d) 11 : 9 (e) None of these

Sol. (c); Required ratio = $\frac{4320}{4545} = \frac{96}{101}$

Directions (56-60): Study the following table carefully to answer the questions that follow.

The table given below provides the incomplete data related to monthly earning & expenditure of five friends. Find the missing value if required to answer the questions.

Income = Expenditure + Savings

Friends	Total Income (In Rs)	Ratio of Savings & Expenditure S : E	Expenditure(in Rs) on		
			Rent	Food	Others
Soha	-	5 : 6	20%	15%	7800
Ruchi	28000	5 : 9	-	16%	62%
Suchi	22000	- : -	2000	2200	58%
Meenu	26000	- : 6	15%	25%	7200
Teena	-	8 : 9	4000	4000	10000

56. Find the annual income of soha.
 (a) Rs 26,400 (b) Rs 2,65,000 (c) Rs 26,500 (d) Rs 2,64,000 (e) None of these

Sol. (d); Total expenditure of soha = $\frac{100}{65} \times 7800 = \text{Rs } 12000$
 $\therefore$ Total annual salary = $\frac{11}{6} \times 12000 \times 12 = \text{Rs } 2,64,000$

57. Find the difference in the monthly savings of Suchi and Meenu.
 (a) Rs 1200 (b) Rs 2200 (c) Rs 2000 (d) Rs 1800 (e) None of these

Sol. (c); Total expenditure of Suchi = $\frac{100}{42} \times 4200 = \text{Rs } 10,000$
 $\therefore$ Savings of Suchi = $22000 - 10000 = \text{Rs } 12,000$
 Total expenditure of Meenu = $\frac{100}{60} \times 7200 = \text{Rs } 12000$
 $\therefore$ Savings of Meenu = $26000 - 12000 = \text{Rs } 14000$
 So, required difference = Rs 2000

58. Expenditure made by Ruchi on rent is what percent of expenditure made by Meenu on food?
 (a) 32% (b) 132% (c) 88% (d) 120% (e) None of these

Sol. (b); Expenditure of Ruhi on rent = $\frac{22}{100} \times \frac{9}{14} \times 28000 = \text{Rs } 3960$
 Expenditure of Meenu on food = $\frac{25}{60} \times 7200 = \text{Rs } 3000$
 $\therefore$ Required percentage = $\frac{3960}{3000} \times 100 = 132\%$

59. The savings of Ruchi is what percent more or less than that of Teena?
 (a) 32% (b) 37% (c) 38% (d) 40% (e) None of these

Sol. (e); Savings of Ruhi = $\frac{5}{14} \times 28000 = \text{Rs } 10000$
 Savings of Teena = $\frac{8}{9} \times 18000 = \text{Rs } 16000$
 $\therefore$ Required percentage = $\frac{6000}{16000} \times 100 = 37.5\%$

60. Find the average of monthly income of Soha, Suchi and Teena.
 (a) Rs 26500 (b) RS 26000 (c) Rs 25600 (d) Rs 22500 (e) None of these

Sol. (b); Required average = $\frac{1}{3} (22000 + 22000 + 34000) = \text{Rs } 26000$

Directions (61 -65): The following table shows the percentage of population of six states below poverty line and the proportion of male and female population in below and above poverty line. Read the following questions and answer them carefully

State	Percentage population below poverty line (in %)	Proportion of male and female	
		Below poverty line M : F	Above poverty line M : F
Ahmedabad	12	3 : 2	4 : 3
Bangalore	15	5 : 7	3 : 4
Chennai	25	4 : 5	2 : 3
Delhi	26	1 : 2	5 : 6
Hyderabad	10	6 : 5	3 : 2
Kolkata	32	2 : 3	4 : 5

61. The number of male in Ahmedabad below poverty line is what percent of the no. of female above poverty line in same state? (approximately)

- (a) 18% (b) 21% (c) 19% (d) 25% (e) 29%

Sol. (c); Let Population = x

According to question,

$$\frac{\frac{12x}{100} \times \frac{3}{5}}{\frac{88x}{100} \times \frac{3}{7}} \times 100 \approx 19\%$$

62. In Delhi, if 780000 people live below poverty line, then what is the ratio of difference between Male and female below poverty line and difference between Male and female above poverty line in same state?

- (a) 143 : 111 (b) 43 : 11 (c) 243 : 222 (d) 43 : 22 (e) None of these

Sol. (a); Given 26% → 780000

$$\therefore \text{Required ratio} = \frac{780000 \times \frac{(2-1)}{3}}{\frac{780000}{26} \times 74 \times \frac{(6-5)}{11}} = \frac{143}{111}$$

63. The number of People above poverty line in Chennai is approximately what percent more or less than the no. of female in Delhi if the no of people in Chennai is 85% of the population in Delhi which is 1,00,0000 ? (approximately)

- (a) 17% (b) 10% (c) 15% (d) 7% (e) 22%

Sol. (b); Population in Chennai = 850000

$$\text{Above poverty line in Chennai} = \frac{3}{4} \times 8,50,000 = 6,37,500$$

$$\text{Population of Delhi} = 10,00,000$$

$$\text{Females} = 2,60,000 \times \frac{2}{3} + 7,40,000 \times \frac{6}{11} \\ \approx 1,73,333 + 403636 \approx 576969 \approx 577000$$

$$\text{Desired\%} = \frac{60500}{577000} \times 100 \approx 10\%$$

64. If in a certain year, the total population of all metros is 8,00,000, then what is the ratio of Male below Poverty line to the Female above Poverty line in all Metro cities ?

- (a) 2 : 5 (b) 5 : 2 (c) 7 : 5 (d) Cannot determined (e) None of these

Sol. (d); Since, individual population of metros is not given we cannot determine the required value.

65. People below poverty line in Kolkata is what percent of Males in Kolkata? (approximately)

- (a) 80% (b) 85% (c) 74% (d) 90% (e) 65%

Sol. (c); Let total population in kolkata be x

$$\text{Given population below poverty line} = 32\%x$$

$$\text{Males in Kolkata} = 32\% \times \frac{2}{5}x + 68\% \times \frac{4}{9}x$$

$$= 12.8\%x + 30.2\%x = 43\%x$$

$$\therefore \text{required percentage} = \frac{32}{43} \times 100$$

$$\approx 74\%$$

Directions (66-70): Read the following table and solve the following questions:

In the following table number of student appeared and percentage of students passed in the given exam (P & Q) in different years.

Years ↓	P		Q	
	Total No. of students appeared	% of students passed	Total No. of students appeared	% of students passed
2011	350	40%	250	32%
2012	250	26%	320	45%
2013	240	30%	280	30%
2014	400	25%	300	25%
2015	320	60%	420	20%

66. What is the average number of students passed in exam 'P' in year 2011, 2013 and 2014.

- (a) 100 (b) 104 (c) 108 (d) 112 (e) 116

Sol. (b); Desired average = $\frac{350 \times \frac{40}{100} + 240 \times \frac{30}{100} + 400 \times \frac{25}{100}}{3} = \frac{140 + 72 + 100}{3} = \frac{312}{3} = 104$

67. What is the ratio of total number of students passed in 2013 in both the exams to total number of students passed in 2014 in both the exam?

- (a) 172 : 153 (b) 153 : 172 (c) 175 : 153 (d) 175 : 156 (e) 156 : 175

Sol. (e); Total no. of students passed in P & Q together in 2013 = $240 \times \frac{30}{100} + 280 \times \frac{30}{100}$
 $= (240 + 280) \times \frac{30}{100} = 52 \times 3 = 156$
 Total no. of students passed in P & Q together in 2014 = $400 \times \frac{25}{100} + 300 \times \frac{25}{100}$
 $= (400 + 300) \times \frac{25}{100} = 700 \times \frac{25}{100} = 175$
 Desired Ratio = $\frac{156}{175}$

68. Find the difference between the number of students passed in exam 'P' in 2011 and 2012 together to number of students passed in exam 'Q' in 2014 and 2015.

- (a) 40 (b) 42 (c) 44 (d) 46 (e) 48

Sol. (d); Total no. of students passed in exam P in 2011 and 2012 = $350 \times \frac{40}{100} + 250 \times \frac{26}{100}$
 $= 140 + 65 = 205$
 Total no. of students passed in exam Q in 2014 and 2015 = $300 \times \frac{25}{100} + 420 \times \frac{20}{100}$
 $= 75 + 84 = 159$
 Desired difference = $205 - 159 = 46$

69. What is the average number of students appeared in exam 'Q'?

- (a) 305 (b) 310 (c) 318 (d) 325 (e) 314

Sol. (e); Desired Average = $\frac{250 + 320 + 280 + 300 + 420}{5} = \frac{1570}{5} = 314$

70. Total number of students appeared in exam 'P' in 2012 and 2013 together is how much percentage more than the number of students appeared in 2014 in exam 'P'?

- (a) 20% (b) 22.5% (c) 25% (d) 27.5% (e) 30%

Sol. (b); Total no. of students appeared in exam P in 2012 and 2013 together = $250 + 240 = 490$
 Total no. of students appeared in exam P in 2014 = 400
 Desired % = $\frac{490 - 400}{400} \times 100 = \frac{90}{400} \times 100 = 22.5\%$

Directions (71 - 75): Refer to the table given below and answer the given questions.

Data related to the number of employees in 5 different companies in December 2008

Company	Total number of employees	Out of total number of employees		
		Percentage of Arts graduates	Percentage of Science graduates	Percentage of commerce graduates
X	-	30%	30%	-
Y	-	-	40%	20%
Z	-	35%	50%	-
K	1000	32%	-	-
L	600	-	42%	30%

Note: Some values are missing, you have find out these value according to the question.

Note: Suppose that all the employees are graduated.

71. What is the difference between the number of commerce graduates employees and Arts graduates employees in company L?
 (a) 12 (b) 18 (c) 10 (d) 22 (e) 15
- Sol.** (a); Number of commerce graduates employees = $30\% \text{ of } 600 = \frac{30}{100} \times 600 = 180$
 Number of arts graduates employees = $28\% \text{ of } 600 = 168$
 $\therefore$ Difference = $180 - 168 = 12$
72. The average number of commerce graduates employees and science graduate employees in company Z was 338. What was the total number of employees in company Z?
 (a) 1020 (b) 1140 (c) 1040 (d) 1240 (e) 940
- Sol.** (c); Average number of commerce graduate employees and science graduate employees in company Z = 338
 Total number of commerce and science graduate employees in company Z = 676
 Total number of employees in Z = $676 \times \frac{100}{65} = 1040$
73. If the respective ratio between the number of science graduate and commerce graduate employees in company K was 10 : 7. What was the number of commerce graduate employees in K?
 (a) 180 (b) 280 (c) 380 (d) 80 (e) 250
- Sol.** (b); Number of Arts graduate employees = $\frac{32}{100} \times 1000 = 320$
 Number of science graduate and commerce graduate employees = $1000 - 320 = 680$
 $\therefore$ Number of commerce graduate employees in K = $680 \times \frac{7}{17} = 280$
74. Total number of employees in company L increased by 20% from December 2008 to December 2009. If 20% of the total number of employees in company L in December 2009 was Arts graduate, what was the number of Arts graduate employees in company L in December 2009?
 (a) 144 (b) 169 (c) 244 (d) 104 (e) 124
- Sol.** (a); Total employees in company L in 2009 = $600 \times \frac{120}{100} = 720$
 Arts Graduate in company L in December 2009 = $\frac{20}{100} \times 720 = 144$
75. Total number of employees in company X was three time the total number of employees in company Y. If the difference between number of commerce graduate employees in company Y and that of science graduate employees in same company was 120, what was the total number of employees in company X?
 (a) 600d (b) 1200 (c) 1800 (d) 3000 (e) 2400
- Sol.** (c); $(40\% - 20\%) \text{ of number of employees in company Y} = 120$
 $\therefore$ Number of employees in company Y = 600
 $\therefore$ Total number of employees in company X = 1800

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PREVIOUS YEAR QUESTIONS

Directions (1-5): Study the table and answer the given questions.

Data related to candidates appeared and qualified from State 'x' in a competitive exam during 5 years

Note: Total appeared candidates = Total qualified candidates + Total Unqualified candidates

Years	No. of appeared candidates	% of appeared candidates who unqualified	Respective ratio of number of qualified male and number of qualified female candidates
2006	700	--	3 : 2
2007	--	50%	2 : 3
2008	480	40%	--
2009	--	70%	4 : 5
2010	900	36%	--

Note: Some values are missing. You have to calculate these value as per data given in the questions.

- In 2007, number of males who qualified were 50% of the number of males who qualified in 2010. Find total number of appeared candidates in 2007 if respective ratio of number of qualified male and female candidates is 13 : 11 in 2010 .
(a) 760 (b) 728 (c) 720 (d) 740 (e) 780
- If the ratio between number of qualified male in 2007 and the number of qualified male in 2009 is 4 : 3 and total number of male qualified in 2007 and 2009 together are 392 then number of candidates appeared in 2007 are what percent of the number of candidates appeared in 2009 ?
(a) $89\frac{1}{3}\%$ (b) $84\frac{3}{4}\%$ (c) $88\frac{8}{9}\%$ (d) $87\frac{3}{7}\%$ (e) $88\frac{7}{9}\%$
- If the ratio between number of qualified female in 2008 and number of qualified female in 2010 is 3 : 7 and the number of qualified male in 2008 is same as number of qualified male in 2010. Then find the total number of qualified female candidates in 2008 and 2010 together?
(a) 756 (b) 688 (c) 725 (d) 720 (e) 696
- If in 2011 Number of appeared candidates are 125% of the number of qualified candidates in 2008 then find the number of unqualified females in 2011 if the ratio of number of unqualified male and number of unqualified female candidates is 7 : 3 . Number of appeared candidates who qualified in 2011 are 376 less than the number of appeared candidates who qualified in 2010
(a) 48 (b) 36 (c) 56 (d) 72 (e) 64
- Number of qualified candidates in 2008 are what percent more/less than the number of qualified candidates in 2010?
(a) 60% (b) 55% (c) 25% (d) 40% (e) 50%

Directions(6-10): Given below is the table which shows the number of appeared and percentage of appeared candidates who qualify the examination from two given states A and B in different years.

Year	State A		State B	
	Number of appeared candidates	% of appeared candidates who qualified	Number of appeared candidates	% of appeared candidates who qualified
2010	900	60%	760	30%
2011	1200	43%	—	40%
2012	—	60%	520	60%
2013	960	70%	400	70%
2014	760	—	660	—

Note: Some values are missing in table. You have to calculate these value if required to answer these question.

- Out of number of qualified candidates from state A in 2012 the ratio of male to female candidates is 7 : 5 and difference between qualified male and qualified female from state A in 2012 is 102 then what is the total number of candidates who appeared from state A in 2012.
(a) 900 (b) 850 (c) 770 (d) 880 (e) 1020
- If number of appeared candidates from state B in 2011 is $33\frac{1}{3}\%$ more than appeared candidates from state B in 2014 and ratio of passed candidates from same state and same years i.e. 2011 and 2014 is 11 : 12 then what is the sum of total passed candidates from same state and same years.
(a) 545 (b) 660 (c) 736 (d) 884 (e) 568
- What is the ratio of candidates passed from state A in 2010, 2011 and 2013 together to the ratio of candidates passed from state B in 2010, 2012 and 2013 together.
(a) 432 : 331 (b) 423 : 205 (c) 432 : 205 (d) 200 : 343 (e) 254 : 255

9. Number of candidates qualified from state A in year 2010 is what percent more or less than number of candidates qualified from state B in year 2013
 (a) $70\frac{2}{3}\%$ (b) $66\frac{2}{3}\%$ (c) $92\frac{6}{7}\%$ (d) $88\frac{3}{5}\%$ (e) $88\frac{1}{3}\%$
10. If from state A sum of candidates who qualified in 2013 and 2014 is 1356 then what percent of candidates remain unqualified from state A in 2014
 (a) 10% (b) 5% (c) 8% (d) 15% (e) 12%

Directions (11-15): Study the following table carefully and answer the following question.

The table represents the cost of production and profit percentages of sports utility companies Adidas and Nike over the years from 2001 to 2005

Year	Adidas		Nike	
	Cost of production (Rs. in lakh)	Profit%	Cost of production (Rs. in lakh)	Profit%
2001	320	40%	—	40%
2002	—	30%	—	25%
2003	420	20%	440	35%
2004	460	45%	470	20%
2005	510	30%	580	—

Note: A few values are missing. It is expected that the candidate should calculate the missing values, if it is required to find answer for the questions given below.

Profit = Sales – cost of production

$$\text{Profit \%} = \left(\frac{\text{Profit}}{\text{Cost of production}} \right) \times 100 \%$$

11. If total sale of Adidas and Nike together in 2001 is 798 lakh, then find the cost of production of Nike ?
 (a) 250 lakh (b) 300 lakh (c) 200 lakh (d) 300 lakh (e) 225 lakh
12. If total sales of Adidas and Nike together in 2002, is 1150 lakh, and cost of production of Adidas to that of Nike is 5 : 4, then find the difference in the cost of production of the two companies in 2002 ?
 (a) 300 lakh (b) 100 lakh (c) 400 lakh (d) 200 lakh (e) 150 lakh
13. The average sales of Adidas in years 2004 and 2005 together is approximately what percent of the average sales of Nike in years 2003 and 2004 together?
 (a) 120% (b) 104% (c) 114% (d) 121% (e) 108%
14. If total sales of Adidas and Nike together in 2005 is 1330 lakh, then find the profit% of Nike in 2005 ?
 (a) 18% (b) 21% (c) 12% (d) 15% (e) 20%
15. What is the ratio of sales of Adidas in 2003 to that of Nike in 2004 ?
 (a) 41 : 43 (b) 42 : 47 (c) 39 : 47 (d) 43 : 41 (e) 43 : 47

Directions (16-20): Given below is the table which shows the total students in 5 classes from which some participate in two cultural activities i.e. Quiz and Painting. It also shows the students who do not participate in cultural activities and ratio of students who participate in Quiz and painting.

Note : Students participate only in these two activities.

Class	Total students	Students who do not participate	Ratio of students in Quiz and Painting
A	420	119	4 : 3
B	330	88	7 : 4
C	240	110	8 : 5
D	125	45	2 : 3
E	390	130	8 : 5

16. What is the total students who participate in Quiz from class B, C and E together
 (a) 370 (b) 394 (c) 268 (d) 352 (e) 270
17. What is the ratio of students who participate in Painting from class B and C together to the students who participate in Quiz from class A and D together.
 (a) 19 : 35 (b) 20 : 37 (c) 23 : 34 (d) 22 : 35 (e) 20 : 33
18. What is the average number of students who participate in Quiz from class A, B and C together
 (a) $117\frac{1}{3}$ (b) $125\frac{1}{3}$ (c) $124\frac{1}{3}$ (d) $130\frac{2}{3}$ (e) $135\frac{1}{3}$

19. Students who participate in both cultural activities from class E is what percent more or less than the students who participate in Quiz from class C and D together.
 (a) $117\frac{1}{7}\%$ (b) $132\frac{1}{7}\%$ (c) $131\frac{2}{7}\%$ (d) $215\frac{3}{7}\%$ (e) $120\frac{1}{7}\%$
20. If students who do not participate in cultural activities from class A is increased by $\frac{300}{17}\%$, then students who participate in Quiz from class A is decreased by what percent. Ratio of students who participate in Quiz and Painting remains same.
 (a) $\frac{200}{43}\%$ (b) $\frac{300}{41}\%$ (c) $\frac{300}{43}\%$ (d) $\frac{400}{47}\%$ (e) $\frac{200}{57}\%$

Directions (21-25): Study the table carefully and answer the given questions. Data related to number of employees in 5 different organisations in April 2013

Companies	Total number of employees	Out of the total number of employees		
		Percentage of science graduate	Percentage of commerce graduates	Percentage of arts graduates
A	—	40%	30%	—
B	—	40%	—	25%
C	900	—	44%	35%
D	1300	48%	—	—
E	—	30%	—	50%

Note :

- (i) Employees of the given companies can be categorized only in three types — Science graduates, Commerce graduates and Arts graduates.
 (ii) Few values are missing in the table (indicated by —). A candidate is expected to calculate the missing value, if it is required to answer the given question, on the basis of the given data and information.
21. The average number of science graduate employees and Commerce graduate employees in Company A was 518. What is the total number of employees in Company A ?
 (a) 1480 (b) 1520 (c) 1560 (d) 1580 (e) 1440
22. Total number of employees in Company E was 3 times the total number of employees in Company B. If the difference between number of Commerce graduate employees in Company E and that in Company B was 300, what was the total number of employees in Company B ?
 (a) 900 (b) 1500 (c) 1200 (d) 1320 (e) 1290
23. If the respective ratio between number of Arts graduate employees and Commerce graduate employees in Company D was 4 : 9, what was the number of Arts graduate employees in Company D ?
 (a) 236 (b) 232 (c) 208 (d) 224 (e) 216
24. Total number of employees in Company C increased by 40% from April, 2013 to April, 2014. If 50% total number of employees in Company C in April, 2014 were Commerce graduates, what was the number of commerce graduate employees in Company C in April 2014 ?
 (a) 650 (b) 630 (c) 590 (d) 570 (e) 510
25. What was the difference between number of Science graduate employees and Arts graduate employees in Company C ?
 (a) 136 (b) 132 (c) 128 (d) 122 (e) 126

Directions (26-30): Study the following table carefully to answer the questions that follow. The table shows the population of four villages and ratio of male to female among those who voted in Panchayat election.

Villages	Total villagers	Total villagers who voted	Ratio of male to female who voted
A	6000	90%	5 : 4
B	4800	60%	2 : 1
C	15000	50%	1 : 1
D	12500	80%	2 : 3

26. Total population of village B and D together is approximately what percent of total population of village C ?
 (a) 120% (b) 125% (c) 115% (d) 154% (e) 118%
27. What is the ratio of females who voted in village B and males who voted in village D ?
 (a) 6 : 25 (b) 7 : 10 (c) 8 : 25 (d) 9 : 20 (e) 25 : 4
28. Total number of villagers who voted in village A and C together is
 (a) 11900 (b) 12900 (c) 10000 (d) 9000 (e) 11000
29. Males who voted in village 'D' are what % of population of village 'A' who did not vote ?
 (a) $654\frac{1}{3}\%$ (b) $666\frac{2}{3}\%$ (c) $625\frac{4}{5}\%$ (d) $525\frac{2}{3}\%$ (e) $620\frac{2}{5}\%$

30. What is the total sum of difference b/w males & females who voted in villages B & A together?
 (a) 1500 (b) 1560 (c) 1600 (d) 1650 (e) 1620

Directions (31-35): Study the following data related to the sales distribution of mobiles phones by five shopkeepers in June, 2017. Total sells of all five shopkeepers in June, 2017 is considered to be 100%.

Shopkeeper	Total sold mobiles	Windows Phone: Android Phone
P	22%	7:11
Q	24%	3:7
R	1350	7:8
S	18%	11:9
T	21%	9:5

31. Find the total number of Windows phone sold by P and Q together ?
 (a) 1814 (b) 1418 (c) 1481 (d) 1148 (e) 1344
32. Android phones sold by S are by what number more or less than the windows phones sold by R (rounded upto two decimal places)?
 (a) 20.05% (b) 12.21% (c) 18.54% (d) 15.71% (e) 23.35%
33. The sells of Q and S increased by $5\frac{5}{9}\%$ and $3\frac{19}{27}\%$ respectively in July 2017 than previous month. Find the total mobiles sold by Q and S in July, 2017 ?
 (a) 3690 (b) 3660 (c) 3960 (d) 3990 (e) 3820
34. Find the ratio of number of Windows phones sold by P to that of Window phones sold by R.
 (a) 11 : 9 (b) 4 : 9 (c) 5 : 9 (d) 5 : 8 (e) 4 : 7
35. Find the difference in the average of Windows phones sold by P and Q and average of Android phones sold by S and T.
 (a) 20 (b) 30 (c) 40 (d) 5 (e) 7

Directions (36-40): Table given below shows the total population of 5 cities in 5 different years. Another table show percentage rise of population in these cities every years. Study the table and solve the given questions:

	2012	2013	2014	2015	2016
A	—	—	—	3,04,175	—
B	—	—	1,45,200	—	—
C	—	1,80,000	—	—	—
D	1,60,000	—	—	—	—
E	—	—	—	—	5,37,824

	A	B	C	D	E
Percentage rise of population every year	15%	10%	20%	25%	40%

Note:-

- Percentage rise is in population is consistent every year.
- Some data is missing in the table. Find the data according to the question.

36. Find the ratio of population of city D in 2013 to population of city A in the same year?
 (a) 23 : 20 (b) 21 : 20 (c) 20 : 21 (d) 19 : 17 (e) 20 : 23
37. Population of city B in 2016 is approximately what percent more than the population of city C in 2012 ?
 (a) 11% (b) 17% (c) 22% (d) 27% (e) 32%
38. Population of city C in 2014 is what percentage of the population of city B in 2012?
 (a) 80% (b) 120% (c) 180% (d) 240% (e) 300%
39. What is the average population of city C, D and E in year 2012 ?
 (a) 1,40,000 (b) 1,45,000 (c) 1,48,000 (d) 1,50,000 (e) 1,52,000
40. What is the ratio of the population of city B and C together in 2012 to the city D and E together in 2013?
 (a) 22 : 15 (b) 15 : 22 (c) 14 : 23 (d) 23 : 14 (e) 15 : 23

Directions (41-45): Study the following table and answer the questions that follow:-

The table shows the data related to investment made by five friends in two schemes A and B for different time periods and at different rate of interest.

Scheme Name	A		B	
	Time (Years)	Rate (PCPA)	Time (Years)	Rate (PCPA)
Sameer	3	5	5	8
Sanjeev	6	x	4	x
Saahil	2	10	4	12
Saket	4	18	3	5
Sarash	2	12	2	10

41. Find the difference in the compound interest and simple interest earned by Saket through scheme B if he invested Rs 12000 in scheme B.
 (a) Rs 89.5 (b) Rs 91.5 (c) Rs 108.2 (d) Rs 102.8 (e) Rs 97.5
42. The amount obtained by Sanjeev by investing same sum of money in scheme A & B at compound interest are in ratio 25 : 16. Find the value of x.
 (a) 15% (b) 10% (c) 20% (d) 25% (e) 18%
43. Saahil invested a sum of money(x) in scheme A at SI while Sarash invested three-seventh of x in scheme B at CI. Total interest earned by Saahil and Sarash is what percent of amount invested by Saahil in scheme A ?
 (a) 31% (b) 29% (c) 23% (d) 28% (e) 20%
44. From a total of Rs 12,000 Sarash invested three-fifth in Scheme A and rest in scheme B at SI. Find the difference in the interest earned.
 (a) Rs 786 (b) Rs 789 (c) Rs 586 (d) Rs 867 (e) Rs 768
45. Saket invested an amount (P) in A at SI and Saahil an amount of Q in B at SI and earn interest in ratio of 5 : 4. Which one of the following is false statement?
 (a) P is $16\frac{2}{3}\%$ less than Q (b) Q is 20% more than P
 (c) Sum of P & Q is 220% of P (d) Difference of P & Q is 22% of P
 (e) None of these

Directions (46-50): Refer to the table given below and answer the given questions. Table shows the total population, percentage of males, females and transgenders of 5 villages in year 2000. Some data are missing, find the missing data to answer the given questions.

Village	Total Population	Percentage of Males	Percentage of Females	Percentage of Transgenders
P	2400	25%	-	-
Q	-	-	40%	20%
R	-	50%	20%	-
S	800	-	-	16%
T	-	-	24%	36%

46. If the ratio of population of females and transgenders in village P in year 2000 is 3 : 7. And females in village P in year 2001 is increased by 20% from that of year 2000. Then find the total number of males and transgenders in village P in year 2001 so that overall population in year 2001 is same as in year 2000?
 (a) 1752 (b) 1852 (c) 2752 (d) 3200 (e) None of these
47. If number of transgenders in village R in year 2000 is 180. And ratio of males and females in village S in year 2000 is 1:2. Then find the difference of males in village R and village S?
 (a) 96 (b) 86 (c) 76 (d) 55 (e) None of these
48. If total population of village Q and village R together in year 2000 is 25% more than the total population of village P in year 2000. And ratio of total population of village Q and village R in year 2000 is 2 : 3. Then find the ratio of males in village Q to transgenders in village R in year 2000?
 (a) 9 : 8 (b) 8 : 9 (c) 2 : 3 (d) 3 : 5 (e) None of these
49. If ratio of males of village S in year 2000 to the females in village P in year 2000 is 2 : 5 and population of transgenders in village P is increased by 20% in year 2001 from year 2000. Then find the total population of transgenders in year 2001 in village P?
 (a) 2000 (b) 1200 (c) 1500
 (d) Cannot be determined (e) None of these
50. If ratio of total population of village R to village T in year 2000 is 5 : 4. Then number of males in village T in year 2000 is approximately what percent more or less than the number of transgenders in village R in year 2000?
 (a) 5.667% (b) 12% (c) 10% (d) 3.334% (e) 6.667%

Directions (51-55): Study the following table carefully and answer the questions that follow.

The table shows the data related top six schools of India. Some values are missing which you need to find out and answer the questions accordingly.

Schools	Number of teachers	Ratio of male to female teachers	Ratio of teachers to students	Percentage of non-teaching staff
Vasant valley	480	15:17	_ : _	15%
Woodstock	450	_ : _	1:6	10%
DPS	—	3:2	3:20	—
Rishi Valley	510	8:7	3:10	—
Pathways	—	2:7	2:5	20%
DAV	—	_ : _	2:7	15%

Note: Total strength of school = Number of teachers + Number of students + Number of non-teaching staff

51. If female teachers of school Vasant valley are 17% of total strength of school, then what is the ratio of teachers to students in that school ?
 (a) 32 : 55 (b) 5 : 11 (c) 32 : 53 (d) 23 : 53 (e) 5 : 13
52. The female students of Pathways school are $15\frac{5}{9}\%$ more than male teachers of school Vasant valley while ratio of male to female students of Pathways school is 14 : 13. Find the number of non-teaching staff of Pathways school.
 (a) 180 (b) 192 (c) 210 (d) 189 (e) None of these
53. If strength of all school is same, then find the number of non-teaching staff of school DAV.
 (a) 420 (b) 525 (c) 580 (d) 630 (e) None of these
54. If total strength and number of students of schools Woodstock and DPS are same, then teachers of DPS are what percent less than students of Woodstock ?
 (a) 80% (b) 85% (c) 72% (d) 92% (e) None of these
55. Find the difference in total strength of school Woodstock and that of Rishi valley.
 (a) 980 (b) 840 (c) 780 (d) can't be determined (e) None of these

Direction (56-60): Study the given table carefully and answer the following questions. Table shown below shows the ratio of literate and illiterate population of five cities

Cities	Percentage of Literate population	Ratio of illiterate Male & Female	Ratio of Literate male and Females
A	48%	8 : 7	7 : 5
B	60%	7 : 5	7 : 8
C	72%	7 : 5	5 : 4
D	64%	5 : 4	9 : 7
E	52%	7 : 9	9 : 4

56. If the number of literate male in city B is 34,440 and that of illiterate female in city D is 24,000. Then by what percentage population of city B is less than population of city D.
 (a) 12% (b) 14% (c) 16% (d) 18% (e) 20%
57. Find the difference between the illiterate male in city C to the literate female in same city if the difference between the literate and illiterate population is 52800.
 (a) 17,800 (b) 18,800 (c) 18,830 (d) 19,300 (e) 20,290
58. By what percentage population of city E is more than population of city A if illiterate male in city A is 31200 whereas literate female in city E is 36000.
 (a) 50% (b) 75% (c) 100% (d) 125% (e) 150%
59. Find the ratio of population of literate male in city C to illiterate female in city E if illiterate female in city C is 14,000 and literate male in city E is 81,000.
 (a) $\frac{81}{64}$ (b) $\frac{64}{81}$ (c) $\frac{93}{67}$ (d) $\frac{67}{93}$ (e) None of these
60. Find the population of illiterate females in city B if the difference between the population of literate male and illiterate males is 5740.
 (a) 24,200 (b) 22,250 (c) 18,750 (d) 20,500 (e) None of these

Directions (61-65): Given below is the table which shows the various items purchased by shopkeeper from wholesaler, % mark up price per kg by shopkeeper, List price per kg marked by shopkeeper and % discount offered by shopkeeper on list price.

Items	Quantity	% mark up	List price per kg	% discount
Wheat	80 Kg	$33\frac{1}{3}\%$	20	15%
Rice	90 kg	40%	35	$\frac{100}{7}\%$
Maize	40 kg	$\frac{200}{7}\%$	45	$\frac{100}{6}\%$
Jowar	60 kg	$33\frac{1}{3}\%$	60	$\frac{100}{6}\%$
Bajra	30 kg	$\frac{400}{9}\%$	52	$\frac{300}{13}\%$

61. What is the ratio of total profit obtained on selling whole quantity of rice to the total profit obtained on selling whole quantity of Maize.
 (a) 3 : 4 (b) 7 : 2 (c) 9 : 2 (d) 5 : 2 (e) 7 : 3
62. If shopkeeper mixes 5 kg of impurity free of cost in Wheat then his total profit increases by what percent.
 (a) $42\frac{2}{7}\%$ (b) $64\frac{2}{3}\%$ (c) $60\frac{1}{3}\%$ (d) $53\frac{1}{8}\%$ (e) $50\frac{1}{7}\%$
63. Cost price of Maize and Bajra together are what percent more or less than the cost price of Rice and Jowar together.
 (a) $\frac{20}{3}\%$ (b) $\frac{10}{7}\%$ (c) $\frac{9}{7}\%$ (d) $\frac{25}{3}\%$ (e) $\frac{50}{3}\%$
64. If 12 kg of Jowar is spoiled then by what percent he should increase his initial List price per kg so that there is no profit and no loss.
 (a) 12.5% (b) 22.5% (c) 8.5% (d) 16.5% (e) 13.75%
65. If shopkeeper uses faulty weight of 800 gm instead of 1 kg while selling then what will be the total profit in selling Jowar
 (a) 850 (b) 625 (c) 1250 (d) 1050 (e) 900

Directions (66-70): Read the given data carefully and answer the given questions.

Given below is the table which shows the different types of work A, B, C, D and E, ratio of efficiency to do these work of a man, a woman and a child respectively and total time taken by a child to complete these different works.

Work	Ratio of Efficiency	Time taken by a child to complete work
A	5 : 3 : 4	20 days
B	4 : 3 : 2	10 days
C	8 : 7 : 6	21 days
D	3 : 2 : 1	15 days
E	6 : 4 : 3	16 days

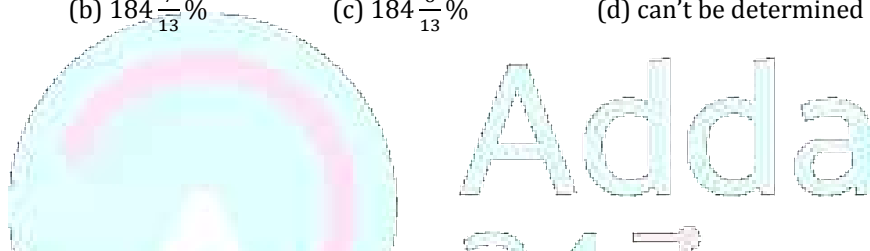
66. Time taken to complete work A by 3 men, 2 women and 1 child together is what % more or less than time taken by 1 man, 3 women and 2 children together to complete work D.
 (a) $\frac{400}{3}\%$ (b) $\frac{200}{3}\%$ (c) $\frac{404}{3}\%$ (d) $\frac{100}{3}\%$ (e) $\frac{100}{6}\%$
67. What is the ratio of time taken by a man and a woman to complete work C together to the time taken by 3 women and a child to complete work E together.
 (a) 20 : 13 (b) 21 : 8 (c) 17 : 18 (d) 12 : 11 (e) 13 : 9
68. If a man, a woman and a child works alternatively starting from child on day 1, man on day 2 and women on day 3 then, in how many days will they complete work E working alternatively.
 (a) 8 days (b) 6 days (c) 9 days (d) 11 days (e) 10 days
69. Time taken by a woman to complete work C is what percent more or less than time taken by man to complete work D.
 (a) 180% (b) 140% (c) 220% (d) 240% (e) 260%
70. Working together they get 4500 Rs. in total for completing work B and 1800 Rs. for completing work D, then share received by man in completing work B is what percent more or less than share received by man in completing work D.
 (a) $98\frac{2}{9}\%$ (b) $122\frac{2}{9}\%$ (c) $110\frac{2}{9}\%$ (d) $115\frac{2}{9}\%$ (e) $125\frac{2}{9}\%$

Directions (71-75): Given below is the percent of number of students from 5 different colleges attended different number of seminars.

	No. of seminars Attended - 1	No. of seminars Attended - 2	No. of seminars Attended - 3	No. of seminars Attended - 4	No. of seminars Attended - 5	No. of seminars Attended - 6
P	19	16	21	9	—	12
Q	—	18	24	—	—	20
R	10	18	29	—	—	—
S	—	—	29	11	24	10
T	16	—	25	—	31	6

Note: Every student attended at least 1 seminar so there is no student from every college who did not attend the seminars.

71. If the number of students from college R who attended at most 3 seminars is equal to number of students from college S who attended at least 3 seminars and the total number of students from College S is 11400. Then find total the number of students from college R.
 (a) 14800 (b) 41900 (c) 15300 (d) 12400 (e) 13000
72. Total no. of students from college P who attended at most 2 seminars is equal to the sum of number of students from college T who attended 1 seminar and the number of students from the same college who attended 6 seminars. Then the total number of students from college P is what percent of total no. of student from college T ?
 (a) $57\frac{1}{7}\%$ (b) $62\frac{6}{7}\%$ (c) $62\frac{4}{7}\%$ (d) $57\frac{6}{7}\%$ (e) $47\frac{3}{7}\%$
73. The number of students from college P who attended more than 2 seminars is approximately what percent less than the number of students from college S who attended at least 3 seminars if the number of student who attended 2 seminars from college P is 48 and the number of students of college S who attended 6 seminars is 48 more than the number of students from college P, who attended 1 seminar?
 (a) 77% (b) 74% (c) 71% (d) 75% (e) 73%
74. If the difference between number of students from college T who attended 3 seminars and students who attended 5 seminars is 60, and the total students from college R is 60% of the total students from college T then find the number of students of college R who are attending 2 seminars.
 (a) 116 (b) 104 (c) 136 (d) 108 (e) 105
75. No. of students who attended at most 2 seminars from college S are what percent more/less than the no. of students who attended at least 3 seminars from the same college ?
 (a) $184\frac{11}{13}\%$ (b) $184\frac{7}{13}\%$ (c) $184\frac{8}{13}\%$ (d) can't be determined (e) None of these



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PREVIOUS YEAR SOLUTIONS

1. (e); No. of male candidates who qualified in 2010

$$= \frac{(100-36)}{100} \times 900 \times \frac{13}{24} = 312$$

No. of males who qualified in 2007

$$= 312 \times \frac{50}{100} = 156$$

Total no. of appeared candidates who qualified in 2007

$$= \frac{156}{2} \times 5 = 390$$

Total no. of appeared candidates in 2007

$$= \frac{390}{50} \times 100 = 780$$

2. (c); No. of qualified male in 2007 = $\frac{4}{7} \times 392 = 224$

$$\text{No. of qualified male in 2009} = \frac{7}{3} \times 392 = 168$$

No. of appeared candidates in 2007

$$= \frac{224}{2} \times 5 \times \frac{1}{50} \times 100 = 1120$$

No. of appeared candidates in 2009

$$= \frac{168}{4} \times 9 \times \frac{1}{30} \times 100 = 1260$$

$$\text{Required \%} = \frac{1120}{1260} \times 100 = 88\frac{8}{9}\%$$

3. (d); Let no. of qualified male in 2008 and 2010 = x
Let no. of qualified female in 2008 and 2010 = 3y and 7y respectively

$$\text{Then, } x + 3y = 288 \quad \dots (i)$$

$$\text{and } x + 7y = 576 \quad \dots (ii)$$

$$\text{from (i) and (ii) } y = 72$$

No. of qualified female candidates in 2008 and 2010 together = $72(3 + 7) = 720$

4. (a); No. of appeared candidates in 2011

$$= \frac{125}{100} \times \frac{60}{100} \times 480 = 360$$

$$\text{No. of appeared candidates who qualified in 2011} = 576 - 376 = 200$$

No. of unqualified females in 2011

$$= \frac{3}{10}(360 - 200) = \frac{3}{10} \times 160 = 48$$

5. (e); Required % = $\frac{576-288}{576} \times 100 = 50\%$

6. (e); Let qualified male from state A in 2012 = 7x
And qualified female from state A in 2012 = 5x
According to question $2x = 102$
 $x = 51$

$$\text{Total appeared candidates} = \frac{12 \times 51}{60} \times 100$$

$$= \frac{12 \times 51 \times 5}{3} = 1020$$

7. (c); Number of appeared candidate from state B in 2011

$$= \frac{4}{3} \times 660 = 880$$

According to question

$$= 880 \times \frac{40}{100} \times \frac{1}{11} \times$$

$$(11+12) = 736$$

8. (c); Required ratio = $\frac{9 \times 60 + 12 \times 43 + 96 \times 7}{76 \times 3 + 52 \times 6 + 4 \times 70}$

$$= \frac{540 + 516 + 672}{228 + 312 + 280}$$

$$= \frac{1728}{820} \Rightarrow = 432 : 205$$

9. (c); Required % = $\frac{9 \times 60 - 4 \times 70}{4 \times 70} \times 100$

$$= \frac{260}{280} \times 100 = \frac{13}{14} \times 100 = 92\frac{6}{7}\%$$

10. (a); Total passed candidate from state A in 2014

$$= 1356 - 96 \times 7 = 684$$

$$\text{Required \%} = \frac{760-684}{760} \times 100 = \frac{76}{760} \times 100$$

$$= 10\%$$

11. (a); Sales of Adidas in 2001 = $320 \times \frac{140}{100} = 448$ lakh

$$\text{Sale of Nike in 2001} = 798 - 448 = 350 \text{ lakh}$$

$$\text{Cost of production of Nike} = 350 \times \frac{100}{140} = 250 \text{ lakh}$$

12. (b); Let cost of production of Adidas be 5x lakh

And cost of production of Nike be 4x lakh

Then,

$$5x \times \frac{130}{100} + 4x \times \frac{125}{100} = 1150$$

$$\text{or, } \frac{1150x}{100} = 1150 \Rightarrow x = 100$$

$$\text{Req. Difference} = 5x - 4x \Rightarrow x = 100$$

13. (c); Req. percentage = $\frac{\frac{1}{2} \times (460 \times \frac{145}{100} + 510 \times \frac{130}{100})}{\frac{1}{2} \times (440 \times \frac{135}{100} + 470 \times \frac{120}{100})} \times 100$

$$\approx 114\%$$

14. (d); Sales of Nike in 2005 = $1330 - 510 \times \frac{130}{100}$

$$= 1330 - 663 = 667 \text{ lakh}$$

$$\text{Profit \% of Nike} = \frac{667-580}{580} \times 100 = \frac{87}{580} \times 100$$

$$= 15\%$$

15. (b); Required Ratio = $\frac{420 \times \frac{120}{100}}{470 \times \frac{120}{100}} = \frac{42}{47}$

16. (b); Students who participate in Quiz from B

$$= (330 - 88) \times \frac{7}{11} = 154$$

Students who participate in Quiz from C

$$= (240 - 110) \times \frac{8}{13} = 80$$

Students who participate in Quiz from E

$$= (390 - 130) \times \frac{8}{13} = 160$$

$$\text{Required sum} = 154 + 80 + 160 = 394$$

17. (c); Students who participate in painting from B and C

$$= (330 - 88) \times \frac{4}{11} + (240 - 110) \times \frac{5}{13}$$

$$= 88 + 50 = 138$$

Students who participate in Quiz from A and D

$$= (420 - 119) \times \frac{4}{7} + (125 - 45) \times \frac{2}{5}$$

$$= 172 + 32 = 204$$

$$\text{Required ratio} = \frac{138}{204} \Rightarrow = \frac{23}{34}$$

18. (e); Students who participate in Quiz from A, B and C together

$$= 172 + 154 + 80 = 406$$

$$\text{Required average} = \frac{406}{3} = 135\frac{1}{3}$$

19. (b); Students who participate in both activities from E = $390 - 130 = 260$

Students who participate in quiz from C and D

$$= (240 - 110) \times \frac{8}{13} + (1250 - 45) \times \frac{2}{5}$$

$$= 80 + 32 = 112$$

$$\text{Required percentage} = \frac{260-112}{112} \times 100$$

$$= 132\frac{1}{7}\%$$

20. (c); Students who do not participate in cultural activities from A after increase

$$= \left(1 + \frac{3}{17}\right) \times 119 = 140$$

Now, students in Quiz from

$$= (420 - 140) \times \frac{4}{7}$$

$$= 280 \times \frac{4}{7} = 160$$

Required percentage

$$= \frac{172-160}{172} \times 100$$

$$= \frac{12}{172} \times 100 = \frac{300}{43} \%$$

21. (a); Required employees = $\frac{1036}{70} \times 100$
= 1480

22. (c); Let total employees in Company B = x
Let total employees in Company E = 3x
 $\therefore (100 - 50 - 30)\%$ of 3x - $(100 - 40 - 25)\%$ of x = 300
x = 1200

23. (c); Arts and Commerce graduate employees
= $\frac{52}{100} \times 1300 = 676$

Arts graduate employees = $\frac{4}{13} \times 676$
= 208

24. (b); Total number of employee in Company C in April 2014 = $\frac{900 \times 140}{100} = 1260$

$\therefore$ Required employees = $\frac{1260}{2} = 630$

25. (e); % of Science graduate employees in company C
= $(100 - 44 - 35)\% = 21\%$
Required difference = $(35 - 21)\%$ of 900 = 126

26. (c); Required % = $\frac{(4800+12500)}{15000} \times 100$
 $\therefore 115.33 \Rightarrow \approx 115\%$

27. (a); Required ratio
= $\frac{4800 \times \frac{60}{100} \times \frac{1}{3}}{12500 \times \frac{80}{100} \times \frac{2}{5}} = 6 : 25$

28. (b); Required no. of villagers
= $\left(6000 \times \frac{90}{100}\right) + \left(15000 \times \frac{50}{100}\right)$
= 5400 + 7500 = 12900

29. (b); Required % = $\frac{12500 \times \frac{80}{100} \times \frac{2}{5}}{6000 \times \frac{10}{100}} \times 100$
= $666\frac{2}{3}\%$

30. (b); Required sum
= $\left[4800 \times \frac{60}{100} \times \frac{2}{3} - 4800 \times \frac{60}{100} \times \frac{1}{3}\right] +$
 $\left[6000 \times \frac{90}{100} \times \frac{5}{9} - 6000 \times \frac{90}{100} \times \frac{4}{9}\right]$
= 960 + 600 $\Rightarrow$ = 1560

31. (b); Required number of windows phones
= $\frac{7}{18} \times \frac{22}{100} \times 9000 + \frac{3}{10} \times \frac{24}{100} \times 9000$
= 770 + 648 = 1418

32. (d); Android phones sold by S = $\frac{9}{20} \times \frac{18}{100} \times 9000 = 729$
Windows phones sold by R = $\frac{7}{15} \times 1350 = 630$
 $\therefore$ Required percentage = $\frac{729-630}{630} \times 100 \approx 15.71\%$

33. (c); Sales of Q in July = $\left(100 + \frac{50}{9}\right)\%$ of $\frac{24}{100} \times 9000$
= $\frac{950}{900} \times \frac{24}{100} \times 9000 = 2280$
Sales of S in July = $\left(100 + \frac{100}{27}\right)\%$ of $\frac{18}{100} \times 9000$
= $\frac{2800}{2700} \times \frac{18}{100} \times 9000 = 1680$

$\therefore$ Total phones sold = 2280 + 1680 = 3960

34. (a); Required ratio = $\frac{\frac{7}{18} \times \frac{22}{100} \times 9000}{\frac{7}{15} \times \frac{24}{100} \times 9000} = 11 : 9$

35. (e); Average of windows phones sold by P and Q
= $\frac{1}{2} \times 1418 = 709$

Average of android phones sold by S & T

$$= \frac{1}{2} \left(729 + \frac{5}{14} \times \frac{21}{100} \times 9000\right)$$

$$= \frac{1}{2} (729 + 675) = 702$$

$\therefore$ Required difference = 709 - 702 = 7

36. (e); Population of city D in 2013 = $1,60,000 \times \frac{125}{100}$
= 2,00,000

Population of city A in 2013 = $3,04,175 \times \frac{100}{115} \times \frac{100}{115}$
= 2,30,000

Desired Ratio = $\frac{2,00,000}{2,30,000} = \frac{20}{23}$

37. (b); Population of city B in 2016

$$= 1,45,200 \times \frac{110}{100} \times \frac{110}{100}$$

$$= 1,75,692$$

Population of city C in 2012

$$= 1,80,000 \times \frac{100}{120} = 1,50,000$$

Desired percentage = $\frac{1,75,692-1,50,000}{1,50,000} \times 100$

$$= \frac{25,692}{1,50,000} \times 100 \Rightarrow = 17.128 \approx 17\%$$

38. (c); Population of city C in 2014

$$= 1,80,000 \times \frac{120}{100} = 2,16,000$$

Population of city B in 2012

$$= 1,45,200 \times \frac{100}{110} \times \frac{100}{110} \Rightarrow = 1,20,000$$

Desired percentage = $\frac{2,16,000}{1,20,000} \times 100 = \frac{216}{120} \times 100$

$$= 180\%$$

39. (d); Population of city C in 2012

$$= 1,80,000 \times \frac{100}{120}$$

$$= 1,50,000$$

Population of city D in 2012 = 1,60,000

Population of city E in 2012

$$= 537824 \times \left[\frac{100}{140}\right]^4 \Rightarrow = 1,40,000$$

Desired average = $\frac{1,50,000+1,60,000+1,40,000}{3}$

$$= 1,50,000$$

40. (b); Population of B = 2012 = 1,20,000

Population of C in 2012 = 1,50,000

Population of B and C together in 2012 = 2,70,000

Population of D in 2013 = $1,60,000 \times \frac{125}{100}$
= 2,00,000

Population of E in 2013 = $5,37,824 \times \left[\frac{100}{140}\right]^3$
= 1,96,000

Population of D and E together in 2013 = 3,96,000

Required Ratio = $\frac{2,70,000}{3,96,000} = \frac{15}{22}$

41. (b); Difference = $\frac{\text{Pr}^2(300+r)}{100^3}$
= $\frac{12000 \times 25 \times 305}{100^3} \Rightarrow = \text{Rs } 91.5$

42. (d); $\frac{P\left(1+\frac{x}{100}\right)^6}{P\left(1+\frac{x}{100}\right)^4} = \frac{25}{16}$

$$\Rightarrow \left(1 + \frac{x}{100}\right)^2 = \frac{25}{16}$$

$$\Rightarrow \frac{x}{100} = \frac{1}{4} \Rightarrow x = 25\%$$

- 43. (b);** Interest earned by Saahil = $\frac{20}{100} \times x = 0.2x$
 Interest earned by Sarash = $\frac{21}{100} \times \frac{3}{7} \times x = 0.09x$
 $\therefore$ Required percentage = $\frac{0.29x}{x} \times 100 = 29\%$
- 44. (e);** Difference in interest
 $= \frac{24}{100} \times \frac{3}{5} \times 12000 - \frac{20}{100} \times \frac{2}{5} \times 12000$
 $= (72 - 40) \times 24 \Rightarrow \text{Rs } 768$
- 45. (d);** $\frac{P \times 18 \times 4}{Q \times 12 \times 4} = \frac{5}{4} \Rightarrow \frac{P}{Q} = \frac{5}{6}$
- 46. (a);** Total population of females and transgenders in village P in 2000 = 75% of 2400 = 1800
 $\therefore$ Number of females in village P in 2000
 $= \frac{3}{10} \times 1800 = 540$
 Females in 2001 in village P = $540 \times \frac{120}{100} = 648$
 $\therefore$ Total males & transgenders in 2001 in village P
 $= 2400 - 648 = 1752$
- 47. (c);** Percentage transgenders in village R in year 2000 = 30%
 $\therefore$ Total population of village R in 2000
 $= \frac{180}{30} \times 100 = 600$
 $\therefore$ males in village R in 2000 = $600 \times 50\% = 300$
 Males in village S in 2000 = $\frac{84}{100} \times 800 \times \frac{1}{3} = 224$
 $\therefore$ Required difference = $300 - 224 = 76$
- 48. (b);** Total population of village Q and Village R in 2000
 $= 2400 \times \frac{125}{100} = 3000$
 $\therefore$ Total population of village Q in 2000 = $\frac{2}{5} \times 3000$
 $= 1200$ and
 total population of village R in 2000
 $= \frac{3}{5} \times 3000 \Rightarrow 1800$
 $\therefore$ Required ratio = $\frac{\frac{40}{100} \times 1200}{\frac{30}{100} \times 1800} = \frac{4 \times 2}{3 \times 3} = \frac{8}{9}$
 $= 8 : 9$
- 49. (d);** Cannot be determined
- 50. (e);** Let the population of R = $5x$
 And the population of T = $4x$
 Required percentage = $\frac{(4x) \times \frac{40}{100} - (5x) \times \frac{30}{100}}{(5x) \times \frac{30}{100}} \times 100$
 $= \frac{(1.6 - 1.5)x}{(1.5)x} \times 100 = \frac{0.1 \times 100}{1.5}$
 $= 6.667\%$
- 51. (c);** Number of female teachers
 $= \frac{17}{32} \times 480 = 255$
 $\therefore$ Total strength of school
 $= \frac{100}{17} \times 255 = 1500$
 Number of students
 $= 1500 - 480 - \frac{15}{100} \times 1500 = 795$
 $\therefore$ Required ratio = $\frac{480}{795} = \frac{32}{53}$
- 52. (d);** Number of female students of Pathways
 $= \frac{1040}{900} \times 225 = 260$
 Number of male students = $260 \times \frac{14}{13} = 280$
 Number of teachers

$$= (260 + 280) \times \frac{2}{5} = 540 \times \frac{2}{5} = 216$$

Required number of non-teaching staff
 $= (540 + 216) \times \frac{20}{80} = 756 \times \frac{20}{80} = 189$

- 53. (b);** Since strength of all schools is same.
 Students in Woodstock = $450 \times 6 = 2700$
 $\therefore$ 90% of strength = $2700 + 450$
 $\therefore$ total strength of Woodstock = $3150 \times \frac{100}{90} = 3500$
 So, required number of non-teaching staff
 $= 15\% \text{ of } 3500 = 525$
- 54. (b);** Total strength = 3500
 Teachers in DPS = $2700 \times \frac{3}{20} = 405$
 $\therefore$ Required percentage = $\frac{2700 - 405}{2700} \times 100 = 85\%$
- 55. (d);** We can't find the strength of school Rishi Valley with given data.
- 56. (d);** Let population of city B = x
 $x \times \frac{60}{100} \times \frac{7}{5} = 34,440$
 $x = 1,23,000$
 Let population of city D = y
 $y \times \frac{36}{100} \times \frac{4}{9} = 24,000$
 $y = 1,50,000$
 Required Percentage = $\frac{1,50,000 - 1,23,000}{1,50,000} \times 100$
 $= \frac{27,000}{1,50,000} \times 100 = 18\%$
- 57. (b);** Let the total population of city C = a
 72% of a - 28% of $a = 52,800$
 44% of $a = 52,800$
 $a = 1,20,000$
 Required difference
 $= \frac{7}{12} \times \frac{28}{100} \times 120,000 \sim \frac{4}{9} \times \frac{72}{100} \times 1,20,000$
 $= 19,600 \sim 38,400 \Rightarrow 18,800$
- 58. (c);** Let total population in city A = x
 $x \times \frac{8}{15} \times \frac{52}{100} = 31,200$
 $x = 1,12,500$
 Let total population of city E = y
 $y \times \frac{4}{13} \times \frac{52}{100} = 36,000$
 $y = 2,25,000$
 Required percentage
 $= \frac{2,25,000 - 1,12,500}{1,12,500} \times 100 \Rightarrow 100\%$
- 59. (b);** Let total population in city C = x
 $x \times \frac{28}{100} \times \frac{5}{12} = 14,000$
 $x = 1,20,000$
 Literate male in city C
 $= 1,20,000 \times \frac{72}{100} \times \frac{5}{9} \Rightarrow 48,000$
 Let total population in city E = y
 $y \times \frac{52}{100} \times \frac{9}{13} = 81,000$
 $y = 2,25,000$
 Illiterate female in city E
 $= 2,25,000 \times \frac{48}{100} \times \frac{9}{16} \Rightarrow 60,750$
 Required Ratio = $\frac{48,000}{60,750} = \frac{64}{81}$
- 60. (d);** Let the total population in city B = x
 $5740 = x \times \frac{60}{100} \times \frac{7}{15} - x \times \frac{40}{100} \times \frac{7}{12}$

$$5740 = \frac{168x - 140x}{28}$$

$$x = \frac{5740 \times 600}{28} = 1,23,000$$

$$\text{Illiterate females in city B} = 1,23,000 \times \frac{40}{100} \times \frac{5}{12}$$

$$= 20,500$$

61. (c); Cost price of Rice = $\frac{35 \times 100}{140} = 25$

$$\text{Selling price of rice} = 35 - 35 \times \frac{1}{7} = 30$$

$$\text{Total profit in selling rice} = 5 \times 90 = 450$$

$$\text{Cost price of Maize} = \frac{45}{\left(\frac{100\% + 200\%}{7}\right)}$$

$$= \frac{45 \times 7}{9} \Rightarrow = 35$$

$$\text{Selling price of Maize} = 45 - 45 \times \frac{1}{6}$$

$$= 37.5$$

$$\text{Total profit in selling Maize} = 2.5 \times 40$$

$$= 100$$

$$\text{Required ratio} = 9 : 2$$

62. (d); Cost price of wheat = $\frac{3}{4} \times 20 = 15$

$$\text{Selling price of wheat} = 20 - \frac{15}{100} \times 20 = 17$$

$$\text{Total Profit} = 80 \times 2 = 160$$

$$\text{New Total profit} = 80 \times 2 + 5 \times 17$$

$$= 160 + 85 = 245$$

$$\text{Required percentage} = \frac{85}{160} \times 100$$

$$= \frac{425}{8} \% \Rightarrow = 53 \frac{1}{8} \%$$

63. (b); Cost price Maize and Bajra together

$$= 45 \times \frac{7}{9} + 52 \times \frac{9}{13}$$

$$= 35 + 36 \Rightarrow = 71$$

$$\text{Cost price of Rice and Jowar} = \frac{35 \times 100}{140} + 60 \times \frac{3}{4}$$

$$= 25 + 45 = 70$$

$$\text{Required \%} = \frac{11}{70} \times 100 \Rightarrow = \frac{10}{7} \%$$

64. (a); Cost price of Jowar = $60 \times \frac{3}{4} = 45$

$$\text{Total cost price} = 60 \times 45 = 2700$$

But 12 kg Jowar is spoiled

Let new List price per kg is x then

$$2700 = 48 \times x \times \frac{5}{6} \Rightarrow x = 67.5$$

$$\text{Required percentage} = \frac{7.5}{60} \times 100 = 12.5\%$$

65. (d); Cost price of Jowar = $60 \times \frac{3}{4} = 45$

$$\text{Total cost price in selling } 60 \times 1000 \text{ gm of Jowar}$$

$$= 45 \times 60 \Rightarrow = 2700$$

Original selling price of Jowar per kg = 50

But due to use of faulty weight

He sells 800 gm for Rs. 50

$$\text{So selling price of } 60 \times 1000 \text{ gm} = \frac{60 \times 1000}{800} \times 50$$

$$= 3750$$

$$\text{Required profit} = 3750 - 2700 = 1050$$

66. (c); Let a child completes 4x unit of work in one day.

Then,

$$\text{In 20 days total units of work} = 80x$$

And

3 men, 2 women and 1 child can complete

$$= (3 \times 5x + 2 \times 3x + 4x) \text{ units/day}$$

Total time taken by them to complete work A

$$= \frac{80x}{25x} \text{ days} \Rightarrow = \frac{16}{5} \text{ days}$$

Similarly,

Time taken by 1 man, 3 women and 2 children to complete work D

$$= \frac{15}{(3+6+2)} \text{ days} \Rightarrow = \frac{15}{11} \text{ days}$$

$$\text{Required \%} = \frac{\frac{16}{5} \times \frac{15}{11}}{\frac{15}{11}} \times 100 \Rightarrow = \frac{404}{3} \%$$

67. (b); Total time taken by a man and a women to Complete work C = $\frac{6 \times 21}{(8+7)} = \frac{126}{15} \text{ days}$

Total time taken by 3 women and a child to complete

$$\text{work E} = \frac{16 \times 3}{(3 \times 4 + 3)} \text{ days} = \frac{16}{5} \text{ days}$$

$$\text{Required ratio} = 21 : 8$$

68. (d); Let child completes 3x units in one day

$$\text{So, in 16 days he will complete} = 16 \times 3x = 48x \text{ units}$$

In 3 days (6x + 4x + 3x) unit will be completed

In 9 days 13x × 3 unit will be completed

On 10th day 3x more units will be completed by child

Remaining 6x units will be completed by man on 11th day

69. (e); Time taken by women to complete work C

$$= \frac{21 \times 6}{7} = 18 \text{ days}$$

$$\text{Time taken by man to compete work D} = \frac{15 \times 1}{3} = 5 \text{ days}$$

$$\text{Required \%} = \frac{18-5}{5} \times 100$$

$$= 13 \times 20 = 260\%$$

70. (b); Share of man in completing work B = $\frac{4500}{9} \times 4 = 2000$

$$\text{Share of man in completing work D} = \frac{1800}{6} \times 3 = 900$$

$$\text{Required \%} = \frac{2000-900}{900} \times 100 \Rightarrow \frac{1100}{9} \%$$

$$= 122 \frac{2}{9} \%$$

71. (a); $\frac{(10+18+29)}{100} \times R = \left(\frac{29+11+24+10}{100}\right) S$

$$\frac{57}{100} R = \frac{74}{100} S$$

$$57R = 74S \Rightarrow \frac{R}{S} = \frac{74}{57}$$

$$\text{Given } 57 \rightarrow 11400$$

$$74 \rightarrow \frac{11400}{57} \times 74 = 14800$$

72. (b); (19 + 16)% of P = (16 + 6)% of T

$$35 \times P = 22 T \Rightarrow \frac{P}{T} = \frac{22}{35}$$

$$\text{Required \%} = \frac{22}{35} \times 100 = 62 \frac{6}{7} \%$$

73. (d); P = $\frac{48}{16} \times 100 = 300$

$$S \rightarrow \left(\frac{48+57}{10}\right) \times 100 = 1050$$

No. of students who attended more than 2 Seminar

$$\text{from college P} = \frac{100-19-16}{100} \times 300 = 195$$

No. of students who attended at least 3 seminars from

$$\text{college S} = \frac{29+11+24+10}{100} \times 1050 = 777$$

$$\text{Required \%} = \frac{777-195}{777} \times 100 \approx 75\%$$

74. (d); $\frac{31-25}{100} \times T = 60$

$$\frac{6}{100} \times T = 60 \Rightarrow T = 1000$$

$$R = \frac{60}{100} \times 1000 = 600$$

$$\text{Required No. of students} = \frac{18}{100} \times 600 = 108$$

75. (c); Required \% = $\frac{74-26}{26} \times 100$

$$= \frac{48}{26} \times 100 = 184 \frac{8}{13} \%$$

PRACTICE SET (LEVEL-I)

Directions (1-5): Given below is a table which gives the number of students who participated in an inter-school competition in different sports.

Sport	Total students	Boys : Girls
Cricket	350	4 : 3
Football	400	11 : 9
Volleyball	250	7 : 3
Table Tennis	200	12 : 13
Badminton	375	9 : 16
Basketball	425	3 : 2

- What is the average of the no. of girls who participated in Cricket, Football and Volleyball?
(a) 135 (b) 145 (c) 132 (d) 156 (e) none of these
- The no. of boys who participated in Basketball is what % more/less than the number of boys who participated in Volleyball?
(a) $44\frac{5}{7}\%$ (b) $45\frac{5}{7}\%$ (c) $35\frac{5}{7}\%$ (d) $45\frac{3}{7}\%$ (e) none of these
- What is the difference between the number of girls who played Volleyball and the number of boys who played Badminton?
(a) 45 (b) 55 (c) 60 (d) 75 (e) none of these
- What is the ratio of average of number of boys who played Football and Cricket to the average of number of girls who played Volleyball, Badminton and Basketball ?
(a) 129 : 31 (b) 129 : 41 (c) 123 : 93 (d) 126 : 97 (e) none of these
- What is the sum of number of boys who played Volleyball, Badminton and Football ?
(a) 530 (b) 430 (c) 320 (d) 640 (e) none of these

Directions (6-10): Read the following table and answer the following question.

The table given bellows gives the total number of children that took birth in different districts in year 2005 and percentage of boys out of these children.

Districts	No. of children that took birth	Percentage of boys out of total children
A	450	30%
B	500	64%
C	470	50%
D	350	36%
E	650	48%
F	525	32%

- Total number of baby boys from district A and B together is how much more/less than total number of baby girls from district E and F together?
(a) 240 (b) 230 (c) 250 (d) 300 (e) None of these
- The average number of children from district E, C and D together is approximately what percent less/more than the no. of baby boys from districts D, E and F together?
(a) 33.33% (b) 19.14% (c) 26.66% (d) 16.66% (e) none of these
- Find the ratio of the baby boys from district D and E together to the baby girls from district D, E and F together?
(a) 410 : 931 (b) 431 : 941 (c) 438 : 919 (d) 419 : 919 (e) None of these
- The no. of baby girls from district C is what percent more/less than the baby boys from district A? (rounded off to nearest integer)
(a) 64.07% (b) 54.07% (c) 44.07% (d) 74.07% (e) None of these
- Find the ratio of no. of children from districts B and E to the no. of baby girls from districts C and A?
(a) 21:43 (b) 21:23 (c) 23:11 (d) 23:12 (e) None of these

Direction (11-15): Study the given table and answer the questions based on it.

Sells of major flowers used in decoration purpose (in kg)

Months	Freesia	Jasmine	Marigold	Orchids	Roses
January	69	91	71	15	100
February	75	88	75	18	120
March	81	97	79	21	102
April	98	107	88	25	131
May	93	110	92	24	143
June	99	116	97	20	154
July	105	122	103	25	163

11. Which flower witnessed the 2nd maximum sell from January to March together?
 (a) Marigold (b) Roses (c) Jasmine (d) Freesia (e) Orchids
12. What is the difference between the average sells (in kg) of Freesia, Marigold and Roses in February, June and May respectively and the average sells of Jasmine and Roses in April and July respectively ?
 (a) 35 (b) 20 (c) 30 (d) 25 (e) None of these
13. What is the ratio of total sells of Marigold in April and July together to that of Jasmine in March and June together ?
 (a) 17:19 (b) 15:17 (c) 3:5 (d) 11:13 (e) None of these
14. Sells of Jasmine in May is what percent of sell of Freesia in February and July together?
 (a) $60\frac{1}{9}\%$ (b) $61\frac{1}{9}\%$ (c) $55\frac{2}{3}\%$ (d) $52\frac{1}{9}\%$ (e) None of these
15. In August the sells of Roses increases by $33\frac{1}{3}\%$ as compared to that in February and the sells of Jasmine decreases by $18\frac{2}{11}\%$ as compared to that of in May. Find the ratio between sells of Roses and Jasmine in August ?
 (a) 16:9 (b) 15:7 (c) 6:9 (d) 5:2 (e) None of these

Directions(16 - 20): Read the following table and answer the questions.

Cities	Population	Percentage of women out of total Population
A	53000	44
B	49000	45
C	65000	40
D	60000	55
E	75000	50

16. What is the ratio of total women from cities D, B and E to the total men from remaining cities?
 (a) 9255 : 6868 (b) 9155 : 6868 (c) 92 : 69 (d) 79 : 64 (e) None of these
17. What is the difference between the total population from top three populous cities to the total no. of Men in those cities?
 (a) 90000 (b) 95500 (c) 65500 (d) 95600 (e) 96500
18. No. of males in Cities D and E is approximately how many times of the no. of females from A and B?
 (a) 1.2 (b) 1.4 (c) 1.6 (d) 2 (e) 2.4
19. No of females in C is what percent more or less than no of females from E?
 (a) 30.66% less (b) 32% more (c) 30.66% more (d) 32% less (e) None of the above
20. What is the two-third of total women population?
 (a) 93580 (b) 92580 (c) 94580 (d) 95580 (e) 90000

Directions (21-25): There are six students A, B, C, D, E and F. The following table shows the marks given to these students from Monday to Friday in a subject. Study the following table and answer the questions given below it.

	Mon	Tue	Wed	Thu	Fri
A	175	205	350	450	355
B	180	285	410	485	395
C	165	190	315	495	575
D	145	210	405	470	675
E	110	295	490	490	650
F	210	145	340	465	910

21. What is percentage increase of marks taken by A from Monday to Friday? (Rounded off to two decimal points)
 (a) 107.68% (b) 102.86% (c) 98.82% (d) 68.71% (e) None of these
22. What is the difference between marks taken by A, B and D together on Wednesday to the average marks taken by C, E and F together on Friday?
 (a) $463\frac{1}{3}$ (b) $453\frac{2}{3}$ (c) $463\frac{2}{3}$ (d) $453\frac{1}{3}$ (e) None of these
23. If on Saturday there is an increase of 20% marks as compared to Monday and 15% marks as compared to Tuesday of B and C respectively. Then find the total marks taken by B and C on Saturday?
 (a) 453.5 (b) 415.25 (c) 434.5 (d) 433.5 (e) None of these
24. What is the ratio of all the marks taken by C from Monday to Thursday to the marks taken by E from Tuesday to Friday?
 (a) 233 : 285 (b) 285 : 233 (c) 11 : 17 (d) 17 : 11 (e) None of these
25. Total marks taken by the students on Wednesday is approximately what percentage more/less than the total marks taken by all of the students on Monday ?
 (a) 135% (b) 140% (c) 125% (d) 130% (e) 150%

Directions (26 - 30): Given below the table which shows the total population of children in (Lakh) who attends school in five different states and percentage of boys in these students. Table also shows the total number of school in these five states. Some values are missing in the table. You have to calculate these values if required to answer the question.

States	Total children who attend school	% of boys	Number of schools
West Bengal	48	$62\frac{1}{2}\%$	960
UP	–	$46\frac{3}{7}\%$	1400
Orissa	–	50%	2000
Bihar	–	$55\frac{5}{9}\%$	–
Karnataka	–	$66\frac{2}{3}\%$	2400

26. If average number of students per school in Orissa is 4000 and for Karnataka is 2500 then find the total number of girls in these two states who attend school.
 (a) 70Lakh (b) 60Lakh (c) 50Lakh (d) 45Lakh (e) 65Lakh
27. Total number of girls who attend school in west Bengal are what percent of total boys in UP who attend school if average number of students per school in UP is 4000.
 (a) $69\frac{3}{13}\%$ (b) $68\frac{4}{13}\%$ (c) $53\frac{4}{13}\%$ (d) $55\frac{11}{12}\%$ (e) None of these
28. By what percent number of schools in Karnataka should be decreased so that number of students per school in Karnataka becomes equal to number of students per school in West Bengal if total students who attended school in Karnataka is 25% more than total students who attend school in west Bengal.
 (a) 60% (b) 40% (c) 45% (d) 55% (e) 50%
29. What is the average number of school in all five states if total boys who attend school in bihar is 40 lakh and average number of students per school in Bihar is 4000.
 (a) 1780 (b) 1630 (c) 1712 (d) 1535 (e) 1820
30. If average student per school in West Bengal, U.P., Orissa and Karnataka are in the ratio 10 : 8 : 8 : 5 and total girls in Bihar who attend school are 32Lakh then find the total students in all five states who attend school.
 (a) 280Lakh (b) 316Lakh (c) 392Lakh (d) 345Lakh (e) 290Lakh

Directions (31-35): Study the given table carefully to answer the following questions.

In the given table there are five colleges in which total student and percentage of engineering students and ratio of arts and commerce students are given.

There are only three types of streams in each college.

Note → some data are missing, calculate the missing data if necessary.

Colleges	Total no. of Students	Percentage of Engineering students	Ratio of arts to commerce students
P	1250	28%	–
Q	–	25%	–
R	–	–	5:8
S	2100	–	5:2
T	1440	–	–

31. If the ratio of boys and girls in college P for commerce student is 2 : 5 and the commerce student are 40% more than arts student. Then find the difference of boys and girls in Commerce?
 (a) 225 (b) 275 (c) 250 (d) 325 (e) 215
32. If the total engineering student in college T is 360 and student in arts are 25% more then the student in commerce and engineering student in college S is 630. Then find ratio of arts student in college S to college T?
 (a) 2 : 3 (b) 4 : 7 (c) 4 : 9 (d) 7 : 4 (e) 7 : 8
33. If Engineering students in college P is 150 less than engineering student in college Q. Then total student in college S is what percent more or less than total student in college Q?
 (a) 1% (b) 3% (c) 9% (d) 7% (e) 5%
34. If total student in college R is 2600 and total engineering student in college R is equal to the total students in arts and commerce. And ratio of boys and girls in college R in engineering 5 : 8. If 20% of boys are transferred to college T. Then find total student in college T?
 (a) 1640 (b) 1840 (c) 1920 (d) 1540 (e) 1640
35. Suppose there is another college A in which engineering students are $\frac{2}{5}$ th of the engineering student of college P and arts student in college A is $33\frac{1}{3}\%$ more than engineering student in college Q. And commerce student in college A is 25% more than commerce student of college S. Then find the total number of students in college A [Given that engineering student in college S is 700 and total student is called Q is 1260].
 (a) 1120 (b) 1020 (c) 1060 (d) 1080 (e) 1050

Directions (36-40): Study the table carefully to answer the questions that follow:

Table below shows the runs scored, Number of matches played, average of runs per innings, total number of 4's and 6's hit and ratio of number of fours and sixes hit by five batsman in Indian Premiere League (IPL) 2016

Note – (No batsman, remained not out in any innings)

Few values are missing in table, a candidate is expected to calculate the missing values if it is required to answer the given questions.

Batsman	Total Runs	No. of Matches played	Average	Total number of 4's and 6's	Ratio of 4's and 6's hit
Virat Kohli	—	16	60	121	—
David Warner	—	20	—	119	—
AB-de-Villiers	848	—	—	—	7:5
G. Gambhir	—	15	—	60	—
S. Dhawan	570	—	—	—	—

36. What is the total runs made by all five batsman in IPL 2016 if average of David warner is $33\frac{1}{3}\%$ less than average of Virat Kohli and runs scored by S. Dhawan is equal to runs scored by G. Gambhir ?
 (a) 3748 (b) 4840 (c) 3642 (d) 3580 (e) 3492
37. What percent of runs are made by AB de Villiers by hitting fours and sixes if total fours and sixes hit by AB-de-Villiers is 60% more than that of total fours and sixes hit by G. Gambhir. (approximately)
 (a) 60% (b) 65% (c) 70% (d) 45% (e) 55%
38. What is the difference in average of runs made by S. Dhawan and AB de Villiers if average of AB de Villiers is $11\frac{4}{6}\%$ less than average of Virat Kohli and number of innings played by G. Gambhir and S. Dhawan are equal.
 (a) 20 (b) 15 (c) 23 (d) 18 (e) 12
39. What is the difference between the number of fours hit by Virat Kohli to the number of fours hit by AB-de-Villier if total four and sixes hit by AB-de-Villier is 60% more than four and sixes hit by G. Gambhir and ratio of number of fours to sixes hit by Virat Kohli is 8 : 3.
 (a) 48 (b) 52 (c) 28 (d) 32 (e) 38
40. What is the sum of percentage of runs made by G. Gambhir and Virat Kohli (except runs made by fours and sixes) if ratio of fours and sixes hit by G. Gambhir and Virat Kohli is 5 : 1 and 8 : 3 respectively and average of G. Gambhir is 38 (approximately)
 (a) 98% (b) 88% (c) 92% (d) 68% (e) 76%

Directions (41-45): Study the table carefully to answer the questions that follow Monthly Bill (in rupees) of paid by three People. Monthly bills paid by three different people in five different months

Month	Monthly Bills											
	Recharge Metro Card			Cab			Mobile Phones			Parking-Charge		
	Ram	Suresh	Raj	Ram	Suresh	Raj	Ram	Suresh	Raj	Ram	Suresh	Raj
January	2340	1900	1130	1450	2450	3150	1930	3230	6500	1440	2340	3450
Feb	1240	2340	3210	2700	2200	1350	1510	1340	3500	1640	2210	3250
March	1560	4320	2110	8600	1500	9800	2320	4420	1320	1430	5320	3320
April	870	1230	1240	1240	1500	1160	2130	3240	1840	2450	1340	1250
May	2210	1040	1560	2350	1030	1310	1430	5320	1430	3240	4320	5430

41. If monthly income of Ram is $10000/3\%$ of his bill on metro card in month of March and monthly income of Suresh is $\frac{20000}{3}\%$ of his bill on Cab in March then find the difference between their monthly income.
 (a) 50,000 (b) 48,000 (c) 52,000 (d) 36,000 (e) 38,000
42. Average bill of Ram in month of April is what percent more or less than the average bill of Raj in month of Feb (approximately)
 (a) 35% (b) 48% (c) 52% (d) 41% (e) 32%
43. If expenditure on these bills for Suresh in February month is 20% of his total expenditure in February and total expenditure in February month is half of his income for February month then his income in February is?
 (a) 72000 (b) 80520 (c) 80900 (d) 64500 (e) None of these

44. What is the difference in total bill of Raj for Recharge of metro and total bill of Ram for February month in these bills.
 (a) 2160 (b) 2450 (c) 2020 (d) 3010 (e) 2260
45. Consider these all bills are for year 2016. If in year 2017 bill for metro card, Cab and mobile phone are increased by 10%, $33\frac{1}{3}\%$ and $\frac{100}{11}\%$ respectively then what will be bill for Raj in march 2017? (approximately)
 (a) 24500 (b) 23200 (c) 19150 (d) 20150 (e) 21150

Direction (46-50): Study the given table carefully to answer the questions that follow:

Number of people Staying in Five Different Localities and the percentage Breakup of Men, Women and Children in Them.

Locality	Total No. of People	Percentage		
		Men	Women	Children
F	5640	55	35	10
G	4850	34	44	22
H	5200	48	39	13
I	6020	65	25	10
J	4900	42	41	17

46. Total number of children staying from locality H and I together are approximately what percent of total number of people staying from all localities together?
 (a) 3% (b) 5% (c) 8% (d) 12% (e) 15%
47. Average number of people from all locality staying together are approximately what percent more than the total children staying from locality G and H together?
 (a) 165% (b) 185% (c) 225% (d) 205% (e) 195%
48. What will be the ratio of number of women and children from locality G and H together to the total number of people from locality I and J together?
 (a) $\frac{1181}{2184}$ (b) $\frac{1221}{1331}$ (c) $\frac{1440}{1443}$ (d) $\frac{3}{5}$ (e) None of these
49. If number of men, women and children who are working are in the ratio 5 : 3 : 2 from locality I and number of women who are working from locality I is 903 then what percentage of people from I are not working?
 (a) 40% (b) 50% (c) 60% (d) 55% (e) 65%
50. If $\frac{1100}{17}\%$ of men from locality G are added to locality F then what will be the percentage of women in locality F (approximately).
 (a) 26% (b) 42% (c) 30% (d) 22% (e) 48%

Directions (51-55): Given below is the table which shows the total number of car in 5 different states, percentage of cars in good condition and number of cars in bad condition

State	Total number of cars	Percentage of cars in Good condition	Cars in Bad condition
A	80,000	60%	20,000
B	90,000	75%	15,000
C	1,20,000	80%	12,000
D	70,000	75%	10,000
E	1,50,000	65%	20,500

Note: Total cars = cars in Good condition + car in Bad condition + others

51. What is the difference between 'other' types of car in state B and E together and cars in bad condition from state A and D together.
 (a) 10500 (b) 9500 (c) 8000 (d) 8500 (e) 9000
52. If $\frac{3}{8}$ of cars in good condition, $\frac{2}{5}$ of cars in bad condition and $\frac{1}{6}$ of cars of other type from state A are driven by females then percentage of cars driven by females in state A is
 (a) 35% (b) 40% (c) 32% (d) 42% (e) 38%
53. What is the difference of average of cars in bad condition from state A, B and D together and average of other types of car from state B, C and E together
 (a) $\frac{5000}{3}$ (b) $\frac{6200}{3}$ (c) $\frac{4850}{3}$ (d) $\frac{6500}{3}$ (e) None of these
54. What is the ratio of cars in good condition from state A and B together to the cars in good condition from state D and E together.
 (a) 231 : 300 (b) 243 : 300 (c) 253 : 257 (d) 197 : 200 (e) 191 : 200

55. What is the average of cars in good condition from all states taken together
 (a) 76800 (b) 65300 (c) 66280 (d) 72300 (e) 75000

Directions (56-60): Study the following table carefully to answer these questions.

Number of workers employed in six units of a factory during the years.						
Unit Year	A	B	C	D	E	F
1998	145	88	115	120	140	135
1999	128	76	122	112	152	132
2000	136	96	132	124	158	140
2001	183	92	125	135	166	126
2002	160	107	140	118	170	146
2003	152	110	148	128	175	150

56. Find the difference between total workers employed by all units in 2001 and total workers employed by all units in 2003.
 (a) 34 (b) 30 (c) 46 (d) 36 (e) None of these
57. Find ratio of workers employed in C and D together in 2001 and workers employed in A and F together in 1998.
 (a) 13 : 14 (b) 14 : 13 (c) 11 : 13 (d) 13 : 11 (e) None of these
58. Total workers employed in unit C throughout years is what approximate percent more or less than the number of workers employed in unit E throughout the years.
 (a) 12.82% (b) 18.6% (c) 11.2% (d) 13 % (e) 14 %
59. In 2000 average number of candidates employed by all the unit together.
 (a) 130 (b) 131 (c) 132 (d) 133 (e) None of these
60. Total number of workers employed by unit B in all the years is approximately what percent of total number of candidates employed in all units in year 2003.
 (a) 66% (b) 70% (c) 71% (d) 69% (e) 85%

Directions (61-65): Study the following table carefully and answer the questions that follow:

Number of different types of models produced in different years of a mobile company and percentage of defective mobile of these models is also given:

	Model A		Model B		Model C		Model D	
	Production	% Defective	Production	% Defective	Production	% Defective	Production	% Defective
2011	18,500	6%	17,000	7.5%	23,200	8%	23,500	5%
2012	21,600	5%	21,900	6%	25,000	5.5%	24,600	4.5%
2013	19,700	9%	27,000	4%	19,200	6%	28,500	3%
2014	25,800	4.5%	26,200	5%	28,000	9.5%	21,200	7%
2015	15,800	9%	21,800	7.5%	25,200	11%	18,400	8%
2016	17,100	11%	24,200	8%	26,800	4%	16,000	6%

61. What is the difference between defective model A in 2016 and defective model C in 2013.
 (a) 537 (b) 729 (c) 1039 (d) 1029 (e) None of these
62. Defective mobiles of model A and model C together in 2012 is what percent of total production of model C in 2012.
 (a) 9.82% (b) 11% (c) 11.36% (d) 10.32% (e) None of these
63. The approximate average number of defective mobiles produced in 2014 of all the models together.
 (a) 1652 (b) 1655 (c) 1654 (d) 1656 (e) 1658
64. Find ratio of defective mobiles of model A in 2014, 2015, 2016 together and defective mobiles of model B in 2011, 2012, 2013 together
 (a) 744 : 601 (b) 1223 : 991 (c) 4 : 3 (d) 3 : 4 (e) None of these
65. In which year defective mobiles of model B is maximum.
 (a) 2016 (b) 2011 (c) 2013 (d) 2015 (e) 2014

Direction (66-70): Study the following table and answer the questions that follow.

Colleges→	A		B		C	
Year ↓	Total Students Registered	Male out of those Students	Total Students Registered	Male out of those Students	Total Students Registered	Male out of those Students
1995	2500	2250	8000	7600	4400	4250
2000	3800	3250	6250	5800	5000	4600
2005	3400	3000	5900	5400	8720	6750
2010	3000	2800	7500	6900	5500	5300
2015	4600	4100	9000	8100	6600	5500

66. If total students registered in 1995 from college B were 25% more than total students registered in 1990. In 1990 there were 20% less female registered in college B than the total female registered in 1995 and 2000 together from same college. Find the total number of registered male in 1990 from college B.
 (a) 2900 (b) 5300 (c) 4800 (d) 5720 (e) 4410
67. Find the ratio between the total number of female registered from college C in 1995 and 2000 together to the total number of female registered from college A in 2005 and 2010 together.
 (a) 13 : 11 (b) 12 : 11 (c) 11 : 12 (d) 11 : 13 (e) none of these
68. Average number of students registered in 2010 and 2015 together from college A is what percent less/more than the average number of students registered in 2005 and 2010 together from college B?
 (a) $43\frac{19}{67}$ (b) $41\frac{53}{67}$ (c) $43\frac{43}{67}$ (d) $43\frac{53}{67}$ (e) none of these
69. Find the difference between the total number of students registered from college C in all of the given years to that of college B?
 (a) 6430 (b) 5575 (c) 6650 (d) 6230 (e) none of these
70. Total female registered from college B in 2015 are what percent of the total female registered from the same college in 2010?
 (a) 100% (b) 50% (c) 125% (d) 140% (e) 150%

Directions (71 – 75) : The following table shows cities and post wise number of candidates appeared in competitive exam conducted by IBPS in 2016, study it carefully and answer the following questions

Post→ Centre↓	Officer	Clerks	Field Officer	Supervisor	Specialist Officer
Bangalore	11000	26750	1290	11150	5995
Delhi	15500	38790	1680	7550	8232
Mumbai	22580	32000	1920	8950	3120
Hyderabad	14900	52525	2125	4385	4822
Kolkata	11360	33225	2375	5795	3980
Lucknow	35500	42650	2570	9725	2282
Chennai	9550	15370	2980	4320	4554

71. The difference between the candidates appeared for clerk in Delhi and Mumbai together and candidates appeared for specialist officer in Bangalore, Delhi and Mumbai together.
 (a) 53543 (b) 45443 (c) 53443 (d) 55443 (e) None of these
72. Find the average of candidates who appeared for field officers in all the cities together. (approx.)
 (a) 2134 (b) 3217 (c) 2029 (d) 2190 (e) 2120
73. Total number of candidates who appeared in exam from Hyderabad is what percent of total number of candidates who appeared from Lucknow approximately.
 (a) 70% (b) 80% (c) 85% (d) 90% (e) 75%
74. Ratio of number of candidates who appeared for field officer in Mumbai and number of candidates who appeared for specialist officer in Kolkata is
 (a) 100 : 199 (b) 96 : 109 (c) 96 : 199 (d) 196 : 99 (e) None of these
75. Find the number of total candidates who appeared for specialist officer in all the given cities.
 (a) 33775 (b) 32985 (c) 31735 (d) 32885 (e) None of these

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PRACTICE SET (LEVEL-I) SOLUTIONS

1. (a); Average = $\frac{350 \times \frac{3}{7} + 400 \times \frac{9}{20} + 250 \times \frac{3}{10}}{3} = \frac{150 + 180 + 75}{3} = \frac{405}{3} = 135$
2. (b); No. of boys who played Basketball = $\frac{3}{5} \times 425 = 255$
No. of boys who played Volleyball = $\frac{7}{10} \times 250 = 175$
Req. % = $\frac{255 - 175}{175} \times 100 = \frac{80}{175} \times 100 = 45\frac{5}{7}\%$
3. (c); No. of girls who played Volleyball = $\frac{3}{10} \times 250 = 75$
No. of who played Badminton = $\frac{9}{25} \times 375 = 135$
Req. Difference = $135 - 75 = 60$
4. (d); Req. ratio = $\frac{\frac{200+220}{2}}{\frac{75+240+170}{3}} = \frac{\frac{420}{2}}{\frac{485}{3}} = \frac{126}{97}$
5. (a); Req. sum = $400 \times \frac{11}{20} + 250 \times \frac{7}{10} + 375 \times \frac{9}{25} = 220 + 175 + 135 = 530$
6. (a); Baby boys from district A & B = $\frac{30}{100} \times 450 + \frac{64}{100} \times 500 = 135 + 320 = 455$
Baby girls from district E and F = $\frac{52}{100} \times 650 + \frac{68}{100} \times 525 = 338 + 357 = 695$
Required Difference = $695 - 455 = 240$
7. (b); Average no. of children from E, C & D = $\frac{650 + 470 + 350}{3} = \frac{1470}{3} = 490$
No. of baby boys from D, E and F = $350 \times \frac{36}{100} + 650 \times \frac{48}{100} + 525 \times \frac{32}{100} = 606$
Required % = $\frac{606 - 490}{490} \times 100 = 19.14\%$
8. (c); No. of Baby boys from D & E = $350 \times .36 + 650 \times .48 = 126 + 312 = 438$
No. of Baby girls from D, E & F = $350 \times .64 + 650 \times .52 + 525 \times .68 = 919$
Ratio = $\frac{438}{919}$
9. (d); No. of Baby boys from A = $450 \times .30 = 135$
No. of Baby girls from C = $470 \times .50 = 235$
Required % = $\frac{235 - 135}{135} \times 100 = 74.07\%$
10. (c); No. of Baby girls from C & A = $470 \times .50 + 450 \times .70 = 235 + 315 = 550$
Required ratio = $\frac{500 + 650}{550} = \frac{1150}{550} = \frac{23}{11}$
11. (c); Sell of Freesia ≈ 220
Sell of Jasmine ≈ 280
Sell of Marigold ≈ 220
Sell of Orchids ≈ 50
Sell of Roses ≈ 320
12. (c); Required difference = $\frac{107+163}{2} - \frac{75+97+143}{3} = 30\text{kg}$
13. (e); Required ratio = $\frac{88+103}{97+116} = \frac{191}{213}$
14. (b); Required percentage = $\frac{110}{180} \times 100 = 61\frac{1}{9}\%$

15. (a); Sells of Roses = $\frac{4}{3} \times 120 = 160$
Sells of Jasmine = $\frac{9}{11} \times 110 = 90$
So, required ratio = 16:9
16. (a); Desired ratio = $\frac{600 \times 55 + 490 \times 45 + 750 \times 50}{530 \times 56 + 650 \times 60} = \frac{33000 + 22050 + 37500}{29680 + 39000} = \frac{92550}{68680} = \frac{9255}{6868}$
17. (e); Top three population cities = E, C, D respectively
Total population in C, D and E = 200000
Total men in those cities = 39000 + 27000 + 37500 = 103500
Difference = 200000 - 103500 = 96500
18. (b); Desired value = $\frac{27000 + 37500}{530 \times 44 + 490 \times 45} = \frac{64500}{45370} \approx 1.4$ times
19. (a); Female in C = $650 \times 40 = 26,000$
Female from E = 37,500
Difference = 11,500
Desired % = $\frac{11500}{37500} \times 100 = 30.66\%$ less
20. (c); Two-third of total women population = $\frac{2}{3} (23320 + 22050 + 26000 + 33000 + 37500) = \frac{2}{3} (141870) = 94580$
21. (b); Required % = $\frac{355 - 175}{175} \times 100 = 102.86\%$
22. (d); Required difference = $(350 + 410 + 405) - \frac{575 + 650 + 910}{3} = 1165 - \frac{2135}{3} = \frac{1360}{3} = 453\frac{1}{3}$
23. (c); Required marks = $\frac{120}{100} \times 180 + \frac{115}{100} \times 190 = 216 + 218.5 = 434.5$
24. (e); Required Ratio = 1165 : 1925 = 233:385
25. (a); Required % = $\frac{2310 - 985}{985} \times 100 = 134.51 \Rightarrow \approx 135\%$
26. (b); Total students in Orissa = $2000 \times 4000 = 8000000 = 80$ lakh
Total students in Karnataka = $2500 \times 2400 = 60$ lakh
Required number of girls = $\frac{1}{2} \times 80 + \frac{1}{3} \times 60 = 60$ lakh
27. (a); Total girls in west Bengal = $\frac{3}{8} \times 48$ L = 18 lakh
Total students in UP = $4000 \times 1400 = 56$ lakh
Total boys in UP = $\frac{325}{7}\%$ of 56 = $\frac{325}{100 \times 7} \times 56 = 26$ lakh
Required percentage = $\frac{18}{26} \times 100 = \frac{9}{13} \times 100 \Rightarrow \frac{900}{13}\%$
= $69\frac{3}{13}\%$

28. (e); Number of students per school in West Bengal

$$= \frac{4800000}{960} \Rightarrow = 5000$$

Total students in Karnataka

$$= \frac{5}{4} \times 48 = 60 \text{ lakh}$$

Let new number of schools = x

$$\text{So, } \frac{60,00,000}{x} = 5000$$

$$x = 1200$$

$$\text{Required percentage} = \frac{2400-1200}{2400} \times 100$$

$$= 50\%$$

29. (c); $\frac{500}{9}\%$ $\rightarrow$ 40 lakh

$$100\% \Rightarrow \frac{40 \times 9}{500} \times 100$$

$$\Rightarrow 72 \text{ lakh}$$

$$\text{Total schools in Bihar} = \frac{7200000}{4000}$$

$$= 1800$$

Required average

$$= \frac{960+1400+2000+1800+2400}{5} = 1712$$

30. (b); Students per school in West Bengal

$$= \frac{4800000}{960} = 5000$$

$$\text{Students per school in UP} = \frac{5000}{10} \times 8 = 4000$$

$$\text{Students per school in orissa} = \frac{5000}{10} \times 8 = 4000$$

$$\text{Students per school in Karnataka} = 2500$$

Total students from all states

$$= 48 + 56 + 80 + 72 + 60$$

$$= 316 \text{ Lakh}$$

31. (a); Total arts and commerce student in college P

$$= 1250 \times \frac{72}{100} = 900$$

Let arts student be x

ATQ,

$$x + \frac{140x}{100} = 900$$

$$240x = 90000 \Rightarrow x = 375$$

$$\therefore \text{Commerce student} = 375 \times \frac{140}{100} = 525$$

$$\therefore \text{Required difference} = \frac{3}{7} \times 525$$

$$= 225$$

32. (d); Percentage of engineering student in college T

$$= \frac{360}{1440} \times 100 = 25\%$$

Let commerce student in college T be x

ATQ,

$$x + \frac{125}{100}x = 1080$$

$$\text{Arts student in school T} = \frac{125}{100} \times 480 = 600$$

Arts student in college S

$$= \frac{5}{7} \times 1470 \Rightarrow = 1050$$

$$\therefore \text{Required ratio} = \frac{1050}{600} = 7 : 4$$

33. (e); Total engineering student in college Q

$$= \frac{28}{100} \times 1250 + 150 = 350 + 150 = 500$$

$\therefore$ Total student in college Q

$$= \frac{500}{25} \times 100 = 2000$$

$\therefore$ Required percentage

$$= \frac{2100-2000}{2000} \times 100 \Rightarrow = 5\%$$

34. (d); Let arts student and commerce student in college R be 5x & 8x

$$\therefore 2 \times 13x = 2600$$

$$x = 100$$

Total engineering student in R = 1300

$$\text{Boys in engineering} = \frac{5}{13} \times 1300 \Rightarrow = 500$$

$\therefore$ Total student in College T

$$= 500 \times \frac{20}{100} + 1440$$

$$= 100 + 1440 = 1540$$

35. (c); Total student in college A

$$= \frac{2}{5} \times \frac{28}{100} \times 1250 + \frac{400}{300} \times \frac{25}{100} \times 1260 + \frac{125}{100} \times 1400 \times \frac{2}{7}$$

$$= 140 + 420 + 500$$

$$= 1060$$

36. (a); Runs made by V. Kohli = 16×60

$$= 960$$

Runs made by David Warner

$$= 20 \times \left(100 - 33\frac{1}{3}\right)\% \times 60$$

$$= 20 \times \left(66\frac{2}{3}\right)\% \times 60$$

$$= 20 \times \frac{200}{3 \times 100} \times 60 \Rightarrow = 800$$

Total runs made by all batsman

$$= 960 + 800 + 848 + 570 + 570$$

$$= 3748$$

37. (e); Total 4's and 6's hit by De Villiers = $\frac{160}{100} \times 60 = 96$

Total runs made by 4's and 6's

$$= 96 \times \frac{7}{12} \times 4 + 96 \times \frac{5}{12} \times 6$$

$$= 224 + 240 \Rightarrow = 464$$

$$\text{Required percentage} = \frac{464}{848} \times 100 \approx 55\%$$

38. (b); Average of De Villiers = $\left(100 - \frac{70}{6}\right)\% \times 60$

$$= \frac{530}{6 \times 100} \times 60 = 53$$

$$\text{Average of Shikhar Dhawan} = \frac{570}{15} = 38$$

$$\text{Required difference} = 53 - 38 = 15$$

39. (d); 4's and 6's hit by de Villiers = $\frac{160}{100} \times 60 = 96$

$$\text{Number of 4's hit by de Villiers} = 96 \times \frac{7}{12} = 56$$

$$\text{Number of 4's hit by Virat Kohli} = 121 \times \frac{8}{11} = 88$$

$$\text{Required difference} = 88 - 56 = 32$$

40. (a); Runs made By V. Kohli through 4's and 6's

$$= \frac{8}{11} \times 121 \times 4 + \frac{3}{11} \times 121 \times 6$$

$$= 352 + 198 = 550$$

% of runs made except runs made by 4's and 6's

$$= \frac{16 \times 60 - 550}{16 \times 60} \times 100 \approx 43\%$$

Runs made by G. Gambhir through 4's and 6's

$$= 60 \times \frac{5}{6} \times 4 + 60 \times \frac{1}{6} \times 6$$

$$= 20 \times 4 + 60 = 260$$

% of runs made except runs made by 4's and 6's

$$= \frac{15 \times 38 - 260}{15 \times 38} \times 100$$

$$\text{Required sum} = 55\% + 43\% = 98\%$$

41. (b); Income of Ram in march = $1560 \times \frac{10000}{3 \times 100}$

$$= 52,000$$

$$\text{Income of Suresh in march} = 1500 \times \frac{20000}{100 \times 3}$$

$$= 100,000$$

Required difference = 100,000 - 52,000

$$= 48,000$$

42. (d); Average bill for Ram in April

$$= \frac{870+1240+2130+2450}{4}$$

$$= 1672.5$$

Average bill of Raj in Feb

$$= \frac{3210 + 1350 + 3500 + 3250}{4} = 2827.5$$

$$\text{Required \%} = \frac{1155}{2827.5} \times 100$$

$$\approx 41\%$$

43. (c); Bill for Suresh in February month

$$= 2340 + 2200 + 1340 + 2210 = 8090$$

$$\text{Total Expenditure in February} = \frac{8090 \times 100}{20}$$

$$= 40450$$

$$\text{Income in February} = 80900$$

44. (a); Total bill of Raj for metro recharge

$$= 1130 + 3210 + 2110 + 1240 + 1560 = 9250$$

Total bill of Ram for February month

$$= 1240 + 2700 + 1510 + 1640 = 7090$$

$$\text{Required difference} = 9250 - 7090 = 2160$$

45. (d); Bill for Raj in march 2017

$$= \frac{110}{100} \times 2110 + \frac{4}{3} \times 9800 + \frac{12}{11} \times 1320 + 3320$$

$$\approx 2321 + 13066 + 1440 + 3320$$

$$\approx 20147 \Rightarrow \approx 20150$$

46. (b); Total children from H and I together

$$= 0.13 \times 5200 + 0.1 \times 6020$$

$$= 676 + 602 \Rightarrow = 1278$$

Total people from all localities

$$= 5640 + 4850 + 5200 + 6020 + 4900$$

$$= 26610$$

$$\text{Required percentage} = \frac{1278}{26610} \times 100 = 4.8\%$$

$$\approx 5\%$$

47. (d); Average of people from all localities together

$$= \frac{26610}{5} = 5322$$

Total children from G and H together

$$= 0.22 \times 4850 + 0.13 \times 5200$$

$$= 1067 + 676 = 1743$$

$$\text{Required percentage} = \frac{5322-1743}{1743} \times 100$$

$$= 205.3\% \Rightarrow \approx 205\%$$

48. (a); Number of women and children from G

$$= .66 \times 4850 \Rightarrow = 3201$$

Number of women and children from H

$$= .52 \times 5200 \Rightarrow = 2704$$

Total people from I and J = 6020 + 4900

$$= 10920$$

$$\text{Required ratio} = \frac{3201+2704}{10920}$$

$$= \frac{1181}{2184}$$

49. (b); Total number of working people from I

$$= \frac{903}{3} \times 10$$

$$= 3010$$

$$\text{Required percentage} = \frac{6020-3010}{6020} \times 100$$

$$= 50\%$$

50. (c); Total people in F after addition

$$= 5640 + \frac{11}{17} \times .34 \times 4850$$

$$= 5640 + 1067 = 6707$$

$$\text{Required percentage} = \frac{0.35 \times 5640}{6707} \times 100$$

$$= 29.43\%$$

$$\approx 30\%$$

$$\begin{aligned} 51. (b); \text{Required difference} &= \left(\frac{25}{100} \times 90,000 - 15,000 \right) + \\ &\left(\frac{35}{100} \times 1,50,000 - 20,500 \right) - (20,000 + 10,000) \\ &= 39500 - 30000 = 9500 \end{aligned}$$

$$52. (a); \text{Total female who drove in state A} = \frac{3}{8} \times \frac{60}{100} \times$$

$$80,000 + \frac{2}{5} \times 20,000 + \frac{1}{6} \left(\frac{40}{100} \times 80,000 - 20,000 \right)$$

$$= 18000 + 8000 + 2000 = 28,000$$

$$\text{Required \%} = \frac{28}{80} \times 100 = 28 \times \frac{5}{4} = 35\%$$

53. (d); Average of cars in bad conditions from state A, B and D

$$= (20,000 + 15,000 + 10,000) \frac{1}{3}$$

$$= 15,000$$

Average of other types of car from state B, C and E together

$$= \frac{1}{3} (25 \times 900 - 15000 + 20 \times 1200 - 12000 +$$

$$35 \times 1500 - 20500) = \frac{51500}{3}$$

$$\text{Required Difference} = \frac{51500}{3} - 15000 = \frac{6500}{3}$$

54. (a); Cars in good condition from A and B together

$$= 800 \times 60 + 75 \times 900$$

$$= 48,000 + 67,500 = 1,15,500$$

Cars in good condition from state D and E together

$$= 75 \times 7,00 + 65 \times 1,500$$

$$= 52,500 + 97,500$$

$$= 1,50,000$$

$$\text{Required ratio} = 231 : 300$$

$$55. (d); \text{Required average} = \frac{1}{5} (48000 + 67500 + 96000 + 52500 + 97500) = 72,300$$

56. (d); Total workers employed in all units in 2001

$$= 183 + 92 + 125 + 135 + 166 + 126 = 827$$

Total workers employed in all units in 2003

$$= 152 + 110 + 148 + 128 + 175 + 150 = 863$$

$$\text{Required difference} = 863 - 827 = 36$$

$$57. (a); \text{Required Ratio} = \frac{125+135}{145+135} = \frac{260}{280} = \frac{13}{14}$$

58. (b); Total workers employed in unit C throughout years

$$= 115 + 122 + 132 + 125 + 140 + 148 = 782$$

Total workers employed in E throughout years

$$= 140 + 152 + 158 + 166 + 170 + 175 = 961$$

$$\text{Required percentage} = \frac{(961-782)}{961} \times 100 \approx 18.6\%$$

$$59. (b); \text{Required average} = \frac{136+96+132+124+158+140}{6} = 131$$

60. (a); Total workers employed by B

$$= 88 + 76 + 96 + 92 + 107 + 110 = 569$$

Total worker employed by all units in 2003

$$= 152 + 110 + 148 + 128 + 175 + 150 = 863$$

$$\text{Required percentage} = \frac{569}{863} \times 100 \approx 66\%$$

$$61. (b); \text{Defective model A in 2016} = 17100 \times \frac{11}{100} = 1881$$

$$\text{Defective model C in 2013} = 19200 \times \frac{6}{100} = 1152$$

$$\text{Required difference} = 1881 - 1152 = 729$$

62. (a); Defective mobiles of model A and model C in 2012

$$= 21600 \times \frac{5}{100} + 25000 \times \frac{5.5}{100}$$

$$= 1080 + 1375 = 2455$$

$$\text{Required percentage} = \frac{2455}{25000} \times 100 = 9.82\%$$

63. (c); Total defective mobiles in 2014 = $25800 \times \frac{4.5}{100} + 26200 \times \frac{5}{100} + 28000 \times \frac{9.5}{100} + 21200 \times \frac{7}{100}$
 $= 1161 + 1310 + 2660 + 1484 = 6615$
 Required average = $\frac{6615}{4} = 1653.75 \approx 1654$

64. (e); Defective mobiles of model A in 2014, 2015 and 2016 together
 $= 25800 \times \frac{4.5}{100} + 15800 \times \frac{9}{100} + 17100 \times \frac{11}{100}$
 $= 4464$
 Defective mobiles of model B in 2011, 2012 and 2013 together
 $= 17000 \times \frac{7.5}{100} + 21900 \times \frac{6}{100} + 27000 \times \frac{4}{100}$
 $= 3669$
 Required Ratio = $4464 : 3669$
 $= 1488 : 1223$

65. (a); It is clearly visible that defective mobile of model B in 2016 is maximum.

66. (d); Let total students registered in 1990 = x
 $\therefore \frac{125x}{100} = 8000$
 $x = 6400$
 In 990, total no. of females registered from College B
 $= \frac{80}{100} (400 + 450) = 680$
 $\therefore$ Total no. of registered male in 1990 from college B
 $= 6400 - 680 = 5720$

67. (c); Required Ratio = $(150 + 400) : (400 + 200)$
 $= 550 : 600$
 $= 55 : 60$
 $= 11 : 12$

68. (a); Average no. of students registered in 2010 and 2015 together from college A
 $= \frac{3000+4600}{2} = \frac{7600}{2} = 3800$
 Average no. of students registered in 2005 and 2010 together from college B = $\frac{5900+7500}{2} = 6700$
 Required percentage = $\frac{6700-3800}{6700} \times 100 = 43 \frac{19}{67}\%$

69. (a); From college B = $(8000 + 6250 + 5900 + 7500 + 9000)$
 $= 36650$
 From college C = $(4400 + 5000 + 8720 + 5500 + 6600)$
 $= 30220$
 Required difference = $36650 - 30220 = 6430$

70. (e); Female registered from college B in 2015
 $= 9000 - 8100 = 900$
 Female registered from college B in 2010
 $= 7500 - 6900 = 600$

$$\text{Required percentage} = \frac{900}{600} \times 100 = 150\%$$

71. (c); Number of candidates appeared for clerk in Delhi and Mumbai = $38790 + 32000 = 70790$
 Number of candidates appeared for specialist officer in Bangalore, Delhi and Mumbai
 $= 5995 + 8232 + 3120$
 $= 17347$
 Required difference = $70790 - 17347$
 $= 53443$

72. (a); Total candidates who appeared for field officer
 $= 1290 + 1680 + 1920 + 2125 + 2375 + 2570 + 2980$

$$= 14940$$

$$\text{Required average} = \frac{14940}{7} \approx 2134$$

73. (c); Total number of candidates who appeared in exam in Hyderabad
 $= 14900 + 52525 + 2125 + 4385 + 4822$
 $= 78757$

Total number of candidates who appeared in Lucknow

$$= 35500 + 42650 + 2570 + 9725 + 2282 = 92727$$

$$\text{Required percentage} = \frac{78757}{92727} \times 100 \approx 85\%$$

74. (c); Candidates who appeared for field officer in Mumbai = 1920

Candidates who appeared for specialist officer in Kolkata = 3980

$$\text{Required Ratio} = \frac{1920}{3980} = 96 : 199$$

75. (b); Required number of candidates
 $= 5995 + 8232 + 3120 + 4822 + 3980 + 2282 + 4554$
 $= 32985$

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PRACTICE SET (LEVEL-II)

Directions (1-5): Table given below show the distribution of diaries of two companies (X and Y) sold by five sellers. In this some are work diaries and remaining are school diaries. Study the data carefully and answer the following question.

Sellers	Total Diaries Sold	Work diaries sold (in %)	X:Y (work diaries)	X:Y (school diaries)
A	14000	55%	6:5	5:4
B	8000	65%	7:6	4 : 3
C	5250	40%	5:9	11:4
D	9000	48%	7:5	5: 8
E	12000	64%	9:7	8:7

- Total work diaries of company 'X' sold by A is what percent more than total school diaries of company 'Y' sold by E ?
(a) $110\frac{1}{3}\%$ (b) $108\frac{1}{3}\%$ (c) $106\frac{2}{3}\%$ (d) $112\frac{1}{3}\%$ (e) $105\frac{1}{3}\%$
- Find the difference between the number of work diaries of company 'X' sold by B and C together to the number of school diaries of same company sold C and D together.
(a) 480 (b) 540 (c) 450 (d) 560 (e) None of these
- Total work diaries sold by A, B together of company Y is approximately what percent more than the total school diaries sold by E, D together of company 'X'.
(a) 51% (b) 48% (c) 37% (d) 41% (e) 44%
- Average number of school diaries of company 'X' sold by A and B together is how much less than average number of work diaries of company 'Y' sold by D and E together?
(a) 60 (b) 15 (c) 30 (d) 45 (e) 10
- If B sold 40% defective work diary and 60% defective school diaries then what is the total number of non-defective diary sold by B.
(a) 4240 (b) 4430 (c) 3680 (d) 3880 (e) 4520

Directions (6-10): Given below is the table which shows total number of people who board on and deboard from train at 6-different railway stations.

Railway station	Board	Deboard
Kurukshetra	22800	16000
Gurugram	18400	12000
Panipat	18300	13000
New Delhi	21000	19000
Agra	32000	28000
Mathura	17000	11000

Percentage of male out of total people who board on and deboard from train at each station.

Railway station	% of male in Boarding	% of male in deboarding
Kurukshetra	50	45
Gurugram	60	55
Panipat	45	65
New Delhi	60	70
Agra	55	45
Mathura	70	60

- Ratio of Sum of number of females deboard from train at Kurukshetra and Panipat to the sum of number of males boarding on train from Mathura and New Delhi is:-
(a) 3283 : 9201 (b) 3329 : 7209 (c) 267 : 490 (d) 113 : 169 (e) None of these
- 'x' is the number of male deboarded at Panipat and 'y' is the number of female boarded at Agra. Then 'x' is what percent of 'y'. (Rounded off to two decimal places)
(a) 61.34% (b) 55.44% (c) 52.46% (d) 58.68% (e) 49.46%
- Number of male who board at Panipat railway station is what percent more than the number of male who deboard from train at Kurukshetra railway station.
(a) 14.375% (b) 6.55% (c) 5.58% (d) 4.49% (e) 10.235%

9. Ratio of difference between the number of females boarding at Kurukshetra and Gurugram to the difference between the numbers of male deboard from train at Panipat and New Delhi.
 (a) 404 : 485 (b) 361 : 371 (c) 264 : 229 (d) 461 : 231 (e) None of these
10. Average of number of female who board on train at Mathura and number of males board on Gurugram railway station is.
 (a) 8360 (b) 8260.5 (c) 8070 (d) 9651.5 (e) 8270

Direction (11-15): Given below is the information about candidates who not appeared and candidates qualified/not qualified out of those who appeared in 2 exams IBPS PO PRE and SBI PO PRE in different years from 2006 to 2010.

Years	IBPS PO PRE			SBI PO PRE		
	No. of Candidate who not appeared	% of candidate who not appeared	% of appeared candidates who Qualified	No. of Candidates who not appeared	% of candidate who not appeared	% of appeared Candidates who are not Qualified
2006	50	10%	60%	–	–	70%
2007	200	25%	43%	–	–	55%
2008	–	–	60%	120	30%	40%
2009	480	50%	70%	450	45%	50%
2010	120	24%	–	100	20%	–

Note: - Few values are missing in table, candidate is expected to calculate the missing values if it is required to answer the given questions on the basis of given information in the question.

11. Out of the number of qualified candidates in IBPS PO pre in 2008, the ratio of male to female candidate is 1 : 7. If the number of Female qualified candidates in IBPS PO pre in 2008 is 126. What is the number of appeared candidates (both Male & Female) in IBPS PO PRE in 2008.
 (a) 144 (b) 236 (c) 240 (d) 250 (e) 244
12. The number of appeared candidates in SBI PO PRE is increased by 100% from 2006 to 2007. If total number of qualified candidates in SBI PO PRE in 2006 and 2007 together is 408 then number of appeared candidates in SBI PO PRE in 2006 is what percent of total number of candidates appeared in SBI PO PRE in 2006 to 2010? {Rounded off to 2 decimal places}
 (a) 15.31% (b) 15.11% (c) 15.51% (d) 15.71% (e) 15.91%
13. If 65% candidates in IBPS PO PRE and 35% candidates in SBI PO PRE qualified in 2010, then find the difference between Candidates qualified in IBPS PO PRE in 2009 and 2010 together and candidates qualified in SBI PO PRE in the same years?
 (a) 248 (b) 348 (c) 448 (d) 254 (e) 168
14. If number of appeared candidates in 2006 in IBPS PO PRE was increased by 25% as compared to previous year (2005) and the percentage of qualified candidates for the same is increased by 20% in 2006 as compared to 2005, then find the ratio of qualified candidates in IBPS PO PRE in 2005 to appeared candidates of the same in 2006.
 (a) 2:3 (b) 3:5 (c) 4:5 (d) 2:5 (e) 1 : 2
15. Qualified candidates in IBPS PO PRE in 2006 and 2007 together is approximately what percent more than qualified candidates in SBI PO PRE in 2008 and 2009 together?
 (a) 19% (b) 29% (c) 17% (d) 23% (e) 26%

Direction- (16-20) Given table shows the number of laptop manufactured & percentage of laptop sold by the company P & Q in difference Months. Read the table & find the solution of the given questions.

	Laptops manufactured By Company P	% sold	Laptop manufactured by Company Q	% sold
Jan	450	20%	–	30%
Feb	300	25%	400	20%
March	–	30%	–	40%
April	540	–	650	20%
May	240	15%	350	–

Note: Some data is missing in the table. You have to find out that data according to question.

16. If the total no. of laptop manufactured by P & Q in March is 1000 and total no. of laptop sold by P & Q in March is 340. Then find the difference b/w the no. of laptop manufactured by company P and company Q in March.
 (a) 100 (b) 150 (c) 200 (d) 250 (e) 300
17. Laptop sold by company P in January is how much % less than the laptop sold by company Q in April?
 (a) $10\frac{3}{13}\%$ (b) $30\frac{10}{13}\%$ (c) $44\frac{4}{9}\%$ (d) $44\frac{1}{9}\%$ (e) $30\frac{9}{13}\%$

18. Find out the no. of laptops manufactured by company Q in March if no. of laptops sold by company Q in March is equal to no. of laptops sold by company P in January & May together?
 (a) 305 (b) 310 (c) 314 (d) 316 (e) 315
19. What is the ratio between no. of laptop sold by company P and Q together in February to no. of laptop sold by company P and Q together in Jan if laptop manufactured by company Q in Jan is 320?
 (a) 6 : 5 (b) 5 : 6 (c) 7 : 5 (d) 5 : 7 (e) 5 : 4
20. If the average no. of laptop manufactured by P from Jan to May is 426 then what is no. of laptops manufactured in March?
 (a) 400 (b) 450 (c) 500 (d) 550 (e) 600

Directions (21-25): Study the following table carefully and answer the given questions.

Company	Employees	Male: Female	Senior: Junior	Indian: foreigner	% of employees who got promoted (among senior employees)
Facebook	1200	14:11	18:7	3:1	50%
TCS	1400	13:15	17:11	5:2	36%
Wipro	1600	27:5	5:3	2:3	45%
HCL	1250	17:8	13:12	7:3	34%
L&T	1525	35:26	14:11	4:1	50%
Oracle	1300	9:4	8:17	8:5	50%
Google	1150	17:29	19:4	16:7	20%

Note- No junior employee got promoted from either of the company.

21. Number of employees who got promoted from Google are approximately what percent more/less than one fifth of the number of male employees from TCS, L&T and Oracle together?
 (a) 69% (b) 61% (c) 58% (d) 55% (e) 65%
22. Half of the number of senior employees who got promoted are how much less than the foreigner employees from Wipro, L&T and Google together?
 (a) 500 (b) 495 (c) 498 (d) 505 (e) 508
23. If out of female employees from HCL, $53\frac{1}{4}\%$ are less than 25 years then find the ratio of the number of female employees who are more than 25 years and number of female employees from Facebook.
 (a) 13:47 (b) 47: 13 (c) 48:17 (d) 17:48 (e) 17 : 53
24. Number of employees from all of the companies together are what % of the Indian employees of all of the companies together? {Rounded up to 2 decimal places}
 (a) 131.31% (b) 156.76% (c) 157.76% (d) 149.24% (e) 151.16%
25. Male employees of all of the companies are how much more than the female employees of all of the companies together?
 (a) 2019 (b) 2557 (c) 2575 (d) 1715 (e) 2757

Directions (26 – 30): Study the table & answer the questions that follow:

	Rs. (in millions)				
	1999	2000	2001	2002	2003
OPBDIT	110	285	395	520	380
Interest	30	80	80	110	145
Depreciation	5	20	50	80	120
Net Profit	75	165	205	245	10
Tax	0	20	60	85	105
Other income	1	2	4	10	6

Net Profit = OPBDIT – Depreciation – Interest – Tax

OPBDIT: Operating profit before depreciation, interest & tax

26. For which year is the operating profit after interest and tax but before depreciation the highest?
 (a) 2002 (b) 2003 (c) 2001 (d) 2000 (e) None of these
27. What is the approximate difference between the percentage increase in net profit for the period 1999-2000 and the OPBDIT for the period 1999-2000?
 (a) 30% (b) 39% (c) 23% (d) 27% (e) 49%
28. Which of the following has witnessed a growth across all the years?
 (a) Depreciation & OPBDIT (b) Depreciation & net profit (c) Tax and Depreciation
 (d) Net profit & other income (e) Tax and net-profit

29. Assuming that the amount depreciated in any year is a percentage of the OPBDIT acquired by the company in the previous year, in which year did the company see the maximum percentage of depreciation?
 (a) 2003 (b) 2001 (c) 2002 (d) 2000 (e) Can't determine
30. The average net profit for the period 1999-2003 is what % of the average interest for the same period? (Calculate up to two decimal points)
 (a) 147.28% (b) 140.35% (c) 126.46% (d) 157.30% (e) 152.3%

Directions (31-35): Percentage of people of different age group who cast their vote in the election of five different states. Some values are missing. You have to calculate these values as per given data:

People → States ↓	Age Group				
	(18-25)	(26-35)	(36-50)	(51-70)	(71-100)
Uttar Pradesh	42%	—	18%	12%	—
Goa	—	38%	—	19%	6%
Uttarakhand	47%	31%	17%	—	4%
Manipur	—	35%	19%	17%	8%
Punjab	40%	22%	—	18%	12%

Note: No other age group cast their vote in these five states.

31. If the no. of people who are more than 50 years of age cast their votes from Goa, are 0.5 times the total number of people who are less than 51 years of age cast their vote from Uttarakhand. Then Total number of people who cast their vote from Goa in all age group are what percent of the total number of people who cast their vote from Uttarakhand in all age group?
 (a) 170% (b) 190% (c) 210% (d) 150% (e) 130%
32. If the number of people who cast their vote from U.P. in the age group of (51-70) years are 36 lakhs and the number of people who cast their vote in the age group of (18-25) years from Punjab are 100 lakh then find the ratio of the number of people who cast their vote from UP in age group (18-25) years and the total number of people who are more than 35 years of age cast their vote from Punjab.
 (a) 126 : 95 (b) 85 : 126 (c) 125 : 13 (d) 13 : 126 (e) 17 : 9
33. If total number of people who cast their vote from Punjab are 425 lakhs then number of people of more than 35 years of age who cast their vote from the Punjab are what percent more/less than the total number of people of less than 36 years of age who cast their vote from Punjab? (calculate upto two decimal points)
 (a) 47.24% (b) 43.53% (c) 44.28% (d) 45.16% (e) 38.71%
34. If the number of people who cast their vote from Manipur in the age group of 18-25 are 10.5 lakh and the no. of people who cast their vote from Goa in the age group of 51-70 are 30% more than the total number of people who cast their vote from Manipur. Then find the approximate total number of people who cast their vote from Goa in all age group ?
 (a) 360 lakh (b) 288 lakh (c) 342 lakh (d) 328 lakh (e) 400 lakh
35. Out of the total number of people who cast their vote from UP, 55% of them are female then find the number of female who cast their vote from UP in the age group of (71-100) if total number of people who cast their vote from UP are 300 lakh ?
 (a) 75 lakhs (b) 85 lakhs (c) 95 lakhs (d) Can't determined (e) None of these

Directions(36-40) : The table below shows the data regarding the production and sale of different crops in India in year 2004.

Crops	Production (in %)	Ratio of amount exported and consumed within India	Ratio of sales of amount exported to sales of amount consumed in India	Total sale (in Rs.)
Wheat	37.5%	7 : 8	—	—
Rice	25%	2 : 3	9 : 11	6,00,000
Maize	12.5%	—	7 : 13	4,00,000
Barley	12.5%	4 : 1	—	5,00,000
Sugar	—	—	4 : 5	—
Jute	6.25%	1 : 1	2 : 3	3,12,500

Note: The amount of production of each crop is given as a percentage of total production of all the given crops. Total production of given crops = 20000 metric tones.

36. If the total sale of wheat in 2004 was Rs. 5719000 and the ratio of selling price of one unit of wheat exported to selling price of one unit of wheat consumed within India is 2 : 3, then Find the selling price of one unit of wheat exported from India.
 (a) Rs. 602 (b) Rs. 402 (c) Rs. 301 (d) Rs. 502 (e) Rs. 903

37. If the selling price of one unit of Barley exported is equal to selling price of one unit of Rice consumed in India, then find the total sale from Barely exported from India ?
 (a) Rs. 2,10,000 (b) Rs. 2,20,000 (c) Rs. 2,50,000 (d) Rs. 2,30,000 (e) Rs. 2,22,000
38. The selling price of one unit of Jute exported is what percent of total sales from Maize exported from India in 2004 ? (approximate)
 (a) 0.28% (b) 0.33% (c) 0.14% (d) 0.66% (e) 0.24%
39. The amount of production of sugar is what percent of the sales from Jute consumed within India ? {Rounded up to 2 decimal places}
 (a) 0.28% (b) 0.33% (c) 0.16% (d) 0.66% (e) 0.86%
40. If amount of Maize exported is 25% of amount of Maize consumed in India then price per tonne of maize consumed in India is what percent of price per tonne of Jute consumed in India.
 (a) $66\frac{2}{3}\%$ (b) $33\frac{1}{3}\%$ (c) $43\frac{1}{3}\%$ (d) $53\frac{1}{3}\%$ (e) $45\frac{1}{3}\%$

Directions (41-45): Study the table given below and answer the following questions. The table gives the details about bowling performances of six bowlers.

Bowlers	Overs	Maidens	Runs conceded	Wickets Taken
Lee	20	5	120	15
Bond	18	4	72	9
Steyn	-	6	-	20
Asif	12	-	-	-
Amir	24	-	-	32
Roch	16	3	96	20

Note: Maiden over is one in which a bowler doesn't not concede any run.

41. If the no. of wickets taken by Asif is $16\frac{2}{3}\%$ of total no. of wickets taken by all the remaining bowlers and he conceded 7 runs per over, then find his strike-rate? {strike-rate is defined as the no. of runs conceded per wicket}
 (a) 5.25 (b) 4.25 (c) 6.25 (d) 7.25 (e) 5.5
42. How many more maiden overs should Roch have bowled in order to equal the economy rate of Bond? (Economy rate is defined as the no. of runs conceded per over)
 (a) 4 (b) 6 (c) 2 (d) 8 (e) 10
43. If Amir bowls a maiden for every 3 overs he bowls and concedes $33\frac{1}{3}\%$ more runs that lee did then find the ratio of his maidens per wicket to the runs conceded by him per overs?
 (a) 5 : 71 (b) 3 : 71 (c) 3 : 80 (d) 5 : 81 (e) 3 : 76
44. If lee and steyn collectively bowled 36 overs and conceded 280 runs, then find the ratio of their strike rates. (use information from previous questions if needed)
 (a) 2 : 1 (b) 1 : 1 (c) 3 : 1 (d) 1 : 4 (e) 1 : 2
45. Strike-rate of Amir is what percent of the economy rate of Roch if Runs conceded by Amir is 150% more than runs conceded by Bond? (use information from previous questions if needed)
 (a) 90% (b) 93.75% (c) 95.25% (d) 98.75% (e) 99.95%

Directions (46-50): Study the table and answer the following questions

Given below is the table which shows the books published and sold by two company X and Y.

	Books Published by Company X	% of books sold	Books published by company Y	% of books sold
Jan	-	25%	-	25%
Feb	-	40%	500	22%
Mar	250	40%	300	35%
April	-	25%	-	20%
May	300	-	240	60%

46. What is the average no. of books sold by company Y in the month of Jan, March & April if total books published in Jan & April by company Y is 540 and books sold by Y in Jan & April is equal?
 (a) 72.5 (b) 77.5 (c) 80 (d) 70 (e) 75
47. If books sold by company X in Feb is equal to books sold by company Y in May. Then, what is total no. of books sold by company X & Y in the month of Feb?
 (a) 250 (b) 252 (c) 254 (d) 256 (e) 258

48. What is the total no. of books sold in the month of March & April by company X & Y if there is an increment of 20% & 25% in books published by X & Y in April relative to the previous month respectively?
 (a) 350 (b) 355 (c) 340 (d) 345 (e) 360
49. What will be the average no. of books sold by company X in Jan, Feb & March if the average no. of books published by X in Jan, Feb & March is 450 and books sold in Jan is equals to books sold in March?
 (a) 150 (b) 155 (c) 165 (d) 160 (e) 170
50. If the total books sold by company Y in March, April & May are 300, then what is the ratio of books published in the month of March & April by company Y?
 (a) 17 : 20 (b) 20 : 17 (c) 20 : 13 (d) 6 : 5 (e) None of these

Directions (51-55): Study the following table carefully and answer the question below it

The table given below shows the total number of people in 6 villages, percentage of female who are literate, number of illiterate female and total number of male in year 2016

Some values are missing in table, you have to calculate these values if required to answer the question given below.

Villages	Total Persons (Male + Female)	% of literate Female	Illiterate Female	Total males
A	22,000	40%	6000	12,000
B	18000	55%	4500	-
C	35000	35%	-	15,000
D	21,000	-	3500	11,000
E	12,000	80%	1000	-
F	-	60%	2000	6000

51. What is the average of males from all the villages taken together?
 (a) $\frac{29500}{3}$ (b) $\frac{19700}{6}$ (c) $\frac{25800}{3}$ (d) $\frac{29000}{6}$ (e) None of these
52. What is the ratio of illiterate female from village A, B and C together to the literate female from village D, E and F together?
 (a) 23 : 43 (b) 43 : 23 (c) 27 : 47 (d) 47 : 27 (e) 53 : 43
53. Total literate female from village D is what percent more or, less than total illiterate female from village C
 (a) 25% (b) 40% (c) 45% (d) 50% (e) 35%
54. Total person (Male + female) from village F and E together are what % more or less than total female from village A, B and C together.
 (a) 32.75% (b) 41.5% (c) 42.5% (d) 44% (e) 40%
55. If the illiterate female from village C decreases by $\frac{200}{13}\%$ and literate female from same village increases by $\frac{150}{7}\%$ in year 2017 but total person (Male + female) remains same in village C in year 2017, then number of male in village C is decreased or increased by what percent.
 (a) $3\frac{1}{3}\%$ (b) $16\frac{2}{3}\%$ (c) $2\frac{6}{7}\%$ (d) $3\frac{3}{4}\%$ (e) None of these

Directions (56-60): Given below is a table which gives details about the students of a school ABC in different years. The table gives the ratio of no. of students who play cricket to that of football. The table also gives the ratio of no. of boys to girls.

Year	Total students	Boys : Girls	Boys	Girls
			Cricket : Football	Cricket : Football
2000	600	5 : 7	3 : 2	3 : 4
2001	700	13 : 12	3 : 1	1 : 3
2002	825	3 : 2	4 : 5	4 : 7
2003	650	7 : 6	3 : 4	7 : 3
2004	550	5 : 6	1 : 1	8 : 7

56. What is the difference between the no. of girls playing cricket in 2001 and the no. of boys playing football in 2002?
 (a) 191 (b) 198 (c) 178 (d) 154 (e) None of these
57. What is the sum of average no. of boys playing cricket in 2000, 2002 and 2003 and the average no. of girls playing football in the same years?
 (a) 350 (b) 340 (c) 360 (d) 370 (e) None of these
58. In 2005, the total no. of students increased by $72\frac{8}{11}\%$ as compared to 2004, and the ratio of boys to girls becomes 9 : 10, then find the no. of girls playing football in 2005 if 40% of the girls played cricket in 2005?
 (a) 450 (b) 400 (c) 300 (d) 350 (e) None of these
59. What is the average no. of boys in the school in those years where no. of boys is less than no. of girls?
 (a) 200 (b) 350 (c) 300 (d) 250 (e) None of these

60. Find the ratio of no. of girls playing cricket in 2000, 2002 & 2004 to the no. of boys playing cricket in 2001 and 2003?
 (a) 430 : 423 (b) 430 : 432 (c) 432 : 423 (d) 419 : 423 (e) None of these

Directions (61-65): The following tables show the total number of candidates appeared (in hundred) and percentage of candidates qualified in the written test of IBPS PO mains from 5 different examination centers in different years. Study the tables carefully and answer the following questions:

Centre	Chandigarh	Delhi	Lucknow	Kanpur	Bhopal
IBPS PO I	351	651	451	511	373
IBPS PO II	172	582	523	502	489
IBPS PO III	248	633	564	523	514
IBPS PO IV	283	703	712	541	523
IBPS PO V	365	692	696	583	554
IBPS PO VI	291	592	641	482	573
IBPS PO VII	324	613	563	491	584

Percentage of candidates qualified in the written test from five centers over the years

Centre	Chandigarh	Delhi	Lucknow	Kanpur	Bhopal
IBPS PO I	12	24	18	17	9
IBPS PO II	10	28	12	21	12
IBPS PO III	15	21	23	25	10
IBPS PO IV	11	27	19	24	8
IBPS PO V	13	23	16	23	13
IBPS PO VI	14	20	21	19	11
IBPS PO VII	16	19	24	20	14

61. What percent of total candidates appeared from Chandigarh in all 7 year PO exams, qualified in the written exam ? (rounded off to two decimal points)
 (a) 13.16% (b) 15.32% (c) 15.16% (d) 14.16% (e) 15.56%
62. If 12.5% of the candidates who qualified in the written exam from Delhi in IBPS PO II, finally selected for the job after qualifying in the interview also. Ratio of candidates finally selected for job in PO II from Delhi to that in PO III from Bhopal is 679 : 257. Then what percent of candidates qualifying in the written test in PO III from Bhopal, finally got the job ?
 (a) 12% (b) 16% (c) 15% (d) 20% (e) 14%
63. Total how many candidates qualified the written test of IBPS PO V from all the cities together ?
 (a) 52405 (b) 51805 (c) 52105 (d) 52408 (e) 52608
64. If the ratio of boys to girls in the candidates qualified from Delhi to PO VI is 23 : 17. Then find the boys qualified from Delhi in PO VI is how much more than the girls qualified from the same ?
 (a) 1777 (b) 1776 (c) 1717 (d) 1771 (e) 1756
65. From which city the total number of candidates qualified in written exam of IBPS PO I, PO II and PO III together is 2nd maximum ?
 (a) Chandigarh (b) Delhi (c) Lucknow (d) Kanpur (e) Bhopal

Directions (66-70): Given below is the percent of number of persons from 5 different countries who attended different number of international summits.

	Percentage of people attending summit-1 times	Percentage of people attending summit-2 times	Percentage of people attending summit-3 times	Percentage of people attending summit-4 times	Percentage of people attending summit-5 times	Percentage of people attending summit-6 times
India	18	14	25	7	—	17
Brazil	—	18	22	—	—	12
Russia	18	24	21	—	—	—
South Africa	—	—	28	11	26	9
China	19	—	21	—	37	4

Note: Some values are missing, you have to calculate these values as per given data in the following questions.

66. Number of people from China who attended 2 summits are $11\frac{1}{9}\%$ more than the number of people from the same country who attended 4 summits then no. of people from china who attended at most 2 summits are what percent more/less than the number of people from China who attended at least 4 summits ?
 (a) 40% (b) 42% (c) 44% (d) 46% (e) 48%
67. If the number of people from Russia who attended at most 3 summits is equal to number of people from South Africa who attended at least 3 summits and the total number of people from South Africa in these summits are 25200. Then find the number of people from Russia who attended 3 summits.
 (a) 6216 (b) 6345 (c) 6298 (d) 6275 (e) 6616
68. Total no. of people from India who attended at most 2 summits is equal to the sum of number of people from China who attended 1 summit and the number of people from the same country who attended 6 summits. Then the total number of people in all these summits from India is what percent of total no. of people from China in all these summits?
 (a) $71\frac{5}{8}\%$ (b) $71\frac{7}{9}\%$ (c) $71\frac{3}{8}\%$ (d) $71\frac{7}{8}\%$ (e) $71\frac{1}{8}\%$
69. The number of people from India who attended more than 2 summits is approximately what percent of the number of people from South Africa who attended at least 3 summits if the number of people who attended 3 summits from India is 4800 and the number of people of South Africa who attended 4 summits is 700 more than the number of people from India who attended 3 summits?
 (a) 37% (b) 44% (c) 30% (d) 38% (e) 35%
70. If the difference between number of people from China who attended 3 summits and people who attended 5 summits is 2400, and the total people from Russia in these summits are 60% more than the total no. of people from China in these summits then find the number of people from Russia who are attended 2 summits.
 (a) 6150 (b) 5760 (c) 5280 (d) 5340 (e) 5660

Directions (71-75): Given data show the gross revenue, amount for commission, amount for discount and net revenue of magazine A during 5 month

Note:

Net revenue = gross revenue – amount allocated for commission – amount allocated for discount and others.
 Some value are missing you have to calculate these values according to question.

Revenue data of Magazine A during 5 month				
Month	Gross revenue (In Rs.)	Amount allocated for commission	Amount allocated for discount and others (in Rs.)	Net Revenue In (In Rs.)
March	3,60,000	31,200		
April	3,20,000		16,000	
May			36,000	3,36,000
June		42,000		3,30,000
July		40,000	28,000	

71. In July, if 40% of the gross revenue of magazine was collected from advertisement, what was the amount of gross revenue collected from advertisement in the july month. Given that Net Revenue in July is $\frac{7}{8}$ th of the Net revenue in May.
 (a) 1,48,000 (b) 1,44,800 (c) 1,44,000 (d) 1,72,000 (e) 144600
72. In March, if Net Revenue of the magazine was 85% of its gross revenue, what was the amount allocated for discount and others?
 (a) 23,200 (b) 24,200 (c) 22,400 (d) 22,800 (e) 21,600
73. Amount collected for commission in March is what percent more or less than Amount allocated for commission in April if net revenue in April is 13.75% less than the gross revenue of April (Approx).
 (a) 11% (b) 13% (c) 15% (d) 20% (e) 10%
74. What is the difference b/w net revenue of magazine in pril and its gross revenue in June, if commission in April is 10% of its Gross revenue in April month and discount allocated in June $\frac{1}{11}$ th of the net Revenue of this june.
 (a) 200000 (b) 130000 (c) 100000 (d) 500000 (e) 135000
75. In May, the respective ratio of amount allocated for commission and amount allocated for discount and other was 4 : 3 then, what was the gross revenue of the magazine in May?
 (a) 4,24,000 (b) 4,40,000 (c) 3,86,000 (d) 4,20,000 (e) 4,39,000

PRACTICE SET (LEVEL-II) SOLUTIONS

1. **(b);** Work diaries of company 'X' sold by A
 $= 14000 \times \frac{55}{100} \times \frac{6}{11} \Rightarrow = 4200$
 School diaries of company 'Y' sold by E
 $= 12000 \times \frac{36}{100} \times \frac{7}{15} \Rightarrow = 2016$
 Required percentage
 $= \frac{4200-2016}{2016} \times 100 = 108\frac{1}{3}\%$
2. **(d);** Work diaries of company 'X' sold by B and C together
 $= 8000 \times \frac{65}{100} \times \frac{7}{13} + 5250 \times \frac{40}{100} \times \frac{5}{14}$
 $= 2800 + 750 \Rightarrow = 3550$
 School diaries of company 'X' sold C and D together
 $= 5250 \times \frac{60}{100} \times \frac{11}{15} + 9000 \times \frac{52}{100} \times \frac{5}{13}$
 $= 2310 + 1800 = 4110$
 Required difference = $4110 - 3550 = 560$
3. **(e);** Total work diaries sold by A, B together of company 'Y'
 $= 14000 \times \frac{55}{100} \times \frac{5}{11} + 8000 \times \frac{65}{100} \times \frac{6}{13}$
 $= 3500 + 2400 = 5900$
 Total school diaries sold by E and D together of company X
 $= 9000 \times \frac{52}{100} \times \frac{5}{13} + 12000 \times \frac{36}{100} \times \frac{8}{15}$
 $= 2304 + 1800 = 4104$
 Required percentage = $\frac{5900-4104}{4104} \times 100$
 $= 43.76\% \approx 44\%$
4. **(c);** School diaries of company 'X' sold by A and B together
 $= 14000 \times \frac{45}{100} \times \frac{5}{9} + 8000 \times \frac{35}{100} \times \frac{4}{7}$
 $= 3500 + 1600 = 5100$
 Work diaries of company 'Y' sold by D and E together
 $= 9000 \times \frac{48}{100} \times \frac{5}{12} + 12000 \times \frac{64}{100} \times \frac{7}{16}$
 $= 1800 + 3360 \Rightarrow = 5160$
 Required difference
 $= \frac{5160-5100}{2} = 30$
5. **(a);** Required value
 $= 8000 \times \frac{65}{100} \times \frac{60}{100} + 8000 \times \frac{35}{100} \times \frac{40}{100}$
 $= 3120 + 1120 \Rightarrow = 4240$
6. **(c);** No. of females deboard from train at Kurkshetra and Panipat
 $= 16000 \times \frac{55}{100} + 13000 \times \frac{35}{100}$
 $= 8800 + 4550 = 13350$
 No. of males Boarding on train from Mathura and New Delhi
 $= \frac{17000 \times 70}{100} + 21000 \times \frac{60}{100}$
 $= 11900 + 12600 = 24500$
 Required ratio = $\frac{13350}{24500} = \frac{267}{490}$
7. **(d);** $x = 13000 \times \frac{65}{100} = 8450$
 $y = 32000 \times \frac{45}{100} = 14400$
 Required percentage = $\frac{8450}{14400} \times 100 = 58.68\%$
8. **(a);** No. of male board at Panipat
 $= \frac{18300 \times 45}{100} = 8235$
 No. of male deboard at Kurukshetra
 $= \frac{16000 \times 45}{100} = 7200$
 Required % = $\frac{(8235-7200)}{7200} \times 100 = 14.375\%$
9. **(a);** Difference between the no. of females boarding at Kurkshetra and Gurugram
 $= \frac{22800 \times 50}{100} - \frac{18400}{100} \times 40 \Rightarrow = 4040$
 Difference between the no. of males deboard at Panipat and New Delhi
 $= 19000 \times \frac{70}{100} - 13000 \times \frac{65}{100}$
 $= 13300 - 8450 = 4850$
 Required ratio = $\frac{4040}{4850} = \frac{404}{485}$
10. **(c);** Required Average
 $= \frac{5100+11040}{2} \Rightarrow = 8070$
11. **(c);** Let number of appeared candidates in IBPS PO PRE = x
 No. of qualified candidates $\frac{60}{100} \times x$
 No. of female candidates who qualified = $\frac{7}{(1+7)} \times \frac{60}{100} \times x$
 $x = 126$
 $x = 126 \times \frac{8}{7} \times \frac{100}{60} = 240$
12. **(b);** Let no. of appeared candidates in 2006 = x
 No. of appeared candidates in 2007 = $\frac{(100+100)}{100} \times x$
 $= 2x$
 $\frac{30}{100}x + \frac{45}{100} \times 2x = 408, x = 340$
 Required percentage = $\frac{340}{(340+680+280+550+400)} \times 100$
 $= 15.11\%$
13. **(e);** Candidates qualified from in IBPS PO PRE in 2009 and 2010 together = $\left(\frac{70}{100} \times 480\right) + \left(\frac{65}{100} \times 380\right)$
 $= 583$
 Candidates qualified in SBI PO PRE in 2009 and 2010 together = $\left(\frac{50}{100} \times 550\right) + \left(\frac{35}{100} \times 400\right)$
 $= 275 + 140 = 415$
 Difference = $583 - 415 = 168$
14. **(d);** No. of appeared candidates in 2005 in IBPS PO PRE
 $= 450 \times \frac{100}{125} = 360$
 % of qualified candidates in 2005 in IBPS PO PRE
 $60 \times \frac{100}{120} = 50\%$
 No. of qualified candidates $\frac{50}{100} \times 360 = 180$
 Required ratio = $\frac{180}{450} = 2 : 5$
15. **(a);** Qualified candidates in 2006 and 2007 in IBPS PO PRE = $\left(450 \times \frac{60}{100}\right) + \left(600 \times \frac{43}{100}\right)$
 $= 270 + 258 = 528$
 Qualified candidates in SBI PO PRE in 2008, and 2009 = $\left(280 \times \frac{60}{100}\right) + \left(\frac{550 \times 50}{100}\right)$
 $= 168 + 275 = 443$

$$\text{Required percentage} = \frac{528-443}{443} \times 100 \approx 19\%$$

- 16. (c);** Let, Laptop manufactured by P, (in March) = x
Laptop manufactured by Q, (in March) = y
Given
 $\rightarrow x + y = 1000$
And, $30\% x + 40\% y = 340$
 $3x + 4y = 3400$... (i)
& $x + y = 1000$... (ii)
On solving (i) & (ii)
We get, $x = 600 = \text{laptop manufactured By P (March)}$
 $y = 400 = \text{laptop manufactured by Q (March)}$
Desired difference = $600 - 400 = 200$
- 17. (b);** Laptop sold by P in Jan = $450 \times 20\%$
 $= 90$
Laptop sold by Q in April = $650 \times 20\% = 130$
Desired% = $\frac{130-90}{130} \times 100 = \frac{40}{130} \times 100 = 30\frac{10}{13}\%$
- 18. (e);** Let, laptop manufactured by Q in march = x
Given data
 $x \times 40\% = 450 \times 20\% + 240 \times 15\%$
 $x \times 40\% = 90 + 36$
 $x \times 40\% = 126$
 $x = \frac{126}{40} \times 100 = 315$
- 19. (b);** Desired ratio = $\frac{300 \times 25\% + 400 \times 20\%}{450 \times 20\% + 320 \times 30\%} = \frac{75+80}{90+96} = \frac{155}{186} = \frac{5}{6}$
- 20. (e);** No. of laptops manufactured in March By P
 $= 426 \times 5 - 450 - 300 - 540 - 240$
 $= 600$
- 21. (b);** No. of employees who promoted from Google
 $= 1150 \times \frac{19}{23} \times \frac{20}{100} = 190$
 $\frac{1}{5}$ th. of male employees from TCS, L&T and Oracle
 $= \left(1400 \times \frac{13}{28} + 1525 \times \frac{35}{61} + 1300 \times \frac{9}{13} \right) \frac{1}{5}$
 $= (650 + 875 + 900) \frac{1}{5}$
 $= \frac{2425}{5}$
 $= 485$
Required % = $\frac{485-190}{485} \times 100$
 $= 60.82\% \approx 61\%$
- 22. (c);** No. of senior employees who got promoted
 $= 1200 \times \frac{18}{25} \times 0.50 + 1400 \times \frac{17}{28} \times 0.36 + 1600 \times \frac{5}{8} \times 0.45$
 $+ 1250 \times \frac{13}{25} \times 0.34 + 1525 \times \frac{14}{25} \times 0.50 +$
 $1300 \times \frac{8}{25} \times 0.5 + 1150 \times \frac{19}{23} \times 0.20$
 $= 432 + 306 + 450 + 221 + 427 + 208 + 190$
 $= 2234$
Foreign employees from Wipro, L&T and Google together = $1600 \times \frac{3}{5} + 1525 \times \frac{1}{5} + 1150 \times \frac{7}{23}$
 $= 960 + 305 + 350$
 $= 1615$
Required employees = $1615 - \frac{2234}{2}$
 $= 1615 - 1117 \Rightarrow = 498$
- 23. (d);** Required ratio = $\frac{(400-213)}{400} \times 1250 \times \frac{8}{25} : 1200 \times \frac{11}{25}$
 $= 187 : 528 \Rightarrow = 17 : 48$
- 24. (e);** No. of employees from all of the companies together
 $= 9425$
Indian employees of all of the companies together
 $= 900 + 1000 + 640 + 875 + 1220 + 800 + 800$

$$= 6235$$

$$\text{Required\%} = \frac{9425}{6235} \times 100 \Rightarrow = 151.16\%$$

- 25. (a);** Male employees from all of the companies = $1200 \times \frac{14}{25} + 1400 \times \frac{13}{28} + 1600 \times \frac{27}{32} + 1250 \times \frac{17}{25} + 1525 \times \frac{35}{61} + 1300 \times \frac{9}{13} + 1150 \times \frac{17}{46}$
 $= 672 + 650 + 1350 + 850 + 875 + 900 + 425$
 $= 5722$
 $\therefore \text{female employees} = 9425 - 5722 \Rightarrow = 3703$
Required no. of employees = 2019
- 26. (a);** Year Operating profit after interest of tax but before depreciation
- | | |
|------|-------------------------|
| 1999 | $110 - 30 - 0 = 80$ |
| 2000 | $285 - 80 - 20 = 185$ |
| 2001 | $395 - 80 - 60 = 255$ |
| 2002 | $520 - 110 - 85 = 325$ |
| 2003 | $380 - 145 - 105 = 130$ |
- 27. (b);** Percentage increase in net profit for the period 1999-2000 = $\frac{165-75}{75} \times 100$
 $= \frac{90}{75} \times 100 \Rightarrow = 120\%$
Percentage increase in net OPBDIT for the period 1999-2000 = $\frac{285-110}{110} \times 100$
 $= \frac{175}{110} \times 100 \Rightarrow = 159.09\%$
Difference = $159.09 - 120 = 39.09\%$
- 28. (c);** It can be clearly observed from the table.
- 29. (e);** Since we don't know the OPBDIT of 1998 so we cannot get the answer.
For other years we can calculate percentage depreciation-
- | Year | % depreciation |
|------|--|
| 2000 | $\frac{20}{110} \times 100 = 18.18\%$ |
| 2001 | $\frac{50}{285} \times 100 = 17.54\%$ |
| 2002 | $\frac{80}{395} \times 100 = 20.25\%$ |
| 2003 | $\frac{120}{520} \times 100 = 23.07\%$ |
- 30. (d);** Average net-profit for period 1999-2003
 $= \frac{75+165+205+245+10}{5}$
 $= \frac{700}{5} = 140$
Average interest for period 1999-2003
 $= \frac{30+80+80+110+145}{5} \Rightarrow = \frac{445}{5} = 89$
Required percentage = $\frac{140}{89} \times 100 = 157.30\%$
- 31. (b);** Let total no. of people who cast their vote from Goa = x
Let total no. of people who cast their vote from Uttarakhand = y
 $\therefore (19 + 6)\% \text{ of } x = \frac{1}{2} (47 + 31 + 17)\% \text{ of } y$
 $\frac{1}{4} \times x = \frac{1}{2} \times \frac{95}{100} \times y$
 $\frac{x}{2} = \frac{95}{100} \times y$
 $100x = 190y$
 $20x = 38y \Rightarrow 10x = 19y$
 $\frac{x}{y} = \frac{19}{10}$
Required% = $\frac{19}{10} \times 100$
 $= 190\%$

32. (a); Total no. of people who cast their vote from U.P.

$$= \frac{360}{12} \times 100 = 300 \text{ lakh}$$

Total no. of people who cast their vote from Punjab

$$= \frac{100}{40} \times 100 = 250 \text{ lakh}$$

$$\text{Required Ratio} = \left(\frac{42}{100} \times 300 \right) : \left(\frac{100-40-22}{100} \right) \times 250$$

$$= (42 \times 3) : 95 \Rightarrow = 126 : 95$$

33. (e); Total no. of people who cast their vote from Punjab = 425 lakhs

$$\text{Required}\% = \frac{(40+22)-(100-40-22)}{(40+22)} \times 100$$

$$= \frac{62-38}{62} \times 100$$

$$= \frac{24}{62} \times 100 \Rightarrow = 38.71\%$$

34. (c); Total no. of people who cast their vote from Goa

$$= \frac{100}{19} \left(\frac{130}{100} \times \frac{10.5}{(100-35-19-17-8)} \times 100 \right)$$

$$= 342.10 \Rightarrow \approx 342 \text{ lakhs}$$

35. (d); Since we don't know the percentage of female in individual age group so we can't determine the required no. of females.

36. (a); Amount of wheat exported = $\frac{7}{15} \times \frac{37.5}{100} \times 20000$
= 3500 mt

$$\text{Amount of wheat consumed} = \frac{8}{15} \times \frac{37.5}{100} \times 20000$$

$$= 4000 \text{ mt.}$$

Let, selling price of one unit of wheat exported be Rs. 2x and that consumed be Rs. 3x

$$\text{Then, } 3500 \times 2x + 4000 \times 3x = 5719000$$

$$\text{or, } 19000x = 5719000$$

$$\text{or, } x = 301$$

Selling price of one unit of wheat

$$\text{Exported from India} = \text{Rs. } 2x = \text{Rs. } 2 \times 301 = \text{Rs. } 602$$

37. (b); Amount of rice consumed = $\frac{3}{5} \times \frac{1}{4} \times 20000$
= 3000 mt.

$$\text{Amount of rice exported} = \frac{2}{5} \times \frac{1}{4} \times 20000 = 2000 \text{ mt.}$$

$$\text{Price of one unit of Rice consumed} = \frac{\frac{11}{20} \times 6000000}{3000}$$

$$= \text{Rs. } 110$$

Total sale from Barley exported

$$= \left(\frac{4}{5} \times \frac{1}{8} \times 20000 \right) \times 110$$

$$= \text{Rs. } 2,20,000$$

38. (c); Total sales from Maize exported = $\frac{7}{20} \times 400000$
Rs. 1,40,000

$$\text{Amount of jute exported} = \frac{1}{2} \times \frac{1}{16} \times 20000 = 625 \text{ mt}$$

Selling price of one unit of Jute exported

$$= \frac{\frac{2}{5} \times 312500}{625} = \text{Rs. } 200$$

$$\text{Req.}\% = \frac{200}{140000} \times 100 = 0.14\%$$

39. (d); Amount of sugar produced = $\frac{1}{16} \times 20000 = 1250 \text{ mt}$

Sales from Jute consumed within India

$$= \frac{3}{5} \times 312500 = \text{Rs. } 187500$$

$$\text{Req.}\% = \frac{1250}{187500} \times 100 = 0.66\%$$

40. (c); Let amount of maize consumed in India = x

$$x + \frac{25}{100}x = \frac{12.5}{100} \times 20,000$$

$$= 2500$$

$$\frac{125x}{100} = 2500 \Rightarrow x = 2000$$

$$\text{Total sale of maize in India} = \frac{65}{100} \times 400000$$

$$= 260,000$$

Per tonne price of maize in consumed in India

$$= \frac{2,60,000}{2,000} = 130 \text{ Rs/ tonne}$$

$$\text{Total sale of Jute in India} = \frac{3}{5} \times 3,12,500$$

$$= 1,87,500$$

Amount of jute consumed in India

$$= \frac{6.25}{100} \times \frac{1}{2} \times 20,000 = 625$$

Price per tonne of jute consumed in India

$$= \frac{1,87,500}{625} = 300 \text{ Rs/ tonne}$$

$$\text{Required percentage} = \frac{130}{300} \times 100$$

$$= 43 \frac{1}{3} \%$$

41. (a); Wickets taken by Asif = $\frac{1}{6} \times 96 = 16$

$$\text{Runs conceded by him} = 12 \times 7 = 84$$

$$\text{Strike rate of Asif} = \frac{84}{16} = 5.25$$

42. (d); Economy rate of Bond = $\frac{72}{18} = 4 \text{ runs/over}$

$$\text{Roch's economy rate} = \frac{96}{16} = 6 \text{ runs/over}$$

To equal Bond's rate, Roch needed to bowl 8 more maidens, as

$$\frac{96}{16+8} = \frac{96}{24} = 4 \text{ runs/over}$$

43. (c); No. of maidens bowled by Amir = $\frac{24}{3} = 8$

$$\text{Runs conceded by Amir} = \frac{4}{3} \times 120 = 160$$

$$\text{Req. Ratio} = \frac{\text{Maidens/Wicket}}{\text{Runs/over}} = \frac{8/32}{160/24} = \frac{3}{80}$$

44. (b); overs bowled by steyn = 36 - 20 = 16

$$\text{Runs conceded by steyn} = 280 - 120 = 160$$

$$\text{Strike rate of Lee} = \frac{120}{15} = 8 \text{ runs/wicket}$$

$$\text{Strike - rate of Steyn} = \frac{160}{20} = 8 \text{ runs/wicket}$$

$$\text{Ratio} = 1 : 1$$

45. (b); Economy rate of Roch = $\frac{96}{16} = 6$

$$\text{Strike rate of Amir} = \frac{72 \times \frac{250}{100}}{32} = 5.625$$

$$\text{Required}\% = \frac{5.625}{6} \times 100 = 93.75\%$$

46. (e); Let, books published in Jan = x

Books published in April = y

$$\text{Given, } x + y = 540 \quad \dots(i)$$

$$\& \ 25\% \text{ of } x = 20\% \text{ of } y$$

$$5x = 4y$$

Put in eq. (i)

$$\frac{4y}{5} + y = 540$$

$$\Rightarrow y = 300$$

$$\Rightarrow x = 240$$

Average books sold

$$\frac{240 \times 25\% + 300 \times 35\% + 300 \times 20\%}{3}$$

$$= \frac{60+105+60}{3} = 75$$

47. (c); Books published in Feb by Company X = a

$$a \times 40\% = 240 \times 60\%$$

$$a = 360$$

Total books sold by X & 4 in Feb = $360 \times 40\% + 500 \times 22\% \Rightarrow = 254$

48. (b); Books published by X in April = $250 + 250 \times \frac{20}{100}$
 $= 300$

Books published by Y in April = $300 + 300 \times \frac{25}{100}$
 $= 375$

Total no. of books sold in March & April by X & Y
 $= 250 \times 40\% + 300 \times 35\% + 300 \times 25\%$
 $+ 375 \times 20\%$
 $= 100 + 105 + 75 + 75 = 355$

49. (d); Books published in Jan = x

Books published in Feb = y

$\frac{x+y+250}{3} = 450$

$x + y = 450 \times 3 - 250 = 1100$

$x \times 25\% = 250 \times 40\%$

$\Rightarrow x = 400$

$\Rightarrow y = 700$

Average = $\frac{400 \times 25\% + 700 \times 40\% + 250 \times 40\%}{3}$
 $= \frac{100 + 280 + 100}{3} = 160$

50. (b); $300 \times 35\% + 240 \times 60\% + x \times 20\% = 300$

$x \times 20\% = 51$

$x = 255$

Ratio = $\frac{300}{255} = \frac{20}{17}$

51. (a); Total male from all villages together

$= 12000 + \left(18000 - \frac{4500}{45} \times 100\right) + 15000 +$

$11000 + \left(12000 - \frac{1000}{20} \times 100\right) + 600$

$= \frac{59000}{6} = \frac{29500}{3}$

52. (d); Illiterate female from village A, B and C together

$= 6000 + 4500 + \left(\frac{65}{100} \times 20000\right)$

$= 6000 + 4500 + 13000$

$= 23500$

Literate female from village D, E and F together

$= (21,000 - 11,000 - 3500) + \frac{1000}{20} \times 80 + \frac{2000}{40} \times 60$

$= 6500 + 4000 + 3000$

$= 13500$

Required ratio = 47 : 27

53. (d); Required % = $\frac{\frac{20000}{100} \times 65 - 6500}{\frac{20000}{100} \times 65} \times 100$

$= \frac{6500}{13000} \times 100$

$= 50\% \text{ less}$

54. (c); Total person in village F & E together

$= 12000 + 6000 + \frac{2000}{40} \times 100$

$= 23000$

Total female in village A, B and C together

$= 10,000 + 10,000 + 20,000$

$= 40,000$

Required % = $\frac{17000}{40000} \times 100$

$= \frac{170}{4} \% = \frac{85}{2} \% = 42.5\%$

55. (a); Total illiterate female in 2017 from village C

$= \left(1 - \frac{2}{3}\right) \times 13000 = 11,000$

Total literate female in C in year 2017

$= \left(1 + \frac{3}{14}\right) \times 7000$

$= \frac{17}{14} \times 7000 = 8500$

Total male in village C in 2017 = $35000 - 19500 = 15500$

Required increase in % = $\frac{500}{15000} \times 100 = \frac{10}{3} \%$

56. (a); No. of girls playing cricket in 2001 = $700 \times \frac{12}{25} \times \frac{1}{4} = 84$

No. of boys playing football in 2002 = $825 \times \frac{3}{5} \times \frac{5}{9} = 275$

Difference = $275 - 84 = 191$

57. (b); Average no. of boys playing cricket in 2000, 2002 & 2003

$= \frac{150 + 220 + 150}{3}$

$= \frac{520}{3}$

Average no. of girls playing football in 2000, 2002 & 2003

$= \frac{200 + 210 + 90}{3}$

$= \frac{500}{3}$

Sum = $\frac{500 + 520}{3} = \frac{1020}{3} = 340$

58. (c); No. of students in 2005 = $550 + 550 \times \frac{800}{1100}$

$= 550 + 400$

$= 950$

No. of girls playing football in 2005 = $\frac{10}{19} \times 950 \times \frac{60}{100}$
 $= 300$

59. (d); Such years are 2000 & 2004

Average no. of boys in these two years

$= \frac{250 + 250}{2} = 250$

60. (a); No. of girls playing cricket in 2000, 2002 & 2004

$= 600 \times \frac{7}{12} \times \frac{3}{7} + 825 \times \frac{2}{5} \times \frac{4}{11} + 550 \times \frac{6}{11} \times \frac{8}{15}$

$= 150 + 120 + 160 = 430$

No. of boys playing cricket in 2001 & 2003

$= 700 \times \frac{13}{25} \times \frac{3}{4} + 650 \times \frac{7}{13} \times \frac{3}{7}$

$= 273 + 150 \Rightarrow = 423$

Required Ratio = 430 : 423

61. (a); Total candidates appeared

$= (351 + 172 + 248 + 283 + 365 + 291 + 324) \times 100$

$= 203400$

Total candidates qualified in written test

$= (351 \times \frac{12}{100} + 172 \times \frac{10}{100} + 248 \times \frac{15}{100} + 283 \times \frac{11}{100} +$

$365 \times \frac{13}{100} + 291 \times \frac{14}{100} + 324 \times \frac{16}{100}) \times 100 = 26768$

Required percentage = $\frac{26768}{203400} \times 100 = 13.16\%$

62. (c); Candidates who got job from Delhi in PO II

$= \frac{12.5}{100} \times \frac{28}{100} \times 58200 = 2037$

Candidates who got job from Bhopal in PO III

$= \frac{257}{679} \times 2037 = 771$

Required percentage = $\frac{771}{5140} \times 100 = 15\%$

63. (d); Total candidates qualified in IBPS PO V

$= (36500 \times \frac{13}{100}) + (69200 \times \frac{23}{100}) + (69600 \times \frac{16}{100}) + (58300 \times \frac{23}{100}) + (55400 \times \frac{13}{100}) = 52408$

64. (b); Total candidate qualified in written test in PO VI from

$$\text{Delhi} = 59200 \times \frac{20}{100} = 11840$$

$$\text{Required difference} = \frac{23-17}{23+17} \times 11840 = 1776$$

65. (d); Total candidates qualified from Chandigarh

$$= (351 \times \frac{12}{100} + 172 \times \frac{10}{100} + 248 \times \frac{15}{100}) \times 100 = 9652$$

From Lucknow

$$= (451 \times \frac{18}{100} + 523 \times \frac{12}{100} + 564 \times \frac{23}{100}) \times 100 = 27366$$

From Delhi

$$= (651 \times \frac{24}{100} + 582 \times \frac{28}{100} + 633 \times \frac{21}{100}) \times 100 = 45213$$

From Kanpur

$$= (511 \times \frac{17}{100} + 502 \times \frac{21}{100} + 523 \times \frac{25}{100}) \times 100 = 32304$$

From Bhopal

$$= (375 \times \frac{9}{100} + 489 \times \frac{12}{100} + 514 \times \frac{10}{100}) \times 100 = 14365$$

66. (b); Let no. of people (as percentage) from China who attended 4 seminar = x%

Then no. of people (as percentage) from China who attended 2 seminar

$$= (100 + 11\frac{1}{9})\% \text{ of } x$$

$$= \frac{1000}{900} \times x = \frac{10}{9} x\%$$

$$\therefore x + \frac{10}{9} x = (100 - 19 - 21 - 37 - 4)$$

$$= \frac{19x}{9} = 19 \Rightarrow x = 9$$

$$\text{Required \%} = \frac{(9+37+4)-(19+10)}{9+37+4} \times 100$$

$$= \frac{50-29}{50} \times 100$$

$$= \frac{21}{50} \times 100 = 42\%$$

67. (a); Let total no. of people from Russia = x

$$(18 + 24 + 21)\% \text{ of } x = \frac{(28+11+26+9)}{100} \times 25200$$

$$\frac{63}{100} \times x = \frac{74}{100} \times 25200$$

$$x = 74 \times 400 \Rightarrow x = 29600$$

$$\text{Required no. of people} = 296 \times 21 = 6216$$

68. (d); Let from India = x

From China = y

$$(18 + 14)\% \text{ of } x = (19 + 4)\% \text{ of } y$$

$$32 \times x = 23y \Rightarrow \frac{x}{y} = \frac{23}{32}$$

$$\text{Required \%} = \frac{23}{32} \times 100 = 71\frac{7}{8}\%$$

69. (e); India = $\frac{4800}{25} \times 100$

$$= 4800 \times 4 = 19200$$

$$\text{South Africa} = \frac{(4800+700)}{11} \times 100 = 5000$$

$$\text{Required \%} = \frac{\frac{(100-32) \times 19200}{100}}{\frac{(28+11+26+9)}{10} \times 50,000} \times 100$$

$$= \frac{13056}{370} \approx 35\%$$

70. (b); China = $\frac{2400}{16} \times 100 = 15000$

$$\text{Russia} = 1.6 \times 15000 \Rightarrow = 24000$$

$$\text{Required no. of people} = \frac{24}{100} \times 24000 = 5760$$

71. (b); Net revenue in July = $\frac{7}{8} \times 336000 = 294000$

Gross revenue in July = Net Revenue + Commission + Discount

$$= 294000 + 40,000 + 28,000 = 362000$$

Gross revenue from ads.

$$= \frac{40}{100} \times 362000 = 144800$$

72. (d); Net revenue

$$= \frac{85}{100} \times 360000 = 306000$$

Amount allocated for discount and others

$$= 360,000 - 306000 - 31200 \Rightarrow = 22800$$

73. (a); Net revenue in april

$$= \frac{320,000 \times 86.25}{100}$$

$$= 276000$$

Commission in April = 28,000

$$\text{Required \%} = \left(\frac{31200-28000}{28000} \right) \times 100$$

$$\approx 11\%$$

74. (b); Net revenue in April

$$= 320000 - 16000 - \frac{10}{100} \times 320000$$

$$= 272000$$

Gross revenue in June = 330000 + 42000 + 30000

$$= 402000$$

Required difference = 130000

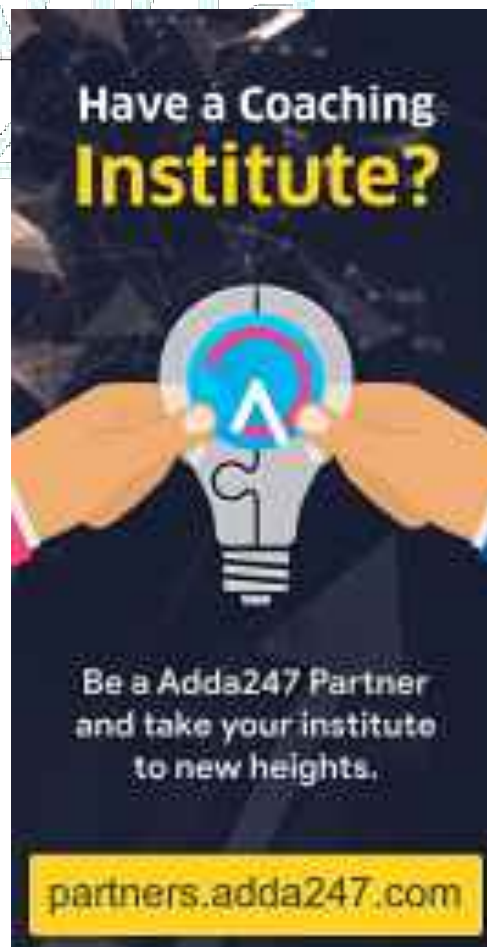
75. (d); Amount allocated for commission in May

$$= \frac{4}{3} \times 36000 = 48,000$$

Gross revenue = 48000 + 36000 + 336000

$$= 420000$$

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Pie Graph

Pie Graph are specific type of data representation where the data is represented in the form of a circle. The circle is divided into various segments or sectors. The circle represents the total value and the different segments or sectors represent certain proportions (degree or percentage value) of the total. The value of each component is in proportion to the circular area representing the component. This chart is used to show the break-up of one variable into its component parts. This chart is less versatile as compared to other representation format like table, bar graph or graph because it can represent only one variable at a time.

This chapter contains:

- Solved Examples
- Previous Year Questions
- Practice Set Level I
- Practice Set Level II

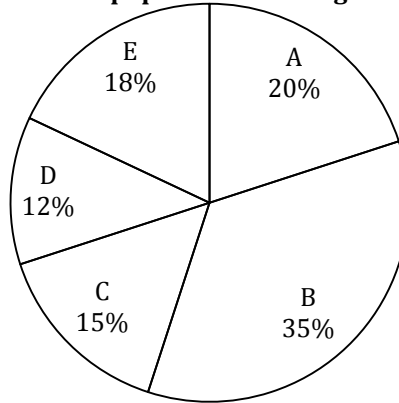
SOLVED EXAMPLES

Directions (1-5): Read the following pie chart carefully and answer the following questions —

In this pie-chart male population of five villages is given.

Note : Ratio between the total male population and total female population of the given villages is 10 : 7.

Total male population of villages = 2500



1. If the ratio between the male population and female population of village B is 7 : 3 then the female population of village B is what percent of the total female population of given villages ?

(a) $19\frac{1}{7}\%$ (b) $21\frac{3}{7}\%$ (c) $23\frac{4}{7}\%$ (d) $27\frac{1}{3}\%$ (e) $21\frac{4}{7}\%$

Sol. (b); Female population of village B = $\frac{3}{7} \times \frac{35}{100} \times 2500 = 375$

$$\text{Required \%} = \frac{375}{1750} \times 100 = 21\frac{3}{7}\%$$

2. Total male population of village A and B together are what percent more/less than the total male population of village C and D together?

(a) $101\frac{2}{19}\%$ (b) $109\frac{1}{17}\%$ (c) $105\frac{7}{17}\%$ (d) $103\frac{19}{27}\%$ (e) None of these

Sol. (d); Required % = $\frac{(35+20)-(15+12)}{(15+12)} \times 100 = \frac{55-27}{27} \times 100 = 103\frac{19}{27}\%$

3. Find the difference between the average male population of village B and C together and the average male population of D and E together?

(a) 150 (b) 200 (c) 250 (d) 300 (e) 280

Sol. (c); Required difference = $\left[\frac{(35+15)}{2} - \frac{(12+18)}{2} \right] \times 25 = (25 - 15) \times 25 = 10 \times 25 = 250$

4. Ratio of female population of village C to that of village D is 2 : 3 and the ratio of female population of village D to male population of village E is 4 : 5 then find the total female population of village D and C together ?

(a) 300 (b) 750 (c) 120 (d) 450 (e) 600

Sol. (e); Female population of village D = $\frac{450}{5} \times 4 = 90 \times 4 = 360$

$$\text{Female population of village C} = \frac{360}{3} \times 2 = 240$$

$$\text{Required population} = 360 + 240 = 600$$

5. Male population of village G is $7\frac{7}{9}\%$ more than the male population of village E then male population of village G is what percent of the total male population of the given villages?

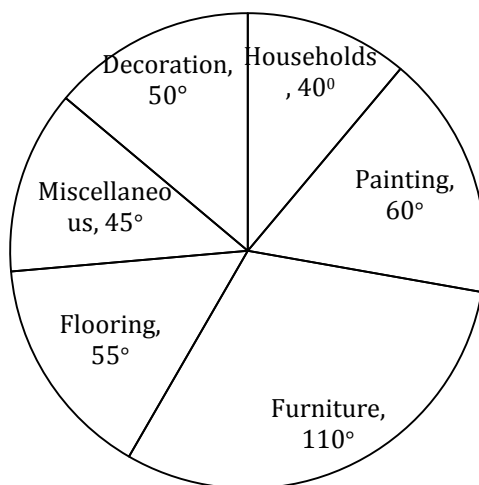
(a) 24.8% (b) 18.1% (c) 21.2% (d) 19.4% (e) 15.2%

Sol. (d); Male population of village G = $107\frac{7}{9}\%$ of 450 = 485

$$\text{Required \%} = \frac{485}{2500} \times 100 = 19.4\%$$

Directions (Q.6-10): Study the following pie chart and answer the questions that follow it
Given below is the pie chart which shows the distribution of expenditure of a man in degree.

Total Expenditure of 2017 = 2,50,000



6. Expenditure on household and flooring is what percent of the total expenditure (approximately)?
(a) 26% (b) 36% (c) 20% (d) 22% (e) 24%

Sol. (a); Central angle of household and flooring together = $40^\circ + 55^\circ = 95^\circ$
Overall central angle = 360°
 $\therefore$ percentage of income spend on household and flooring together = $\frac{95^\circ}{360} \times 100\% = 26.33 \approx 26\%$

7. What is the ratio of expenditure on miscellaneous and painting together to the expenditure on flooring and furniture together?

(a) 33 : 21 (b) 7 : 11 (c) 7 : 8 (d) 23 : 31 (e) 11 : 7
Sol. (b); Angle of miscellaneous and painting together = $45^\circ + 60^\circ = 105^\circ$
Angle of flooring and furniture together = $55^\circ + 110^\circ = 165^\circ$
 $\therefore$ ratio = $\frac{105^\circ}{165^\circ} = \frac{7}{11}$

8. What is difference between the expenditure on decoration and furniture together to the expenditure on households?

(a) Rs 83,333.33 (b) Rs 84,333.33 (c) Rs 86,333.33 (d) Rs 82,333.33 (e) Rs. 85333.33
Sol. (a); Angle of decoration & furniture together = $50 + 110 = 160^\circ$
Angle of household = 40°
Difference of angle = $160^\circ - 40^\circ = 120^\circ$
 $\therefore$ expenditure = $\frac{120^\circ}{360^\circ} \times 250,000 = \text{Rs } 83,333.333$

9. If expense on painting is Rs 30,000 in 2018 then what is the percentage of expenditure on painting to total expenditure if total expenditure remains the same?

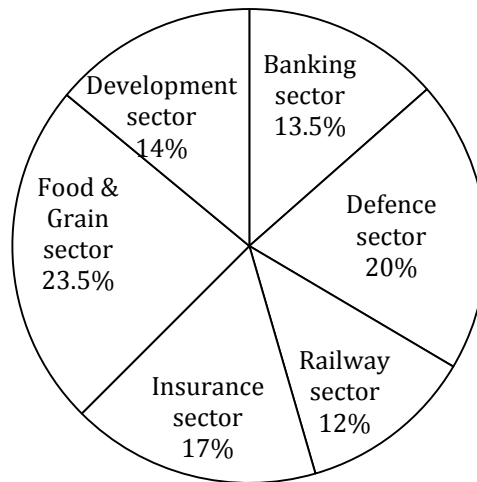
(a) 10% (b) 16% (c) 12% (d) 18% (e) 15%
Sol. (c); Expenditure on painting = Rs 30,000
Total expenditure = Rs 2,50,000
 $\therefore$ Percentage of expenditure on painting to total = $\frac{30,000}{2,50,000} \times 100 = 12\%$

10. If the expenditure on each decoration & household is increased by 20% in 2018, then what is the total expenditure on decoration and household in 2018?

(a) Rs 63,500 (b) Rs 62,500 (c) Rs 75,000 (d) Rs 66,500 (e) None of these
Sol. (c); Angle of decoration & household together = 90°
Expenditure on decoration and household together in 2017 = $\frac{90^\circ}{360^\circ} \times 2,50,000 = \text{Rs } 62,500$
For 20% increase = $62,500 \times \frac{20}{100} = \text{Rs } 12,500$
 $\therefore$ Total expenditure on decoration and household in 2018 = $62500 + 12500 = 75,000$

Directions (11-15): Pie-chart given below shows the investment of a government entity (in crore) in different sectors. Study the pie-chart and answer the following.

Total Investment = 1,000 crores



11. What is the ratio of average investment on banking and defence sector to the average investment on insurance, railway and development sector.

(a) 172 : 201 (b) 201 : 172 (c) 67 : 86 (d) 86 : 67 (e) 65 : 86

Sol. (b); Average investment on Banking and defence = $\frac{135+200}{2} = \frac{335}{2}$ crore

Average investment on Insurance, Railway & Development = $\frac{170+120+140}{3} = \frac{430}{3}$ crore

Desired Ratio = $\frac{\frac{335}{2}}{\frac{430}{3}} = \frac{335}{2} \times \frac{3}{430} = \frac{201}{172}$

12. What is the percentage of investment in banking sector to the investment in Railway sector:

(a) 110% (b) 114% (c) 112% (d) 116% (e) None of these

Sol. (e); Desired% = $\frac{135}{120} \times 100 = 112.5\%$

13. What is the central angle for Food and Grain sector ?

(a) 84° (b) 84.2° (c) 84.4° (d) 84.6° (e) 84.8°

Sol. (d); Investment on Food & Grain in % = $\frac{235}{1000} = 23.5\%$

In Central Angle = $23.5 \times \frac{18}{5} = 84.6^\circ$

14. If the investment in Railway and Defence sector is increases by 15% and 25% respectively then how much percentage increase in total investment(in percentage) ?

(a) 6.8% (b) 7% (c) 7.2% (d) 6.6% (e) 6.4%

Sol (a); Increase in Railway = $120 \times \frac{15}{100} = 18$ Crore

Increase in defence = $200 \times \frac{25}{100} = 50$ Crore

Total Increase = 68 Cr.

Desired % = $\frac{68}{1000} \times 100 = 6.8\%$ increase

15. If the government reduces the investment on Defence sector by 20% and distributed this money on Railway and insurance sector in the ratio 5 : 3 then investment in insurance sector changes by what percentage ?(approximately).

(a) 8% decrease (b) 8.8% increase (c) 10.2% increase (d) 11% increase (e) 10.5% decrease

Sol. (b); Reduction on Defence sector = $200 \times \frac{20}{100} = 40$ Crore

Investment in Railway and Insurance sector be 5x and 3x respectively

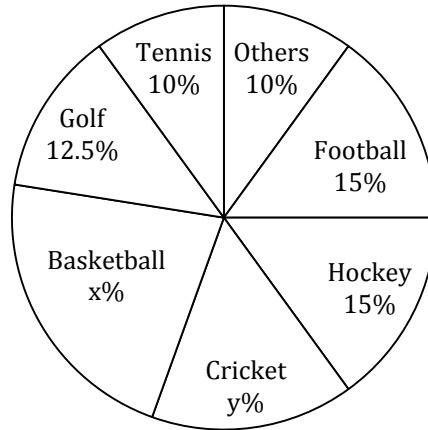
Total = $8x = 40 \Rightarrow x = 5$

Increase in Insurance sector = $3 \times 5 = 15$ Crore

Effect on insurance sector = $\frac{15}{170} \times 100 \approx 8.8\%$ increase

Directions (16-20): Given below is the pie chart which shows the percentage expenditure issued by government on different sports in a state in year 2016

Total Expenses = 500 Lakhs



16. What is the ratio of expenditure on Football and Golf together to the expenditure on Hockey and Tennis together ?
 (a) 11 : 10 (b) 9 : 10 (c) 10 : 11 (d) 11 : 12 (e) 5 : 6

Sol. (a); Required ratio = $\frac{\frac{27.5}{100} \times 500}{\frac{25}{100} \times 500} = \frac{27.5}{25} = 11 : 10$

17. What is the difference between average of expenditure on sports Golf, Football together to the average of expenditure on sport tennis and Hockey together ?
 (a) 4.25L (b) 10.25L (c) 6.25L (d) 5.5L (e) 7.25L

Sol. (c); Average expenditure on football and Golf together = $\frac{27.5 \times 5}{2} = 68.75$
 Average of expenditure Tennis and Hockey = $\frac{25 \times 5}{2} = 62.5$
 Required difference = 6.25 L

18. If in year 2017 expenditure on Cricket and basketball increases by 20% and 12% than the previous year respectively and ratio of expenditure between these two sports in 2016 is 2 : 1 then find the total expenditure of these two sports in 2017.
 (a) 180 L (b) 310 L (c) 285 L (d) 220 L (e) 170 L

Sol. (d); Expenditure on Cricket in 2016 = $\left\{ \frac{(100\% - 62.5\%) \times 500}{3} \right\} \times 2$
 $\frac{37.5 \times 5}{3} \times 2 = 12.5 \times 5 \times 2 = 125 L$
 Expenditure on Basketball in 2016 = $\frac{37.5 \times 5}{3} = 12.5 \times 5 = 62.5 L$
 Required expenditure in 2017 = $\frac{120}{100} \times 125 + \frac{112}{100} \times 62.5$
 $= 150 + 70 = 220$

19. If expenditure on football and Hockey increases 20% and 25% in 2017 than that in 2016 respectively then what is the total expenditure for these two sports in 2017 ?
 (a) 120 .65 L (b) 170.50 (c) 183.75 (d) 190.00 (e) 201.5

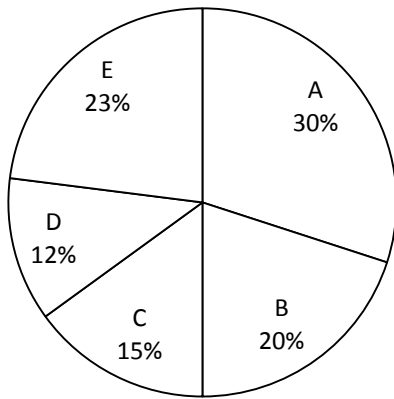
Sol. (c); Required expenditure = $15 \times 5 \times \frac{120}{100} + 15 \times 5 \times \frac{125}{100}$
 $= 90 + 93.75 = 183.75$

20. Total expenditure of Tennis and football together in 2016 is what percent of total expenditure on all, sports in 2017 if in 2017 expenditure on all sports increases by 20% than that in 2016 ?

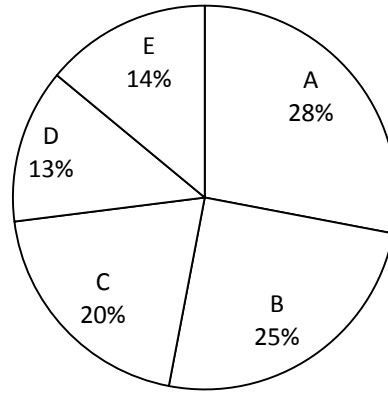
(a) $20\frac{5}{6}\%$ (b) $30\frac{2}{3}\%$ (c) $13\frac{7}{9}\%$ (d) $14\frac{2}{7}\%$ (e) $16\frac{2}{3}\%$

Sol. (a); Required percentage = $\frac{(10\% + 15\%)500}{120\% \times 500} \times 100 = 20\frac{5}{6}\%$

Directions (21-25): Given below are two pie charts, first pie chart shows the percentage distribution of total population of five cities in 2016 and second pie chart shows the percentage distribution of male population in these five cities in 2016



Population of five cities in 2016
Total population 6,00,000



Male population of five cities in 2016
Total population = 4,00,000

21. What is the total number of females in city C and E together ?

- (a) 1,00,000 (b) 98,000 (c) 96,000 (d) 94,000 (e) 92,000

Sol. (e); Total population in city C and E = 38% of 6,00,000 = 2,28,000
Total male population in city C and E = 34% of 4,00,000 = 1,36,000
Total no. of females in city C and E = 2,28,000 – 1,36,000 = 92,000

22. What is the ratio between males in city B and D together to females in city A and B together ?

- (a) 11 : 7 (b) 7 : 11 (c) 11 : 14 (d) 19 : 11 (e) 14 : 9

Sol. (d); Males in B and D together = 38% of 4,00,000 = 1,52,000
Total population in A and B = 50% of 6,00,000 = 3,00,000
Total male population in city A and B = 53% × 4,00,000 = 2,12,000
Total female population in A and B in = 88,000
Ratio = $\frac{1,52,000}{88,000} = \frac{19}{11}$

23. If in 2017, population of city D and C increases by 10% and 15% respectively over previous year and the male population is increased by 15% and 20% respectively over previous year, then find the ratio between number of females in city D to number of females in city C in year 2017?

- (a) 191 : 75 (b) 75 : 194 (c) 194 : 75 (d) 75 : 191 (e) None of these

Sol. (c); Population of D in 2017 = $6,00,000 \times \frac{12}{100} \times \frac{110}{100} = 79,200$
Male population of D in 2017 = $4,00,000 \times \frac{13}{100} \times \frac{115}{100} = 59,800$
Female population in D in 2017 = 79,200 – 59,800 = 19,400
Population of C in 2017 = $6,00,000 \times \frac{15}{100} \times \frac{115}{100} = 1,03,500$
Male population in C in 2017 = $4,00,000 \times \frac{20}{100} \times \frac{120}{100} = 96,000$
Female population in C in 2017 = 1,03,500 – 96,000 = 7,500
Ratio = $\frac{19,400}{7,500} = \frac{194}{75}$

24. What was the population of city C in 2014, if population increase at the rate of 20% annually.

- (a) 62,000 (b) 62,500 (c) 63,000 (d) 64,000 (e) 64,500

Sol. (b); Let, population in city C in 2014 = x
 $x \times \left[1 + \frac{20}{100}\right] \left[1 + \frac{20}{100}\right] = 6,00,000 \times 15\%$
 $x \times \frac{120}{100} \times \frac{120}{100} = 90,000 \Rightarrow x = 62,500$

25. Number of females in city C is how much percent more or less than the males in City D [approximately]?

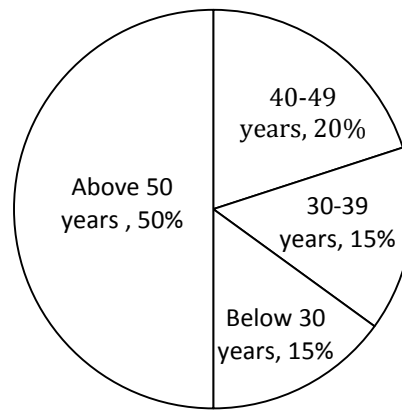
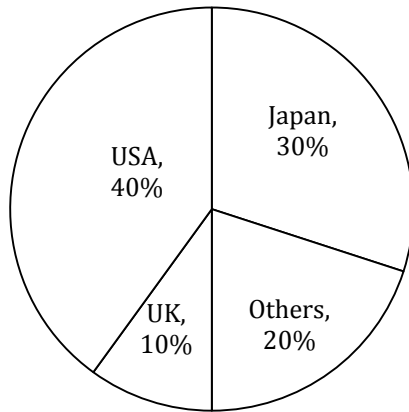
- (a) 19% less (b) 81% more (c) 81% less (d) 19% more (e) 27% more

Sol. (c); No. of females in city C = $6,00,000 \times \frac{15}{100} = 90,000$
No. of males in D = $4,00,000 \times \frac{20}{100} = 80,000$
% = $\frac{90,000 - 80,000}{80,000} \times 100 = \frac{10,000}{80,000} \times 100 = 12.5\% \approx 13\%$
No. of females in city C is 13% more than the number of males in city D.

Direction(26-30)- Study the given pie charts and answer the following questions.

In the first pie chart distribution of overseas tourist traffic from India to different countries is given and in the second pie chart distribution of overseas tourist traffic from India according to age wise is given.

Distribution of Overseas Tourist Traffic from India



26. If the tourist traffic from India to USA is 165000 more than that of UK then overseas tourist traffic in the age group of (40-49) years are how much (in lakh) more/less than the overseas traffic from India in the age group of (30 - 39) years?
 (a) 0.725 lakh (b) 0.275 lakh (c) 0.55 lakh (d) 0.527 lakh (e) 0.42 lakh

Sol. (b); Given
 $30\% \rightarrow 165000$
 $1\% \rightarrow 5500$
 $100\% \rightarrow 550000$
 $\therefore$ Total overseas tourist from India = 550,000
 Then,
 $(20 - 15) = 5\%$ of 550,000 = $5 \times 5500 = 27500 = 0.275$ lakh

27. The ratio of the number of Indian tourists that went to USA to the number of Indian tourists who were below 40 years of age is :
 (a) 2 : 1 (b) 8 : 3 (c) 3 : 8 (d) Cannot be determined (e) 4 : 3

Sol. (e); Required Ratio = $\frac{40}{15+15} = \frac{40}{30} = 4 : 3$

28. If amongst other countries, Switzerland, accounted for 25% of the Indian tourist traffic, and it is known from official Swiss records that a total of 25 lakh Indian tourists had gone to Switzerland during the year, then find the number of 30-39-year-old Indian tourists who went abroad in that year.
 (a) 18.75 lakh (b) 25 lakh (c) 50 lakh (d) 75 lakh (e) 80 lakh

Sol. (d); $25\% \rightarrow 25$ lakh
 $100\% \rightarrow 100$ lakh
 $\therefore$ total overseas tourist from India = $\frac{100}{20} \times 100 = 500$ lakh
 Then required no. of overseas tourist = $\frac{15}{100} \times 500 = 75$ lakh

29. If amongst other countries, Switzerland, accounted for 25% of the Indian tourist traffic, and it is known from official Swiss records that a total of 25 lakh Indian tourists had gone to Switzerland during the year, then what was the volume of traffic of Indian tourists in the US?
 (a) 150 lakh (b) 125 lakh (c) 200 lakh (d) 225 lakh (e) 230 lakh

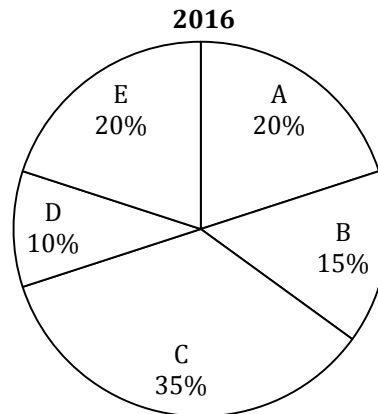
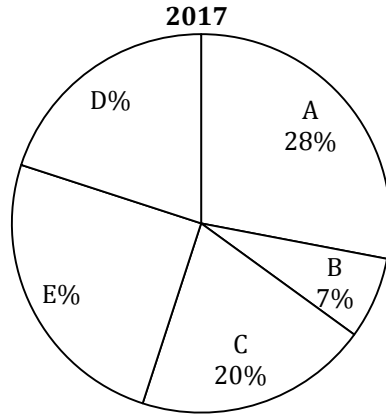
Sol. (c); Total overseas Indian tourist = 500 lakh
 $\therefore$ Required No. of tourist = $\frac{40}{100} \times 500 = 200$ lakh

30. If tourist of age group above 50 years are 3.6 lakh more than the tourist of age group 40-49 years then what is the total no. tourist of age below 40 years ?
 (a) 4.8 lakh (b) 4.2 lakh (c) 3.6 lakh (d) 4.5 lakh (e) 3.2 lakh

Sol. (c); $30\% \rightarrow 3,60,000$
 $100\% \rightarrow \frac{3,60,000}{30} \times 100$
 $\rightarrow 12,00,000$
 $\therefore$ Required answer = $\frac{15+15}{100} \times 12,00,000 = 3.6$ lakh

Directions (31-35): Read the given data carefully and answer the following questions.

Given below are two pie charts which show the percentage distribution of Income of 5 firms in 2016 and in 2017. Percentage distribution for some firms is not given. You have to calculate these value if required



Note: Total income of all firm in 2016 and in 2017 is in the ratio 5 : 7

$$\text{Profit \%} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$

31. What is the ratio of income of A & B together in 2016 to the income of B and C together in 2017.

- (a) 25 : 27 (b) 23 : 25 (c) 12 : 17 (d) 12 : 13 (e) 19 : 20

Sol. (a); Let total income in 2016 = $5x$

Total income in 2017 = $7x$

Income of A & B in 2016 = $35 \times \frac{5x}{100}$

Income of B & C in 2017 = $27 \times \frac{7x}{100}$

Required ratio = $\frac{35 \times 5x}{27 \times 7x} = 25 : 27$

32. If income of D & E in 2017 is in the ratio 4 : 5, then income of E in 2017 is what percent more or less than income of B in 2016.

- (a) $16\frac{2}{3}\%$ (b) $33\frac{1}{3}\%$ (c) $133\frac{1}{3}\%$ (d) $14\frac{2}{7}\%$ (e) None of these

Sol. (c); Income of E in 2017 = $\frac{45}{9} \times 5 \times \frac{7x}{100} = \frac{7x}{4}$

Required % = $\frac{\frac{7x}{4} - \frac{7x}{10}}{\frac{7x}{10}} \times 100 = 133\frac{1}{3}\%$

33. If profit percentage of A in 2016 is equal to profit percentage of C in 2017 and expenditure of A in 2016 is 20 L then what is the expenditure of C in 2017.

- (a) 15 L (b) 18 L (c) 21 L (d) 35 L (e) 28 L

Sol. (e); According to question, $\frac{20 \times \frac{5x}{100} - 20}{20} = \frac{20 \times \frac{7x}{100} - E}{E} \Rightarrow \frac{x-20}{20} = \frac{\frac{7x}{5} - E}{E}$

$Ex - 20E = \frac{140x}{5} - 20E \Rightarrow E = 28 \text{ L}$

34. If difference between income of D in 2016 and income of B in 2017 is 1 L then what is the total income of D and E in 2017.

- (a) 280 L (b) 210 L (c) 315 L (d) 140 L (e) 215 L

Sol. (c); According to question $\Rightarrow 10 \times \frac{5x}{100} - 7 \times \frac{7x}{100} = 1 \Rightarrow x = 100$

Total income of D & E in 2017 = $\frac{45}{100} \times 7 \times 100 = 315 \text{ L}$

35. Income of A and B together in 2016 is what percent of income of C in 2017.

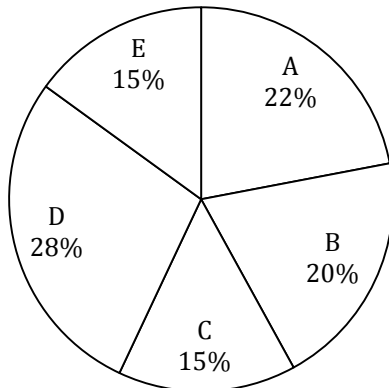
- (a) $66\frac{2}{3}\%$ (b) 80% (c) 100% (d) $133\frac{1}{3}\%$ (e) 125%

Sol. (e); Income of A & B in 2016 = $35 \times \frac{5x}{100}$

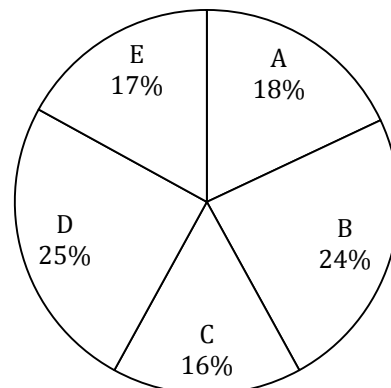
Income of C in 2017 = $20 \times \frac{7x}{100}$

Required percentage = $\frac{35 \times 5x}{20 \times 7x} \times 100 = 125\%$

Directions (36-40): In the following pie-chart Income and expenses of five employees is given. Read the given data & answer the following questions.



Net Income = 4,00,000



Net Expense = 3,00,000

Note- Income = expenditure + saving

36. What is the average saving of B and C together?

- (a) 8,000 (b) 9,000 (c) 10,000 (d) 11,000 (e) 12,000

Sol. (c); $\text{Income of B} = \frac{20}{100} \times 4 \text{ Lakh} = 80,000$

$$\text{Expenses of B} = \frac{24}{100} \times 3 \text{ Lakh} = 72,000$$

$$\text{Saving of B} = 8,000$$

$$\text{Income of C} = \frac{15}{100} \times 4,00,000 = 60,000$$

$$\text{Expenses of C} = \frac{16}{100} \times 3,00,000 = 48,000$$

$$\text{Saving of C} = 12,000$$

$$\text{Average of saving} = \frac{12,000 + 8,000}{2} = 10,000$$

37. If Income of Employee A decreases by 10% and its expenses increase by 20%. Then, his saving changes by what percent? (approximately)

- (a) 42% (b) 50% (c) 58% (d) 62% (e) 68%

Sol. (c); $\text{Income of A} = \frac{22}{100} \times 4,00,000 = 88,000$

$$\text{Expense of A} = \frac{18}{100} \times 3,00,000 = 54,000$$

$$\text{Present saving of A} = 34,000$$

$$\text{Income after decrement} = 88,000 \times \frac{90}{100} = 79,200$$

$$\text{Expenses after increment} = 54,000 \times \frac{120}{100} = 64,800$$

$$\text{Saving after changes} = 14,400$$

$$\% \text{ changes in saving} = \frac{34,000 - 14,400}{34,000} \times 100 = 57.64\% \approx 58\%$$

38. If the income of D increases by 10% and income of E decreases by 20% then what will be the effect on net income of five employees?

- (a) Rs. 800 increase (b) Rs. 1000 Increase (c) Rs. 1000 decrease (d) Rs. 800 decrease (e) None of these

Sol. (d); $\text{Changes in D's salary} = \frac{28}{100} \times 4,00,000 \times \frac{10}{100} = 11,200 (\uparrow)$

$$\text{Changes in E's salary} = \frac{15}{100} \times 4,00,000 \times \frac{20}{100} = 12,000 (\downarrow)$$

$$\text{Effect on Net Income} = \text{Rs. 800 } (\downarrow) = \text{Rs. 800 decrease}$$

39. What is the ratio of saving of A to B ?

- (a) 17 : 4 (b) 4 : 17 (c) 17 : 15 (d) 15 : 17 (e) 12 : 7

Sol. (a); Saving of B = 8,000 [see question 1]

$$\text{Saving of A} = 34,000 \text{ [see question 2]}$$

$$\text{Ratio} = \frac{34,000}{8,000} = \frac{17}{4}$$

40. If expenses of C increase by 15% then to how much percentage of increment is necessary in his income to keep his saving same as before?

- (a) 10% (b) 12% (c) 15% (d) 8% (e) 6%

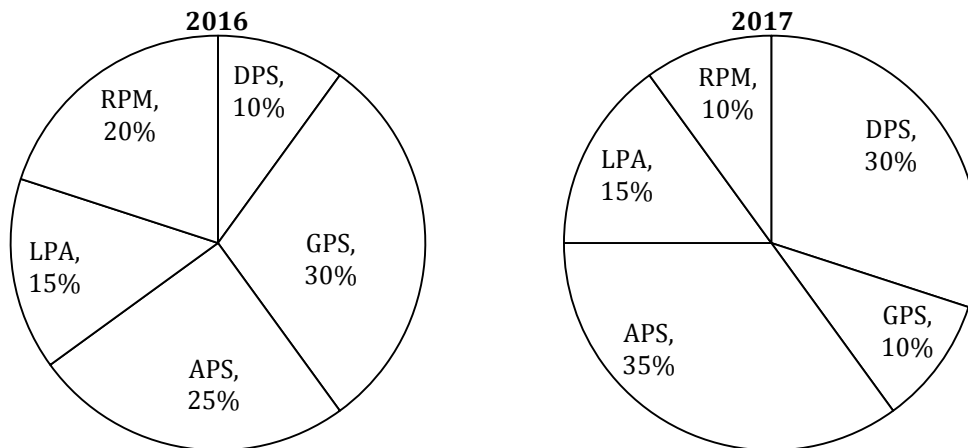
Sol. (b); Expenses of C = $\frac{16}{100} \times 3,00,000 = 48,000$

Increment in expense = $48,000 \times \frac{15}{100} = 7200$

% increment necessary on salary = $\frac{7200 \times 100}{15\% \times 4,00,000} = \frac{7200 \times 100}{60,000} = 12\% \Rightarrow 12\% \text{ increment}$

Directions (41-45): Study the following pie-chart and answer the questions that follow it.

Given below are the two pie charts which shows the percentage distribution of admission of students in five different schools in year 2016 and 2017.



41. Total number of admission in 2016 and 2017 are 2000 and 2500 respectively. Number of students in DPS in 2016 is what percent less or more than number of students in GPS in 2017?

- (a) 35% (b) 25% (c) 40% (d) 20% (e) 15%

Sol. (d); Number of students in DPS in 2016 = $\frac{10}{100} \times 2000 = 200$

Number of student in GPS in 2017 = $\frac{10}{100} \times 2500 = 250$

$\therefore$ Percentage = $\frac{250-200}{250} \times 100 = \frac{50}{250} \times 100 = 20\%$

42. If the total number of students in 2016 is 4000 and ratio of boys and girls in RPM is 2 : 3. Then the difference between boys and girls in APS is?

- (a) 500 (b) 600 (c) Cannot be determined
(d) 400 (e) 160

Sol. (c); Since ratio of number of boys and girls in 2016 in APS is not given.

43. If the total number of admission in 2017 is 5000 and 500 students left DPS in 2017 and taken admission in RPM in 2017 then number of admissions in RPM increases by what percent?

- (a) 80% (b) 100% (c) 120% (d) 60% (e) 10%

Sol. (b); Number of student in DPS in 2017 = $\frac{30}{100} \times 5000 = 1500$

Number of student in DPS in 2017 after 500 left = $1500 - 500 = 1000$

Number of student in RPM in 2017 = $\frac{10}{100} \times 5000 = 500$

Number of student after 500 student joined = $500 + 500 = 1000$

$\therefore$ Percentage increase = $\frac{500}{500} \times 100 = 100\%$

44. If total student taking admission in 2017 is 6000 and in 2016 is 4000. Then find the ratio of total student taking admission in RPM and LPA in 2016 and total student taking admission in GPS and LPA in 2017?

- (a) 14 : 15 (b) 15 : 14 (c) 12 : 14 (d) 12 : 15 (e) 13 : 15

Sol. (a); Total student taking admission in RPM and LPA in 2016 = $\frac{20}{100} \times 4000 + \frac{15}{100} \times 4000$
 $= 800 + 600 = 1400$

Total student taking admission in GPS and LPA in 2017 = $\frac{10}{100} \times 6000 + \frac{15}{100} \times 6000$
 $= 600 + 900 = 1500$

Therefore, ratio = $\frac{1400}{1500} = 14 : 15$

45. If total student taking admission in 2016 and 2017 is 8000 and 10,000 respectively. And number of boys in GPS is 400 in 2016 and number of girls in DPS in 2017 is 1000. Then find the ratio of number of girls in GPS in 2016 to the number of boys in DPS in 2017 is ?

(a) 1 : 2 (b) 2 : 1 (c) 1 : 3 (d) 1 : 4 (e) 1 : 1

Sol. (e); Total number of student in GPS in 2016 = $\frac{30}{100} \times 8000 = 2400$
 Number of girls in GPS in 2016 = $2400 - 400 = 2000$
 Total number of student in DPS in 2017 = $\frac{30}{100} \times 10,000 = 3000$
 Number of boys in DPS in 2017 = $3000 - 1000 = 2000$
 $\therefore$ Ratio = $\frac{2000}{2000} = 1 : 1$

Directions (46-50): Study the following table carefully and answer the questions given below

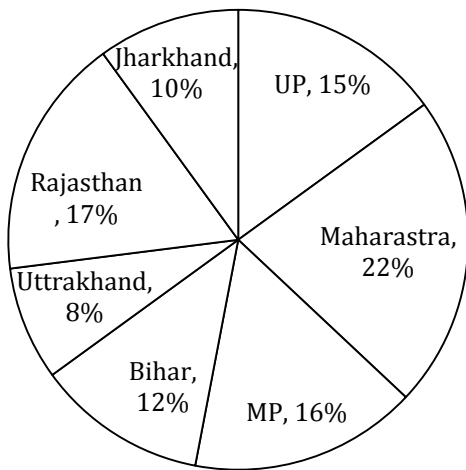
Ist pie chart shows distribution of candidates applied for NIACL Assistant exam 2017 from 7 different states.

IInd pie chart shows distribution of candidates who qualified the NIACL assistant pre exam from these 7 states.

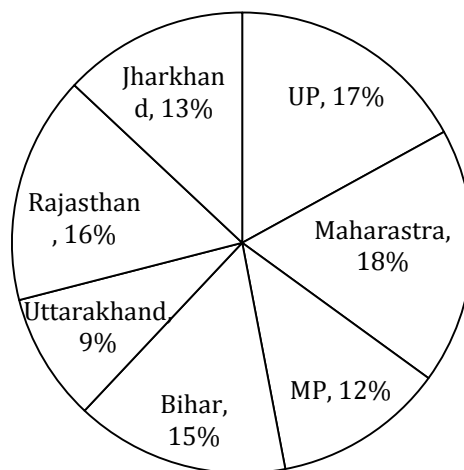
Total applied candidates from these seven states = 7,50,000

Candidates who qualified pre exam = 38000

Pie Chart I



Pie chart II



46. What percentage of candidates applied from Maharashtra state have qualified the pre exam of NIACL assistant (approximately)

(a) 8% (b) 10% (c) 7% (d) 4% (e) 6%

Sol. (d); Number of candidates applied from Maharashtra state = $22 \times 7500 = 165,000$
 Number of candidates qualified = $18 \times 380 = 6840$
 Required percentage = $\frac{6840}{165000} \times 100 \approx \frac{680}{165} \approx 4\%$

47. What is the difference between total number of failed candidates from state MP and Bihar together and the failed candidates from state Maharashtra and Jharkhand together. (Consider all candidates who have applied have given exam.)

(a) 30383 (b) 28480 (c) 25680 (d) 19720 (e) 12320

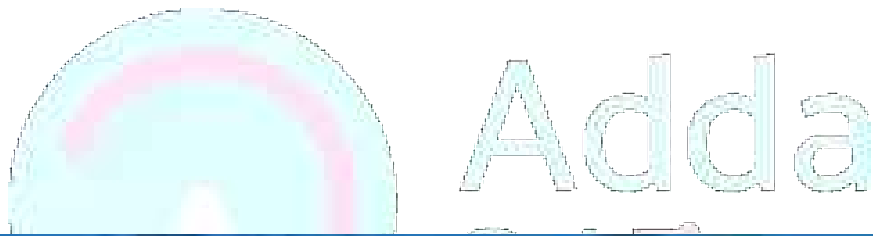
Sol. (b); Candidates failed from MP = $16 \times 7500 - 12 \times 380 = 120,000 - 4560 = 115440$
 Candidate failed from Bihar = $12 \times 7500 - 15 \times 380 = 90,000 - 5700 = 84300$
 Total failed from MP and Bihar = $115440 + 84300 = 199740$
 Candidates failed from Maharashtra = $22 \times 7500 - 18 \times 380 = 165000 - 6840 = 158160$
 Candidate failed from Jharkhand = $10 \times 7500 - 13 \times 380 = 75000 - 4940 = 70060$
 Total failed from Maharashtra and Jharkhand = $158160 + 70060 = 228220$
 Required difference = $228220 - 199740 = 28480$

48. 15% of candidates who have applied from state Rajasthan did not appear for the exam then what percent of the appeared candidates from Rajasthan pass the exam. (Approximately)

(a) 3% (b) 5.6% (c) 8% (d) 10% (e) 12%

Sol. (b); Candidate passed from Rajasthan = $16 \times 380 = 6080$
 Appeared candidates = $\frac{85}{100} \times 17 \times 7500 = 108375$
 Required percentage = $\frac{6080}{108375} \times 100 \approx \frac{610}{108} \approx 5.6\%$

49. If ratio of male to female who applied from state UP is 7 : 5 and ratio of male to female candidates who qualified the pre exam from state UP is 2 : 3 then what is the number of female who failed in the exam (Consider all candidates who have applied have appeared for the exam.)
 (a) 42999 (b) 53620 (c) 41200 (d) 39500 (e) 24242
49. (a); Total females who have applied from UP = $15 \times 7500 \times \frac{5}{12} = 46875$
 Total qualified females from UP = $17 \times 380 \times \frac{3}{5} = 3876$
 Total failed female candidates from UP = $46875 - 3876 = 42999$
50. What is the difference between the central angle of candidates who applied from state Uttarakhand and Jharkhand and the central angle of candidates who qualified exam from Maharashtra and Bihar?
 (a) 58° (b) 60° (c) 48° (d) 38° (e) 54°
- Sol. (e); Central angle for state Uttarakhand and Jharkhand for applied candidates = $\frac{18 \times 18}{5} = \frac{324}{5} = 64.8$
 Central angle for qualified candidates from state Maharashtra and Bihar = $\frac{18 \times 33}{5} = 118.8^\circ$
 Required difference = $118.8 - 64.8 = 54^\circ$



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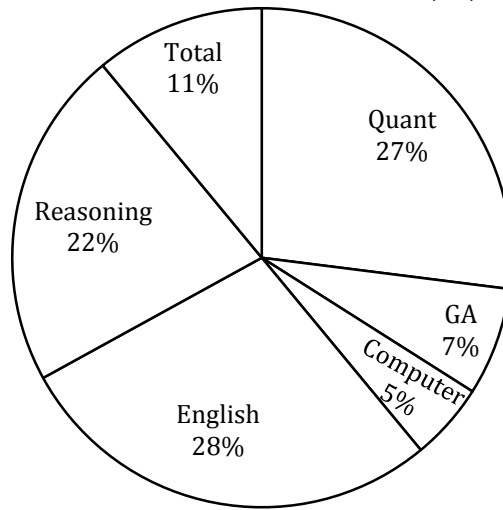
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PREVIOUS YEAR QUESTIONS

Directions (1-5): The given Pie chart represents the percentage of failed students in different sections of IBPS Mains Exam 2016.

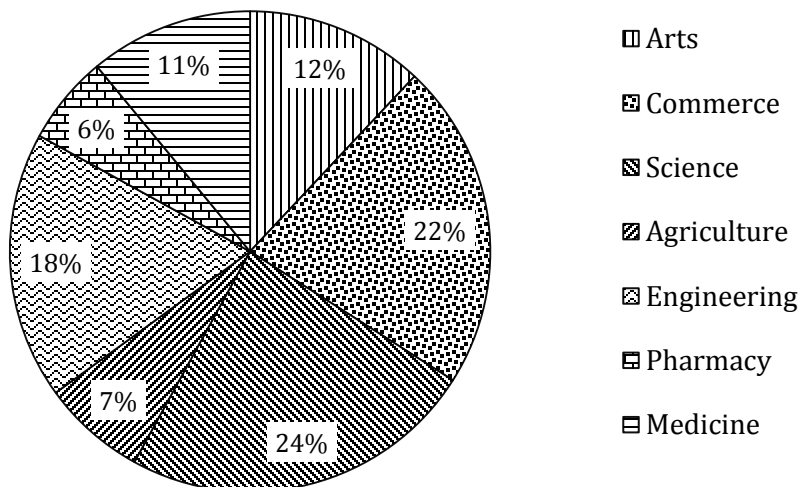
Total No of Failed Candidates = 1,44,000



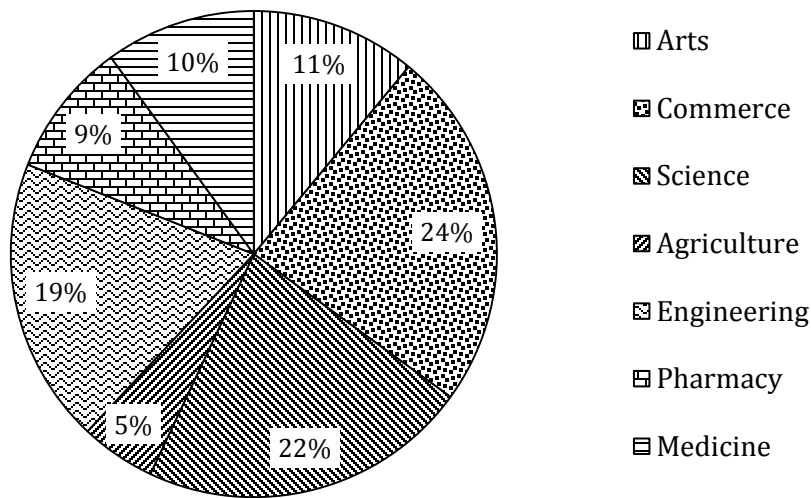
- If total candidates failed in IBPS Mains exam is only 24% of the total number of candidates appeared for IBPS PO pre exam, then total number of candidates appeared in IBPS PO pre exam are approximately how many times the number of candidates failed in Quant section of Mains Exam?
(a) 23 (b) 15 (c) 20 (d) 12 (e) 18
- What is the difference between the number of candidates failed in Reasoning & Total section together and number of candidates failed in Quant Section?
(a) 8640 (b) 8600 (c) 9000 (d) 8000 (e) 7775
- Total number of candidates failed in IBPS mains exam is 75% of the number of candidates appeared for same exam. If final selection will be one-third of the candidates who qualified in Mains Exam, then the number of finally selected candidates are :
(a) 16600 (b) 16000 (c) 17000 (d) 19000 (e) 18000
- What is the ratio of number of candidates who failed in English and Reasoning together to the number of candidates failed overall in Mains exam?
(a) 2 : 3 (b) 2 : 1 (c) 1 : 2 (d) 3 : 2 (e) None of these
- What is the central angle of the candidates who passed in English and Quant ?
(a) 85.6 (b) 90 (c) 80.8 (d) Can't be determined (e) None of the above

Directions (6 - 10): The following pie chart shows the percentage distribution of the choices made in various discipline by CBSE XII passout students in year 2016 and 2017 respectively. Study the following pie-diagrams carefully and answer the questions given below.

Total Students in year 2016 = 4,40,000



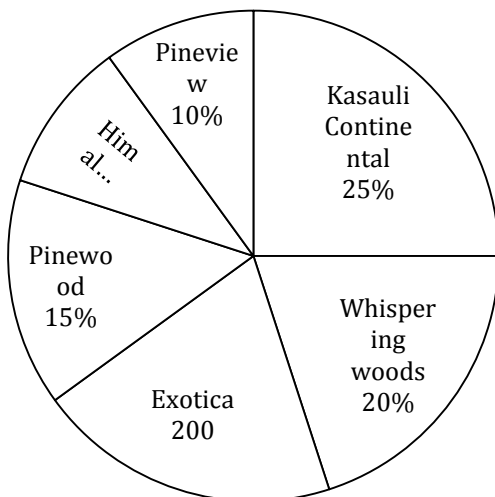
Total students in year 2017 = 5,20,000



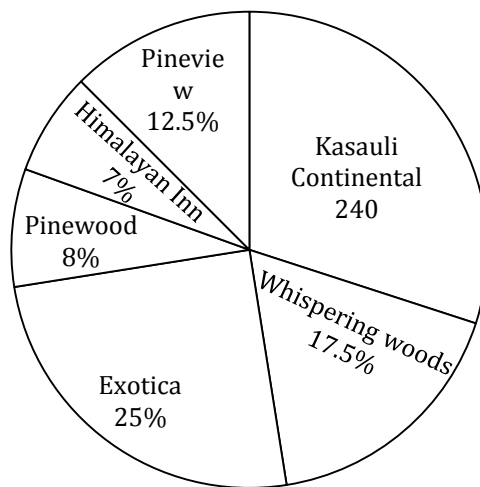
6. In which discipline there was decrease in the number of students from 2016 to 2017?
 (a) Science (b) Agriculture (c) Pharmacy (d) Engineering (e) Medicine
7. What is the ratio between the number of students whose choices is pharmacy in the years 2016 and 2017 respectively?
 (a) 22 : 39 (b) 22 : 37 (c) 23 : 29 (d) 17 : 29 (e) None of these
8. What was the approximate percentage increase in the number of students of Engineering from the year 2016 to 2017?
 (a) 25% (b) 21% (c) 18% (d) 26% (e) 17%
9. In the year 2016, the number of students whose choice is arts and commerce together is what percent of the number of students whose choice are these subjects together in 2017? (approximately)
 (a) 85% (b) 75% (c) 80% (d) 82% (e) 87%
10. In which of the following discipline the percent increase in the number of students was maximum from year 2016 to 2017?
 (a) Medical (b) Pharmacy (c) Commerce (d) Science (e) Agriculture

Directions (11-15): Study the following pie-graphs and answer the questions based on the information given in it.
 (Note: A person books a single room for himself/herself unless stated otherwise)
 Some values are given as absolute data and same as given as percentage.

No. of people who booked rooms in different hotels



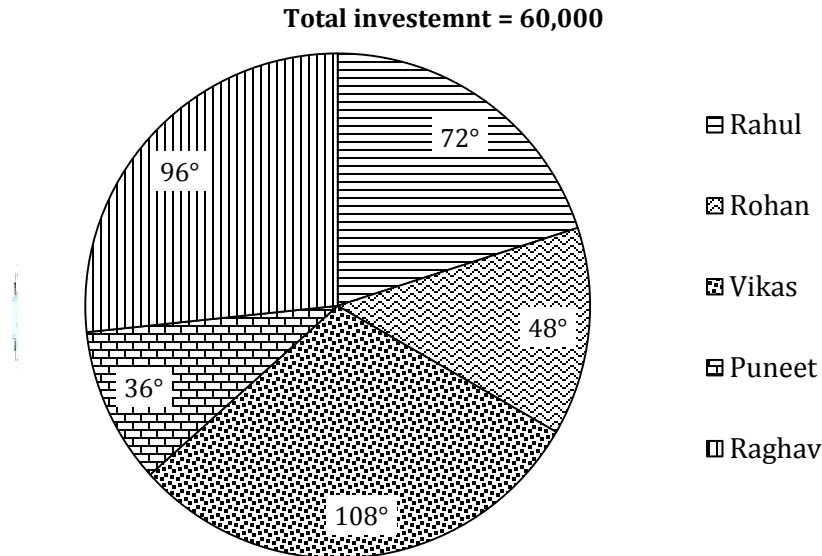
No. of persons who actually showed up in the given hotels



11. What is the ratio of the number of persons who didn't show up in Kasauli continental to that of Whispering woods?
 (a) 1 : 6 (b) 2 : 7 (c) 1 : 8 (d) 3 : 5 (e) 5 : 3

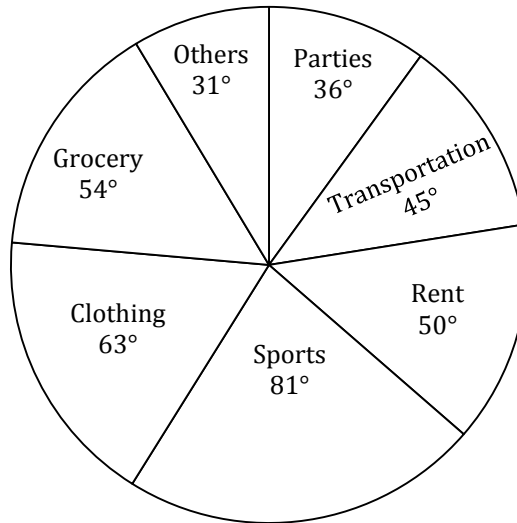
12. If the cost of stay per person in Himalayan Inn was Rs. 3000 and there was no provision of refund in case of no show up then calculate the profit made by Himalayan Inn on account of the persons who didn't show up ? (All the persons paid for the booking in advance)
 (a) Rs. 152000 (b) Rs. 132000 (c) Rs. 141000 (d) Rs. 140000 (e) Rs. 145000
13. If in Whispering woods, only couples booked the rooms, then find the number of couples who didn't show up there as the percentage of total number of persons who didn't show up in Pinewood ? (A couple booked only a room)(Calculate nearby value)
 (a) 40% (b) 24% (c) 35% (d) 28% (e) 50%
14. Hotel Exotica charges Rs. 5000 per person and Rs. 500 extra for the rooms with balcony. If 30% of the persons who booked Exotica booked rooms without balcony & rest booked the rooms with balcony, then find overall revenue for Exotica ?
 (a) Rs. 12,10,000 (b) Rs. 10,80,000 (c) Rs. 9,48,000 (d) Rs. 10,70,000 (e) Rs. 11,55,000
15. Number of person who showed up in Hotel Himalayan Inn & Pinewood together is what percent of the number of person who booked the room in same hotel together ?
 (a) 42% (b) 38% (c) 48% (d) 52% (e) 56%

Directions (16-20): The following pie-graph shows the investments made by five friends in a business for a year. Answer the questions based on the information given in it.



16. If Rahul earned a profit of Rs. 7200, then find the difference between the average profit earned by Puneet & Vikas & the average profit made by Rohan & Raghav?
 (a) Rs. 0 (b) Rs.10 (c) Rs. 5 (d) Rs. 8 (e) Rs. 12
17. Vikas invested the profit earned by him in another business and got a profit of 20%. If his profit from second business in Rupees was Rs. 3600 then find the profit percentage for Puneet from original Business?
 (a) 150% (b) 100% (c) 125% (d) 200% (e) 220%
18. Due to some reasons Rahul withdrew half of his initially planned amount after 4 months. If Rohan invested more amount equal to that withdrawn by Rahul for rest of year , then find the percentage increase in the profit of Rohan at the end of the year:
 (a) 25% (b) 28% (c) 50% (d) 35% (e) 48%
19. All the friends decided to withdraw $\frac{1}{3}$ rd of their amount and thus, at the end of year Puneet & Rohan shared an average profit of Rs. 1400. What is the ratio of profit earned by Rahul & Raghav together to the investment made by Vikas?
 (a) 9 : 30 (b) 7 : 22 (c) 11 : 23 (d) 7 : 15 (e) 15 : 7
20. If Vikas earned Rs. 6000 more profit than Rahul then profit earned by Rohan is what percent more or less than profit earned by Raghav ?
 (a) 50% more (b) 50% less (c) 45% more (d) 45% less (e) 60% less

Directions (21-25): The circle graph shows the spendings of a man in various terms during a particular year. Study the graph carefully and answer the questions that follow.

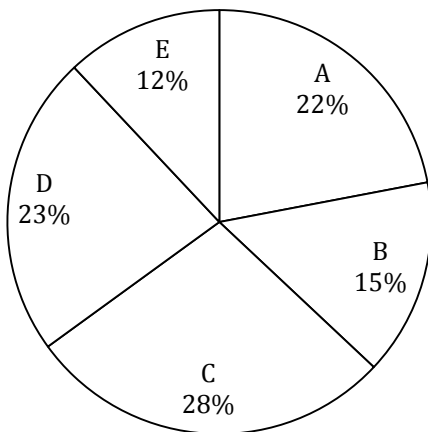


21. What is difference between percentage spending on sports and Grocery?
 (a) 7.5% (b) 9.5% (c) 8.5% (d) 8% (e) 9%
22. Spendings on parties, Sports and Grocery are what percent of spending on others, Rent and Clothing.
 (a) 121.35% (b) 120% (c) 118.75% (d) 112.75% (e) 111.75%
23. If 12% spending of Rent is added in Grocery then find percentage increase in spending of Grocery.
 (a) $9\frac{1}{11}\%$ (b) $11\frac{1}{9}\%$ (c) 12.5% (d) $13\frac{1}{3}\%$ (e) $11\frac{4}{9}\%$
24. If spending on transportation is Rs. 1350, find spending on Sports, Clothing and other together.
 (a) Rs. 6150 (b) Rs. 3750 (c) Rs. 5250 (d) Rs. 6250 (e) Rs. 7150
25. What is the total spending of man, if amount spend by him on sports is Rs. 11,520 more than amount spend by him on clothing?
 (a) 2,30,400 (b) 2,20,200 (c) 2,35,400 (d) 2,40,600 (e) 2,45,500

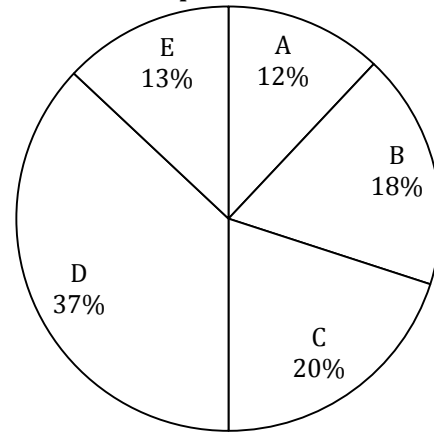
Directions (26-30): The following pie charts show the income of 5 different companies and profit of these 5 companies in 2016. Study it carefully and answer the following questions.

Income - Expenditure = Profit

Total income = 250 lakh

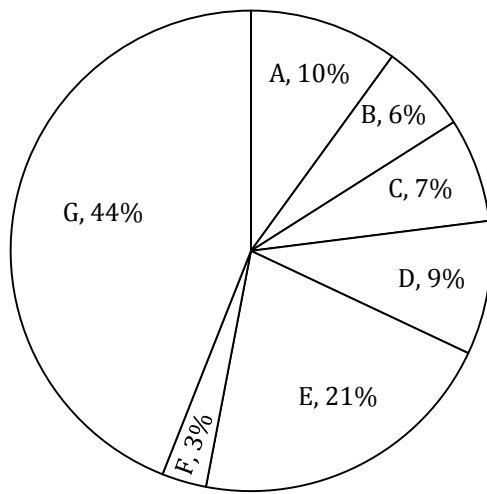


Total profit = 80 lakh

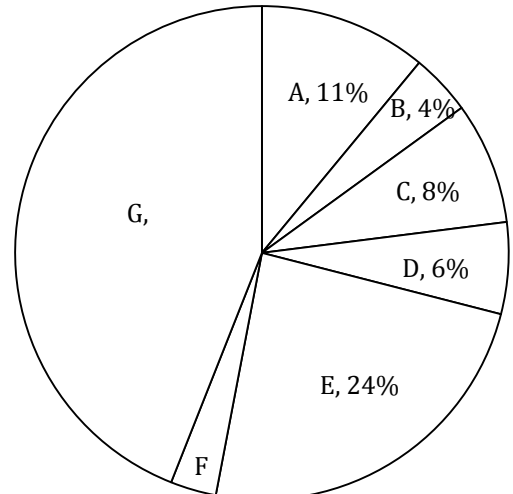


26. Find the expenditure (in lakh) of company B ?
 (a) 32.1 (b) 23.8 (c) 23.1 (d) 33.3 (e) 22.1
27. Find the profit (in lakh) of company B, D and E together.
 (a) 54.4 (b) 45.5 (c) 54.8 (d) 45.2 (e) 60.1
28. Find the difference (in lakh) between the income of company D and company B.
 (a) 18.2 (b) 25.5 (c) 24 (d) 20 (e) 22
29. Income of company E is how much more than profit of company A and B together ?
 (a) 8.5 lakh (b) 8 lakh (c) 9.2 lakh (d) 10 lakh (e) 6 lakh
30. Find the ratio of total income of companies A and E together to total profit of companies B and C together.
 (a) 11:4 (b) 17:9 (c) 9:4 (d) 13:6 (e) 425:152

Directions (31-35): Percentage distribution of Income of 7 firms in year 2015 and 2016 is given below in pie charts. Percentage distribution of some firms are not given in the chart. You have to calculate these values if required to answer the question.



2015

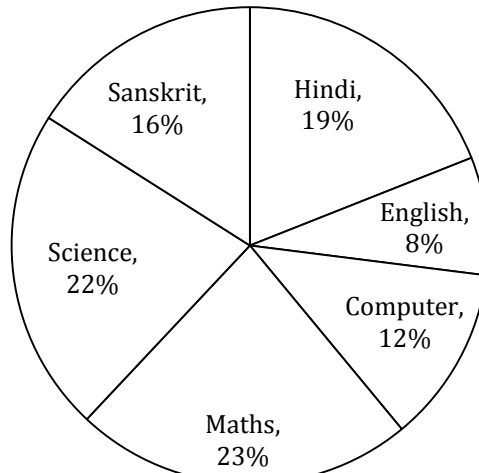
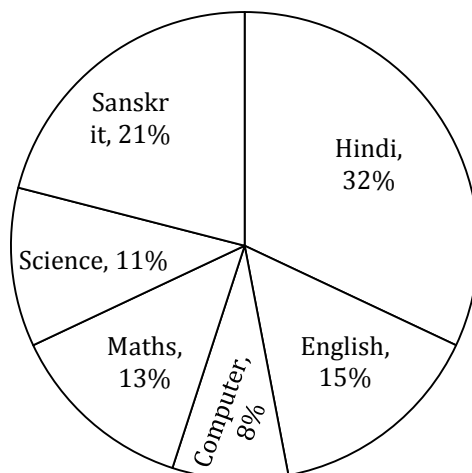


2016

Ratio of Total Income of all 7 firms in 2015 to total income of all seven firms in 2016 is 5 : 7.

31. If profit percent earned by company C in 2015 and profit percent earned by company D in 2016 are equal and income of company D in 2016 is 10 million and expenditure of company D in 2016 is 8 million then what will be profit of C in 2015?
(a) $\frac{7}{6}$ million (b) $\frac{5}{3}$ million (c) $\frac{8}{7}$ million (d) $\frac{2}{3}$ million (e) 3 million
32. If total income of all firm in 2015 is 13860 million then what is the difference between the income of firm E in 2015 and income of A in 2016.(approx.)
(a) 776 million (b) 820 million (c) 720 million (d) 810 million (e) 800 million
33. What will be the ratio of income of firm D in 2015 to the income of firm G in 2016 if income of G and F in 2016 is in the ratio of 24 : 23?
(a) $\frac{21}{28}$ (b) $\frac{33}{35}$ (c) $\frac{15}{59}$ (d) $\frac{15}{56}$ (e) $\frac{15}{16}$
34. If difference between the total income of all firms in 2015 and 2016 is N then what will be the difference between the average of income of firm A, B and C together in 2015 and average of income of firm B, C and D together in 2016
(a) $\frac{23N}{600}$ (b) $\frac{11N}{600}$ (c) $\frac{23N^3}{600}$ (d) $\frac{11N^2}{600}$ (e) None of these
35. If income of company G in 2016 is $\frac{100}{11}\%$ more than income of company G in 2015 then what is the percentage distribution of income for firm F in 2016.
(a) $\frac{89}{7}\%$ (b) $33\frac{1}{3}\%$ (c) $66\frac{2}{3}\%$ (d) $16\frac{2}{3}\%$ (e) None of these

Direction (36-40)-Percentage of students interested in studying different subjects (Hindi, English, Computer, Maths, Science, Sanskrit) in Pie chart I & percentage of girls interested in studying these subjects in pie chart II.



RATIO OF BOYS : GIRL = 5:3
TOTAL STUDENTS = 48,000

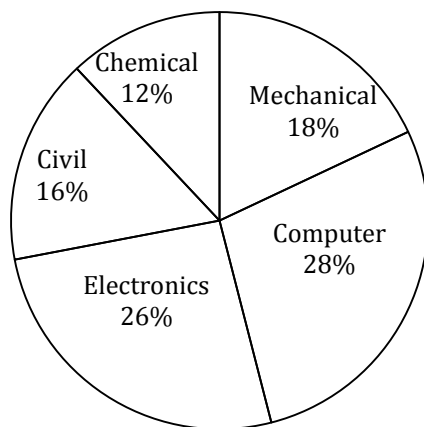
36. For which of the subject, the ratio of percentage of student interested in that subject to the percentage of girls interested in that subject is minimum?
 (a) Science (b) Computer (c) Maths (d) English (e) Sanskrit
37. What is the difference between the no. of girls interested in studying computer and that of science?
 (a) 1.5 thousand (b) 2.2 thousand (c) 1.8 thousand (d) 1.9 thousand (e) 2.4 thousand
38. What is the ratio of the no. of boys interested in studying Computer and English together to that of girls interested in studying Sanskrit and Maths together?
 (a) 124 : 117 (b) 128 : 119 (c) 19 : 17 (d) 23 : 19 (e) 5 : 3
39. What is the ratio of the no. of students interested in studying maths and Sanskrit together to that interested in Hindi and Science together?
 (a) 23 : 32 (b) 34 : 43 (c) 101 : 130 (d) 11 : 32 (e) None of these
40. No. of girls studying Hindi and English together is approximately what percent of the no. of boys studying the same subject?
 (a) 27% (b) 30% (c) 17% (d) 23% (e) 21%

Directions (41-45): Study the pie charts carefully to answer the questions that follow.

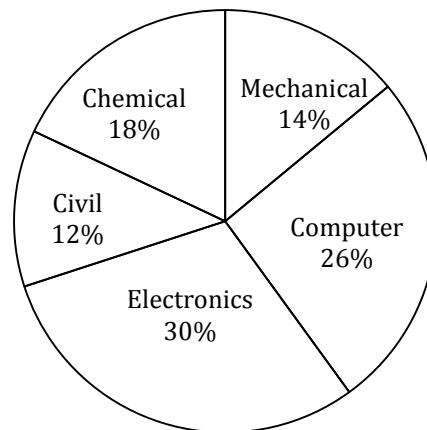
Percentage wise distribution of students and female students in five different streams in an Engineering College.

Total number of students in the college is 5400, out of which number of female students is 2400.

Distribution of students

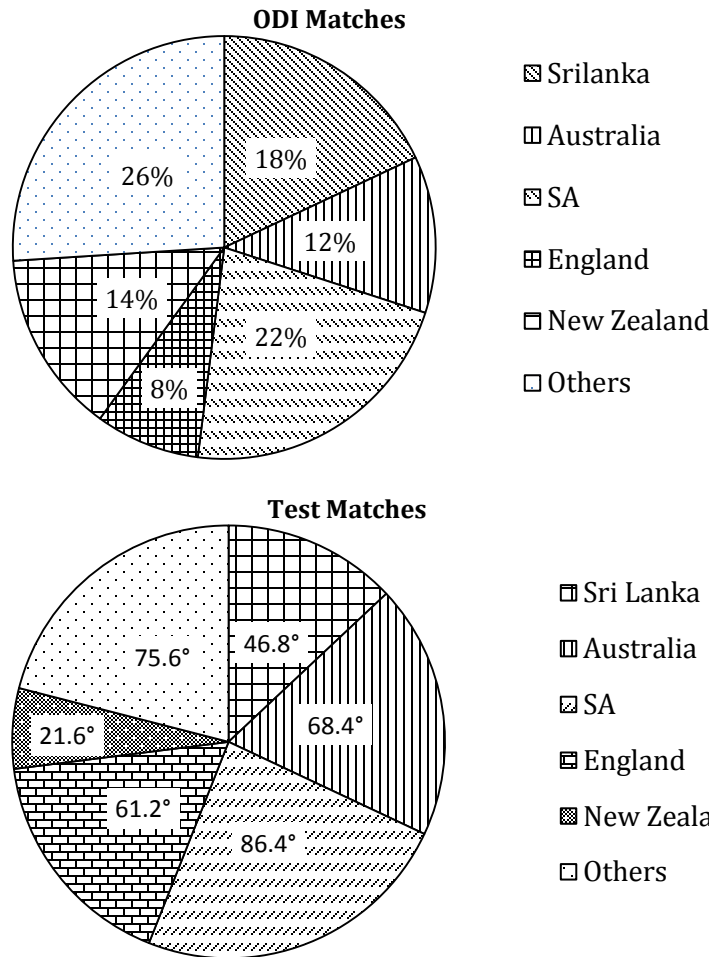


Distribution of female students



41. The number of female students in Chemical Engineering is what percent of the number of male students in Civil Engineering?
 (a) 120% (b) 80% (c) 125% (d) 75% (e) 60%
42. What is the ratio of the number of female students in Mechanical and Computer Engineering together to the number of male students in Electronics and Civil Engineering together?
 (a) 21 : 16 (b) 16 : 21 (c) 5 : 7 (d) 7 : 5 (e) None of these
43. In a new stream Biotechnology, the total number of students is $8\frac{1}{3}\%$ less than that in Chemical Engineering. If the ratio of the number of male and female students in Biotechnology is 5 : 6 then what is the difference between the number of female students in Biotechnology and Chemical Engineering?
 (a) 54 (b) 135 (c) 81 (d) 108 (e) 180
44. What is the total number of male students in Computer, Electronics and Chemical Engineering?
 (a) 1888 (b) 1776 (c) 1788 (d) 1876 (e) 1728
45. What is the difference between the average of number of male students in Mechanical and Electronics Engineering together and the average of the number of male students in rest of the streams?
 (a) 100 (b) 60 (c) 120 (d) 80 (e) 70

Directions (46-50) : Given below are the pie charts showing the distribution of runs scored by MS Dhoni against different teams in ODI matches and test matches. The total runs scored by him in ODI matches is 25500 and in test matches is 11200.



46. If $44\frac{4}{9}\%$ of the runs scored against Sri Lanka in ODI's and $\frac{5}{14}$ of the runs scored against the same team in test matches, are scored in India. Then find the difference between runs scored against Sri Lanka in test matches outside India and the runs scored against the same team in ODI's outside India.
 (a) 1516 (b) 1614 (c) 3419 (d) 1450 (e) 1416
47. Total runs scored by M.S. Dhoni in ODI's against Sri Lanka and SA together are what percent less/more than total runs scored by him in tests against New Zealand and England together ?
 (a) 194% (b) 196% (c) 294% (d) 296% (e) 264%
48. Total number of runs scored by him in ODI's against all of the team excluding others are how many times the runs scored by him in tests against SA ?
 (a) 8.23 (b) 7.14 (c) 7.02 (d) 6.95 (e) 8.02
49. What is the ratio between of the runs scored by Dhoni in ODI's against England and New Zealand together and runs scored by him in tests against Sri Lanka and Australia together ?
 (a) 2805 : 1792 (b) 2905 : 1792 (c) 2805 : 1799 (d) 2875 : 1292 (e) 1792 : 1801
50. If the runs scored in ODI matches against west indies is 20% of the total runs scored in ODI matches against "others", then find the difference between runs scored against West Indies in ODI matches and the total runs scored against England in test matches and ODI matches together?
 (a) 2541 (b) 2455 (c) 2461 (d) 2375 (e) 2618

PREVIOUS YEAR SOLUTIONS

1. (b); 24% → 1,44,000

∴ 100% → 600000 = Total no. of candidate appeared in Pre Exam

Candidates failed in Quant section in Mains = $1440 \times 27 = 38,880$

∴ Desired value will be $\frac{6,00,000}{38,880} \approx \frac{6,00,000}{40,000} = 15$ times

2. (a); No. of candidate failed in (Reasoning & Total section) =
- $(22 + 11) \times 1440 = 47520$

No. of candidate failed in Quant section = $27 \times 1440 = 38,880$

Difference = $47,520 - 38,880 = 8,640$

3. (b);
- $144000 = x \times \frac{3}{4}$

∴ x = Total candidates appeared for main exam = 1,92,000

Candidates Qualified in Mains Exam = $1,92,000 - 1,44,000 = 48,000$

∴ Final Selection = $48,000 \times \frac{1}{3} = 16,000$

4. (c);
- $\frac{\text{Failed in (Eng+Reasoning)}}{\text{Total Failed}} = \frac{(28+22)1440}{144000} = \frac{1}{2} = 1:2$

5. (d); As per given value in graph, Data is not sufficient to calculate the desired value.

6. (b);

	2016	2017
Arts	52,800	57,200
Commerce	96,800	1,24,800
Science	1,05,600	1,14,400
Agriculture	30,800	26,000
Pharmacy	26,400	46,800
Medicine	48,400	52,000
Engineering	79,200	98,800

Clearly in Agriculture, there was decrease in number of students from 2016 to 2017.

7. (a); Ratio =
- $26400 : 46800 = 22 : 39$

8. (a); % increase in Engineering students from

2016 to 2017 = $\frac{98800 - 79200}{79200} \times 100 \approx 25\%$

9. (d); Arts & Commerce together in 2016 =
- $52800 + 96800 = 1,49,600$

Arts & commerce together in 2017 = $57,200 + 1,24,800 = 1,82,000$

Desired % = $\frac{1,49,600}{1,82,000} \times 100 \approx 82\%$

10. (c); Clearly in commerce % increase was maximum

11. (a); No. of persons who booked rooms in Kasauli Continental

= $300 \times \frac{100}{30} \times \frac{25}{100} = 250$

No. of persons who showed up in Kasauli Continental = 240

No. of persons who booked rooms in Whispering woods = $300 \times \frac{100}{30} \times \frac{20}{100} = 200$

No. of persons who showed up in Whispering woods = $240 \times \frac{100}{30} \times \frac{17.5}{100} = 140$

Req. Ratio = $\frac{250-240}{200-140} = \frac{10}{60} = \frac{1}{6}$

12. (b); No. of persons who booked in Himalayan Inn = 100

No. of persons who showed up Himalayan Inn

= $240 \times \frac{100}{30} \times \frac{7}{100} = 56$

Profit made on account of those who didn't show up = $(100 - 56) \times 3000 = \text{Rs. } 132000$

13. (c); No. of persons who didn't show up in Whispering woods =
- $200 - 140 = 60$

No. of couples = $\frac{60}{2} = 30$

No. of persons who didn't show up in Pinewood

= $1000 \times \frac{15}{100} - 800 \times \frac{8}{100}$

= $150 - 64 = 86$

Req. % = $\frac{30}{86} \times 100 \approx 35\%$

14. (d); Overall revenue for Exotica

= $\frac{3}{10} \times 200 \times 5000 + \frac{7}{10} \times 200 \times 5500$

= Rs. 10,70,000

15. (c); No. of person who booked hotel in Pinewood

= $\frac{300}{30} \times 15 = 150$

Required percentage = $\frac{\frac{240}{30} \times 15}{250} \times 100 = \frac{12000}{250} = 48\%$

16. (a); Ratio of their investments = 72 : 48 : 108 : 36 : 96

= 6 : 4 : 9 : 3 : 8 (starting from Rahul & going clockwise)

Total profit = $7200 \times \frac{30}{6} = \text{Rs. } 36000$

Average profit of Puneet & Vikas = $\frac{1}{2} \times \frac{12}{30} \times 36000 = 7200$

Average profit of Rohan & Raghav = $\frac{1}{2} \times \frac{12}{30} \times 36000 = 7200$

Difference = 0

17. (b); Amount invested by Vikas in another business

= $3600 \times \frac{100}{20} = \text{Rs. } 18000$

This is equal to profit earned by him from original business.

Profit earned by Puneet = $18000 \times \frac{30}{9} \times \frac{3}{30} = 6000$

Profit % for Puneet = $\frac{6000}{\frac{3}{30} \times 60000} \times 100 = 100\%$

18. (c); New ratio of investment/profit

= $(6 \times 4 + 3 \times 8) : (4 \times 12 + 3 \times 8) : 9 \times 12 : 3 \times 12 : 8 \times 12$

= $48 : 72 : 9 \times 12 : 3 \times 12 : 8 \times 12$

= 4 : 6 : 9 : 3 : 8

Rohan's earlier profit = $\frac{4}{30}$

$$\text{Rohan's new profit} = \frac{6}{30}$$

$$\% \text{ increase} = \frac{\frac{6}{30} - \frac{4}{30}}{\frac{4}{30}} \times 100 = 50\%$$

19. (d); Total investment = $\frac{2}{3} \times 60000 = 40,000$

$$\text{Total profit} = 2800 \times \frac{30}{7} = 12000$$

$$\text{Req. ratio} = \frac{\frac{14}{30} \times 12000}{\frac{9}{30} \times 40000} = \frac{7}{15}$$

20. (b); Ratio of their profits = 72 : 108 = 6 : 9

$$\text{Let profit earned by Vikas} = 9x$$

$$\& \text{ that by Rahul} = 6x$$

$$\therefore 3x = 6000 \Rightarrow x = 2000$$

$$\therefore \text{Rahul's profit} = 12000$$

$$\therefore \text{Total profit} = 12000 \times \frac{30}{6} = 60,000$$

$$\therefore \text{Required percentage} = \frac{16000 - 8000}{16000} \times 100 = 50\% \text{ less}$$

21. (a); Required percentage difference = $\left(\frac{81-54}{360}\right) \times 100$
 $= \frac{27}{360} \times 100 = 7.5\%$

22. (c); Spending on Parties, Sports and Grocery in terms of degree = $36^\circ + 81^\circ + 54^\circ = 171^\circ$

$$\text{Spending on others, Rent and Clothing in degree} = 31^\circ + 50^\circ + 63^\circ = 144^\circ$$

$$\text{Required percentage} = \frac{171}{144} \times 100 = 118.75\%$$

23. (b); Required percentage increase = $\left(\frac{12 \times 50}{54}\right) \times 100$
 $= \frac{6}{54} \times 100 = 11\frac{1}{9}\%$

24. (c); Spending on Sport, Clothing and other together
 $= \frac{1350}{45} \times (81 + 63 + 31) = \text{Rs. } 5,250$

25. (a); $81^\circ - 63^\circ = 11,520$

$$18^\circ = 11,520$$

$$360^\circ = \frac{11,520}{18} \times 360 = 2,30,400$$

26. (c); Income of B = $\frac{15}{100} \times 250 = 37.5 \text{ lakh}$

$$\text{Profit of B} = \frac{18}{100} \times 80 = 14.4 \text{ lakh}$$

$$\therefore \text{Expenditure of B} = 37.5 - 14.4 = 23.1 \text{ lakh}$$

27. (a); Profit of B, D and E together = $(18 + 37 + 13)\%$ of 80
 $= 54.4 \text{ lakh}$

28. (d); Required difference = $(23 - 15)\%$ of 250
 $= \frac{8 \times 250}{100} = 20 \text{ lakh}$

29. (e); Income of E = $\frac{12}{100} \times 250 = 30 \text{ lakh}$

$$\text{Profit of A and B together} = \frac{30}{100} \times 80 = 24 \text{ lakh}$$

$$\therefore \text{Income of E is more than profit of A and B together by } 30 - 24 = 6 \text{ lakh}$$

30. (e); Required ratio = $\frac{\frac{34 \times 250}{100}}{\frac{38 \times 80}{100}} = \frac{34 \times 25}{38 \times 8} = \frac{425}{152}$

31. (b); Profit % of company D in 2016 = $\frac{2}{8} \times 100 = 25\%$

$$\text{Income of C in 2015} = \frac{10}{6} \times \frac{100}{7} \times \frac{5}{100} \times 7$$

$$= \frac{350}{42} \text{ million} = \frac{25}{3}$$

$$\text{Total income} = \frac{25}{3}$$

$$\text{Profit \%} = 25\%$$

$$\text{Expenditure} = \frac{25}{3} \times \frac{100}{125} = \frac{20}{3}$$

$$\text{Profit of C} = \frac{25}{3} - \frac{20}{3} = \frac{5}{3} \text{ million}$$

32. (a); Income of E in 2015 = $\frac{13860}{100} \times 21 = 2910.6$

$$\text{Income of A in 2016} = \frac{13860 \times 7 \times 11}{5 \times 100} = 2134.44$$

$$\text{Required difference} = 776.16 \approx 776 \text{ million}$$

33. (d); Required Ratio = $\frac{\frac{5x}{100} \times 9}{\frac{7x}{100} \times 24} = \frac{15}{56}$

34. (b); Total Income of all firms in 2015 = $\frac{5}{2}N$

$$\text{Total Income of all firms in 2016} = \frac{7}{2}N$$

$$\text{Average of firm A, B and C in 2015} = \frac{5N \times 23}{2 \times 100 \times 3}$$

$$\text{Average of firm B, C and D in 2016} = \frac{7N \times 18}{2 \times 100 \times 3}$$

$$\text{Required difference} = \frac{7N \times 18}{2 \times 100 \times 3} - \frac{5N \times 23}{2 \times 100 \times 3}$$

$$= \frac{N}{600} (7 \times 18 - 5 \times 23)$$

$$= \frac{N}{600} (126 - 115) = \frac{11N}{600}$$

35. (a); Income of G in 2015 = $\frac{5x}{100} \times 44$

$$\text{Income of G in 2016} = \frac{5x}{100} \times 44 \times \frac{12}{11}$$

$$\text{Percentage distribution of G for 2016}$$

$$= \frac{\frac{5x \times 12 \times 4}{100}}{7x} \times 100 = \frac{240}{7}\%$$

$$\text{Percentage distribution of F for 2016}$$

$$= 100 - \left(\frac{240}{7} + 11 + 4 + 8 + 6 + 24\right)$$

$$= 100 - \left(\frac{240}{7} + 53\right) = \frac{89}{7}\%$$

36. (a); Required ratio = $\frac{\text{Percentage of total student}}{\text{Percentage of girls}}$

$$\text{For science} = \frac{11}{22} = 0.5$$

$$\text{For computer} = \frac{8}{12} = 0.66$$

$$\text{For Maths} = \frac{13}{23} = 0.565$$

$$\text{For English} = \frac{15}{8} = 1.875$$

$$\text{For Hindi} = \frac{32}{19} = 1.684$$

$$\text{For Sanskrit} = \frac{21}{16} = 1.3125$$

$$\therefore \text{Ratio for science is minimum}$$

37. (c); Total no. of girls studying = $\frac{3}{8} \times 48 = 18 \text{ thousands}$

$$\text{Difference between no. of girls interested in studying computer and that of science.}$$

$$= 10\% \text{ of } 18 \text{ thousand} = 1.8 \text{ thousand}$$

38. (a); Boys interested in computer

$$= \frac{8}{100} \times 48 - \frac{12}{100} \times 18 = 1.68 \text{ thousand}$$

$$\text{Boys interested in English}$$

$$\frac{15}{100} \times 48 - \frac{8}{100} \times 18 = 5.76 \text{ thousand}$$

$$\text{Girls interested in Sanskrit}$$

$$= \frac{16}{100} \times 18 = 2.88 \text{ thousand}$$

$$\text{Girls interested in Maths} = \frac{23}{100} \times 18$$

$$= 4.14 \text{ thousands}$$

$$\therefore \text{Required ratio} = \frac{5.76 + 1.68}{2.88 + 4.14} = \frac{124}{117}$$

39. (b); Required ratio = $\frac{21+13}{11+32} = 34 : 43$

40. (a); No. of boys studying Hindi = $\frac{32}{100} \times 48 - \frac{19}{100} \times 18$

$$= 15.36 - 3.42 = 11.94$$

No. of boys studying English = $\frac{15}{100} \times 48 - \frac{8}{100} \times 18$

$$= 7.20 - 1.44 = 5.76$$

∴ required percentage

$$= \frac{4.86}{17.7} \times 100 \approx 27.457\% \approx 27\%$$

41. (d); Number of female students in Chemical Engineering

$$= 18\% \text{ of } 2400 = 432$$

Number of male students in Civil Engineering

$$= 16\% \text{ of } 5400 - 12\% \text{ of } 2400$$

$$= 864 - 288 = 576$$

$$\text{Required \%} = \frac{432}{576} \times 100 = 75\%$$

42. (b); Number of female students in Mechanical and Computer Engineering together

$$= (14 + 26)\% \text{ of } 2400 = 40\% \text{ of } 2400 = 960$$

Number of male students in Electronics and Civil Engineering together

$$= (26 + 16)\% \text{ of } 5400 - (30 + 12)\% \text{ of } 2400$$

$$= 42\% \text{ of } 5400 - 42\% \text{ of } 2400$$

$$= 2268 - 1008 = 1260$$

$$\text{Required ratio} = \frac{960}{1260} = \frac{16}{21}$$

43. (d); Number of female students in Biotechnology

$$= \frac{6}{11} \times \frac{11}{12} \times \frac{12}{100} \times 5400 = 324$$

Number of female students in Chemical Engineering

$$= 18\% \text{ of } 2400 = 432$$

$$\text{Required Difference} = 432 - 324 = 108$$

44. (c); Total number of male students in Computer, Electronics and Chemical Engineering

$$= (28 + 26 + 12)\% \text{ of } 5400 - (26 + 30 + 18)\% \text{ of } 2400$$

$$= 66\% \text{ of } 5400 - 74\% \text{ of } 2400$$

$$= 3564 - 1776 = 1788$$

45. (a); Number of male students in Mechanical and Electronics Engineering together

$$= (18 + 26)\% \text{ of } 5400 - (14 + 30)\% \text{ of } 2400$$

$$= 44\% \text{ of } 5400 - 44\% \text{ of } 2400$$

$$= 2376 - 1056 = 1320$$

Number of male students in Computer, Civil and Chemical Engineering together

$$= (28 + 16 + 12)\% \text{ of } 5400 - (26 + 12 + 18)\% \text{ of } 2400$$

$$= 56\% \text{ of } 5400 - 56\% \text{ of } 2400$$

$$= 3024 - 1344 = 1680$$

$$\text{Difference between averages} = \frac{1320}{2} - \frac{1680}{3}$$

$$= 660 - 560 = 100$$

46. (b); Required difference

$$= \left[\left(100 - 44\frac{4}{9} \right) \% \text{ of } 4590 - \left(1 - \frac{5}{14} \right) \% \text{ of } 1456 \right]$$

$$= 2550 - 936 = 1614$$

47. (d); Required % = $\frac{(4590+5610)-(1904+672)}{1904+672} \times 100$

$$= \frac{7624}{2576} \times 100 \approx 296\%$$

48. (c); Required fraction = $\frac{18870}{2688} \approx 7.02$

49. (a); Required Ratio = $(2040 + 3570) : (1456 + 2128)$

$$= 5610 : 3584 = 2805 : 1792$$

50. (e); Required difference = $(1904 + 2040) - \frac{20}{100} \times 6630$

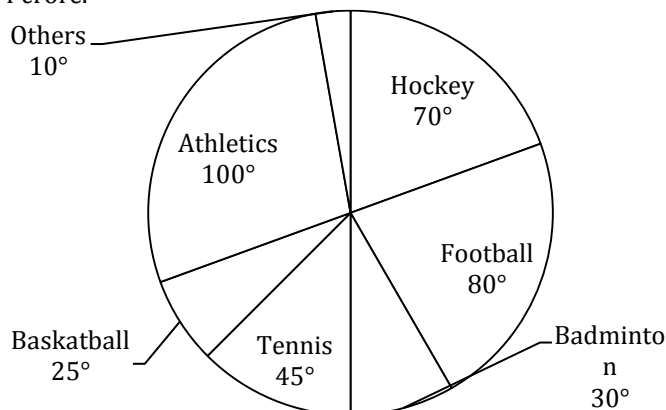
$$= 3944 - 1326 = 2618$$



PRACTICE SET (LEVEL-I)

Directions (1-5): Study the following pie chart showing the distribution of sports budget among various sports to answer the questions that follow.

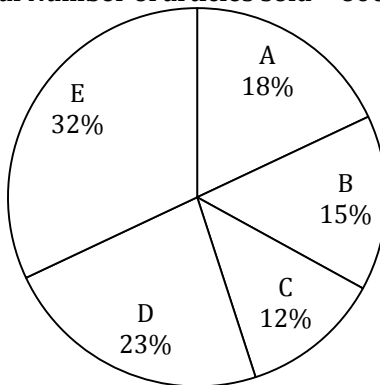
Spending of total Sport budget 144 crore.



- What percentage of total spending is spent on Tennis?
(a) 12.5% (b) 18% (c) 14% (d) 10% (e) 8%
- How much more is spent on Hockey than on Basketball?
(a) 15 cr (b) 12 cr (c) 18 cr (d) 20 cr (e) 10 cr
- If total spending on Hockey is increased by 20% while that of football is decreased by 30%, what will be difference between the spending on hockey and football now?
(a) 11.8 Cr (b) 11.2 Cr (c) 10.9 Cr (d) 12.1 Cr (e) 12.8 Cr
- What is the ratio of amount spent on Hockey, football and Tennis together to Rs. 84 crore?
(a) $\frac{11}{14}$ (b) $\frac{13}{14}$ (c) $\frac{3}{7}$ (d) $\frac{4}{9}$ (e) $\frac{5}{2}$
- If for another game, billiards with additional budget of 17 crore is allocated, what is the difference between amounts spend on Others & Billiards?
(a) 12cr (b) 10 cr (c) 18 cr (d) 13 cr (e) 20 cr

Directions (6-10): Given below is the pie chart which shows the percentage distribution of number of article sold by 5 shopkeepers A, B, C, D and E in a year 2016

Total Number of articles sold = 6000

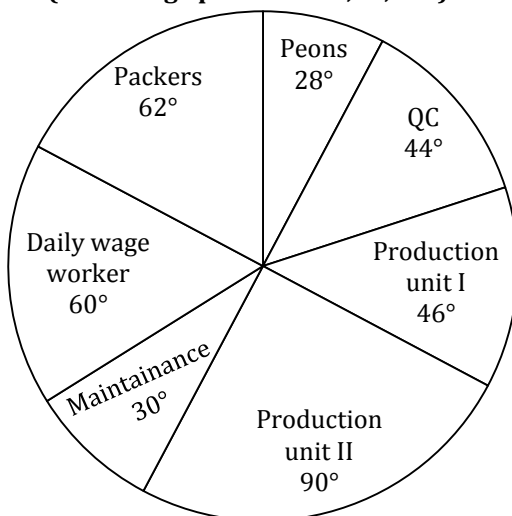


- What is the difference between the central angle for articles sold by B and D.
(a) 25.5° (b) 23.6° (c) 28.8° (d) 26° (e) 27°
- What is the ratio of number of articles sold by B and C together to 150% of total number of articles sold by A and E together
(a) 2 : 5 (b) 9 : 14 (c) 9 : 25 (d) 16 : 25 (e) 13 : 20
- If in 2017 articles sold by A and C increased by $\frac{100}{3}\%$ and $\frac{100}{6}\%$ respectively then what is the sum of articles sold by them in 2017.
(a) 2280 (b) 1990 (c) 2250 (d) 1970 (e) 2200
- What is the difference between the average of articles sold by A and B together and average of articles sold by C and D together.
(a) 65 (b) 60 (c) 58 (d) 48 (e) 54

10. If total number of articles sold by all increases by $\frac{100}{3}\%$ then what is the difference between number of articles sold by A and D together and number of articles sold by B and E together. (Percentage distribution remains same)
 (a) 180 (b) 175 (c) 256 (d) 480 (e) 216

Directions (11– 15): The following pie- chart shows the distribution of wages paid in an organization in 2015 -16 for seven different departments. Read the questions carefully and answer accordingly.

Wages paid for 2015-16
(Total wage paid - Rs. 10,80,000)

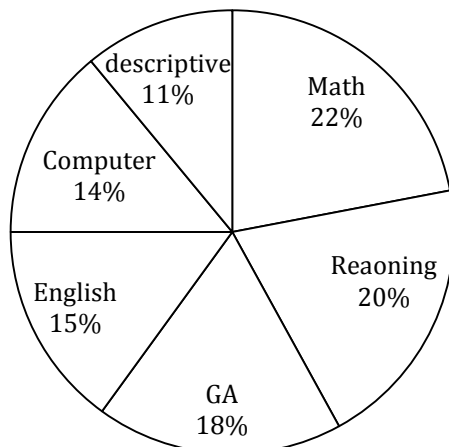


11. What is the difference between the wages paid to the QC staff & Maintenance-worker ?
 (a) Rs. 42000 (b) Rs. 44000 (c) Rs. 48000 (d) Rs. 54000 (e) Rs. 46500
12. For which of the following, were the wages paid exactly equal to Production Unit II ?
 (a) Maintenance (b) Daily wage (c) Production unit I (d) Sum of (a) and (b) (e) Sum of (a) and (c)
13. Which one of the following combinations of workers was paid Rs. 270000 ?
 (a) Production unit II (b) Maintenance and Daily wage workers together
 (c) Packers and Maintenance workers together (d) Both (a) and (b) (e) Both (a) and (c)
14. The management decided to give an ex-gratia grant to daily wage workers. The grant was 20% of the total wages paid to daily wage workers. If a daily wage worker works for 48 days in a year and get Rs. 900 per day, what was the number of such workers in the period 2015-16 ?
 (a) 6 (b) 10 (c) 5 (d) 11 (e) None of these
15. If the management increased the total wages to be paid by 10% (across the board) for the year 2016-17, what was the difference between the wages paid to packers and those paid to peons for the year 2016-17 ? (wage distribution remain same)
 (a) Rs. 114400 (b) Rs. 112200 (c) Rs. 122800 (d) Rs. 120000 (e) Rs. 121200

Directions (16 – 20): Study the following pie chart carefully and answer the questions given below.

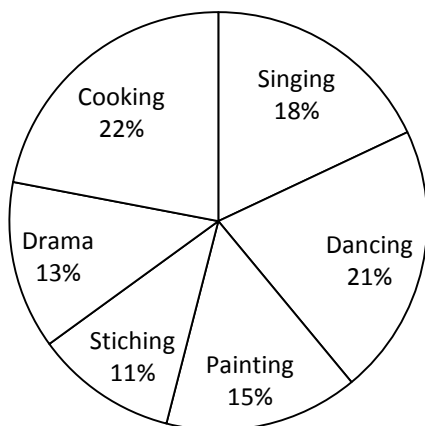
A student scored 60% of the total marks in an exam. Percentage distribution of marks scored by the student in various subjects with respect to total marks obtained.

Total Marks Obtained = 1200



16. The marks scored in GA, Computer and Descriptive when added together, represents what percentage of the total marks?
 (a) 24.1% (b) 23.7% (c) 25.8% (d) 19.9% (e) 33.3%
17. The marks scored in Math and Reasoning added together, exceed the marks scored in GA and Descriptive added together, by how many marks?
 (a) 156 (b) 278 (c) 148 (d) 198 (e) 238
18. In which subject did the student score 168 marks?
 (a) Math (b) Reasoning (c) Computer (d) GA (e) None of these
19. The marks scored in English and Computer together is how much per cent of marks scored in GA and Math together?
 (a) 82% (b) 68.5% (c) 75% (d) 72.5% (e) 77.5%
20. The approximate average marks scored by student in each subject are
 (a) 140 (b) 100 (c) 150 (d) 200 (e) 155

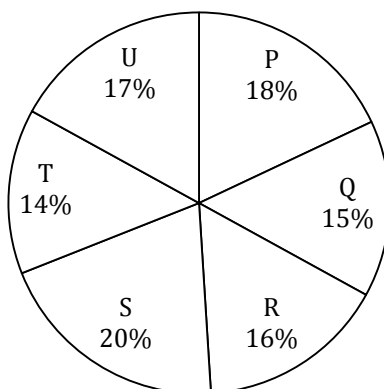
Directions (21-25): Given below is the pie chart which shows the percentage of students enrolled in different Hobby classes in a school in year 2016



Total Enrolled students = 7200

21. What is the difference between average of number of students enrolled in Dancing and sticking hobby together and average of students enrolled in Painting and cooking hobby together
 (a) 160 (b) 180 (c) 175 (d) 165 (e) 190
22. If in year 2017 total students who were enrolled in singing and painting increases by $\frac{100}{3}\%$ and 20% respectively, then find the total number of enrolled students in singing and painting in 2017.
 (a) 3334 (b) 3245 (c) 3525 (d) 3600 (e) 3024
23. What is the ratio of total students enrolled in Dancing and drama together to the 160% of total students enrolled in cooking and painting together.
 (a) 85 : 148 (b) 80 : 83 (c) 33 : 43 (d) 11 : 22 (e) 11 : 27
24. What is the average of number of students enrolled in Singing, Dancing and Painting together.
 (a) 1331 (b) 941 (c) 1296 (d) 1225 (e) 1025
25. If total students who are enrolled in painting and Singing together in 2017 are $\frac{100}{3}\%$ more than those enrolled in these two hobby in 2016 and in 2017 total enrolled boys in these hobby are 20% more than enrolled girls in these hobby in 2017. Find number of enrolled girls in 2017 in these hobbies together?
 (a) 1320 (b) 1225 (c) 1520 (d) 1580 (e) 1440

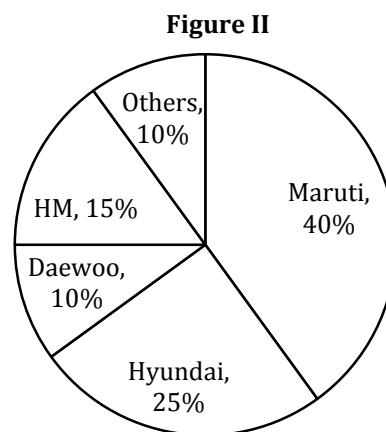
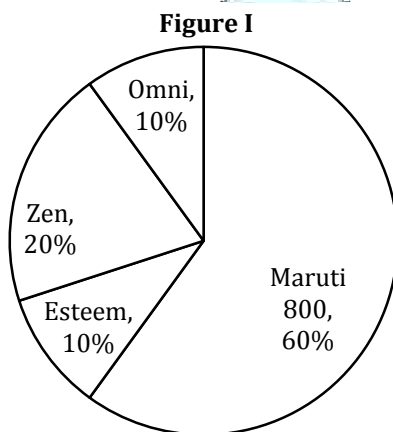
Directions (26-30): Given below is the pie-chart which shows the percentage distribution of total population of Six villages in year 2016



Total population = 85000

26. If ratio of male to female in village P and village Q is 2 : 3 and 4 : 1 respectively then females in village P are what percent of females in village Q.
 (a) 200% (b) 175% (c) 350% (d) 360% (e) 275%
27. If in the year 2017 population of village R and S increases by 20% and 25% respectively what is the sum of population in these villages in 2017.
 (a) 43280 (b) 35275 (c) 37570 (d) 32250 (e) 39450
28. If in village S ratio of male to female is 3 : 2 and 20% males and 10% females are illiterate the what percent of population in village S is literate
 (a) 81% (b) 82% (c) 83% (d) 84% (e) 85%
29. What is the ratio of population of village P, Q and S together to the population of village R, T and U together.
 (a) 53 : 47 (b) 48 : 53 (c) 47 : 42 (d) 30 : 23 (e) 33 : 32
30. If female from village R and U together equals to total population of village Q and female in village R equals to $\frac{200}{3}$ % of females of village U then number of males in village U is
 (a) 7650 (b) 6500 (c) 7800 (d) 6800 (e) 7000

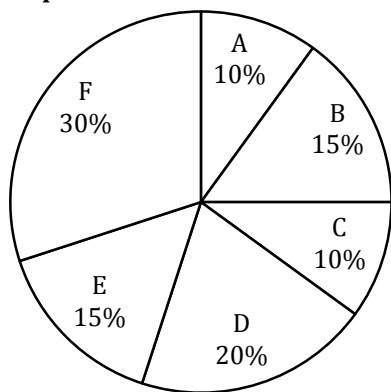
Directions (31-35): The break-up of the volume share of different cars sold by Maruti Car Company in Delhi for year 2002 is shown in figure I. The break up for the year 2003 for maruti car company in delhi was same as that for the year 2002. The breakup of the car market according to market share by volume possessed by different car manufacturing in Delhi for 2002 is shown in figure II.



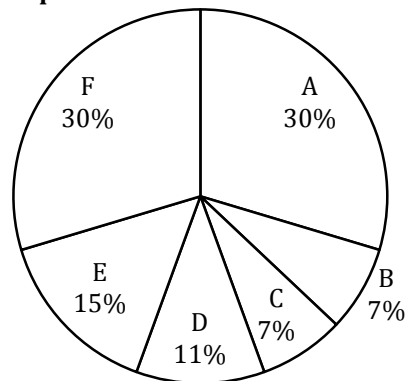
31. If in 2002, the total no. of cars sold in Delhi was 2.5 lakh, then what is the ratio of no. of cars sold by company HM in 2002 to the no. of cars sold by Maruti of models Omini, Zen and Esteem together in the same year?
 (a) 15 : 16 (b) 12 : 13 (c) 17 : 15 (d) 16 : 15 (e) 15 : 17
32. If total no. of cars sold in Delhi in 2002 is 4 lakhs, then how many of the cars manufactured by Maruti remained unsold?
 Given: no. of cars sold by Maruti in 2002 is $88\frac{8}{9}$ % of the no. of cars manufactured by it.
 (a) 0.6 lakh (b) 0.2 lakh (c) 0.4 lakh (d) 0.25 lakh (e) 0.35 lakh
33. What is the no. of Santro sold in Delhi if Santro accounts for 75% of the Hyundai sales in 2002? The no. of Zen sold in 2002 is equal to 3.0 lakh.
 (a) 232489 (b) 730125 (c) 703125 (d) 264489 (e) 710325
34. If the total sales of cars in 2002 in India was 24 lakh and Delhi constituted 5% of India's total car sales then what was the sales of Maruti 800 in Delhi?
 (a) 26400 (b) 24200 (c) 27000 (d) 28800 (e) 28600
35. If total sales of cars in Delhi in 2002 was 3 lakh and it increased by 20% in 2003, then find percentage increase in the sales of Omni in 2003?
 (a) Can't be determined (b) 25% (c) 20% (d) 22% (e) 14.25%

Directions (36-40): The pie chart given below shows the percentage distribution of the production of various models of a car manufacturing company in 2007 and 2008. The total production in 2007 was 35 lakh cars and in 2008 the production was 44 lakh. Study the chart and answer the following questions.

Total production in 2007 = 35 Lakhs



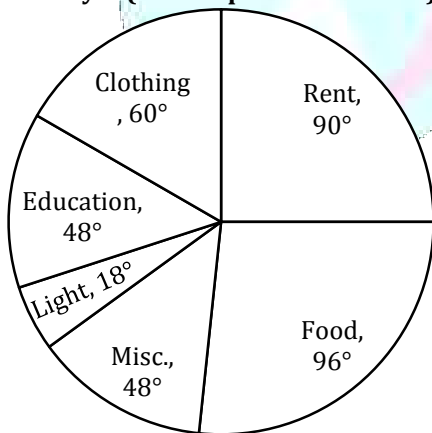
Total production in 2008 = 44 Lakhs



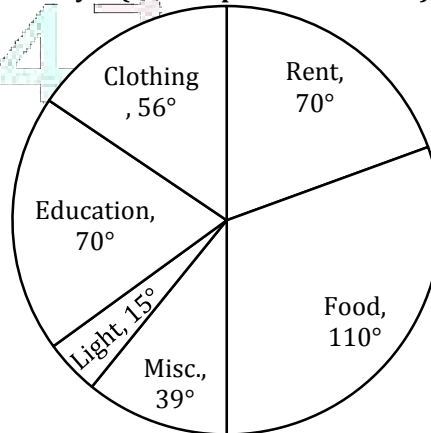
36. Total number of cars of models A, B and E manufactured in 2007 was
 (a) 2450000 (b) 22750000 (c) 1400000 (d) 1925000 (e) 225000
37. For which models the difference between number of cars produced in 2007 and 2008 is maximum?
 (a) Model A (b) model B (c) Model C (d) Model D (e) None of these
38. What was the difference in the number of B type cars produced in 2007 and 2008?
 (a) 217000 (b) 270000 (c) 225000 (d) 175000 (e) 200000
39. The production of A type cars in 2008 is approximately what percent of the production of A type cars in 2007?
 (a) 360% (b) 377% (c) 365% (d) 362% (e) 370%
40. If 85% of D type cars produced in each year were sold by the company, then how many D type cars remained unsold in total?
 (a) 76500 (b) 93500 (c) 177600 (d) 122500 (e) 196000

Directions (41-45): The two pie diagrams given below provide relative expenses of two families A and B.

Family A (total expense= Rs. 4800)



Family B (total expense= Rs. 7200)



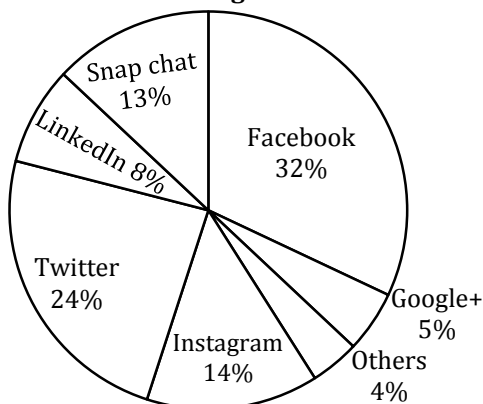
41. If 60% of expenses on misc. of family A are on 'repair' and 30% of expenses on misc. of family B were on 'travel' then what is the ratio of expenses on education of family A to the sum of expenses on repair of family A and that of travel of family B?
 (a) 310 : 309 (b) 330 : 319 (c) 320 : 309 (d) 330 : 312 (e) 309 : 311
42. If total expenses of family A are tripled, then expense on education, clothing and rent together of family B is what percentage of expenses on clothing, rent and Food together of family A? (Rounded off to two decimal places)
 (a) 36.75% (b) 42.23% (c) 38.43% (d) 39.83% (e) 32.98%
43. The expenses on education of families A and B together is what percent more/less than the expense on rent of both the families? (approximately)
 (a) 22% (b) 28% (c) 18% (d) 32% (e) 16%
44. If the total expenses of A and B are doubled, then what is the ratio of expenditure on light of family A to that of misc. of family B?
 (a) 37 : 43 (b) 4 : 13 (c) 42 : 11 (d) 11 : 42 (e) 13 : 4
45. The item showing the least difference in expenditure between family A and family B is
 (a) Light (b) Clothing (c) Misc. (d) Education (e) None of these

Directions (46-50) : Study the following information carefully and answer the following questions.

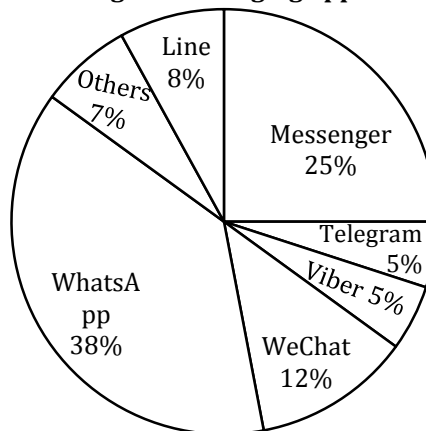
The pie-charts show the percentage of users of different social networking sites and the percentage of users of different messaging apps.

The ratio of the total number of social networking sites to the total number of messaging apps is 4: 5.

Percentage of users of social networking sites



Percentage of messaging apps users



46. If the total number of Facebook users is 192 million then what is the difference between the total number of Instagram users and the total number of WhatsApp users? (in million)
 (a) 201 (b) 200 (c) 185 (d) 211 (e) 241
47. If the total number of WhatsApp users is 285 million then what is the average number of messaging apps users (in million)? (approximately)
Note: Assume Others to be a single messaging app.
 (a) 120 (b) 125 (c) 118 (d) 115 (e) 107
48. If the total number of people who use LinkedIn networking site is 64 million then what is the difference between the total number of people who use social networking sites and the total number of people who use messaging apps?
 (a) 200 million (b) 241 million (c) 215 million (d) 214 million (e) 220 million
49. If the total number users of Twitter are 192 million then the users of Snapchat are what per cent of the Instagram and Google+ users together? (rounded off to two decimal places)
 (a) 62.42% (b) 68.42% (c) 64.82% (d) 70.28% (e) 78.48%
50. If the total number of Line users is 68 million then what is the sum of Facebook and WeChat users?
 (a) 501.5 million (b) 319.6 million (c) 532.1 million (d) 502.6 million (e) None of these

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PRACTICE SET (LEVEL-I) SOLUTIONS

1. (a); Total spending on Tennis = $\frac{45}{360} \times 100 = 12.5\%$
2. (c); Desired value = $\frac{(70-25)}{360} \times 144 = 18 \text{ cr.}$
3. (b); Increase in Hockey = $144 \times \frac{70}{360} \times \frac{120}{100} = 33.6$
Decrease in football = $144 \times \frac{80}{360} \times \frac{70}{100} = 22.4$
Difference in spending = $33.6 - 22.4 = 11.2$
4. (b); Amount spent on Tennis, Football and Hockey
= $\frac{195}{360} \times 144 = 19.5 \times 4 = 78 \text{ cr.}$
Desired ratio = $\frac{78}{84} = \frac{13}{14}$
5. (d); Amount spend on others = $\frac{10}{360} \times 144 = 4 \text{ cr.}$
Difference = $17 \text{ cr} - 4 \text{ cr} = 13 \text{ cr.}$
6. (c); Required difference = $\frac{18 \times (23-15)}{5} = \frac{18 \times 8}{5} = \frac{144}{5} = 28.8$
7. (c); Required ratio = $(15 + 12)\% : \frac{150}{100}(18 + 32)\%$
= $27 \times 2 : 3 \times 50 = 9 : 25$
8. (a); Sum of articles sold by both A and C in 2017
= $18 \times 60 \times \frac{4}{3} + 12 \times 60 \times \frac{7}{6}$
= $18 \times 20 \times 4 + 2 \times 60 \times 7 = 1440 + 840 = 2280$
9. (b); Required difference
= $\frac{1}{2} \times \frac{6000}{100} [(12 + 23) - (18 + 15)]$
= $30[35 - 33] = 60$
10. (d); Total articles sold now = $\frac{4}{3} \times 6000 = 8000$
Required difference = $\frac{8000}{100} [(15 + 32) - (18 + 23)]$
= $\frac{8000}{100} (47 - 41) = 80 \times 6 = 480$
11. (a); Required difference = $\frac{14}{360} \times 1080000 = \text{Rs. } 42000$
12. (d); Clearly wages paid to Daily wage workers & Maintenance staffs are equal to production unit II members.
13. (d); $270000 = \frac{1}{4} \text{ of } 1080000 = \frac{90}{360} \text{ of } 1080000$
According to given pie-chart 90° belongs to production unit II.
Also $90^\circ = 60^\circ + 30^\circ$ which belongs to maintenance staff and Daily wage workers
Hence, option d is correct.
14. (c); Wages paid to daily wage workers = $\frac{1}{6} \text{ of } 1080000$
= 180000
After ex-gratia grant,
Total wages paid = $180000 \times \frac{120}{100} = 216000$
No. of such workers = $\frac{216000}{48 \times 900} = 5$
15. (b); For year 2016-17,
Total wages to be paid = $10,80,000 \times \frac{110}{100}$
= $11,88,000$

- $$\text{Required difference} = \frac{62-28}{360} \times 1188000 = \text{Rs. } 11,2200$$
16. (c); Total marks obtained in GA, Computer and Descriptive together
= $\frac{(18+14+11)}{100} \times 1200 = 43 \times 12 = 516$
And total marks in an examination
= $\frac{1200}{60} \times 100 = 2000$
Required % = $\frac{516}{2000} \times 100 = 25.8\%$
 17. (a); Required marks = $\frac{[(22+20)-(18+11)]}{100} \times 1200$
= $13 \times 12 = 156$
 18. (c); 168 marks = $\frac{168}{1200} \times 100 = 14\%$
⇒ Marks obtained in Computer
 19. (d); Required % = $\frac{15+14}{22+18} \times 100 = \frac{29}{40} \times 100 = 72.5\%$
 20. (d); Required average = $\frac{1200}{6} = 200$
 21. (b); Required difference
= $\frac{1}{2} [(15 + 22)\% - (21 + 11)\%] 7200$
= $\frac{1}{2} [5\%] 7200 = 180$
 22. (e); Total enrolled students in singing and painting in 2017
= $(\frac{4}{3} \times 18\% + \frac{6}{5} \times 15\%) 7200 = 42 \times 72 = 3024$
 23. (a); Required ratio
= $(21\% + 13\%) : 160\%(22\% + 15\%)$
= $5 \times 34 : 8 \times 37 = 170 : 296 = 85 : 148$
 24. (c); Required average = $\frac{1}{3} (18 + 21 + 15)\% \times 7200$
= $18 \times 72 = 1296$
 25. (e); Total enrolled students in Painting and Singing in 2017 = $\frac{4}{3} \times (15\% + 18\%) \times 7200 = 3168$
Let total enrolled girls in 2017 in Painting & Singing = x
So,
 $3168 = (x + \frac{6}{5}x)$ Or, $x = \frac{3168 \times 5}{11} = 1440$
 26. (d); In this question, we don't have to calculate values.
Let total population of all 6 villages be x
So, female in village P = $\frac{3}{5} \times 18\% x$
Female in village Q = $\frac{1}{5} \times 15\% x$
Required percentage = $\frac{\frac{3}{5} \times 18}{\frac{1}{5} \times 15} \times 100 = 360\%$
 27. (c); Sum of population of R and S in 2017
= $\frac{6}{5} \times \frac{16}{100} \times 85000 + \frac{5}{4} \times \frac{20}{100} \times 85000$
= $850 (\frac{6}{5} \times 16 + \frac{5}{4} \times 20) = 37570$
 28. (d); Literate male in village S = $\frac{3}{5} \times \frac{20}{100} \times 85000 \times \frac{80}{100}$

$$\text{Literate female in village S} = \frac{2}{5} \times \frac{20}{100} \times 85000 \times \frac{90}{100}$$

$$\text{Require \%} = \frac{20 \times 850 \left(\frac{3}{5} \times \frac{4}{5} + \frac{2}{5} \times \frac{9}{10} \right)}{20 \times 850} \times 100$$

$$= \left(\frac{12}{25} + \frac{18}{50} \right) \times 100 = \frac{(24+18)}{50} \times 100 = 84\%$$

$$29. \text{ (a); Required ratio} = \frac{(18\%+15\%+20\%) \times 85000}{(16\%+14\%+17\%) \times 85000} = 53 : 47$$

$$30. \text{ (d); Total females in village R \& U} = \frac{15}{100} \times 85000$$

$$= 12750$$

Let females in village U = x

So,

$$\left(x + \frac{2}{3}x \right) = \frac{15}{100} \times 85000$$

$$\frac{5}{3}x = 12750$$

$$x = 7650$$

$$\text{Males in village U} = 17 \times 850 - 7650 = 6800$$

$$31. \text{ (a); Required ratio} = \frac{\frac{15}{100} \times 2.5}{\frac{(10+20+10)}{100} \times \frac{40}{100} \times 2.5} = \frac{15}{16}$$

$$32. \text{ (b); No. of cars sold by Maruti} = \frac{40}{100} \times 4 = 1.6 \text{ lakh}$$

$$\text{No. of cars Manufactured by Maruti} = 1.6 \times \frac{9}{8}$$

$$= 1.8 \text{ lakh}$$

$$\text{No. of cars remains unsold} = 1.8 - 1.6 = 0.2 \text{ lakh}$$

$$33. \text{ (c); Total no. of maruti cars sold in Delhi} = 3 \times \frac{100}{20}$$

$$= 15 \text{ lakh}$$

$$\text{Total no. of cars sold in Delhi} = 15 \times \frac{100}{40} = 37.5 \text{ lakh}$$

$$\text{No. of Santro sold} = \frac{75}{100} \times \frac{25}{100} \times 37.5 = 7.03125 \text{ lakh}$$

$$= 703125$$

$$34. \text{ (d); No. of Maruti 800 cars sold} = \frac{60}{100} \times \frac{40}{100} \times \frac{5}{100} \times 24$$

$$= 28800$$

35. (a); Since market share for the year 2003 is not given, the answer can't be determined.

$$36. \text{ (c); Required Answer} = \frac{35}{100} (10 + 15 + 15) \text{ lakh}$$

$$= \frac{35 \times 40}{100} = 14 \text{ lakh}$$

37. (a); it can be clearly seen from the chart that maximum variation is in production of model A = (30% of 44 lakh - 10% of 35 lakh) = 9.7 lakh

$$38. \text{ (a); Required difference} = \frac{35 \text{ lakh} \times 15}{100} - \frac{44 \text{ lakh} \times 7}{100}$$

$$= 2.17 \text{ lakh}$$

$$= 217000$$

$$39. \text{ (b); Required \%} = \frac{13.2}{3.5} \times 100 = 377 \% (\text{approx.})$$

$$40. \text{ (c); Required No.}$$

$$= 35 \text{ lakh} \times \frac{20}{100} \times \frac{15}{100} + 44 \text{ lakh} \times \frac{11}{100} \times \frac{15}{100}$$

$$= 1.776 \text{ lakh}$$

$$= 177600$$

$$41. \text{ (c); Req. Ratio} = \frac{\frac{48}{360} \times 4800}{\frac{60}{100} \times \frac{48}{360} \times 4800 + \frac{30}{100} \times \frac{39}{360} \times 7200}$$

$$= \frac{640}{384+234} = \frac{640}{618} = \frac{320}{309}$$

$$42. \text{ (d); Req. Percentage} = \frac{\frac{196}{360} \times 7200}{\frac{246}{360} \times 14400} \times 100 = 39.83\%$$

$$43. \text{ (a); Expense on education} = \frac{48}{360} \times 4800 + \frac{70}{360} \times 7200$$

$$= 2040$$

$$\text{Expense on rent} = \frac{90}{360} \times 4800 + \frac{70}{360} \times 7200$$

$$= 2600$$

$$\text{Req. percentage} = \frac{(2600-2040)}{2600} \times 100$$

$$= 21.53\% \approx 22\%$$

$$44. \text{ (b); Req. ratio} = \frac{\frac{18}{360} \times 4800}{\frac{39}{360} \times 7200} = \frac{4}{13}$$

(Note: Doubled expenses don't alter the ratio)

45. (a);

Item	Difference
Clothing	1120 - 800 = 320
Rent	1400 - 1200 = 200
Food	2200 - 1280 = 920
Misc.	780 - 640 = 140
Light	300 - 240 = 60
Education	1400 - 640 = 760

$$46. \text{ (a); Required difference} = \frac{192 \times 100 \times 5 \times 38}{32 \times 4 \times 100} - \frac{192}{32} \times 14$$

$$= 285 - 84 = 201 \text{ million}$$

$$47. \text{ (e); Required average} = \frac{1}{7} \times \frac{285}{38} \times 100$$

$$= 107.14 \text{ million}$$

$$48. \text{ (a); Required difference} = \frac{800}{4} \times 5 - \frac{64}{8} \times 100$$

$$= 1000 - 800 = 200 \text{ million}$$

$$49. \text{ (b); Required \%} = \frac{\frac{192}{24} \times 13}{\frac{192}{24} \times (14+5)} \times 100$$

$$= \frac{104}{152} \times 100 = 68.42\%$$

$$50. \text{ (b); Required users} = \frac{68 \times 100 \times 4 \times 32}{8 \times 100 \times 5} + \frac{68}{8} \times 12$$

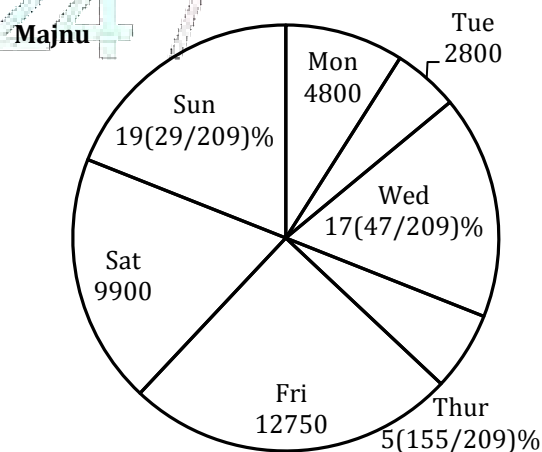
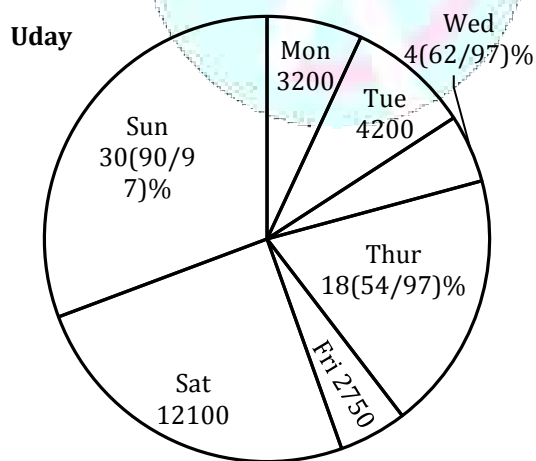
$$= 217.6 + 102 = 319.6 \text{ million}$$



PRACTICE SET (LEVEL-II)

Directions (1-5): Study the following pie charts and answer the following questions carefully :-

Uday and Majnu are working in a restaurant, they serve different no. of customers on different days in a week. No. of customers or percentage of the customers are given in both pie charts from Monday to Sunday

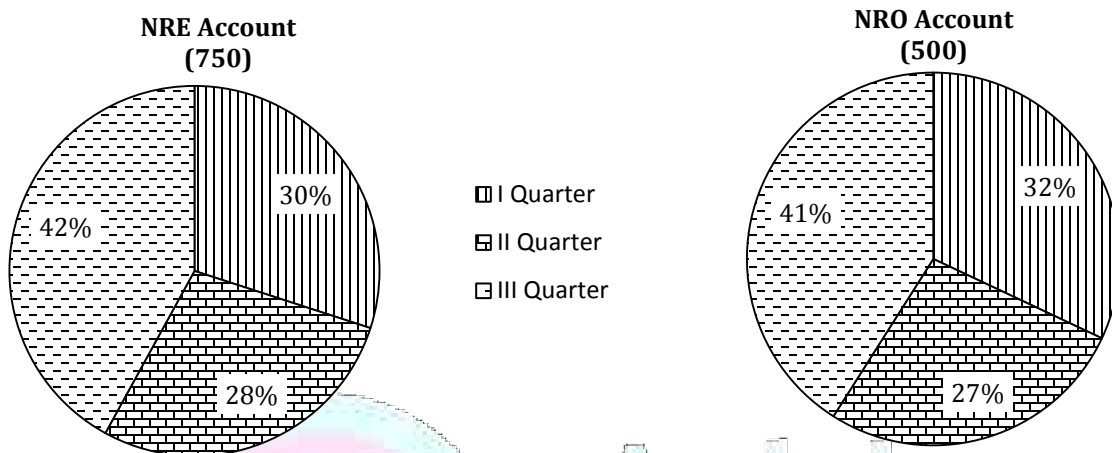


- Customers served by Majnu on Saturday are what % more/less than that the customers served by Uday on Monday and Tuesday together ? (Rounded off to two decimal points)
 (a) 33.66% (b) 33.78% (c) 33.87% (d) 33.58% (e) 32.78%
- Total customers served by Uday during the whole week are approximately what % of the total customers served by both together during the whole week ?
 (a) 42% (b) 48% (c) 52% (d) 45% (e) 53%
- What is the ratio of the customer served by Uday on Tuesday and Sunday together to the customer served by Majnu on Wednesday and Friday together ?
 (a) 63 : 64 (b) 51 : 53 (c) 147 : 148 (d) 178 : 179 (e) 128 : 145

4. The customers served by Majnu from Mon to Wednesday are how much less that the customers served by Uday from Thursday to Sunday?
 (a) 23,209 (b) 25,666 (c) 22,250 (d) 22,102 (e) 21,935
5. On which day of the week Majnu served the maximum customers ?
 (a) Sunday (b) Monday (c) Thursday (d) Friday (e) Saturday

Directions (6-10): In the given pie chart, In state Bank of India there are two types of accounts NRE account and NRO account which can be opened by a foreigner.

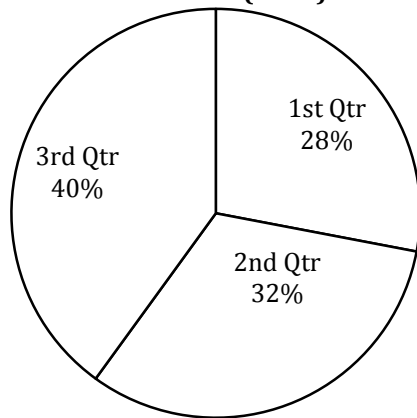
These pie charts show the percentage wise breakup of these accounts spend in a given year. There are 4 quarters in a year and graph shown the information about three quarters.



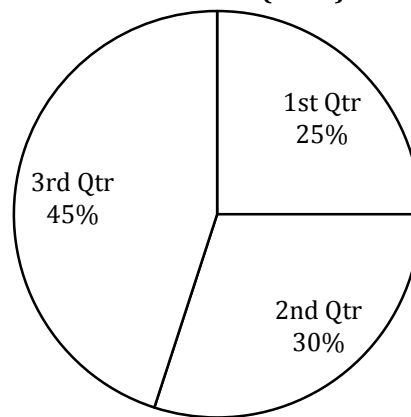
6. If we include the 4th quarter in the given year, percentage of NRO accounts opened in 2nd quarter will become $16\frac{7}{8}\%$ of the total NRO accounts opened during the whole year. Then what is the number of NRO accounts opened in 4th quarter ?
 (a) 450 (b) 300 (c) 350 (d) 250 (e) 260
7. Total number of NRE accounts opened in the 4th quarter are $42\frac{6}{7}\%$ more than the NRE accounts opened in 3rd quarter, then find the average number of NRE accounts opened during whole year if the NRE accounts opened in 4th quarter are 50% more than the NRO accounts opened in 4th quarter?
 (a) 325 (b) 200 (c) 350 (d) 250 (e) 300
8. If the total number of NRE accounts opened in the whole year are 50% more than the NRO account opened in the whole year then find the ratio of the NRE accounts opened in 4th quarter to NRO accounts opened in 4th quarter ?
 (a) 2 : 3 (b) 2 : 5
 (c) 3 : 2 (d) Can't be determined
 (e) None of these
9. If the NRE accounts opened in IVth quarter is 240 more than the NRE accounts opened in 2nd quarter, then NRE accounts opened in 4th quarter is what percent of the total NRE accounts opened in the whole year ?
 (a) 37.5% (b) 39.5% (c) 35.5% (d) 36% (e) 34.6%
10. If 16 % NRE account holders and 18% NRO account holders close their account then total no. of NRE accounts in 2nd and 3rd quarter is approximately what percent more than the total no. of NRO accounts in these quarters respectively ?
 (a) 36 % (b) 45 %
 (c) 52 % (d) Can't be determined
 (e) None of these

Directions(11-15)- The given pie graphs show the percentage wise breakup of production of cricket bats and cricket balls in a given year. There are 4 quarters in a year and graph shows the information for three quarters.

Cricket bats (2000)

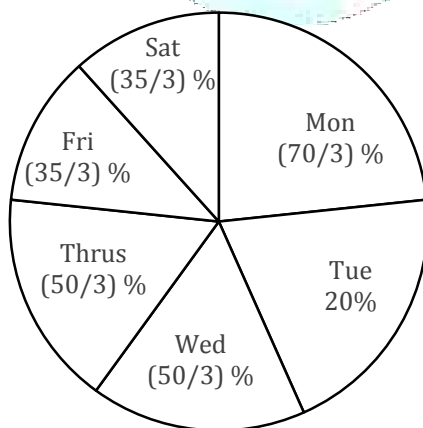


Cricket balls (5000)

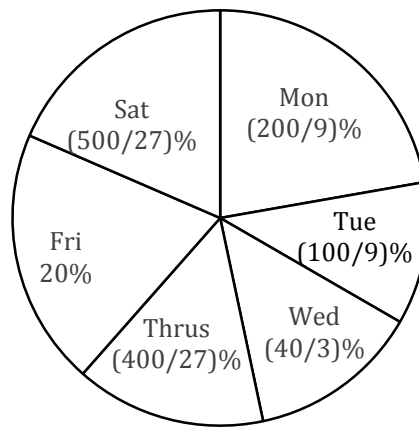


11. If we include the 4th quarter of the year, percentage of cricket bats in 1st quarter will become 25% of the total cricket bats produced during the whole year. Then what is the number of cricket bats in 4th quarter?
(a) 260 (b) 230 (c) 280 (d) 240 (e) 250
12. If the Cricket balls produced in 4th quarter is $\frac{1}{3}$ less than the Cricket balls produced in 2nd quarter. Then Cricket balls produced in 4th quarter is what percent of total number of Cricket balls produced. (up to 2 decimal places).
(a) 25% (b) 16.67% (c) 14.28% (d) 33.33% (e) 22.22%
13. Total no of cricket bats produced in 2nd and 3rd quarter is what percent of total no. of the Cricket balls produced in 2nd and 3rd quarter?
(a) 39.23% (b) 38.4% (c) 37.6% (d) 33.33% (e) 41.15%
14. Average number of cricket bats produced in 1st and 2nd quarter is how much percent more or less than the number of cricket balls produced in 3rd quarter? (up to 2 decimal places).
(a) 73.33% (b) 72.16% (c) 26.67% (d) 27.84% (e) 71.84%
15. If the production of cricket balls in 4th quarter is 35% more than that of cricket bats in 2nd quarter and production of cricket bats in 4th quarter is 25% less than the production of cricket balls in 4th quarter, then the production of cricket bats in 2nd quarter is what % of total production of cricket bats taking all the quarter together. (up to 2 decimal points)
(a) 23.52% (b) 24.17% (c) 26.37% (d) 23.92% (e) 24.96%

Directions (16-20): Given below are the two pie charts which shows the percentage distribution of people who travel a certain distance in Delhi metro and Mumbai metro on six different days of the week starting from Monday to Saturday.



Delhi metro



Mumbai metro

Note:

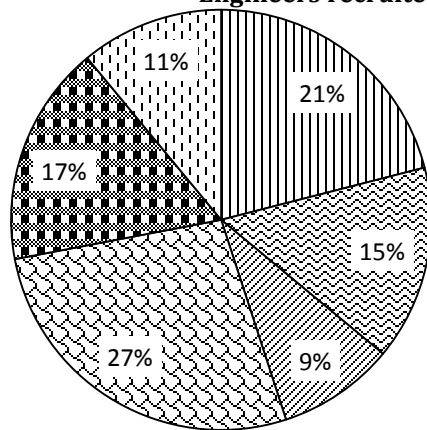
1. Ratio of total person travelling in these six days in Delhi metro to Mumbai metro is 10 : 9
2. Difference between person travelling in Delhi metro and Mumbai metro on Wednesday is 70.
16. If fare per person in Delhi metro and Mumbai metro on all days for the particular distance is Rs. 18 and Rs. 20 respectively then what is the difference between total fare obtained by both metro on Saturday.
(a) 1375 (b) 1750 (c) 1850 (d) 1700 (e) 1650
17. If in both metro, number of people travelling on Sunday of same week decreases by 'x' with respect to people travelling on Saturday then the ratio of people travelling in Delhi metro to Mumbai metro on Sunday is 2 : 3, then find the value 'x'
(a) 20 (b) 30 (c) 22 (d) 24 (e) 25

18. Number of people travelling in Delhi metro on Wednesday and Thursday together is what percent of people travelling in Mumbai metro on Monday and Saturday together ?
- (a) $90\frac{10}{11}\%$ (b) $89\frac{10}{11}\%$ (c) $90\frac{2}{11}\%$ (d) $92\frac{8}{11}\%$ (e) $95\frac{5}{11}\%$
19. If fare per person of Delhi metro to Mumbai metro is 10 : 9 on all days and sum of fare obtained from both metro on Tuesday is Rs. 4350, then total fare obtained from Delhi metro on Monday is what percent more or less than total fare obtained from Mumbai metro on Saturday.
- (a) $46\frac{4}{9}\%$ (b) $45\frac{4}{9}\%$ (c) $55\frac{5}{9}\%$ (d) $54\frac{4}{9}\%$ (e) $52\frac{5}{9}\%$
20. If on Sunday of same week, person who travel by Delhi metro and Mumbai metro are increased by 20% and 30% respectively over Saturday, then total people who travelled by both metro on Sunday is what percent of total people who travelled by both metro on Monday.
- (a) $81\frac{5}{17}\%$ (b) $87\frac{2}{13}\%$ (c) $93\frac{4}{15}\%$ (d) $82\frac{4}{13}\%$ (e) $78\frac{3}{13}\%$



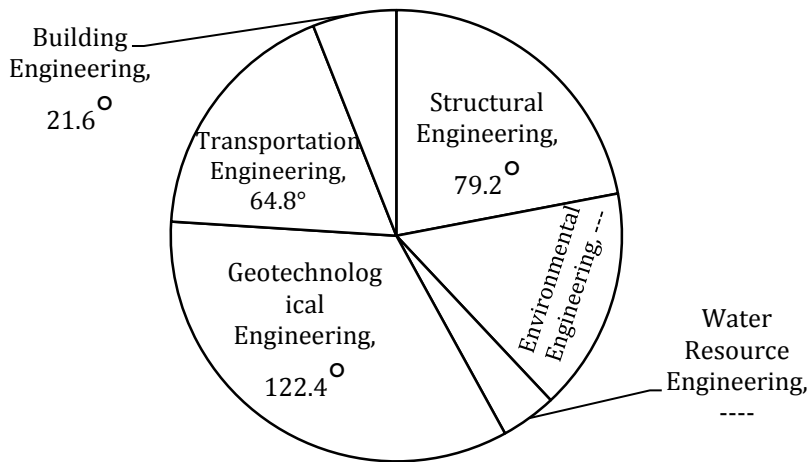
Direction (21-25): L&T pvt limited recruited civil engineers for Infrastructure Project in different streams from all over the India. But it fixes some seats for engineers from IITs. The information regarding this is given below

Engineers recruited all over the world Total=12600



- ☐ Structural Engineering
- ☒ Environmental Engineering
- ☒ Water Resource Engineering
- ☒ Geotechnological Engineering
- ☒ Transportation Engineering
- ☐ Building Engineering

Engineers recruited from IITs Total=6800

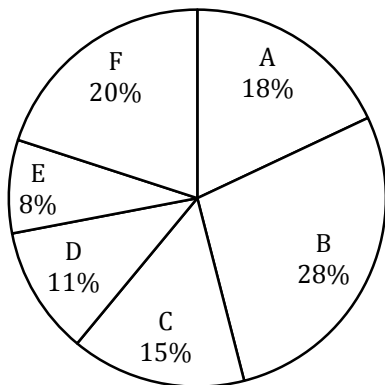


NOTE:- SOME VALUE ARE MISSING. YOU HAVE TO CALCULATE THESE VALUES ACCORDING TO QUESTION.

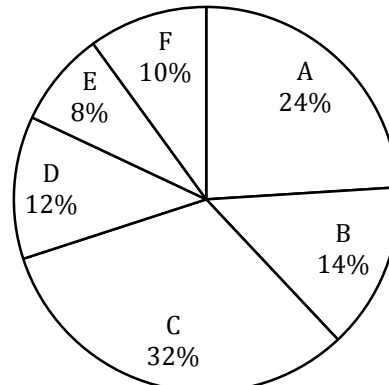
21. Engineers Recruited by company except from IITs in stream Geotechnological Engineering 30% are master degree holders and Engineers recruited from IITs in Building Engineering 25% are master degree holders. Then find the ratio of engineers recruited from non-IIT in Geotechnological Engineering having master degree to Engineers recruited from IITs in building Engineering having master degree?
 (a) 106 : 39 (b) 107 : 35 (c) 109 : 34 (d) 101 : 31 (e) 105 : 32
22. If the number of engineers recruited from IITs in Environmental and water Resource Engineering are in ratio of 4 : 1, then how many Engineers recruited in Environmental Engineering are non-IITians ?
 (a) 804 (b) 802 (c) 799 (d) 796 (e) 792
23. Number of Transport Engineers recruited by the company from IIT by what percent more than the number of non - IITian Transport Engineers recruited by company?
 (a) $33\frac{1}{3}\%$ (b) $33\frac{2}{3}\%$ (c) $32\frac{1}{3}\%$ (d) $32\frac{2}{3}\%$ (e) 34%
24. If in structural Engineering, the ratio of Engineers from IITs and non-IITs should be 1 : 1. Then by what percent less it should recruit Engineers from IITs in structural Engineering.
 (a) 14.29% (b) 12.78% (c) 10.31% (d) 11.56% (e) 15.29%
25. Average number of non-IITian Engineers recruited in Geotechnological and Transportation Engineering, taking together?
 (a) 1298 (b) 1004 (c) 1678 (d) 928 (e) 1238

Directions (26-30): Study the pie-charts carefully and answer the following question.
 Distribution of Laptops (HP and Asus) sold by six sellers

Total number of laptops sold = 48,000



Number of Asus laptops sold = 28,000

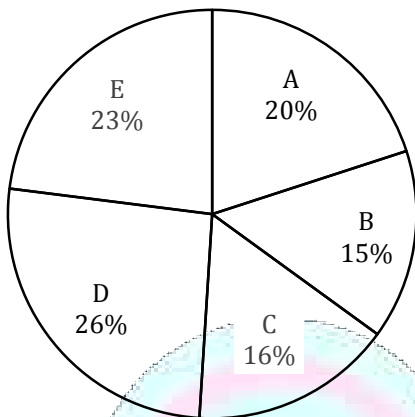


26. Number of HP Laptops sold by A and B together is approximately what percentage more or less than the number of Asus laptop sold by E and F together?
 (a) 145% (b) 227% (c) 127% (d) 245% (e) 97%

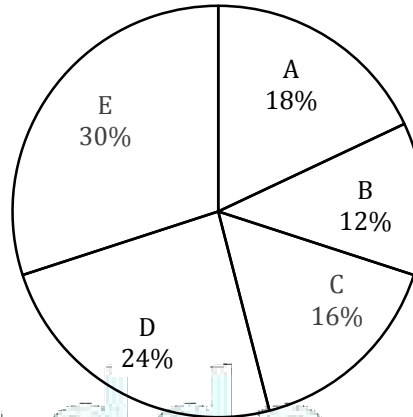
27. Find the difference between the number of Asus laptop sold by D and E together to the number of HP laptop sold by A, B and F together?
 (a) 10,450 (b) 12,460 (c) 10,540 (d) 12,640 (e) 12,540
28. What is ratio of number of HP laptop sold by D and F together to HP laptop sold by B and E together?
 (a) $\frac{139}{109}$ (b) $\frac{109}{139}$ (c) $\frac{113}{127}$ (d) $\frac{127}{113}$ (e) $\frac{121}{147}$
29. Find the total profit earned by seller A if seller A earns Rs. 40 and Rs. 70 per laptop on HP and Asus respectively.
 (a) Rs. 5,47,200 (b) Rs. 5,27,500 (c) Rs. 5,47,900 (d) Rs. 4,77,200 (e) Rs. 5,77,200
30. If seller E sells Asus laptop at a loss of Rs. 30 each so at how much profit he should sell each HP laptop to gain Rs. 3,200 in total.
 (a) Rs.39 (b) Rs.41 (c) Rs.43 (d) Rs.45 (e) Rs. 44

Directions (31-35): Following pie charts show the percentage distribution of males and females in five companies. Study the charts carefully to answer the questions that follow.

Distribution of males



Distribution of females

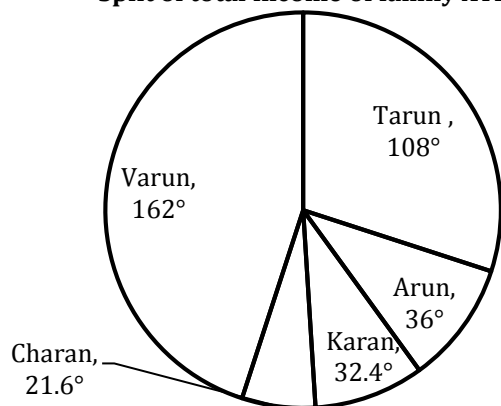


31. If ratio of number of females to number of males in company B is 32 : 45 then number of males in company E is what percent more than the number of females in company C. (approx.)
 (a) 53% (b) 62% (c) 65% (d) 80% (e) 75%
32. If the average of total males and females of all the company together is 21000 and difference between male to female in company C is 320 (no. of males > no. of females) then find the total females in company A.
 (a) 3422 (b) 3500 (c) 3420 (d) 2000 (e) 3600
33. Find the average of number of males in company D and company C and females in company B if ratio of total number of males to total number of females is 3:2 and number of males in the company D is 3900.
 (a) 2500 (b) 1500 (c) 1600 (d) 1700 (e) 2600
34. If ratio between total females to total males is $x^3 : (x + 1)^2$ (x is a whole number), then find the minimum possible value of x , given that number of males in company A is 25% more than the number of females in the same company.
 (a) 4 (b) 1 (c) 6 (d) 3 (e) 2
35. Ratio between average of number of male of company A, B and C to the average of number of females of company A, B and E is 51 : 65 then total females are how much percent more than total males.
 (a) 8.45% (b) 7.34% (c) 7.14% (d) 8.33% (e) 9.33%

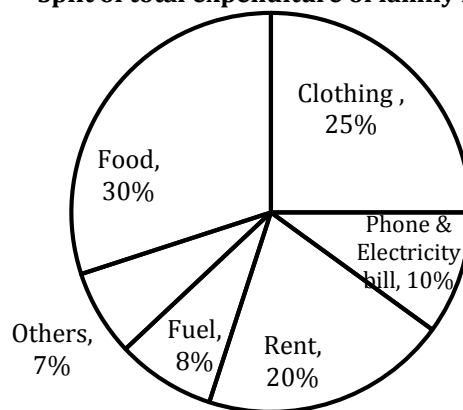
Direction(36-40): Answer the questions on the basis of the following information.

The following pie charts gives the breakup of the income of all the five members – Varun, Tarun, Arvind, karan and Charan of family XYZ and the breakup of the total family expenditure under different heads.

Split of total income of family XYZ



Split of total expenditure of family XYZ



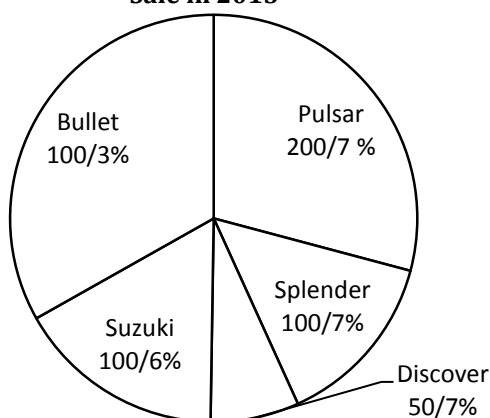
Note: The total income of the family is equal to the total expenditure and the family has no other sources of income.

In question head means individual part of expenditure i.e. clothing, rent, fuel etc.

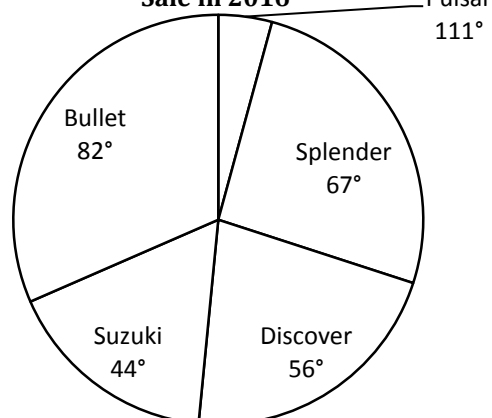
36. If Varun did not pay for "others", then his income can fully account for expenses under at most how many heads?
 (a) 2 (b) 3 (c) 4 (d) 5 (e) 6
37. Whenever possible, if all the expenses under one head is paid by a single person, the number of heads under which more than one person shared the expenses is at least
 (a) 1 (b) 2 (c) 3 (d) 4 (e) 5
38. If Varun does not spend any amount on food, then the expenditure of Varun on clothing and rent as a percentage of the total expenditure on rent and clothing cannot be less than
 (a) 33.33% (b) 44.44% (c) 25% (d) 66.66% (e) 54.44%
39. If at most 40% of the income of each person is paid for food, then the number of persons who did not pay for food is at most
 (a) 1 (b) 2 (c) 3 (d) 4 (e) 6
40. If at least 5% of the total expenses under each head is paid from Karan's income, then the percentage share of Karan's payment under any head can be a maximum of
 (a) 22.5% (b) 90% (c) 62.14% (d) $66\frac{2}{3}$ % (e) 61.24%

Directions (41-45): Given below are the two pie charts. Pie chart I shows the percentage distribution of different models of bike sold in year 2015 and pie chart II shows the sale of these models of bike in 2016 in degree.

Sale in 2015



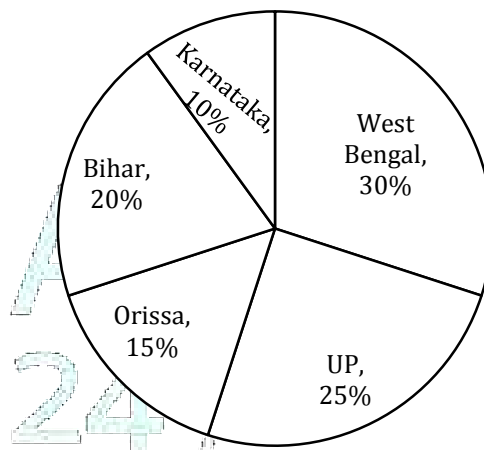
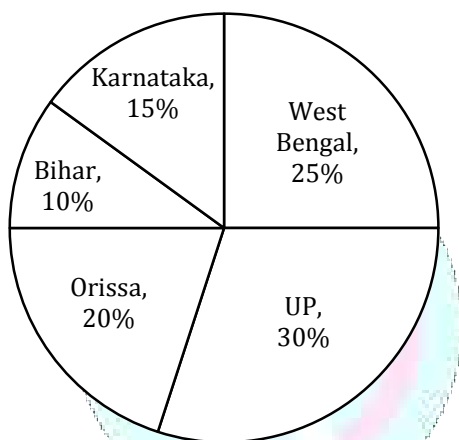
Sale in 2016



41. If ratio of total sale in 2015 to the total sale in 2016 is 4 : 5 then sale of pulsar in 2016 increases or decreases by what percentage in comparison to sale of Pulsar in 2015. ?

- (a) 35% (b) 30% (c) 43% (d) 52% (e) 24%
42. If total sale in 2015 is 84000 and total sale of bikes in 2016 is increased by 20% then find the difference in sale of bike splendor in 2015 and 2016.
(a) 5530 (b) 6250 (c) 6760 (d) 5230 (e) None of these
43. If the ratio of sale of pulsar in 2015 to sale of Bullet in 2016 is 3 : 7 then sale of discover in 2016 is what percent of sale of Suzuki in 2015 (approximately)
(a) 238% (b) 242% (c) 217% (d) 273% (e) 222%
44. If selling price of Suzuki per bike in 2015 is 45000 and selling price of Suzuki per bike in 2016 is 54000 then what is the ratio of total selling price of Suzuki in 2015 to the total selling price of Suzuki in 2016. If ratio of total sale in 2015 to total sale in 2016 is 2 : 3.
(a) $\frac{25}{33}$ (b) $\frac{36}{37}$ (c) $\frac{24}{29}$ (d) $\frac{18}{23}$ (e) None of these
45. If total sale of all bikes in 2015 and in 2016 are equal then sale of Bullet in 2015 is what percent more or less than sale of Bullet in 2016 (approximately)
(a) 40% (b) 32% (c) 38% (d) 43% (e) 46%

Directions (46-50) : Study the pie charts given below and answer the following questions . First Pie chart represent percentage distribution of 'Population of children' and second pie chart represent percentage distribution of 'no. of schools' in five Indian states – Karnataka, Orissa, Bihar, UP and West Bengal.



Note: All the children attend school.

46. Which state has got the maximum no. of children per school?
(a) Karnataka (b) Bihar (c) Orissa (d) UP (e) none of these
47. By how much percent should the no. of schools in Orissa be increased so that the no. of students per school in Bihar becomes equal to the no. of students per school in Orissa?
(a) $133\frac{1}{3}\%$ (b) $166\frac{2}{3}\%$ (c) $120\frac{2}{3}\%$ (d) $140\frac{1}{3}\%$ (e) $162\frac{2}{3}\%$
48. If Orissa has 1 lakh children and UP has 500 schools then find the ratio of average no. of children in Karnataka and west Bengal to the no. of schools in all the states excluding Karnataka?
(a) 200:9 (b) 300:9 (c) 500:9 (d) 400:9 (e) none of these
49. If total area of all the schools in Bihar is 5000000 square feet and all the schools in Bihar are identical and the no. of children in UP is 5 lakh more than that in West Bengal then find the no. of children per school in UP? (Area of a school in Bihar is 2500 sq. ft)
(a) 3000 (b) 2000 (c) 2500 (d) 1200 (e) 1500
50. If 30% of the students in Orissa drop out of school after 5th standard, then find the difference between the no. of students per school till 5th standard & no. of students per school after 5th standard for the state of Orissa? (No. of children in Orissa = 1 lakh and total no. of schools in all the five states = 2000)
(a) 90 (b) 85 (c) 100 (d) 80 (e) 120

PRACTICE SET (LEVEL-II) SOLUTIONS

Directions (1-5):**UDAY:**

We have to find the value which are missing like on Wed = 4

$$\frac{62}{97} \% ; \text{THU} = 18 \frac{54}{97} \% ; \text{SUN} = 30 \frac{90}{97} \%$$

$$\therefore 4 \frac{62}{97} \% + 18 \frac{54}{97} \% + 30 \frac{90}{97} \% = 54 \frac{12}{97} \% = ?$$

$$\text{Given : } (100 - 54 \frac{12}{97}) \% = 12100 + 2750 + 4200 + 3200 = 22250$$

$$\therefore 1 \% = \frac{22250 \times 97}{4450} = 485$$

Hence, Total customers for Uday = 100% = 48500

$$\text{Customers on Wednesday} = 485 \times 4 \frac{62}{97} = 2250$$

$$\text{Customers on Thursday} = 485 \times 18 \frac{54}{97} = 9000$$

$$\text{Customers on Sunday} = 485 \times 30 \frac{90}{97} = 15000$$

MANJU :

We have to find the value which are missing like on Wed = 17

$$\frac{47}{209} \% ; \text{Thu} = 5 \frac{155}{209} \% ; \text{Sun} = 19 \frac{29}{209} \%$$

$$\therefore 17 \frac{47}{209} \% + 5 \frac{155}{209} \% + 19 \frac{29}{209} \% = 42 \frac{22}{209} \% = ?$$

$$\text{Given : } (100 - 42 \frac{22}{209}) \% = 4800 + 2800 + 12750 + 9900 = 30250$$

$$\therefore 1 \% = 522.5$$

Or, 100% = Total customers for Manju = 52250

$$\text{Hence, Customers on Wednesday} = 522.5 \times 17 \frac{47}{209} = 9000$$

$$\text{Customers on Thursday} = 522.5 \times 5 \frac{155}{209} = 3000$$

$$\text{Customers on Sunday} = 522.5 \times 19 \frac{29}{209} = 10000$$

Therefore, accumulated table as follows:

	UDAY	MAJNU
Mon	3200	4800
Tue	4200	2800
Wed	2250	9000
Thu	9000	3000
Fri	2750	12750
Sat	12100	9900
Sun	15000	10000
Total	48500	52250

$$1. \text{ (b); Required \%} = \frac{(9900-7400)}{7400} \times 100 = \frac{2500}{74} = 33.78\%$$

$$2. \text{ (b); Required \%} = \frac{48500}{100750} \times 100 \approx 48\%$$

$$3. \text{ (e); Required Ratio} = 19200 : 21750 = 128 : 145$$

$$4. \text{ (c); Required difference} = 38850 - 16600 = 22250$$

$$5. \text{ (d); Clearly from the graph, it's on Friday}$$

$$6. \text{ (b); Let total NRO accounts opened in the whole year} = x$$

$$\therefore 16 \frac{7}{8} \% \text{ of } x = \frac{27}{100} \times 500$$

$$x = 800$$

$$\text{Required accounts opened} = (800 - 500) = 300$$

$$7. \text{ (e); NRE accounts opened in 4th quarter}$$

$$= 142 \frac{6}{7} \% \text{ of } \left(\frac{42}{100} \times 750 \right) = 450$$

$$\text{Required average} = \frac{450+750}{4}$$

$$= \frac{1200}{4} = 300$$

$$8. \text{ (d); Let}$$

$$\text{NRE accounts} = 3x$$

$$\text{NRO accounts} = 2x$$

$$(3x - 750) = (2x - 500)1.5x$$

We can't determine the value of x .

$$9. \text{ (a); IV quarter (NRE accounts)} = 240 + \frac{28}{100} \times 750 = 210 + 240 = 450$$

$$\text{Required \%} = \frac{450}{1200} \times 100 = 37.5\%$$

$$10. \text{ (d); Since we don't know total no. of accounts. Hence we can't give the required answer}$$

$$11. \text{ (d); Production of Cricket bats in 1st quarter} = 2000 \times \frac{28}{100}$$

According to the question,

$$560 = 25\% \text{ of total production of bats}$$

$$\text{Total production} = \frac{560}{25} \times 100 = 2240$$

$$\text{Production of 4th quarter} = 2240 - 2000 = 240$$

$$12. \text{ (b); Cricket- balls produced in 4th quarter}$$

$$= \left(5000 \times \frac{30}{100} \right) \times \left(1 - \frac{1}{3} \right) = 1000$$

$$\text{Total production including 4th quarter} = 5000 + 1000 = 6000$$

$$\text{Required answer} = \frac{1000}{6000} \times 100 = 16.67\%$$

$$13. \text{ (b); Total production of Cricket- bats in 2nd and 3rd quarter}$$

$$= \frac{2000 \times (40+32)}{100} = 1440$$

$$\text{Total production of Cricket- balls in 2nd and 3rd quarter}$$

$$= \frac{5000 \times (30+45)}{100} = 3750$$

$$\text{Required answer} = \frac{1440}{3750} \times 100 = 38.4\%$$

$$14. \text{ (a); Total production of Cricket- bats in 1st and 2nd quarter}$$

$$= \frac{2000 \times (28+32)}{100} = 1200$$

$$\text{Average production} = \frac{1200}{2} = 600$$

$$\text{Production of Cricket- balls in 3rd quarter} = \frac{5000 \times 45}{100} = 2250$$

$$\text{Required answer} = \frac{(2250 - 600)}{2250} \times 100 = 73.33\% \text{ less}$$

$$15. \text{ (b); Production of Cricket- bats in 2nd quarter} = \frac{2000 \times 32}{100}$$

$$= 640$$

$$\text{Production of Cricket- balls in 4th quarter}$$

$$= \frac{640 \times (100 + 35)}{100} = 864$$

$$\text{Production of 4th quarter Cricket- bats} = \frac{864 \times (100 - 25)}{100}$$

$$= 648$$

$$\text{Total production of Cricket- bats all the quarter together} = 2000 + 648 = 2648$$

$$\text{Required answer} = \frac{\left(2000 \times \frac{32}{100} \right) \times 100}{2648} = 24.17\%$$

- 16. (c);** Let total person travelling through Delhi metro in all six days = $10x$

So total person travelled through Mumbai metro in all six days = $9x$

According to condition

$$\frac{10x}{100} \times \frac{50}{3} - \frac{9x}{100} \times \frac{40}{3} = 70$$

$$500x - 360x = 70 \times 300$$

$$140x = 70 \times 300$$

$$x = 150$$

Total person travelled through Delhi metro in all six days = 1500

Total person travelled through Mumbai metro in all six days = 1350

Total fare of Delhi metro on Saturday

$$= \frac{35}{300} \times 1500 \times 18 = \text{Rs. } 3150$$

Total fare of Mumbai metro on Saturday

$$= \frac{5}{27} \times 1350 \times 20 = \text{Rs. } 5000$$

Required difference = 1850

17. (e); $\frac{\frac{35}{300} \times 1500 - x}{\frac{5}{27} \times 1350 - x} = \frac{2}{3}$

$$\frac{175 - x}{250 - x} = \frac{2}{3}$$

$$525 - 3x = 500 - 2x$$

$$x = 25$$

- 18. (a);** Total people travelling in Delhi metro on Wednesday and Thursday

$$= \frac{10x}{100} \times \left(\frac{50}{3} + \frac{50}{3} \right) = \frac{10x}{3}$$

Total people travelling in Mumbai metro on Monday and Saturday together

$$= \frac{9x}{100} \times \left(\frac{200}{9} + \frac{500}{27} \right) = \frac{11x}{3}$$

$$\text{Required percentage} = \frac{\frac{10x}{3}}{\frac{11x}{3}} \times 100$$

$$= \frac{1000}{11} \% = 90 \frac{10}{11} \%$$

- 19. (c);** $20 \times 15 \times 10x + \frac{100}{9} \times 13.5 \times 9x = 4350$

$$3000x + 1350x = 4350$$

$$4350x = 4350$$

$$x = 1$$

$$\therefore \text{Required percentage} = \frac{15 \times \frac{70}{3} \times 10 - 13.5 \times \frac{500}{27} \times 9}{13.5 \times \frac{500}{27} \times 9} \times 100$$

$$= \frac{3500 - 2250}{2250} \times 100 = \frac{1250}{2250} \times 100$$

$$= \frac{500}{9} \% = 55 \frac{5}{9} \%$$

Alternate Method

Person travelled on Delhi metro on Monday

$$= 1500 \times \frac{70}{300} = 350$$

Person travelled on Mumbai metro on Saturday

$$= 1350 \times \frac{500}{2700} = 250$$

Let fare per person of Delhi metro and Mumbai metro is $10x$ and $9x$ respectively

Required percentage

$$= \frac{350 \times 10x - 250 \times 9x}{250 \times 9x} \times 100 = \frac{1250x}{2250x} \times 100 = 55 \frac{5}{9} \%$$

- 20. (d);** Total person travelling both metro on Sunday

$$= \frac{120}{100} \times 15 \times \frac{35}{3} + \frac{130}{100} \times 13.5 \times \frac{500}{27}$$

$$= 210 + 325 = 535$$

Total people travelled by both metro on Monday

$$= 15 \times \frac{70}{3} + 13.5 \times \frac{200}{9}$$

$$= 350 + 300 = 650$$

$$\text{Required percentage} = \frac{535}{650} \times 100 = 82 \frac{4}{13} \%$$

- 21. (c);** Required ratio = $\frac{\frac{30}{100} \times 1090}{\frac{25}{100} \times 408} = \frac{30 \times 1090}{25 \times 408} = \frac{109}{34}$

- 22. (b);** Number of Environmental Engineers recruited from

$$\text{IITs} = \frac{4}{5} \times \frac{72^\circ}{360^\circ} \times 6800 = 1088$$

Number of non-IITians Environmental Engineers

$$\text{recruited} = \frac{15}{100} \times 12600 - 1088$$

$$= 1890 - 1088 = 802$$

- 23. (a);** Number of Transport Engineers from IITs

$$= \frac{64.8^\circ}{360^\circ} \times 6800$$

$$\text{Total number Transport Engineers} = \frac{17}{100} \times 12600$$

$$= 2142$$

$$\text{Non-IITian Transport Engineers} = 2142 - 1224$$

$$= 918$$

$$\text{Required percent} = \frac{1224 + 918}{918} \times 100$$

$$= \frac{306}{918} \times 100 = 33 \frac{1}{3} \%$$

- 24. (d);** Total number recruited structural Engineers

$$= \frac{21}{100} \times 12600 = 2646$$

Number of Engineers should be recruited from IITs

$$= \frac{1}{2} \times 2646 = 1323$$

Initially number of structural Engineers recruited

$$\text{from IITs} = \frac{79.2}{360} \times 6800 = 1496$$

$$\text{Required percent} = \frac{1496 - 1323}{1496} \times 100 = 11.56\%$$

- 25. (b);** Average = $\frac{918 + 1090}{2} = \frac{2008}{2} = 1004$

- 26. (c);** HP laptop sold by A = $\frac{18}{100} \times 48,000 - \frac{24}{100} \times 28,000$

$$= 8640 - 6720 = 1920$$

$$\text{HP laptop sold by B} = \frac{28}{100} \times 48,000 - \frac{14}{100} \times 28,000$$

$$= 13,440 - 3,920 = 9,520$$

HP laptop sold by A and B together = 11,440

$$\text{Asus laptop sold by E and F together} = \frac{18}{100} \times 28,000$$

$$= 5040$$

$$\text{Required percentage} = \frac{11,440 - 5040}{5040} \times 100$$

$$= \frac{6400}{5040} \times 100 \approx 127\%$$

- 27. (d);** Asus laptop sold by D and E together = $\frac{20}{100} \times 28,000$

$$= 5600$$

$$\text{HP laptop sold by A and B} = 11,440$$

$$\text{HP laptop sold by F} = \frac{20}{100} \times 48,000 - \frac{10}{100} \times 28,000$$

$$= 9600 - 2800 = 6800$$

$$\text{Desired difference} = 11,440 + 6,800 - 5,600 = 12,640$$

28. (b); HP laptop sold D and F together

$$= \frac{(20+11)}{100} \times 48,000 - \frac{(12+10)}{100} \times 28,000$$

$$= 14,880 - 6,160 = 8,720$$

HP laptop sold by B and E together

$$= \frac{(28+8)}{100} \times 48,000 - \frac{(14+8)}{100} \times 28,000$$

$$= 17,280 - 6,160 = 11,120$$

$$\text{Desired Ratio} = \frac{8,720}{11,120} = \frac{109}{139}$$

29. (a); Asus laptop sold by A = $\frac{24}{100} \times (28,000) = 6,720$

HP laptop sold by A

$$= \frac{18}{100} (8,000) - \frac{24}{100} (28,000) = 1,920$$

$$\text{Total profit of A} = 40 \times 1,920 + 70 \times 6,720$$

$$= 76,800 + 4,70,400 = \text{Rs. } 5,47,200$$

30. (e); Number of Asus laptop sold by E = $\frac{8}{100} \times 28,000$

$$= 2240$$

$$\text{Total loss} = 2,240 \times 30 = 67,200$$

$$\text{Total S.P. of HP laptop} = 67,200 + 3,200 = 70,400$$

Total no. of HP laptop sold by E

$$= \frac{8}{100} \times 48,000 - \frac{8}{100} \times 28,000$$

$$= 3,840 - 2240 = 1600$$

$$\text{Profit should be gain on HP laptop} = \frac{70,400}{1,600}$$

$$= \text{Rs. } 44 \text{ each}$$

31. (b); Let total no. of males = x

total no. of females = y

ATQ,

$$\frac{12 \times y}{100} \times \frac{100}{15 \times x} = \frac{32}{45}$$

$$\frac{y}{x} = \frac{8}{9}$$

Number of males and females 8a, 9a

$$\text{No. of males in E} \rightarrow \frac{23 \times 9a}{100} = 2.07a$$

$$\text{No. of females in C} = \frac{16 \times 8a}{100} = 1.28a$$

$$\text{Required \%} = \frac{0.79a}{1.28a} \times 100 \approx 62\%$$

32. (e); Total males and females = $21000 \times 2 = 42000$

Now let total no. of males is 'x' and total no. of females is 'y'

ATQ,

$$\frac{16 \times x}{100} - \frac{16 \times y}{100} = 320$$

$$x - y = 2000 \quad \dots(i)$$

$$x + y = 42000 \quad \dots(ii)$$

Solving (i) and (ii)

$$x = 22000$$

$$y = 20000$$

$$\text{Female in company A} = \frac{18 \times 20000}{100} = 3600$$

33. (a); Total no. of males = $\frac{3900 \times 100}{26} = 15000$

$$\text{Total no. of females} = \frac{15000 \times 2}{3} = 10,000$$

$$\text{Required average} = \frac{3900+2400+1200}{3} = 2500$$

34. (e); Let number of females in company A = 4y

So, number of males in company A = 5y

$$\text{Total number of males} = \frac{5y}{20} \times 100 = 25y$$

$$\text{Total number of females} = \frac{4y}{18} \times 100 = \frac{200}{9}y$$

ATQ,

$$\frac{200y}{9 \times 25y} = \frac{x^3}{(x+1)^2}$$

$$\frac{8}{9} = \frac{x^3}{(x+1)^2}$$

$$\text{Value of } x = 2$$

35. (d); Let total no. males = x

total no. of females = y

So, average no. of male in company A, B and C

$$= \frac{(20+16+15)x}{100 \times 3} = \frac{17x}{100}$$

Average no. of females of company A, B and E

$$= \frac{(18+12+30)y}{100 \times 3} = \frac{20y}{100}$$

Ratio $\rightarrow$

$$\frac{17x}{100} : \frac{20y}{100} = 51 : 65$$

$$\frac{x}{y} = \frac{12}{13}$$

$$\text{Required percentage} = \frac{1}{12} \times 100 = 8.33\%$$

36. (b); If Varun did not pay for 'others' he can fully pay for fuel (8%), phone & electricity bill (10%) and rent (20%) or clothing (25%).

37. (a); To get the least number of heads of expenses paid by more than one person, Varun (45%) must pay for clothing (25%) and rent (20%), Tarun (30%) must pay for food (30%), Arun (10%) must pay for the phone and electricity bill (10%) and Karan (9%) must pay for fuel.

Only 'others' (7%) is paid by Charan (6%) and Karan (9%)

38. (b); If Varun does not spent any amount on food, his expenditure will be only on the remaining items. As remaining items constitute 70% out of which 45 percent points are contributed by Varun. If Varun fully contributes to fuel, phone and electricity bill and others, then his contribution on rent and clothing will become the least.

$$\therefore \text{The required percentage} = \frac{45 - (10 + 8 + 7)}{45} \times 100 = 44.44\%$$

39. (c); The bill for food is 30%, and at most 40% of each person's income can be paid for food. If we use 40% of each person's income, we get 40% of the total. As we need only 30%, i.e., 75% of 40%, 25% of the total income need not be used. As the sum of the incomes of Arun, Karan and Charan is 25%, if we use 40% of incomes of only Varun and Tarun, all expenses of food can be accounted for.

40. (c); Assuming exactly 5% of the total expenses under each head is paid from Karan's income, it will account for 5% of the total income. As Karan's income is 9%

of the total income, the remaining = 4% of total income. For the percentage share of karan's payment under any head to be maximum, he should contribute all his remaining income for the head under which the expenditure is the least, i.e. 'others'. As he has already paid for 5% of the expenses under that head, together with the remaining 4%, his share for payment under the head 'others' would be $5\% + \frac{4}{7} \times 100 = 62.14\%$.

41. (a); Let Total sale in 2015 = $4x$

Let total sale in 2016 = $5x$

$$\text{Sale of pulsar in 2015} = \frac{4x}{100} \times \frac{200}{7} = \frac{8x}{7}$$

$$\text{Sale of pulsar in 2016} = \frac{5x}{360} \times 111 = \frac{37x}{24}$$

$$\text{Required percentage} = \frac{\frac{37x}{24} - \frac{8x}{7}}{\frac{8x}{7}} \times 100$$

$$= \frac{67}{24 \times 8} \times 100 \Rightarrow \frac{67}{192} \times 100 \approx 35\% \text{ increase}$$

42. (c); Total sale of bike in 2016 = $\frac{120}{100} \times 84000 = 100800$

$$\text{Sale of splendor in 2015} = \frac{84000}{100} \times \frac{100}{7} = 12000$$

$$\text{Sale of splendor in 2016} = \frac{100800}{360} \times 67 = 18760$$

$$\text{Required difference} = 18760 - 12000 = 6760$$

43. (d); Let sale of Pulsar in 2015 = $3x$

Let sale of Bullet in 2016 = $7x$

$$\text{Sale of Suzuki in 2015} = \frac{3x \times 7}{200} \times \frac{100}{6} = \frac{7x}{4}$$

$$\text{Sale of discover in 2016} = \frac{7x}{82} \times 56$$

$$\text{Required \%} = \frac{\frac{7x \times 56}{82} - \frac{7x}{4}}{\frac{7x}{4}} \times 100$$

$$= \frac{112}{41} \times 100 \approx 273\%$$

44. (a); Let total sale in 2015 = $2x$

Let total sale in 2016 = $3x$

$$\text{Sale of Suzuki in 2015} = \frac{2x}{100} \times \frac{100}{6} = \frac{x}{3}$$

$$\text{Sale of Suzuki in 2016} = \frac{3x}{360} \times 44 = \frac{11x}{30}$$

$$\text{Required ratio} = \frac{\frac{x}{3} \times 45000}{\frac{11x}{30} \times 54000} = \frac{25}{33}$$

45. (e); Let total sale for both years = x

$$\text{Sale of Bullet in 2015} = \frac{x}{100} \times \frac{100}{3} = \frac{x}{3}$$

$$\text{Sale of Bullet in 2016} = \frac{x}{360} \times 82$$

$$\text{Required \%} = \frac{\frac{x}{3} - \frac{82x}{360}}{\frac{82x}{360}} \times 100 = \frac{19x}{41x} \times 100 \approx 46\%$$

46. (a); Let, no. of children in all five states be 'x'

And no. of schools in all five states be 'y'

Ratio of students to no. of schools in Karnataka

$$= \frac{\frac{15}{100} \times x}{\frac{10}{100} \times y} = \frac{3}{2} \left(\frac{x}{y} \right)$$

This is max. among all the states.

47. (b); Students per school in Bihar = $\frac{10x}{20y} = \frac{1x}{2y}$

$$\text{Students per school in Orissa} = \frac{20x}{15y} = \frac{4x}{3y}$$

No. of schools in Orissa should be increased by '5y', so that

$$= \frac{4x}{(3y+5y)} = \frac{4x}{8y} = \frac{1x}{2y}$$

$$\text{Percentage increase} = \frac{5y}{3y} \times 100 = 166.66\%$$

$$= 166\frac{2}{3}\%$$

48. (c); Average no. of children in Karnataka & West Bengal

$$= \frac{100000 \times \frac{100}{20} \times \frac{(15+25)}{100}}{2} = \frac{500000 \times 40}{200} = 100000$$

No. of schools in all states excluding Karnataka

$$= 500 \times \frac{100}{25} \times \frac{90}{100} = 1800$$

$$\text{Ratio} = \frac{100000}{1800} = \frac{500}{9}$$

49. (d); No. of schools in Bihar = $\frac{5000000}{2500} = 2000$

$$\text{No. of schools in UP} = 2000 \times \frac{100}{20} \times \frac{25}{100} = 2500$$

Let, total no. of children be 'x', then

$$= \frac{30x}{100} - \frac{25x}{100} = 5,00,000$$

$$\text{or, } x = 1,00,00,000$$

$$\text{Children per school in U.P.} = \frac{\frac{30}{100} \times 100000000}{2500} = 1200$$

50. (c); No. of students per school till 5th standard in Orissa

$$= \frac{100000}{\frac{15}{100} \times 2000}$$

No. of students per school after 5th standard in Orissa

$$= \frac{\frac{7}{10} \times 100000}{\frac{15}{100} \times 2000}$$

$$\text{Difference} = \frac{100000}{300} - \frac{70000}{300} = 100$$



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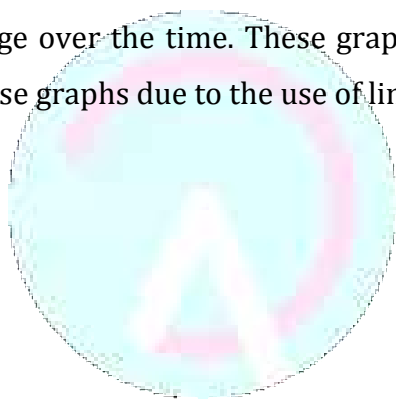
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Line Graph

Line Graphs are very useful in representing the data related to time-series and frequency distribution. These graphs are also very useful in determining trends, rate of change and for illustrating comparisons with respect to some time series. A time series is an arrangement of data in chronological order. Line graphs are drawn by lines connecting the dots which show the value of a variable. It indicates the variation of one parameter with respect to another. It determines trends and rate of change over the time. These graphs are easier to interpret as we can easily see data movement in these graphs due to the use of lines.



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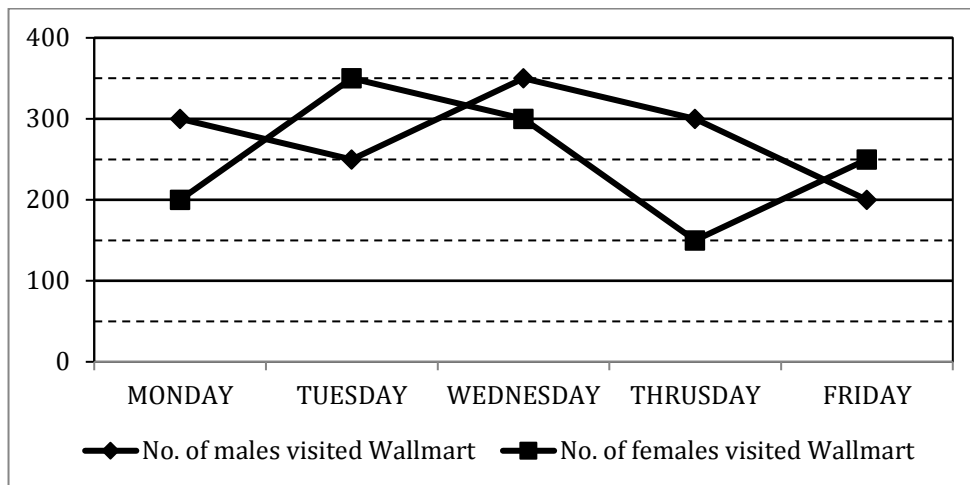
This chapter contains:

- Solved Examples
- Previous Year Questions
- Practice Set Level I
- Practice Set Level II

Solved Examples

Directions (1-5): Read the following line graph and answer the following questions

In the following line graph number of males and number of females visited Walmart on different days is shown.



1. What is the average number of males visited Walmart on Tuesday, Thursday and Friday?

(a) 220 (b) 240 (c) 260 (d) 280 (e) 250

Sol. (e); Desired Average = $\frac{250+300+200}{3} = \frac{750}{3} = 250$

2. What is the difference between the numbers of males who visited Walmart on Monday, Tuesday and Friday together and number of females who visited Walmart on Wednesday, Tuesday and Friday together?

(a) 125 (b) 150 (c) 175 (d) 200 (e) 225

Sol. (b); No. of males visited Walmart on Monday, Tuesday and Friday = $300 + 250 + 200 = 750$
 No. of females visited Walmart on Wednesday, Tuesday and Friday = $300 + 350 + 250 = 900$
 Desired difference = $900 - 750 = 150$

3. What is the percentage increases in total number of males and females who visited Walmart on Wednesday over the total number of males and females who visited on Monday?

(a) 25% (b) 28% (c) 35% (d) 30% (e) 40%

Sol. (d); On Wednesday = 650

On Monday = 500

$$\% \text{ increase} = \frac{650-500}{500} \times 100 = \frac{150}{5} \% = 30\%$$

4. If number of males visited on Friday is increased by 25% on Saturday and number of females visited on Friday is decreased by 10% on Saturday, then what will be the total number of males and females visited Walmart on Saturday?

(a) 400 (b) 425 (c) 450 (d) 475 (e) 500

Sol. (d); No. of males on Saturday = $200 \times \frac{125}{100} = 250$

$$\text{No. of females on Saturday} = 250 \times \frac{90}{100} = 225$$

$$\text{Total no. of males and females visited Walmart on Saturday} = 250 + 225 = 475$$

5. Find the ratio between the average number of females visited Walmart to average number of males visited Walmart?

(a) 28 : 25 (b) 25 : 28 (c) 27 : 23 (d) 23 : 27 (e) None of these

Sol. (b); Average no. of females = $\frac{200+350+300+150+250}{5} = 250$

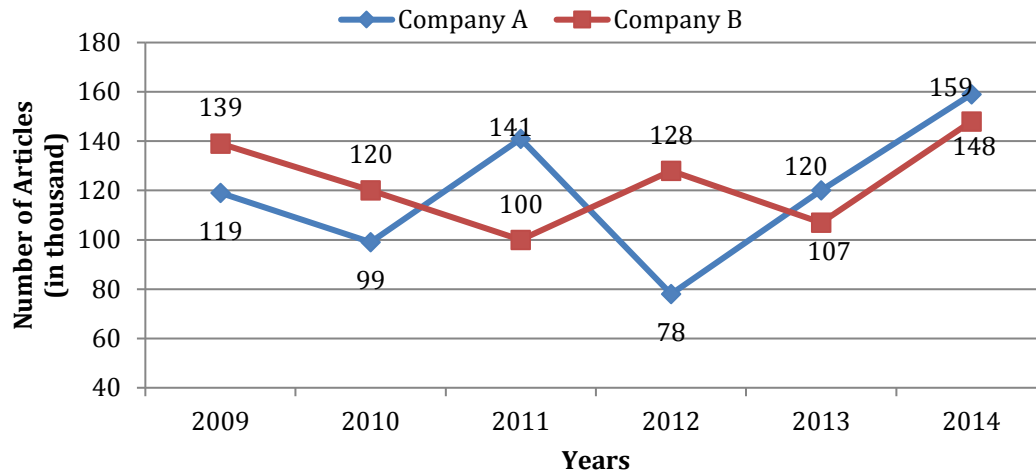
$$\text{Average no. of males} = \frac{300+250+350+300+200}{5} = 280$$

$$\text{Desired Ratio} = \frac{250}{280} = \frac{25}{28}$$

Directions (6-10): Study the following line graph and answer the questions based on it.

Given below is the line graph which shows the number of articles sold by two companies A and B over the years

Number of Articles (in thousand) sold by two companies over the years



6. If 20% and 15% of articles sold by company A in 2012 and 2014 respectively are defective then defective article sold by A in 2012 and 2014 together are what percent of total articles sold by A in 2012 and 2014 together.

(a) $14\frac{53}{79}\%$ (b) $12\frac{51}{79}\%$ (c) $16\frac{51}{79}\%$ (d) $19\frac{47}{69}\%$ (e) 22%

Sol. (c); Defective articles sold by A in 2012 and 2014 = $\left(\frac{20}{100} \times 78 + \frac{15}{100} \times 159\right) \times 1000$

$$= 15600 + 23850 = 39450$$

$$\text{Required \%} = \frac{39450}{(78+159) \times 1000} \times 100$$

$$= \frac{39.45}{237} = \frac{1315}{79} = 16\frac{51}{79}\%$$

7. What is the ratio of articles sold by company A in 2011, 2012 and 2013 together to the articles sold by B in 2009, 2010 and 2011 together.

(a) 339 : 359 (b) 249 : 250 (c) 331 : 336 (d) 125 : 139 (e) None of these

Sol. (a); Required ratio = $\frac{141+78+120}{139+120+100} = \frac{339}{359}$

8. Number of articles sold by A in 2009 and 2011 together is what percent more or less than articles sold by B in 2012 and 2013 together (approximately)

(a) $12\frac{30}{47}\%$ (b) $10\frac{30}{47}\%$ (c) $14\frac{46}{47}\%$ (d) $7\frac{29}{47}\%$ (e) 5%

Sol. (b); Required % = $\frac{(119+141)-(128+107)}{(128+107)} \times 100 = 10\frac{30}{47}\%$

9. If number of articles sold by A in 2008 is 120% more than difference between articles sold by A and B in 2009, then articles sold by A in 2008 is what percent more or less than articles sold by B in 2009.

(a) $64\frac{65}{139}\%$ (b) $68\frac{48}{139}\%$ (c) $63\frac{63}{139}\%$ (d) 70% (e) $65\frac{65}{139}\%$

Sol. (b); Articles sold by A in 2008 = $\frac{220}{100} \times (139 - 119)$

$$= \frac{220}{100} \times 20 = 44$$

$$\text{Required \%} = \frac{139-44}{139} \times 100$$

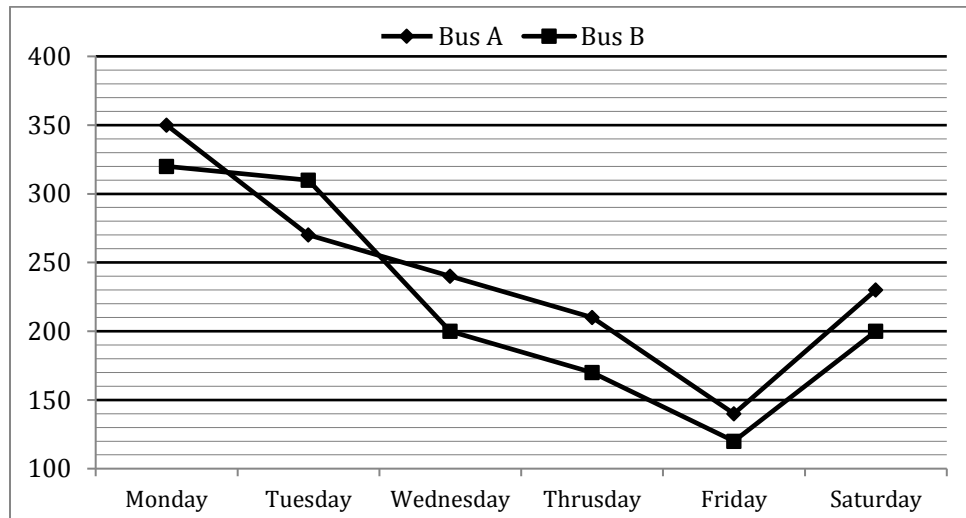
$$= \frac{95}{139} \times 100 = 68\frac{48}{139}\%$$

10. What is the difference between the average of articles sold by A over all years except year 2009 and average of article sold by B over all the years except year 2014 (in thousands).

(a) 1 (b) 0.6 (c) 0.8 (d) 0.4 (e) 0.5

Sol. (b); Required difference = $\frac{1}{5}(597 - 594) = \frac{3}{5} = 0.6$ thousand

Directions (11-15): Given below is the graph which shows the number of people who travelled from Delhi to Bhopal by Bus A and Bus B on six different days.



11. The number of people who travelled by Bus B on Sunday of the same week are 20% more than those of who travelled by the same bus on Saturday. What is the ratio of the number of people who travelled on Sunday to those of who travelled on Tuesday by the same bus B.

(a) 24 : 31 (b) 23 : 32 (c) 33 : 41 (d) 11 : 19 (e) 5 : 7

Sol. (a); People who travelled by bus B on Sunday = $\frac{120}{100} \times 200 = 240$

$$\text{Required ratio} = \frac{240}{310} = 24 : 31$$

12. What is the difference between total people who travelled on Monday and Tuesday together by bus A and total people who travelled on Friday and Saturday by bus B.

(a) 250 (b) 300 (c) 350 (d) 200 (e) 400

Sol. (b); Total people who travelled on Monday and Tuesday together by bus A = $350 + 270 = 620$

$$\text{Total people who travelled on Friday and Saturday by Bus B} = 120 + 200 = 320$$

$$\text{Required difference} = 300$$

13. What is the difference between average number of people who travelled by bus A on Tuesday, Wednesday and Thursday and average number of people who travelled by bus B on Wednesday, Monday and Friday.

(a) $26\frac{2}{3}$ (b) $33\frac{1}{3}$ (c) $14\frac{2}{7}$ (d) $66\frac{2}{3}$ (e) $28\frac{1}{3}$

Sol. (a); Average of people who travelled by bus A on Tuesday, Wednesday and Thursday

$$= \frac{270+240+210}{3} = \frac{720}{3} = 240$$

$$\text{Average of people travelled by bus B on Wednesday, Monday and Friday}$$

$$= \frac{(320+200+120)}{3} = \frac{640}{3}$$

$$\text{Required difference} = 240 - \frac{640}{3} = \frac{80}{3} = 26\frac{2}{3}$$

14. If on Sunday people who travel by bus A and who travel by bus B are increased by 10% and $\frac{25}{2}\%$ respectively over Saturday then the total people travelling on Sunday by both bus is what percent of total people who travel on Monday by both bus.

(a) $37\frac{2}{35}\%$ (b) $72\frac{7}{68}\%$ (c) $71\frac{23}{67}\%$ (d) $48\frac{5}{6}\%$ (e) None of these

Sol. (c); Total people who travel by both bus (A & B) on Sunday

$$= \frac{110}{100} \times 230 + \left(\frac{25}{200} \times 200 + 200 \right)$$

$$= 253 + 225 = 478$$

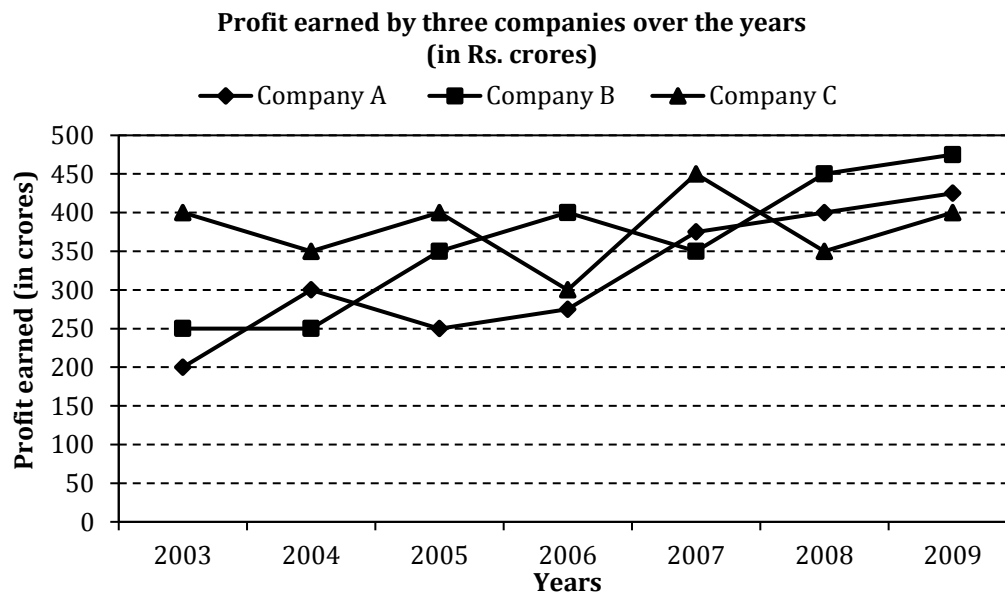
$$\text{Required percentage} = \frac{478}{670} \times 100 = 71\frac{23}{67}\%$$

15. If fare per person of Bus B is 90% of fare per person of Bus A on all days and difference between total fare of bus A and bus B on Monday is Rs. 1240 then find the total fare (in Rs.) of both bus on Friday.

(a) 2025 (b) 5550 (c) 4960 (d) 5354 (e) 3885

Sol. (c); Let fare per person of Bus A = x
 Then, fare per person of Bus B = $0.9x$
 According to question
 $350x - 320 \times 0.9x = 1240$
 $62x = 1240$
 $x = 20$
 Required fare = $140 \times 20 + 120 \times 18$
 $= 2800 + 2160 = 4960$

Directions (16-20): Study the following graph carefully and answer the questions given below:



16. What is the ratio of total profit earned by all companies in year 2007 to the total profit earned by all companies in year 2009.

- (a) 23 : 22 (b) 47 : 52 (c) 44 : 45 (d) 33 : 34 (e) None of these

Sol. (b); Required ratio = $\frac{350+375+450}{400+425+475} = \frac{1175}{1300} = 47 : 52$

17. In which of the following years was the difference between the profits earned by company B and company A maximum?
 (a) 2003 (b) 2004 (c) 2005 (d) 2008 (e) 2009

Sol. (c); Difference in 2003 = 50
 Difference in 2004 = 50
 Difference in 2005 = 100 (maximum among given options)
 Difference in 2008 = 50
 Difference in 2009 = 50

18. If profit earned by company A in 2010 increased by $\frac{20}{17}\%$ over previous year and profit earned by company B is increased by $\frac{300}{19}\%$ in 2010 over previous year then what is the sum of profit (in crores) earned by company A and B in 2010.

- (a) 650 (b) 780 (c) 980 (d) 825 (e) 725

Sol. (c); Profit of A in 2010 = $\left(100 + \frac{20}{17}\right)\% \times 425$
 $= \frac{1720}{100 \times 17} \times 425 = 430$
 Profit of B in 2010 = $\left(100 + \frac{300}{19}\right)\% \text{ of } 475$
 $= \frac{2200}{19 \times 100} \times 475 = 550$
 Required sum = $430 + 550 = 980$

19. Highest total profit earned by all 3 companies together for any year is what percent of lowest total profit of all 3 companies together for any year.

- (a) $152\frac{16}{17}\%$ (b) $107\frac{2}{7}\%$ (c) $105\frac{3}{8}\%$ (d) $95\frac{3}{17}\%$ (e) None of these

Sol. (a); Total profit in 2003 = 850
 Total profit in 2004 = 900
 Total profit in 2005 = 1000
 Total profit in 2006 = 975
 Total profit in 2007 = 1175
 Total profit in 2008 = 1200
 Total profit in 2009 = 1300
 Required % = $\frac{1300}{850} \times 100 = 152\frac{16}{17}\%$

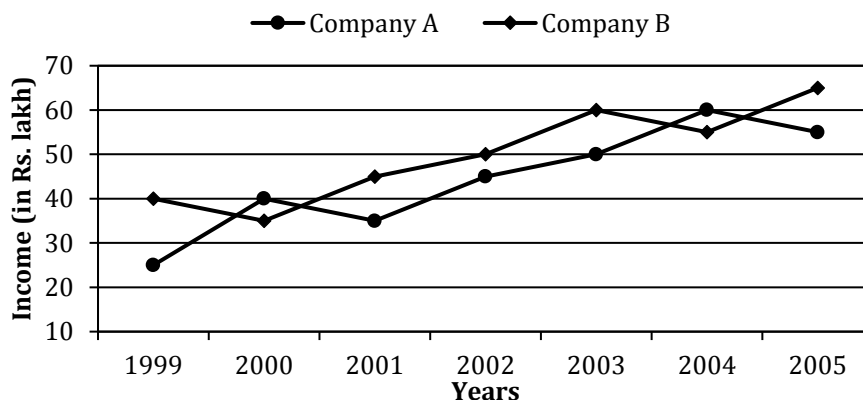
20. If in year 2002 ratio of profit earned by company A, B and C are 3 : 2 : 5 and profit of C in 2002 is 25% less than profit of C in 2003 then profit of B in 2009 increases by approximately what % percent over profit of B in 2002.

- (a) 225% (b) 235% (c) 262.5% (d) 222.5% (e) 296%

Sol. (e); Profit of C in 2002 = $\frac{75}{100} \times 400 = 300$
 Profit of B in 2002 = $\frac{300}{5} \times 2 = 120$
 Required % = $\frac{475-120}{120} \times 100 \approx 296\%$

Directions (21-25): Given below is the graph showing the income of two companies A and B. Study the graph carefully and answer the questions given below.

$$\text{profit percent} = \frac{\text{income} - \text{expenditure}}{\text{expenditure}} \times 100$$



21. Profit percent for company A in 1999 is $\frac{25}{2}\%$ and profit percent for company B in 2003 is $\frac{50}{3}\%$ and if profit percent is calculated on income then what is the ratio of expenditure for A in 1999 and B in 2003?

- (a) 7 : 16 (b) 5 : 9 (c) 7 : 8 (d) 9 : 13 (e) 8 : 15

Sol. (a); For company A in 1999
 Let expenditure of A in 1999 is E_A
 Then $E_A + \frac{25}{200} \times 25 = 25$
 $E_A = \frac{175}{8}$
 For company B in 2003
 Let expenditure of B in 2003 is E_B
 $E_B + \frac{1}{6} \times 60 = 60$
 $E_B = 50$
 Required ratio = $\frac{175}{8 \times 50} = 7 : 16$

22. For which of the following combinations of company and year, the percentage increase in income from previous year is the maximum among all the given combinations?

- (a) A, 2000 (b) B, 2003 (c) B, 2001 (d) A, 2001 (e) B, 2005

Sol. (a); For A in 2000 = $\frac{15}{25} \times 100 = 60\%$
 For B in 2003, = $\frac{10}{50} \times 100 = 20\%$
 For B in 2001 = $\frac{10}{35} \times 100 = \frac{200}{7}\% = 28\frac{4}{7}\%$
 For A in 2001, there is a decrease in income from the previous year.
 For B in 2005 = $\frac{10}{55} \times 100 = \frac{200}{11}\% = 18\frac{2}{11}\%$

23. Income of A in 2006 increases by 20% over the previous year and income of B in 2007 is x. Income of B in 2006 is 75% more than income of B in 1999 and income of B in 2006 and 2007 are in ratio 7 : 8 then income of A in 2006 is what percent more or less than Income of B in 2007?

(a) 12.5% (b) 8.5% (c) 13.25% (d) 17.5% (e) 19.25%

Sol. (d); Income of A in 2006 = 66

Income of B in 2006 = 70

So income of B in 2007 = 80

Required percentage = $\frac{14}{80} \times 100 = 17.5\%$

24. Average income of company A over all given years is what percent more or less than average of income of company B over all years?

(a) $\frac{50}{3}\%$ (b) $26\frac{2}{7}\%$ (c) $\frac{80}{7}\%$ (d) $\frac{25}{2}\%$ (e) $\frac{50}{7}\%$

Sol. (c); Required percentage

$$= \frac{\frac{350}{6} - \frac{310}{6}}{\frac{350}{6}} \times 100 = \frac{40}{350} \times 100 = \frac{80}{7}\%$$

25. If profit percent earned by both companies in 2005 are equal and expenditure of B in 2005 is 40 lakh then expenditure of A in 2005 is what % more or less than expenditure of B in 2005? (approximately)

(a) 12 (b) 18 (c) 15 (d) 9 (e) 13

Sol. (c); According to the question,

$$\frac{65 - 40}{40} = \frac{55 - x}{x}, \text{ where } x \text{ is expenditure of A in 2005}$$

$$25x = 2200 - 40x$$

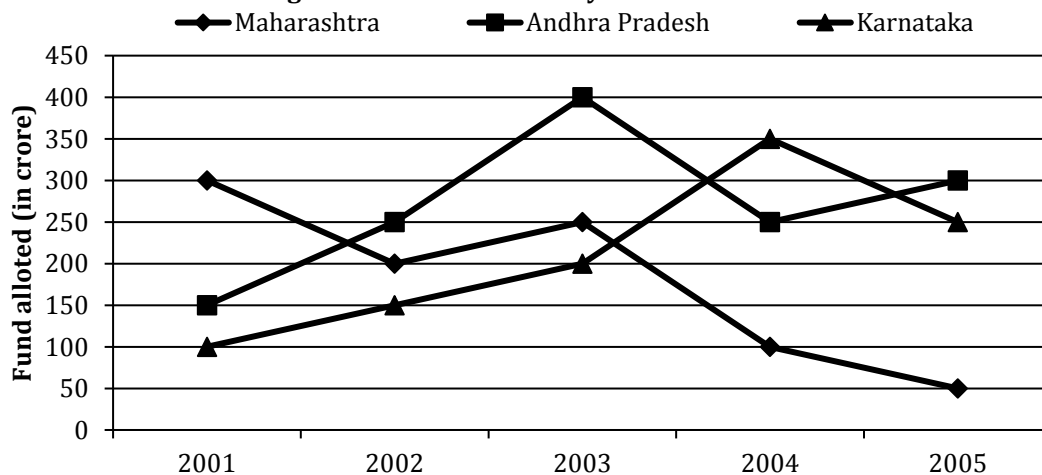
$$65x = 2200$$

$$= 33.84 \approx 34$$

$$\text{required percentage} \approx \frac{40 - 34}{40} \times 100 = 15\%$$

Directions (26-30): Study the following line-graph and answer the following questions.

The graph below shows the funds allotted to three states by the government in different years.



26. What is the ratio of average funds allotted to Karnataka in years 2002, 2004 and 2005 to the average funds allotted to Andhra Pradesh in 2001, 2003 and 2005?

(a) 15:17 (b) 12:17 (c) 15:16 (d) 13:14 (e) 17:15

Sol. (a); Req. ratio = $\frac{\frac{150+350+250}{3}}{\frac{150+400+300}{3}} = \frac{750}{850} = \frac{15}{17}$

27. Total funds allotted to these three states in 2002 is what percent less than the total funds allotted to the three states in 2005?

(a) 5% (b) 0% (c) 4% (d) 6% (e) 2.5%

Sol. (b); Total funds in 2002 = 250 + 200 + 150 = 600 crore

Total funds in 2005 = 300 + 250 + 50 = 600 crore

Both are equal, Hence 0%

28. If in 2006, the funds allotted to Maharashtra, Andhra Pradesh and Karnataka increased by 10%, 20% and 40% respectively as compared to 2005, then find the average funds allotted to three states in 2006.
 (a) 200 crores (b) 240 crores (c) 255 crores (d) 260 crores (e) 235 crores

Sol. (c); Fund allocated to Maharashtra in 2006 = $50 \times 1.1 = 55$ crore
 Fund allocated to Andhra Pradesh in 2006 = $300 \times 1.2 = 360$ crore
 Fund allocated to Karnataka in 2006 = $250 \times 1.4 = 350$ crore
 Req. average = $\frac{55+360+350}{3} = \frac{765}{3} = 255$ crore

29. Funds allotted to Maharashtra in 2001, 2002 and 2003 is what percent less/more than funds allotted to Karnataka in 2003, 2004 and 2005 ?
 (a) 3.25% (b) 7.25% (c) 4.25% (d) 6.25% (e) 6%

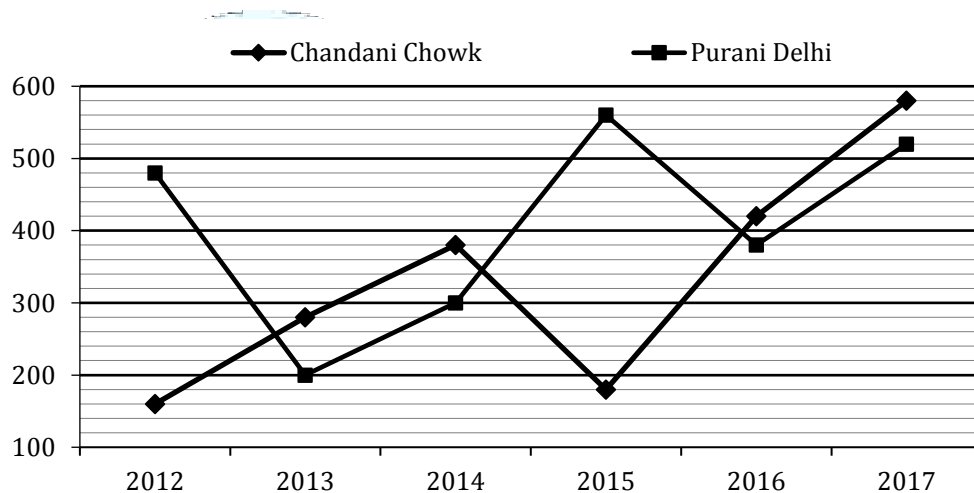
Sol. (d); Fund allocated to Maharashtra in 2001, 2002, 2003 = $300 + 200 + 250 = 750$
 Fund allocated to Karnataka in 2003, 2004, 2005 = $200 + 350 + 250 = 800$
 Req. % = $\frac{(800-750)}{800} \times 100 = \frac{50}{8} = 6.25\%$

30. What is the average funds (in crore) allotted to the given three states in 2002 and 2005 ?
 (a) 200 (b) 300 (c) 400 (d) 250 (e) 210

Sol. (a); Req. Average = $\frac{\frac{600}{3} + \frac{600}{3}}{2} = 200$ crores

Direction(31-35):- Refer the graph and answer the following questions.

Data related to the number of votes polled in two constituencies Chandani Chowk and Purani Delhi during six years.



31. If in 2012, $\frac{100}{9}\%$ of total registered voters did not poll vote from purani delhi and 20% of registered voters did not poll from chandani chowk then what is the sum of total registered voters in purani Delhi & Chandani chowk in 2012.
 (a) 740 (b) 820 (c) 6240 (d) 520 (e) 550

Sol. (a); Total registered voters from chandani chowk = $\frac{160 \times 100}{80} = 200$
 Total registered voters from purani Delhi = $\frac{480 \times 100}{800} = 540$
 Required sum = 740

32. In 2015 if 20% of total votes polled from constituencies were invalid and valid votes from chandini chowk and Purani delhi are in the ratio 3 : 5 in 2015 then what is the number of invalid votes from Purani Delhi in 2015.
 (a) 150 (b) 165 (c) 180 (d) 170 (e) 190

Sol. (e); Sol. Total polled votes from both constituencies in 2015
 = $560 + 180 = 740$
 Valid votes from Purani Delhi = $740 \times \frac{80}{100} \times \frac{5}{8} = 370$
 Invalid votes from purani Delhi = $560 - 370 = 190$

33. Number of votes Polled in purani delhi in 2012 is what percent of no. of votes polled of purani Delhi in 2017.
 (a) $80\frac{2}{13}\%$ (b) $87\frac{3}{13}\%$ (c) $92\frac{4}{13}\%$ (d) $72\frac{9}{13}\%$ (e) $89\frac{3}{13}\%$

Sol. (c); Required % = $\frac{480}{520} \times 100 = 92\frac{4}{13}\%$

34. What is the ratio of votes Polled from chandani chowk in 2012, 2013 and 2015 together to the votes polled from purani delhi in 2014, 2015, 2017 together.

- (a) 30 : 67 (b) 31 : 69 (c) 28 : 67 (d) 29 : 69 (e) 31 : 68

Sol. (b); Required ratio = $\frac{160+280+180}{300+560+520}$
= 31: 69

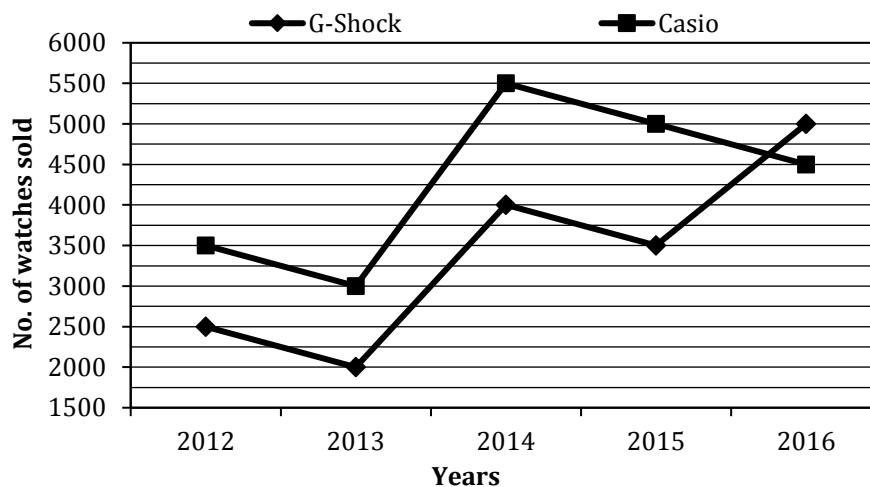
35. Number of votes polled in 2017 from purani Delhi decreases by what % over year 2015.

- (a) $\frac{100}{3}\%$ (b) $\frac{100}{6}\%$ (c) $\frac{100}{7}\%$ (d) $\frac{50}{7}\%$ (e) $\frac{50}{17}\%$

Sol. (d); Required percentage = $\frac{560-520}{560} \times 100$
= $\frac{40}{560} \times 100 = \frac{50}{7}\%$

Directions (36-40): Read the following line graph and answer the following questions given below it.

There are two watch manufacturing companies G-Shock and Casio. The sale of watches by these two different companies in different years is given in the graph below.



36. What is the ratio of total sales of company G-Shock in 2012 and that of company Casio in 2014 together to the total sales of company Casio in 2012 and that of company G-Shock in 2015 together?

- (a) 8:7 (b) 11:9 (c) 9:7 (d) 13:10 (e) 10:9

Sol. (a); Required ratio = $\frac{2500 + 5500}{3500 + 3500} = \frac{8000}{7000} = \frac{8}{7}$

37. What is the difference between the sales of company G-Shock in 2017 and that of company Casio in 2017 if the sales of company G-Shock and Casio increase by 20% and 10% respectively in 2017 as compared to 2016?

- (a) 1340 (b) 1050 (c) 1080 (d) 1300 (e) 1500

Sol. (b); Sales of company G-Shock in 2017 = $1.2 \times 5000 = 6000$
Sales of company Casio in 2017 = $1.1 \times 4500 = 4950$
Required Difference = $6000 - 4950 = 1050$

38. The total sales of both companies in 2015 is what percent more than the total sales of both the companies in 2013?

- (a) 65% (b) 80% (c) 70% (d) 55% (e) 170%

Sol. (c); Sales of both the companies in 2015 = $3500 + 5000 = 8500$
Sales of both the companies in 2013 = $3000 + 2000 = 5000$
Required % = $\frac{(8500 - 5000)}{5000} \times 100 = \frac{3500}{5000} \times 100 = 70\%$

39. Find the difference between the total sales of company G-Shock from 2012 to 2014 and that of company Casio from 2013 to 2015?

- (a) 7500 (b) 5500 (c) 6000 (d) 5000 (e) 6500

Sol. (d); Total sales of G-shock from 2012 to 2014 = $2500 + 2000 + 4000 = 8500$
Total sales of casio from 2013 to 2015 = $3000 + 5500 + 5000 = 13500$
Required Difference = $13500 - 8500 = 5000$

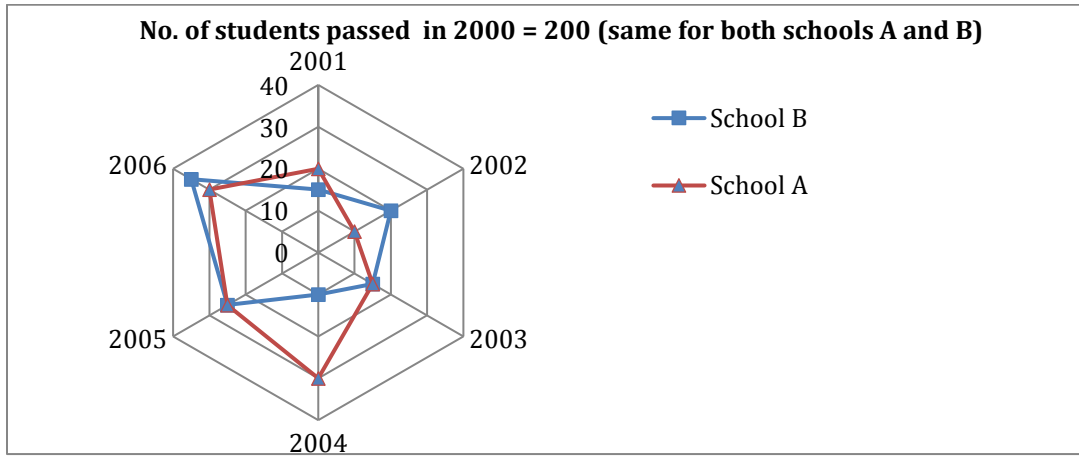
40. If the sales of company G-Shock increased by 25 % in 2012 over its sales in 2011, then find the percent increase in the sales of company G-Shock in 2015 with respect to the sales in 2011?

(a) 65% (b) 75% (c) 55% (d) 60% (e) 72%

Sol. (b); Sales of G-shock in 2011 = $2500 \times \frac{100}{125} = 2000$

$$\text{Required percentage increase} = \frac{(3500 - 2000)}{2000} \times 100 = \frac{1500}{2000} \times 100 = 75\%$$

Directions (41-45): Study the Radar graph given below and answer the following questions.



The Radar graph shows the percentage increase in the number of students passing out from schools A and B with respect to the number of students passed in 2000.

41. If the ratio of boys to girls (who passed out) in 2002 from school B was 6 : 4 and the ratio of students with Science background to those with non-science background was 7 : 3, then find the number of girls who had non-Science background from school B in 2002 ? (Given: boys with Science background were $85\frac{5}{7}\%$ of the students with Science background)

(a) 72 (b) 48 (c) 96 (d) 108 (e) 82

Sol. (a); Total no. of girls who passed out in 2002 from school B = $\frac{4}{10} \times 200 \times \frac{120}{100} = 96$

$$\text{No. of students with Science background} = \frac{7}{10} \times 240 = 168$$

$$\text{Girls with Science background} = \frac{1}{7} \times 168 = 24$$

$$\text{Hence, girls without Science background} = 96 - 24 = 72$$

42. If the ratio of boys to girls (who passed) in 2000 from school A was 12 : 8 and it was the same in 2004 as well for same school, then find the percentage increase in the number of girls passing out in 2004 from school A with respect to that of 2000 from the same school ?

(a) 35% (b) 30% (c) 45% (d) 60% (e) None of these

Sol. (b); No. of girls in 2000 = 80

$$\text{No. of girls in 2004} = \frac{8}{20} \times \frac{130}{100} \times 200 = 104$$

$$\text{Percentage \%} = \frac{104-80}{80} \times 100 = \frac{24}{80} \times 100 = 30\%$$

43. What is ratio of the average number of students passed from school A in 2001, 2002 and 2004 to that of from school B in 2002, 2004 and 2005 ?

(a) 73 : 71 (b) 72 : 67 (c) 72 : 71 (d) 75 : 71 (e) 71 : 72

Sol. (c); Required ratio = $\frac{\frac{240+220+260}{3}}{\frac{240+220+250}{3}} = \frac{720}{710} = \frac{72}{71}$

44. If $77\frac{7}{9}\%$ of the students who passed out from school B in 2006 went on to pursue engineering and the ratio of students who pursued engineering from school B in 2006 to that of from school A in 2005 is 3 : 2, then find the number of students who didn't pursue engineering from school A in 2005 ?

(a) 140 (b) 112 (c) 120 (d) 110 (e) 180

Sol. (d); No. of students from school B who pursued engineering in 2006 = $\frac{7}{9} \times \frac{135}{100} \times 200 = 210$

$$\text{No. of students from school A pursued engineering in 2005} = \frac{2}{3} \times 210 = 140$$

$$\text{No. of students from school A who didn't pursue engineering in 2005} = \frac{125}{100} \times 200 - 140 = 250 - 140 = 110$$

45. What is the difference of the number of student who passed of school B in year 2004, 2005 and 2006 and number of student who passed of school A in same year ?

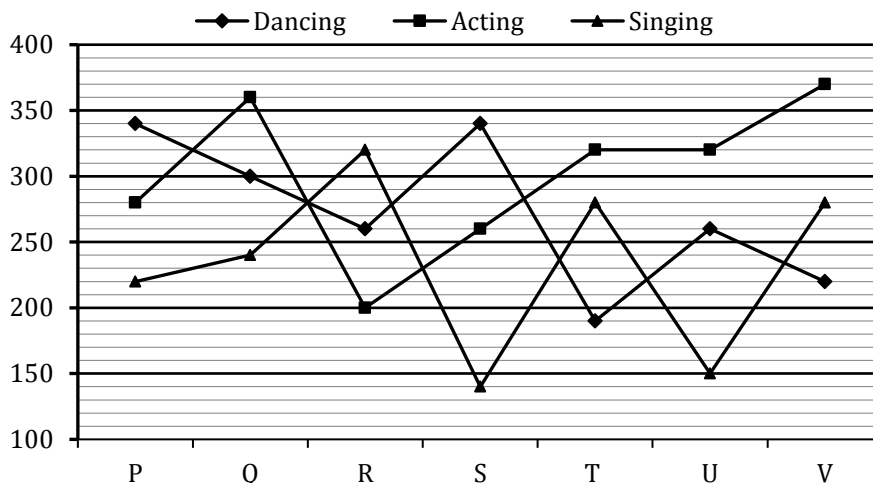
(a) 50 (b) 40 (c) 20 (d) 30 (e) 80

Sol. (d); Required difference

$$= \left[200 \times \frac{110}{100} + 200 \times \frac{125}{100} + 200 \times \frac{135}{100} \right] \sim \left[200 \times \frac{130}{100} + 200 \times \frac{125}{100} + 200 \times \frac{130}{100} \right]$$

$$= [220 + 250 + 270] \sim [260 + 250 + 260] = 30$$

Directions (46-50): The following graph shows the number of students participated in annual talent show of Kurukshetra university from different colleges. Study the graph carefully to answer the given questions.



46. Find the ratio of the total number of students participated in dancing from college P, T & V together to the total students participated in singing from college P, Q & R together.

(a) 28 : 27 (b) 25 : 26 (c) 25 : 29 (d) 27 : 29 (e) None of these

Sol. (b); Required ratio = $\frac{340+190+220}{240+320+220} = 25 : 26$

47. What is the average number of student participating in acting from Q and R and number of student participating in singing from T and V?

(a) 250 (b) 240 (c) 260 (d) 280 (e) 320

Sol. (d); Required average = $\frac{360+200+280+280}{4} = \frac{1120}{4} = 280$

48. If 40% of the total students of college S who participated in acting, are doing solo acting and other are in group acting. $66\frac{2}{3}\%$ of the students participated in group acting are a part of comedy dramas. Then find the total number of students who are acting in comedy dramas.

(a) 104 (b) 120 (c) 110 (d) 108 (e) 140

Sol. (a); Total number of students participating in Acting of college S = 260

$$\text{Number of students who are acting in comedy drama} = \frac{60}{100} \times \frac{2}{3} \times 260 = 104$$

49. The difference between the no. of students participating in dancing from college P and R and that in singing from T and U ?

(a) 165 (b) 170 (c) 82 (d) 94 (e) 190

Sol. (b); Total number of students participating in dance from P and R = 340 + 260 = 600

$$\text{Number of students participating in singing from T and U} = 280 + 150 = 430$$

$$\text{Required Difference} = 600 - 430 = 170$$

50. The total number of students participated from college S is what percent more/less than that from college R? (Rounded off to two decimal places)

(a) 5.23% (b) 5.13% (c) 5.03% (d) 4.93% (e) 4.73%

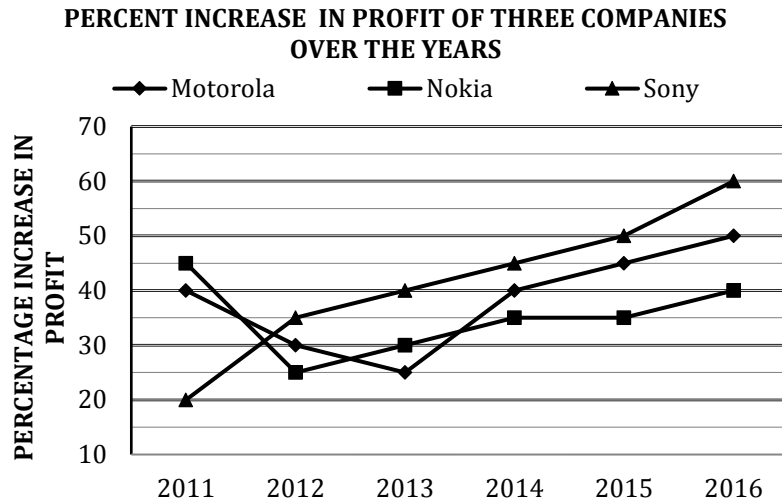
Sol. (b); Total number of students from college S = 340 + 260 + 140 = 740

$$\text{Total number of students from college R} = 260 + 200 + 320 = 780$$

$$\text{Required percentage} = \frac{780-740}{780} \times 100 \approx 5.13\%$$

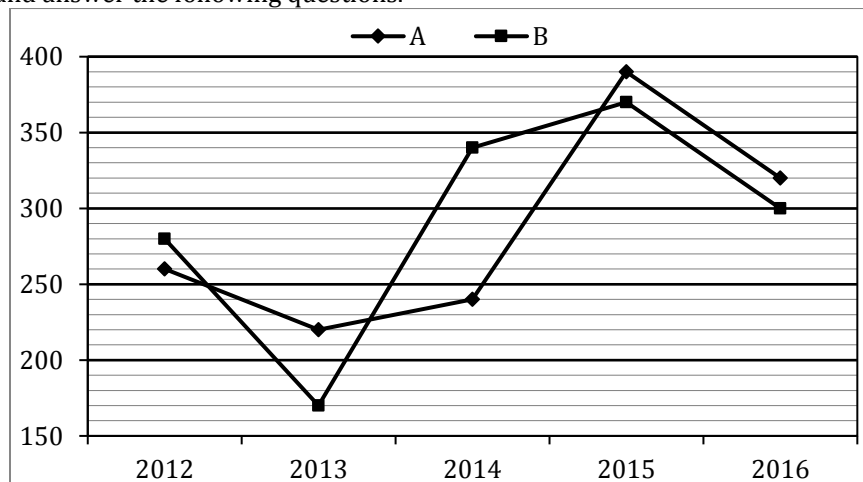
PREVIOUS YEAR QUESTIONS

Directions (1-5): Study the graph carefully to answer the questions that follow.



- If profit for company Nokia in 2012 is 2000 and expenditure in 2013 for company Nokia is 50,000, then what is the total revenue in 2013 for Nokia? Give that total revenue = expenditure + profit.
(a) Rs. 52600 (b) Rs. 54200 (c) Rs. 53280 (d) Rs. 55800 (e) Rs. 56020
- If profit in year 2015 for company Sony is 3000 and profit of company Motorola in 2013 is equal to profit of company Sony in 2014 then what is the profit of company Motorola in 2013 ?
(a) Rs. 1500 (b) Rs. 4000 (c) Rs. 3500 (d) Rs. 2000 (e) Rs. 2500
- What is the average percentage increase in profit for company Nokia over all the years?
(a) 49% (b) 32% (c) 23% (d) 38% (e) 35%
- What was the percentage increase in percent increase of profit of company Motorola in the year 2014 over its previous year ?
(a) 60% (b) 65% (c) 55% (d) 50% (e) 70%
- If profit earned by company Nokia in 2014 is 27,000 and by company Sony in 2014 is 43500 then what is the total profit earned by them in year 2013?
(a) Rs. 25,000 (b) Rs. 35,000 (c) Rs. 40,000 (d) Rs. 50,000 (e) Rs. 45,000

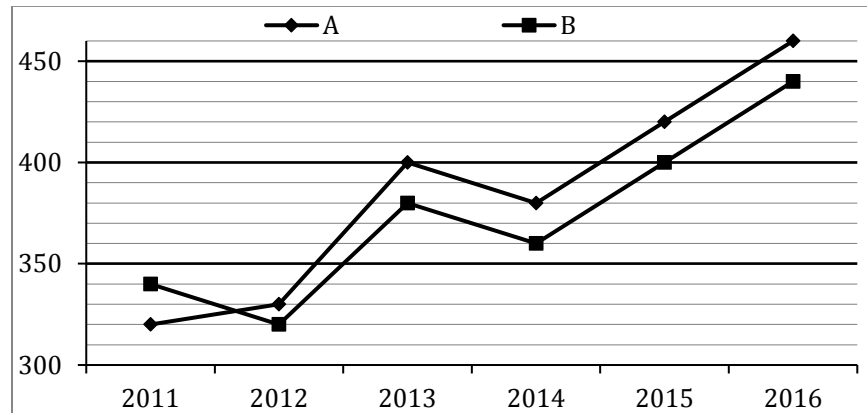
Directions (6-10): The following line graph shows the number of visitors (in hundreds) in 2 museums A and B, in different years. Study it carefully and answer the following questions.



- Number of visitors in museum B in 2015 is approximately what percent of sum of number in visitors of museum A in 2012 and that of museum B in 2016?
(a) 60% (b) 52% (c) 66% (d) 58% (e) 72%
- If in 2012, no. of visitors of museum A decreases by 25% and that of museum B increases of 20% then find the total number of visitors in 2012 in both of the museums.
(a) 51300 (b) 53100 (c) 60500 (d) 63000 (e) 51550

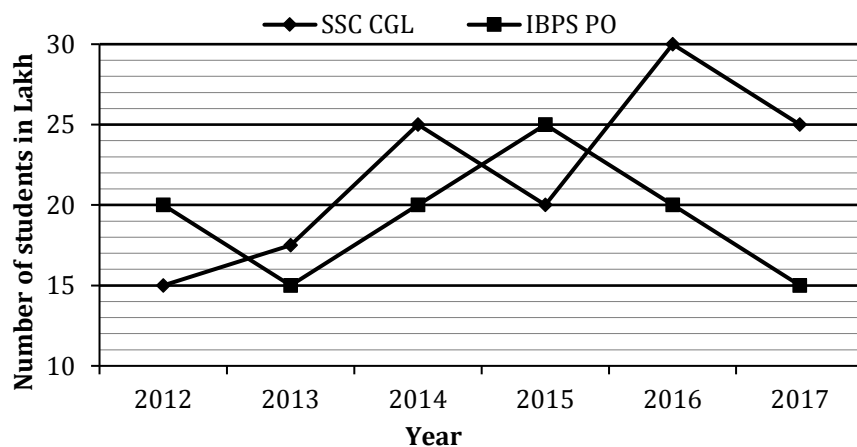
8. If in 2014, 40% of the visitors in museum A are female and 60% of the visitors in museum B are male. Then find the total number of female visitors in both museums in 2014.
 (a) 21200 (b) 23200 (c) 22000 (d) 22400 (e) 22300
9. Find the total number of visitors in museum B over the given years.
 (a) 146000 (b) 144000 (c) 164000 (d) 125000 (e) 136000
10. Find the ratio of total visitors in museum B in 2012 and 2013 together to that of museum A in 2013 and 2016 together.
 (a) 1 : 2 (b) 5 : 6 (c) 2 : 3 (d) 4 : 5 (e) 5 : 7

Directions (11-15): The graph shows the no. of students in two classes A and B in five different years. Read the following graph and answer accordingly.



11. What is the sum of differences between number of students through all these years in A and B?
 (a) 120 (b) 130 (c) 125 (d) 110 (e) 105
12. Total number of students in 2012 and 2015 together is approximately what percent of the total number of student from A in all these years?
 (a) 64% (b) 68% (c) 70% (d) 62% (e) 60%
13. What is the ratio of Number of children in Class B for all the years to the total number of student in class A for all the years?
 (a) 31 : 33 (b) 32 : 35 (c) 32 : 33 (d) 29 : 33 (e) 33 : 35
14. What is the sum total of student for class A in 2011 and 2013 and total number of students in 2015 & 2016 for class B?
 (a) 1560 (b) 1400 (c) 1500 (d) 1460 (e) 1650
15. Total students in class B for all the years is approximately what percent less than total no of students in both schools for all these years?
 (a) 55% (b) 58% (c) 60 % (d) 50.8% (e) 62.2%

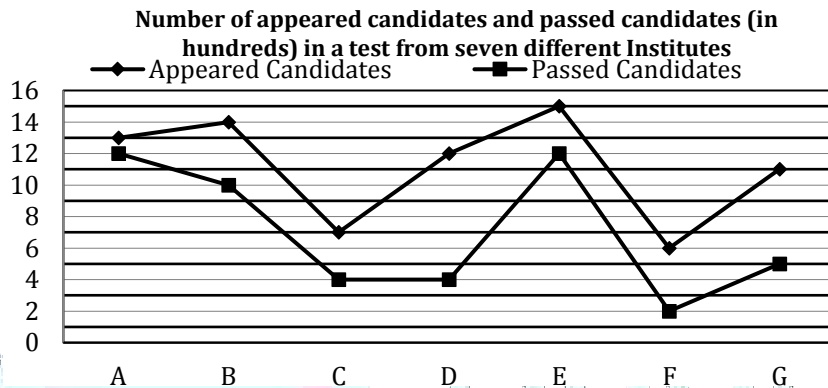
Directions (16-20): Given below is the line graph which shows the number of students who applied for SSC CGL and IBPS bank PO exams over different years.



16. What is the ratio of number of students who applied for SSC CGL in 2013, 2015 and 2016 together to the number of students who applied for IBPS PO in 2012, 2013 and 2017 together?
 (a) 35 : 41 (b) 27 : 20 (c) 22 : 34 (d) 51 : 32 (e) 43 : 53

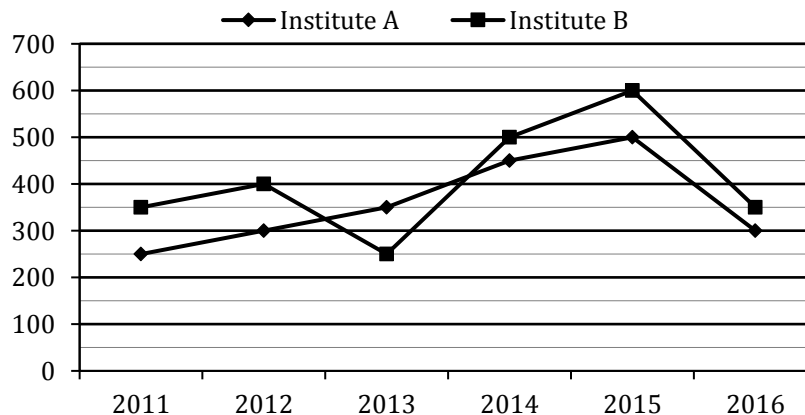
17. If in year 2014, 20% of students who applied for IBPS PO also applied for SSC CGL then the number of students who applied only for SSC-CGL are what percent of number of students who applied only for IBPS PO in year 2014?
 (a) $131\frac{1}{4}\%$ (b) $133\frac{1}{4}\%$ (c) $123\frac{1}{3}\%$ (d) $110\frac{2}{3}\%$ (e) $99\frac{1}{10}\%$
18. If number of students who applied for SSC CGL in 2011 are $\frac{100}{3}\%$ less than students who applied for SSC CGL in 2012 then, number of students who applied for SSC CGL in 2011 are what percent of number of students who applied for IBPS PO in 2015?
 (a) 30% (b) 25% (c) 35% (d) 40% (e) 50%
19. What is the difference between average of number of students who applied for SSC-CGL over all years and average of number of students who applied for IBPS-PO over all years? (approximately)
 (a) 3.5L (b) 4L (c) 3L (d) 2.5L (e) 2L
20. If 45% of students who applied for SSC-SGL in 2015 are girls and ratio of number of girls who applied for SSC CGL in 2015 to the number of girls who applied for IBPS PO in 2017 is 4 : 5 then find the number of boys who applied for IBPS PO in 2017.
 (a) 8L (b) 7L (c) 9.5L (d) 4L (e) 3.75L

Directions (21-25): Study the graph carefully to answer the questions that follow



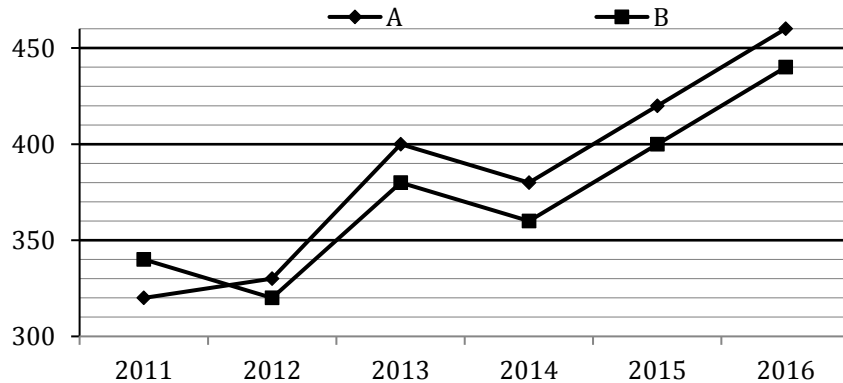
21. What is the difference between the number of candidates appeared from institutions B, C, D and F together and number of candidates passed from institutions A, E and G together?
 (a) 1100 (b) 900 (c) 1000 (d) 1200 (e) 800
22. What is the average number of candidates passed from all the institutions together?
 (a) 700 (b) 490 (c) 350 (d) 630 (e) 560
23. Number of candidates passed from institutions C and E together is approximately, what per cent of the total number of candidates appeared from institutions A and G together?
 (a) 72% (b) 62% (c) 54% (d) 75% (e) 67%
24. From which institution, the difference between the number of appeared candidates and passed candidates is maximum?
 (a) B (b) G (c) D (d) F (e) C
25. What is the respective ratio between the number of candidates who have failed from institution B and the number of candidates who have appeared from institution F?
 (a) 2 : 5 (b) 2 : 3 (c) 4 : 3 (d) 1 : 3 (e) 5 : 3

Directions (26-30): Given below is the graph showing the number of students taking admission to two different institutes in the given years.



26. What is the ratio of 150% of number of student taking admission in year 2013 from both institutes to the 125% of number of students taking admission in year 2015 from both institutes?
 (a) 36 : 55 (b) 55 : 57 (c) 53 : 57 (d) 46 : 53 (e) 51 : 57
27. In which year, total number of students taking admission in both institutes together is second highest?
 (a) 2015 (b) 2013 (c) 2014 (d) 2012 (e) 2016
28. Number of students taking admission in institute A in years 2010 and 2012 together are what percent of number of students taking admission in institute B in years 2013 and 2014 together, if the number of students taking admission in institute A in 2010 is 20% more than the number of students taking admission in institute A in 2011?
 (a) 60% (b) 65% (c) 85% (d) 90% (e) 80%
29. Number of students taking admission in institute A in year 2011, 2013 and 2014 together are what percent more or less than the number of students taking admission in institute B in year 2012, 2013 and 2015 together?
 (a) 20% (b) 16% (c) 19% (d) 13% (e) 15%
30. What is the ratio of total number of students taking admission in institute B to the total number of students taking admission in institute A over all years?
 (a) 23 : 27 (b) 53 : 54 (c) 20 : 23 (d) 49 : 43 (e) 53 : 57

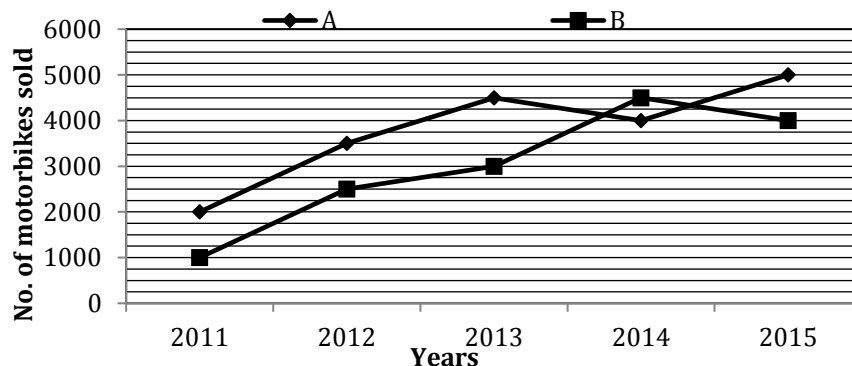
Directions(31-35): The graph shows the no. of students in two classes A and B in five different years. Read the following graph and answer accordingly.



31. What is the average of difference between number of students in A and B (no. of students in A-no. of students in B) through all these years?
 (a) 12.67 (b) 13.67 (c) 11.67 (d) 15 (e) 14
32. Total number of students in 2012 and 2015 is what percent of the total number of student from A in all these years?
 (a) $63\frac{7}{11}\%$ (b) $65\frac{5}{11}\%$ (c) 70% (d) 62% (e) $67\frac{3}{11}\%$
33. What is the ratio of Number of children in Class B for all the years to the total number of student in class A for all the years?
 (a) 31 : 33 (b) 32 : 35 (c) 32 : 33 (d) 29 : 33 (e) 33 : 35
34. What is the sum of total of student for A in 2011, 2013 and total number of students in 2015 & 2016 for B?
 (a) 1560 (b) 1400 (c) 1500 (d) 1460 (e) 1550
35. Total students in class B for all the years is approximately what percent less than total no of students in both schools for all these years?
 (a) $54\frac{9}{13}\%$ (b) 52% (c) $60\frac{10}{13}\%$ (d) $50\frac{10}{13}\%$ (e) $50\frac{11}{13}\%$

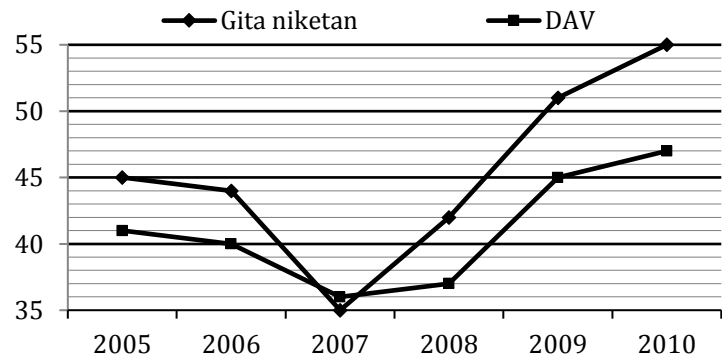
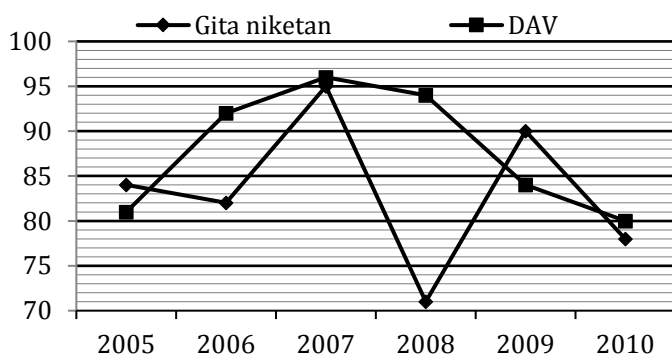
Directions (36-40): Read the following line graph and answer the following questions given below it –

There are two motorbike manufacturing companies A and B. The sale of motorbikes by these two different companies in different years is given in the graph below.



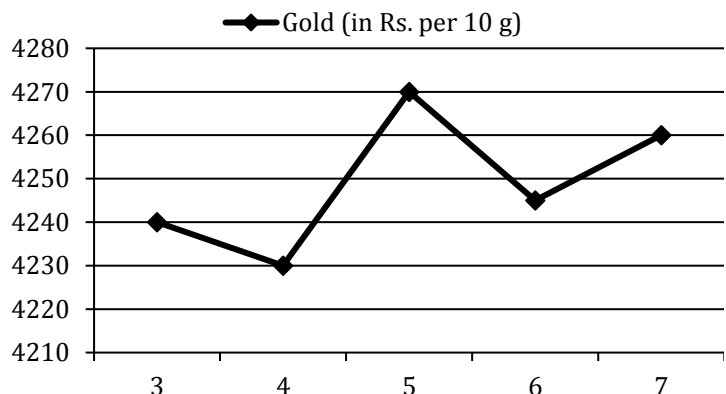
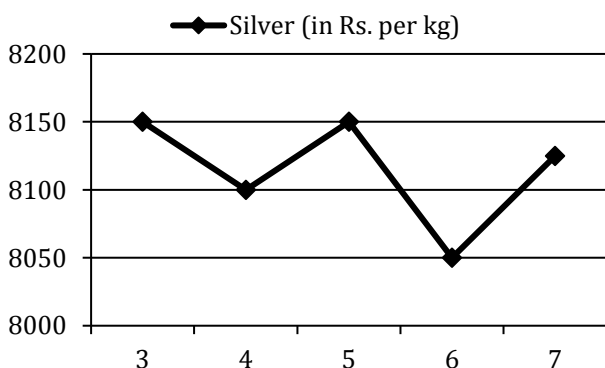
36. What is the ratio of total sales of company B in 2012 and that of company A in 2014 to the total sales of company A in 2011 and that of company B in 2015?
 (a) 13:12 (b) 11:9 (c) 12:7 (d) 13:10 (e) 12 : 13
37. What is the difference between the sales of company A in 2016 and that of company B in 2016 if the sales of company A and B increase by 20% and 10% respectively in 2016 as compared to 2015?
 (a) 1700 (b) 1600 (c) 1800 (d) 2100 (e) 1400
38. The total sales of both companies in 2015 is what percent more than the total sales of both the companies in 2011?
 (a) 280% (b) 180% (c) 200% (d) 250% (e) 220%
39. Find the difference between the total sales of company A from 2012 to 2014 and that of company B from 2013 to 2015?
 (a) 750 (b) 500 (c) 600 (d) 400 (e) 550
40. If the sales of company A increased by 33.33% in 2011 over its sales in 2010, then find the percent increase in the sales of company A in 2015 with respect to the sales in 2010?(up to two decimal places)
 (a) $233\frac{1}{3}\%$ (b) $210\frac{1}{3}\%$ (c) $333\frac{1}{3}\%$ (d) $272\frac{3}{4}\%$ (e) None of these

Directions(41-45): Given below is the line graphs, first showing number of students participated (in hundreds) in NTSE (National Talent Search Exam) from 2 different schools from 2005-2010, the second line graph shows the corresponding percentage of girls participated in this exam. Read the graphs carefully and answer the following questions:



41. If no. of boys participated from Greenfield public school in 2009 is 10% less than the total no. of girls participated from DAV and Geeta Niketan in that year and the boys participated in 2009 from Greenfield was 45% of the total students participated from greenfield in that year, then find the no. of girls participated from greenfield school in 2009?
 (a) 9428 (b) 8294 (c) 9211 (d) 9207 (e) 9084
42. The difference between total number of boys participated and total number of girls participated from gita niketan in all years together is what percent of the total students participated from gita niketan in all years?
 (a) 9.7% (b) 9.1% (c) 10.6% (d) 8.4% (e) 8.7%
43. Girls participated from DAV in 2007 is approximately what percent less/more than the boys participated from Gita Niketan in 2009 and 2010 together?
 (a) 56% (b) 42% (c) 50% (d) 44% (e) 66%
44. Find the difference between average no. of students participated from the 2 Schools over the years.
 (a) 4.5 (b) 45 (c) 415 (d) 465 (e) 450
45. Find the total number of boys participated from Gita Niketan in all years together
 (a) 23225 (b) 27425 (c) 28525 (d) 29625 (e) None of these

Directions (46-50): The following graphs show the price of gold (in Rs. per 10 g) and silver (in Rs. per kg) on 3rd, 4th 5th, 6th and 7th of August 2010 in Mumbai. Study the graphs and answer the questions that follow.



46. On 8th August, the price of silver (in Rs. per kg) is increased by 12% as compared to previous day and the price of gold (in Rs. 10 g) is decreased by 15% as compared to previous day then find the ratio of the average price of silver (in Rs. per kg) from 4th to 8th August to the average price of gold (in Rs. per 10 g) from 5th to 8th August.
 (a) 1491 : 3020 (b) 8305 : 4099 (c) 4017 : 1213 (d) 1213 : 4017 (e) None of these
47. On 2nd August the ratio between the price of silver (in Rs. per kg) and gold (in Rs. per 10 g) is 51 : 25 and the price of gold on 3rd August was 6% more than that of 2nd August then Find the average price of silver (in Rs. per kg) from 2nd August to 6th August ?
 (a) 8212 Rs. (b) 8132 Rs. (c) 8130 Rs. (d) 8120 Rs. (e) 8122 Rs.
48. By how much per cent the rate of silver is less than the rate of gold on 6th August, 2010 ?
 (a) 92% (b) 98% (c) 108% (d) Can't be determined (e) 88%
49. What is difference of average price of gold (in Rs/10gm) and average price of silver (in Rs/kg) ?
 (a) 3866 (b) 4866 (c) 3226 (d) 3846 (e) 3626
50. What is the average price of silver (in Rs./kg) for the given dates ?
 (a) 8217 (b) 8007 (c) 8120 (d) 8140 (e) 8115



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PREVIOUS YEAR SOLUTIONS

1. (a); Profit in 2013 = $2000 \times \frac{130}{100} = \text{Rs.}2600$
Total revenue = $50,000 + 2600 = \text{Rs.}52600$
2. (d); Profit of company Motorola in 2013 = $\frac{3000 \times 100}{150} = \text{Rs.}2000$
3. (e); Required average = $\frac{45+25+30+35+35+40}{6} \times \frac{210}{6} = 35\%$
4. (a); Required percentage = $\frac{40-25}{25} \times 100 = \frac{15}{25} \times 100 = 60\%$
5. (d); Profit earned by Company Nokia in 2013 = $\frac{27000 \times 100}{135} = \text{Rs.}20,000$
Profit earned by company Sony in 2013 = $\frac{43500 \times 100}{145} = \text{Rs.}30,000$
Total profit = Rs.50,000
6. (c); Required percentage = $\frac{370}{560} \times 100 \approx 66\%$
7. (b); $260 \times \frac{75}{100} + 280 \times \frac{120}{100} = 195 + 336 = 531$ hundreds = 53100
8. (b); Female in museum A = 9600
Female in museum B = 13600
Total female visitors = 23200
9. (a); $280 + 170 + 340 + 370 + 300 = 1460$ hundred = 146000
10. (b); Required ratio = $\frac{450}{540} = 5:6$
11. (d); Sum of differences = $20 + 10 + 20 + 20 + 20 + 20 = 110$
12. (a); Total students in 2012 & 2015 = $650 + 820 = 1470$
Total students from A in all given years = 2310
Desired % = $\frac{1470}{2310} \times 100 \approx 64\%$
13. (c); No. of children for Class B in all years = 2240
No of children for class A in all years = 2310
Desired ratio = $\frac{2240}{2310} = 32:33$
14. (a); Total desired sum = $(320 + 400) + (400 + 440) = 1560$
15. (d);
Class B = 2240
Class A = 2310
Total = 4550
Desired value = $\frac{4550-2240}{4550} \times 100 \approx 50.8\%$
16. (b); Required ratio = $(17.5 + 20 + 30) : (20 + 15 + 15) = 67.5 : 50 = 27 : 20$
17. (a); Number of students applied only for SSC-CGL in 2014 = $25 - \frac{20}{100} \times 20 = 25 - 4 = 21$ L
Required % = $\frac{21}{20-4} \times 100 = \frac{21}{16} \times 100 = 131\frac{1}{4}\%$
18. (d); Number of students who applied for SSC-CGL in 2011 = $\frac{2}{3} \times 15 = 10$ L
Required % = $\frac{10}{25} \times 100 = 40\%$
19. (c); Required difference = $\frac{1}{6} [(15 + 17.5 + 25 + 20 + 30 + 25) - (20 + 15 + 20 + 25 + 20 + 15)]$
20. (e); Number of girls who applied for SSC CGL in 2015 = $20 \times \frac{45}{100} = 9$ L
Number of girls who applied for IBPS PO in 2017 = $9 \times \frac{5}{4} = 11.25$ L
Number of boys who applied for IBPS PO in 2017 = $15 - 11.25 = 3.75$ L
21. (c); Required difference = $(14+7+12+6) - (12+12+5) = 39 - 29 = 10$ hundred = 1000
22. (a); Required average = $\frac{12+10+4+4+12+2+5}{7} = \frac{49}{7} = 7$ hundred = 700
23. (e); (Required % = $\frac{(4+12)}{(13+11)} \times 100 = \frac{16}{24} \times 100 = 66.67\% \approx 67\%$)
24. (c); Difference for institute A = $(13 - 12) = 1$
B = $(14 - 10) = 4$
C = $(7 - 4) = 3$
D = $(12 - 4) = 8$
E = $(15 - 12) = 3$
F = $(6 - 2) = 4$
G = $(11 - 5) = 6$
 $\therefore$ Required institute = D
25. (b); Required Ratio = $\frac{(14-10)}{6} = \frac{4}{6} = 2:3$
26. (a); Required ratio = 150% of $(350 + 250) : 125\%$ of $(500 + 600) = 36:55$
27. (c); While observing the graph carefully, we find that the number of students are more in year 2014 and 2015 than any other year. So, the number of students taking admission in both institute is second highest in 2014.
28. (e); Number of students taking admission in institute A in 2010 = $\frac{6}{5} \times 250 = 300$
Required % = $\frac{300+300}{250+500} \times 100 = \frac{600}{750} \times 100 = 80\%$
29. (b); Required percentage = $\frac{(400+250+600)-(250+350+450)}{400+250+600} \times 100 = \frac{200}{1250} \times 100 = 16\%$
30. (d); Required Ratio = $2450 : 2150 = 49 : 43$
31. (c); Difference = $-20 + 10 + 20 + 20 + 20 + 20 = 70$
Avg. = $\frac{70}{6} \approx 11.67$
32. (a); Total students in 2012 & 2015 = $650 + 820 = 1470$
Total students from A in all given years = 2310
Desired % = $\frac{1470}{2310} \times 100 = 63\frac{7}{11}\%$
33. (c); No. of children for Class B in all years = 2240
No of children for class A in all years = 2310
Desired ratio = $\frac{2240}{2310} = 32:33$
34. (a); Total desired sum = $(320 + 400) + (400 + 440) = 1560$
35. (d);
Class B = 2240
Class A = 2310
Total = 4550

$$\text{Desired value} = \frac{4550-2240}{4550} \times 100 = 50\frac{10}{13}\%$$

36. (a); Total sales of company B in 2012 and that of company A in 2014 = 2500 + 4000 = 6500

Total sales of company A in 2011 and that of company B in 2015 = 2000 + 4000 = 6000

$$\text{Ratio} = \frac{6500}{6000} = \frac{13}{12}$$

37. (b); Sales of company A in 2016 = $5000 \times \frac{120}{100} = 6000$

$$\text{Sales of company B in 2016} = 4000 \times \frac{110}{100} = 4400$$

$$\text{Difference} = 6000 - 4400 = 1600$$

38. (c); Total sales in 2011 = 2000 + 1000 = 3000

$$\text{Total sales in 2015} = 5000 + 4000 = 9000$$

$$\text{Req.}\% = \frac{9000-3000}{3000} = 200\%$$

39. (b); Sales of company A from 2012 to 2014 = 3500 + 4500 + 4000 = 12000

$$\text{Sales of company B from 2013 to 2015} = 3000 + 4500 + 4000 = 11500$$

$$\text{Difference} = 500$$

40. (a); Sales of company A in 2010 = $2000 \times \frac{3}{4} = 1500$

$$\text{Percentage}\% = \frac{5000-1500}{1500} \times 100$$

$$= \frac{3500}{1500} \times 100 = 233\frac{1}{3}\%$$

41. (d); Girls participated in 2009 = $\frac{45}{100} \times 8400 + \frac{51}{100} \times 9000$
 $= 3780 + 4590 = 8370$

boys participated from green field public school

$$= \frac{90}{100} \times 8370 = 7533$$

$$\text{total no. of students of green field} = 7533 \times \frac{100}{45} = 16740$$

$$\text{no. of girls} = 16740 - 7533 = 9207$$

42. (a); Total no. of girls participated = 22575

$$\text{Total no. of boys participated} = 27425$$

$$\text{required percentage} = \frac{27425-22575}{50000} \times 100 = 9.7\%$$

43. (a); Girls participated from DAV in 2007

$$= 9600 \times \frac{36}{100} = 3456$$

boys participated from Gita Niketan in 2009 and 2010 together = $9000 \times \frac{49}{100} + 7800 \times \frac{45}{100} = 7920$

$$\text{percentage} = \frac{7920-3456}{7920} \times 100 \approx 56\%$$

44. (e); required difference = $\frac{527}{6} - \frac{500}{6} = \frac{2700}{6}$ hundred
 $= \frac{2700}{6} = 450$

45. (b); Total no. of boys

$$= 84 \times \frac{55}{100} + 82 \times \frac{56}{100} + \frac{65}{100} \times 95 + \frac{58}{100} \times 71$$

$$+ \frac{49}{100} \times 90 + \frac{45}{100} \times 78 = 27425 \text{ boy}$$

46. (b); Average price of silver from 4th to 8th August

$$= \frac{8100+8150+8050+8125+1.12 \times 8125}{5} = \frac{41525}{5}$$

Average price of Gold from 5th to 8th August

$$= \frac{4270+4245+4260+0.85 \times 4260}{4}$$

$$= \frac{16396}{4}$$

$$\therefore \text{Required Ratio} = \frac{41525 \times 4}{16396 \times 5} = 8305 : 4099$$

47. (e); Price of gold on 2nd August = $\frac{100}{106} \times 4240 = 4000$ Rs.

$$\text{Price of silver on 2nd August} = \frac{4000}{25} \times 51 = 8160$$

$$\text{Required average price} = \frac{8160+8150+8100+8150+8050}{5}$$

$$= 8122 \text{ Rs.}$$

48. (b); Rate of 1 kg silver on 6th August = Rs. 8050

Rate of 1 kg gold on 6th August = Rs. 424500

$$\text{Therefore, required percentage} = \frac{424500-8050}{424500} \times 100$$

$$= \frac{416450}{424500} \times 100 \approx 98\%$$

49. (a); Average price of gold per 10gm for given five days

$$= \frac{(4240+4230+4270+4245+4260)}{5} = \frac{21245}{5} = \text{Rs. } 4249$$

Average price of silver per kg for given five days =

$$\frac{8150+8100+8150+8050+8125}{5} = 8115$$

$$\therefore \text{Required difference} = 8115 - 4249 = 3866$$

50. (e); Average price of silver = $\frac{8150+8100+8150+8050+8125}{5}$

$$= \frac{40575}{5} = \text{Rs. } 8115 \text{ per kg}$$

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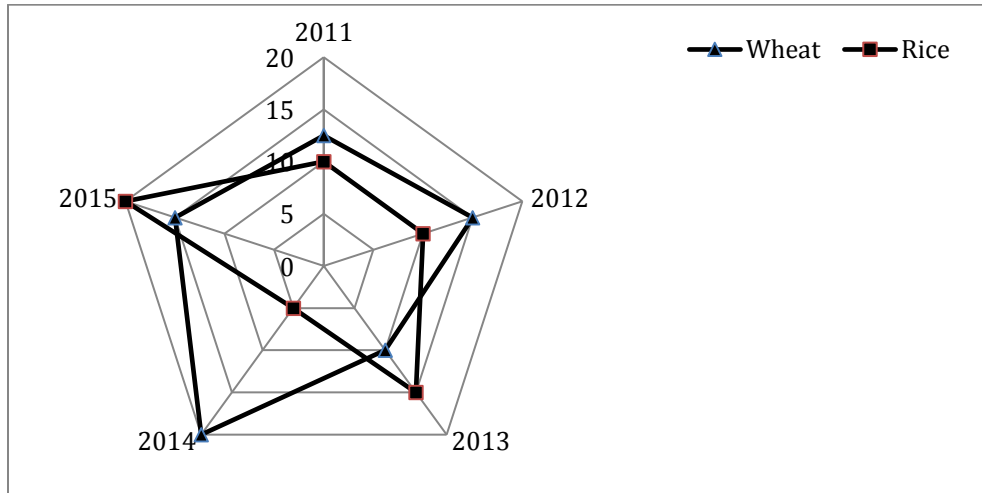


FROM STAR FACULTIES OF BANKERSADDA

PRACTICE SET (LEVEL-I)

Directions (1-5): Study the following Radar graph and answer the questions based on it.

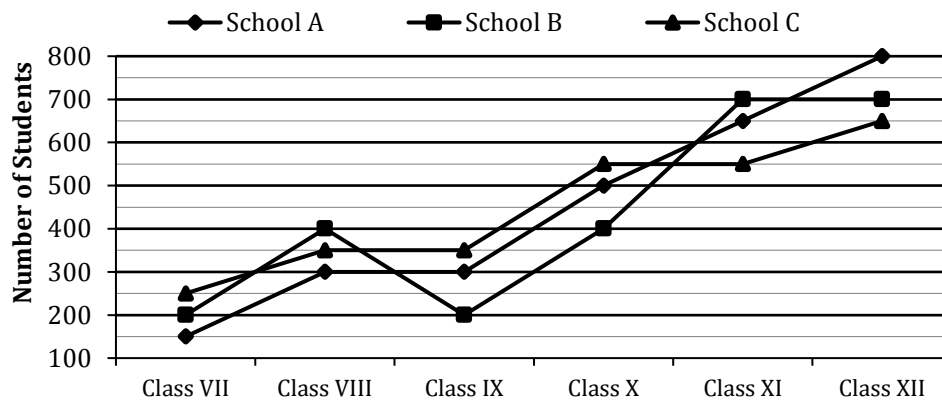
Given below is the Radar graph which shows the percentage rise in price of Wheat & Rice over the given years.



- If ratio between price of rice & wheat in 2014 is 3 : 4 then what will be their ratio of price in 2015
(a) 20 : 23 (b) 19 : 21 (c) 18 : 23 (d) 23 : 28 (e) 17 : 19
- If price of wheat in year 2011 is 7200 Rs /Quintal then what will be its price in year 2013
(a) 8420 (b) 9012 (c) 10500 (d) 83250 (e) 9108
- What is the effective percentage increase in price of wheat from year 2011 to year 2013
(a) 30% (b) 22% (c) 23.5% (d) 26.5% (e) 32.75%
- If a person expends Rs 4140 in buying rice at the rate of 120 Rs/kg in year 2012 then he has to reduce his consumption of rice by how many kg in year 2013 for the same expenditure of 4140.
(a) 4.5 kg (b) 3 kg (c) 2 kg (d) 2.5 kg (e) 4 kg
- If the price of wheat in 2013 is 132 Rs/kg then what will be total cost of 25 kg of wheat in 2012.
(a) 1250 Rs (b) 3000 Rs (c) 1500 Rs (d) 2000 Rs (e) 2500 Rs

Directions (6-10): Study the following graph carefully to answer the questions given below.

Number of students in different grades in three schools in year 2015



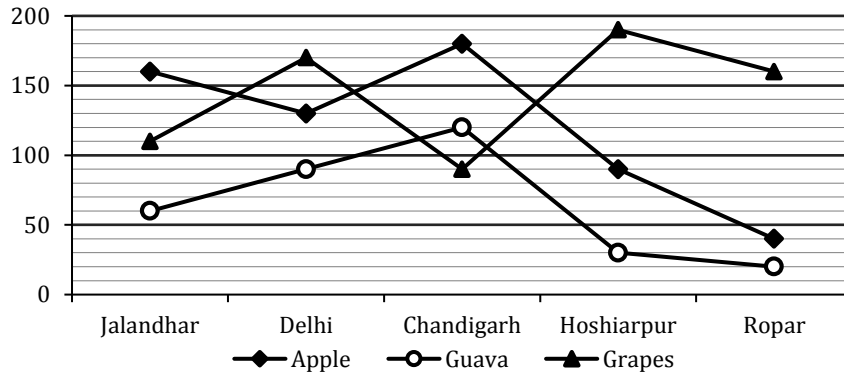
- In year 2016 the number of students of class XII in three schools A, B and C increase by 5%, 10% and 20% respectively with comparison to the last year in same class. Find the ratio of students in class XII of all schools in 2016 ?
(a) 84:78:77 (b) 84:77:78 (c) 88:77:78 (d) 8:7:9 (e) 78 : 84 : 77
- By what percent the number of students in class IX in school C less than total students in class XII in all schools together ?
(a) $83\frac{31}{43}\%$ (b) $67\frac{31}{43}\%$ (c) $74\frac{31}{43}\%$ (d) $71\frac{21}{43}\%$ (e) 58%
- What is average number of students in school A in all grades taken together ?
(a) 456 (b) 465 (c) 450 (d) 460 (e) 445
- What is respective ratio of the total students in all the grades in all 3 schools A, B and C ?
(a) 27 : 26 : 27 (b) 23 : 13 : 9 (c) 9 : 13 : 26 (d) 3 : 13 : 11 (e) None of these

10. The number of class VIII students in school B is what percent of total students in same school ?

- (a) $15\frac{5}{13}\%$ (b) $16\frac{2}{13}\%$ (c) $17\frac{5}{13}\%$ (d) $15\frac{8}{13}\%$ (e) $12\frac{7}{13}\%$

Directions (11-15): Study the following graph carefully to answer the questions that follow:

Given below is the graph showing the Cost of three different fruits (in rupees per kg.) in five different cities.

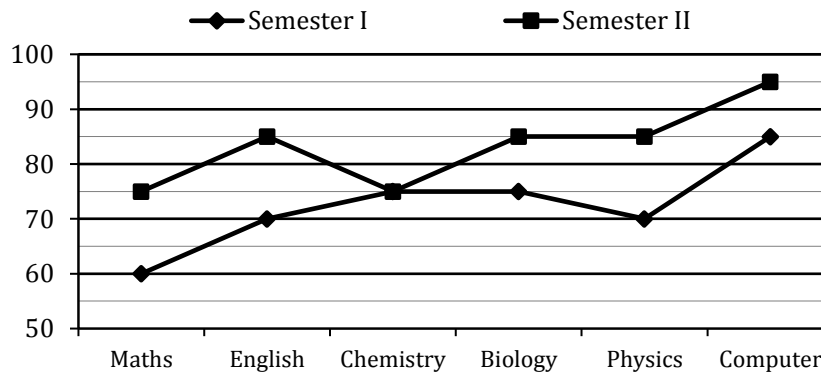


11. In which city difference between cost of one kg of apple and cost of one kg of Grapes is second lowest
(a) Delhi (b) Chandigarh (c) Jalandhar (d) Ropar (e) Hosiarpur
12. Average of cost of all three fruit in Ropar is what percent of average of all three fruits is Jalandhar.
(a) $33\frac{1}{3}\%$ (b) $66\frac{2}{3}\%$ (c) 62.5% (d) $20\frac{1}{4}\%$ (e) None of these
13. What is the ratio of sum of cost price of apple in Delhi, Chandigarh and Ropar together to the sum of cost price of Guava in Jalandhar, Delhi and Hoshiarpur together.
(a) 35 : 18 (b) 23 : 19 (c) 27 : 14 (d) 36 : 29 (e) 24 : 17
14. A shopkeeper from Jalandhar purchases apple and Guava at the rate given in table. At what price should he sell 2 kg of apple and 3 kg of Guava so, that on selling all quantity he gains overall profit of 35% percent
(a) 625 (b) 575 (c) 600 (d) 620 (e) 675
15. Cost of Guava in Jalandhar and Ropar together are what percent more or less than cost of Grapes from Delhi and Chandigarh together.
(a) $69\frac{3}{13}\%$ (b) $72\frac{5}{7}\%$ (c) $67\frac{3}{13}\%$ (d) $70\frac{5}{7}\%$ (e) None of these

Direction (16-20): Refer the following line graph and answer the questions based on it.

The line graph shows the percentage of students passed in different subjects in a class in two semesters.

Total number of student in the class = 600 (Same for both the semesters)

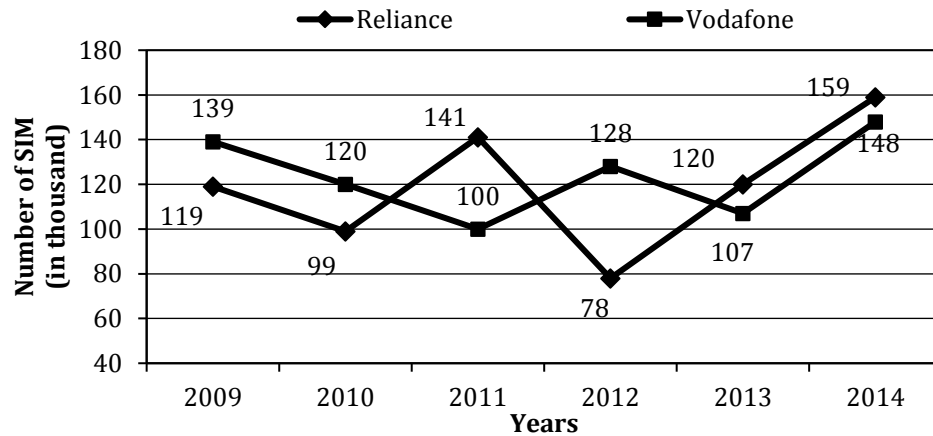


16. What is the average number of students passed in semester II in Chemistry, Biology and Physics?
(a) 500 (b) 490 (c) 480 (d) 510 (e) 450
17. What is the ratio of number of students not passed in Physics in semester I to the number of students passed in Computer in semester II?
(a) 19 : 14 (b) 14 : 19 (c) 19 : 6 (d) 6 : 19 (e) 12 : 19
18. In semester II, the number of students passed in Computer is how much percent more than the number of students passed in Maths in semester II ?
(a) $26\frac{2}{3}\%$ (b) $27\frac{1}{2}\%$ (c) $23\frac{1}{3}\%$ (d) $33\frac{1}{3}\%$ (e) $26\frac{1}{3}\%$

19. After the revision of result for semester I, it is found that the number of students passed in Chemistry is increased by 4% to the previous number. What is the number of students passed in Chemistry after the revision of the results?
 (a) 472 (b) 470 (c) 468 (d) 466 (e) 486
20. What is the total number of students passed in Maths or English in semester I, if the number of students passed in both the subjects is 245?
 (a) 540 (b) 530 (c) 525 (d) 535 (e) Cannot be determined

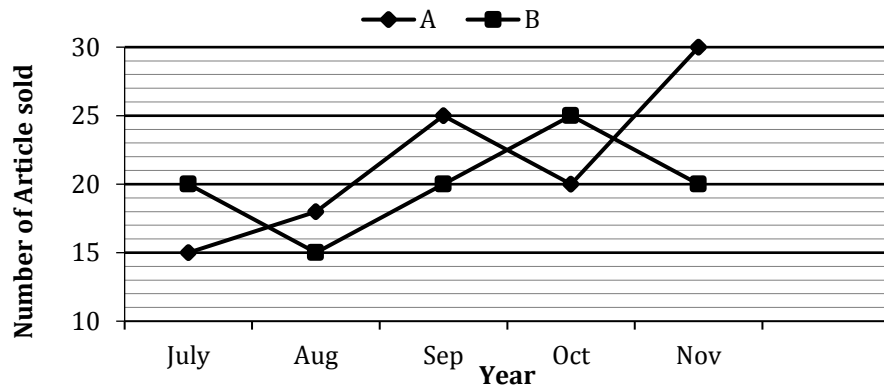
Directions (21-25): Study the following line graph and answer the questions based on it.

Number of Sim (in thousand) sold by two companies over the years



21. What is the difference between the number of sim sold by company Vodafone in 2011 and 2012 ?
 (a) 50000 (b) 42000 (c) 33000 (d) 28000 (e) 27000
22. What is the difference between the total sim sold by the two companies in the given years ?
 (a) 19000 (b) 22000 (c) 26000 (d) 28000 (e) 23000
23. What is the average numbers of sim sold by Reliance company over the given period ? (rounded off to nearest integer)
 (a) 119333 (b) 113666 (c) 112778 (d) 111223 (e) 191333
24. In which of the following year, the difference between the number of sims sold by Reliance company and Vodafone is the maximum in the given years ?
 (a) 2009 (b) 2010 (c) 2011 (d) 2012 (e) 2013
25. The number of sims sold by company Vodafone in 2011 was what per cent of the number of sims sold by Reliance company in the same year ?
 (a) $73\frac{130}{141}\%$ (b) 111% (c) $80\frac{130}{141}\%$ (d) $82\frac{121}{141}\%$ (e) $70\frac{130}{141}\%$

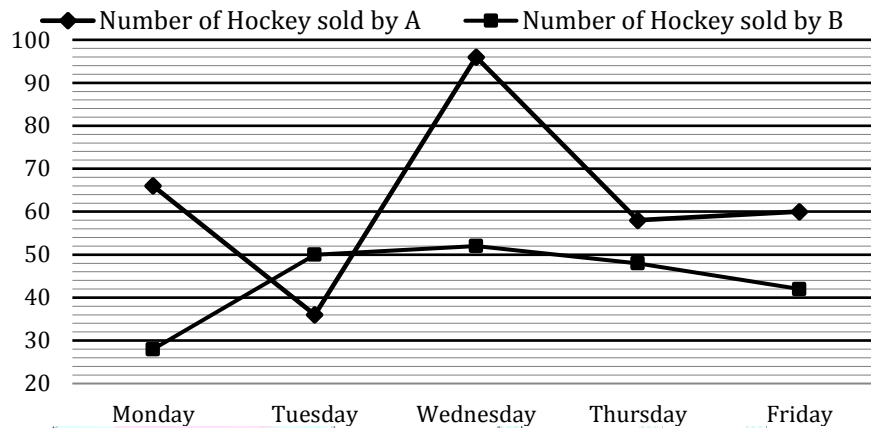
Directions (26-30): Given below is the line graph which shows the number of article sold by two shopkeepers in five different months



26. What is the ratio of total articles sold by A and B together in Sep to the total articles sold by A in July and Aug together.
 (a) 15 : 17 (b) 15 : 11 (c) 19 : 17 (d) 20 : 13 (e) 19 : 12
27. If number of articles sold by A in December month of same year is increased by $23\frac{1}{3}\%$ over previous month, then what will be the average number of articles sold by A in Sep, Oct and December
 (a) $20\frac{1}{3}$ (b) $18\frac{2}{3}$ (c) $26\frac{2}{3}$ (d) $27\frac{1}{3}$ (e) 24

28. Find the ratio of number of articles sold by A in August and November together to the number of articles sold by B in July and Oct together.
 (a) 12 : 11 (b) 20 : 19 (c) 16 : 15 (d) 17 : 16 (e) 13 : 12
29. If Articles sold By A in June of same year is $33\frac{1}{3}\%$ more than that of sold in July of same year then articles sold by B in Aug and Sep. together are what percent more or less than articles sold by A in June.
 (a) 80% (b) 87.5% (c) 62.5% (d) 50% (e) 75%
30. What is the difference between average of articles sold by A is July, Oct and Nov to the average of articles sold by B in Aug, Sep and Oct.
 (a) $2\frac{2}{3}$ (b) $1\frac{2}{3}$ (c) $2\frac{1}{3}$ (d) $4\frac{1}{3}$ (e) $3\frac{1}{3}$

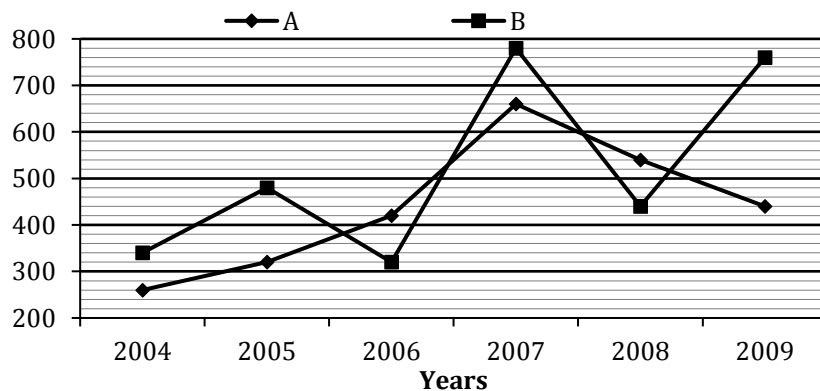
Direction (31-35): Bar graph is given below which shows the number of Hockey sold by seller A and seller B on five days.



31. Total number of Hockey sold by A and B together on Wednesday is how much percentage more than the number of Hockey sold by A and B together on Monday?
 (a) $15\frac{2}{3}\%$ (b) $8\frac{1}{3}\%$ (c) $\frac{2700}{47}\%$ (d) $16\frac{2}{3}\%$ (e) $21\frac{3}{7}\%$
32. If number of Hockey sold on Tuesday by A is increases by $33\frac{1}{3}\%$, then what will be the average no. of Hockey sold on Monday, Tuesday and Friday by A?
 (a) 85 (b) 58 (c) 56 (d) 82 (e) 52
33. Find the number of Hockey sold on Saturday by A and B together, if number of Hockey sold on Saturday is $5\frac{15}{17}\%$ more than the hockey sold on Friday by A and B together?
 (a) 110 (b) 114 (c) 116 (d) 108 (e) 120
34. What is the difference between the number of Hockey sold on Monday and Wednesday by B to the number of Hockey sold on Friday by both A & B together?
 (a) 22 (b) 12 (c) 14 (d) 21 (e) 24
35. A sold 70% defective Hockey on Friday and B sold 50% defective Hockey on Tuesday. Then find the number of Hockey sold by A on Friday and B on Tuesday that are not defected?
 (a) 25 (b) 43 (c) 18 (d) 32 (e) 40

Direction(36-40): Study the following table and answer the questions that follow.

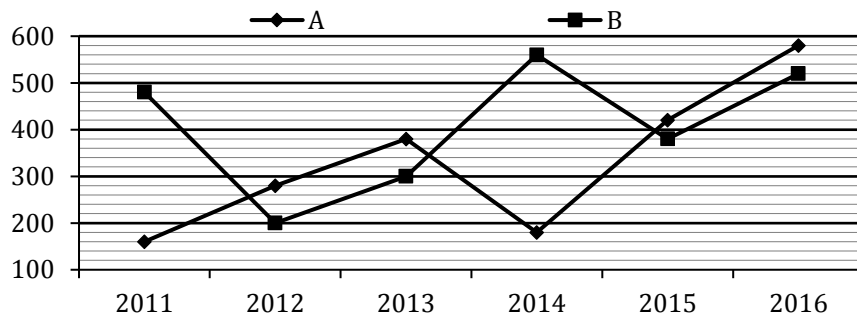
Given line graph shows the number of students appeared from state A and state B in an examination.



36. Number of students appeared from state B in 2009 is about what percent of total students appeared from state A all over the years?(approx.)
 (a) 32 (b) 30 (c) 33 (d) 28 (e) 22
37. What is the difference between the total number of students from state A in 2004 and 2005 together and those of state B in 2008 and 2009 together?
 (a) 520 (b) 580 (c) 620 (d) 720 (e) 680
38. What is the ratio of number of students appeared in examination from state B in 2004, 2006 and 2008 to the number of students appeared from state A in 2005, 2007 and 2009?
 (a) 73:55 (b) 55:71 (c) 79:15 (d) 75:13 (e) 13:85
39. If in 2010 the number of students appeared from state A is increase by 10% and those from state B increased by 15% as compared to the number of students from respective states in year 2009, then what is the ratio of number of students from state A and state B in 2010?
 (a) 287:439 (b) 285:437 (c) 289:437 (d) 433:189 (e) 242:437
40. What is the difference between average number of students from state A and state B all over the years?
 (a) 90 (b) 60 (c) 80 (d) 70 (e) 110

Directions (41-45): Refer the graph and answer the given questions.

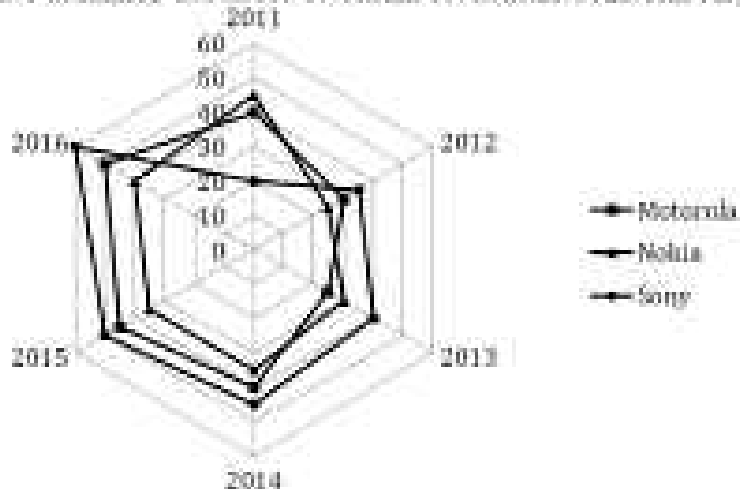
The following line graph shows the number of votes polled in two constituencies A and B of a city during six years.



41. What is the difference between number of votes polled in constituency A and B together in 2013 and number of votes polled in both constituencies together in 2011?
 (a) 50 (b) 30 (c) 60 (d) 40 (e) 70
42. Find ratio of votes polled in constituency A in 2014 and polled votes in constituency B in 2016.
 (a) 3 : 8 (b) 9 : 26 (c) 2 : 3 (d) 1 : 3 (e) 2 : 5
43. Number of votes polled in constituency A and B together in 2015 is what percent less or more than the number of votes polled in constituency B in 2015 and 2016 together?
 (a) $11\frac{1}{9}\%$ (b) $9\frac{1}{11}\%$ (c) $16\frac{2}{3}\%$ (d) $14\frac{1}{7}\%$ (e) $13\frac{2}{3}\%$
44. If 10% of total votes polled in constituency A in 2012 is invalid and 5% of votes polled in constituency B in 2013 is invalid, then find the average of valid votes in constituency A in 2012 and B in 2013.
 (a) 268.5 (b) 267.5 (c) 283.5 (d) 272.5 (e) 265.4
45. Find average number of votes polled in constituency A in all the given years.
 (a) 343.33 (b) 333.33 (c) 233.33 (d) 330.33 (e) 353.33

Directions (46-50): Study the graph carefully to answer the questions that follow.

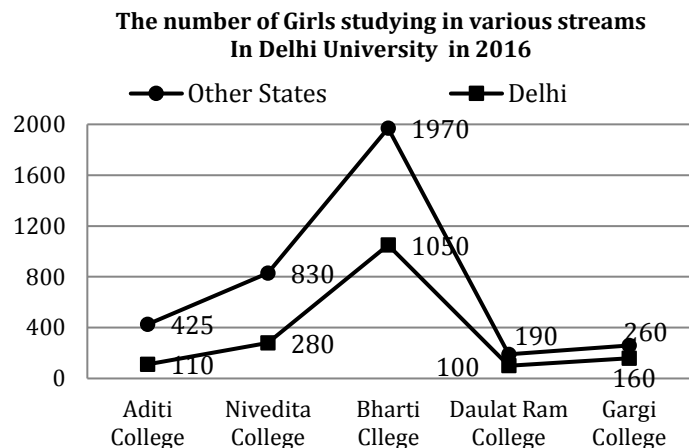
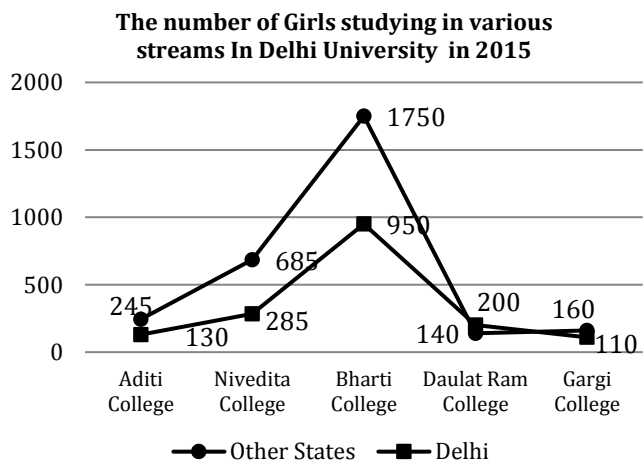
PERCENT INCREASE IN PROFIT OF THREE COMPANIES OVER THE YEARS



46. If profit for company Nokia in 2012 is 2000 and expenditure in 2013 for company Nokia is 50,000, then what is the total revenue in 2013 for Nokia? Give that total revenue = expenditure + profit.
 (a) Rs. 52600 (b) Rs.54200 (c) Rs.53280 (d) Rs.55800 (e) Rs.56020
47. If profit in year 2015 for company Sony is 3000 and profit of company Motorola in 2013 is equal to profit of company Sony in 2014 then what is the profit of company Motorola in 2013 ?
 (a) Rs.1500 (b) Rs.4000 (c) Rs.3500 (d) Rs.2000 (e) Rs.2500
48. What is the average percentage increase in profit for company Nokia over all the years?
 (a) 49% (b) 32% (c) 23% (d) 38% (e) 35%
49. What was the percentage increase in percent increase of profit of company Motorola in the year 2014 over its previous year ?
 (a) 60% (b) 65% (c) 55% (d) 50% (e) 70%
50. If profit earned by company Nokia in 2014 is 27,000 and by company Sony in 2014 is 43500 then what is the total profit earned by them in year 2013?
 (a) Rs. 25,000 (b) Rs. 35,000 (c) Rs. 40,000 (d) Rs. 50,000 (e) Rs. 45,000

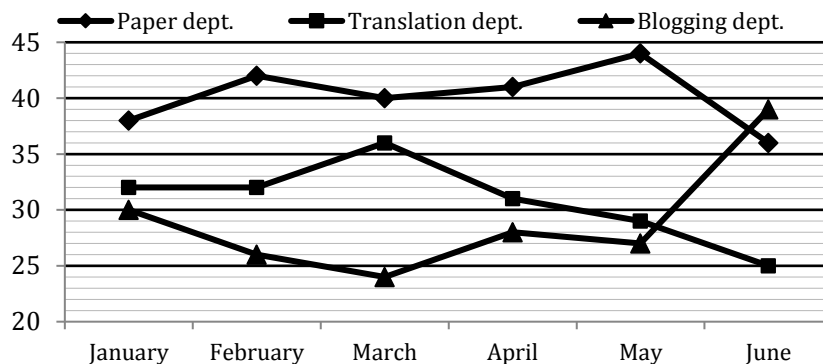
Directions (51-55): Study the following graph carefully and answer the questions that follow:

The line graph show the number of Girls who study in various streams in Delhi University from Delhi and other states in two different years.



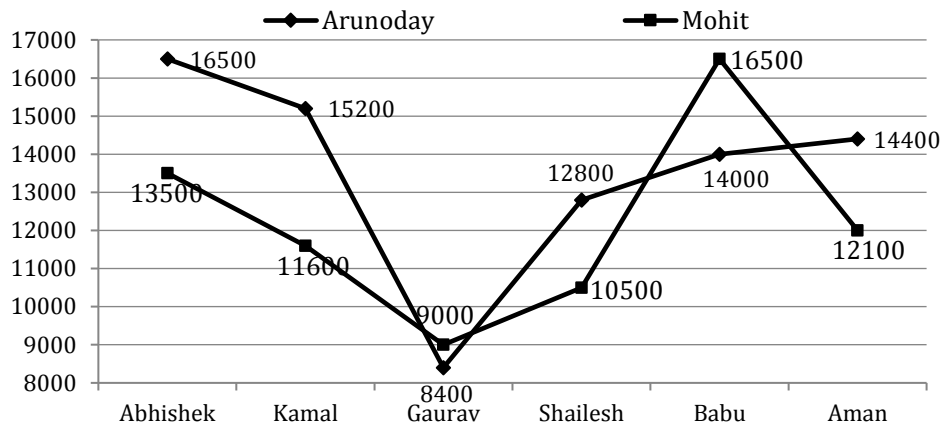
51. What is the difference between the average number of all the girls who study in various streams in Delhi University in 2015 from Other states and the average number of girls who study in Delhi University in 2016 from Delhi?
 (a) 269 (b) 359 (c) 296 (d) 356 (e) 256
52. The number of girls who study in various streams in Delhi University from other-states in 2015 from Daulat Ram College and Gargi College is what per cent of the number of girls from same college from other-states in 2016? (Rounded off to two decimal places)
 (a) 62.23% (b) 66.67% (c) 72.34% (d) 78.67% (e) 59.23%
53. What is the ratio of the total number of girls who study in various streams of Delhi University from Delhi in 2015 to the number of girls who study in various streams of Delhi University from other-States in 2016?
 (a) 67 : 147 (b) 137 : 67 (c) 77 : 147 (d) 57 : 147 (e) 137 : 147
54. The number of girls who study in various streams of Delhi University from Other-states in 2016 is approximate what percent of the number of Girls who study in various streams of Delhi University from Delhi in 2016?
 (a) 98% (b) 116% (c) 188% (d) 86% (e) 216%
55. What is the sum of the average number of girls who study in various streams from Delhi in 2015 and the average number of girls who study in various streams from other-states in 2016?
 (a) 1070 (b) 3410 (c) 1680 (d) 2680 (e) 1080

Directions (56-60): The following line graph shows the number of reams (packets of A4 size paper) in terms of percentage used by three departments of Career Power i.e. Paper dept., Translation dept. and Blogging dept. Provided that total number of reams used per month is 1200 and it remains consistent for all the months. There is no other department using these reams.



56. Find the difference in total number of reams used by paper dept. from January to March and that of by Blogging dept. from April to May?
 (a) 700 (b) 718 (c) 780 (d) 468 (e) None of these
57. In May, Babu working in Paper dept. used 25% of the reams from which he wasted $66\frac{2}{3}\%$ reams. Find the number of reams not wasted by Babu.
 (a) 68 (b) 88 (c) 32 (d) 44 (e) 27
58. In July Translation dept. demanded 25% more reams than that provided to paper dept. in June by stationary supervisor Mr. Vinod, which he denied. But he provided them the reams $16\frac{2}{3}\%$ less than that provided to paper dept. in June. Find the ratio of number of reams demanded and actually provided to translation dept.
 (a) 4 : 3 (b) 3 : 2 (c) 13 : 12 (d) 5 : 3 (e) 5 : 8
59. By approximately what percent the total number of reams used by Blogging dept. is less than that of by paper team throughout the given months?
 (a) 36% (b) 32% (c) 20% (d) 22% (e) 28%
60. Total number of reams used by paper dept. in February, blogging dept. in April and translation dept. in June respectively is what percent of total reams used by Translation dept. in given 6 months?
 (a) $49\frac{3}{8}\%$ (b) $51\frac{13}{37}\%$ (c) $51\frac{15}{37}\%$ (d) 50.35% (e) None of these

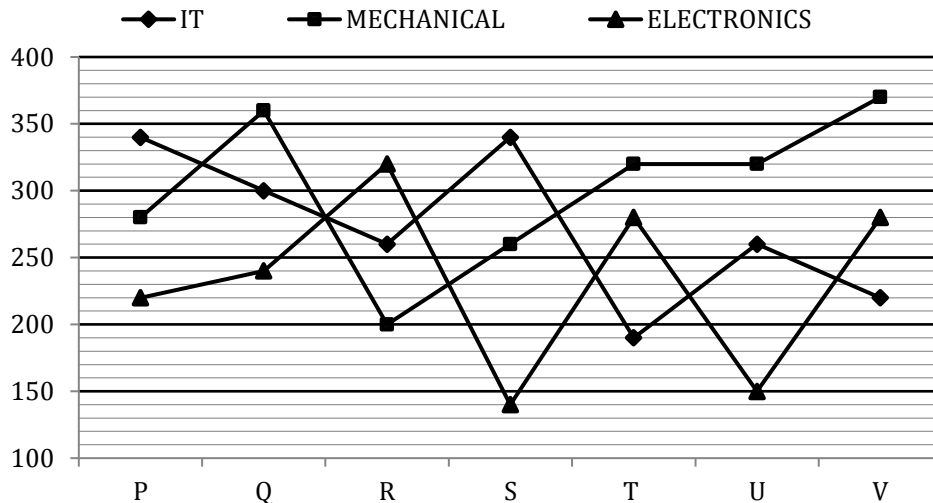
Direction (Q61-65): Study the following line graph carefully to answer the questions based on it. The graph shows the amount lent by Mohit and Arunoday to six persons.



61. What is the difference between the average amount lent out by Arunoday and Mohit?
 (a) Rs 1530 (b) Rs 1350 (c) Rs 1050 (d) Rs 1320 (e) None of these
62. What is the interest earned by Arunday from Abishek and Shailesh if he lent them money at 10% and 11% p.a. respectively, interest being compound annually for 2 years?
 (a) Rs 6435 (b) Rs 6345.88 (c) Rs 6435.88 (d) Rs 6453.88 (e) Rs 6525
63. Babu returned Rs 16633.4 to Arunday after 2 years. Find the rate pcpa at which he borrowed the amount at compound interest compounded annually?
 (a) 19% (b) 12% (c) 10% (d) 9% (e) 8%
64. If Aman borrowed same amount as Babu from Arunday at same rate of interest for 3 years at simple interest while Babu borrowed at compound interest compounded annually at same rate and same time period. If the difference between their interest is Rs 992.25 then find the rate pcpa.
 (a) 15% (b) 10% (c) 5% (d) 12% (e) 18%
65. Find the amount earned by Mohit by lending money to given persons at 10% pa in 1 year?
 (a) Rs 7235 (b) Rs 7330 (c) Rs 7230 (d) Rs 7320 (e) Rs 7420

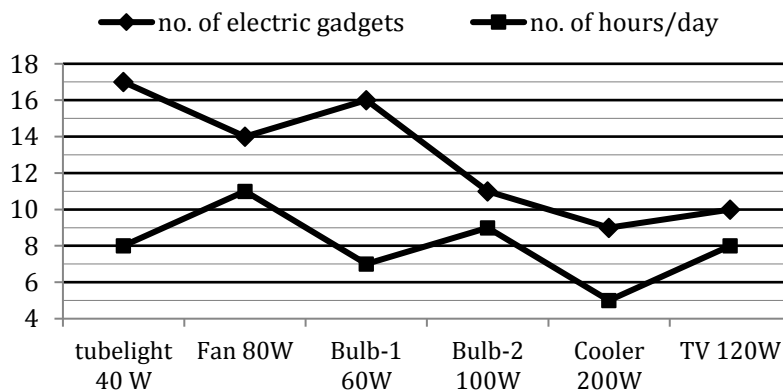
Directions (66-70): Study the following graph carefully to answer the given questions.

The following line graph shows the total no. of students in seven different institutes with three different specialization in 2012.



66. If the number of students with Mechanical specialization in each institute increased by 20% and the number of students with Electronics specialization in each institute decreased by 15% from 2012-2013, total number of students with Mechanical from all the institutes in 2013 is what per cent of the total number of students with Electronics specialization from all the institutes in 2013 ?
 (a) 122% (b) 116% (c) $162\frac{29}{163}\%$ (d) $181\frac{28}{163}\%$ (e) $194\frac{28}{163}\%$
67. What is the ratio between total number of students in institute R and V, respectively?
 (a) 39 : 44 (b) 26 : 29 (c) 29 : 26 (d) 44 : 39 (e) 38 : 43
68. What is the difference between total number of students with IT specialization from all the institutes together and the total number of students with Mechanical specialization from all the institutes together ?
 (a) 260 (b) 240 (c) 280 (d) 200 (e) 250
69. If the number of students in institutes P, Q and R with IT specialisation increased by 15%, 22% and 10% respectively from 2012 to 2013, what was the total number of students with IT specialisation in the three institutes together in 2013 ?
 (a) 1028 (b) 1056 (c) 1043 (d) 1142 (e) 1145
70. If out of the total number of students for all three specializations together in institute Q number of students having liking for Music, Painting and Cricket are in the ratio 5 : 6 : 7 respectively, then what is the number of students liking Music from institute Q ?
 (a) 250 (b) 300 (c) 350 (d) 360 (e) 280

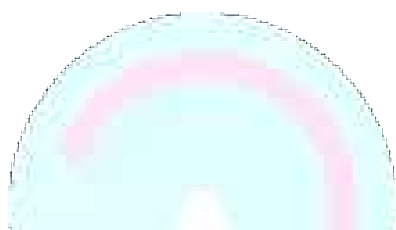
Directions (71-75): The following graph shows the number of different electric gadgets and their hours of use per day in a jeweler's shop. Study the graph carefully and answer the following questions:-



NOTE: If an 80W fan used for 2 hours then it consumes 160watt hours energy and 1000W hour=1Unit.

71. Which of the following consumes 2nd highest amount of electricity in the shop?
 (a) Tube light (b) Cooler (c) bulb-2 (d) TV (e) Fan

72. If 1 electric unit costs Rs. 2.7 then what money is paid to electricity department for using fans and tube lights for the month of August?
 (a) Rs. 1438.56 (b) Rs. 1486.512 (c) Rs. 1495.64 (d) Rs. 1498.37 (e) Rs. 1485.612
73. Electricity consumption by all the fans is what percent of the consumption by Bulb I and Bulb II together?
 (a) $72\frac{106}{831}\%$ (b) $75\frac{106}{831}\%$ (c) $71\frac{601}{831}\%$ (d) $74\frac{106}{831}\%$ (e) None of these
74. Find the total consumption of energy in the shop in kwh (kilowatt hours) in 1 week.
 (a) 367.63 (b) 354.52 (c) 384.85 (d) 370.86 (e) 348.85
75. If the coolers are of power 175W instead of 200W then how much less units it would consume in the whole month (30 days)?
 (a) 33.75 units (b) 43.25 units (c) 41.85 units (d) 34.95 units (e) 37.35 units



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PRACTICE SET (LEVEL-I) SOLUTIONS

1. (c); Let ratio of price of rice to price of wheat in 2014 be $3x$ and $4x$
Ratio of price of rice to price of wheat in 2015 = $(120\% \text{ of } 3x) : (115\% \text{ of } 4x) = 18 : 23$
2. (e); Price of wheat in 2013 = $7200 \times \frac{115}{100} \times \frac{110}{100} = 9108$
3. (d); Let price of wheat in 2011 = 100
so, price of wheat in 2013 = $\frac{115}{100} \times \frac{110}{100} \times 100 = 126.5$
So, effective increases equals = 26.5%
4. (a); In 2012 person buys = $\frac{4140}{120}$ kg of Rice
So, in 2013 person buys = $\frac{4140}{\frac{115}{100} \times 120}$ kg of rice
So decrease in consumption = $34.5 - 30 = 4.5$ kg
5. (b); Price per kg of wheat in 2012 = $\frac{132 \times 100}{110} = 120$
Total cost for 25 kg of wheat in 2012 = $120 \times 25 = 3000$ Rs
6. (b); In 2017, number of students in class XII of
School A = $\frac{105}{100} \times 800 = 840$
School B = $\frac{110}{100} \times 700 = 770$
School C = $\frac{120}{100} \times 650 = 780$
So, required ratio = $840:770:780 = 84:77:78$
7. (a); Required percentage = $\frac{2150-350}{2150} \times 100 = 83\frac{31}{43}\%$
8. (c); Required average = $\frac{1}{6}(150 + 300 + 300 + 500 + 650 + 800)$
 $= \frac{1}{6} \times 2700 = 450$
9. (a); Students in school A = $(150 + 300 + 300 + 500 + 650 + 800) = 2700$
Students in school B = $(200 + 400 + 200 + 400 + 700 + 700) = 2600$
Students in school C = $(250 + 350 + 350 + 550 + 550 + 650) = 2700$
So, ratio = $27:26:27$
10. (a); Required percentage = $\frac{400}{2600} \times 100 = 15\frac{5}{13}\%$
11. (c); The difference between costs of 1 kg of apple and 1 kg of grapes is second lowest in Jalandhar.
12. (b); Average of cost of all three fruits in Ropar = $\frac{1}{3}(20 + 40 + 160) = \frac{220}{3}$
Average of cost of all three fruits in Jalandhar = $\frac{60+110+160}{3} = \frac{330}{3}$
Required percentage = $\frac{220/3}{330/3} \times 100 = \frac{22}{33} \times 100 = 66\frac{2}{3}\%$
13. (a); Required ratio = $(130 + 180 + 40) : (60 + 90 + 30) = 350 : 180 = 35 : 18$
14. (e); Cost of 2 kg apple and 3 kg of Guava in Jalandhar = $160 \times 2 + 3 \times 60 = 320 + 180 = 500$
Required selling price = $\frac{135}{100} \times 500 = 675$
15. (a); Cost of guava in Jalandhar & Ropar together = $60 + 20 = 80$
Cost of Grapes in Delhi and Chandigarh together = $170 + 90 = 260$
Required % = $\frac{260-80}{260} \times 100 = \frac{180}{260} \times 100 = 69\frac{3}{13}\%$
16. (b); Required average = $\frac{(75 + 85 + 85)}{3 \times 100} \times 600 = 490$
17. (d); Required Ratio = $(100 - 70) : 95 = 30 : 95 = 6 : 19$
18. (a); Required Percentage = $\frac{(95 - 75)}{75} \times 100 = 26\frac{2}{3}\%$
19. (c); Number of students passed in Chemistry in semester I after the revision of the result
 $= \frac{75}{100} \times \frac{100+4}{100} \times 600 = 468$
20. (d); Number of students passed in Maths or English in semester I
 $= \left(\frac{60+70}{100} \times 600\right) - 245 = 780 - 245 = 535$
21. (d); Difference = $128 - 100 = 28$ thousands = 28000
22. (c); Total number of sim sold by Reliance in the given year = $119 + 99 + 141 + 78 + 120 + 159 = 716$ thousand
Total number of sim sold by Vodafone in the given year = $139 + 120 + 100 + 128 + 107 + 148 = 742$ thousand
Difference = $742 - 716 = 26$ thousand = 26000
23. (a); Average number of sim sold by Reliance Over the given period = $\frac{716}{6} = 119.333$ thousand = 119333
24. (d); Difference in 2009 = $139 - 119 = 20$ thousand
In 2010 = $120 - 99 = 21$ thousand
In 2011 = $141 - 100 = 41$ thousand
In 2012 = $128 - 78 = 50$ thousand (maximum)
In 2013 = $120 - 107 = 13$ thousand
In 2014 = $159 - 148 = 11$ thousand
25. (e); Required percentage = $\frac{100}{141} \times 100 = 70\frac{130}{141}\%$
26. (b); Required ratio
 $= \frac{25+20}{15+18} = \frac{45}{33} = 15 : 11$
27. (d); Articles sold by A in December
 $= \left(1 + \frac{7}{30}\right) 30 = 37$
Required average
 $= \frac{25+20+37}{3} = \frac{82}{3} = 27\frac{1}{3}$
28. (c); Required ratio
 $= \frac{18+30}{20+25} = \frac{48}{45} = 16 : 15$
29. (e); Articles sold by A in June
 $= \frac{4}{3} \times 15 = 20$
Articles sold by B in Aug and Sep = $15 + 20 = 35$
Required percentage = $\frac{35-20}{20} \times 100 = 75\%$
30. (b); Average of articles sold by A in July, Oct and Nov
 $= \frac{15+20+30}{3} = \frac{65}{3}$
Average of articles sold by B in Aug, Sep and Oct
 $= \frac{15+20+25}{3} = \frac{60}{3}$
Required difference
 $\frac{65}{3} - \frac{60}{3} = \frac{5}{3} = 1\frac{2}{3}$
31. (c); Required percentage
 $= \frac{148-94}{94} \times 100 = \frac{5400}{94} = \frac{2700}{47}\%$

32. (b); Hockey on Tuesday after increase by A.

$$= \frac{4}{3} \times 36 = 48$$

$$\therefore \text{Required average} = \frac{66+48+60}{3} = \frac{174}{3} = 58$$

33. (d); Required no. of Hockey sold on Saturday by A and B

$$= \frac{18}{17} \times 102 = 108$$

34. (a); Required difference = $80 \sim 102 = 22$

$$35. (b); = \frac{30}{100} \times 60 + \frac{50}{100} \times 50$$

$$= 18 + 25 = 43$$

$$36. (d); \text{Required percent} = \frac{760}{2640} \times 100 = 28.78\% \approx 28\%$$

37. (c); Total students from state A in 2004 and 2005

$$= 260 + 320 = 580$$

Total students from state B in 2008 and 2009

$$= 440 + 760 = 1200$$

$$\text{Difference} = 1200 - 580 = 620$$

$$38. (b); \text{Required Ratio} = \frac{340+320+440}{320+660+440} = \frac{1100}{1420} = \frac{55}{71}$$

$$39. (e); \text{Required ratio} = \frac{\frac{110}{100} \times 440}{\frac{115}{100} \times 760} = \frac{484}{874} = \frac{242}{437}$$

$$40. (c); \text{Difference} = \frac{3120}{6} - \frac{2640}{6} = 520 - 440 = 80$$

41. (d); Number of votes polled in both constituencies in

$$2013 = 300 + 380 = 680$$

Number of votes polled in both constituencies in

$$2011 = 160 + 480 = 640$$

$$\text{Required difference} = 680 - 640 = 40$$

$$42. (b); \text{Required ratio} = \frac{180}{520} = \frac{9}{26}$$

43. (a); Number of total votes polled in both constituencies in

$$2015 = 420 + 380 = 800$$

Number of votes polled in constituency B in 2015 and

$$2016 = 380 + 520 = 900$$

$$\text{Required percentage} = \frac{900-800}{900} \times 100 = 11\frac{1}{9}\%$$

44. (a); Valid votes of constituency A in 2012 = $280 \times \frac{100}{95} = 252$

$$\text{Valid votes of constituency B in 2013} = 300 \times \frac{100}{95} = 285$$

$$\text{Average of valid votes} = \frac{285+252}{2} = 268.5$$

45. (b); Required Average

$$= \frac{160+280+380+180+420+580}{6} = 333.33$$

$$46. (a); \text{Profit in 2013} = 2000 \times \frac{130}{100} = \text{Rs.2600}$$

$$\text{Total revenue} = 50,000 + 2600 = \text{Rs.52600}$$

$$47. (d); \text{Profit of company Motorola in 2013} = \frac{3000 \times 100}{150}$$

$$= \text{Rs.2000}$$

$$48. (e); \text{Required average} = \frac{45+25+30+35+35+40}{6} = \frac{210}{6} = 35\%$$

$$49. (a); \text{Required percentage} = \frac{40-25}{25} \times 100 = \frac{15}{25} \times 100 = 60\%$$

$$50. (d); \text{Profit earned by Company Nokia in 2013} = \frac{27000 \times 100}{135}$$

$$= \text{Rs.20,000}$$

$$\text{Profit earned by company Sony in 2013} = \frac{43500 \times 100}{145}$$

$$= \text{Rs.30,000}$$

$$\text{Total profit} = \text{Rs.50,000}$$

51. (e); Average of girls from other states in 2015

$$= \frac{245+685+1750+140+160}{5} = \frac{2980}{5} = 596$$

Average of girls from Delhi in 2016

$$= \frac{110+280+1050+100+160}{5} = \frac{1700}{5} = 340$$

$$\text{Difference} = 596 - 340 = 256$$

52. (b); Number of girls from other states in 2015

$$= 140 + 160 = 300$$

Number of girls from other states in 2016

$$= 190 + 260 = 450$$

$$\text{Required percent} = \frac{300}{450} \times 100 = 66.67\%$$

$$53. (a); \text{Required Ratio} = \frac{130+285+950+200+110}{425+830+1970+190+260} = \frac{1675}{3675} = \frac{67}{147}$$

$$54. (e); \text{Required percent} = \frac{3675}{1700} \times 100 \approx 216.17\%$$

55. (a); Average of girls from Delhi in 2015

$$= \frac{130+285+950+200+110}{5} = \frac{1675}{5} = 335$$

$$\text{Sum} = 335 + 735 = 1070$$

56. (c); Required difference

$$= [(38 + 42 + 40) - (27 + 28)] \times \frac{1}{100} \times 1200 = 780$$

57. (d); Number of reams not wasted by Babu = $132 - 88 = 44$

58. (b); Reams demanded = $1.25 \times 0.36 \times 1200 = 540$

$$\text{Reams provided} = 432 - 72 = 360$$

$$\therefore \text{Required ratio} = 3 : 2$$

$$59. (e); \text{Required percentage} = \frac{2892-2088}{2892} \times 100 \approx 28\%$$

$$60. (b); \text{Required percentage} = \frac{\frac{95 \times 1200}{100} \times 100}{\frac{185 \times 1200}{100}} = 51\frac{13}{37}\%$$

61. (b); Average amount lent by Arunoday

$$= \frac{1}{6} \times (165 + 152 + 84 + 128 + 140 + 144) \times 100$$

$$= 13550$$

Average amount lent by Mohit

$$= \frac{1}{6} \times (135 + 116 + 90 + 105 + 165 + 121) \times 100$$

$$= 12200$$

$$\therefore \text{Required difference} = 1350 \text{ Rs}$$

62. (c); Required interest = 21% of $16500 + 23.21\%$ of 12800

$$= 3465 + 2970.88 = \text{Rs } 6435.88$$

$$63. (d); 16633.4 = 14000 \left(1 + \frac{r}{100}\right)^2$$

$$\text{or, } \frac{16633.4}{14000} = \left(1 + \frac{r}{100}\right)^2$$

$$\text{or, } \frac{11881}{10000} = \left(1 + \frac{r}{100}\right)^2$$

$$\text{or, } \frac{109}{100} = 1 + \frac{r}{100}$$

$$\text{or, } r = 9\%$$

$$64. (a); 992.25 = 14000 \left(\frac{r}{100}\right)^2 \left(\frac{300+r}{100}\right)$$

$$\text{or, } \frac{567}{8} = \frac{r^2(300+r)}{1000}$$

$$\text{or, } 70875 = r^2(300 + r)$$

By using option,

$$r = 15\%$$

$$65. (d); \text{Required earning} = 1350 + 1160 + 900 + 1050 + 1650 + 1210 = \text{Rs } 7320$$

66. (e); Total number of students with mechanical specialisation in 2013

$$= 336 + 432 + 240 + 312 + 384 + 384 + 444 = 2532$$

Total number of students with Electronics specialisation in 2013 = $176 + 192 + 256 + 112 +$

$$224 + 120 + 224 = 1304$$

$$\therefore \text{Required \%} = \frac{2532}{1304} \times 100 = 194\frac{28}{163}\%$$

$$67. (b); \text{Total number of students in institute R} = 320 + 260 + 200 = 780$$

$$\text{Total number of students in institute V} = 370 +$$

$$290 + 220 = 870$$

$$\text{Required Ratio} = 780 : 870 = 26 : 29$$

68. (d); Total number of students with IT specialisation from all the institutes together
 $= 340 + 300 + 260 + 340 + 190 + 260 + 220 = 1910$

Total number of students with Mechanical specialisation from all the institutes together =
 $280 + 360 + 200 + 260 + 320 + 320 + 370 = 2110$
 Required difference = $2110 - 1910 = 200$

69. (c); In 2013, total no. of students with IT specialisation in institute P = $340 \times 1.15 = 391$

$$Q = 300 \times 1.22 = 366$$

$$R = 260 \times 1.1 = 286$$

$$\therefore \text{Required no. of students} = 391 + 366 + 286 = 1043$$

70. (a); Total no. of students in institute Q = $360 + 300 + 240 = 900$

From these, No. of students liking Music

$$= \frac{5}{5+6+7} \times 900 = \frac{5}{18} \times 900 = 250$$

71. (c); Electricity consumption per day tube lights = $17 \times 8 \times 40 = 5440$ w hours

$$\text{Of fans} = 14 \times 11 \times 80 = 12320 \text{ w hours}$$

$$\text{Of bulb I} = 6720 \text{ w hours}$$

$$\text{Of bulb II} = 9900 \text{ w hours}$$

$$\text{Of coolers} = 9000 \text{ w hours}$$

$$\text{Of TVs} = 9600 \text{ w hours}$$

2nd highest consumption is by bulb - II

72. (b); Required amount to be paid

$$= \frac{(5440+12320)31 \times 2.7}{1000} = \text{Rs. } 1486.512$$

73.(d); Required percentage = $\frac{12320}{16620} \times 100 = 74.13\%$

74.(d); Total consumption in 1 week = $\frac{52980 \times 7}{1000} = 370.86$ kwh

75.(a) units consumed by cooler (200w)

$$= \frac{9000 \times 30}{1000} = 270 \text{ units}$$

Units consumed by cooler (175 w)

$$= \frac{7875 \times 30}{1000} = 236.25 \text{ units}$$

$$\text{Required difference} = 270 - 236.25 = 33.75 \text{ units}$$



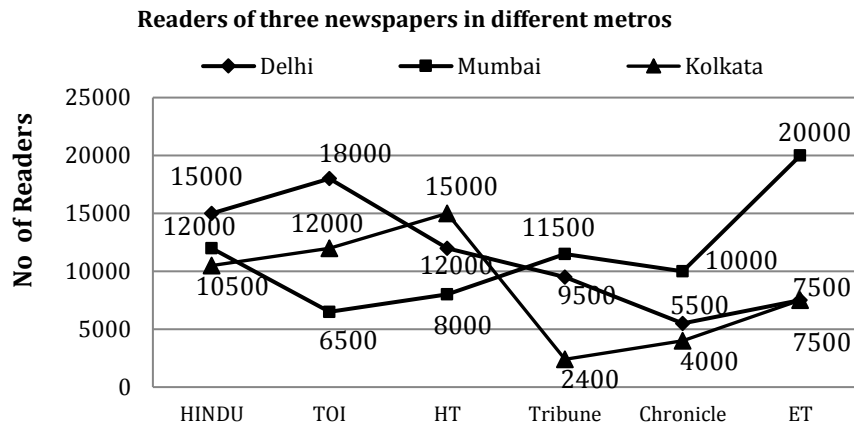
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PRACTICE SET (LEVEL-II)

Directions (1-5); Read the following graph and answer the questions that follow.



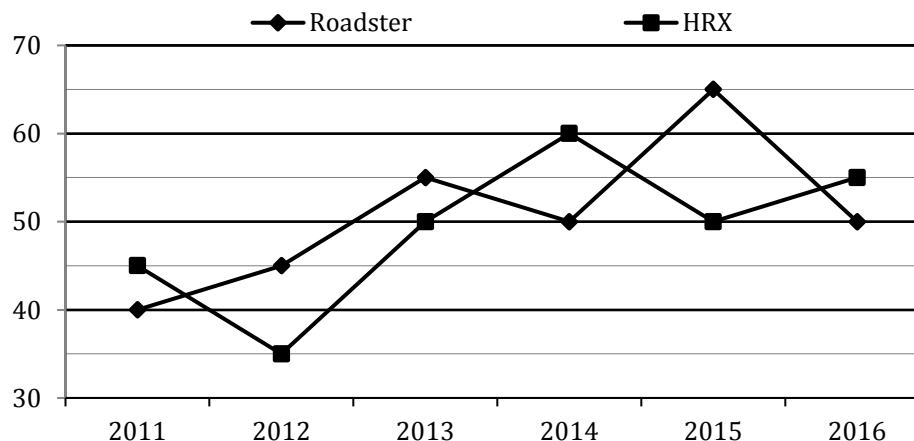
- What is the difference between TOI reader in Mumbai and average of Tribune readers in all three Metros?
(a) 1350 (b) 1300 (c) 1750 (d) 1450 (e) None of these
- Total readers in Delhi except ET are approximately what percent of total readers in Mumbai except Chronicle and ET together?
(a) 140% (b) 150% (c) 165% (d) 160% (e) 170%
- How many readers are there in all three Metros for all newspaper?
(a) 186900 (b) 168900 (c) 189600 (d) 188600 (e) None of these
- What is the ratio between one third of readers of Mumbai and three times the Hindu Reader in all three metro together?
(a) 7 : 11 (b) 17 : 31 (c) 13 : 77 (d) 4 : 25 (e) 136 : 675
- If total no of readers in Chennai are 30% less than in Kolkata but no of Tribune readers in Chennai are 25% more than the ET reader in Delhi, What is the total no of readers except Tribune in all four metro city now?
(a) 190015 (b) 109105 (c) 190105 (d) 180100 (e) 198105

Directions (Q.6-10): Study the following graph carefully and answer the questions that follow.

Profit = Income – Expenditure

Profit % is defined as percentage of total income

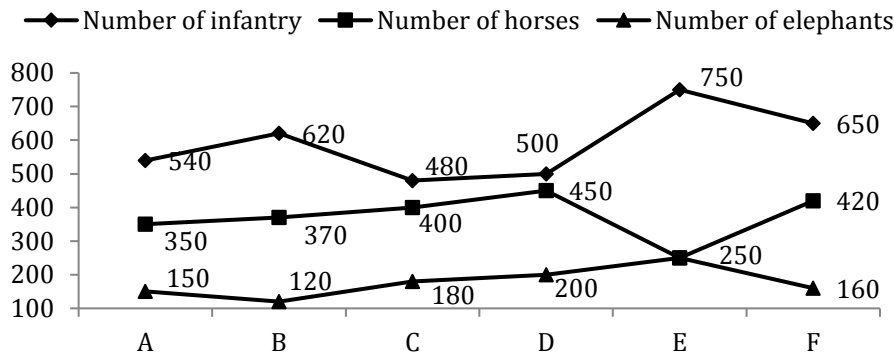
Line graph given below shows the percentage profit of two companies in six different years.



- In year 2012, Roadster earned a profit of Rs. 2,16,000 while HRX's income is 10% less as compared to Roadster in the same year. Find the difference (in Rs.) between their expenditures.
(a) 18600 (b) 16800 (c) 15500 (d) 14800 (e) 16400
- The company HRX made a total expenditure of 2,50,000 in year 2013. If its income in year 2015 remains same as that in 2013, find its profit (in Rs.) in 2015.
(a) 2,45,000 (b) 2,00,000 (c) 2,50,000 (d) 2,25,000 (e) 3,50,000
- In year 2016, both companies have same income. Find the ratio between expenditure of company HRX to that of company Roadster.
(a) 10 : 9 (b) 9 : 11 (c) 9 : 10 (d) 5 : 7 (e) 8 : 9

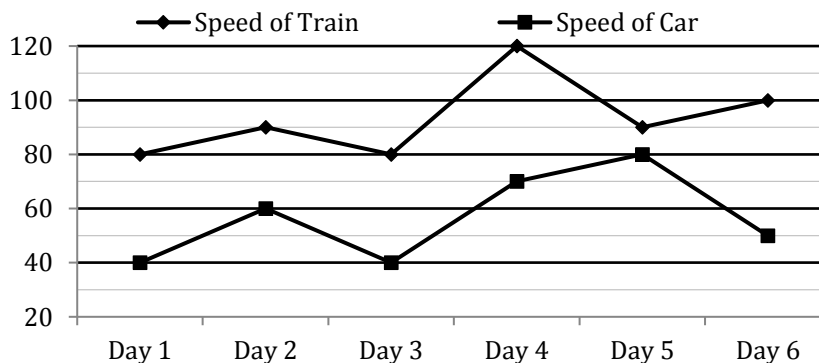
9. The company HRX has the second lowest percentage change in the profit percent over the previous year in which of the following year ?
 (a) 2016 (b) 2015 (c) 2012 (d) 2014 (e) None of these
10. The company Roadster shows its expenditure in 2013 as Rs. 90,000 and declares that income in 2013 is equal to the expenditure in 2014. Find the percentage change in income for year 2014.
 (a) 10% (b) 200% (c) 50% (d) 100% (e) None of these

Directions (11-15): Study the following line graph carefully to answer the questions based on it. The graph shows the number of infantry, horses and elephants in each army unit of six kings in a certain year. Horses and elephants are being ridden by 1 and 4 soldiers respectively.



11. If $16\frac{2}{3}\%$ of the infantry of one army unit of king C is transferred to infantry of one army unit of king B. Then new infantry of one army unit of king C is what percent of the new infantry of one army unit of king B?
 (a) $54\frac{3}{7}\%$ (b) $51\frac{2}{3}\%$ (c) $54\frac{2}{7}\%$ (d) 51% (e) 61%
12. In a war, $83\frac{1}{3}\%$ of total number of one army unit of king F died. If the ratio of alive infantry, horses and elephants (with soldiers) is 6 : 5 : 2, then find the number of infantry left alive.
 (a) 95 (b) 105 (c) 86 (d) 90 (e) 80
13. By what percent approximately the total number of soldiers in one army unit of king A is more or less than that of King E ?
 (a) 25% (b) 25.5% (c) 23.5% (d) 24% (e) 26%
14. King D gave his 1 army unit to king A as dowry. King A sent that army unit along with his one army unit to king E. King E divided the infantry, horses and elephants of the two gifted units equally into his 5 army units. Find number of elephants in his new army unit. (Given that all unit has same number of three elements)
 (a) 325 (b) 300 (c) 290 (d) 320 (e) 420
15. What is the ratio of average number of soldiers of 1 army unit of King F to the average number of infantry, horses and elephants of 1 army unit of king B ?
 (a) 5 : 3 (b) 7 : 5 (c) 11 : 7 (d) 13 : 9 (e) 57 : 37

Direction (16-20): Refer the following line graph and answer the questions based on it. The line graph shows the speeds (in km/h) of car and train on six days in a week.

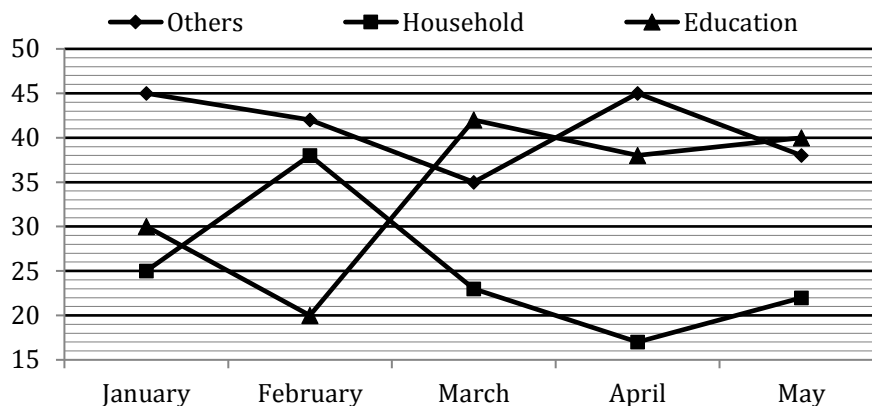


16. What is the average speed of the train on Day 1, Day 3 and Day 4 together if the train travelled same distance on each of these days?
 (a) 95 km/h (b) 90 km/h (c) 85 km/h (d) 100 km/h (e) None of these
17. On day 5, the train covered a distance of 300 km while the car covered 240 km. What is the ratio of time taken by the car to the time taken by the train on that day?
 (a) 6 : 7 (b) 7 : 6 (c) 10 : 9 (d) 9 : 10 (e) 8 : 9

18. If the time taken by the car is thrice to the time taken by the train on day 3, then the distance covered by the car is how much percent more than the distance covered by the train on that day?
 (a) 50% (b) 40% (c) 60% (d) 30% (e) 55%
19. For which day, the percentage increase/decrease in the speed of train from the previous day is the maximum?
 (a) Day 4 and 5 (b) Day 5 (c) Day 4 (d) Day 2 and 4 (e) None of these
20. If both the train and the car travelled for 3 hours 20 minutes each on day 2, then what is the difference of the distances travelled by the train and the car on that day?
 (a) 60 km (b) 120 km (c) 90 km (d) 100 km (e) Cannot be determined

Direction (21-25): - Study the following line graph and answer the questions that follow.

The graph shows the percentage distribution of money spent by Avanish on Education, Household and Others in five different months. He spent only 60% of his monthly salary.



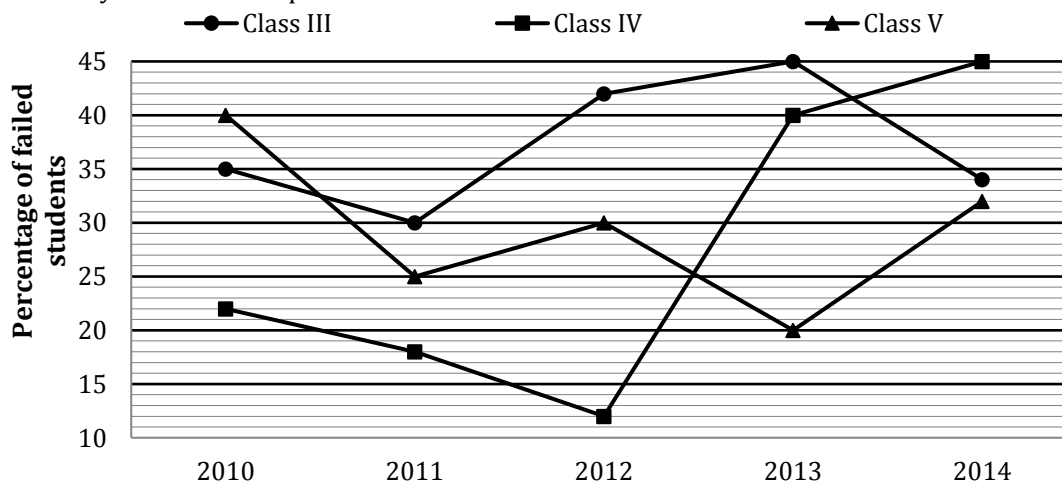
Note: Monthly income throughout the year remains constant.

MONTHLY SALARY:- 50,000

21. In June, Avanish spent 25% less amount on others as compared to what he spent on others in May. Find the percentage contribution of money spent on others in June in monthly salary of June.
 (a) 20.5% (b) 22.5% (c) 17.1% (d) 23.2% (e) 34.6%
22. Avanish invested 10% of the money spent on others in January in a business which gives him 10% interest per annum on amount invested. Find the interest earned by him in one year?
 (a) Rs. 125.5 (b) Rs. 152 (c) Rs. 150 (d) Rs. 135 (e) Rs. 168
23. What is the difference in amount saved and amount spent on education by Avanish in April?
 (a) Rs. 12100 (b) Rs. 8600 (c) Rs. 7200 (d) Rs. 8500 (e) Rs. 7850
24. Find the total expenditure made in February except education?
 (a) Rs. 24,000 (b) Rs. 24,400 (c) Rs. 22,400 (d) Rs. 20,400 (e) None of these
25. Find the ratio of money spent on Education in February to monthly salary in February?
 (a) 3 : 22 (b) 4 : 27 (c) 5 : 29 (d) 8 : 35 (e) 3 : 25

Directions (Q.26-30): The following graph shows the percentage of failed students in three different classes of a school in five different years.

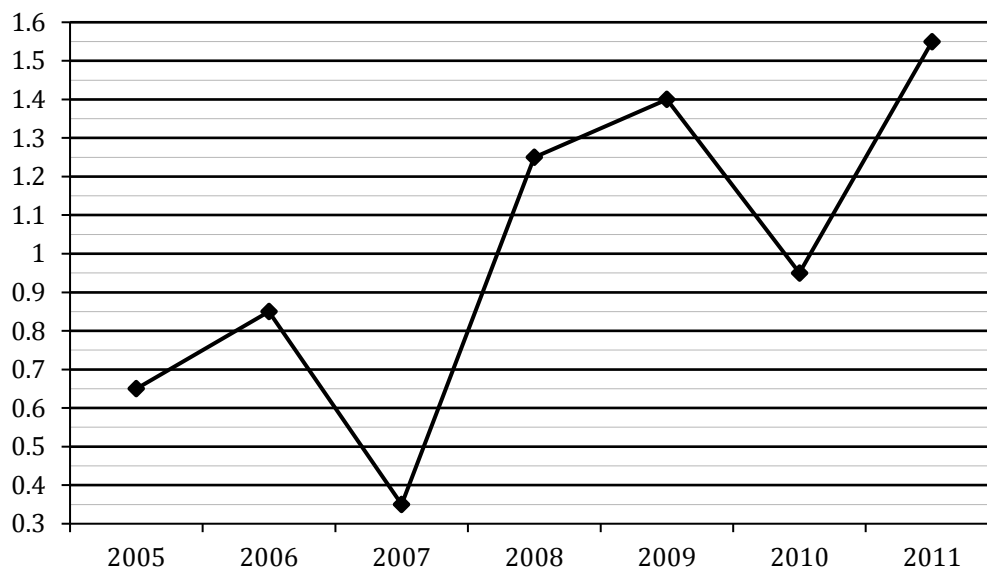
Study the graph carefully to answer the questions based.



26. In 2012, in class V, 140 students passed which is equal to 175% of failed students of same class in 2013. Find the ratio of failed students and passed students in 2012 & 2013 respectively in class V.
 (a) 5 : 16 (b) 3 : 16 (c) 4 : 17 (d) 3 : 17 (e) 16 : 3
27. If the total strength of students in 2010 in class IV was 150 and it increases every year by 50, then find the number of passed students in 2013.
 (a) 182 (b) 160 (c) 180 (d) 170 (e) 200
28. The ratio of girls to boys who failed in class III in 2014 was 7 : 10. One-seventh i.e. 5 of these girls passed when their answer sheet were re-evaluated which made the number of passed girls i.e. 84 in 2014, 20% more than those in 2010 in same class. Find the number of passed boys in 2010 in class III. Total students in class III in 2014 and 2010 are in ratio 5 : 4.
 (a) 60 (b) 65 (c) 70 (d) 62 (e) 78
29. In 2012, 105 students failed in class III, while 198 students passed in class IV. The number of passed boys is 17 more than the passed girls in class III and the number of failed girls is 15 less than the failed boys in class IV. Find the difference in failed boys of class IV and passed girls of class III.
 (a) 45 (b) 50 (c) 52 (d) 43 (e) 67
30. Find the average of percentage of passed students of class IV throughout the given years
 (a) 76.2% (b) 70.8% (c) 72.6% (d) 70.2% (e) 56%

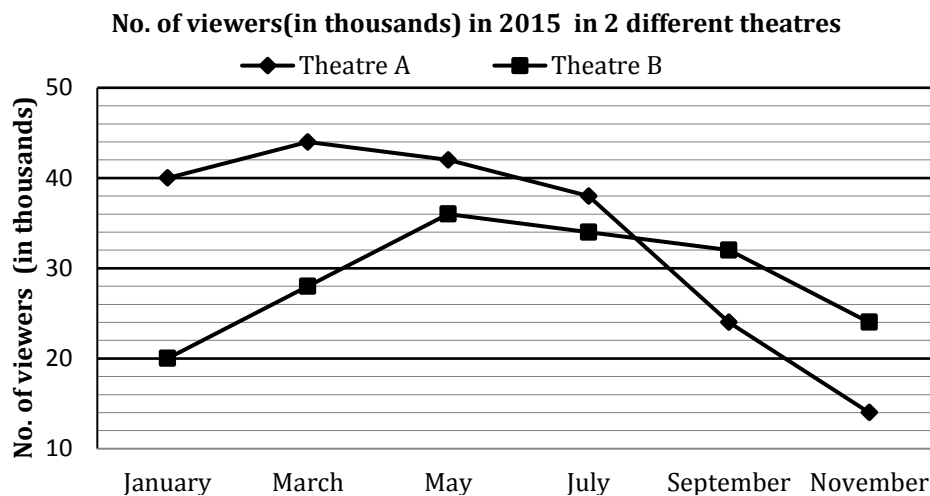
Directions (31-35): The following line graph gives the ratio of the amounts of imports by a company to the amount of exports from that company over the period from 2005 to 2011.

Ratio of value of imports to exports by a company over the years



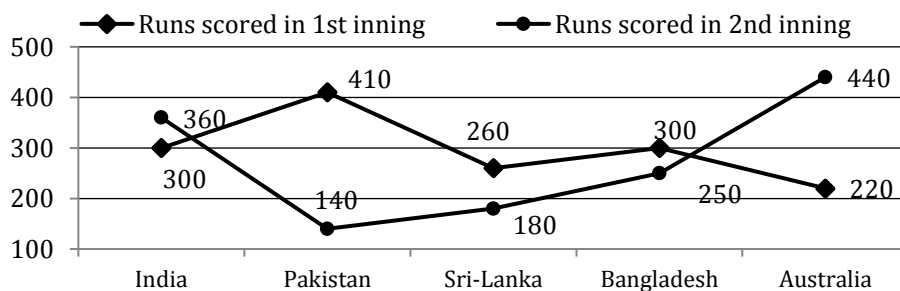
31. If the imports in 2008 was Rs. 250 crore and the total exports in the years 2008 and 2009 together was Rs. 500 crore, then the imports in 2009 was
 (a) Rs. 250 crore (b) Rs. 300 crore (c) Rs. 357 crore (d) Rs. 420 crore (e) Rs. 480 crore
32. What was the percentage increase in imports from 2007 to 2008 ?
 (a) 72% (b) 56% (c) 28% (d) Data inadequate (e) None of these
33. If the imports in 2010 are 40% of the export in 2009 then total imports and exports in 2010 is what percent of the total imports and exports in 2009 ? (calculate up to two decimal points)
 (a) 34.21% (b) 39.62% (c) 36.26% (d) 39.92% (e) 42.12%
34. IF TOTAL IMPORTS AND EXPORTS IN 2011 IS 255 CR. AND THE TOTAL IMPORTS IN 2005 IS 35% LESS THAN THE EXPORTS IN 2011 THEN FIND THE TOTAL IMPORTS AND EXPORTS IN 2005 ?
 (a) 155 cr (b) 165 cr (c) 320 cr (d) 210 cr (e) 175 cr
35. If the export in year 2011 is 400 crore. The import in year 2012 is 50% of the import of the year 2011 and export in 2012 is $\frac{2}{5}$ th of the export of 2011. Then find the ratio of import to export in year 2012 ?
 (a) 3 : 2 (b) 7 : 5 (c) 31 : 17 (d) 16 : 31 (e) 31 : 16

Directions (36-40): Study the following line graph carefully and answer the following questions.



36. If out of the total number of viewers from both of the theatres in January, the ratio of male to female is 7 : 5 and out of the total number of viewers from both of the theaters in November, the ratio of male to female is 5: 3 then male viewers from both of the theatres in January are approximately what percentage of the female viewers from both of the theaters in November ?
 (a) 200% (b) 246% (c) 150% (d) 220% (e) 225%
37. Find the ratio between the average number of viewers from January and July from theater A to the average number of viewer from July, September and November from theatre B ?
 (a) 7 : 5 (b) 5 : 7 (c) 10 : 13 (d) 13 : 10 (e) 12 : 11
38. If number of viewers of theatre A in January 2016 increases by 20% and of theatre B by 10% as compared to the corresponding no. of viewers of these theatres in January in 2015. Then find the difference between no. of viewers of theatre A and theatre B in January 2016.
 (a) 20000 (b) 22000 (c) 25000 (d) 26000 (e) 24500
39. The number of viewers of theatre B in October is equal to average number of the viewers of same theatre in September and November. Also the viewers of theatre A in October is $\frac{5}{7}$ of the viewers of theatre B in the same month. Find the number of viewers of theatre A in October.
 (a) 24000 (b) 22000 (c) 25000 (d) 20000 (e) 48000
40. The total number of viewers in March 2016 increased by 40% as compared to that in March 2015. If the viewers of theatre A in March 2016 are 25% more than that in 2015. Then find the difference between number of viewers of theatre B in March 2016 and in March 2015.
 (a) 15800 (b) 19800 (c) 17800 (d) 18800 (e) 18700

Directions (41-45): Given below is the line graph which shows the runs scored by five teams in first and second innings in different test matches. Read the data and solve the questions.

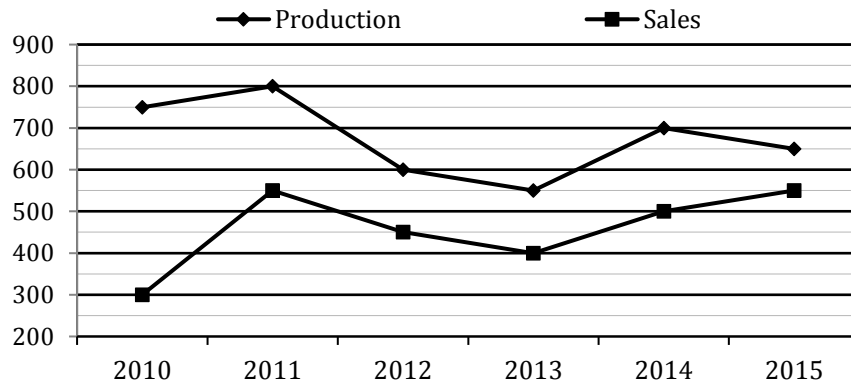


41. What is the average runs scored by Sri-Lanka, Bangladesh and Australia in 2nd innings?
 (a) 260 (b) 270 (c) 280 (d) 290 (e) 300
42. If there is 10% and 20% increase in runs scored in 1st and 2nd innings respectively by both teams, India and Pakistan, then what will be the ratio of total runs scored by India to the total runs scored by Pakistan?
 (a) 762 : 619 (b) 756 : 646 (c) 756 : 619 (d) 762 : 646 (e) 647 : 761
43. In a match between Bangladesh and New-Zealand, if New-Zealand scored 30% more runs in 1st inning and 10% less runs in 2nd inning as compared to Bangladesh then how many more runs did New-Zealand score with respect to Bangladesh?
 (a) 45 (b) 55 (c) 65 (d) 75 (e) 85

44. If Sri Lanka has to score runs equal to the runs scored by India in both innings then by what percent Sri Lanka has to increase their total score of both innings?
 (a) $\frac{100}{3}\%$ (b) 50% (c) $66\frac{2}{3}\%$ (d) $55\frac{5}{9}\%$ (e) 55%
45. If there is a match between Pakistan & Bangladesh and Pakistan & Bangladesh both score 10% & 30%, more runs in 1st and 2nd innings respectively, then which team will score more and how many more runs than the other team?
 (a) Pakistan, 22 runs (b) Pakistan, 25 runs (c) Match Draw
 (d) Bangladesh, 25 runs (e) Bangladesh, 22 runs

Directions (46-50): Study the following graph carefully and answer the following question

The graph below represents the production (in tonnes) and sales (in tonnes) of a company X from 2010-2015

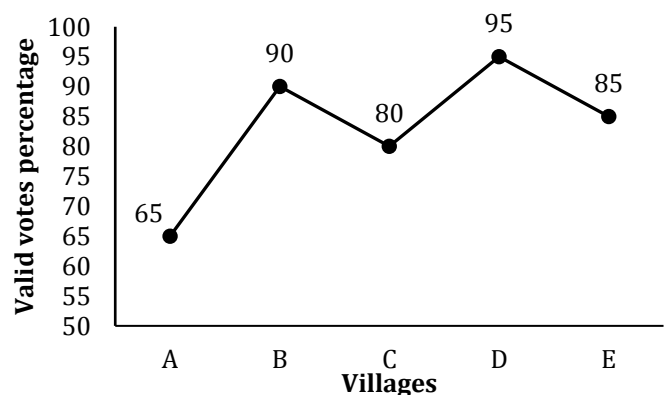


46. If production of company X and another company Y is in the ratio 14 : 13 in year 2014, then production of company Y in 2014 is what percent more or less than production of company X in 2010?
 (a) $13\frac{1}{3}\%$ (b) $33\frac{1}{3}\%$ (c) $66\frac{2}{3}\%$ (d) $16\frac{2}{3}\%$ (e) $6\frac{2}{3}\%$
47. If production of company X in 2016 is 120% of its production in 2015 then what is the ratio of sales company X in 2010 to the production of company X in 2016?
 (a) $\frac{7}{9}$ (b) $\frac{13}{20}$ (c) $\frac{20}{13}$ (d) $\frac{5}{13}$ (e) $\frac{7}{13}$
48. If production cost is Rs.1,500 per tonne and sale is at the rate of Rs.2,800 per tonne over all years then what is the ratio of profit or loss of company X in 2013 to the profit or loss of the company in year 2014? (Profit = Income through sales - Production cost)
 (a) $\frac{59}{70}$ (b) $\frac{20}{23}$ (c) $\frac{53}{94}$ (d) $\frac{27}{38}$ (e) $\frac{37}{47}$
49. If production cost in year 2013 is Rs.150 per tonne and production cost increases by 10% every year after 2013 then what is the average production cost of company X over all years after year 2013?
 (a) 1,23,764.5 (b) 1,16,737.5 (c) 2,22,467 (d) 1,33,647 (e) 1,22,373.5
50. If 35% of production of company X in 2010 is added to the sale of company X in 2012 then total sale of company X in 2012 is what percent of the total sale of company X over all the years now? (approximately)
 (a) 14% (b) 18% (c) 35% (d) 28% (e) 24%

Directions (51-55): Given below are two line graphs, First line graph shows the percentage of voters who polled votes out of total voters from five different villages in the elections held in year 2016. Second line graph shows the percentage of valid votes polled out of total votes polled in these villages.

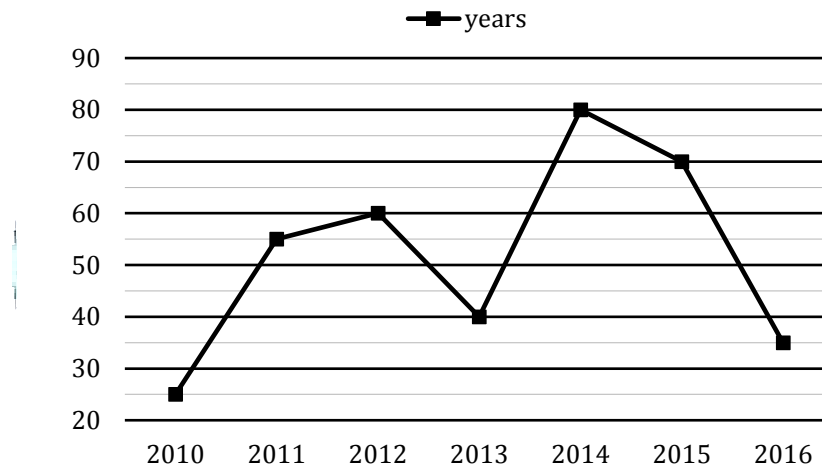
Note → Total voters = voters who polled votes + voters who did not poll votes

Total votes polled = valid votes polled + invalid votes polled



51. If difference between the total votes polled and valid votes polled from village C is 3740 and sum of total votes polled and invalid votes polled from village D is 9450 then find the total voters from village C and D together.
 (a) 35500 (b) 36000 (c) 33000 (d) 32000 (e) 34000
52. If ratio of total voters from village E to village B is 7 : 9 and ratio of males to female in the invalid votes of village E and village B is 4 : 3 and 3 : 2 respectively then females who cast invalid votes from village E are what percent of males, who cast invalid votes from village B? (approximately)
 (a) 96% (b) 88% (c) 80% (d) 75% (e) 82%
53. In village A, only two contestants participated in the election and winner got $52\frac{1}{2}\%$ of the total valid votes and won by 390 votes. Find the total voters in village A.
 (a) 13,500 (b) 12,000 (c) 15,000 (d) 14,000 (e) 10,000
54. In village D, if it was found that percentage of votes polled which are valid is 80% instead of 95% (as given in line graph) then total valid votes are decreased by 1350. If ratio of total voters from village D to village E is 6 : 7 then, find total valid votes polled in village E.
 (a) 11305 (b) 11200 (c) 10805 (d) 9500 (e) 10985
55. If total voters in village A is 15000 out of which 60% are males and ratio of males to females in total valid votes for same village is 9 : 13, then how many females are there who did not poll the votes?
 (a) 870 (b) 1420 (c) 1320
 (d) Can't be determined (e) 2200

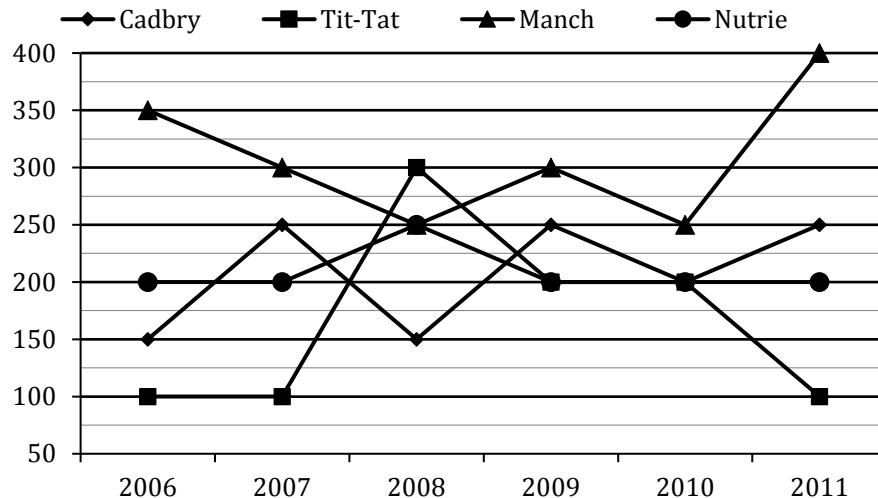
Directions (56 – 60): The following line graph shows the percentage of the number of candidates who qualified an examination out of the total number of candidates who appeared for the examination over period of seven years from 2010 to 2016.



56. If the number of qualified boys in year 2011 is 1210 and ratio of number of qualified boys and girls is 11 : 9, then find the total number of students who are not qualified in year 2011.
 (a) 2500 (b) 1800 (c) 1300 (d) 1900 (e) 1600
57. If the ratio of total number of students who appeared in year 2016 and 2017 is 7 : 9 and ratio of number of boys to girls who qualified in year 2017 is 4 : 5, then find the difference in number of unqualified boys and girls in year 2017. [Given that total number of students appeared in year 2016 is 1400].
 (a) 70 (b) 90 (c) 120
 (d) Cannot be determined (e) 180
58. If the total number of students in year 2013 is 40% more than the total number of students in year 2015, then the number of students who are not qualified in year 2013 is what percent more than number of students who are not qualified in year 2015?
 (a) 180% (b) 160% (c) 135% (d) 125% (e) 170%
59. If the number of qualified boys in year 2009 is 770 which is 35% of the total number of qualified students in year 2010 and ratio of total number of qualified students in year 2009 to that in year 2010 is 7 : 11, then find the ratio of total number of students in year 2009 to that in year 2010. (Number of qualified students in year 2009 is 20% of the total number of students appeared in that year).
 (a) 44 : 31 (b) 31 : 44 (c) 35 : 44 (d) 44 : 35 (e) 35 : 41
60. If total number of student appeared in year 2012 is 4200 which is 80% of the number of students appeared in year 2014, then find the total number of unqualified girls in year 2012 and unqualified boys in year 2014. [Given that number of unqualified boys are 50% more than number of unqualified girls for both years].
 (a) 1502 (b) 1202 (c) 1402 (d) 1302 (e) 1602

Directions(61-65) : Answer the questions on the basis of the information given below.

Sales (by volume) of chocolates by different companies (in lakh units)



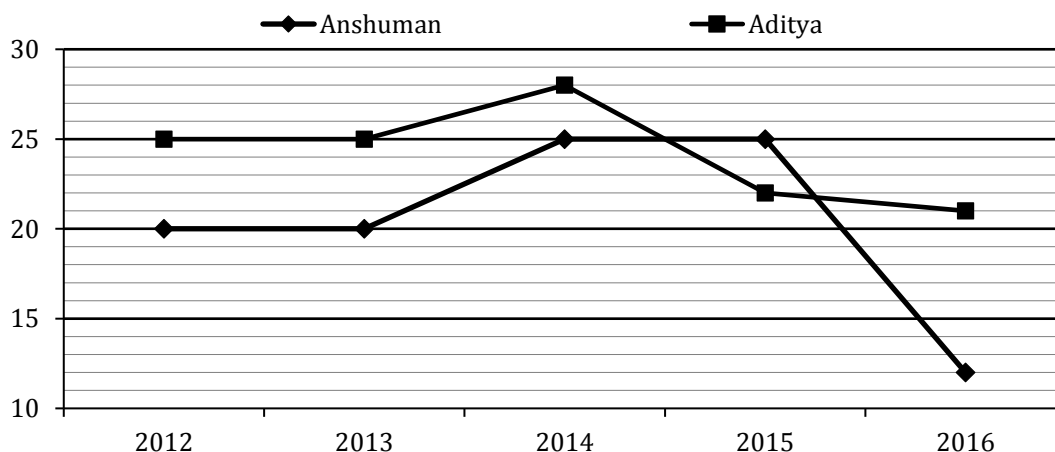
Revenue = Sales (by volume) × Selling price of each chocolate

Profit = Revenue – Expenditure

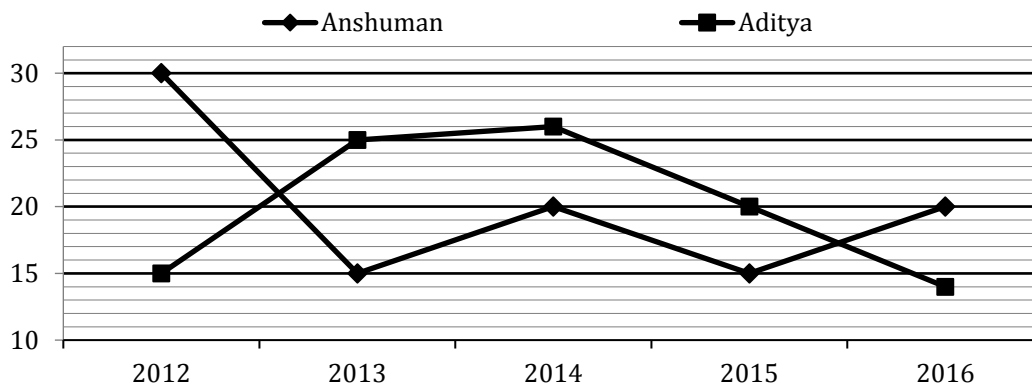
Profit percent = $\frac{\text{Revenue} - \text{Expenditure}}{\text{Expenditure}} \times 100$

61. The market share of a company is defined as the volume of the sales of the company as a percentage of the total sales volume of all the four given companies. In which year was the market share of Manch the highest?
 (a) 2011 (b) 2008 (c) 2006 (d) 2009 (e) 2010
62. In the year 2010, if the profit percent on selling each Cadbury chocolate is 25%, and the selling price of each Cadbury chocolate is Rs. 10, what was the expenditure incurred by Cadbury in making chocolates?
 (a) Rs. 12 crore (b) Rs. 14 crore (c) Rs. 16 crore (d) Rs. 10 crore (e) Rs. 8 crore
63. In the year 2009, the expenditures of Cadbury, Tit-Tat, Manch and Nutrie are in ratio 3 : 2 : 6 : 8. Which company had the highest profit percentage in 2009?
 (a) Cadbury (b) Manch (c) Nutrie
 (d) Cannot be determined (e) Tit-Tat
64. Which company had the highest growth rate for the period 2006 to 2010?
 (a) Tit-Tat (b) Cadbury (c) Nutrie (d) Manch (e) none of these
65. Total sales (by volume) of cadbury from 2007 to 2010 are what percent more/less than the total sales (by volume) of nutrie from 2008 to 2011?
 (a) 100% (b) 50% (c) 150% (d) 200% (e) 0%

Directions (66-70): First line graphs shows the percentage profit of 2 brothers from year 2012 to 2016 and the second line graph shows the percentage distribution of total income of each in different years. Study the data carefully and answer the following questions.



Total income of Anshuman = 1400 thousand and Aditya = 1600 thousand



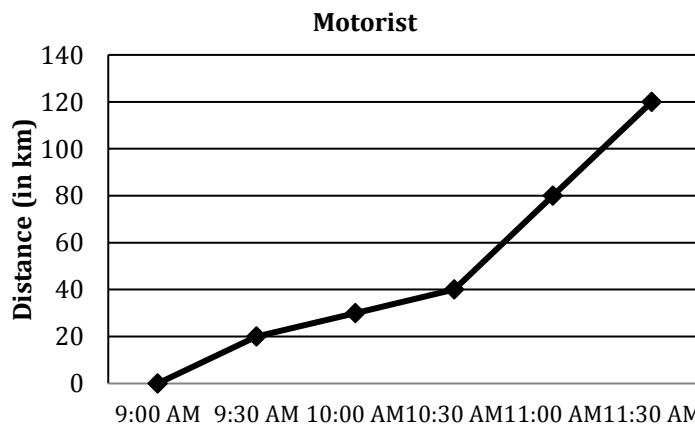
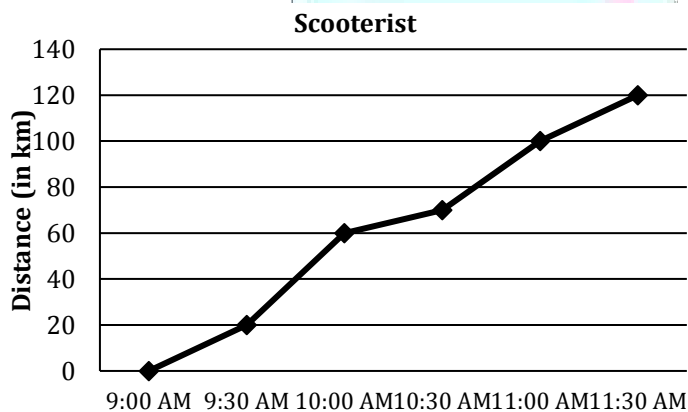
Note: $\% \text{profit} = \frac{\text{income} - \text{expenditure}}{\text{expenditure}} \times 100$

66. Find the difference (in thousand rupees) in the expenditures of both in 2014.
(a) 110 (b) 103 (c) 101 (d) 100 (e) 112
67. Expenditure of Aditya in 2013 is what percent more or less than that of Anshuman in 2016?
(a) 26% (b) 25% (c) 23% (d) 28% (e) 32%
68. Find the ratio of income of Anshuman in 2012 and expenditure of Aditya in 2014.
(a) 8 : 5 (b) 65 : 84 (c) 84 : 65 (d) 5 : 8 (e) None of these
69. Find the average expenditure (in thousand rupees) of Anshuman from 2012 to 2016.
(a) 233 (b) 232.4 (c) 235 (d) 333.4 (e) 342
70. If in 2011 the expenditure of Aditya was 50% of income of Anshuman in 2012 and his income in the year 2011 was 50% more than expenditure of Anshuman in 2015, then find profit or loss percentage of Aditya in 2011.
(a) 22% (b) 20% (c) 25% (d) 26% (e) 30%

Directions (71-75): Read the following line graphs carefully and answer the following questions :

A Scooterist and a motorist cover a distance of 120 km. They both start their journey at 9 : 00 AM. In the first line graph the distance time-slots of 30 minutes is given for the scooterist and the same is given in second graph for the motorist.

Note: Both scooterist and mototrist run with a constant speed in the given intervals of 30 minutes.



71. Average speed of the scooterist to cover first 110 km distance is what percent more/less than the average speed of the motorist to cover first 100 km distance ?
(a) 5% (b) 10% (c) 15% (d) 20% (e) 18%
72. At 10 : 30 am, a car starts chasing the scooterist with the speed of 140 km/hr from the point where scooterist started his journey , then find the time at which the car will catch the scooterist ?
(a) 11 : 18 am (b) 11 : 08 am (c) 11 : 15 am (d) 11 : 32 am (e) Can't be determined
73. If the speed of the scooterist and the motorist is increased by 10% and 20% respectively throughout the journey then find the difference in time taken by the scooterist and motorist to cover the given total distance ?
(a) $\frac{11}{23}$ hr (b) $\frac{7}{22}$ hr (c) $\frac{4}{11}$ hr (d) $\frac{25}{132}$ hr (e) None of these
74. At what time they will be 30 km apart from each other if they choose the same road to cover 120 km of distance ?
(a) 10 : 20 am (b) 11 : 15 am (c) 11 : 09 am (d) 10 : 00 am (e) None of these
75. Distance covered by scooterist in first two hours is what percent of the distance covered by motorist in same time?
(a) 35% (b) 25% (c) 40% (d) 20% (e) 45%

PRACTICE SET (LEVEL-II) SOLUTIONS

1. (b); TOI Mumbai : 6500

$$\text{Avg. of Tribune riders} = \frac{1}{3}[9500 + 11500 + 2400] = \frac{23400}{3} = 7800$$

$$\text{Difference} = 7800 - 6500 = 1300$$

2. (d); Delhi : (except ET) $15000 + 18000 + 12000 + 9500 + 5500 = 60000$
 Mumbai (except ET & Chronicle) : $12000 + 6500 + 8000 + 11500 = 38000$
 Desired % = $\frac{60000}{38000} \times 100 \approx 160\%$

3. (a); Delhi : $60000 + 7500 = 67500$
 Mumbai : $38000 + 30000 = 68000$
 Kolkata : $10500 + 12000 + 15000 + 2400 + 4000 + 7500 = 51400$
 Total users = $67500 + 68000 + 51400 = 186900$

4. (e); $\frac{1}{3}$ (Mumbai Reader) = $\frac{68000}{3}$
 3 (Hindu reader) = $3[15000 + 12000 + 10500] = (37500 \times 3)$
 Ratio = $\frac{68000}{37500 \times 9} = 136 : 675$

5. (c); Readers Chennai = $\frac{70}{100}(51400) = 35,980$
 Tribune in Chennai = $7500 \times \frac{5}{4} = \frac{37500}{4} = 9375$
 Total reader except Tribune in all 4 metro city now = $(35980 + 67500 + 68000 + 51400) - (9375 + 9500 + 11500 + 2400) = 222880 - 32775 = 190105$

6. (b); Income of Roadster in 2012 = $\frac{216000}{45} \times 100 = \text{Rs. } 4,80,000$
 Income of HRX in 2012 = $\frac{90}{100} \times 4,80,000 = \text{Rs. } 4,32,000$
 Expenditure of Roadster in 2012 = $(48 - \frac{45 \times 48}{100}) \times 10000 = \text{Rs. } 2,64,000$
 Expenditure of HRX in 2012 = $(432 - \frac{432 \times 35}{100}) \times 1000 = \text{Rs. } 2,80,800$
 $\therefore$ Required difference = Rs. 16,800

7. (c); Income of HRX in 2013 = $\frac{2,50,000 \times 100}{50} = \text{Rs. } 5,00,000$
 $\therefore$ Required profit = $\frac{50 \times 5,00,000}{100} = \text{Rs. } 2,50,000$

8. (c);

$$\frac{E_{\text{HRX}}}{E_{\text{Roadster}}} = \frac{\left(\frac{100-55}{100}\right) \times I}{\left(\frac{100-50}{100}\right) \times I} = \frac{9}{10}$$

9. (b); Percentage change in 2012 = $\frac{45-35}{45} \times 100 = 22.22\%$
 Percentage change in 2013 = $\frac{15}{35} \times 100 \approx 42.86\%$
 Percentage change in 2014 = $\frac{10}{50} \times 100 = 20\%$
 Percentage change in 2015 = $\frac{10}{60} \times 100 \approx 16.67\%$
 Percentage change in 2016 = $\frac{5}{50} \times 100 = 10\%$
 $\therefore$ 2nd lowest percentage change is in year 2015

10. (d); Income in year 2013 = $\frac{100 \times 90,000}{100-55} = \text{Rs. } 2,00,000$
 And Income in year 2014 = $\frac{100 \times 2,00,000}{100-50} = \text{Rs. } 4,00,000$

$$\therefore \text{Required percentage change} = \frac{2,00,000}{2,00,000} \times 100 = 100\%$$

11. (c); Required percentage = $\frac{400}{700} \times 100 = \frac{400}{7} = 54\frac{2}{7}\%$

12. (d); Number of soldiers left = $16\frac{2}{3}\%$ of $(650 + 420 + 160 \times 4) = 285$

Then, according to question,

$$6x + 5x + (2 \times 4)x = 285$$

$$\Rightarrow x = \frac{285}{19} = 15$$

$$\therefore \text{Required number of soldiers} = 15 \times 6 = 90$$

13. (b); Total soldiers of A = $540 + 350 + 150 \times 4 = 1490$

$$\text{Total soldiers of E} = 750 + 250 + 250 \times 4 = 2000$$

$$\therefore \text{Required percentage} = \frac{510}{2000} \times 100 = 25.5\%$$

14. (d); Required number of elephants = $250 + \frac{200+150}{5} = 320$

15. (e); Required ratio = $\frac{\frac{1}{3} \times (650 + 420 + 160 \times 4)}{\frac{1}{3} \times (620 + 370 + 120)}$
 $= \frac{1710}{1110} = \frac{57}{37}$

16. (b); Average Speed = $\frac{3}{\left(\frac{1}{80} + \frac{1}{80} + \frac{1}{120}\right)} = \frac{3 \times 240}{8} = 90 \text{ km/h}$

17. (d); Required Ratio = $\frac{240}{80} : \frac{300}{90} = 9 : 10$

18. (a); Let the time taken by the train be t hours
 Then, the time taken by the car = $3t$ hours
 Distance covered by the train = $80 \times t = 80t \text{ km}$
 Distance covered by the car = $40 \times 3t = 120t \text{ km}$
 Required Percentage = $\frac{(120t - 80t)}{80t} \times 100 = 50\%$

19. (c); Percentage Increase = $\frac{(\text{Today's Speed} - \text{Previous Day's Speed})}{\text{Previous Day's Speed}} \times 100$

$$\text{Percentage increase for day 2} = \frac{(90 - 80)}{80} \times 100 = 12.5\%$$

Similarly,

$$\text{Percentage decrease for day 3} = 11.11\%$$

$$\text{Percentage increase for day 4} = 50\%$$

$$\text{Percentage decrease for day 5} = 25\%$$

$$\text{Percentage increase for day 6} = 11.11\%$$

Hence, maximum percentage increase/decrease is for Day 4

20. (d); Required Difference = $(90 - 60) \times \frac{10}{3} = 100 \text{ km}$

$$\text{Money spent} = \frac{60}{100} \times 50000 = 30,000$$

21. (c); Money spent by Avanish on others in June

$$= \frac{75}{100} \times \frac{38}{100} \times 30000 = 8550$$

$$\therefore \% \text{ contribution} = \frac{8550}{50000} \times 100 = 17.1\%$$

22. (d); Money invested by Avanish = $\frac{10}{100} \times \frac{45}{100} \times 30,000 = \text{Rs. } 1350$

$$\therefore \text{Interest earned by him} = \frac{10}{100} \times 1350 = \text{Rs. } 135$$

23. (b); Amount saved in April = $\frac{40}{100} \times 50000 = \text{Rs. } 20,000$

$$\text{Amount spent on education} = \frac{38}{100} \times 30,000$$

$$= \text{Rs. } 11,400$$

$$\therefore \text{Required difference} = \text{Rs. } 8600$$

24. (a); Asked expenditure in February except education =

$$\frac{(100-20)}{100} \times 30000 = \text{Rs. } 24,000$$

25. (e); Required ratio = $\frac{\frac{20}{100} \times 30,000}{50,000} = \frac{3}{25}$

26. (b); Total students of class V in 2012 = $\frac{140}{70} \times 100 = 200$

$$\text{Failed students of class V in 2013} = 140 \times \frac{100}{175} = 80$$

$$\therefore \text{Required ratio} = \frac{\frac{30}{100} \times 200}{\frac{80}{100} \times \frac{80}{20} \times 100} = \frac{3}{16}$$

27. (c); Total strength in 2013 = $150 + 150 = 300$

$$\therefore \text{Number of passed students} = \frac{60}{100} \times 300 = 180$$

28. (a); Failed girls in 2014 = $7 \times 5 = 35$

$$\text{Failed boys in 2014} = 35 \times \frac{10}{7} = 50$$

$$\therefore \text{Total students in 2014} = 85 \times \frac{100}{34} = 250$$

$$\text{Total students in 2010} = 250 \times \frac{4}{5} = 200$$

$$\text{Number of passed girls in 2010} = 84 \times \frac{100}{120} = 70$$

$$\therefore \text{Number of passed boys in 2010} = \frac{65}{100} \times 200 - 70 = 60$$

29. (d); Passed students in class III = $\frac{105}{42} \times 58 = 145$

$$\text{Failed students in class IV} = \frac{198}{88} \times 12 = 27$$

$$\text{Passed girls in class III} = \frac{145 - 17}{2} = 64$$

$$\text{Failed boys in class IV} = \frac{27 + 15}{2} = 21$$

$$\therefore \text{Required difference} = 43$$

30. (c); Average = $\frac{1}{5} (78 + 82 + 88 + 60 + 55) = \frac{1}{5} \times 363 = 72.6\%$

31. (d); The ratio of imports to exports for the years 2008 and 2009 are 1.25 and 1.40, respectively.

Let the exports in the year 2008 = Rs. x crore

Then, the exports in the year 2009 = Rs. (500 - x) crore

$$\therefore 1.25 = \frac{250}{x} \Rightarrow x = \frac{250}{1.25} = 200 \text{ (using ratio for 2008)}$$

Thus, the exports in the year 2009

$$= \text{Rs. } (500 - 200) \text{ crore} = \text{Rs. } 300 \text{ crore}$$

Let the imports in the year 2009 = Rs. y crore

$$\text{Then, } 1.40 = \frac{y}{300} \Rightarrow y = (300 \times 1.40) = 420$$

$$\therefore \text{Imports in the year 2009} = \text{Rs. } 420 \text{ crore}$$

32. (d); The graph gives only the ratio of imports to exports for different years. To find the percentage increase in imports from 2007 to 2008, we require more details such as the value of imports or exports during these years.

Hence, the data is inadequate to answer this question.

33. (a); Let $I_{2009} = 7x$
and $I_{2010} = 19y$

$$E_{2009} = 5x$$

$$E_{2010} = 20y$$

$$19y = \frac{40}{100} \times 5x$$

$$2x = 19y$$

$$\boxed{x = \frac{19}{2}y}$$

$$\text{Total imports and exports in 2009} = 12x = 12 \times \frac{19y}{2} = 114y$$

$$\text{Total imports and exports in 2010} = 39y$$

$$\text{Required percentage} = \frac{39y}{114y} \times 100 = 34.21\%$$

$$34. (b); \frac{I_{2011}}{E_{2011}} = \frac{155}{100} = \frac{31}{20}$$

$$\therefore I_{2011} = \frac{31}{51} \times 255 = 155 \text{ cr}$$

$$E_{2011} = \frac{20}{51} \times 255 = 100 \text{ cr}$$

$$I_{2005} = 65 \text{ cr}$$

$$E_{2005} = \frac{65}{13} \times 20 = 100 \text{ cr}$$

$$\therefore \text{Imports and Exports in 2005} = 100 + 65 = 165 \text{ cr}$$

35. (e); Import of year 2011 = $\frac{400}{20} \times 31 = 620$ crore

$$\therefore \text{Import of year 2012} = 620 \times \frac{1}{2} = 310 \text{ crore}$$

$$\text{And export of year 2012} = 400 \times \frac{2}{5} = 160 \text{ crore}$$

$$\therefore \text{Required ratio} = \frac{310}{160} = 31 : 16$$

36. (b); Male viewers from both the theatres in January =

$$\frac{7}{12} \times 60,000 = 35,000$$

female viewers from both the theatres in November

$$= \frac{3}{8} \times 38,000 = 14,250$$

$$\therefore \text{Required percentage} = \frac{35,000}{14,250} \times 100 \approx 246\%$$

37. (d); Required Ratio = $\frac{(40+38)}{2} : \frac{(34+32+24)}{3}$
= $39 : 30 = 13 : 10$

38. (d); Required difference = $48,000 - 22,000 = 26,000$

39. (d); No. of viewers of theatre A in October = $\frac{5}{7} \times$

$$\left(\frac{32+24}{2}\right) = 20 \text{ thousand}$$

40. (c); Total viewers in March 2016 = $\frac{140}{100} \times 72,000 = 100,800$

$$\text{Viewers of theatre A in March 2016} = \frac{125}{100} \times 44,000 = 55,000$$

$$\text{Viewers of theatre B in March 2016} = 100,800 - 55,000 = 45,800$$

$$\text{Required difference} = 45,800 - 28,000 = 17,800$$

41. (d); Average = $\frac{180+250+440}{3} = \frac{870}{3} = 290$

42. (a); India's run after increment = $300 \times \frac{110}{100} + 360 \times$

$$\frac{120}{100} = 762$$

$$\text{Pakistan's Run after increment} = 410 \times \frac{110}{100} +$$

$$140 \times \frac{120}{100} = 619$$

$$\text{Ratio} = \frac{762}{619}$$

43. (c); New-Zealand Scored in

$$1\text{st inning} = 300 \times \frac{130}{100} = 390$$

$$2\text{nd inning} = 250 \times \frac{90}{100} = 225$$

$$\text{Total score of New-Zealand} = 615$$

$$\text{Total score of Bangladesh} = 550$$

New-Zealand scored 65 more runs than Bangladesh.

44. (b); India's score in both innings = 660

$$\text{Sri-Lanka's score in both innings} = 440$$

'220' more runs required.

$$\% \text{ of runs needed by Sri-Lanka} = \frac{220}{440} \times 100 = 50\%$$

more runs needed

45. (e); Pakistan's score = $410 \times \frac{110}{100} + 140 \times \frac{130}{100} = 633$

$$\text{Bangladesh's score} = 300 \times \frac{110}{100} + 250 \times \frac{130}{100} = 655$$

Bangladesh will score 22 more runs.

46. (a); Production of company Y in 2014 = $\frac{700}{14} \times 13 = 650$ tonnes

$$\begin{aligned}\text{Required percentage} &= \frac{100}{750} \times 100 \\ &= \frac{40}{3} \% = 13\frac{1}{3} \% \text{ less}\end{aligned}$$

47. (d); Production of company X in 2016 = $\frac{120}{100} \times 650 = 780$ tonnes

$$\text{Required ratio} = \frac{300}{780} = \frac{5}{13}$$

48. (a); Cost of production in 2013 = $1500 \times 550 = \text{Rs.}8,25,000$

Total Income through sales in 2013 = $2800 \times 400 = \text{Rs.}11,20,000$

Profit in 2013 = $11,20,000 - 8,25,000 = \text{Rs.}2,95,000$

Cost of production in 2014 = $\text{Rs.}1500 \times 700 = \text{Rs.}10,50,000$

Total Income through sales in 2014 = $\text{Rs.}2800 \times 500 = \text{Rs.}14,00,000$

Profit in 2014 = $\text{Rs.}3,50,000$

$$\text{Required ratio} = \frac{295}{350} = \frac{59}{70}$$

49. (b); Total production cost in 2014 and 2015 = $165 \times 700 + 181.5 \times 650$
 $= 1,15,500 + 1,17,975$
 $= \text{Rs.}2,33,475$

$$\text{Required average} = \frac{2,33,475}{2} = 1,16,737.5$$

50. (e); Total sale of company X in 2012 = $450 + \frac{35}{100} \times 750 = 712.5$ tonnes

$$\begin{aligned}\text{Required percentage} &= \frac{712.5}{\frac{300+550+450+400+500+550+262.5}{712.5}} \times 100 \\ &= \frac{712.5}{3012.5} \times 100 = 23.65\% \sim 24\%\end{aligned}$$

51. (e); Let total voter of village C = x

And total voter of village D = y

$$\frac{85}{100}x - \frac{85}{100}x \times \frac{80}{100} = 3740$$

$$\frac{17}{20}x \left(1 - \frac{4}{5}\right) = 3740$$

$$x = 22000$$

$$\frac{75}{100}y + \frac{75}{100}y \times \frac{5}{100}y = 9450$$

$$\frac{3}{4}y \left(1 + \frac{1}{20}\right) = 9450$$

$$y = 12000$$

$$\text{Required sum} = 12000 + 22000 = 34000$$

52. (b); Let total voters from village E = $70000x$

And total voters from village B = $90000x$

Votes polled in E = $66,500x$

Invalid votes in E = $9975x$

Invalid votes cast by females in village E

$$= \frac{3}{7} \times 9975x = 4275x$$

Votes polled in B = $900x \times 90 = 81000x$

Invalid votes in B = $810x \times 10 = 8100x$

Males who cast invalid votes in B

$$= \frac{3}{5} \times 8100x = 4860x$$

Required percentage

$$= \frac{4275x}{4860x} \times 100 \approx 88\%$$

53. (c); Let total voters in A = $10000x$

According to question

Total valid votes in A = $5200x$

$$\frac{5}{100} \times 5200x = 390$$

$$260x = 390$$

$$x = \frac{3}{2}$$

total voters = 15000

54. (a); Let total votes polled = x

So

$$\frac{15}{100}x = 1350$$

$$x = 9000$$

Total voters from village D

$$= 9000 \times \frac{100}{75}$$

$$= 12000$$

Total voters in E

$$= \frac{12000}{6} \times 7 = 14000$$

Total valid votes polled in E

$$= 140 \times 95 \times \frac{85}{100} = 11,305$$

55. (d); Since we cannot determine number of males and females who vote so value cannot be determined

56. (b); Total number of qualified student in year 2011 =

$$\frac{1210}{11} \times 20 = 2200$$

Number of student who are not qualified in year

$$2011 = \frac{45}{55} \times 2200 = 1800$$

57. (d); Cannot be determined since percentage of qualified candidates and ratio of unqualified boys to girls from the year 2017 is not given.

58. (a); Let total number of student in year 2015 be 100

Then total number of student in year 2013 is 140

$$\text{Required percentage} = \frac{140 \times 60 - 30 \times 100}{30 \times 100} \times 100 = 180\%$$

59. (c); Total number of qualified student in year 2010 =

$$\frac{770}{35} \times 100 = 2200$$

Total number of qualified student in year 2009 =

$$\frac{2200}{11} \times 7 = 1400$$

$$\text{Total number of student in year 2009} = \frac{1400}{20} \times 100 = 7000$$

$$\text{Total number of student in year 2010} = \frac{2200}{25} \times 100 = 8800$$

$$\therefore \text{Required ratio} = \frac{7000}{8800} = 35 : 44$$

60. (d); Total number of appeared student in year 2014 =

$$\frac{4200}{80} \times 100 = 5250$$

Total number of unqualified student in year 2012 =

$$\frac{40}{100} \times 4200 = 1680$$

Total number of unqualified student in year 2014 =

$$\frac{20}{100} \times 5250 = 1050$$

Let the number of unqualified girls in year 2012 are x

$$\therefore 2.5x = 1680$$

$$x = 672$$

And,

Let the number of unqualified girls in year 2014 is y

$$\therefore 2.5y = 1050$$

$$y = 420$$

$\therefore$ Required total

$$= 672 + (1050 - 420)$$

$$= 672 + 630 = 1302$$

61. (c); By observation we can say that the sales volume of Manch is more in 2006 compared to 2007 and also 2010 and total sales volume is less in 2006 compared to 2007 and 2010. So the market share of Manch is not the highest for years 2007 and 2010. In the same way the market share of Manch is not the highest in 2009.

$$\text{In the year 2006, market share of Manch} = \frac{350}{800} = \frac{7}{16}$$

$$\text{In the year 2011, market share of Manch} = \frac{400}{950} = \frac{8}{19}$$

Since $\frac{7}{16}$ is more than $\frac{8}{19}$, then market share is the highest in 2006.

62. (c); Given that profit percent

$$= \frac{\text{Sales revenue} - \text{Expenditure}}{\text{Expenditure}} \times 100$$

$$25 = \frac{(10 \times 200) \times 10^5 - \text{expenditure}}{\text{expenditure}} \times 100$$

$$\therefore \text{expenditure} = \text{Rs. } 16 \times 10^7 = \text{Rs. } 16 \text{ crore.}$$

63. (d); We need the sales revenue and expenditure. Now we do not know the sales revenue as selling prices of the chocolates are not known. We cannot answer the question.

64. (a); Before doing the calculation to check if there is any possibility to answer the question by observation. By observation we can say that the sales of Tit-Tat in year 2010 are two times the sales of 2006, but for other companies, it is less than double. So the average annual growth rate is the highest for Tit-Tat from year 2006 to 2010.

$$\text{65. (e); Required percent} = \frac{850 - 850}{850} \times 100 = 0\%$$

$$\text{66. (c); Expenditure of Anshuman} = \frac{280}{1.25} = 224 \text{ thousands}$$

$$\text{Expenditure of Aditya} = \frac{416}{1.28} = 325 \text{ thousands}$$

$$\therefore \text{Required difference} = 325 - 224 = 101 \text{ thousands}$$

$$\text{67. (d); Expenditure of Aditya in 2013} = \frac{400}{1.25} = 320 \text{ thousands}$$

$$\text{Expenditure of Anshuman in 2016} = \frac{280}{1.12} = 250 \text{ thousands}$$

$$\therefore \text{Required percentage} = \frac{320 - 250}{250} \times 100 = 28\%$$

$$\text{68. (c); Required ratio} = \frac{\frac{30}{100} \times 1400}{\frac{416}{1.28}} = \frac{420}{325} = \frac{84}{65}$$

$$\text{69. (e); Expenditure of Anshuman in 2012} = \frac{420}{1.20}$$

$$= 350 \text{ thousand}$$

$$\text{Expenditure of Anshuman in 2013} = \frac{210}{1.20}$$

$$= 175 \text{ thousand}$$

$$\text{Expenditure of Anshuman in 2014} = \frac{280}{1.25}$$

$$= 224 \text{ thousand}$$

$$\text{Expenditure of Anshuman in 2015} = \frac{210}{1.25}$$

$$= 168 \text{ thousand}$$

$$\text{Expenditure of Anshuman in 2016} = \frac{280}{1.12}$$

$$= 250 \text{ thousand}$$

$$\therefore \text{Required average} = \frac{1}{5} \times 1167 = 233.4 \text{ thousand}$$

$$\text{70. (b); In 2011, expenditure of Aditya} = \frac{1}{2} \times 420 = 210 \text{ thousand}$$

$$\text{In 2011, income of Aditya} = 1.5 \times \frac{210}{1.25} = 252 \text{ thousand}$$

$$\therefore \text{Required profit \%} = \frac{252 - 210}{210} \times 100 = 20\%$$

$$\text{71. (b); Avg. speed of scooterist} = \frac{110}{2 + \frac{1}{4}} = \frac{110 \times 4}{9}$$

$$= \frac{440}{9} \text{ km/hr}$$

$$\text{Avg. speed of motorist} = \frac{100}{2 + \frac{1}{4}} = \frac{400}{9}$$

$$\text{Required \%} = \frac{\frac{440}{9} - \frac{400}{9}}{\frac{400}{9}} \times 100 = \frac{40}{400} \times 100 = 10\%$$

72. (a); At 10 : 30 distance covered by scooterist = 70 km
At 11 : 00 am distance covered by scooterist = 100 km

$$\text{At 11 : 00 am distance covered by car} = \frac{140}{2} = 70 \text{ km}$$

$$\text{Relative speed} = 140 - 40 = 100 \text{ km/hr}$$

$$\text{Required time} = \frac{(100 - 70)}{100} = \frac{30}{100} = \frac{3}{10} \text{ hr} = 18 \text{ min}$$

$\therefore$ At 11 : 18 am the car will catch the scooterist

73. (d); Time taken by scooterist to cover the total distance with increased speed = $\frac{120}{\frac{11}{10} \times 48} = \frac{1200}{11 \times 48}$

Time taken by motorist to cover the total distance

$$\text{with increased speed} = \frac{120}{\frac{12}{10} \times 48} = \frac{1200}{12 \times 48}$$

$$\text{Difference in time} = \frac{1200}{48} \left(\frac{1}{11} - \frac{1}{12} \right) = \frac{1200}{48 \times 132} = \frac{25}{132} \text{ hr.}$$

74. (e); There will be more than one possibility to be 30 km apart.

$$\text{75. (b); Required percentage} = \frac{100 - 80}{80} \times 100 = 25\%$$

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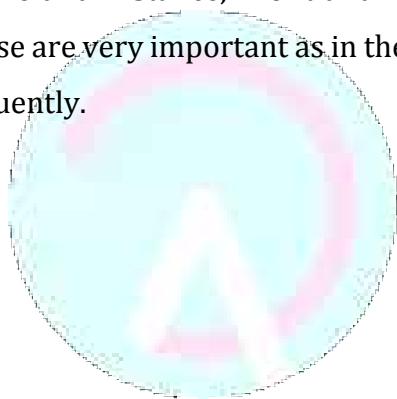
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Arithmetic DI are the type of representation in which the values of variables are represented in proportion with the distances with respect to a central point. This Graph is based on Arithmetic Concepts like Time and Distance, Profit and Loss, Time and Work, Boat and Stream, SI and CI and other topics. These are very important as in the recent Examinations we have seen these types of DI being asked frequently.



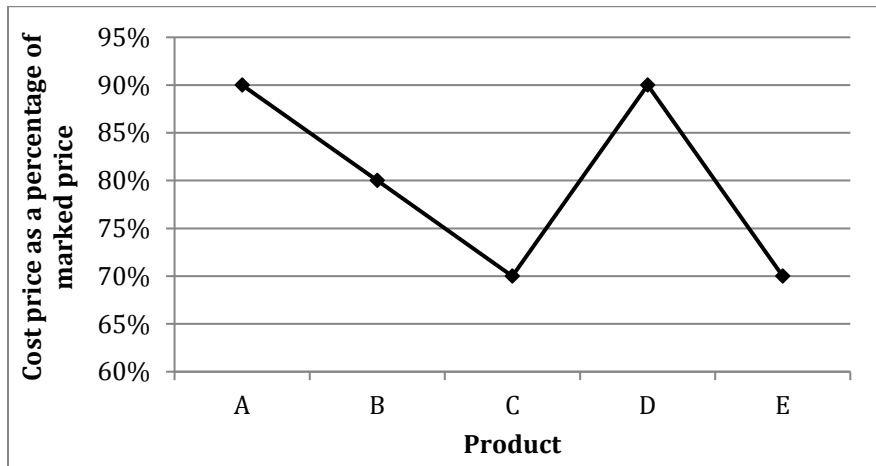
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This chapter contains:

- Solved Examples
- Previous Year Questions
- Practice Set Level I
- Practice Set Level II

SOLVED EXAMPLES

Directions (1-5): Given below is the line graph which shows the cost price of five products as a percentage of marked price. Table shows the quantity of these five products sold. Marked price = Discount on marked price + selling price
Selling price = Cost price + profit on cost price



Product →	A	B	C	D	E
Quantity sold → (in Kg)	8	12	9	20	15

1. If a discount of 8% is applied on product A then there is a profit of 300 Rs. on selling 1 kg of product A. Find the total selling price of product A.

(a) 1,00,250 (b) 1,10,400 (c) 1,20,500 (d) 98,250 (e) 90,500

Sol. (b); Let marked price of product A = x

$$\text{So, } \frac{92}{100}x - \frac{90}{100}x = 300$$

$$x = 15000$$

$$\text{Total selling price of product A} = 8 \times 0.92 \times 15000 = 1,10,400$$

2. If 4 kg of impurity free of cost is mixed in product B and product B is sold at 10% discount then, what will be the percentage profit in selling the mixture

(a) 48% (b) 60% (c) 50% (d) 40% (e) 55%

Sol. (c); Let marked price of product B = $100x$

So cost price of product B will be = $80x$

Selling price of product B = $90x$

Total cost price = $80x \times 12 = 960x$

Total selling price = $16 \times 90x = 1440x$

$$\text{Percentage profit} = \frac{1440x - 960x}{960x} \times 100 = \frac{480}{960} \times 100 = 50\%$$

3. If marked price per kg of product C and product D are in the ratio 4 : 5, then what will be the ratio of total cost price for product C to total cost price for product D.

(a) 5 : 36 (b) 7 : 31 (c) 3 : 19 (d) 7 : 25 (e) 4 : 27

Sol. (d); Let marked price per kg of product C and D be 400 and 500

$$\text{Cost price per kg of product C} = \frac{70}{100} \times 400 = 280$$

$$\text{Cost price per kg of product D} = \frac{90}{100} \times 500 = 450$$

$$\text{Required ratio} = 9 \times 280 : 20 \times 450 = 7 : 25$$

4. If cost price of product C and product D is same and there is a profit of 10% on both product then what will the ratio of discount offered on product C to discount offered on product D.

(a) 207 : 7 (b) 198 : 3 (c) 203 : 2 (d) 502 : 3 (e) 403 : 2

Sol. (a); Let Marked price product C and Product D be x and y respectively

So,

$$0.7x = 0.9y$$

$$\frac{x}{y} = \frac{9}{7}$$

Let cost price product C = $0.7 \times 9 = 6.3$

and cost price of product D = $0.9 \times 7 = 6.3$

Selling price of C will be = 6.93

Selling price of D will be = 6.93

Required ratio = $(9 - 6.93) : (7 - 6.93) = 2.07 : 0.07 = 207 : 7$

5. If marked price of product A, B and C are in the ratio 10 : 15 : 14 the what is the average of cost price of product A, B and C together. Cost price of product A is 900 Rs.

(a) $1251\frac{2}{3}$

(b) $1028\frac{1}{3}$

(c) $1155\frac{2}{3}$

(d) $928\frac{1}{3}$

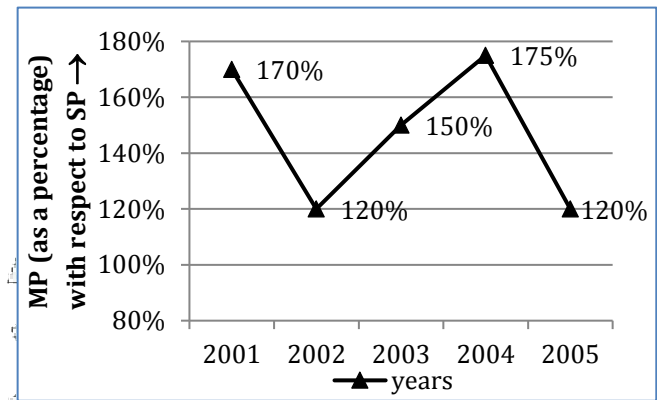
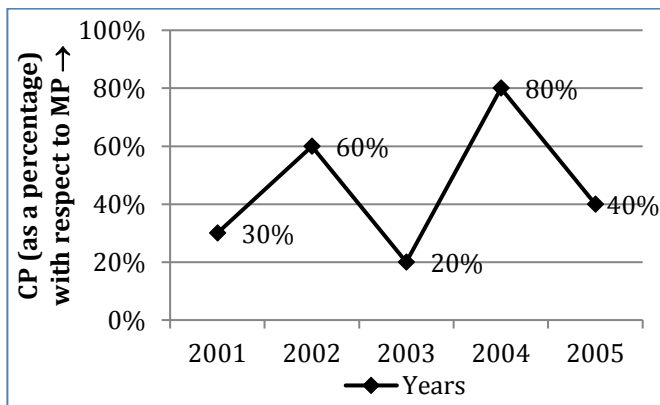
(e) $1026\frac{2}{3}$

Sol. (e); Let marked price of product A, B and C be $100x$, $150x$ and $140x$

Cost price of A, B and C will be = $90x$, $120x$ and $98x$

$$\text{Required average} = \frac{900 + 1200 + 980}{3} = \frac{3080}{3} = 1026\frac{2}{3}$$

Directions (6-10): First line graph shows the Cost Price as a percentage of mark price of a product and second line graph shows the mark price as a percentage with respect to the selling price of a product in five different years from 2001 to 2005.



6. If the mark price of the product in the year 2002, 2003 and 2005 was same then the CP of the product in year 2002 and 2003 together was what percent more than the cost price of the product in the year 2005?

(a) 100%

(b) 125%

(c) 150%

(d) 200%

(e) 50%

Sol. (a);

	2002	2003	2005
Let MP	100	100	100
CP	60	20	40

$$\text{Required \%} = \frac{80 - 40}{40} \times 100 = 100\%$$

7. If the CP of the product in year 2006 was $\frac{3}{8}$ th times of the CP of the product in year 2005 and there was same profit (in Rs.) in year 2006 as in year 2005, then find the selling price (in Rs.) of the product in year 2006 if selling price of the product in year 2005 is 100 Rs.

(a) 50 Rs

(b) 60 Rs.

(c) 70 Rs

(d) 40 Rs.

(e) 80 Rs

Sol. (c); Given

SP of the product in 2005 = 100

$\therefore$ MP of the product in 2005 = 120

$\therefore$ CP of the product in 2005 = $\frac{40}{100} \times 120 = 48$

Profit (in Rs) = $100 - 48 = 52$ Rs

Now, CP of the product in 2006 = $\frac{3}{8} \times 48 = 18$ Rs

$\therefore$ Required SP = $18 + 52 = 70$ Rs.

8. If the cost price of the product in year 2001, 2002 and 2003 were same then find the ratio of mark price of the product in the same year?

(a) 1:3:2

(b) 3:2:1

(c) 1:2:3

(d) 2:1:3

(e) 1:4:2

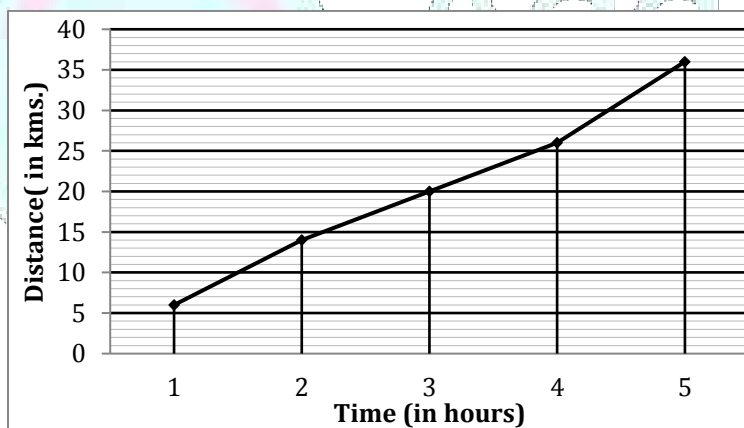
Sol (d); Let marked price of Product in year 2001 = M_1
 Let marked price of Product in year 2002 = M_2
 Let marked price of Product in year 2003 = M_3
 According to question,
 $30\% \text{ of } M_1 = 60\% \text{ of } M_2 = 20\% \text{ of } M_3$
 So,
 $M_1 : M_2 : M_3 = \frac{1}{3} : \frac{1}{6} : \frac{1}{2} \quad \text{or} \quad M_1 : M_2 : M_3 = 2 : 1 : 3$

9. If the mark price of the product in year 2003 was 1050 Rs and cost price of the product in year 2004 was 700 Rs then find the average selling price of the product in year 2003 and 2004 together?
 (a) Rs. 560 (b) Rs. 350 (c) Rs. 450 (d) Rs. 800 (e) Rs. 600

Sol. (e); Given
 MP of the product in year 2003 = 1050
 $\therefore \text{SP of the product in year 2003} = \frac{100}{150} \times 1050 = 700 \text{ Rs}$
 And, Given CP of the product in year 2004 = 700 Rs
 $\text{MP in 2004} = 700 \times \frac{100}{80} = 875$
 $\therefore \text{Sp of the product in 2004} = \frac{100}{175} \times 875 = 500 \text{ Rs}$
 $\therefore \text{Average Selling price} = (700 + 500) \times \frac{1}{2} = 600 \text{ Rs.}$

10. Total CP of the products in all of the given years together was what % less then the total MP of the product in all of the given years?
 (a) 64% (b) 52% (c) 62% (d) 65% (e) Can't be determined
Sol. (e); Since we don't know exact values of M.P. and C.P. Hence the answer can't be determined.

Directions (11-15): The graph here shows a car following a linear path. Study the graph and answer the questions that follow.



11. What is the average speed of car for the duration 3rd and 4th hours ?
 (a) 6 km/hr (b) 9 km/hr (c) 10 km/hr (d) 12 km/hr (e) None of these

Sol. (a); Average speed = $\frac{\text{Total Distance}}{\text{Total time taken}} = \frac{12}{2} = 6 \text{ km/hr}$

12. What is the difference between the average speed of car for 3rd to 5th hour and that of 2nd to 4th hour?
 (a) $\frac{1}{3}$ km/hr (b) $\frac{2}{3}$ km/hr (c) $\frac{5}{3}$ km/hr (d) $\frac{4}{3}$ km/hr (e) None of these

Sol. (b); Average speed for hours 3rd to 5th = $\frac{22}{3}$
 Average speed for hours 2nd to 4th = $\frac{20}{3}$
 Difference = $\frac{22}{3} - \frac{20}{3} = \frac{2}{3} \text{ km/hr}$

13. At what speed should car run if it has to cover equal distance every hour ?
 (a) 8.5 km/hr (b) 9 km/hr (c) 7.2 km/hr (d) 6.5 km/hr (e) None of these

Sol. (c); To cover equal distance every hour, the car must move at its average speed for the overall journey.
 Req. speed = $\frac{36}{5} = 7.2 \text{ km/hr}$

14. If the car travels another 28 kms in 4 hours after the fifth hour, then what is the ratio of average speed of car for 4th to 8th hour to that of 6th to 9th hour?

(a) 35 : 31 (b) 42 : 31 (c) 43 : 34 (d) Can not determined (e) None of these

Sol. (d); Average speed for hours 6th to 9th = $\frac{28}{4} = 7$ km/hr

But, distance covered for 6th, 7th or 8th hour individually is unknown. Hence the average speed of car for hours 4th to 8th can not be calculated. Hence we can't determine the desired ratio.

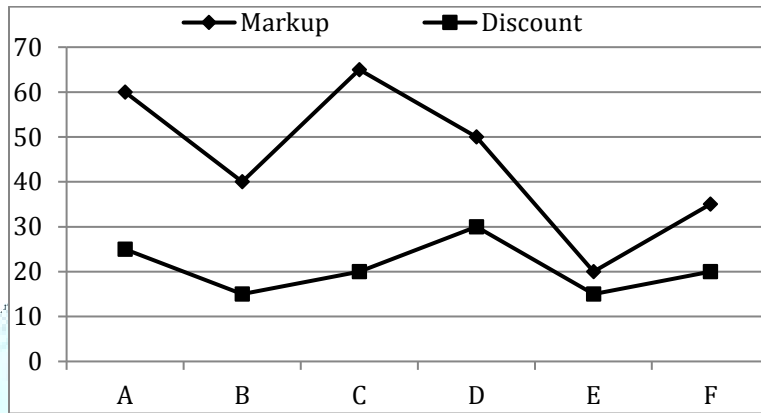
15. How much time will the car take to travel a distance of 72 kms. running at a speed equal to $\frac{3}{2}$ of its average speed for the given data ?

(a) $6\frac{2}{3}$ hrs. (b) $7\frac{4}{5}$ hrs. (c) 6 hrs (d) $8\frac{2}{3}$ hrs. (e) None of these

Sol. (a); Required time = $\frac{72}{\frac{3}{2} \times 7.2} = \frac{72}{10.8} = \frac{20}{3} = 6\frac{2}{3}$ hrs.

Direction (16-20): Refer the following line graph and answer the questions based on it.

The line graph shows the percentages of markup on CP and discount offered on MP of six different articles sold by a shopkeeper.



16. Selling price of article C is Rs.528 and the marked prices of articles C and E are the same. What is the average selling price of both the articles?

(a) Rs.534.5 (b) Rs.544.5 (c) Rs.554.5 (d) Rs.524.5 (e) Rs.527.5

Sol. (b); Marked price of article C = $528 \times \frac{100}{(100 - 20)} = \text{Rs.}660$

Then, marked price of article E = Rs.660

Selling price of article E = $660 \times \frac{(100 - 15)}{100} = \text{Rs.}561$

Required average = $\frac{(528 + 561)}{2} = \text{Rs.}544.5$

17. Cost Price of article A is Rs.450 and that of article D is Rs.600. What is the ratio of marked price of both the articles?

(a) 5 : 6 (b) 3 : 4 (c) 3 : 5 (d) 4 : 5 (e) 3 : 7

Sol. (d); Required Ratio = $450 \times \frac{(100 + 60)}{100} : 600 \times \frac{(100 + 50)}{100} = 4 : 5$

18. If the cost price of article E is double to that of article A, the marked price of article E is how much percent more than that of article A?

(a) 50% (b) 40% (c) 60% (d) 30% (e) 45%

Sol. (a); Let the cost price of article A be Rs.100

Then, the cost price of article E = Rs.200

Marked price of article A = $100 + 60\%$ of 100 = 160

Marked price of article E = $200 + 20\%$ of 200 = 240

Required Percentage = $\frac{(240 - 160)}{160} \times 100 = 50\%$

19. For which article, the profit percent is the minimum?

(a) D and E (b) D (c) E (d) E and F (e) None of these

Sol. (c); Profit % = $(100 + \text{Markup}\%) \times \frac{(100 - \text{Discount}\%)}{100} - 100$

Profit % for article A = $(100 + 60) \times \frac{(100 - 25)}{100} - 100 = 20\%$

Similarly,

Profit % for article B = 19%

Profit % for article C = 32%

Profit % for article D = 5%

Profit % for article E = 2%

Profit % for article F = 8%

Hence, minimum profit % is for article E

20. If the cost price of each of the article B and F is Rs.800, then what is the total profit earned by shopkeeper on both the articles?

(a) Rs.152

(b) Rs.248

(c) Rs.184

(d) Rs.216

(e) Cannot be determined

Sol. (d); Profit % = $(100 + \text{Markup}\%) \times \frac{(100 - \text{Discount}\%)}{100} - 100$

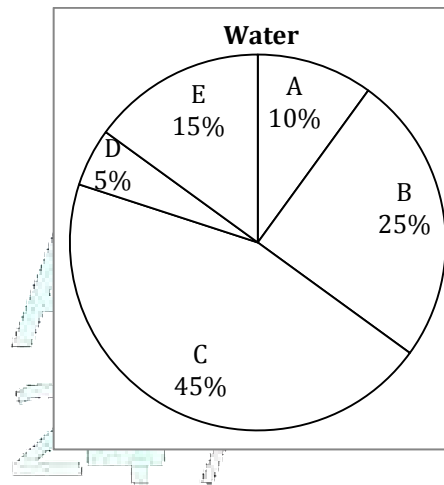
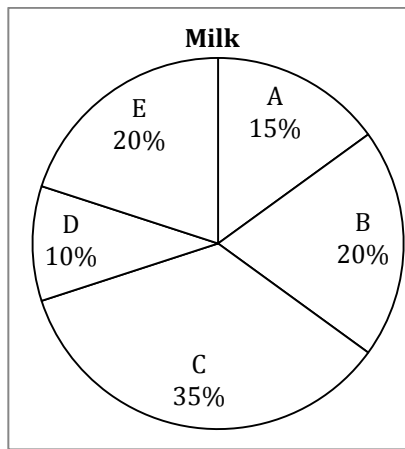
Profit % for article B = 19%

Profit % for article F = 8%

Total Profit = 19% of 800 + 8% of 800 = 27% of 800 = Rs.216

Directions (21-25): Read the given data carefully and answer the given question

Pie chart shows percentage distribution of total quantity of milk and water in five different vessel namely A, B, C, D and E



Total milk = 500 Lt

Total Water = 250 Lt

21. What is ratio of milk and water in F. If the mixture of vessel A and B is poured in the other vessel F.

(a) 1 : 2

(b) 3 : 5

(c) 2 : 1

(d) 4 : 3

(e) None of these

Sol. (c); Total quantity of milk in vessel A = $500 \times \frac{15}{100} = 75 \text{ Lt}$

Total quantity of milk in vessel B = $500 \times \frac{20}{100} = 100$

Total quantity of Water in vessel A = $\frac{250 \times 10}{100} = 25$

Total quantity of Water in vessel B = $\frac{250 \times 25}{100} = 62.5$

Required ratio = $(100 + 75) : (25 + 62.5) = 2:1$

22. With the intention of cheating a shopkeeper sells mixture of vessel-D showing that he is taking only 4% profit over cost price. Find his actual profit. If he professes to sell pure milk and cost price is due to milk only

(a) 40%

(b) 30%

(c) 25%

(d) 35%

(e) None of these

Sol. (b); Quantity of milk in D = $500 \times \frac{10}{100} = 50$

Quantity of Water in D = $250 \times \frac{5}{100} = 12.5$

Total quantity = $50 + 12.5 = 62.5$

Let CP of 1 Lt = Rs. 10

Total cost for customer = $62.5 \times 10 = 625$

SP = $625 \times \frac{26}{25} = 650$

Actual cost = $50 \times 10 = 500$

Profit% = $\left(\frac{650 - 500}{500} \right) \times 100 = 30\%$

23. If 40% from vessel A and 50% from vessel E is taken out mixed together. Find the quantity of water in resultant mixture.
 (a) 28.75 (b) 29.25 (c) 27.50 (d) 25 (e) None of these

Sol. (a); Water in A = $\frac{250 \times 10}{100} = 25 \text{ Lt.}$

$$\text{Water in E} = \frac{250 \times 15}{100} = 37.5 \text{ Lt.}$$

$$\text{Required answer} = 25 \times \frac{40}{100} + 37.5 \times \frac{50}{100} = 10 + 18.75 = 28.75 \text{ Lt}$$

24. What will be the final ratio of milk and water if all the vessels are poured into the bigger vessel except from vessel C only 50% of mixture is taken out

(a) 2 : 1 (b) 66 : 31 (c) 3 : 2 (d) 55 : 21 (e) None of these

Sol. (b); Required AM = $500 - \left(\frac{500 \times 35}{100}\right) \times \frac{1}{2} : 250 - \left(250 \times \frac{45}{100}\right) \times \frac{1}{2}$
 $= 412.5 : 193.75 \Rightarrow 66 : 31$

25. Find profit% after selling mixture at cost price from vessel B, C and D together. If one professes to sell pure milk and cost price is due to milk only (approximately)

(a) 54% (b) 60% (c) 63% (d) 58% (e) 62%

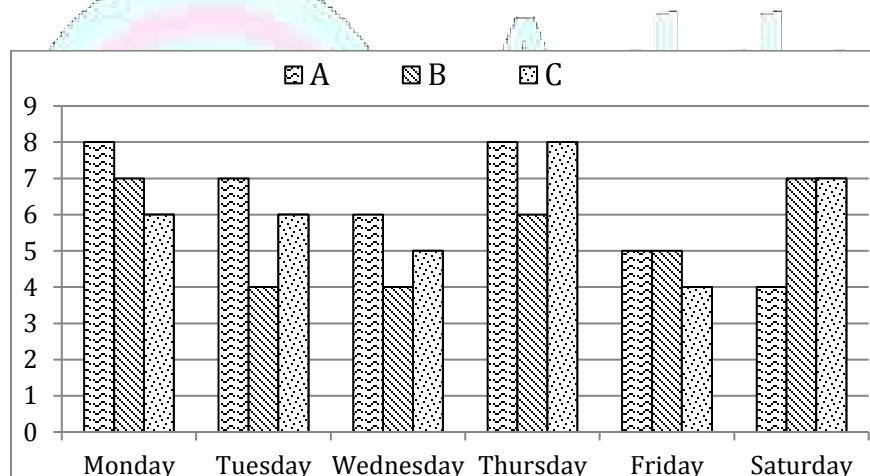
Sol. (d); Milk in B, C and D together = $65 \times \frac{500}{100} = 325 \text{ Lt.}$

$$\text{Water in B, C and D together} = 250 \times \frac{75}{100} = 187.5 \text{ Lt.}$$

$$\text{Total mixture} = 512.5 \text{ Lt}$$

$$\% \text{ profit} = \frac{187.5}{325} \times 100 \cong 57.6\% = 58\%$$

Directions (26 – 30): The following data shows the working hours for three employees in week days to complete a task. The working hours also indicates the time taken by them to complete the same task alone. Read the following questions and answer carefully.



26. What is the difference between average of working hours of A on Monday and Saturday together and average of working hours B on Friday and Saturday together?

(a) 0 (b) 1 (c) 0.5 (d) 2 (e) 2.5

Sol. (a); A : $\frac{8+4}{2} = 6$

$$B : \frac{5+7}{2} = 6$$

$$\text{Difference} = 0$$

27. On Tuesday, If A takes half the working hour time to complete a work while B takes equal time as A and C takes 50% more time to complete the same work alone. What will be the ratio of their working efficiency to complete a work?

(a) 7 : 7 : 18 (b) 7 : 18 : 18 (c) 18 : 18 : 7 (d) 18 : 7 : 18 (e) 9 : 9 : 4

Sol. (c); A : $\frac{7}{2} \text{ hrs}$

$$B : \frac{7}{2} \text{ hrs}$$

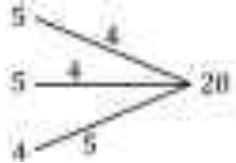
$$C : 6 + 3 = 9 \text{ hrs}$$

$$\text{Ratio} = 18 : 18 : 7$$

28. Working hours on Friday by A, B and C are the time taken by them to complete a work alone. How much time will they take to complete thrice the same work working together?

(a) $6\frac{2}{13}$ days (b) $4\frac{8}{13}$ days (c) $3\frac{8}{13}$ days (d) 4 days (e) $4\frac{3}{13}$ days

Sol. (b); On Friday

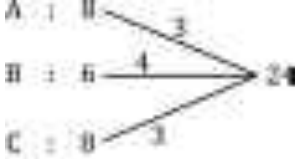


$$\text{Time taken to complete thrice the work} = \frac{20 \times 3}{13} = 4\frac{8}{13} \text{ days}$$

29. On Thursday, A was supposed to complete the task in time. Due to illness, he stopped when only one-third of the task was complete. Half of the remaining work was completed by B alone and rest by C alone. Find the total time taken by them to complete the task.

(a) 8 days (b) $7\frac{1}{3}$ days (c) 7 days (d) $7\frac{2}{3}$ days (e) $7\frac{3}{4}$ days

Sol. (b); Thursday:



$$\text{By A} \rightarrow \frac{1}{3}(24) = 8 \text{ units in } \frac{8}{3} \text{ hrs}$$

$$\text{By B} \rightarrow \frac{24-8}{2} = 8 \text{ units in } \frac{8}{4} \text{ hrs}$$

$$\text{By C} \rightarrow 8 \text{ units in } \frac{8}{3} \text{ hrs}$$

$$\text{Total time taken} = \frac{8}{3} + 2 + \frac{8}{3} = 7\frac{1}{3} \text{ days}$$

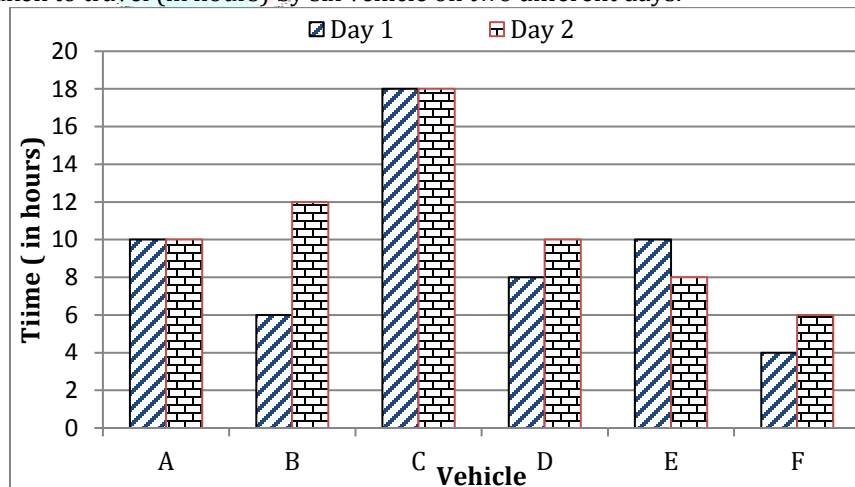
30. What is the ratio of total work hour of A and C together throughout the week to that of B throughout the week?

(a) 7 : 5 (b) 5 : 2 (c) 31 : 13 (d) 37 : 15 (e) 74 : 33

Sol. (e); $A + C = 74 \Rightarrow B = 33$

So, required ratio = 74 : 33

Direction(31-35)- Study the following bar-graph and data given in the table carefully and answer the questions asked. Bar-graph shows Time taken to travel (in hours) by six vehicle on two different days.



Distance covered (in km) by six vehicle on each day.

Vehicle	Day One	Day Two
A	850	840
B	520	500
C	450	630
D	400	800
E	640	320
F	320	480

31. What is difference between the average speed of A on day one and day two together to the average of speed of F on Day one and Day two together? (in km/hr)

(a) 4.5 (b) 6.5 (c) 5.5 (d) 2.5 (e) 3.5

Sol. (a); Average speed of A on day one and Day two together = $\frac{\text{total distance}}{\text{total time}}$
 $= \frac{1690}{20} = 84.5 \text{ km/hr}$

Average speed of F on Day one and Day two together = $\frac{\text{total distance}}{\text{total time}}$
 $= \frac{800}{10} = 80 \text{ km/hr}$

$\therefore$ Difference = $84.5 - 80 = 4.5 \text{ km/hr}$

32. Which of the following vehicles travelled at the same speed on both the days?

(a) Vehicle A (b) Vehicle E (c) Vehicle C (d) Vehicle D (e) Vehicle F

Sol. (e); From the table and graph

Speed of F on day one = $\frac{320}{4} = 80 \text{ km/hr}$

Speed of F on day two = $\frac{480}{6} = 80 \text{ km/hr}$

$\therefore$ Speed on both days is same by vehicle F.

33. What is the ratio of average speed of vehicle D on day one and day two together to the average speed of vehicle E on day one and day two together?

(a) 65 : 53 (b) 5 : 4 (c) 4 : 5 (d) 50 : 63 (e) 7 : 8

Sol. (b); Average speed by vehicle D on day one and day two together = $\frac{\text{total distance}}{\text{total time}} = \frac{200}{3} \text{ km/hr}$

Average speed by vehicle E on day one and day two together = $\frac{\text{total distance}}{\text{total time}}$
 $= \frac{640+320}{18} = \frac{960}{18} \text{ km/hr} = \frac{160}{3} \text{ km/hr}$

$\therefore$ Ratio = $200 : 160 = 5/4$

34. Which vehicle have maximum speed on day two?

(a) Vehicle D (b) Vehicle F (c) Vehicle C (d) Vehicle A (e) None of these

Sol. (d); From the table and graph it is clear that vehicle A has maximum speed on day two

35. The distance travelled on day two by vehicle A, D and F is what percent more than that of the distance travelled on day one by vehicle A and D?

(a) 69.6% (b) 62.6% (c) 65.6% (d) 59.6% (e) 67.5%

Sol. (a); Distance travelled on day two by vehicle A, D and F = $840 + 800 + 480 = 2120 \text{ km}$

Distance travelled on day one by vehicle A and D = $850 + 400 = 1250 \text{ km}$

$\therefore$ percentage = $\frac{2120-1250}{1250} \times 100 = \frac{87000}{1250} = 69.6\%$

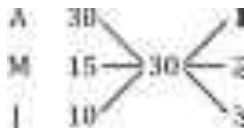
Directions (36-40): The table given below shows the number of days taken by six workers to complete five different tasks in a field individually. Study the table carefully to answer the questions based on it.

Workers	Digging	Watering	Ploughing	Seeding	Leveling
Mayank	15	8	14	8	16
Faisal	12	20	35	6	12
Rakesh	20	10	20	12	18
Jatin	10	15	24	10	20
Lokesh	8	24	10	8	14
Ashfaq	30	12	16	20	16

36. If Ashfaq started digging alone and was coordinated by Mayank and Jatin respectively on alternate days from next day, find the number of days taken by them to complete the work?

(a) $10\frac{2}{3}$ days (b) $11\frac{3}{4}$ days (c) $9\frac{1}{3}$ days (d) $9\frac{3}{4}$ days (e) None of these

Sol. (c);



Day 1 $\rightarrow$ A : 1 unit

Day 2 + Day 3 work = $(A + M) + (A + J) = (1 + 2) + (1 + 3) = 7$ unit

2 Days $\rightarrow$ 7 unit

8 Days $\rightarrow$ 28 unit

Total required days = $1 + 8 + \frac{1}{3} = 9\frac{1}{3}$ days

37. Lokesh, Faisal and Jatin started watering in the field. But after 2 days Lokesh left and Jatin also left 2 days before completion of work. Find total number of days Faisal worked?

(a) 7 days (b) 9 days (c) 8 days (d) 10 days (e) None of these

Sol. (b); $L - 24$
 $F - 20$ 120 6
 $J - 15$ 8

Work done in 2 days = 38 units

Remaining work = $120 - 38 = 82$ units

$\therefore$ Time taken to complete remaining work = $\frac{82+16}{8+6} = 7$ days

So, Faisal work for $7 + 2$ ie 9 days

38. What is the difference in the number of days taken by Rakesh and Mayank together for seeding and that of by Faisal and Jatin together for watering?

(a) $3\frac{3}{4}$ days (b) $4\frac{27}{35}$ days (c) $3\frac{27}{31}$ days (d) $3\frac{27}{35}$ days (e) None of these

Sol. (d); Total time for seeding = $\frac{1}{\frac{1}{12} + \frac{1}{8}} = \frac{24}{5}$ days

Total time for watering = $\frac{1}{\frac{1}{20} + \frac{1}{15}} = \frac{60}{7}$ days

So, required difference = $3\frac{27}{35}$ days

39. What is the ratio of the average number of working days of Rakesh and that of Faisal for all the activities?

(a) 17 : 16 (b) 16 : 17 (c) 17 : 18 (d) 19 : 20 (e) None of these

Sol. (b); Required ratio = $\frac{\frac{1}{5} \times 80}{\frac{1}{5} \times 85} = \frac{16}{17}$

40. Mayank, Faisal and Rakesh together started ploughing while at the same time Jatin, Lokesh and Ashfaq started watering. What is the time difference in the completion of the two works?

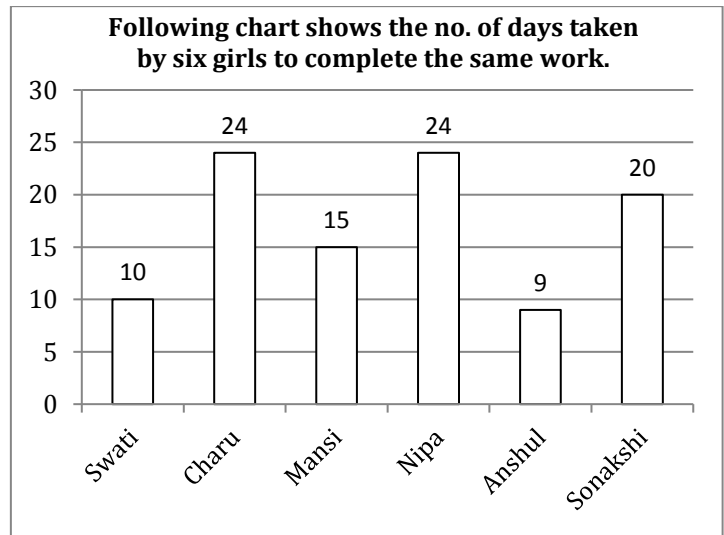
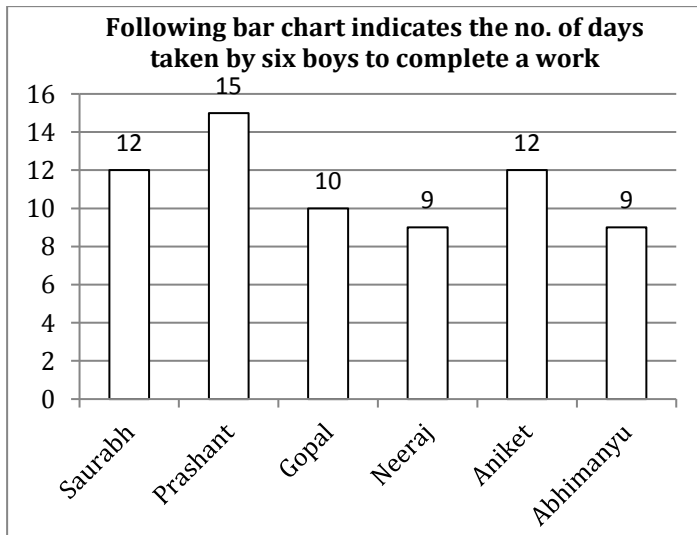
(a) $3\frac{31}{67}$ days (b) $3\frac{3}{7}$ days (c) $1\frac{3}{7}$ days (d) $1\frac{31}{67}$ days (e) $1\frac{31}{69}$ days

Sol. (e); Time for ploughing = $\frac{1}{\frac{1}{14} + \frac{1}{35} + \frac{1}{20}} = \frac{20}{3}$ days

Time for watering = $\frac{1}{\frac{1}{15} + \frac{1}{24} + \frac{1}{12}} = \frac{120}{23}$ days

$\therefore$ Required difference = $1\frac{31}{69}$ days

Directions (41-45): Read the following bar chart carefully and answer the following questions –



41. If Neeraj and Abhimanyu started working and after 2 days they were replaced by Charu, Anshul and Nipa and then they worked for 2 days together. How much work is still left?

(a) $1/10$ (b) $1/6$ (c) $1/9$ (d) $1/12$ (e) None of these

Sol. (b); 2 days' work of Neeraj and Abhimanyu $= \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$
 2 days work of Charu, Anshul and Nipa $= \frac{2}{24} + \frac{2}{9} + \frac{2}{24} = \frac{6+16+6}{72} = \frac{28}{72} = \frac{7}{18}$
 Total work done in 4 days $= \frac{4}{9} + \frac{7}{18} = \frac{8+7}{18} = \frac{15}{18}$
 Remaining work $= 1 - \frac{15}{18} = \frac{3}{18} = \frac{1}{6}$

42. If all the boys work together then time taken by them will be how much less than the time taken by Swati, Mansi, Charu and Nipa together to do the work?

(a) 2 days (b) 2.1 days (c) 2.2 days (d) 2.3 days (e) None of these

Sol. (c); Work done by all the boys in one day $= \frac{1}{12} + \frac{1}{15} + \frac{1}{10} + \frac{1}{9} + \frac{1}{12} + \frac{1}{9}$
 $= \frac{15+12+18+20+15+20}{180} = \frac{100}{180}$
 Time taken by the boys = 1.8 days
 Work done by Swati, Mansi, Charu and Nipa in one day $= \frac{1}{10} + \frac{1}{15} + \frac{1}{24} + \frac{1}{24} = \frac{12+8+5+5}{120} = \frac{30}{120} = \frac{1}{4}$
 Time taken by Swati, Mansi, Charu and Nipa = 4 days
 Desired difference = 4 - 1.8 = 2.2 days

43. Aniket and Gopal started working, Aniket worked at 120% of his efficiency and Gopal worked at $5/6$ at his efficiency. They started working on alternate days starting with Aniket. How many days will be taken by them to complete the work?

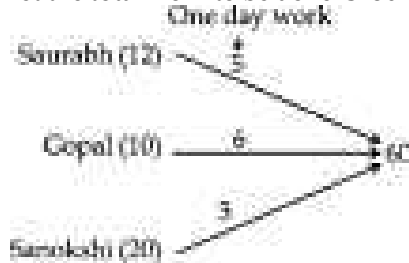
(a) $10\frac{5}{6}$ days (b) $11\frac{5}{6}$ days (c) $10\frac{2}{3}$ days (d) $10\frac{1}{6}$ days (e) None of these

Sol. (a); Aniket one day work with 120% efficiency $= \frac{1}{12} \times \frac{120}{100} = \frac{1}{10}$
 Gopal one day work with $5/6$ of his efficiency $= \frac{5}{6} \times \frac{1}{10} = \frac{1}{12}$
 2 days' work $= \frac{1}{10} + \frac{1}{12} = \frac{11}{60}$
 10 days's work starting with Aniket $= \frac{55}{60}$
 Remaining work $= 1 - 55/60 = \frac{5}{60} = \frac{1}{12}$
 Time taken by Aniket to complete the remaining work $= \frac{1}{12} \div \frac{1}{10} = \frac{10}{12} = \frac{5}{6}$
 Total time taken = $10\frac{5}{6}$ days

44. How many days will it take to complete the work if saurabh does the 25% of the work alone, 10% is done by Gopal alone, the remaining work is done by Sonakshi and only one of them is working at a time?

(a) 19 days (b) 16 days (c) 13 days (d) 17 days (e) None of these

Sol. (d); Let the total work to be done is '60' units.



Time taken by Saurabh $= \frac{25\% \times 60}{5} = 3$ days

Time taken by Gopal $= 10\% \times \frac{60}{6} = 1$ days

Time taken by Sonakshi $= 65\% \times \frac{60}{3} = 13$ days

Total time taken = 3 + 1 + 13 = 17 days

Or

Time taken by saurabh to do 25% of work = 25% × 12 days = 3 days

Time taken by Gopal to do 10% of work = 10% of 10 days = 1 day

Time taken by Sonakshi to do 65% of work = 65% × 20 days = 13 days

Total time taken = 3 + 1 + 13 = 17 days

45. Prashant started working alone but after 5 days of working he found that he has done only 25% work so to complete the remaining work on time he was expected to complete the work, by how much percentage he should increase his efficiency?

(a) 52.5% (b) 55% (c) 50% (d) 45% (e) None of these

Sol. (c); 25% of the work is done in 5 days. Then, total work will be done in 20 days.

$$\text{Efficiency/day} = \frac{1}{20}$$

Remaining 75% of the work has to be done in 10 days (15 – 5 days). Total work will be done in $10 \times \frac{100}{75}$ days with the same efficiency.

$$\text{New Efficiency/day} = \frac{1}{10} \times \frac{100}{75} = \frac{3}{40}$$

$$\text{Percentage increase in efficiency to do the work in time} = \frac{\frac{3}{40} - \frac{1}{20}}{\frac{1}{20}} \times 100 = \frac{\frac{1}{40}}{\frac{1}{20}} \times 100 = 50\%$$

Directions (46-50): Study the following table carefully to answer the questions that are based on it. The table shows the dimensions of various shapes of iron blocks.

Blocks	Length (cm)	Breadth (cm)	Height (cm)	Radius (cm)
Cube	20	20	20	—
Cuboid	12	8	5	—
Sphere	—	—	—	35
Cylinder	—	—	14	8
Cone	—	—	28	21

46. Find the number of cones (of given dimension) that can be formed by melting 18 spheres of given dimension.

(a) 252 (b) 265 (c) 250 (d) 270 (e) 280

Sol. (c); Let n number of cones are formed.

So,

Volume of n cones = Volume of 18 spheres

$$\Rightarrow n \times \frac{1}{3} \times \pi \times 21^2 \times 28 = 18 \times \frac{4}{3} \times \pi \times 35^3 \Rightarrow n = 250$$

47. Find the cost of painting 630 conical blocks of given dimension provided that only the curved surface area is to be painted at Rs. 5 per m². (approx.)

(a) Rs. 777 (b) Rs. 727 (c) Rs. 780 (d) Rs. 725 (e) Rs. 790

Sol. (b); Slant height of cone = $\sqrt{21^2 + 28^2} = 35$ cm

$$\therefore \text{Required cost of painting} = 630 \times \frac{22}{7} \times 21 \times 35 \times \frac{5}{100} \approx 727$$

48. Find the total number of cubical blocks of given dimension required to raise the water level to the brim of a cylindrical container having dimensions 10 times than that of the given cylinder. Initially container is half filled.

(a) 176 (b) 80 (c) 220 (d) 128 (e) 180

Sol. (a); Volume of container = $\frac{22}{7} \times 80^2 \times 140 = 2816000 \text{ cm}^3$

$\therefore$ Volume of required number of cubical blocks should be equal to half the volume of container

$$\text{So, } x \times 20^3 = \frac{2816000}{2} \Rightarrow x = 176$$

49. Find the ratio of total surface area of cylinder, cone and sphere.

(a) 88 : 294 : 1225 (b) 575 : 352 : 89 (c) 18 : 119 : 54 (d) 2 : 8 : 13 (e) 5 : 17 : 7

Sol. (a); Required ratio = $2\pi \times 8(8+14) : \pi \times 21(21+35) : 4\pi \times 35^2$
 $= 352 : 1176 : 4900 = 88 : 294 : 1225$

50. By what percent the total surface area of cuboid is more or less than the lateral surface area of cube ?

(a) 70.5% (b) 65% (c) 75.5% (d) 80% (e) None of these

Sol. (c); T.S.A. of cuboid = $2(12 \times 8 + 8 \times 5 + 5 \times 12) = 392 \text{ cm}^2$

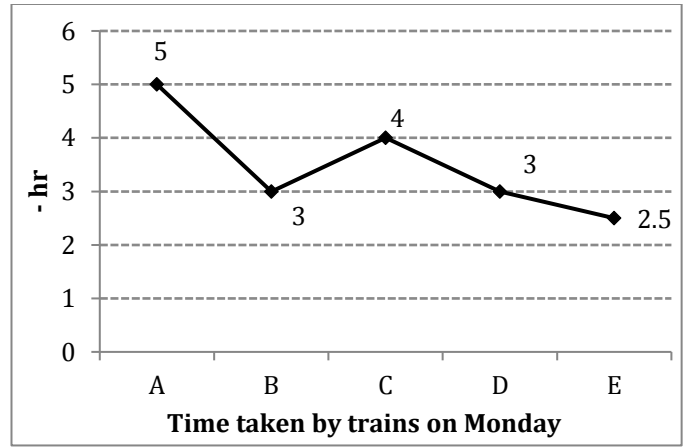
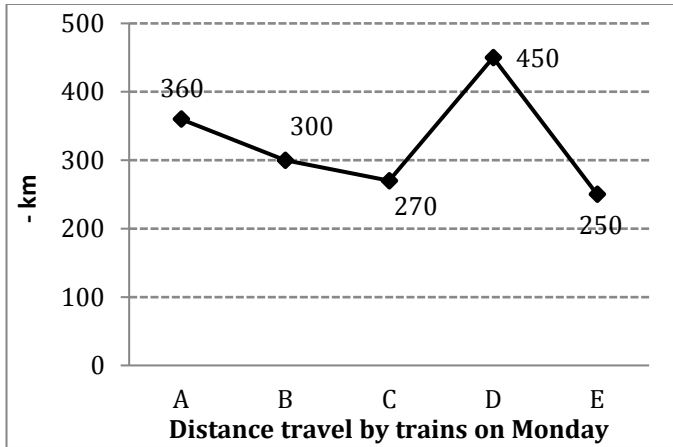
$$\text{L.S.A. of cube} = 4 \times 20^2 = 1600 \text{ cm}^2$$

$$\therefore \text{Required percentage} = \frac{1208}{1600} \times 100 = 75.5\%$$

PREVIOUS YEAR QUESTIONS

Directions (1-5): Study the graph and answer the following questions

Given below are two line graphs. First line graph shows the distance travelled by different trains on Monday and second line graph shows the time taken by different trains on Monday to cover the distance given in first line graph.



- If speed of the train A is increased by 20% every day, then what was the speed of train on last or previous Saturday?
(a) 45 km/hr (b) 55 km/hr (c) 50 km/hr (d) 57.5 km/hr (e) 60 km/hr
- What is the approximate speed (in km/hr) of train C, if the distance travel by C is increased by $11\frac{1}{9}\%$ and time taken is reduced by 12.5%?
(a) 83 (b) 89 (c) 86 (d) 80 (e) 75
- What is the ratio of the sum of speed of train A & B together to the sum of speed of train C & E together.
(a) $\frac{414}{527}$ (b) $\frac{344}{335}$ (c) $\frac{314}{329}$ (d) $\frac{327}{551}$ (e) None of these
- The speed of the slowest train is what percent less than the speed of fastest train.
(a) 55% (b) 56% (c) 60% (d) 45% (e) 57.5%
- What will be the Average speed (in km/hr) of the all 5 trains, if train B and D travel at 20% less than the planned distance.
(a) 88.7 (b) 87.9 (c) 85 (d) 86 (e) None of these

Directions (6-10): In the following table, the Investment and profit of three persons from different states is given.

State	Investment (in Rs.)			Profit (in Rs.)		
	A	B	C	A	B	C
UP	15000	—	25000	—	8000	12500
Rajasthan	—	7000	8000	—	—	14000
Gujarat	4000	5000	4500	—	—	—
Tamil Nadu	9000	10000	—	4500	6000	—
West Bengal	—	—	17000	20000	30000	40000

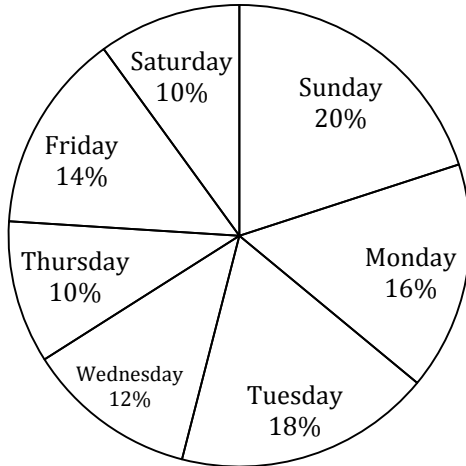
Note : Some values are missing. You have to calculate these values as per data given in the questions: -

- If Person A invested his amount in UP state for 9 months and person C invested his amount in the same state for 10 months then find the total profit made by all of them from UP state ?
(a) 29250 Rs. (b) 24250 Rs. (c) 27250 Rs. (d) 31200 Rs. (e) 28360 Rs.
- If the total profit earned from Rajasthan by all of them is 32375 Rs. and each invested for 9 months then find the ratio of investment of A in Rajasthan state to the profit of B from UP state ?
(a) 16 : 7 (b) 7 : 16 (c) 8 : 13 (d) 13 : 8 (e) 5 : 17
- If A, B and C invested in Gujarat state for 5 months, 8 months and 6 months respectively then profit earned by C from Gujarat state is what % of the profit earned by A and B together from the same state, if total profit earned by all of them from Gujarat state is 8700 Rs.
(a) 45% (b) 50% (c) 55% (d) 40% (e) 60%
- In West Bengal state total Investment of A and B is 85000 Rs, A and B invested their amount for 4 months and 6 months respectively in the same state, then find the number of months that C invested his amount ?
(a) 8 months (b) 9 months (c) 20 months (d) Can't be determined (e) 15 months

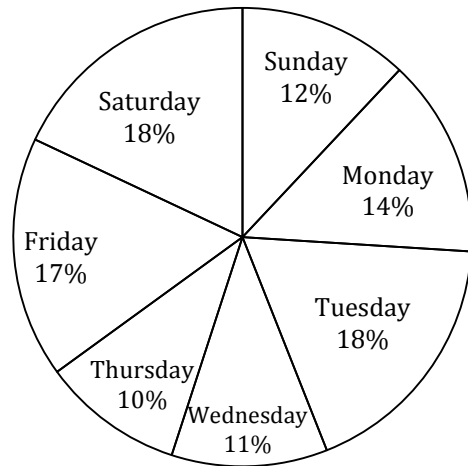
10. Average Investment made by all of them in Tamil Nadu is 10,000 Rs. and average profit earned by all of them from the same state is 6000 Rs. , then profit earned by C in the same state is what percent more/less than the amount invested by C in the same state ?
- (a) $35\frac{1}{3}\%$ (b) $37\frac{6}{7}\%$ (c) $32\frac{7}{11}\%$ (d) $33\frac{7}{11}\%$ (e) $31\frac{9}{11}\%$

Directions (11-15): Study the data given below and answer the following questions. The pie charts shown below shows the distance covered by a boat moving upstream and downstream on different days of a week. And the table shows the speed of stream in km/hr. on different days of a week.

Total distance covered upstream = 4800 km



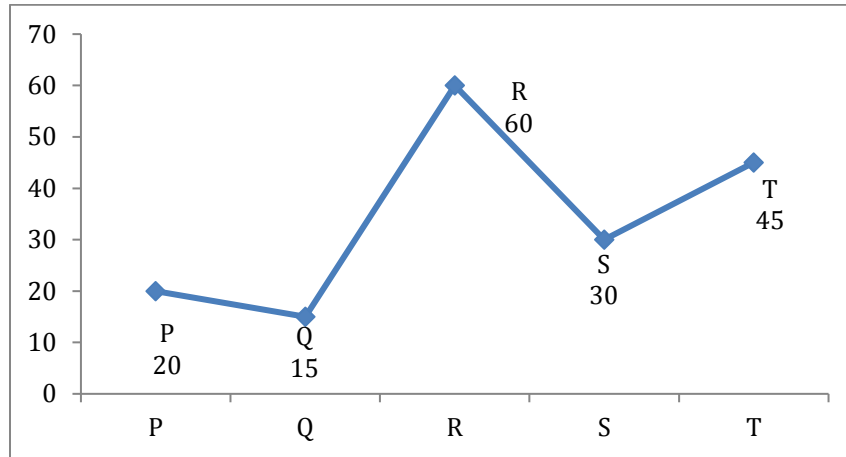
Total distance covered downstream = 2400 km



Day	Speed of stream (km/hr)
Monday	5
Tuesday	2
Wednesday	6
Thursday	—
Friday	1
Saturday	—
Sunday	3

11. If the time taken by boat to travel upstream on Wednesday is $\frac{6}{7}$ times than the time taken to travel downstream on Monday and the speed of boat in still water on Monday is 15 kmph then find the speed of boat in still water on Wednesday? (speed of boat in still water is different for different days)
- (a) 52 kmph (b) 62 kmph (c) 42 kmph (d) 48 kmph (e) 46 kmph
12. If the time taken by boat to travel upstream on Monday is $27\frac{1}{5}$ hrs. more than the time taken by it to travel downstream on the same day, then find the speed of boat in still water on Monday ?
- (a) 25 kmph (b) 18 kmph (c) 20 kmph (d) 15 kmph (e) 10 kmph
13. If the speed of boat in still water on Saturday was 27 km/hr and the speed of boat in still water on Wednesday was $66\frac{2}{3}\%$ more than that of Saturday and time taken to travel upstream on Wednesday is $\frac{16}{13}$ times than time taken by it to travel downstream on Saturday, then find the speed of stream (in kmph) on Saturday?
- (a) 2 (b) 4 (c) 9 (d) 8 (e) 7
14. The speed of boat in still water on Saturday was 21 km/hr. and that on Sunday was $28\frac{4}{7}\%$ more than that on Saturday and if the time taken by boat to travel upstream on Saturday is $2\frac{1}{2}$ times of the time taken to travel downstream on Sunday, then find the time taken by the boat to cover a distance of 125 km upstream on Saturday?
- (a) 6 hrs. 45 min. (b) 2 hrs. 45 min. (c) 4 hrs. 30 min. (d) 6 hrs. 15 min. (e) None of these
15. If the time taken by boat to travel upstream on Friday is 30 hours more than the time taken by it to travel downstream on Wednesday and the speed of boat in still water on Friday is 17 kmph, then find the upstream speed of boat on Wednesday? (speed of boat in still water is different on different days)
- (a) 27 kmph (b) 22 kmph (c) 20 kmph (d) 25 kmph (e) 10 kmph

Directions (16-20): P, Q, R, S and T are five persons employed to complete a job A. Line graph shows the data regarding the time taken by these persons to complete the job A. Table 2 shows the actual time for which every one of them worked on the job A.



Person	Time (in Days)
P	4
Q	–
R	10
S	–
T	4.5

Note 1: Q and S worked on the job A for 'whole number' days.

Note 2: Two jobs B and C are similar to job A and require same effort as required by job A.

16. R and S worked together on job B for 10 days. Q then worked for 'x' days and T worked for 'x + 3' days. If $\frac{1}{45}$ of the job was still remained, then find the value of 'x'?
- (a) $\frac{37}{8}$ days (b) $\frac{21}{8}$ days (c) 4 days (d) 3 days (e) 1 day
17. T worked on job 'C' for 15 days. Then P, R and S worked for 15 days working on alternate days starting with P followed by R and S in that order. If Q is supposed to complete the remaining work in $1\frac{1}{2}$ days, then by how much percent must Q increase his efficiency in order to complete the work in given time?
- (a) $33\frac{1}{3}\%$ (b) 45% (c) 37.5% (d) $66\frac{2}{3}\%$ (e) None of these
18. If P, Q and R worked on job C for 9 days in the following manner: P and Q work together on day 1, then Q and R work together on day 2 and then R and P work together on day 3, then find the no. of days for which T worked if the ratio of no. of days for which S and T worked to complete the remaining job is 4:3.
- (a) 5 days (b) 4 days (c) 3 days (d) 6 days (e) None of these
19. If all five person worked on job A for the number of days as given in table and the ratio of number of days for which Q and S worked on job A 5 : 6, then find the sum of the number of days for which Q and S worked ?
- (a) 39 (b) 26 (c) 11 (d) 52 (e) None of these
20. If P and R worked on job B with $\frac{4}{3}$ times their given efficiency and they are assisted by Q every 3rd day, then find the time taken by both of them to complete the job B together?
- (a) 13 days (b) $22\frac{1}{6}$ days (c) $13\frac{1}{2}$ days (d) 3 days (e) 22 days

Directions (21-25): There are five shop owners A, B, C, D and E. They are selling four different items given in the table. In the table, Discount (as a percentage) is given on mark price of these four products by different sellers. Study the table and answer the following questions:

	Item I	Item II	Item III	Item IV
A	22%	16%	32%	—
B	18%	—	23%	30%
C	—	32%	24%	10%
D	25%	18%	7%	—
E	—	9%	—	4%

Note: 1.
2.

Some values are missing. You have to calculate these values as per data given in the questions.
Mark price of a particular item is same for all of the shop owners.

21. If the profit percentage of seller A after selling item II is $s\%$ and that of seller C for the same item is $(3s - 2)\%$ and the ratio of cost price of item II by seller A and seller C is $21 : 17$ then find the value of s ?
 (a) 2 (b) 3 (c) 4 (d) 1 (e) none of these
22. For seller D, difference between the selling price of item II and that of item III is 4810 Rs. if the sum of the mark price of item II and item III by the same seller is 8000 then find the Mark price (in Rs.) of item II is how much more (in Rs.) than that of item III for the same seller? (Selling price of item II is greater than that of item III)
 (a) 5000 (b) 6000 (c) 6500 (d) 5500 (e) 4500
23. Average SP of item II by seller A and B is Rs 4400. Average SP of item II by seller B and C is Rs 3600. Find the SP (in Rs.) of item II by seller C.
 (a) 4500 (b) 3600 (c) 5400 (d) 4000 (e) 6800
24. If the cost price of item I and selling price of item III by seller E are in the ratio of $5 : 6$. If the seller earned a profit of 25% which is Rs. 750 on item I and 20% on item III then find the total profit (in Rs.) by selling item I and item III together by the same seller?
 (a) 1250 (b) 2000 (c) 1750 (d) 1350 (e) 1500
25. If Cost price of item III is 5400 Rs. for all of the sellers and all of them marked the same product at 100% higher than the cost price, then to get a total profit of 13500 Rs. by all of the five sellers after selling item III, what is the minimum discount (as a percentage) should be provided by seller E on item III.
 (a) 15% (b) 13% (c) 11% (d) 20% (e) None of these

Directions (26-30): Given below is the table of five persons in a business, time for which investment made, share of profit and percentage of profit. Some values are missing in the table, you have to calculate these value if necessary to answer the questions.

Note- Profit percent is calculated on total profit made by all.

Person	Invest-ment (in Rs.)	Time (in month)	Share of profit	Percentage of profit
A	—	8	—	$\frac{3600}{211}\%$
B	20,000	—	—	—
C	—	—	10,000	—
D	—	12	—	$\frac{3600}{211}\%$
E	24,000	—	10,800	—

26. What is the sum of profit of A and C together if investment of A and C together is 215% of investment of B and investment of A is 28% less than investment of C. A invested for same time as C invested.
 (a) 17200 (b) 18900 (c) 19400 (d) 14200 (e) None of these
27. What is the total profit of all 5 person if profit percentage of E is 50% more than profit % of D.
 (a) 40500 (b) 43500 (c) 42200 (d) 53200 (e) 38500
28. What is the total investment of c and e if e invested for one month more than c and ratio between the time taken by both i.e. C and e is $8 : 9$.
 (a) 48000 (b) 47000 (c) 46000 (d) 49000 (e) 50000
29. Profit of A is what % of profit of C if profit of C is $\frac{350}{9}\%$ of more than profit of A.
 (a) 72% (b) 75% (c) 60% (d) 48% (e) 55%
30. What is the difference between the time for which investment is made by d and e if profit of d is 28% less than profit of c and investment of d is 50% of investment of e.
 (a) 3 (b) 4 (c) 5 (d) 6 (e) 7

Directions (31-35): Given below is the table which shows five different schemes and rate of simple interest (S.I.) and rate of compound interest (C.I.) offered on these schemes.

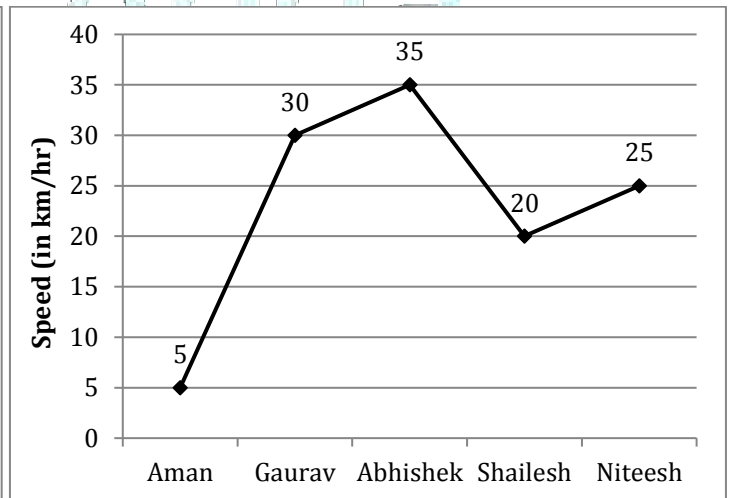
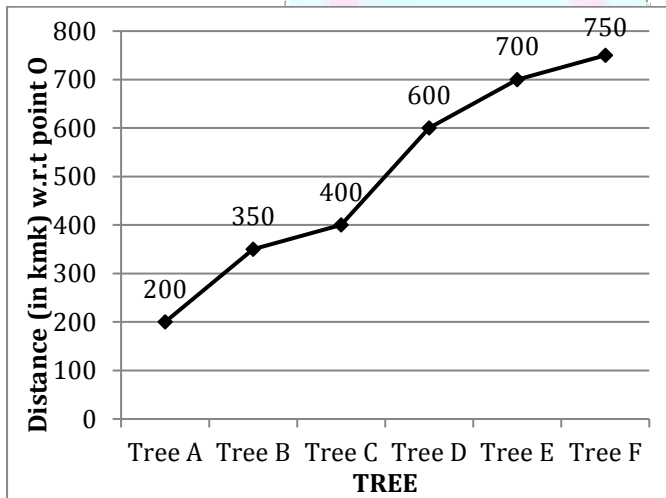
Note : All the interest is calculated annually.

SCHEME	S.I.	C.I.
A	—	8%
B	12%	—
C	—	40%
D	15%	—
E	10%	20%

31. If a sum is invested in scheme B at C.I, then amount obtained after 2 year from this scheme is 1.44 times the sum invested. Rate of simple interest for scheme A is half of the rate of compound interest for scheme B. Find out the interest earned when 8000 was invested for 2 years in Scheme A at S.I and in Scheme B at C.I for 2 years.
 (a) 5120 (b) 5000 (c) 4800 (d) Can't be determined (e) None of these
32. A man invested 10,000 in scheme D at S.I. for 6 years, the interest he obtained is divided into equal halves and invested in two different schemes i.e. scheme B and scheme C for 4 year each at S.I. If the ratio of interest obtained in both scheme is 3 : 2, then find out the rate of interest in C scheme.
 (a) 10% (b) 8% (c) 11% (d) 5% (e) None of these
33. A sum is invested in scheme E at S.I. for 2 year and then whole amount obtained is invested at C.I. in same scheme for 2 more years. If same sum would have been invested in scheme D for 4 year with S.I. then, what would have been the ration of amount obtained from scheme E to the amount obtained from scheme D.
 (a) 27 : 25 (b) 21 : 23 (c) 40 : 49 (d) Can't be determined (e) None of these
34. A man invests equal sum in two different schemes, D and E at S.I. for 4 year each. The total interest he got is invested in the scheme A for 3 year at C.I. Due to some reason instead of getting interest from scheme A, the scheme is flopped and sum invested in scheme A is depreciated each year with same rate, and he got Rs. 778688 after 3 year. Find the amount he invested in both scheme initially.
 (a) 30,00,000 (b) 40,00,000 (c) 20,00,000 (d) 10,00,000 (e) None of these
35. A sum is invested in scheme C for 5 years at S.I. and then the amount received from it is invested in same scheme for 2 years at C.I. Total amount received after 7 years is 194% more than the sum invested initially. Find out the rate of interest in scheme C for S.I.
 (a) 10% (b) 12.5% (c) 15% (d) 5% (e) None of these

Directions (36-40): Read the following line graphs and answer the following questions-

Various trees are placed in a straight line in the jungle. Distance of each tree is given with respect to point O in the jungle and speed of different persons is also given :-



36. At 8 : 00 A.M. Aman started running from tree E. At 11:00 A.M. a lion, who was at tree E saw Aman and chased him. Aman was running away from the lion and after 10 min. he increased his speed by 100%. At what time will the lion catch Aman? (speed of lion is 20 km/hr)
 (a) 12 : 30 pm (b) 12 : 25 pm (c) 01 : 00 pm (d) 01 : 25 pm (e) None of these
37. Gaurav and Abhishek start at same time from tree B to tree D, after reaching tree D they turned to tree B. At approximately how much distance from tree B they meet 1st time?
 (a) 235 km (b) 225 km (c) 230 km (d) 215 km (e) 150 km
38. Nitesh covers a distance from point O to tree E, if he stops 30 min after reaching every tree. Find the total time to cover the distance by Nitesh?
 (a) 30 hrs (b) 28 hrs (c) 20 hrs (d) 32 hrs (e) 34 hrs
39. If Gaurav is standing at tree D and Shailesh is standing at tree F. In how much time will they meet if they walk towards each other ?
 (a) 1 hr (b) $3\frac{1}{2}$ hrs (c) $2\frac{1}{2}$ hrs (d) 2 hrs (e) 3 hrs

40. Gaurav covers a distance from point O to tree C then he reduced his speed by 40% and further reach to point F. Find the average speed of Gaurav during his whole journey?

(a) $22\frac{52}{59}$ km/hr (b) $21\frac{51}{59}$ km/hr (c) $19\frac{52}{59}$ km/hr (d) $17\frac{53}{59}$ km/hr (e) $23\frac{51}{59}$ km/hr

Directions (41-45): Study the table and answer the following question

Given below is the table shows five types of laptops sold by two seller (X and Y). Table shows cost price, Profit percentage and Market price of the laptops.

Seller → Laptops ↓	X			Y		
	CP	Profit%	MP	CP	Profit%	MP
Dell	—	—	—	—	25%	—
Lenovo	—	20%	25,000	—	12%	—
HP	—	—	—	—	—	28,000
Asus	—	20%	—	—	—	—
Sony	—	35%	—	16,000	30%	—

Note: (i) Some data is missing, you have to calculate that data if it is required to answer the question.

(ii) Selling price may or may not be equal to M.P.

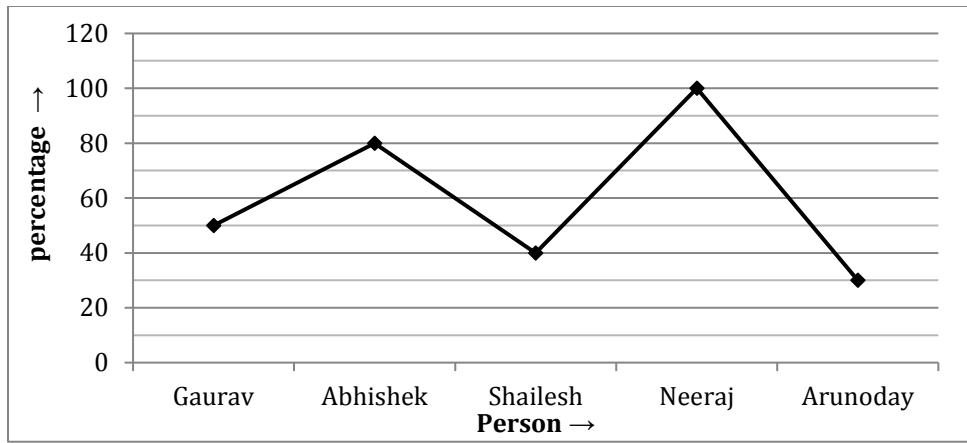
41. By what percentage C.P. of Lenovo laptop sold by seller X is less than MP of Sony laptop sold by seller Y. If X gave 10% discount on Lenovo laptop while seller Y gave 20% discount on Sony Laptop on M.P.? (approximately)
(a) 20% (b) 28% (c) 35% (d) 45% (e) 72%
42. What is the ratio between C.P. of Asus sold by seller X to C.P. of Dell sold by seller Y if M.P. of Asus sold by X is 44% more than the M.P. of Lenovo sold by X and M.P. of Dell sold by Y is 56.25% more than the M.P. of Sony sold by Y. [Take S.P. equals to M.P.]
(a) 7 : 13 (b) 14 : 15 (c) 15 : 14 (d) 13 : 15 (e) 15 : 13
43. If seller Y sells HP laptop at 20% discount, he got Rs. 2400 as profit and if he give 30% discount, he losses Rs. 400. Then what will be the profit percentage if in total 8 laptop sold by seller Y, '2' laptop at 20% discount and '6' Laptop at 30% discount.
(a) 1.5% (b) 15% (c) 5% (d) 8% (e) 12%
44. If the ratio between S.P. of Lenovo and M.P. of HP sold by Y is 3 : 4 then what is the average of Cost price of 2 laptops of Lenovo bought by X and 6 laptops of Lenovo bought by Y if X gave 10% discount of M.P.
(a) 18,000 (b) 19,000 (c) 18,500 (d) 18,705 (e) None of these
45. If the average C.P. of Asus and Sony bought by 'X' is 14,000 and average S.P. of Asus and Sony by 'X' is 18,000 then what will be the difference between the C.P of Asus and Sony laptop bought by seller 'X' ?
(a) 3,500 (b) 4,000 (c) 4,500 (d) Can't be determined (e) None of these

Directions (46-50): Read the following table and line graph carefully and answer the following questions.

Following table shows the time taken by five persons to complete a work on Monday and Ratio of Time taken by these five persons to complete the work on Monday to the time taken to complete the work on Wednesday is also given.

Line graph shows the efficiency (as a percentage) of these five persons on Tuesday with respect to that on Monday.

Person	Time taken to complete the work on Monday	Ratio of Time taken to complete the work on Monday to the time taken to complete the work on Wednesday
Gaurav	25 min.	5 : 4
Abhishek	20 min.	4 : 5
Shailesh	50 min.	10 : 7
Neeraj	10 min.	5 : 13
Arunoday	150 min.	3 : 5



46. Gaurav, Abhishek and Neeraj work in a rotation to complete the job on Tuesday with only 1 person working in a minute. Who should start the job so that the job is completed in the least possible time?
 (a) Gaurav (b) Abhishek (c) Neeraj (d) Any one of three (e) Can't determine
47. On Tuesday, Gaurav and Arunoday started the work and they worked for 5 minutes then Gaurav is replaced by Abhishek. In how many minutes Abhishek and Arunoday complete the remaining work?
 (a) $20\frac{3}{7}$ min. (b) $21\frac{4}{21}$ min. (c) $21\frac{5}{21}$ min. (d) $20\frac{4}{17}$ min. (e) None of these
48. On Tuesday, Abhishek, Shailesh and Neeraj work in a rotation in this order to complete the job with only 1 person working in a minute. They earned a total of 875 Rs. Find the share of Shailesh.
 (a) 41 Rs. (b) 31 Rs. (c) 51 Rs. (d) 49 Rs. (e) None of these
49. On Tuesday, Aman who is half as efficient as Shailesh, worked for 50 minutes on the same day then he left. In how many minutes Neeraj and Abhishek together will complete the remaining work?
 (a) $5\frac{2}{9}$ mins. (b) $4\frac{3}{7}$ mins. (c) $5\frac{3}{7}$ mins. (d) $4\frac{1}{7}$ mins. (e) $5\frac{5}{7}$ mins.
50. On Wednesday, all of them started the work together. After working for 2 minutes Gaurav left. All except Gaurav worked for another 3 minutes and then all left except Arunoday. In how much time Arunoday will complete the remaining work? (find the approximate value)
 (a) 86 minutes (b) 81 minutes (c) 96 minutes (d) 56 minutes (e) 79 minutes

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PREVIOUS YEAR SOLUTIONS

1. (c); Speed of train A on Monday = $\frac{360}{5} = 72$ km/h

Now speed increase by 20% everyday so,

Speed on Sunday = $\frac{72}{120} \times 100 = 60$ km/h

Speed on Saturday = $\frac{60}{120} \times 100 = 50$ km/h

2. (c); Distance travel by C = 270 km

Distance increases by = $270 \times \frac{100}{9 \times 100} = 30$ km

New Distance travel by C = 300 km

Actual time taken = 4 hr

Time reduced by = $4 \times \frac{125}{1000} = \frac{1}{2}$ hr

New time taken by C = $3\frac{1}{2}$ hr

Speed = $\frac{300 \times 2}{7} \approx 86$ km/hr

3. (b); Speed of A + B = $72 + 100 = 172$ km/h

Speed of C + E = $67.5 + 100 = 167.5$

Ratio = $\frac{172}{167.5} = \frac{344}{335}$

4. (a); Speed of Slowest train is C = $\frac{270}{4} = 67.5$ km/h

Fastest train is D = $\frac{450}{3} = 150$ km/h

Percentage % = $\frac{(150-67.5) \times 100}{150} = 55\%$

5. (b); Reduced distance travel by B = $300 - \frac{300 \times 20}{100} = 240$ km

Reduced distance travel by D = $450 - \frac{450 \times 20}{100} = 360$ km

New Speed of B = $\frac{240}{3} = 80$ km/hr, New Speed of D = $\frac{360}{3} = 120$ km/hr

Average of speed = $\frac{72+80+67.5+120+100}{5} = 87.9$ km/hr

6. (c); $\frac{15000 \times 9}{25000 \times 10} = \frac{x}{12500}$

$\frac{27}{50} = \frac{x}{12500}$

$x = 6750$ Rs.

$\therefore$ Required profit = $6750 + 8000 + 12500 = 27250$ Rs.

7. (b); $\frac{7000}{8000} = \frac{P_B}{14000}$

$P_B = 12250$ Rs.

$P_A = 32375 - 12250 - 14000$

$P_A = 6125$

Let Investment of A in Rajasthan = x

$\therefore \frac{x}{7000} = \frac{6125}{12250}$

$x = 3500$ Rs.

Required Ratio = $(3500) : (8000) = 7 : 16$

8. (a); Profit $\begin{matrix} A & : & B & : & C \\ (4000 \times 5) & : & (5000 \times 8) & : & (4500 \times 6) \\ 20 & : & 40 & : & 27 \end{matrix}$

$\therefore P_A = \frac{20}{87} \times 8700 = 2000$ Rs.

$P_B = \frac{40}{87} \times 8700 = 4000$ Rs.

$P_C = 2700$ Rs.

Required % = $\frac{2700}{6000} \times 100 = 45\%$

Trick :

Required value =

$\frac{27}{40+20} \times 100 = 45\%$

9. (c); $\frac{x \times 4}{(85,000 - x)6} = \frac{20,000}{30,000}$

$\frac{2x}{3(85,000 - x)} = \frac{2}{3}$

$6x = 2 \times 3 \times 85000 - 6x$

$12x = 6 \times 85000$

$x = 42500$ Rs.

$I_A = 42500$ Rs.

$I_B = 42500$ Rs.

Let Required months = y

$\therefore \frac{42500 \times 6}{17,000 \times y} = \frac{30,000}{40,000}$

$y = 20$ months

10. (e); $I_C = 30000 - 9000 - 10000 = 11000$ Rs.

$P_C = 18000 - 4500 - 6000 = 7500$ Rs.

Required % = $\frac{11000 - 7500}{11000} \times 100 = 31\frac{9}{11}\%$

11. (e); $\frac{12 \times 48}{x-6} = \frac{14 \times 24}{15+5} \times \frac{6}{7}$

$x - 6 = 40 \Rightarrow x = 46$ km/hr

12. (a); $\frac{16 \times 48}{x-5} = \frac{14 \times 24}{x+5} + 27\frac{1}{5}$

By option if we put $x = 25$

Then L.H.S. = R.H.S.

13. (c); Given

Speed of boat in still water on Saturday = 27 km/hr

and Speed of boat in still water on Wednesday

= $27 + 18 = 45$ km/hr

Now, $\frac{12 \times 48}{45-6} = \frac{18 \times 24}{27+x} \times \frac{16}{13}$

Solving, $x = 9$ kmph

14. (d); Speed of boat in still water on Saturday = 21 km/hr

Speed of boat in still water on Sunday = $21 + 6 = 27$ km/hr

$\frac{10 \times 48}{21-x} = \frac{5}{2} \times \frac{12 \times 24}{27+3}$

$21 - x = 20 \Rightarrow x = 1$ km/hr

Required time = $\frac{125}{21-1} = \frac{125}{20} = 6$ hrs 15 min

15. (e); $\frac{14 \times 48}{17-1} = 30 + \frac{11 \times 24}{x+6}$

$x + 6 = 22 \Rightarrow x = 16$ km

Upstream speed on Wednesday = $16 - 6 = 10$ km/hr

16. (a); As per given condition

$\frac{10}{60} + \frac{10}{30} + \frac{x}{15} + \frac{x}{45} + \frac{3}{45} = \left(1 - \frac{1}{45}\right)$

$\frac{4x}{45} = \frac{44}{45} - \frac{17}{30}$

$x = \frac{45}{4} \left(\frac{88-51}{90}\right) = \frac{37}{8}$ days.

17. (d); Part of work completed by T = $\frac{15}{45} = \frac{1}{3}$

3 day work by (P + R + S) = $\frac{1}{20} + \frac{1}{60} + \frac{1}{30}$

= $\frac{3+1+2}{60} = \frac{1}{10}$

15 day work = $(5P + 5R + 5S) = \frac{1}{2}$

Remaining work = $1 - \frac{5}{6} = \frac{1}{6}$

Now, this work can be done by Q in $\frac{1}{6} \times 15 = \frac{5}{2}$ days

But it has to be completed in $3\frac{1}{2}$ days.

Ratio of new efficiency to old efficiency = 5:3.

Hence increase in efficiency = $\frac{5-3}{3} \times 100 = 66\frac{2}{3}\%$

18. (c); Suppose total units of job C be 180 units.

Then, units done by P, Q, R, S and T per day is 9, 12, 3, 6 and 4 respectively.

Units completed by P, Q and R in 9 days

$$= 3(21+15+12)=144$$

Remaining $180-144 = 36$ units are completed by S and T.

ATQ,

$$6 \times 4x + 4 \times 3x = 36 \Rightarrow x = 1$$

Hence, the answer is 3 days.

19. (c); According to question

$$\frac{4}{20} + \frac{5x}{15} + \frac{10}{60} + \frac{6x}{30} + \frac{4.5}{45} = 1$$

$$\Rightarrow \frac{1}{5} + \frac{x}{3} + \frac{1}{6} + \frac{x}{5} + \frac{1}{10} = 1$$

$$\frac{6+10x+5+6x+3}{30} = 1$$

$$16x + 14 = 30 \Rightarrow x = 1$$

Required sum = $5x + 6x = 5 + 6 = 11$

20. (d); With new efficiency P will complete the job in $\frac{3}{4} \times 20 = 15$ days

And R will complete the job in $\frac{3}{4} \times 60 = 45$ days

3 days work of P and R and 1 day work of Q

$$= 3\left(\frac{1}{15} + \frac{1}{45}\right) + \frac{1}{15} = \frac{1}{3}$$

Hence, Days required = 3 days

21. (d); Let marked price for article II is $100x$

Cost price of item II for seller A

$$= \frac{100}{100+s} \times (100x - 16x) = \frac{100}{100+s} (84x)$$

Cost price of item II for seller C

$$= \frac{100}{100+3s-2} (100x - 32x) = \frac{100}{98+3s} (68x)$$

Given

$$\frac{100}{100+s} 84x = \frac{21}{17}$$

$$\frac{100}{98+3s} 68x = \frac{21}{17}$$

$$\frac{98+3s}{100+s} \times \frac{21}{17} = \frac{21}{17}$$

$$98 + 3s = 100 + s$$

$$2s = 2 \Rightarrow s = 1$$

22. (b); Let marked price for item II be $100x$

Let marked price for item III be $100y$

Let for seller D, Selling price of item II = $82x$

Let for seller D, selling price of item III = $93y$

Given

$$82x - 93y = 4810 \quad \dots (i)$$

According to question

$$100x + 100y = 8000$$

$$x + y = 80 \quad \dots (ii)$$

From eqn. (i) and (ii)

$$y = 10$$

$$x = 70$$

$$\text{MP of item II} = 100x = 7000$$

$$\text{MP of item III} = 100y = 1000$$

$$\text{Required difference} = 7000 - 1000 = 6000 \text{ Rs.}$$

23. (e); Let marked price of Item II be $100x$

SP of item II by seller A = $100x - 16x = 84x$

Let SP of item II by seller B = y

$$\therefore 84x + y = 8800 \quad \dots (i)$$

And S.P. of item II by seller C = $68x$

$$\text{Now, } y + 68x = 7200 \quad \dots (ii)$$

From eqn. (i) and (ii)

$$16x = 1600 \Rightarrow x = 100$$

$$\text{S.P. of item II by seller C} = 68x = 6800 \text{ Rs.}$$

24. (d); Let Cost price of item I by seller E = x

Let selling price of item III by same seller = y

$$\text{Given, } \frac{x}{y} = \frac{5}{6}$$

$$\text{C. P. of item I by seller E} = \frac{750}{25} \times 100 = 3000 \text{ Rs.}$$

$$\text{S. P. of item III by seller E} = \frac{6x}{5} = \frac{6 \times 3000}{5} = 3600 \text{ Rs.}$$

$$\text{C. P. of item III by seller E} = \frac{100}{100+20} \times 3600$$

$$= \frac{100}{120} \times 3600 = 3000 \text{ Rs.}$$

$$\text{Profit on item III by seller E} = 3600 - 3000 = 600 \text{ Rs.}$$

Total profit by selling item I and III together

$$= 750 + 600 = 1350 \text{ Rs.}$$

25. (c); Here total profit earned by five sellers

$$= \frac{13500}{5400 \times 5} \times 100 = 50\%$$

Let C.P. of each item = 100

Then MP of item III by each seller = $100 + 100 = 200$

$$\text{SP of item III by seller A} = \frac{68}{100} \times 200 = 136$$

$$\text{SP of item III by seller B} = 154$$

$$\text{SP of item III by seller C} = 152$$

$$\text{SP of item III by seller D} = 186$$

Total profit excluding that of seller E

$$= 36 + 54 + 52 + 86 = 228 \text{ Rs.}$$

To make 50% profit, minimum profit of item III by seller E = $(250 - 228) = 22$

$$\text{Required profit} = \frac{22}{200} \times 100 = 11\%$$

26. (a); Investment of A and C together = $215 \times 200 = 43000$

Investment of A = 18000

Investment of C = 25000

Ratio in which profit between A and C is shared

$$(18 \times 8) : (25 \times 8)$$

$$\text{Profit of A} = \frac{18 \times 10,000}{25} = 7200$$

$$\text{Sum of profit of A \& C} = 7200 + 10,000 = 17,200$$

27. (c); Profit % of E = $\frac{3}{2} \times \frac{3600}{211} = \frac{5400}{211}\%$

$$\frac{5400}{211}\% \rightarrow 10800$$

$$1\% \rightarrow \frac{10800 \times 211}{5400}$$

$$\text{So } 100\% \rightarrow 2 \times 211 \times 100 \rightarrow 42200$$

28. (d); Let both take $8x$ and $9x$ moth.

So C take 8 month and E take 9 month

$$\frac{10000}{10800} = \frac{y \times 8}{24000 \times 9} \text{ where } y \text{ is C's investment}$$

$$y = 25000$$

$$\text{Total of C and E} = 49000$$

29. (a); Let profit of A = x

$$\frac{25}{18}x = 10,000 \Rightarrow x = 7200$$

$$\text{Required \%} = \frac{7200}{10000} \times 100 = 72\%$$

30. (a); Profit of D = 7200

Investment of D = 12000

So,

$$\frac{72}{108} = \frac{12000 \times 12}{24000 \times x} \text{ Where } x \rightarrow \text{time investment of E}$$

$$\frac{2}{3} = \frac{6}{x}$$

$$x = 9 \text{ months}$$

$$\text{Required difference} = 12 - 9 = 3 \text{ months}$$

31. (a); Let sum invested in B with C.I. = x

$$\text{Acc. to question} = 1.44x = x \left(1 + \frac{r}{100}\right)^2$$

r = rate of interest of C.I. in B = 20%

Rate of interest of S.I. in A = 10%

$$\text{Interest} = \frac{8000 \times 2 \times 10}{100} + 8000 \left[\left(1 + \frac{20}{100}\right)^2 - 1 \right] = 5120$$

32. (b); Interest accrued = $\frac{10000 \times 6 \times 15}{100} = 9000$

First half 4500 on scheme B for 4 years with S.I.

$$\text{Interest} = \frac{4500 \times 12 \times 4}{100} = 2160$$

Now ratio of interest received = 3 : 2

$$\text{Interest received in scheme C} = \frac{2160}{3} \times 2 = 1440$$

$$\text{Rate of interest in Scheme C} = \frac{1440 \times 100}{4500 \times 4} = 8\%$$

33. (a); Let sum invested in each scheme = 100x

In scheme E

$$\text{Amount after 2 year at S.I} = 100x + \frac{100x \times 2 \times 10}{100} = 120x$$

$$\text{Then in C.I.} = 120x \left(1 + \frac{20}{100}\right)^2 = \frac{864}{5}x$$

In scheme D

Amount after 4 years at S.I.

$$= \frac{100x \times 4 \times 15}{100} + 100x = 160x$$

$$\text{Required ratio} = \frac{864x}{5} : 160x = 27 : 25$$

34. (c); Let amount he invested in scheme A with x

Now

$$778688 = x \left(1 - \frac{8}{100}\right)^3 \Rightarrow x = 10,00,000$$

Now this amount is the interest received from scheme D and E with S.I.

Let amount invested in both scheme = y

Total interest earn in 4 years from both scheme

$$10,00,000 = \frac{y \times 15 \times 4}{100} + \frac{y \times 10 \times 4}{100} \Rightarrow y = 10,00,000$$

sum he invested = 20,00,000

35. (a); Let Initial sum = 100x

$$\text{After 7 year Amount} = 100x + \frac{194}{100} \times 100x = 294x$$

In scheme C with C.I.

Rate of interest = 40%

Time = 2 year

Now,

$$294x = y \left(1 + \frac{40}{100}\right)^2$$

y = sum invested in scheme C with C.I.

$$y = 150x$$

amount get from scheme (with S.I.)

$$\text{Interest} = 180x - 100x = 80x$$

$$50x = \frac{100x \times R \times 15}{100} \Rightarrow R = 10\%$$

R = rate of interest for scheme C in S.I.

36. (b); Distance covered by Aman in three hours = 3×5

$$= 15 \text{ km}$$

$$\text{In 10 min. relative distance} = (20 - 5) \times \frac{10}{60} = 15 \times \frac{10}{60}$$

$$= \frac{10}{4} = 2.5 \text{ km}$$

$$\text{Remaining distance} = 15 - 2.5 = 12.5 \text{ km}$$

$$\text{Now, } \frac{12.5}{t} = (20 - 10)$$

$$\frac{12.5}{t} = 10$$

$$t = \frac{12.5}{10} = 1.25 \text{ hrs} = 1 \text{ hr } 15 \text{ min.}$$

$$\therefore \text{Required time} = 11:00 + 0:10 + 1:15 = 12:25 \text{ pm.}$$

37. (c); Ratio of their speed of Abhishek and Gaurav = 35 : 30 = 7 : 6

If Abhishek covers 250 km then at the same time

$$\text{Gaurav covers} = \frac{250}{7} \times 6 = \frac{1500}{7}$$

After Abhishek reach at tree D,

$$\text{Distance between them} = 250 - \frac{1500}{7} = \frac{250}{7}$$

To cover $\frac{250}{7}$ km, time required to meet them

$$= \frac{\frac{250}{7}}{7 \times (35 + 30)} = \frac{250}{455} = \frac{50}{91}$$

$$\text{Required distance} = \left(250 - \frac{35 \times 50}{91}\right)$$

$$= \frac{22750 - 1750}{91} \approx 230 \text{ km}$$

38. (a); To reach at tree E, time taken by Nitesh = $\frac{700}{25} = 28 \text{ hr}$

$$\text{Stopping time} = 4 \times 30 \text{ min} = 120 \text{ min} = 2 \text{ hr}$$

$$\text{Total time} = 28 + 2 = 30 \text{ hrs}$$

39. (e); Required time = $\frac{750 - 600}{(30 + 20)} = \frac{150}{50} = 3 \text{ hrs.}$

40. (a); Average speed = $\frac{750}{\frac{400}{30} + \frac{350}{18}} = \frac{750}{32\frac{5}{9}} = 22\frac{52}{59} \text{ km/hr}$

41. (b); S. P. of Lenovo sold by X = $25,000 \left[1 - \frac{10}{100}\right]$

$$= \text{Rs. } 22,500$$

$$\text{S. P.} = 22,500 = \left[1 + \frac{20}{100}\right] \times (\text{C. P.})_{\text{Lenovo}}$$

$$\Rightarrow (\text{C. P.})_{\text{Lenovo}} = \text{Rs. } 18,750$$

$$\text{S. P. of Sony sold by Y} = 16,000 \left[1 + \frac{30}{100}\right] = 20,800$$

$$(\text{M.P.})_{\text{Sony}} \times \left[1 - \frac{20}{100}\right] = 20,800$$

$$(\text{M. P.})_{\text{Sony}} = 26,000$$

Desired percentage

$$= \frac{26,000 - 18,750}{26,000} \times 100 \approx 27.88\% \approx 28\%$$

42. (e); Desired Ratio = $\frac{\text{C.P. of Asus by X}}{\text{C.P. of Dell by Y}}$

$$\text{M. P. of Asus} = \left[1 + \frac{44}{100}\right] \times 25,000 = 25,000 \times \frac{144}{100}$$

$$\text{C. P. of Asus} = 25,000 \times \frac{144}{100} \times \frac{100}{120} = 30,000$$

$$\begin{aligned} \text{M. P. of Dell} &= 16,000 \left[1 + \frac{30}{100} \right] \left[1 + \frac{56.25}{100} \right] \\ &= 16,000 \times \frac{130}{100} \times \frac{156.25}{100} \\ \text{C. P. of Dell} &= 16,000 \times \frac{130}{100} \times \frac{156.25}{100} \times \frac{100}{125} = 26,000 \\ \text{Desired Ratio} &= \frac{30,000}{26,000} = \frac{15}{13} \end{aligned}$$

43. (a); M. P. after 20% discount = $28,000 \left[1 - \frac{20}{100} \right]$
= 22,400

C.P. = 22,400 - 2,400 = 20,000

Or,

M. P. after 30% discount = $28,000 \left[1 - \frac{30}{100} \right]$
= 19,600

C.P. = 19,600 + 400 = 20,000

Net profit = $2 \times 2400 - 6 \times 400 = 2400$

Profit % = $\frac{2400}{8 \times 20,000} \times 100 = 1.5\%$

44. (e); S. P. of Lenovo by Y = $28,000 \times \frac{3}{4} = 21,000$

C. P. of Lenovo bought by X = $25,000 \times \frac{90}{100} \times \frac{100}{120}$
= 18,750

C. P. of Lenovo bought by Y = $21,000 \times \frac{100}{112} = 18,750$

Desired average = $\frac{2 \times 18,750 + 6 \times 18,750}{8} = 18,750$

45. (b); Let, C.P. of Asus = x

C.P. of Sony = y

According to question,

$$\frac{x+y}{2} = 14,000$$

$$\Rightarrow x + y = 28,000 \quad \dots (i)$$

$$\frac{x \times 1.2 + y \times 1.35}{2} = 18,000$$

$$1.2x + 1.35y = 36,000 \quad \dots (ii)$$

On solving (i) and (ii)

$$y = 16,000 \Rightarrow x = 12,000$$

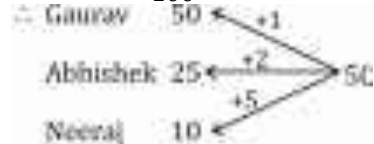
Desired difference = $16,000 - 12,000 = 4,000$

46. (c); On Tuesday

Gaurav = $\frac{25 \times 100}{50} = 50$ minutes

Abhishek = $\frac{20 \times 100}{80} = 25$ minutes

Neeraj = $\frac{10 \times 100}{100} = 10$ minutes



Clearly on Tuesday, the efficiency of Neeraj is maximum. So he should start the job so that the job is completed in the least possible time.

47. (b); On Tuesday

Gaurav = 50 minutes

Arunoday = $\frac{150 \times 100}{30} = 500$ minutes

Abhishek = 25 minutes

(Gaurav + Arunoday)'s 5 minutes work

$$= \frac{5}{50} + \frac{5}{500} = \frac{1}{10} + \frac{1}{100} = \frac{11}{100}$$

$$\text{Remaining work} = 1 - \frac{11}{100} = \frac{89}{100}$$

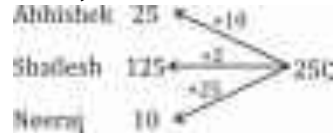
$$\text{Required time} = \frac{\frac{89}{100}}{\frac{1}{10} + \frac{1}{25}} = 21 \frac{4}{21} \text{ minutes}$$

48. (d); On Tuesday —

Abhishek = 25 min.

Shailesh = $\frac{50 \times 100}{40} = 125$ min.

Neeraj = 10 min.



(Abhishek + Shailesh + Neeraj)'s 1 minute work
= $10 + 2 + 25 = 37$ units

Shailesh will work on this job for 7 minutes.

$$\therefore \text{Share of Shailesh} = \frac{7 \times 2}{250} \times 875 = 49 \text{ Rs.}$$

49. (e); On Tuesday —

Aman = $125 \times 2 = 250$ min.

Neeraj = 10 min.

Abhishek = 25 min.

Aman's 50 min. work = $\frac{50}{250} = \frac{1}{5}$

Remaining work = $1 - \frac{1}{5} = \frac{4}{5}$

Required time = $\frac{\frac{4}{5}}{\frac{1}{10} + \frac{1}{25}} = 5 \frac{5}{7}$ minutes

50. (a); Let Arunoday worked for x minutes

$$\therefore \frac{2}{20} + \frac{5}{25} + \frac{5}{35} + \frac{5}{26} + \frac{x}{250} = 1$$

$$\frac{x}{250} = 1 - \frac{578}{910}$$

$$x \approx 91 \text{ minutes}$$

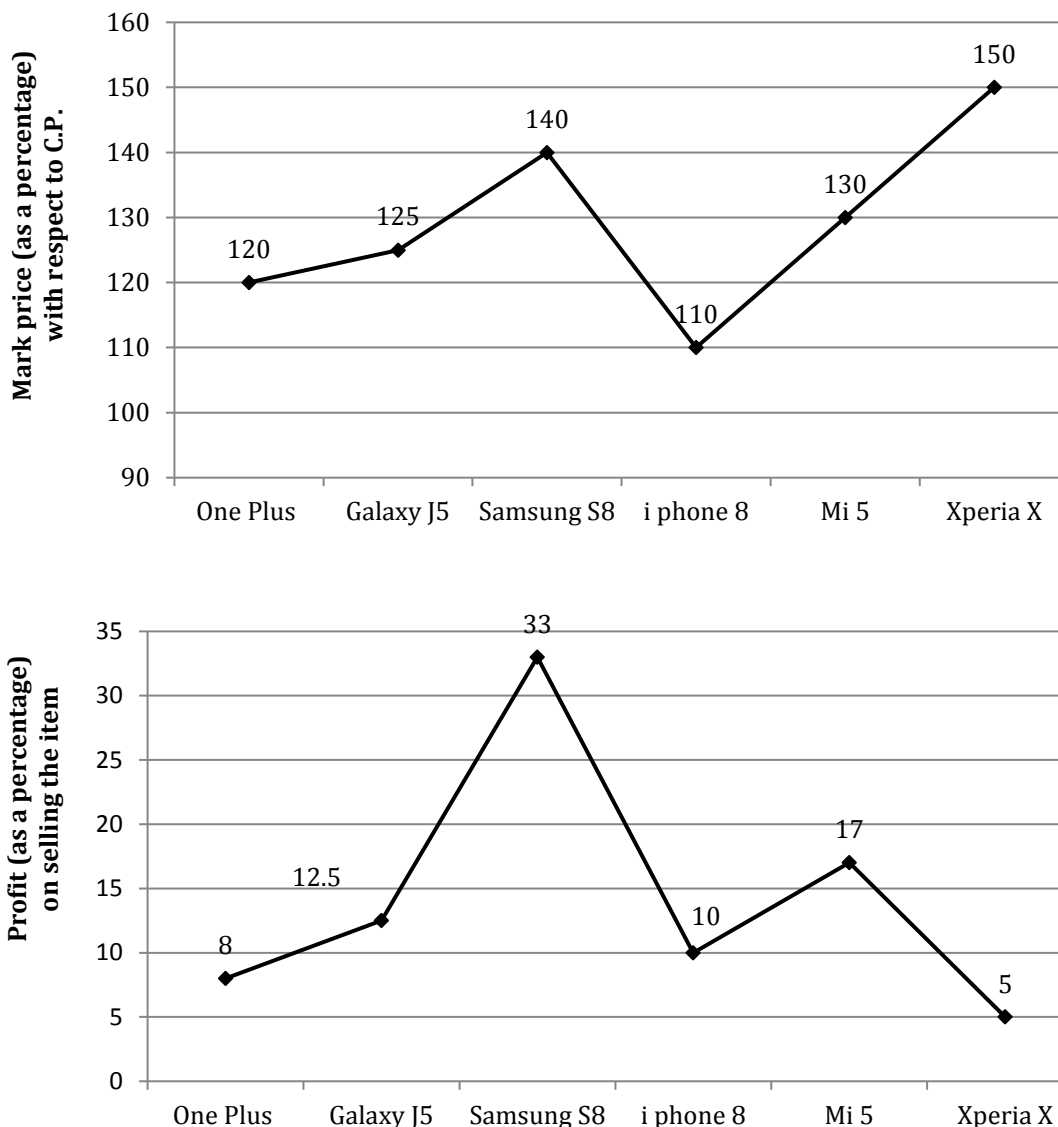
$\therefore$ Required time = $91 - 5 = 86$ minutes

Adda247



PRACTICE SET (LEVEL-I)

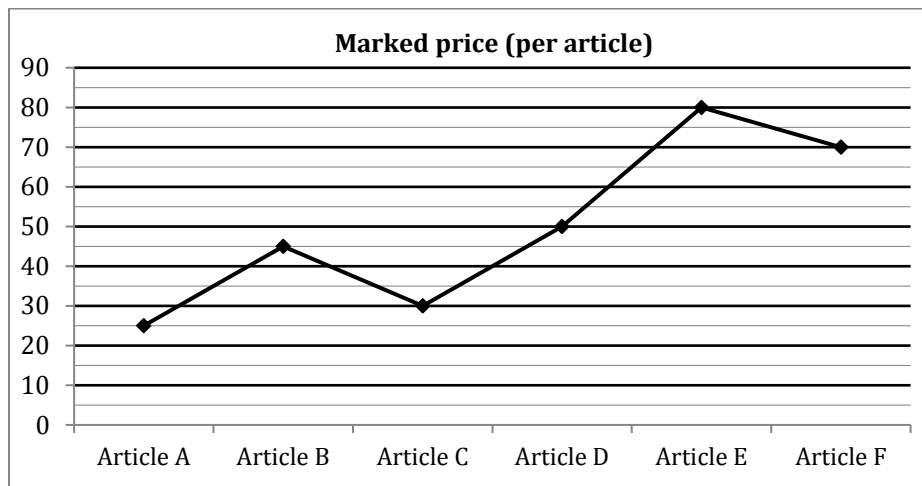
Directions (1-5): Given below are two line graphs, first line graph shows the Mark Price as a percentage of cost price and second line graph shows the profit percentage on selling the phones.



- Selling price of One plus phone is what percent more/less than the selling price of MI 5 if cost price of One plus is 50% more than the cost price of MI 5 phone ?
 (a) 39.23% (b) 38.46% (c) 38.42% (d) 39.27% (e) 33.16%
- Find the ratio between the cost price of Galaxy J5 and Xperia X mobile phones if the ratio of selling price between Galaxy J5 and Xperia X is 700 : 500 ?
 (a) 77 : 91 (b) 75 : 98 (c) 98 : 75 (d) 91 : 77 (e) None of these
- If a Redmi A5 mobile phone is to be launched in the market. Cost price of Redmi A5 is $33\frac{1}{3}\%$ more than that of MI 5 and the selling price of Redmi A5 is 30% more than that of MI 5 then find the profit percent on selling the mobile phone Redmi A5 ?
 (a) 23.015% (b) 19.040% (c) 16.075% (d) 13.050% (e) 14.075%
- If the average cost price of iphone 8 and Samsung Galaxy S8 is 5000 Rs. and cost price of i-phone 8 is 50% more than that of Samsung Galaxy S8 then find the ratio of the selling prices of i-phone 8 and Samsung Galaxy S8?
 (a) 73 : 161 (b) 161 : 73 (c) 133 : 165 (d) 165 : 133 (e) None of these
- If marked price of Samsung S8 and Xperia X is same then find the ratio of selling price of Samsung S8 and Xperia X
 (a) 133 : 88 (b) 133 : 98 (c) 144 : 133 (d) 144 : 137 (e) 169 : 143

Directions (6-10): Study the graph carefully to answer the following questions:

Given below is the Line graph which shows the Marked price per article of six different articles and table shows the number of six different kinds of articles sold.



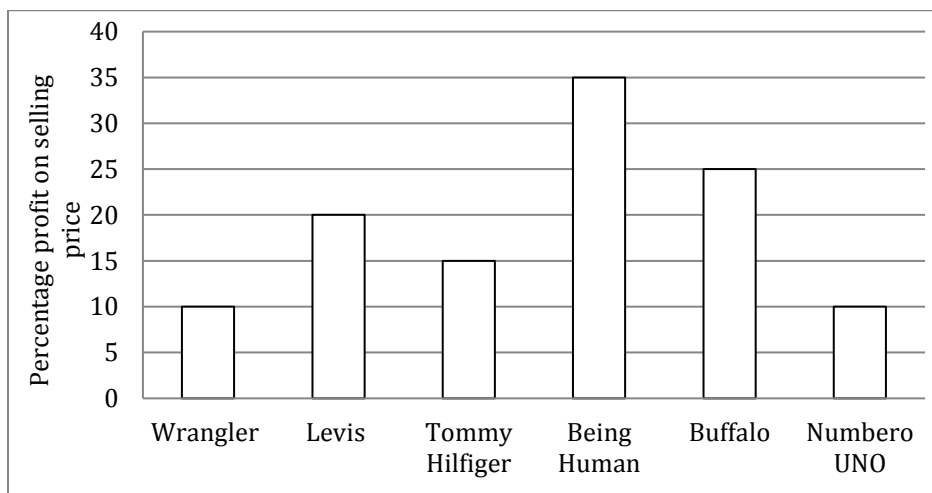
Article →	A	B	C	D	E	F
Number of Articles sold	680	560	640	440	360	160

6. If selling price per article A is 125% of its cost price and article A is sold at 25% discount on marked price then what is the total profit on all article of A.
 (a) 2550 (b) 2680 (c) 2420 (d) 2320 (e) 2590
7. If Article E is sold at 25% discount on marked price and number of article sold is doubled. Find the ratio of old profit (article sold at M.P.) to new profit if cost price is 62.5% of marked price.
 (a) 8:7 (b) 3:5 (c) 5:2 (d) 3:2 (e) 9:7
8. On selling of article F at $\frac{100}{7}\%$ discount on marked price there is the profit of 20%. If $\frac{3}{4}$ of article F are sold at $\frac{100}{7}\%$ discount and remaining at $\frac{50}{7}\%$ discount then what is the total profit on selling all articles of F at these rates.
 (a) 2400 (b) 1200 (c) 1500 (d) 1700 (e) 1800
9. Total selling price of article B is what percent more or less then total selling price all article D (all articles are sold at marked price)
 (a) $\frac{170}{12}\%$ (b) $\frac{160}{11}\%$ (c) $\frac{180}{11}\%$ (d) $\frac{150}{12}\%$ (e) $\frac{130}{11}\%$
10. Total selling price of which article is maximum (all articles are sold at marked price)
 (a) A (b) B (c) C (d) D (e) E

Directions (11-15): Study the following graphs carefully and answer the questions

Given below is the bar graph which shows the percentage profit made on the selling price on selling a Jeans of different brand by a shopkeeper.

Table shows the cost price per Jeans of different brands. Some values are missing in the table. If needed, you have to calculate these value.

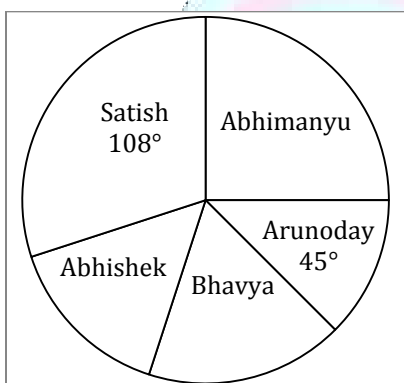


Brand	Cost price of one unit (in Rs.)
Wrangler	4500
Levis	—
Tommy Hilfiger	5100
Being Human	5200
Buffalo	—
Numero UNO	—

11. Profit of Being Human per jeans is what percent more or less than profit per jeans of Buffalo, if cost price per jeans of Buffalo is $41\frac{3}{17}\%$ less than cost price per jeans of Tommy Hilfiger.
- (a) 150% (b) 180% (c) 160% (d) 125% (e) 190%
12. What will be the ratio of profit on selling 3 jeans of tommy Hilfiger to profit on selling 4 jeans wrangler.
- (a) 23 : 18 (b) 24 : 19 (c) 25 : 19 (d) 27 : 20 (e) 28 : 23
13. If ratio of profit on selling one wrangler jeans to the profit on selling one Levis jeans is 2 : 3 then, what is the cost price of one Levis je
- (a) 3000 (b) 4500 (c) 3600 (d) 2700 (e) 2500
14. If total of cost price per jeans of Buffalo and cost price per jeans of Numero Uno is 6600 then what is the cost price of Numero Uno. Given that selling price per unit for both jeans is same.
- (a) 4500 (b) 5400 (c) 3600 (d) 2700 (e) 3000
15. If total profit on selling 'n' unit of Wrangler jeans is 7500 then what is the value of 'n' ?
- (a) 10 (b) 18 (c) 15 (d) 12 (e) 8

Directions (16-20): Given below is the pie chart which shows the distribution of distance travelled by five people out of total distance of 3200 km. Table shows the average speed and time taken by these five people in covering the distance.

Note: some value in pie chart and table are missing you have to calculate these values if required to answer the question

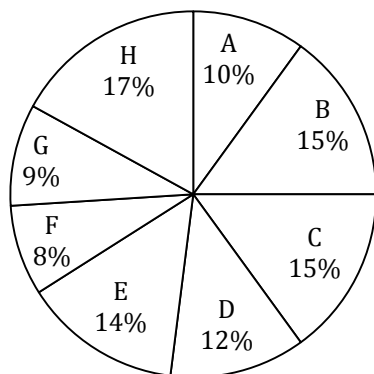


Person	Speed (in km/hr)	Time (in hrs)
Abhimanyu	—	9
Arunoday	—	5
Bhavya	64	10
Abhishek	120	—
Satish	—	20

16. If speed of Abhimanyu is 50% more than speed of Arunoday then distance travelled by Abhimanu is what percent more or less than that of Bhavya.
- (a) 65% (b) 52.25% (c) 68.75% (d) 22.5% (e) 62.5%
17. Find the time of Abhishek if it is known that the distance travelled by Abhimanyu is 87.5% of the distance travelled by Bhavya
- (a) $\frac{27}{4}$ h (b) $\frac{16}{3}$ h (c) $\frac{15}{2}$ h (d) $\frac{22}{3}$ h (e) $\frac{8}{3}$ h
18. Time taken by Abhishek is how much more less than the time taken by Arunoday if it is known that the speed of Abhimanyu is same as the speed of Arunoday.
- (a) 1 hour (b) 2 hour (c) 2.5 hour (d) 1.5 hour (e) 3 hour
19. A train, which is at a distance of 2400 km from Satish starts approaching towards Satish with a speed of x km/hr. After 5 hours from the start of train, Satish also starts to move towards train at his normal speed. If they meet after 6 hours after Satish had started to move, then find the value of x
- (a) 180 km/hr (b) 164 km/hr (c) 72 km/hr (d) 144 km/hr (e) 192 km/hr
20. If Bhavya had travelled half of distance at a speed of y km/hr and remaining at a speed of 80 km/hr then find the value of y
- (a) $\frac{180}{7}$ km/hr (b) $\frac{160}{3}$ km/hr (c) $\frac{150}{9}$ km/hr (d) $\frac{140}{9}$ km/hr (e) $\frac{170}{9}$ km/hr

Directions (21-25): Pie chart shows the % time taken by 8 different persons namely A, B, C, D, E, F, G and H to reach a particular point but the starting point is not same and total time taken by all of them together is 500 hr. (partly by car and remaining by train)

The table shows % distance travelled by car out of total distance travelled by them individually.



% Distance travelled by car	
A	70
B	50
C	30
D	45
E	60
F	75
G	40
H	60

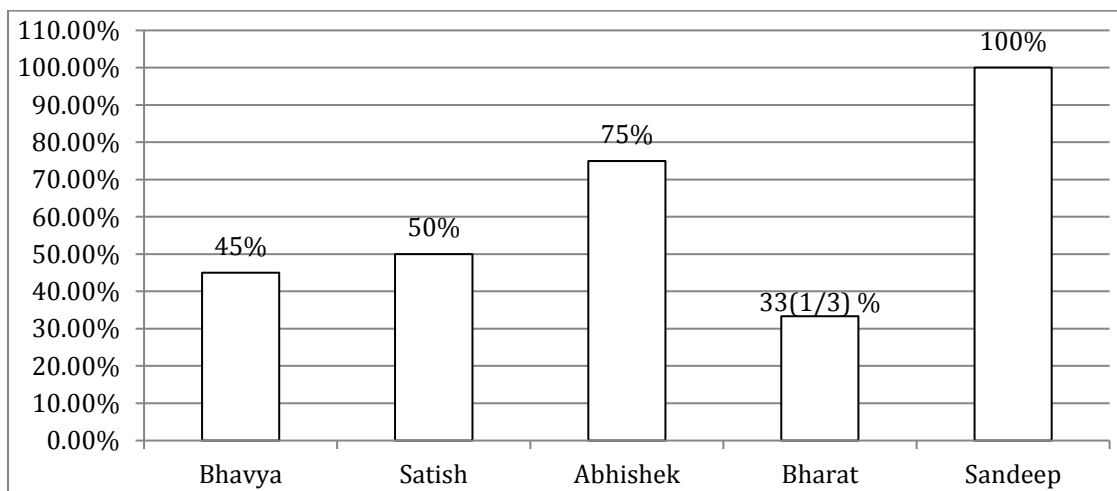
21. Find speed of F by train, if his speed by car is 60 km/hr and time taken by car and train are in the ratio of 3 : 2.
 (a) 50 km/hr (b) 45 km/hr (c) 30 km/hr (d) 20 km/hr (e) 25 km/hr
22. Find the speed of C by car, if his speed by train is 22 km/hr and time taken by car is 50% less than that by the train.
 (a) $19\frac{2}{7}$ km/hr (b) $18\frac{5}{7}$ km/hr (c) $18\frac{6}{7}$ km/hr (d) 18 km/hr (e) 48 km/hr
23. If the distance travelled by B, C and D together and E, F, G and H together is same, then find the ratio of average speed of B, C and D together and E, F, G and H together.
 (a) 6 : 5 (b) 8 : 7 (c) 7 : 8 (d) 5 : 6 (e) 7 : 9
24. Average speed of A and C are in the ratio of 4 : 3 and C have travelled 500 km more than A. Find the speed of A by car if the speed of A by train is 120 km/hr.
 (a) 80 km/hr (b) 60 km/hr (c) 65 km/hr (d) 75 km/hr (e) None of these
25. Total distance travelled by A is 2250 and the speed by train is 45 km/hr. Find the speed of A by car.
 (a) 50 km/hr (b) 45 km/hr (c) 70 km/hr (d) 40 km/hr (e) None of these

Directions (26-30): Read the following table and bar graph carefully and answer the following questions-

Five persons are travelling on different days of a week. Table shows the speed of the person on Monday and Ratio of time taken on Monday to the time taken on Tuesday for same distance.

Bar graph shows the time (as of percentage) taken by these five persons on Wednesday with respect to that on Monday for same distance.

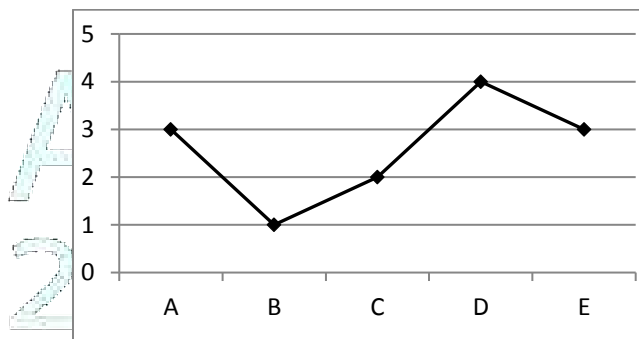
Person	Speed on Monday (in km/hr)	Ratio of time taken on Monday to time taken on Tuesday for same distance
Bhavya	45 km/h	4 : 9
Satish	60 km/h	11 : 12
Abhishek	120 km/h	2 : 3
Bharat	15 km/h	7 : 5
Sandeep	45 km/h	13 : 9



26. Bhavya covers a certain distance from his house to his office on Tuesday then he is late by 10 min. but if he travels the same distance on Wednesday then he reaches his office 5 min earlier. Find the distance from his home to his office.
 (a) $6\frac{3}{4}$ km (b) $6\frac{1}{4}$ km (c) $5\frac{3}{4}$ (d) $5\frac{1}{4}$ km (e) None of these
27. Abhishek goes to a certain place on Monday in a certain time and back to his initial point on Tuesday in a certain time find his average speed during the entire journey.
 (a) 96 km/hr (b) 108 km/hr (c) 82 km/hr (d) 72 km/hr (e) None of these
28. Bharat goes Delhi to his home on Monday in certain hours and return on Wednesday in certain hours. If he takes overall 10 hrs during the journey. Find the distance from Delhi to his home?
 (a) 150 km (b) 110 km (c) 125 km (d) 112.5 km (e) None of these
29. Arunoday daily starts from his home at a certain time with a certain speed to pick up his girlfriend from office at 6 : 00 PM. One day his girlfriend left his office at 4 : 00 pm and starts walking towards Arunoday home with a speed of 40 km/hr and meet Arunoday in the way who left his home at his usual time. They reached Arunoday home 40 min. earlier than their usual time. Then speed of Arunday is what percentage of the speed of Abhishek on Wednesday?
 (a) 75% (b) 100% (c) 150% (d) 175% (e) None of these
30. To cover a certain distance on Wednesday, Sandeep takes 15 hrs more than Satish to reach the destination. Find the time taken by Satish to reach the destination?
 (a) 5 hrs (b) 7 hrs (c) 9 hrs (d) 11 hrs (e) None of these

Directions (31-35): Given below is the table which shows the ratio of Efficiency of X and Y in completing five different works and time taken by X alone to complete these five works. Line Graph shows the number of days Y actually worked on these five works.

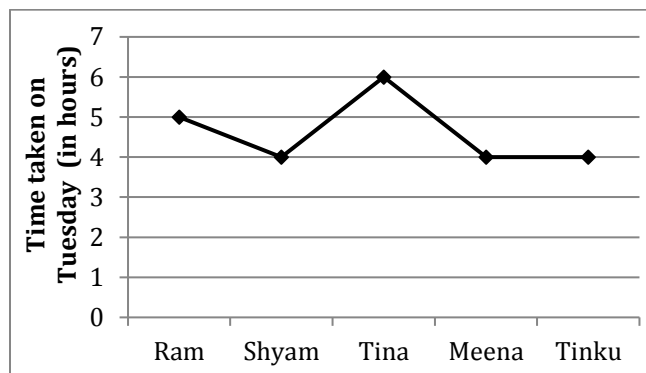
Work	Ratio of Efficiency of X and Y	Time taken by X alone to complete work (in days)
A	3 : 2	6
B	4 : 5	5
C	1 : 2	8
D	7 : 6	6
E	3 : 4	6



31. After Y had worked for the given number of days on work C and work D, X completes the remaining of work C and work D. Time taken by X in completing the remaining of work C is what percent more or less than time taken by him in completing the remaining of work D.
 (a) $\frac{100}{9}\%$ (b) $\frac{500}{9}\%$ (c) $\frac{200}{3}\%$ (d) $\frac{100}{6}\%$ (e) $\frac{200}{9}\%$
32. X and Y together started working on work D but both left after working for 2 days. Remaining work is completed by M and N together in 4 days. If ratio of efficiency of X and M in completing work D is 7 : 3, then in how much time N alone will complete the work D.
 (a) 14 days (b) 28 days (c) 21 days (d) 42 days (e) 35 days
33. If percentage of work C completed by X in 4 days is equal to the percentage of work C completed by 4 women in 5 days and ratio of efficiency of a woman and a child in completing work C is 5 : 3, then in how much time work C will be completed by 6 children.
 (a) $\frac{100}{9}$ days (b) $\frac{100}{6}$ days (c) $\frac{50}{7}$ days (d) $\frac{200}{3}$ days (e) $\frac{20}{3}$ days
34. If another person Z can complete work B in (Q - P) days where P and Q are the times taken by X and Y together to complete work B and C respectively, then what is the ratio of efficiency of Y and Z in completing work B?
 (a) 2 : 9 (b) 1 : 9 (c) 9 : 17 (d) 5 : 23 (e) 2 : 5
35. After Y had worked for the given numbers of days on work C, D and E, what is the sum of times taken by X in completing the remaining of work C, D and E?
 (a) $\frac{90}{7}$ days (b) $\frac{25}{7}$ days (c) $\frac{50}{7}$ days (d) $\frac{100}{7}$ days (e) $\frac{60}{7}$ days

Directions (36-40): Given below is the table which shows the ratio of distance travelled on Monday to Tuesday by five persons and ratio of speed of these persons on Monday to Tuesday. There is also a line graph which shows the time taken by these persons to cover the given distance on Tuesday with the speed of Tuesday.

Person	Ratio of Distance travelled on Monday to Tuesday	Ratio of speed on Monday to Tuesday
Ram	3 : 4	3 : 4
Shyam	5 : 4	1 : 4
Tina	7 : 9	2 : 3
Meena	13 : 11	13 : 22
Tinku	9 : 7	9 : 7



36. If distance travelled on Tuesday by Shyam is $45\frac{5}{11}\%$ more than the distance travelled by Meena on Tuesday, then find the ratio of speed of Shyam on Monday to speed of Meena on Tuesday.
 (a) 4 : 11 (b) 11 : 4 (c) 5 : 12 (d) 12 : 5 (e) 6 : 7
37. If Ram and Tinku are 600 km apart and they start moving towards each other with the speed of Tuesday, then they meet after 4 hours. If Ram covered 300 km on Monday then find the distance covered by Tinku on Monday.
 (a) 2700 km (b) 300 km (c) 360 km (d) 450 km (e) 500 km
38. Time taken by Tina on Monday is what percent more or less than time taken by Meena on Monday.
 (a) $33\frac{1}{3}\%$ (b) $66\frac{2}{3}\%$ (c) $14\frac{2}{7}\%$ (d) $12\frac{1}{2}\%$ (e) $15\frac{2}{5}\%$
39. What is the ratio of distance covered by Ram and Shyam on Monday if difference between Total distance covered by Ram on Monday and Tuesday together and Shyam on Monday and Tuesday together is 740 km and speed of Ram of Tuesday is 20 km/hr more than speed of Monday.
 (a) 9 : 11 (b) 7 : 8 (c) 3 : 7 (d) 5 : 3 (e) 3 : 8
40. If Shyam had travelled 800 km on Monday and Tinku had covered 360 km on Monday, then find the ratio of speed of Shyam on Monday to speed of Tinku on Monday.
 (a) 9 : 4 (b) 4 : 9 (c) 3 : 2 (d) 2 : 3 (e) 1 : 2

Directions (41-45): In the following table, the investments and profits of three businessmen in different sectors are given. Study the table carefully and solve the following questions:

	Investment (in Rs.)			Profit (in Rs.)		
	Aditya	Veer	Sushant	Aditya	Veer	Sushant
Energy	—	—	15000	—	132000	165000
Finance	—	17000	—	105000	85000	—
Technology	18000	—	—	144000	90000	—
Industrial	—	—	8000	—	30000	24000
Telecom	—	—	6000	—	—	75000

Note: 1. Apart from Telecom and Technology sector, businessmen invested the amount for same period.
 2. Some values are missing, you have to calculate these values as per given data.

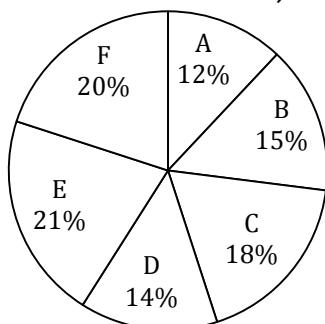
41. If the total profit in Industrial sector is Rs.81000, then find the ratio of investment by Aditya in Finance to Industrial Sector.
 (a) 3 : 7 (b) 7 : 3 (c) 5 : 9 (d) 9 : 5 (e) None of these
42. In telecom sector profit earned by Aditya, Veer and Sushant is in the ratio 4 : 5 : 3. Total amount invested by Aditya and Veer is Rs.14000 but Aditya invested for 8 month and Veer invested for 10 months. Find the period that Sushant invested his amount?
 (a) 7 Months (b) 5 Months (c) 6 Months (d) 9 Months (e) Can't be determined
43. If the average of total profit earned in Energy sector by all three businessmen is Rs.132000, then amount invested by Aditya is what percentage of the total money invested by all three businessmen in Energy Sector?
 (a) $41\frac{2}{3}\%$ (b) $33\frac{1}{3}\%$ (c) 25% (d) 18.67% (e) Can't be determined

44. In technology sector, ratio of time given by Aditya, Veer and Sushant is 4 : 3 : 5 and average of total investment is Rs.19000, then what will be the average profit made by all three businessmen?
 (a) Rs.148000 (b) Rs.185000 (c) Rs.160000 (d) None of these (e) Can't be determined
45. Total investment made by Aditya in Finance is what percentage more than the investment made by Veer in Energy sector?
 (a) 175% (b) $133\frac{1}{3}\%$ (c) 75% (d) $33\frac{1}{3}\%$ (e) None of these

Directions (46-50): Pie-chart shown below shows percentages of markers sold by six sellers.

Table shows ratio three type of marker out of total markers sold by different sellers. Study the data carefully and solve the following questions.

Total Markers sold = 15,000



Type of markers →	X	Y	Z
A	4	3	2
B	3	4	3
C	9	7	9
D	6	4	5
E	3	2	1
F	4	5	3

46. Seller 'A' fixed his selling price of markers at 40% above the cost price but at the time of selling he gave 40%, 20% and 10% discount on X, Y and Z respectively. Find the total profit or loss percentage if cost price of all the markers is same?
 (a) $2\frac{1}{3}\%$ (b) $1\frac{2}{3}\%$ (c) $3\frac{1}{3}\%$ (d) $2\frac{2}{3}\%$ (e) $1\frac{1}{3}\%$
47. Seller 'E' and 'F' keep the S.P. of each X, Y and Z markers same and total S.P. of X, Y, Z sold by E is Rs.47250. Find the total S.P. of all the markers sold by F if E kept the SP of each X, Y, Z marker in the ratio 1 : 1.5 : 3.
 (a) Rs. 48250 (b) Rs. 51250 (c) Rs. 54520 (d) Rs. 57520 (e) Rs. 45500
48. Seller 'C' sold all the markers for a certain sum and there was a loss of $11\frac{1}{9}\%$. Had it been sold for Rs.9000 more, there would have been a gain of $11\frac{1}{9}\%$. If seller 'C' wants to earn 20% profit then what would be the total S.P. of Y marker if S.P. of each marker is in the ratio 2 : 3 : 4 respectively.
 (a) Rs.13680 (b) Rs.12680 (c) Rs.13608 (d) Rs.12608 (e) None of these
49. There are two customers, Satish and Veer. Seller 'B' sells 60% of X marker to Satish, and remaining to Veer, B also sells 40% of Y marker to Satish and remaining to veer. Find the S.P. of each Y marker if Satish and Veer pays Rs.8370 and Rs.9180 for X and Y marker respectively.
 (a) Rs.10 (b) Rs.12 (c) Rs.14 (d) Rs.16 (e) Rs.18
50. Out of six sellers, which seller sells maximum number of X type of marker?
 (a) B (b) C (c) D (d) F (e) None of these

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PRACTICE SET (LEVEL-I) SOLUTIONS

1. **(b);** Let CP of MI 5 = 100
 $\therefore$ CP of One plus = 150
 SP of MI 5 = 117
 $\text{SP of One plus} = \frac{108}{100} \times 150 = 162$
 $\text{Required \%} = \frac{162-117}{117} \times 100 = \frac{45}{117} \times 100 = 38.46\%$
2. **(c);** Let SP of Galaxy J5 = 700
 $\therefore$ SP of Xperia X = 500
 $\therefore$ CP of Galaxy J5 = $\frac{100}{112.5} \times 700$
 $\text{CP of Xperia X} = \frac{100}{105} \times 500$
 $\therefore$ Required ratio = $\frac{7}{112.5} : \frac{5}{105}$
 $= 735 : 562.5 = 98 : 75$
3. **(e);** Let CP of MI 5 = 300
 $\therefore$ CP of Redmi A5 = 400
 $\text{SP of Redmi A5} = \frac{130}{100} \times 351 = 456.3$
 $\text{Required profit \%} = \frac{456.3-400}{400} \times 100 = 14.075\%$
4. **(d);** CP of iPhone 8 = 6000 Rs.
 CP of Samsung S8 = 4000 Rs.
 $\therefore$ Required Ratio = $\frac{110}{100} \times 6000 : \frac{133}{100} \times 4000$
 $= 110 \times 60 : 133 \times 40$
 $= 110 \times 6 : 133 \times 4 = 165 : 133$
5. **(b);** Let cost price of Samsung S8 and Xperia X be x and y respectively
 So,
 $140x = 150y$
 $\frac{x}{y} = \frac{15}{14}$
 So, ratio of selling price is = $\frac{133 \times 15}{105 \times 14} = 133 : 98$
6. **(a);** Let cost price per article of A = x
 Selling price per article of A = $(100\% - 25\%) \times 25$
 $= \frac{3}{4} \times 25 = 18.75$
 and
 $\frac{125}{100}x = 18.75 \Rightarrow x = 15$
 Total profit of A = $(18.75 - 15) \times 680 = 2550$
7. **(d);** Selling price per article of E = $75\% \times 80 = 60$
 Cost price per article of E = $62.5\% \text{ of } 80 = \frac{5}{8} \times 80 = 50$
 $\text{Required ratio} = \frac{(80-50) \times 360}{(60-50) \times 360 \times 2} = \frac{30}{20} = 3 : 2$
8. **(e);** Selling price of article F after discount = $\frac{6}{7} \times 70 = 60$
 Cost price of article F = $60 \times \frac{5}{6} = 50$
 Total profit
 $= \frac{3}{4} \times 160 \times (60 - 50) + \frac{1}{4} \times 160 \times (65 - 50)$
 $= 1200 + 600 = 1800$
9. **(b);** Required percentage = $\frac{45 \times 560 - 50 \times 440}{50 \times 440} \times 100$
 $= \frac{3200}{50 \times 440} \times 100 = \frac{160}{11} \%$
10. **(e);** Selling price of all article of A = 17,000
 Selling price of all article of B = 25,200

Selling price of all article of C = 19,200
 Selling price of all article of D = 22,000
 Selling price of all article of E = 28,800
 Selling price of all article of F = 11,200
 So selling price of all articles is maximum for E.

11. **(b);** Cost price per Jeans of Buffalo
 $= \left(100\% - \frac{700}{17}\%\right) \times 5100$
 $= \frac{1000}{100 \times 17} \times 5100 = 3000$
 Let selling price of Buffalo and Being Human per unit be x and y
 So,
 $3000 + \frac{25}{100}x = x \Rightarrow \frac{3}{4}x = 3000 \Rightarrow x = 4000$
 and
 $5200 + \frac{35}{100}y = y \Rightarrow 5200 = \frac{13}{20}y$
 $y = 8000$
 $\text{Required percentage} = \frac{(8000-5200)-(4000-3000)}{(4000-3000)} \times 100$
 $= \frac{2800-1000}{1000} \times 100 = 180\%$
12. **(d);** Let selling price of Tommy Hilfiger = x
 So,
 $5100 + \frac{15}{100}x = x$
 $1500 = \frac{17}{20}x \Rightarrow x = 6000$
 Profit per unit of Tommy Hilfiger = $6000 - 5100 = 900$
 Let selling price of Wrangler = y
 So, $4500 + \frac{10}{100}x = x \Rightarrow x = 5000$
 Profit on selling price per unit of Wrangler = 500
 Required ratio = $3 \times 900 : 4 \times 500 = 27 : 20$
13. **(a);** Profit on selling price per unit of Wrangler = 500 (solved in previous question)
 So profit of Levis = $\frac{500}{2} \times 3 = 750$
 Let selling price per unit of Lewis = x
 So,
 $\frac{20}{100}x = 750 \Rightarrow x = 3750$
 So cost price per unit of Levis = $3750 - 750 = 3000$
14. **(c);** Let selling price per unit for both jeans be = x
 So for Buffalo je
 $\text{Cost price} + \frac{25}{100}x = x$
 $\text{Cost price of Buffalo} = \frac{3}{4}x$
 Similarly,
 $\text{Cost price of Numero Uno} = \frac{9}{10}x$
 Ratio of cost price of Buffalo to cost price of Numero Uno
 $\text{Uno} = \frac{\frac{3}{4}x}{\frac{9}{10}x} = 5 : 6$
 Cost price of Numero Uno = $\frac{6}{11} \times 6600 = 3600$

- 15. (c);** Let selling price per unit of wrangler = x
 So,
 $4500 + \frac{10}{100}x = x \Rightarrow x = 5000$
 Profit per unit = 500
 So, $n = \frac{7500}{500} = 15$ unit
- 16. (c);** Distance covered by Arunoday = $\frac{3200}{360} \times 45 = 400$ km
 Speed of Arunoday = $\frac{400}{5} = 80$ km/hr
 Speed of Abhimanyu = 120 km/hr
 Required percentage = $\frac{120 \times 9 - 64 \times 10}{64 \times 10} \times 100$
 $= 68.75\%$
- 17. (b);** Distance travelled by Abhimanyu = $\frac{7}{8} \times 64 \times 10$
 $= 560$ km
 Distance travelled by Arunoday = 400 km
 Distance travelled by Bhavya = 640 km
 Distance travelled by Satish = 960 km
 Distance travelled by Abhishek = $3200 - 2560 = 640$
 Time of Abhishek = $\frac{640}{120} = \frac{16}{3}$ hours
- 18. (a);** Speed of Arunoday = 80 km/hr
 So, speed of Abhimanyu = 80 km/hr
 Distance covered by Abhishek
 $= 3200 - 80 \times 9 - 80 \times 50 - 640 - 960 = 480$
 Time of Abhishek = $\frac{480}{120} = 4$ hours
 Required time = $5 - 4 = 1$ hour
- 19. (e);** In 5 hours train would have covered = $5x$ km
 Speed of Satish = $\frac{960}{20}$
 So, $\frac{2400 - 5x}{48 + x} = 6$
 $2400 - 5x = 288 + 6x \Rightarrow x = 192$ km/hr
- 20. (b);** Average speed of Bhavya = 64 km/hr
 So,
 $\frac{640}{\frac{320}{y} + \frac{320}{80}} = 64$
 $y = \frac{160}{3}$ km/hr
- 21. (c);** Total time taken by F = $500 \times \frac{8}{100} = 40$ hr
 Time taken by car = $40 \times \frac{3}{5} = 24$ hr
 Distance travelled by car = $24 \times 60 = 1440$
 Distance travelled by train = $\frac{1440}{75} \times 100 \times \frac{25}{100}$
 $= 480$ km
 Time taken by train = $40 \times \frac{2}{5} = 16$
 Speed of train = $\frac{480}{16} = 30$ km/hr
- 22. (c);** Total time taken by C = $500 \times \frac{15}{100} = 75$ hr
 Now
 ATQ,
 $x + \frac{x}{2} = 75$ where x = time taken by train
 $\Rightarrow x = 50$
 Time taken by car = $75 - 50 = 25$ hr
 Distance travelled by train = $22 \times 50 = 1100$ km

$$\text{Distance travelled by car} = \frac{1100}{70} \times \frac{30}{100} \times 100$$

$$= \frac{3300}{7} \text{ km}$$

$$\text{Speed of C by car} = 18\frac{6}{7} \text{ km/hr}$$

$$\text{23. (b); Time taken by B, C and D together} = 500 \times \frac{42}{100}$$

$$= 210 \text{ hr}$$

$$\text{Time taken by E, F, G and H together} = 500 \times \frac{48}{100}$$

$$= 240 \text{ hr}$$

$$\text{Let Avg. speed of B, C and D} = x$$

$$\text{Let Avg. Speed of E, F, G and H} = y$$

$$\text{ATQ, } 210 \times x = 240 \times y$$

$$\frac{x}{y} = \frac{240}{210} = \frac{8}{7}$$

$$\text{Required answer} = 8 : 7$$

$$\text{24. (e); Time taken by A} = 500 \times \frac{10}{100} = 50 \text{ hr}$$

$$\text{Time taken by C} = 500 \times \frac{15}{100} = 75 \text{ hr}$$

$$\text{Let Avg. speed of A} = 4x$$

$$\text{Let Avg. speed of B} = 3x$$

$$\text{ATQ,}$$

$$75 \times 3x - 50 \times 4x = 500 \Rightarrow x = 20$$

$$\text{Avg. speed of A} = 4 \times 20 = 80 \text{ km/hr}$$

$$\text{Total distance travelled by A} = 80 \times 50 = 4000 \text{ km}$$

$$\text{Let the speed of A by car} = x \text{ km/hr}$$

$$\text{ATQ,}$$

$$\frac{4000}{\frac{2800}{x} + \frac{1200}{120}} = 80 \Rightarrow x = 70 \text{ km/hr}$$

$$\text{25. (b); Time taken by A} = 500 \times \frac{10}{100} = 50$$

$$\text{Avg. speed of A} = \frac{2250}{50} = 45 \text{ km/hr}$$

$$\text{Distance travelled by train by A} = 2250 \times \frac{30}{100} = 675$$

$$\text{Time taken by train} = \frac{675}{45} = 15 \text{ hr}$$

$$\text{Time taken by car} = 50 - 15 = 35 \text{ hr}$$

$$\text{Distance travelled by car} = 2250 - 675 = 1575$$

$$\text{Speed of A by car} = \frac{1575}{35} = 45 \text{ km/hr}$$

$$\text{26. (b); Speed of Bhavya on Tuesday} = \frac{45}{9} \times 4 = 20 \text{ km/h}$$

$$\text{Ratio of time taken by Bhavya on Monday to Wednesday is } 100:45.$$

$$\text{So, Ratio of speed on Monday to Wednesday is } 45:100.$$

$$\text{Speed of Bhavya on Wednesday} = \frac{45}{45} \times 100$$

$$= 100 \text{ km/h}$$

$$\text{Let distance} = d \text{ km}$$

$$\therefore \frac{d}{20} - \frac{d}{100} = \frac{15}{60}$$

$$\frac{4d}{100} = \frac{1}{4}$$

$$d = \frac{25}{4} = 6\frac{1}{4} \text{ km}$$

$$\text{27. (a); Speed of Abhishek on Monday} = 120 \text{ km/hr}$$

$$\text{Speed of Abhishek on Tuesday} = \frac{120}{3} \times 2 = 80 \text{ km/hr}$$

$$\text{Required average speed} = \frac{2 \times 80 \times 120}{(120 + 80)}$$

$$= \frac{2 \times 80 \times 120}{200} = 96 \text{ km/hr}$$

- 28. (d);** Speed of Bharat on Monday = 15 km/hr
 Speed of Bharat on Wednesday = $15 \times 3 = 45$ km/hr
 Let distance from his home to Delhi = d
 $\therefore \frac{d}{15} + \frac{d}{45} = 10$
 $\frac{4d}{45} = 10$
 $d = 45 \times 2.5 \Rightarrow d = 112.5$ km

29. (e);



Today Arunoday don't have to go from A to O or O to A. Hence he saved 20 minutes each on one way and they meet at 5 : 40 pm on the way

	Time	Speed	
Girlfriend	100	1	$\xrightarrow{40}$ 40 km/hr
Arunoday	20	5	$\xrightarrow{40}$ 200 km/hr

Speed of Abhishek on Wednesday = $\frac{120}{3} \times 4$
 $= 160$ km/hr

Required % = $\frac{200}{160} \times 100 = \frac{5}{4} \times 100 = 125\%$

- 30. (c);** Time taken by Satish = $\frac{1}{5} \times 45 = 9$ hrs

- 31. (b);** Y alone will complete work C in = $\frac{8 \times 1}{2} = 4$ days

Y alone will complete work D in = $\frac{6 \times 7}{6} = 7$ days

Part of work C and work D completed by Y in given time = $\frac{2}{4} + \frac{4}{7}$

Remaining of work C and work D is completed by X
 So,

X will complete remaining of work C = $\left(1 - \frac{2}{4}\right) \times 8$
 $= 4$ days

X will complete remaining of work D = $\left(1 - \frac{4}{7}\right) \times 6$
 $= \frac{18}{7}$ days

Required percentage = $\frac{4 - \frac{18}{7}}{\frac{18}{7}} \times 100 = \frac{500}{9} \%$

- 32. (d);** Y can complete work D in = $\frac{6 \times 7}{6} = 7$ days

Part of work D completed by X and Y in 2 days = $\frac{2}{6} + \frac{2}{7}$
 $= \frac{1}{3} + \frac{2}{7} = \frac{7+6}{21} = \frac{13}{21}$

Time taken by M in completing work D = $\frac{6}{3} \times 7$
 $= 14$ days

So,

In 4 days M will complete = $\frac{4}{14} = \frac{2}{7}$ part

M and N together complete = $\left(1 - \frac{13}{21}\right) = \frac{8}{21}$

But M completes $\frac{2}{7}$ of work D.

Remaining $\left(\frac{8}{21} - \frac{2}{7} = \frac{2}{21}\right)$ is completed by N in 4 days

So, N alone will complete work D in = $4 \div \frac{2}{21} = 42$ days

- 33. (a);** Percentage of work C completed by X in 4 days

$= \frac{4}{8} \times 100 = 50\%$

This is equal to work C completed by 4 women in 5 days

So, one woman will complete it in = 40 days

One child will complete it in $\frac{40}{3} \times 5$

6 children will complete it in = $\frac{40 \times 5}{3 \times 6} = \frac{100}{9}$ days

- 34. (b);** Y will complete work B in = $\frac{4 \times 5}{5} = 4$ day

Y will complete work C in = $\frac{8 \times 1}{2} = 4$ day

$P = \frac{5 \times 4}{9} = \frac{20}{9}$ days $\Rightarrow Q = \frac{8 \times 4}{12} = \frac{8}{3}$ days

Z will complete work B in = $\frac{8}{3} - \frac{20}{9} = \frac{24-20}{9} = \frac{4}{9}$ days

Ratio of time taken by Y and Z in completing work B
 $= 4 : \frac{4}{9} = 9 : 1$

Ratio of efficiency will be = 1 : 9

- 35. (e);** Y alone will complete work C, D and E in 4, 7 and $\frac{9}{2}$ days respectively

Part of work of C, D and E done by Y is $\frac{2}{4}, \frac{4}{7}$ and $\frac{2}{3}$ days respectively

Remaining of work C, D and E is completed by X in $\frac{1}{2} \times 8, \frac{3}{7} \times 6$ and $\frac{1}{3} \times 6$ days respectively

Required sum = $4 + \frac{18}{7} + 2 = 6 + \frac{18}{7} = \frac{60}{7}$ days.

- 36. (a);** Time taken by Shyam on Tuesday = 4 h

Let distance covered by Shyam on Monday and Tuesday be 5x and 4x respectively

And speed of Shyam on Monday and Tuesday be y and 4y respectively.

So, $\frac{4x}{4y} = 4 \Rightarrow x = 4y$

Let distance covered by Meena on Monday and Tuesday be 13 m and 11 m

And speed of Meena on Monday and Tuesday be 13n and 22n

$\frac{11m}{22n} = 4 \Rightarrow m = 8n$

According to question

$4x = \left(1 + \frac{5}{11}\right) 11m$

$4x = \frac{16}{11} \times 11m \Rightarrow x = 4m$

Or

$x = 4 \times 8n \Rightarrow x = 32n$

Required ratio = $y : 22n = \left(\frac{x}{4}\right) : \left(22 \times \frac{x}{32}\right)$

$= \frac{x}{4} : \frac{11x}{16} = 4 : 11$

- 37. (c);** Let speed of Ram and Tinku on Tuesday is 4y and 7n respectively

So,

$\frac{600}{4y+7n} = 4$

$4y + 7n = 150$

... (i)

Let distance covered by Ram on Monday and Tuesday be $3x$ and $4x$

$$\frac{4x}{4y} = 5$$

$$x = 5y \quad \dots (ii)$$

$$\text{But } 3x = 300 \Rightarrow x = 100$$

Putting x in eq. (ii)

$$y = \frac{x}{5} = \frac{100}{5} \Rightarrow y = 20$$

Putting value of y in (i)

$$4 \times 20 + 7n = 150$$

$$7n = 150 - 80 \Rightarrow n = 10$$

$$\text{Distance covered by Tinku on Tuesday} = 7 \times 10 \times 4 = 280 \text{ km}$$

Distance covered by Tinku on Monday

$$= \frac{280}{7} \times 9 = 360 \text{ km}$$

38. (d); Let Distance covered by Tina on Monday & Tuesday be $7x$ and $9x$

And speed of Tina on Monday and Tuesday be $2y$ and $3y$

$$\text{so } \frac{9x}{3y} = 6 \Rightarrow x = 2y$$

$$\begin{aligned} \text{Time taken by Tina on Monday} &= \frac{7x}{2y} \\ &= \frac{7 \times 2y}{2y} = 7 \text{ hours} \end{aligned}$$

Similarly time taken by Meena on Monday = 8 hour

$$\text{Required percentage} = \frac{8-7}{8} \times 100 = \frac{100}{8} \% \Rightarrow 12\frac{1}{2} \%$$

39. (e); Speed of Ram on Monday and Tuesday will be 60 km/hr, 80 km/hr respectively

$$\begin{aligned} \text{Distance covered by Ram on Tuesday} &= 80 \times 5 \\ &= 400 \text{ km} \end{aligned}$$

$$\begin{aligned} \text{Distance covered by Ram on Monday} &= \frac{400}{4} \times 3 \\ &= 300 \end{aligned}$$

According to question,

$(400 + 300)$ difference, distance covered by Shyam on both days = 740

Distance covered by Shyam on Monday

$$= \frac{740+700}{9} \times 5 = \frac{1440}{9} \times 5 = 800 \text{ km}$$

Required ratio = 3 : 8

40. (b); Distance travelled by Shyam on Tuesday = $\frac{800}{5} \times 4 = 640 \text{ km}$

$$\begin{aligned} \frac{640}{4} &= 4y \text{ when } (4y \text{ is speed of Shyam on Tuesday}) \\ y &= 40 \end{aligned}$$

$$\begin{aligned} \text{Distance travelled by Tinku on Tuesday} &= \frac{360}{9} \times 7 \\ &= 280 \text{ km} \end{aligned}$$

$$\frac{280}{4} = 7y \text{ (where } 7y \text{ speed of Tinku on Tuesday)}$$

$$y = 10$$

Speed of Tinku on Monday = $9y = 90 \text{ km/hr}$

Required ratio = 4 : 9

41. (b); Profit of Aditya in Industrial sector = $81000 - 30000 - 24000 = \text{Rs.}27000$

Let, the investment of Aditya in Industrial sector be Rs.x

$$\frac{24000}{8000} = \frac{27000}{x} \Rightarrow x = 9000$$

Let, the investment of Aditya in finance sector be Rs.y

$$\frac{85000}{17000} = \frac{105000}{y} \Rightarrow y = 21000$$

$$\text{Desired Ratio} = \frac{21000}{9000} = \frac{7}{3}$$

42. (a); Profit of Veer = $\frac{75000}{3} \times 5 = 125000$

$$\text{Profit of Aditya} = \frac{75000}{3} \times 4 = 100000$$

Let, the amount invested by Aditya and Veer be Rs.x and Rs.y respectively

According to the question,

$$\frac{x \times 8}{y \times 10} = \frac{100000}{125000}$$

$$\frac{x}{y} = \frac{1}{1}$$

And,

$$x + y = 14000 \Rightarrow x = 7000, y = 7000$$

Let, the time period for which Sushant invested his amount be z months

$$\therefore \frac{75000}{6000 \times z} = \frac{100000}{7000 \times 8} \Rightarrow z = 7 \text{ months}$$

43. (c); Profit of Aditya = $396000 - 132000 - 165000 = \text{Rs.}99000$

$$\begin{aligned} \text{Amount invested by Aditya} &= \frac{15000}{165000} \times 99000 \\ &= \text{Rs.}9000 \end{aligned}$$

$$\begin{aligned} \text{Amount invested by Veer} &= \frac{15000}{165000} \times 132000 \\ &= \text{Rs.}12000 \end{aligned}$$

$$\begin{aligned} \text{Desired Percentage} &= \frac{9000}{12000+9000+15000} \times 100 \\ &= \frac{9000}{36000} \times 100 = 25\% \end{aligned}$$

44. (d); Let, the time invested by Aditya, Veer Sushant $4x$, $3x$ and $5x$ respectively

Let, Amount invested by Veer = Rs.y

$$\therefore \frac{144000}{18000 \times 4x} = \frac{90000}{y \times 3x} \Rightarrow y = \text{Rs.}15000$$

Amount invested by Sushant

$$= 3 \times 19000 - 15000 - 18000 = \text{Rs.}24000$$

Let, profit of Sushant = Rs.a

$$\frac{144000}{18000 \times 4x} = \frac{a}{24000 \times 5x} \Rightarrow a = \text{Rs.}240000$$

$$\text{Average profit} = \frac{144000+90000+240000}{3} = \text{Rs.}158000$$

45. (c); Investment of Aditya in Finance = $\frac{17000}{85000} \times 105000 = \text{Rs.}21000$

$$\begin{aligned} \text{Investment of Veer in Energy} &= \frac{15000}{165000} \times 132000 \\ &= \text{Rs.}12000 \end{aligned}$$

$$\text{Desired Percentage} = \frac{21000-12000}{12000} \times 100 = 75\%$$

46. (d); Total markers sold by A = $12\% \times 15,000 = 1800$

$$\text{X marker sold by A} = \frac{1800}{9} \times 4 = 800$$

$$\text{Y marker sold by A} = \frac{1800}{9} \times 3 = 600$$

$$\text{Z marker sold by A} = \frac{1800}{9} \times 2 = 400$$

Let C.P. of one marker = 'x'

$$\text{S. P. of X marker} = \frac{140}{100} \times x \times \frac{60}{100} = 0.84x$$

$$\text{S. P. of Y marker} = \frac{140}{100} \times x \times \frac{80}{100} = 1.12x$$

$$\begin{aligned}\text{S. P. of Z marker} &= \frac{140}{100} \times x \times \frac{90}{100} = 1.26x \\ \text{Total C.P.} &= [800 + 600 + 400]x = 1800x \\ \text{Total S.P.} &= 800 \times 0.84x + 600 \times 1.12x + 400 \times 1.26x \\ &= 672x + 672x + 504x = 1848x \\ \text{Total Profit Percentage} &= \frac{1848x - 1800x}{1800x} \times 100 \\ &= \frac{48x}{1800x} \times 100 = 2\frac{2}{3}\%\end{aligned}$$

47. (b); Total markers sold by E = $\frac{21}{100} \times 15000 = 3150$

$$\begin{aligned}\text{X, Y and Z sold by E} &= 3 : 2 : 1 \\ &= 1575; 1050; 525\end{aligned}$$

$$\begin{aligned}\text{Let S.P. of each marker sold by E} &= x, 1.5x, 3x \\ \text{Total S.P.} &= x \times 1575 + 1.5x \times 1050 + 3x \times 525 \\ &= 4725x = 47250 \Rightarrow x = 10\end{aligned}$$

$$\text{S.P. of x, y, z} = 10, 15, 30$$

$$\text{Total marker sold by F} = \frac{20}{100} \times 15000 = 3000$$

$$\text{X, Y and Z sold by F} = 4 : 5 : 3 = 1000; 1250; 750$$

$$\begin{aligned}\text{Total S.P. of markers sold by F} \\ &= 10 \times 1000 + 15 \times 1250 + 30 \times 750 \\ &= 10,000 + 18,750 + 22,500 = \text{Rs. } 51250\end{aligned}$$

48. (c); Let, total C.P. = x

ATQ

$$x \times \frac{10}{9} - \left[x \times \frac{8}{9} \right] = 9000$$

$$\frac{2}{9}x = 9000 \Rightarrow x = 40,500$$

$$\begin{aligned}\text{Total S.P. of marks if C wants to earn 20\% profit} \\ &= 40500 \times \frac{120}{100} = 48600\end{aligned}$$

$$\text{Let, S.P. of each marker} = 2x, 3x, 4x$$

$$\text{Total marker sold by C} = \frac{18}{100} \times 15000 = 2700$$

$$\begin{aligned}\text{X, Y and Z marker sold by C} &= 9 : 7 : 9 \\ &= 972; 756; 972\end{aligned}$$

$$\text{Total S.P.} = 972 \times 2x + 756 \times 3x + 972 \times 4x = 8100x$$

$$\text{Total S.P. of Y marker} = \frac{756 \times 3x \times 48600}{8100x} = \text{Rs. } 13608$$

49. (b); Total markers sold by 'B' = $\frac{15}{100} \times 15000 = 2250$

$$\begin{aligned}\text{X, Y and Z markers sold by B} &= 3 : 4 : 3 \\ &= 675; 900; 675\end{aligned}$$

	Satish	Veer
X markers sold =	60%	40%
	= 405;	270

Y markers sold =	40%	60%
	= 360;	540

$$\text{Let S.P. of each X and Y marker} = x, y$$

ATQ

$$405x + 360y = 8370 \quad \dots (i)$$

$$270x + 540y = 9180 \quad \dots (ii)$$

By solving (i), and (ii)

$$x = 10, y = 12$$

50. (e); X type of Marker sold by A = $\frac{4}{9} \times \frac{12}{100} \times 15000 = 800$

$$\begin{aligned}\text{X type of Marker sold by B} &= \frac{3}{10} \times \frac{15}{100} \times 15000 \\ &= 675\end{aligned}$$

$$\begin{aligned}\text{X type of Marker sold by C} &= \frac{9}{25} \times \frac{18}{100} \times 15000 \\ &= 972\end{aligned}$$

$$\begin{aligned}\text{X type of Marker sold by D} &= \frac{6}{15} \times \frac{14}{100} \times 15000 \\ &= 840\end{aligned}$$

$$\begin{aligned}\text{X type of Marker sold by E} &= \frac{3}{6} \times \frac{25}{100} \times 15000 \\ &= 1575\end{aligned}$$

$$\begin{aligned}\text{X type of Marker sold by F} &= \frac{4}{12} \times \frac{20}{100} \times 15000 \\ &= 1000\end{aligned}$$

E sold maximum number of X type of markers

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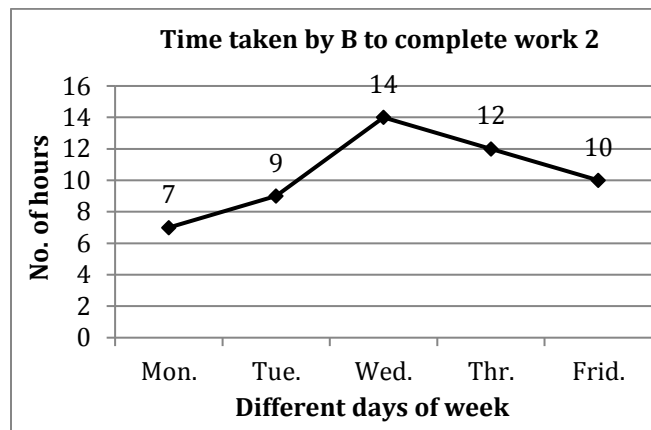
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PRACTICE SET (LEVEL-II)

Directions (1-5): Given below is table which shows the ratio of efficiency of both A and B on different days and total time taken by A and B to complete the work 1 if they complete whole work with the efficiency of different days. There is also the line graph which shows the time taken by B to complete work 2 if it complete whole work with efficiency of different days.

Days	Efficiency of A & B	Time taken by both to complete work 1(hours)
Mon.	3 : 2	3
Tue.	3 : 2	4
Wed.	7 : 9	6
Thr.	8 : 9	5
Frid.	5 : 4	8

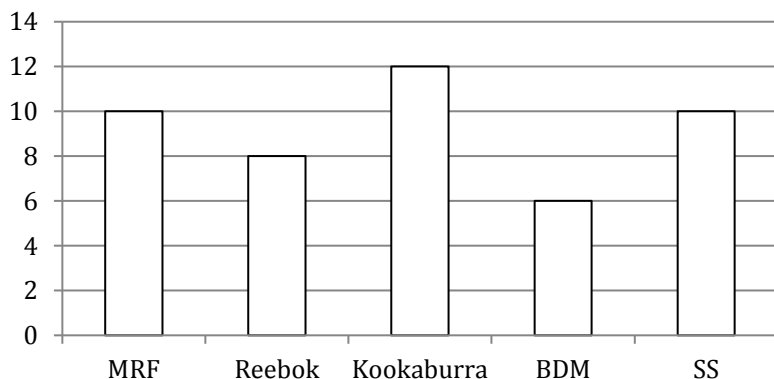


Note: The ratio of efficiency of A to B to do work 2 on different days is same as data given in the table for work 1.

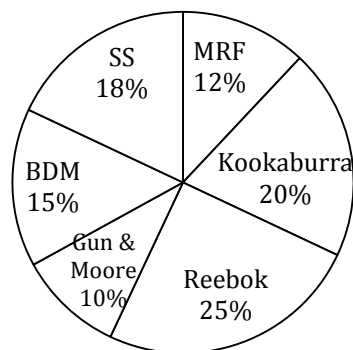
- A and B both started to complete work 1 on Tuesday but A left after working for 2 hours. Another person C whose efficiency is 60% of the efficiency of A (as of Tuesday) joins B. B leaves 2 hours before the completion of work then C alone finishes the remaining work. What is the total time in which work 1 is completed.
 (a) $\frac{105}{2}$ hours (b) $\frac{107}{13}$ hours (c) $\frac{108}{19}$ hours (d) $\frac{110}{19}$ hours (e) $\frac{110}{13}$ hours
- If a part of work 2 completed by 4 women in 5 hours equals to the part of work 2 done by B on Wednesday in 7 hours and ratio of efficiency of a women and a children to complete work 2 is 5 : 3 then in what time work 2 will be completed by 3 children.
 (a) $\frac{100}{9}$ hours (b) $\frac{200}{9}$ hours (c) $\frac{100}{11}$ hours (d) $\frac{200}{11}$ hours (e) $\frac{150}{11}$ hours
- x can complete a work in (n - m) hours while y can complete the same work in (n + m) hours where m is the time taken by A to complete work 2 on Tuesday and n is time taken by A to complete work 2 on Friday. Find the time in which x and y together can complete the work.
 (a) $\frac{3}{2}$ hours (b) $\frac{7}{4}$ hours (c) $\frac{7}{5}$ hours (d) $\frac{81}{3}$ hours (e) $\frac{9}{5}$ hours
- A and B started to complete work 1, alternatively starting from A on first hour on Monday then time taken by A and B in completing 80% of work 1, alternatively on Monday is what percent more or less than time taken by A and B together to complete work 2 together on Friday.
 (a) 3% (b) 5% (c) 8% (d) 15% (e) 6%
- If B with another person C works on work 2 on Friday for 2 hours then 80% of work 2 is completed then, time taken by C alone to finish work 2 is what percent to time taken by B to finish work 1 with efficiency of Friday —
 (a) $\frac{500}{27}$ % (b) $\frac{400}{13}$ % (c) $\frac{300}{17}$ % (d) $\frac{400}{21}$ % (e) $\frac{500}{21}$ %

Directions (6-10): The graph given below shows the number of bats & balls available in a store of different brands. Bar graph shows the number of ball and pie chart shows the number of bats

Note: There are balls of five brands and bats of six brands



Total no. of Bats = 100



NOTE- There are balls of five brands and bats of 6 brands

6. Rahul wants to buy a pair of bat and ball of same company. What is the probability that he buys a pair of his choice, if he has to pick up a pair from a bag which contains a bat and a ball of each company.
 (a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{5}{6}$ (d) $\frac{1}{7}$ (e) $\frac{1}{5}$
7. Rvinder has to choose balls of three companies i.e. MRF, BDM and SS. If he has to choose 6 balls, then in how many ways can he choose 6 balls so that there are 4 balls of BDM company.
 (a) 2430 (b) 1500 (c) 2530 (d) 2650 (e) 2470
8. The storekeeper puts all the bats in a bag. What is the probability of picking up two bats of Gun & Moore without replacement if a person wants to pick up 3 bats?
 (a) $\frac{33}{1078}$ (b) $\frac{43}{1078}$ (c) $\frac{27}{1078}$ (d) $\frac{55}{1078}$ (e) $\frac{47}{1008}$
9. Among the balls of Reebok and SS there were 4 and 6 balls defective respectively. The storekeeper offers a boy 3 chances to pick up a ball of these two companies. What is the probability of boy picking up a defective ball?
 (a) $\frac{5}{32}$ (b) $\frac{5}{34}$ (c) $\frac{5}{36}$ (d) $\frac{5}{38}$ (e) none of these
10. Banti first chooses three balls of Kookaburra and 3 bats of SS and then arranges them in a row so that no two bats or balls are together. In how many ways can he do that?
 (a) $36 \times {}^{12}C_3 \times {}^{18}C_3$ (b) $72 \times {}^{12}C_3 \times {}^{18}C_3$ (c) $42 \times {}^{12}C_3 \times {}^{18}C_3$ (d) $68 \times {}^{12}C_3 \times {}^{18}C_3$ (e) none of these

Directions (11-15): The following table shows different plans offered by a lender, type of interest and rates of interest applicable during first, second and third years.

(Note: Some values are missing, you need to calculate those values if required.)

Plans	Type of Interest	Rate of Interest		
		First Year	Second Year	Third Year
A	Simple Interest	-	$6\frac{2}{3}\%$	$3\frac{2}{3}\%$
B	Compound Interest	-	$6\frac{1}{4}\%$	-
C	Simple Interest	$8\frac{3}{4}\%$	$5\frac{1}{4}\%$	-
D	Compound Interest	$7\frac{1}{2}\%$	-	$4\frac{3}{4}\%$
E	Simple Interest	-	$5\frac{4}{5}\%$	$4\frac{3}{5}\%$

11. If two persons borrows an equal amount of Rs.12000 under plan B and plan E respectively and rate of interest for the first year under plan B and D is same, then what is the difference between second year's interests alone paid by each of them?
 (a) Rs.105.25 (b) Rs.110.25 (c) Rs.115.25 (d) Rs.120.25 (e) Cannot be determined
12. A person borrows Rs.20480 under plan C. After completion of the loan tenure of three years under plan C, he extends the tenure for further two years under plan D on the amount payable at that time. He settles his loan by paying Rs.27778. What is the rate of interest for the second year under plan D if rate of interest for the third year under plan C and D is same?
 (a) $5\frac{3}{4}\%$ (b) $5\frac{1}{4}\%$ (c) $6\frac{1}{4}\%$ (d) $4\frac{3}{4}\%$ (e) $6\frac{3}{4}\%$
13. If the amounts borrowed by a person under plan B and C are in the ratio 16 : 13 and rate of interest applicable during the first year under plan B and D is same, then what is ratio of interests payable under these plans at the end of second year.
 (a) 5 : 6 (b) 3 : 5 (c) 3 : 4 (d) 5 : 4 (e) None of these
14. The lender decides to offer a fixed rate of interest at $6\frac{2}{3}\%$ per year under plan C. By how much percent the interest payable will increase from the interest payable previously under the old plan for the period of three years if rate of interest for the third year under old plan C and plan D is same?
 (a) $6\frac{1}{3}\%$ (b) $6\frac{1}{4}\%$ (c) $6\frac{2}{3}\%$ (d) $6\frac{2}{5}\%$ (e) Cannot be determined
15. Rates of interest for the first year under plan A and E are $8\frac{2}{3}\%$ and $7\frac{3}{5}\%$ respectively. A person borrows a total of Rs.30000 partially under plan A and E and pays a total interest of Rs.5540 at the end of third year. How much amount does he borrow under plan A?
 (a) Rs.14000 (b) Rs.18000 (c) Rs.16000 (d) Rs.12000 (e) Rs.20000

Directions (16-20): Given below is the table which shows the percentage of profit obtained on five different products sold by a shopkeeper in 5 different months.

Note:

Some values are missing from the table, you have to calculate these values if required.

Assume that the cost price of different products can be different and the cost price of same product in different months can also be different as well.

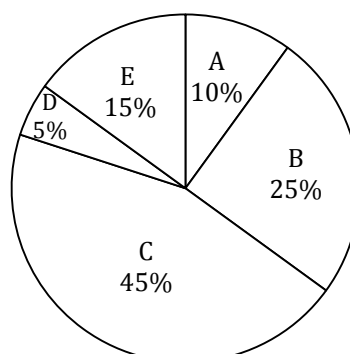
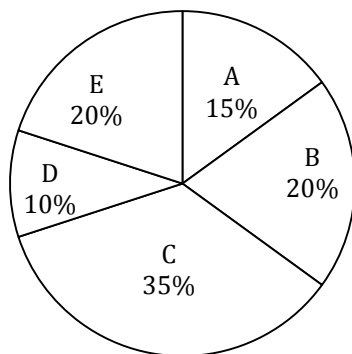
Marked Price = Selling Price + Discount on Marked Price

And, Selling Price = Cost Price + Profit

Month	% profit on A	% profit on B	% profit on C	% profit on D	% profit on E
Jan	15	20	30	25	—
Feb	22	—	30	8	—
March	10	—	—	24	20
April	—	25	13	10	25
May	25	30	15	20	10

16. If ratio of cost price of products D and E in January is 5 : 6 and sum of profit on D and E in January is 150% of profit on E in January, then what is the percentage profit of E in January? (If profit of E is calculated on selling price).
 (a) $\frac{250}{19}\%$ (b) $\frac{500}{17}\%$ (c) $\frac{600}{19}\%$ (d) $\frac{700}{13}\%$ (e) None of these
17. Average of profit obtained on products B, C, D and E in January is equal to average of profit obtained on products A, B, C and E in February. If total profits in January on all five products is Rs.4900 more than that in February and cost price of A in January and cost price of D in February is same, then what is the profit of A in January?
 (a) Rs.10200 (b) Rs.7500 (c) Rs.8500 (d) Rs.9800 (e) Rs.10500
18. Cost price of A in January is equal to cost price of B in May and percentage discounts on marked price of A in January and B in May are $2n\%$ and $(25 + n)\%$ respectively. If ratio of marked price of A in January and marked price of B in May is 23 : 26, then find the value of n.
 (a) 20 (b) 22 (c) 23 (d) 25 (e) 18
19. If out of 10 kg of product D bought by the shopkeeper in March, 2 kg got spoiled and has to be sold at half the selling price and remaining quantity is sold at the normal selling price, then what will be the overall profit or loss percentage?
 (a) 11.6% (b) 10.8% (c) 15.4% (d) 13.5% (e) 8.5%
20. If cost price of product D in April and May is same and cost price of product E in April and May is same; and average of selling price of D and E in April and average of selling price of D and E in May are Rs.650 and Rs.630 respectively, then find the cost price of E in May.
 (a) Rs.500 (b) Rs.600 (c) Rs.550 (d) Rs.400 (e) Rs.650

Directions (21-25): Given below are two pie charts. Pie chart I shows the percentage distribution of milk in five vessels out of the total milk in these five vessels. Pie chart II shows the percentage distribution of water in same five vessels out of total quantity of water in these five vessels.



Note: Ratio of total milk to total water in these five containers is 2 : 1.

21. A shopkeeper pours the mixture of vessel A and B into another vessel F. Vessel F contains water only which is equal to 25% of water of vessel B. If shopkeeper professes to sell the whole mixture at the cost price of pure milk and cost price for shopkeeper is due to milk only, then find the percentage profit of shopkeeper in selling whole mixture.
 (a) $58\frac{13}{14}\%$ (b) $3\frac{13}{15}\%$ (c) $54\frac{13}{15}\%$ (d) $53\frac{13}{14}\%$ (e) $55\frac{20}{21}\%$

22. Mixture of vessel A and C are mixed into another vessel M. If 62 liters of the mixture M is taken out and replaced with 17 L of water, the ratio of milk to water in M becomes 6 : 5. Find the quantity of milk in vessel B.
 (a) 60 L (b) 20 L (c) 40 L (d) 45 L (e) 50 L
23. All the contents of mixture from all vessels except C is poured into bigger vessel and from vessel C, only 115 liters of mixture is taken out and poured into bigger vessel, then ratio of milk and water in bigger vessel becomes 9 : 4. Find the total quantity of water in all five vessels.
 (a) 550 L (b) 500 L (c) 600 L (d) 650 L (e) 700 L
24. Some quantity of mixture from vessel D and E are taken out and ratio of quantity taken out from vessel D and E is in ratio 3 : 2, then find the ratio of milk and water of the final mixture.
 (a) $\frac{211}{47}$ (b) $\frac{202}{59}$ (c) $\frac{220}{89}$ (d) $\frac{112}{57}$ (e) $\frac{212}{63}$
25. If total quantity of milk and water in vessel C is 115 L, then the quantity of milk and water is vessel B is how much more or less than quantity of milk and water in vessel E?
 (a) $15\frac{2}{11}\%$ (b) $14\frac{5}{7}\%$ (c) $15\frac{5}{6}\%$ (d) $18\frac{2}{11}\%$ (e) $17\frac{2}{11}\%$

Directions (26-30): In this table, number of leaps taken by the different animal in one minute and ratio of distance covered by the animal in one leap to the distance covered by Lion in one leap are given.

Animals	No. of leaps in one minutes	Ratio of distance covered by the animal in one leap to the distance covered by Lion in one leap
Cat	5	2 : 5
Monkey	6	3 : 4
Dog	7	4 : 7
Jackal	4	5 : 8
Rabbit	6	1 : 5

26. Dog sees a stationary cat and estimates that it will catch the cat in 1 min and starts moving towards it. At the same time cat also starts moves away from the dog. In how much time the dog will catch the cat?
 (a) 1 min. (b) 2 min. (c) 4 min. (d) $1\frac{1}{2}$ min. (e) None of these
27. Jackal sees a Rabbit and finds that it is 2 minute away from him. Jackal starts to run towards Rabbit and at the same time rabbit also starts to run away from Jackal. In which minute Jackal will catch the Rabbit?
 (a) Jackal could not catch the rat (b) In 4th min (c) In 5th min.
 (d) In 6th min. (e) None of these
28. What is the ratio of the speed of Dog to Cheetah, if Cheetah covers 50% of more distance in one leap than lion and sum of leaps taken by Cat and Monkey in one minute is equal to the number of leaps taken by Cheetah in one minute?
 (a) 2 : 5 (b) 5 : 8 (c) 11 : 13 (d) 8 : 33 (e) None of these
29. When a Cat climb up a tree its speed is reduced by 20% and when Monkey climb up a tree its speed is increased by 20%. Cat's speed on climbing up a tree is what percent of speed of Monkey when it climbs the tree. (Approximately)
 (a) 30% (b) 40% (c) 50% (d) 10% (e) 15%
30. What is the sum of distance cover by all the 5 animals in 2 minutes if distance cover by Jackal in 1 minute is 10 meters?
 (a) 80 m (b) 70.8 m (c) 68.3 m (d) 16 m (e) None of these

Directions (31-35): Study the following table carefully to answer the questions that are based on it.

A toymaker makes different types of toys by joining various solids as given in table below. Some values are missing, you have to calculate these values if required to answer the question.

Dimensions→ Type of solids ↓	Diameter (in cm)	Length (in cm)	Breadth (in cm)	Height (in cm)
Cylinder	—	—	—	12
Cube	—	—	—	—
Cuboid	—	24	—	10
Cone	14	—	—	—
Sphere	21	—	—	—
Hemisphere	—	—	—	—

31. The toymaker makes a toy in which the cone is mounted on the base of the hemisphere. If the total surface area of the toy is 858 cm^2 , then find the volume of the toy? (bases of hemisphere and cone are equal)
 (a) $1950\frac{2}{3} \text{ cm}^3$ (b) $1250\frac{2}{3} \text{ cm}^3$ (c) 1400 cm^3 (d) 1500 cm^3 (e) $1950\frac{1}{3} \text{ cm}^3$

32. Toymaker mounted the cube on the cylinder such that cylinder top is exactly in the middle of the face of the cube and all sides of bottom face of the cube touch the circumference of top face of the cylinder. Find the total surface of the toy formed, if the height of formed toy is twice the height of cylinder and curved surface area of cylinder is 66 times the height of cylinder
 (a) 3125 cm^2 (b) 2794.5 cm^2 (c) 4112 cm^2 (d) 5123 cm^2 (e) 1656 cm^2
33. If given sphere is cut into two hemisphere and these hemispheres are mounted on both ends of the cylinder, then find out the ratio of volumes of toy formed by joining both hemispheres on cylinder, cylinder and sphere. (radius of sphere and cylinder is equal)
 (a) 7 : 6 : 13 (b) 6 : 13 : 7 (c) 13 : 6 : 7 (d) 13 : 7 : 6 (e) 13 : 8 : 7
34. Volume of the cuboid is approximately what percent more or less than the volume of cone if slant height of cone is 25 cm and the breadth of the cuboid is 25% of the height of cone.
 (a) 7% (b) 11% (c) 14% (d) 17% (e) 21%
35. Find the cost of painting a toy at the rate of Rs. 25 per cm^2 , if the toy is formed by joining the cone and the hemisphere on two ends of cylinder. (Given that radius of all three bodies are equal and height of cone is 24 cm)
 (a) Rs. 38650 (b) Rs. 34650 (c) Rs. 34560 (d) Rs. 36560 (e) Rs. 32550

Directions (36-40): Study the given graph carefully to answer the questions that follow:

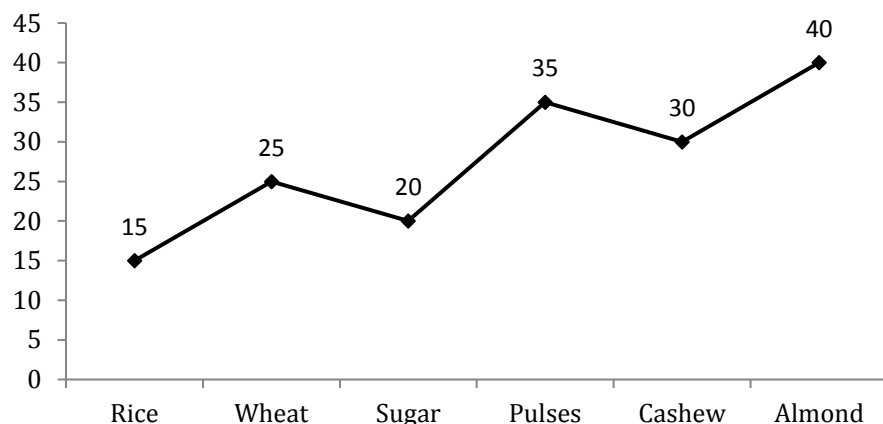
Table below shows different items sold by shopkeeper, Quantity of items sold in k.g., discount offered on list price, percentage mark up price/kg and list price /kg.

Some value are missing. You need to calculate these values if required.

Items	Quantity	Discount offered %	Markup % per kg	List price per kg
Tea	12	-	$16\frac{2}{3}\%$	-
Coffee	-	20%	-	-
Wheat	70	-	-	42
Rice	80	-	$33\frac{1}{3}\%$	-
Honey	-	$26\frac{2}{3}\%$	-	1500

36. List price per kg of wheat is what percent less than the list price per kg of tea if selling price of 2.5 kg of tea is 900 Rs. and discount offered on list price is $\frac{400}{49}\%$.
 (a) 80% (b) 83% (c) 93% (d) 90% (e) 85%
37. What is the ratio of percentage discount offered on wheat to the percentage discount offered on Rice if shopkeeper gains 1330 Rs. on selling whole quantity of wheat and % markup price of wheat is 110% and percentage of discount offered on rice is half the percentage discount offered on Honey.
 (a) 15 : 28 (b) 18 : 23 (c) 24 : 29 (d) 23 : 21 (e) None of these
38. What is the average of all quantities (in kg) sold by the shopkeeper if total profit in selling Honey is Rs.1800 and ratio of difference between List price and selling price to the difference between selling price and cost price of Honey is 4 : 3 and quantity of coffee sold is $33\frac{1}{3}\%$ less than quantity of tea sold.
 (a) 35.2 (b) 36.4 (c) 28.2 (d) 32.2 (e) 30.5
39. 8 kg of rice got spoiled then what is the total profit or loss made by the shopkeeper in selling remaining quantity of rice if a discount of $16\frac{2}{3}\%$ is given and list price per kg of rice is 60 Rs.
 (a) 15% (b) 20% (c) 18% (d) No profit or No loss (e) None of these
40. What is the ratio of cost price per kg of Tea to the cost price per kg of coffee if profit of 12% is obtained on selling Coffee and List price per kg of both items is same.
 (a) 6 : 5 (b) 4 : 5 (c) 3 : 2 (d) 2 : 3 (e) 7 : 4

Directions (41-45): Study the graph and table given below and answer the following questions. The line graph shows the listed price per kg of various items in a wholesale store.

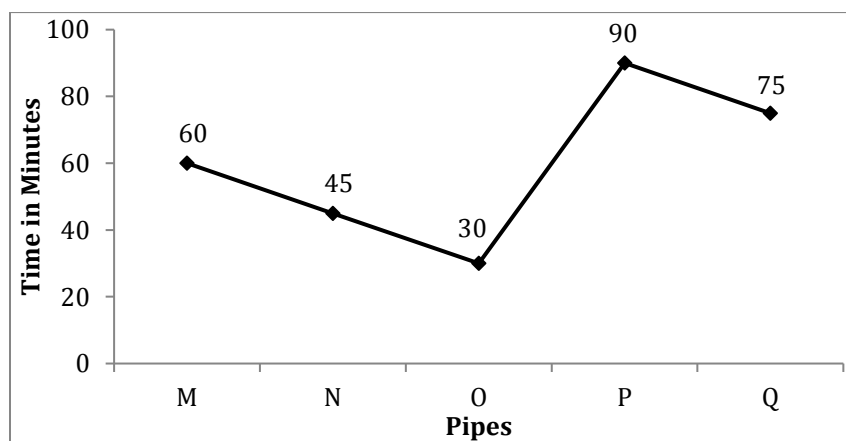


The table given below shows the amount of items bought by a retailer from the wholesale store. The table also shows the discount % offered by the wholesaler on the list price and total cost incurred by the retailer.

Items	Quantity (in kgs)	Discount (in %)	Total (in Rs.)
Rice	20	10	-
Wheat	30	-	675
Sugar	15	-	240
Pulses	18	30	-
Cashew	40	-	900
Almond	25	15	-

41. Calculate the profit earned by retailer on selling 20 kgs of wheat purchased by him to a customer at a discount of 5% on the listed price ?
 (a) Rs. 25 (b) Rs. 45 (c) Rs. 75 (d) Rs. 50 (e) None of these
42. The retailer sold all the cashew bought by him to a customer at a price 25% more than the listed price. Calculate his overall profit percent.
 (a) 33.33% (b) 66.66% (c) 55.55% (d) 42.64% (e) 77.77%
43. If 50% of the rice bought by the retailer got spoiled, then at what price/kg must he sell the remaining amount of rice to be at a situation of no loss-no gain ?
 (a) Rs. 40 (b) Rs. 19 (c) Rs. 27 (d) Rs. 22 (e) None of these
44. The retailer sold all the pulses he bought at a price that is 30% more than the listed price and offered 2 kgs of Almond free with it. Find overall profit% of the retailer in this bargain ? (approximate)
 (a) 50% (b) 40% (c) 35% (d) 61% (e) 45%
45. The retailer mixed 6 kgs. of impurity (free of cost) with all the sugar he had and sold the mixture at a discount which is 25% less than that discount (in percentage) offered by the wholesaler. Find the profit % on the sale of all of the amount of this mixture?
 (a) 52.50% (b) 46.15% (c) 48.75% (d) 57.50% (e) None of these

Directions (46-50): Given below is the line graph which shows the time taken by five pipes M, N, O, P and Q to fill a tank individually in minutes. Table shows the pipes which remain open to fill the same tank on different days of week



Days	Pipe which remain on different days.
Mon	M, P
Tue	N, O, P
Wed	O, M
Thrus	N, O, P
Frid	M, Q

46. If on Monday pipe M works with a efficiency of 120 % and both pipe M and P on Monday remained open for 7 minutes but alternatively on each minute starting with M. Remaining part of the tank is filled on next day. What is the total time for which pipe P remained open on both days if next day all pipes filled the tank together?
 (a) 12 min 15 sec (b) 16 min 18 sec (c) 18 min 15 sec (d) 17 min 12 sec (e) 20 min 10 sec
47. On Wednesday tank is filled by both pipes working simultaneously but on Thursday all pipes work alternatively on each minute starting from pipe N then O and then P. Find the difference in time taken to fill the tank on Wednesday and the time taken be fill tank on Thursday.
 (a) 5 min (b) 18 min (c) 15 min (d) 20 min (e) 25 min
48. If on Friday 36 litre of water per minute is filled by both the pipes then, amount of water filled by pipe P on Monday is what percent of amount of water filled by pipe Q on Friday.
 (a) 80% (b) 95% (c) 90% (d) 80% (e) 75%
49. On Friday M worked with 120% of its efficiency and Q with 75% of its efficiency and they together can fill 162 litre of water in 12 min. On Monday if both the pipes (M and P) are working with a different efficiency then both pipes working together can fill $\frac{7}{30}$ part of the tank in 8 min and if M is opened for 8 min and P is opened for 15 min then they can fill 157.5 litre of water. Find the ratio of time taken by M alone and time taken by P alone to fill tank according to new efficiency on Monday.
 (a) 2 : 3 (b) 4 : 3 (c) 3 : 4 (d) 5 : 3 (e) 3 : 5
50. If rate of flow of pipe N is 18 litre/min, and cost incurred in filling 1 litre of water in the tank by pipe N, O and P is 12 Rs./L, 15 Rs./L and 10 Rs./L respectively, then find the total cost incurred in filling the tank on Tuesday if all the pipes filled the tank simultaneously.
 (a) 10,665 Rs. (b) 11,552 Rs. (c) 12,666 Rs. (d) 9,848 Rs. (e) 8,440 Rs.

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PRACTICE SET (LEVEL-II) SOLUTIONS

1. (c); Let A and B can do $3x$ and $2x$ unit of work 1 in one hour respectively.

So, total work 1 done by both = $(3x + 2x)4 = 20x$

A alone will complete work 1 = $\frac{20x}{3x} = \frac{20}{3}$ hours

B alone will complete work 1 = $\frac{20x}{2x} = 10$ hours

Ratio of efficiency of A and C = $5 : 3$

Ratio of time taken by A and C = $3 : 5$

C alone will complete work 1 = $\frac{20}{3 \times 3} \times 5$ hours

= $\frac{100}{9}$ hours

Let total time taken in completing work 1 is y

So,

$$\frac{2}{\frac{20}{3}} + \frac{(y-2)}{10} + \frac{(y-2)}{\frac{100}{9}} = 1$$

$$\frac{(y-2)}{10} + \frac{9(y-2)}{100} = \frac{7}{10}$$

$$10y - 20 + 9y - 18 = 70$$

$$y = \frac{108}{19} \text{ hours}$$

2. (b); Part of work 2 done by B on Wednesday in 7 hours = $\frac{1}{2}$

This part of work done by 4 women in 5 hours.

So whole work will be completed by 4 women in = 10 hours

One women will complete it in = 40 hours

3 children will complete it in = $\frac{40 \times 5}{3 \times 3} = \frac{200}{9}$ hours

3. (b); Ratio of efficiency A and B on Tuesday = $3 : 2$

Let A and B does $3x$ and $2x$ work in one hour

And B completes work 2 in 9 hours

So, total work = $9 \times 2x = 18x$

A will complete work 2 in $\frac{18x}{3x} = 6$ hours.

So, $m = 6$

Similarly

$$n = \frac{10 \times 4x}{5x} = 8$$

Total x and y will complete the work in

$$= \frac{(8-6)(8+6)}{(8-6)+(8+6)} = \frac{28}{16} = \frac{7}{4} \text{ hours}$$

4. (b); Let A and B can do $3x$ and $2x$ work in one hour on Monday

Then 80% of total work 1 = $\frac{4}{5}(3x + 2x) \times 3 = 12x$

In 4 hours $10x$ work 1 is completed working alternatively and remaining $2x$ is complete by A on 5th hour

So total time = $\left(4 + \frac{2x}{3x}\right)$ hours = $\frac{14}{3}$ hours

Ratio of efficiency on Friday is $5 : 4$

Ratio of time taken to complete work will be $4 : 5$

But B completes work 2 in 10 hours on Friday

So, A will complete work 2 in 8 hours on Friday

$\therefore$ Together they will complete work 2 in = $\frac{8 \times 10}{18}$

$$= \frac{40}{9} \text{ hours}$$

$$\text{Required percentage} = \frac{\left(\frac{14}{3} - \frac{40}{9}\right)}{\frac{40}{9}} \times 100$$

$$= \frac{\frac{42-40}{9}}{\frac{40}{9}} \times 100 = \frac{2}{40} \times 100 = 5\%$$

5. (a); Let C complete work 2 in x hours

According to question.

$$\frac{2}{10} + \frac{2}{x} = \frac{4}{5}$$

$$\frac{2}{x} = \frac{4}{5} - \frac{1}{5}$$

$$\frac{2}{x} = \frac{3}{5}$$

$$x = \frac{10}{3}$$

Time taken by B to finish work 1 on Friday

$$= \frac{(5+4)8}{4} = 18 \text{ hours}$$

$$\text{Required percentage} = \frac{10}{3 \times 18} \times 100 = \frac{500}{27}\%$$

6. (a); Since, there are bats of six companies & balls of five companies,

Hence, possible no. of pairs = $6 \times 5 = 30$

No. of favourable pairs = 5

$$\text{Req. probability} = \frac{5}{30} = \frac{1}{6}$$

7. (b); Possible ways are :

MRF BDM SS

1 4 1

Hence no. of ways = $10c_1 \times 10c_1 \times 6c_1 = 1500$

8. (c); Possible ways :

(i) First two Gun & Moore, last one diff.

(ii) First & third G & M, second one diff.

(iii) First diff., Last two G & M

$$\text{Req. probability} = 3 \times \frac{10 \times 9 \times 90}{100 \times 99 \times 98} = \frac{27}{1078}$$

9. (b); Since out of $8 + 10 = 18$ balls, $4 + 6 = 10$ are defective.

$$\text{Hence, req. probability} = \frac{{}^{10}C_3}{{}^{18}C_3} = \frac{120}{816} = \frac{5}{34}$$

10. (b); No. of ways of choosing 3 balls of Kookabura and 3 bats of SS = ${}^{12}C_3 \times {}^{18}C_3$

No. of ways of arranging them = $2 \times 3! \times 3! = 72$

Therefore, req. answer = $72 \times {}^{12}C_3 \times {}^{18}C_3$

11. (b); Difference between second year's interests

$$= 12000 \left(1 + 7\frac{1}{2}\%\right) \left(6\frac{1}{4}\%\right) - 12000 \left(5\frac{4}{5}\%\right)$$

$$= 12000 \left(\frac{43}{40}\right) \left(\frac{1}{16}\right) - 12000 \left(\frac{29}{5}\%\right)$$

$$= 806.25 - 696 = \text{Rs.}110.25$$

12. (c); Loan of Rs.20480 is settled by paying Rs.27778 after five years. Simple interest is applicable for first three years while compound interest is applicable for next two years.

$$27778 = 20480 \left(1 + \left(8\frac{3}{4}\% + 5\frac{1}{4}\% + 4\frac{3}{4}\%\right)\right) \left(1 + 7\frac{1}{2}\%\right) \left(1 + \frac{x}{100}\right)$$

$$27778 = 20480 \left(1 + \left(18\frac{3}{4}\%\right)\right) \left(1 + 7\frac{1}{2}\%\right) \left(1 + \frac{x}{100}\right)$$

$$27778 = 20480 \left(\frac{19}{16}\right) \left(\frac{43}{40}\right) \left(1 + \frac{x}{100}\right)$$

$$x = 6\frac{1}{4}\%$$

- 13. (d);** Let the amounts borrowed under plan B and C be $16x$ and $13x$ respectively.

$\therefore$ Ratio of interests

$$= 16x \times \left(\left(1 + 7\frac{1}{2}\% \right) \left(1 + 6\frac{1}{4}\% \right) - 1 \right) : 13x \times \left(8\frac{3}{4}\% + 5\frac{1}{4}\% \right)$$

$$= 16x \times \left(\left(\frac{43}{40} \right) \left(\frac{17}{16} \right) - 1 \right) : 13x \times \left(\frac{14}{100} \right)$$

$$= 16x \times \frac{91}{640} : 13x \times \frac{14}{100} = 5 : 4$$

- 14. (c);** Effective rate of interests for three years:

For old plan C = $8\frac{3}{4}\% + 5\frac{1}{4}\% + 4\frac{3}{4}\% = 18\frac{3}{4}\%$

For new plan C = $3 \times 6\frac{2}{3}\% = 20\%$

% Increase in interest = % Increase in effective interest rate for three years

$$= \frac{(20 - 18\frac{3}{4})}{18\frac{3}{4}} \times 100 = 6\frac{2}{3}\%$$

- 15. (a);** Let the amount borrowed under plan A be Rs.x
Effective rate of interests for three years:

For plan A = $8\frac{2}{3}\% + 6\frac{2}{3}\% + 3\frac{2}{3}\% = 19\%$

For plan E = $7\frac{3}{5}\% + 5\frac{4}{5}\% + 4\frac{3}{5}\% = 18\%$

Total interest = 19% of x + 18% of (30000 - x)

$$\Rightarrow 5540 = 18\% \text{ of } 30000 + 1\% \text{ of } x$$

$$\Rightarrow x = \text{Rs.}14000$$

- 16. (b);** Let cost price of product D and E in January be 500 and 600 respectively

Acc. to question,

$$\frac{25}{100} \times 500 + \text{Profit of E} = 150\% \text{ of Profit of E}$$

$$\Rightarrow 50\% \text{ of Profit of E} = 125$$

$$\Rightarrow \text{Profit of E} = 250$$

$$\text{Required percentage profit} = \frac{250}{850} \times 100 = \frac{500}{17}\%$$

- 17. (e);** Let the cost price of A in January and D in February be Rs.x.

Let profit of all products in January be P_A, P_B, P_C, P_D and P_E

And, profit of all products in February be Q_A, Q_B, Q_C, Q_D and Q_E

According to question,

Average of P_B, P_C, P_D and P_E = Average of Q_A, Q_B, Q_C and Q_E

$$\therefore P_B + P_C + P_D + P_E = Q_A + Q_B + Q_C + Q_E$$

And,

Total profits in January –

$$\text{Total profits in February} = 4900$$

$$(P_A + P_B + P_C + P_D + P_E) - (Q_A + Q_B + Q_C + Q_D + Q_E) = 4900$$

$$\therefore P_B + P_C + P_D + P_E = Q_A + Q_B + Q_C + Q_E$$

$$\Rightarrow P_A - Q_D = 4900$$

$$\frac{15}{100}x - \frac{8}{100}x = 4900$$

$$\frac{7x}{100} = 4900$$

$$x = 70,000$$

$$\text{Profit of A in January} = 15 \times 700 = \text{Rs.}10500$$

- 18. (d);** Let cost price of A in January and B in May be $100x$

So, selling price of A in January = $115x$

And, selling price of B in May = $130x$

Marked price – Discount = selling price

$$MP_1 - 2n\% \text{ of } MP_1 = 115x$$

$$MP_1 = \frac{115x}{(100-2n)\%}$$

Similarly

$$MP_2 = \frac{130x}{(100-25-n)\%}$$

$$\frac{MP_1}{MP_2} = \frac{\frac{115x}{(100-2n)\%}}{\frac{130x}{(100-25-n)\%}}$$

Or

$$\frac{\frac{115x}{(100-2n)\%}}{\frac{130x}{(75-n)\%}} = \frac{23}{26} \Rightarrow \frac{(75-n)\%}{(100-2n)\%} = \frac{1}{1}$$

$$n = 25$$

- 19. (a);** Let cost price per kg for shopkeeper is Rs.100

Total cost price = $10 \times 100 = \text{Rs.}1000$

Profit per kg on selling at 24% profit = Rs.24

Selling price per kg = Rs.124

Total selling price = $124 \times 8 + 62 \times 2 = \text{Rs.}1116$

$$\text{Overall profit \%} = \frac{1116-1000}{1000} \times 100 = 11.6\%$$

- 20. (b);** Let cost price of D in April and May be $100x$

And, cost-price of E in April and May be $100y$

According to question

$$110x + 125y = 650 \times 2 = 1300 \quad \dots (i)$$

$$120x + 110y = 630 \times 2 = 1260 \quad \dots (ii)$$

On solving equations

$$x = 5 \Rightarrow y = 6$$

So, cost price of E in May = $100y$

$$= 100 \times 6 = \text{Rs.}600$$

- 21. (a);** Let total quantity of milk = $200x$ L

And total quantity of water = $100x$ L

Total milk in A and B = $(20\% + 15\%) 200x$

$$= 35 \times 2x = 70x \text{ L}$$

Total water in A and B = $35 \times x$

$$\text{Total water in F} = 35x + \frac{25}{100} \times \frac{25}{100} \times 100x$$

$$= 35x + 6.25x = 41.25x \text{ L}$$

Let cost price of milk per liter be Rs.10

So, cost price of $(70x + 41.25x)$ L of mixture

$$= 70x \times 10 = \text{Rs.}700x$$

Selling price of $(70x + 41.25x)$ L of mixture

$$= 111.25x \times 10 = \text{Rs.}1112.5x$$

$$\% \text{ profit} = \frac{1112.5x - 700x}{700x} \times 100 = \frac{412.5}{7} = \frac{825}{14} = 58\frac{13}{14}\%$$

Or we can say that profit is due the quantity of water in the mixture.

$$\text{So we can directly write \% profit} = \frac{41.25x}{70x} \times 100 = 58\frac{13}{14}\%$$

- 22. (c);** Milk in vessel A and C = $\frac{50}{100} \times 2x = x$

$$\text{Water in vessel A and C} = \frac{55}{100} \times x = 0.55x$$

Ratio of milk and water in M = $x : 0.55x = 20 : 11$

According to question,

$$\Rightarrow \frac{x - \frac{20}{31} \times 62}{55x - \frac{11}{31} \times 62 + 17} = \frac{6}{5} \Rightarrow \frac{x - 40}{55x - 5} = \frac{6}{5}$$

$$\Rightarrow 5x - 200 = 3.30x - 30 \Rightarrow x = 100$$

$$\text{Quantity of milk in vessel B} = \frac{20}{100} \times 2 \times 100 = 40 \text{ L}$$

23. (b); Let total milk in all 5 vessel = 200x

And total water in all 5 vessel = 100x

So,

$$\text{Total milk in all vessel except C} = \frac{65}{100} \times 200x = 130x$$

$$\text{Total water in all vessel except C} = \frac{55}{100} \times 100x = 55x$$

And

$$\text{Ratio of milk and water in vessel C} = 35 \times 2x : 45x$$

$$= 70x : 45x = 14 : 9$$

According to question,

$$\frac{130x + \frac{14}{23} \times 115}{55x + \frac{9}{23} \times 115} = \frac{9}{4}$$

$$\frac{130x + 70}{55x + 45} = \frac{9}{4}$$

$$520x + 280 = 495x + 405$$

$$25x = 125 \Rightarrow x = 5$$

$$\text{Total quantity of water in all five vessel} = 100x = 500 \text{ L}$$

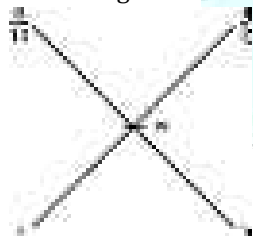
24. (e); Ratio of milk to water in vessel-D

$$= \frac{10}{100} \times 2x : \frac{5}{100} \times x = 4 : 1$$

$$\text{Ratio of milk to water in vessel E} = \frac{20}{100} \times 2x : \frac{15}{100} \times x$$

$$= 8 : 3$$

From allegation



$$\frac{2}{3} = \frac{\frac{4}{5} - w}{w - \frac{8}{11}}$$

$$2w - \frac{16}{11} = \frac{12}{5} - 3w$$

$$5w = \frac{12}{5} + \frac{16}{11}$$

$$5w = \frac{132 + 80}{5 \times 11}$$

$$w = \frac{212}{275}$$

$$\text{Required ratio} = \frac{212}{63}$$

25. (d); Quantity of milk and water in vessel C

$$= \frac{35}{100} \times 2x + \frac{45}{100} \times x$$

$$= 0.7x + 0.45x = 1.15x$$

$$1.15x = 115 \Rightarrow x = 100$$

$$\text{Milk and water in B} = \frac{20}{100} \times 200 + \frac{25}{100} \times 100$$

$$= 40 + 25 = 65$$

$$\text{Milk and water in E} = \frac{20}{100} \times 200 + \frac{15}{100} \times 100$$

$$= 40 + 15 = 55$$

$$\text{Required \%} = \frac{65 - 55}{55} \times 100 = \frac{10}{55} \times 100 = 18\frac{2}{11}\%$$

26. (b); Ratio of distance covered by cat, lion and dog in one leap will be

Cat	Lion	Dog
2	5	
	7	4
14	35	20

Let they cover 14x, 35x and 20x distance in one leap

So, in one min dog covers = 20x × 7 = 140x distance

This distance is equal to distance between dog and cat

In one min cat covers = 14x × 5 = 70x distance

$$\text{Required time} = \frac{\text{Distance between cat and dog}}{\text{Relative speed between dog and cat}}$$

$$= \frac{140x}{140x - 70x} = 2 \text{ min.}$$

Another method:

According to dog, cat was 1 minute away from him.

Now, cat complete 5 leaps in 1 minute

distance { 5 leaps of cat = 2 leaps of lion
10 leaps of cat = 4 leaps of lion

⇒ this takes 2 minute of cat.

Now, dog's 7 leap = 4 leaps of lion

7 leaps of dog = 10 leaps of cat

And, dog take 1 minute to cover 4 leaps of lion but in the same time cat moves away 2 leaps of lion. In 1 more min

Dog cover 4 leaps of lion and again cat moves away 2 leaps of lion. So finally dog will catch cat in 2 mins

Total time to catch the cat = 1 + 1 minute = 2 minute.

27. (b); According to Jackal

Jackal is 2 min away from Rat.

Distance ⇒ 2 min → 8 leaps of Jackal → 5 leaps of lion

$$4 \text{ leaps of Jackal} \Rightarrow \frac{5}{2} = 2.5 \text{ leaps of lion}$$

$$\text{In 1 min} \rightarrow 6 \text{ leaps of Rabbit} \Rightarrow \frac{1}{5} \times 6 \text{ leaps of lion}$$

$$\rightarrow \frac{6}{5} \text{ leaps of lion} = 1.2 \text{ leaps}$$

$$\text{Resultant velocity} = 2.5 - 1.2 = 1.3 \text{ leaps of lion.}$$

$$\rightarrow \text{time} = \frac{5}{1.3} = 3.84 \text{ min}$$

In 4th min.

28. (d); Speed of dog

Dog ⇒ 7 leap in 1 min.

7 leap distance of dog = 4 leap of lion.

Speed = Distance of 4 leap of lion/min

Cheetah.

No. of leaps in 1 min. = (no. of leaps of cat + no. of leap of monkey) in 1 min

$$= (5 + 6) \rightarrow 11 \text{ leap in 1 min}$$

Distance

2 leaps of Cheetah = 3 leaps of lion.

1 leaps of cheetah = 3/2 leaps of lion

In 11 leap of cheetah = 33/2 leaps of lion

Ratio

Speed of Dog : Speed of cheetah

$$= 4 : \frac{33}{2}$$

$$= 8 : 33$$

29. (a); Cat →

Cat takes 5 leaps – 1 minute

5 leaps of cat = 2 leaps of lion

Cat's speed = Distance covered in 2 leaps of Lion/min

Reduced speed when climbing

$$= 2 \text{ leaps of lion/m} \times \frac{80}{100}$$

$$= \frac{8}{5} \text{ leaps of lion/m}$$

Monkey →

Monkey take = 6 leaps – 1 min.

4 leaps of monkey = 3 leaps of lion

6 leaps of monkey = 4.5 leaps of lion

Speed → distance cover in 4.5 leaps of lion/min

Increased speed when climbing

$$= \frac{4.5 \times 20}{500} \text{ leaps of lion/m}$$

$$= 5.4 \text{ leaps of lion/m}$$

$$\text{Required \%} = \frac{8 \times 100}{5 \times 5.4} \% = 29.629\% \approx 30\%$$

30. (b); Jackal – 1 minute distance = 10 meter

Jackal 8 leaps = 10 meter

Lions 5 leaps = 10 meter

1 leap of lion = 2 meter

Jackal 2 min distance = $10 \times 2 = 20$ meter

Cat's 2 min distance = $(5 \times 2) = (3 \times 2)$ of Lion's leap
 $= 6 \times 2 = 12$ meter

Monkey 2 min distance = $6 \times 2 = 12$ leaps

= leaps of lion = 18 meter

Dog's 2 min distance = 14 leaps = 8 leaps of lion = 16 meter.

Rabbit 2 min distance = 12 leaps = 2.4 leaps of lion = 4.8 meter

$$\text{Sum} = (20 + 12 + 18 + 16 + 4.8) \text{ m} = 70.8 \text{ m}$$

31. (a); Total surface area of the toy = C.S.A of cone + C.S.A of Hemisphere

Let, slant height of cone be l cm

$$\therefore \pi r l + 2\pi r^2 = 858 \text{ cm}^2$$

$$\pi r(l + 2r) = 858 \text{ cm}^2$$

$$l = 25 \text{ cm}$$

$$\text{height of cone} = \sqrt{l^2 - r^2} = \sqrt{25^2 - 7^2} = 24 \text{ cm}$$

volume of the toy

$$= \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3 = \frac{1}{3} \pi r^2 (h + 2r) = 1950 \frac{2}{3} \text{ cm}^3$$

32. (e); Height of cylinder = 12 cm

Height of toy is double the height of cylinder = 24 cm

Edges of cube = $24 - 12 = 12$ cm

C.S.A of cylinder = $2\pi rh = 66 \times h$

$$r = 10.5 \text{ cm}$$

Total surface area of toy = $(6a^2 - \pi r^2) + 2\pi rh + \pi r^2$
 ($-\pi r^2$, area subtracted due to alignment)

$$= 6 \times 12 \times 12 + 2 \times \frac{22}{7} \times 12 \times 10.5 = 1656 \text{ cm}^2$$

33. (c); Sphere radius = $\frac{21}{2}$

So, cylinder radius = $\frac{21}{2}$

height of cylinder = 12

$$\text{required ratio} = \frac{4}{3} \pi r^3 + \pi r^2 h : \pi r^2 h : \frac{4}{3} \pi r^3$$

$$\frac{4}{3} r + h : h : \frac{4}{3} r$$

$$13 : 6 : 7$$

34. (d); Height of cone = $\sqrt{25^2 - 7^2} = 24$ cm

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 24 = 1232 \text{ cm}^3$$

$$\text{Volume of cuboid} = 24 \times 10 \times 25\% \text{ of } 24 = 1440 \text{ cm}^3$$

$$\text{Difference} = 1440 - 1232 = 208 \text{ cm}^3$$

$$\text{Required \%} = \frac{208}{1232} \times 100 = 16.88\% \approx 17\%$$

35. (b); Slant height of cone = $\sqrt{h^2 + r^2} = \sqrt{24^2 + 7^2} = 25$ cm

$$\therefore \text{Total surface area of toy} = 2\pi r^2 + 2\pi rh + \pi rl$$

$$= \frac{22}{7} \times 7(2 \times 7 + 2 \times 12 + 25) = 1386 \text{ cm}^2$$

So, required cost of painting = 1386×25

$$= \text{Rs. } 34650$$

36. (d); Selling price per kg of Tea = $\frac{900}{2.5} = 360$ Rs.

Let cost price per kg of Tea is x

$$\text{Selling price of Tea per kg} = \left(1 - \frac{4}{49}\right) \times \frac{7}{6} x = 360$$

$$x = 336$$

$$\text{List price of Tea} = \frac{7}{6} x = \frac{7}{6} \times 336 = 392$$

$$\text{Required \%} = \frac{392 - 360}{392} \times 100 = \frac{32}{392} \times 100 \approx 8\%$$

37. (a); Profit per kg on wheat = $\frac{1330}{70} = 19$ Rs.

$$\text{Cost price of Wheat} = \frac{42 \times 100}{210} = 20 \text{ Rs. per kg}$$

$$\% \text{ discount offered on Wheat} = \frac{3}{42} \times 100 = \frac{50}{7} \%$$

$$\% \text{ discount offered on Rice} = \frac{80}{6} \%$$

$$\text{Required ratio} = \frac{50}{7} \times \frac{6}{80} = 15 : 28$$

38. (a); List price of Honey = 1500

$$\text{Selling price of Honey} = \left(100 - \frac{80}{3}\right) \% 1500$$

$$= \text{Rs. } 1100 \text{ per kg}$$

Given

(List price – selling) : (selling price – cost price)

$$\frac{(1500 - 1100)}{(1100 - x)} = \frac{4}{3}$$

$$1200 = 4400 - 4x$$

$$4x = 3200$$

$$x = 800 \text{ (cost price)}$$

So profit per kg = 300

$$\text{Total quantity} = \frac{1800}{300} = 6 \text{ kg}$$

Quantity of coffee = 8 kg

$$\text{Required average} = \frac{12 + 8 + 70 + 80 + 6}{5} = \frac{176}{5} = 35.2 \text{ kg}$$

39. (d); Selling price of 1 kg of rice = $\left(100\% - \frac{100}{6}\right) \% 60$

$$= \frac{5}{6} \times 60 = 50 \text{ Rs.}$$

$$\text{Cost price of 1 kg of rice} = \frac{(3 \times 60)}{4} = 45 \text{ Rs.}$$

$$\text{Loss} = 8 \times 45 = 360 \text{ Rs.}$$

$$\text{Profit} = 72 \times 5 = 360$$

So overall no profit no loss

40. (a); Let cost price per kg of tea = CP_1

And cost price per kg of coffee = CP_2

And List price of both = x

$$\text{So, } \frac{7}{6} CP_1 = x \quad \dots (i)$$

And

$$\frac{112}{100} CP_2 = \frac{80}{100} x$$

$$7CP_2 = \frac{80}{100} x$$

$$7CP_2 = \frac{80}{100} x$$

$$x = \frac{7CP_2}{5} \quad \dots (ii)$$

From (i) and (ii)

$$\frac{7}{6} CP_1 = \frac{7CP_2}{5}$$

$$\text{Required ratio } \frac{CP_1}{CP_2} = \frac{6}{5}$$

41. (a); Cost price of 20 kg of wheat for retailer

$$= 20 \times 25 \times \frac{90}{100} = \text{Rs. } 450$$

Price at which he sold this amount of wheat to

$$\text{customer} = 20 \times 25 \times \frac{95}{100} = \text{Rs. } 475$$

$$\text{Profit} = 475 - 450 = 25$$

42. (b); Cost price of cashew for retailer = Rs. 900

$$\text{Price at which he sold all the cashew} = 40 \times 30 \times \frac{125}{100}$$

$$= \text{Rs. } 1500$$

$$\text{Profit \%} = \frac{1500-900}{900} \times 100 = 66.66\%$$

43. (c); Cost price of Rice for Retailer = $20 \times 15 \times \frac{90}{100}$

$$= \text{Rs. } 270$$

To be in a situation of no loss -no gain, he must sell remaining 50% at Rs. 270.

$$\text{Price per kg} = \frac{270}{10} = \text{Rs. } 27$$

44. (d); Cost price of pulses for the retailer = $18 \times 35 \times \frac{70}{100}$

$$= \text{Rs. } 441$$

$$\text{Cost price of 2 kgs of Almond} = 2 \times 40 \times \frac{85}{100} = \text{Rs. } 68$$

$$\text{Total CP} = 441 + 68 = \text{Rs. } 509$$

$$\text{Total SP} = 18 \times 35 \times \frac{130}{100} = \text{Rs. } 819$$

$$\text{Profit \%} = \frac{819-509}{509} \times 100 = \frac{310}{509} \times 100 \approx 61\%$$

45. (c); Cost price per kg of sugar = $\frac{240}{15} = 16$

$$\text{Discount offered by wholesaler} = \frac{4}{20} \times 100 = 20\%$$

Discount offered by retailer to customer

$$= \frac{75}{100} \times 20\% = 15\%$$

$$\text{Selling price of mixture} = (6 + 15) \times 20 \times \frac{85}{100} = 357$$

$$\text{Profit \%} = \frac{357-240}{240} \times 100 = \frac{117}{240} \times 100 = 48.75\%$$

46. (b); With 120% efficiency pipe M alone will fill the tank in $= \frac{60}{6} \times 5 = 50$ min.

Part of tank filled in 7 minutes working alternatively

$$= \frac{4}{50} + \frac{3}{90} = \frac{4}{50} + \frac{1}{30} = \frac{12+5}{150} = \frac{17}{150}$$

$$\text{Part of tank filled in 1 min on Tuesday} = \frac{1}{45} + \frac{1}{30} + \frac{1}{90}$$

$$= \frac{2+3+1}{90} = \frac{1}{15}$$

So, remaining part of tank will be filled in

$$= 15 \times \left(1 - \frac{17}{150}\right) = 15 \times \frac{133}{150} = \frac{133}{10} \text{ min}$$

$$= 13 \text{ min } 18 \text{ sec}$$

$$\text{Required time} = (3 + 13) \text{ min } 18 \text{ sec} = 16 \text{ min } 18 \text{ sec}$$

47. (e); Time taken to fill tank on Wednesday = $\frac{30 \times 60}{90}$

$$= 20 \text{ min}$$

Part of tank filled in 3 min on Thursday

$$= \frac{1}{45} + \frac{1}{30} + \frac{1}{90} = \frac{2+3+1}{90} = \frac{1}{15} \text{ min}$$

$$\text{Total time to fill tank on Wednesday} = 15 \times 3 = 45 \text{ min.}$$

$$\text{Required difference} = 45 - 20 = 25 \text{ min.}$$

48. (c); Total time taken on Friday to fill the tank = $\frac{60 \times 75}{135}$

$$= \frac{100}{3} \text{ min.}$$

$$\text{Total capacity of tank} = \frac{100}{3} \times 36 = 1200 \text{ litre}$$

$$\text{Ratio of efficiency of pipe M and Q} = 5 : 4$$

Amount of water filled on Friday by Q

$$= \frac{4}{9} \times 1200 = \frac{4800}{9} \text{ Litre}$$

$$\text{Ratio of efficiency of pipe M to P} = 3 : 2$$

Amount of water filled on Monday by pipe P

$$= \frac{2}{5} \times 1200 = 480 \text{ litre}$$

$$\text{Required percentage} = \frac{480}{\frac{4800}{9}} \times 100 = 90\%$$

49. (b); Time taken to fill the tank by M alone with increased efficiency = 50 min.

Time taken to fill the tank by Q alone with decreased efficiency = 100 min.

$$\text{Now, Together they can fill the tank in} = \frac{100 \times 50}{150}$$

$$= \frac{100}{3} \text{ min.}$$

$$\text{Capacity of tank} = \frac{162}{12} \times \frac{100}{3} = 450 \text{ litre}$$

Let M and P can fill tank alone with different

efficiency in x min and y min respectively

$$\text{So, } \frac{8}{x} + \frac{8}{y} = \frac{7}{30} \quad \dots (i)$$

$$\text{and } \frac{8}{x} + \frac{15}{y} = \frac{7}{20} \quad \dots (ii)$$

Solving (i) and (ii)

$$\frac{7}{y} = \frac{7}{20} - \frac{7}{30}$$

$$\frac{7}{y} = \frac{21-14}{60} \Rightarrow y = 60 \text{ min}$$

$$\text{So, } x = 80 \text{ min.}$$

$$\text{Required ratio} = 4 : 3$$

50. (a); Rate of flow of pipe N = 18 ℓ /min.

$$\text{Capacity of tank} = 18 \times 45 = 810 \text{ litre}$$

$$\text{Rate of flow of pipe O} = \frac{810}{30} = 27 \ell/\text{min.}$$

$$\text{Rate of flow of pipe P} = \frac{810}{90} = 9 \ell/\text{min.}$$

$$\text{Part of tank filled in one min} = \frac{1}{45} + \frac{1}{30} + \frac{1}{90} = \frac{1}{15}$$

Time taken to fill tank by all = 15 min

Total cost incurred in filling tank

$$= 15 \times 18 \times 12 + 15 \times 27 \times 15 + 15 \times 9 \times 10$$

$$= 15 (216 + 405 + 90) = 10665 \text{ Rs.}$$

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Mixed Graph

Mixed graphs are a combination of two or more graphs. Sometimes, the data that need to be represented contains numerous variables which are hard to represent through a single representation format. In other cases, the data need to be segregated into small parts for effective representation. Hence, the data is segregated and represented through two or more than two suitable graphs. These graphs may or may not represent similar variables. If the variables represented by these graphs are not similar, we have to understand the relationships between these variables which are described through some additional statements.

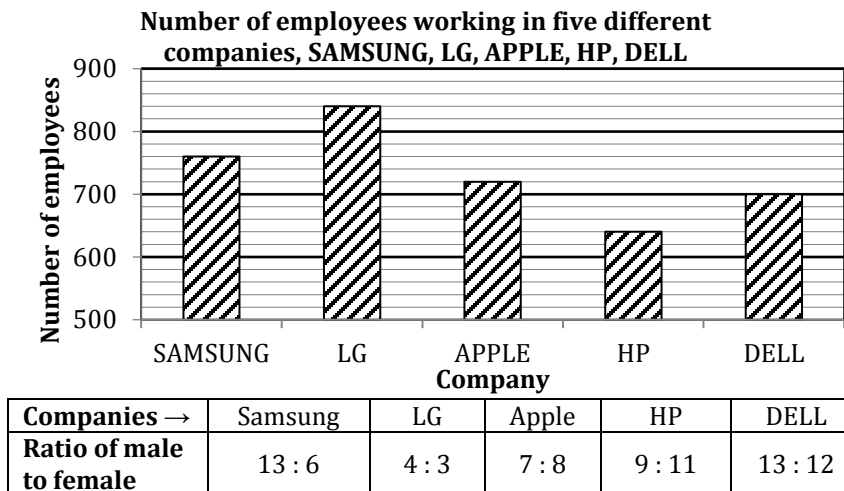
This chapter contains:

- Solved Examples
- Previous Year Questions
- Practice Set Level I
- Practice Set Level II

Solved Examples

Directions (1 - 5): Study the following bar diagram and table carefully to answer the questions :

Bar graph shows number of total employee working in five different company and table shows Ratio of male to female in these five company.



1. What is the ratio of female employees in company Samsung and H.P. together to the females in company DELL and Apple together.

(a) 43 : 53 (b) 54 : 59 (c) 37 : 45 (d) 23 : 27 (e) 20 : 23

Sol. (c); Required ratio = $\frac{\frac{6}{19} \times 760 + \frac{11}{20} \times 640}{\frac{12}{25} \times 700 + \frac{8}{15} \times 720} = \frac{240 + 352}{336 + 384} = \frac{592}{720} = 37 : 45$

2. Males from company Samsung and HP together is what percent of total employees in company Apple. (approximately)

(a) $125\frac{2}{9}\%$ (b) $112\frac{2}{9}\%$ (c) $130\frac{2}{9}\%$ (d) $135\frac{2}{9}\%$ (e) $138\frac{2}{9}\%$

- Sol.** (b); Males from company Samsung and HP together

$$= 760 \times \frac{13}{19} + 640 \times \frac{9}{20} = 520 + 288 = 808$$

$$\text{Required percentage} = \frac{808}{720} \times 100 = 112\frac{2}{9}\%$$

3. If 20% females from LG company resigns and 12.5% females resigns from company H.P. then what is the ratio of remaining employees in LG to remaining employees in HP.

(a) 192 : 149 (b) 153 : 129 (c) 72 : 73 (d) 53 : 42 (e) 57 : 49

Sol. (a); Required ratio = $\frac{840 - \frac{3}{7} \times 840 \times \frac{20}{100}}{640 - \frac{11}{20} \times 640 \times \frac{12.5}{100}} = \frac{840 - 72}{640 - 44} = 768 : 596 = 192 : 149$

4. What is the difference between average of males from Samsung and HP together to the average of females from company Apple and DELL together.

(a) 40 (b) 42 (c) 44 (d) 36 (e) 28

- Sol.** (c); Average of males from Samsung and HP

$$= \left(760 \times \frac{13}{19} + 640 \times \frac{9}{20} \right) \frac{1}{2} = (520 + 288) \frac{1}{2} = 404$$

Average of females from Apple and DELL

$$= \left(720 \times \frac{8}{15} + 700 \times \frac{12}{25} \right) \frac{1}{2} = (384 + 336) \frac{1}{2} = 360$$

Required difference

$$= 404 - 360 = 44$$

5. If ratio of number of females at present to the number of females next year in company Apple is 8 : 11 so, what should be increase or decrease in number males in Apple so that overall number of employees in Apple next year is same as present total number of employees in LG.

(a) 24 (b) 6 (c) 12 (d) 20 (e) 16

- Sol.** (a); Number of females in company Apple next year

$$= 720 \times \frac{8}{15} \times \frac{1}{8} \times 11 = 528$$

Male employees in Apple next year

$$= 840 - 528 = 312$$

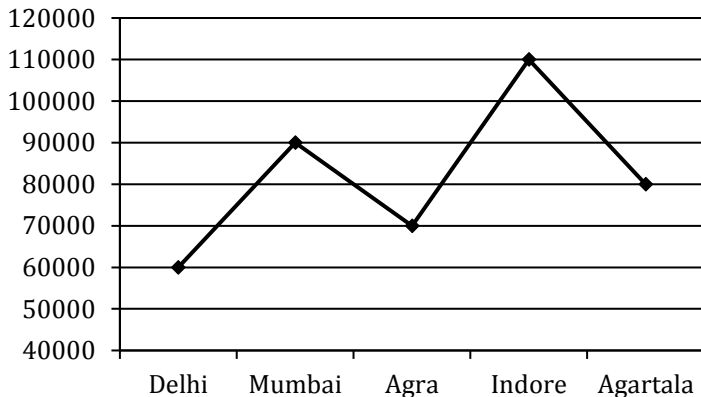
$$\text{Males at present in Apple} = 720 \times \frac{7}{15} = 336$$

$$\text{So, In next year male decreases by} = 336 - 312 = 24$$

Directions (6-10): Read the data given below and answer the following questions.

Given below is the line graph which shows the population of five cities in 2016 and table shows the ratio of male to female in these five cities.

NOTE- Some values are missing in the table, you have to calculate these values if necessary to answer the questions



Cities	M : F
Delhi	3 : 5
Mumbai	7 : 5
Agra	3 : 4
Indore	6 : 5
Agartala	- : 5

6. If there is an increase of 25% and 15% population of male & female live in Delhi in year 2017 with respect to previous year. Then what will be total percentage rise in the population of Delhi in 2017 with respect to previous year?

(a) 18.85% (b) 18.65% (c) 19% (d) 18% (e) 18.75%

Sol. (e); In 2017

$$= 60,000 \times \frac{3}{8} \times \frac{125}{100} + 60,000 \times \frac{5}{8} \times \frac{115}{100}$$

$$28,125 + 43,125 = 71,250$$

$$\text{Total increased population} = 71,250 - 60,000 = 11,250$$

$$\% \text{ increase} = \frac{11,250}{60,000} \times 100 = 18.75\%$$

7. In Agartala $\frac{1}{8}$ th of person affected from Dengue then what is ratio of un-affected females to unaffected male in Agartala. If no. of un-affected males in Agartala is same as no. of males in Delhi

(a) 9 : 19 (b) 19 : 9 (c) 17 : 9 (d) 9 : 17 (e) 17 : 9

Sol. (b); Unaffected person = $\frac{7}{8} \times 80,000 = 70,000$

No. of un-affected males in Agartala = No. of males in Delhi

$$= 60,000 \times \frac{3}{8} = 22,500$$

No. of unaffected females = 47,500

$$\text{Ratio} = \frac{47,500}{22,500} = \frac{475}{225} = \frac{19}{9}$$

8. What was the difference. of male & female in Mumbai city in 2014 if there is an increase in 20% of population every year?

(a) 10417 (b) 10217 (c) 10317 (d) 10400 (e) Cannot be determined

Sol. (e); Cannot be determined since ratio of population of males and females in Delhi in 2014 is not given

9. Males of Mumbai is what percent less or more than the male of Agartala if the average number of males in Delhi, Agra & Agartala is 27,500?

(a) 42.86% less (b) 75% less (c) 75% more (d) 42.86% more (e) 62.5% more

Sol. (c); Males in Delhi + Agra + Agartala = $27,500 \times 3$

$$= 82,500$$

$$\text{Males in Agartala} = 82,500 - \frac{3}{8} \times 60,000 - \frac{3}{7} \times 70,000$$

$$= 82,500 - 22,500 - 30,000$$

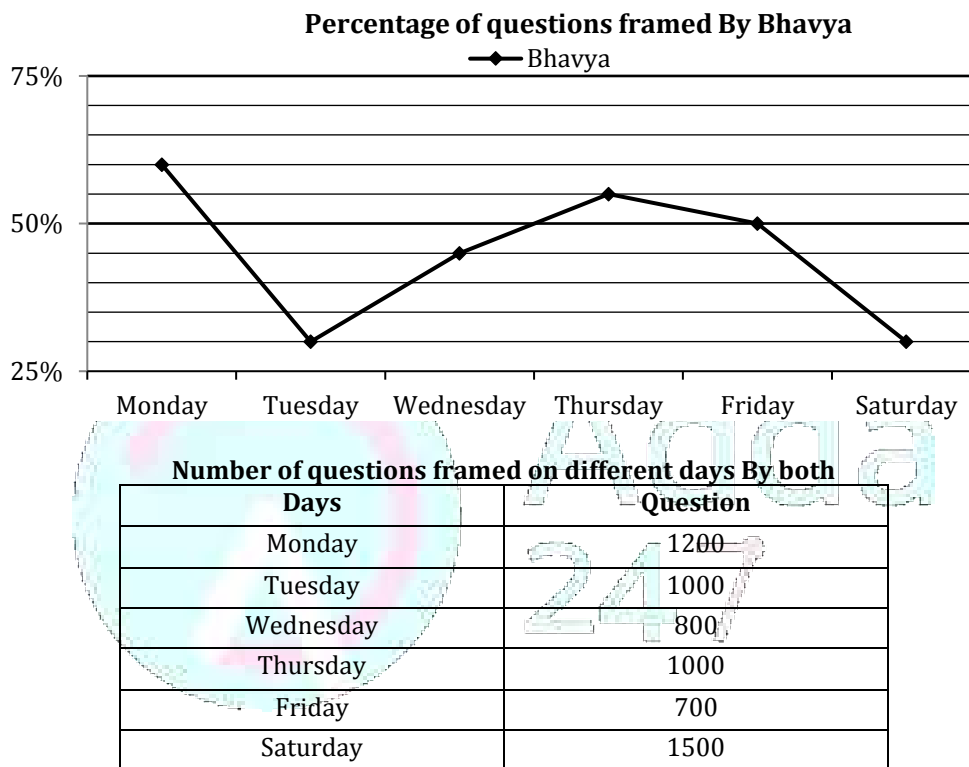
$$= 30,000$$

$$\text{Males in Mumbai} = \frac{7}{12} \times 90,000 = 52,500$$

$$\text{required \%} = \frac{52,500 - 30,000}{30,000} = 75\% \text{ more}$$

10. If $\frac{1}{4}$ th of male and $\frac{1}{5}$ of female of Indore leave the city and all those who leave Indore came to Delhi & Mumbai in the ratio of 3 : 2 respectively then what is the percent increase of the population in Mumbai.
- (a) $33\frac{1}{3}\%$ (b) $11\frac{1}{9}\%$ (c) $12\frac{1}{9}\%$ (d) $16\frac{2}{3}\%$ (e) $8\frac{1}{3}\%$
10. (b); Total person who leave indore
 $= \frac{1}{4} \times \frac{6}{11} \times 1,10,000 + \frac{1}{5} \times \frac{5}{11} \times 1,10,000 = 25,000$
 Person come to Mumbai $= 25,000 \times \frac{2}{5} = 10,000$
 $\%$ increase of population in Mumbai $= \frac{10,000}{90,000} \times 100 = 11\frac{1}{9}\%$

Directions (11-15): Bhavya and Abhishek are two persons who works in career power and frame different number of questions on six different days of a week. The line graph shows the percentage of questions framed by Bhavya out of total questions framed by both on different days of a week and table shows the total number of questions framed on different days by both.



11. Questions framed by Bhavya on Monday and Tuesday together is what percent of question framed by Abhishek on Thursday?

(a) $\frac{685}{3}\%$ (b) $\frac{680}{3}\%$ (c) $\frac{690}{3}\%$ (d) $\frac{670}{3}\%$ (e) $\frac{685}{3}\%$

Sol. (b); Questions solved by Bhavya on Monday & Tuesday

$$= \frac{60}{100} \times 1200 + \frac{30}{100} \times 1000$$

$$= 720 + 300$$

$$= 1020$$

Questions solved by Abhishek on Thursday

$$= \frac{45}{100} \times 100 = 450$$

$$\therefore \text{Percent} = \frac{1020}{450} \times 100 = \frac{680}{3}\%$$

12. On Which day of week Abhishek framed minimum number of questions?

(a) Monday (b) Wednesday (c) Thursday (d) Friday (e) Tuesday

Sol. (d); No. of questions solved by Abhishek

$$\text{Monday} = \frac{40}{100} \times 1200 = 480$$

$$\text{Tuesday} = \frac{70}{100} \times 1000 = 700$$

$$\text{Wednesday} = \frac{55}{100} \times 800 = 440$$

$$\text{Thursday} = \frac{45}{100} \times 1000 = 450$$

$$\text{Friday} = \frac{50}{100} \times 700 = 350$$

$$\text{Saturday} = \frac{70}{100} \times 1500 = 1050$$

∴ Minimum no. of questions solved by Abhishek is on Friday

13. The average no. of questions framed by Abhishek on Monday and Saturday together is what percent more or less than average number of questions framed by Bhavya on Monday, Tuesday and Saturday together?

(a) $\frac{2750}{49}\%$ (b) $\frac{2780}{49}\%$ (c) $\frac{2750}{7}\%$ (d) $\frac{2770}{49}\%$ (e) $\frac{2750}{51}\%$

Sol. (a); Avg. no. of questions solved by Abhishek on Monday and Saturday

$$= \frac{1}{2} \left(\frac{40}{100} \times 1200 + \frac{70}{100} \times 1500 \right) = \frac{480+1050}{2} = 765$$

Avg. No. of questions solved by Bhavya on Monday Wednesday and Saturday

$$= \frac{1}{3} \left(\frac{60}{100} \times 1200 + \frac{30}{100} \times 1000 + \frac{30}{100} \times 1500 \right) = \frac{720+300+450}{3} = 490$$

$$\therefore \text{Percentage} = \frac{765-490}{490} \times 100 = \frac{2750}{490} = \frac{2750}{49}\%$$

14. If questions framed by Abhishek on Tuesday of next week is increased by 50% over the tuesday of given week and total number of questions framed on tuesday of next week by both of them is 1800. Then find ratio of number of questions framed by Bhavya to Abhishek on Tuesday of next week?

(a) 10 : 12 (b) 8 : 12 (c) 12 : 10 (d) 5 : 7 (e) 7 : 5

Sol. (d); Total number of questions solved by Abhishek on tuesday of next week = 150% of 70% of 1000 = 1050

Required ratio = (1800 - 1050) : 1050 = 5 : 7

15. Find the ratio of number of questions framed by Bhavya on Wednesday and Friday together to the number of questions framed by Abhishek on Monday and Wednesday together ?

(a) 72 : 92 (b) 70 : 91 (c) 71 : 92 (d) 92 : 71 (e) 71 : 93

Sol. (c); No. of questions solved by Bhavya on Wednesday and Friday

$$= \frac{45}{100} \times 800 + \frac{50}{100} \times 700 = 360 + 350 = 710$$

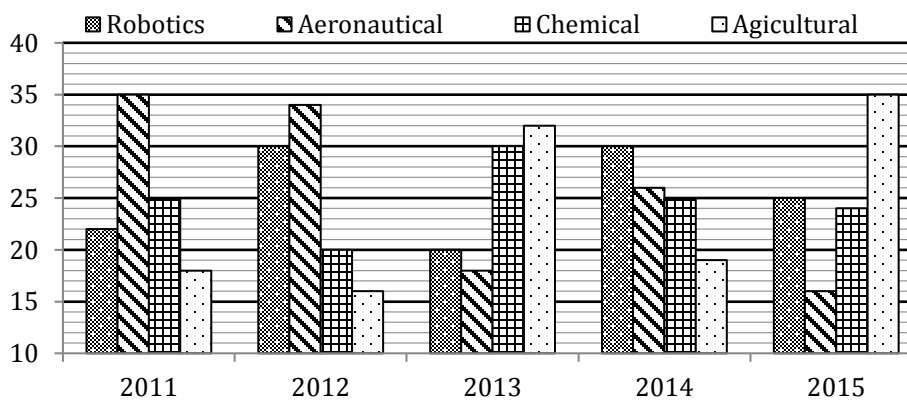
No. of questions solved by Abhishek on Monday and Wednesday

$$= \frac{40}{100} \times 1200 + \frac{55}{100} \times 800 = 480 + 440 = 920$$

$$\therefore \text{Ratio} = \frac{710}{920} = 71 : 92$$

Directions (16-20): Read the following graph and table carefully and answer the questions given below.

Percentage distribution of admitted students in four different disciplines in a college from 2011 to 2015. Assume that these colleges take admission in given disciplines only.



Total number of admitted students in these years

Year	No. of students admitted
2011	2500
2012	4800
2013	3800
2014	6000
2015	1200

16. What is the average number of students admitted in Aeronautical from 2013 to 2015?
 (a) 821 (b) 812 (c) 824 (d) 828 (e) 842

Sol. (b); Required average = $\frac{1}{3} \left(3800 \times \frac{18}{100} + 6000 \times \frac{26}{100} + 1200 \times \frac{16}{100} \right) = \frac{1}{3} (684 + 1560 + 192) = 812$

17. The number of students admitted in Robotics in 2014 is what percent of the number of students admitted in Agricultural in 2012?

(a) $278\frac{1}{3}\%$ (b) $279\frac{3}{8}\%$ (c) $234\frac{3}{8}\%$ (d) $263\frac{5}{8}\%$ (e) $231\frac{3}{8}\%$

Sol. (c); Required percentage = $\frac{30 \times 60}{16 \times 48} \times 100 = 234\frac{3}{8}\%$

18. What is the difference between the number of students admitted in Chemical in 2014 and number of admitted students of Robotics in 2015?

(a) 1200 (b) 1250 (c) 1220 (d) 1230 (e) 1210

Sol. (a); Required difference = $\left[\left(6000 \times \frac{25}{100} \right) - \left(1200 \times \frac{25}{100} \right) \right] = 1500 - 300 = 1200$

19. Find the ratio between number of Agricultural students admitted in 2015 to total number of students admitted in 2011?

(a) 13:107 (b) 21:123 (c) 7:50 (d) 21:125 (e) 125 : 21

Sol. (d); Required Ratio = $\frac{1200 \times \frac{35}{100}}{2500} = \frac{420}{2500} = 21 : 125$

20. Find the total number of students admitted in Robotics in 2014 if in that year 5% of students admitted in Agricultural were transferred in Robotics?

(a) 1587 (b) 1867 (c) 1857 (d) 1757 (e) 1957

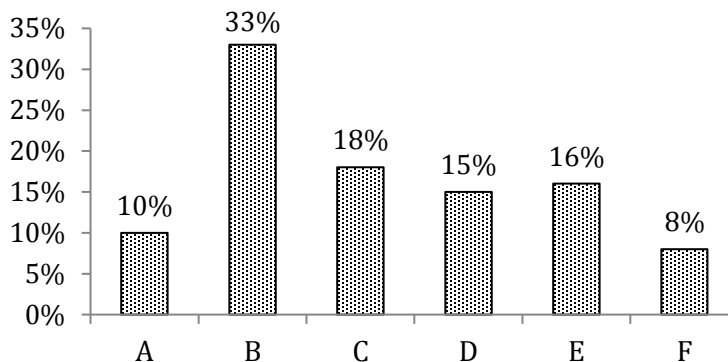
Sol. (c); Required no. of students admitted in Robotics in 2014

$$= 6000 \times \frac{30}{100} + 6000 \times \frac{19}{100} \times \frac{5}{100} \\ = 1800 + 57 = 1857$$

Direction (21–25): Read the following bar graph and table carefully and answer the questions given below.

Bar graph shows Percentage distribution of flats constructed by six different Real Estate groups in Delhi.

Number of total flats = 8000



Ratio of 2 BHK and 3 BHK flats

Company	Ratio 2 BHK : 3 BHK
A-Group	3 : 2
B-Group	11 : 5
C-Group	9 : 7
D-Group	5 : 3
E- group	7 : 8
F- Group	3 : 1

21. What is the respective ratio between the number of 2 BHK flats in C group and number of 3 BHK flats in B group?

(a) 18 : 23 (b) 11 : 15 (c) 17 : 28 (d) 27 : 16 (e) 54 : 55

Sol. (e); Required ratio = $\frac{18 \times 8000}{100} \times \frac{9}{16} : \frac{33 \times 8000}{100} \times \frac{5}{16} = 54 : 55$

22. The number of 2 BHK flats in C group is what percent more than the number of 3 BHK flats in D group?

(a) 80 (b) 75 (c) 78 (d) 83 (e) 76

Sol. (a); Required percentage = $\frac{8000 \times \frac{18}{100} \times \frac{9}{16} - 8000 \times \frac{15}{100} \times \frac{3}{8}}{8000 \times \frac{15}{100} \times \frac{3}{8}} \times 100 = \frac{810 - 450}{450} \times 100 = 80\%$

23. What is approximate average number of 2 BHK flats in D and E group?

- (a) 680 (b) 682 (c) 640 (d) 620 (e) 674

Sol. (e); Average = $\frac{8000 \times \frac{15}{100} \times \frac{5}{8} + 8000 \times \frac{16}{100} \times \frac{7}{15}}{2} = \frac{750 + 597.33}{2} = 673.665 \approx 674$

24. The number of 2 BHK flats in F group is approximately what percent of the number of 3 BHK flats in C group?

- (a) $75\frac{4}{21}\%$ (b) $76\frac{4}{21}\%$ (c) $77\frac{5}{21}\%$ (d) $78\frac{3}{19}\%$ (e) $76\frac{2}{21}\%$

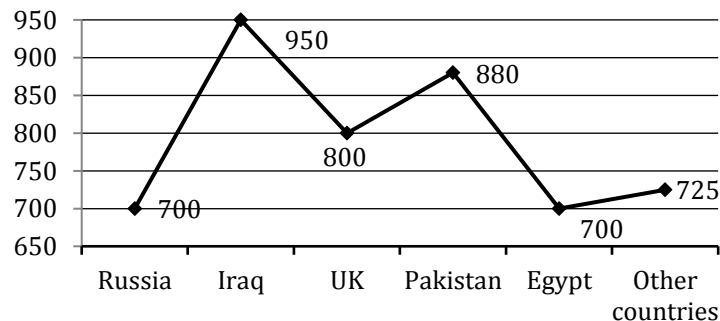
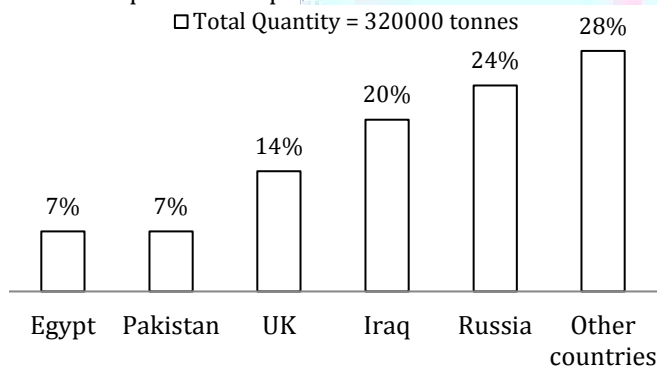
Sol. (b); Required % = $\frac{8000 \times \frac{8}{100} \times \frac{3}{4}}{8000 \times \frac{18}{100} \times \frac{7}{16}} \times 100 = \frac{480}{630} \times 100 = 76\frac{4}{21}\%$

25. What is the difference between the number of 2 BHK flats in D group and number of 3 BHK flats in B group?

- (a) 80 (b) 75 (c) 88 (d) 85 (e) 72

Sol. (b); Required Difference = $8000 \times \frac{33}{100} \times \frac{5}{16} - 8000 \times \frac{15}{100} \times \frac{5}{8} = 825 - 750 = 75$

Directions (26-30): The Bar-graph shows the percentage distribution share of export of Tea from India and the Line-graph shows the price of Tea per tonne in different countries in December 2016?



(Note: The price of Tea is given in \$ per tonne.)

26. What is the difference between the average of the export of Tea to Russia and Iraq and the average of the export of Tea to other countries, Pakistan and Egypt? (in tonnes)

- (a) 25400 (b) 26500 (c) 27600 (d) 24600 (e) 25600

Sol. (e); Difference = $\left(\frac{20+24}{2} - \frac{28+7+7}{3} \right) \% \text{ of } 320000$
 $= 8\% \text{ of } 320000 = 25600$

27. If 40% of the export of Tea to other countries goes to the Australia and the export price for the Australia is \$900 per tonne then what is the total price of Tea exported to the Australia? (in \$ thousand)

- (a) 36525 (b) 35256 (c) 32256 (d) 32265 (e) 35525

Sol. (c); Total price of Tea exported to the Australia = $\frac{40}{100} \times \frac{28}{100} \times 320000 \times 900 = 32256 \$ \text{ thousand}$

28. If the export of Tea to Pakistan is 20% less in September compared to that in December and the price of export of Tea to Pakistan is 10% more in September compared to that in December, then find the total price of Tea exported to Pakistan in September (in \$ thousand).

- (a) 1770.6560 (b) 1734.8560 (c) 1634.7560 (d) 2283.8560 (e) 17346.560

Sol. (e); Total price of tea exported to Pakistan = $\frac{80}{100} \times \frac{7}{100} \times 320000 \times 880 \times \frac{110}{100} = 17346.560 \$ \text{ (thousand)}$

29. What is the ratio of total price of exported Tea to Egypt to the total price of exported Tea to Other countries?
 (a) 7 : 31 (b) 7 : 19 (c) 7 : 27 (d) 7 : 29 (e) 7 : 25

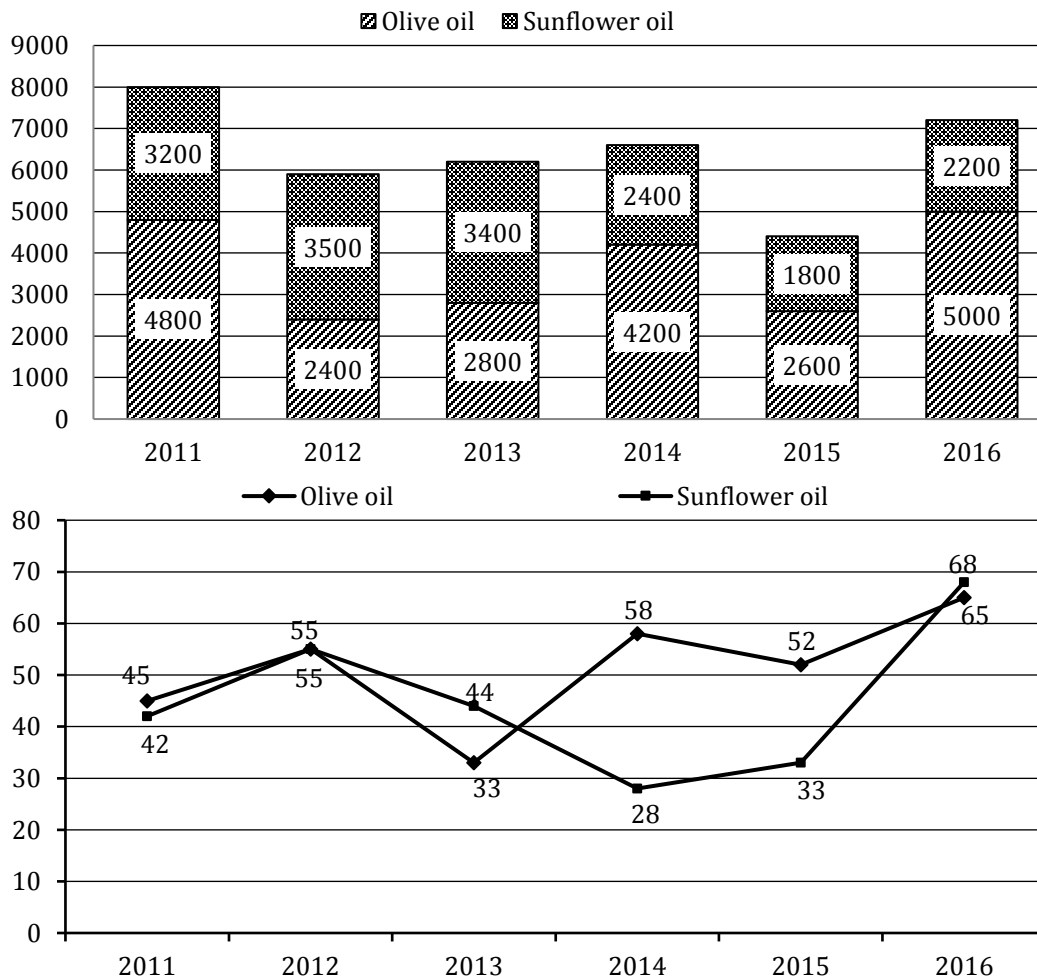
Sol. (d); Required Ratio = $\frac{7 \times 700}{28 \times 725} = \frac{7}{29}$

30. What is the difference between the total price of tea exported to Russia and the total price of tea exported to UK and Egypt together? (in \$ thousand)
 (a) 2240 (b) 2400 (c) 2248 (d) 2420 (e) 2244

Sol. (a); Total price of tea exported to Russia = $\frac{24}{100} \times 320000 \times 700 = 53760$ \$ (thousand)
 Total price of tea exported to UK and Egypt
 $= \left(\frac{14}{100} \times 800 + \frac{7}{100} \times 700 \right) \times 320000 = (112 + 49) \times 320000 = 51520$ \$ (thousand)
 Difference = $53760 - 51520 = 2240$ \$ thousand

Directions (31-35): Read the following graph carefully and answer the questions given below.

Bar graph shows the production (in litres) of Olive oil and Sunflower oil by a firm in 6 different years and the line graph shows the percentage of these two oils exported in respective years.



31. Find the quantity of Olive oil exported throughout the given years?
 (a) 11342 l (b) 11440 l (c) 11442 l (d) 11242 l (e) 11042 l

Sol. (c); Olive oil exported = $4800 \times \frac{45}{100} + 2400 \times \frac{55}{100} + 2800 \times \frac{33}{100} + 4200 \times \frac{58}{100} + 2600 \times \frac{52}{100} + 5000 \times \frac{65}{100}$
 $= 2160 + 1320 + 924 + 2436 + 1352 + 3250 = 11442$ l

32. What is the ratio of sunflower oil exported in 2011 to Olive oil exported in 2014 ?
 (a) 16 : 31 (b) 15 : 29 (c) 16 : 29 (d) 29 : 16 (e) 16 : 27

Sol. (c); Required ratio = $\frac{0.42 \times 3200}{0.58 \times 4200} = \frac{1344}{2436} = \frac{16}{29}$

33. What is the approximate percentage increase in the export of Olive oil during year 2015-16?
 (a) 135% (b) 140% (c) 130% (d) 132% (e) 148%

Sol. (b); Required percentage increase = $\frac{5000 \times \frac{65}{100} - 2600 \times \frac{52}{100}}{2600 \times \frac{52}{100}} \times 100 = \frac{3250 - 1352}{1352} \times 100 \approx 140\%$

34. What is the difference in the average quantity, of the two oils produced in all the years together?
 (a) $883\frac{1}{3}$ (b) $883\frac{2}{3}$ (c) 883.57 l (d) 883 l (e) $883\frac{2}{5}$

Sol. (a); Average of Sunflower oil produced = $\frac{1}{6} \times 16500 = 2750$ l

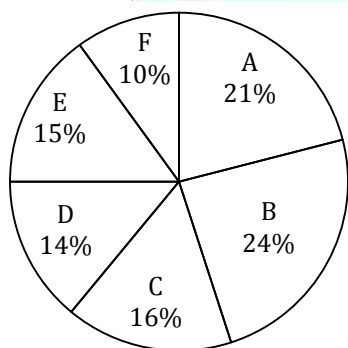
Average of Olive oil produced = $\frac{1}{6} \times 21800 = 3633\frac{1}{3}$

$\therefore$ Required difference = $883\frac{1}{3}$

35. Find the quantity of Sunflower oil that was not exported from year 2012 to 2014.
 (a) 5007 l (b) 5107 l (c) 5200 l (d) 5207 l (e) 5720 l

Sol. (d); Sunflower oil, which not exported from year 2012-2014 = $3500 \times \frac{45}{100} + 3400 \times \frac{56}{100} + 4200 \times \frac{24}{100}$
 Required quantity = 1575 + 1904 + 1728 = 5207 l

Directions (36-40): The following pie chart shows the distribution of the total population of six cities and the table shows the percentage of adults in these cities and the ratio of males to females among these adult populations. Total population of six cities together is 8.5 lakh.



City	% Adult	Males : Females
A	72	7 : 5
B	65	8 : 5
C	75	3 : 2
D	80	9 : 7
E	70	4 : 3
F	60	7 : 5

36. The number of adults population of City A is how many times the adult population of city D ?
 (a) 0.85 (b) 1.35 (c) 1.75 (d) 1.45 (e) 2

Sol. (b); City D: $\frac{8.5 \times 14 \times 80}{100} = \frac{20}{27}$
 City A: $\frac{8.5 \times 21 \times 72}{100} = \frac{27}{20}$
 Required value = $\frac{27}{20} = 1.35$ times

37. What is the difference between total Adult population of cities C and D together and total male (adults) from C, D and F together ?
 (a) 52700 (b) 52000 (c) 57000 (d) 52900 (e) 57500

Sol. (a); Required difference = $8.5 \times \left(\frac{16}{100} \times \frac{75}{100} + \frac{14 \times 80}{10000} \right) - 8.5 \left(\frac{16 \times 75 \times 3}{10000 \times 5} + \frac{14 \times 80}{10000} \times \frac{9}{16} + \frac{10 \times 60 \times 7}{10000 \times 12} \right)$
 = $85[1200 + 1120] - 85[720 + 630 + 350] = 52700$

38. What is the ratio between the adult females of city A and B together to the adult male population of city D and E together?
 (a) 1 : 2 (b) 1 : 3 (c) 1 : 1 (d) 1 : 4 (e) 2 : 1

Sol. (c); $\frac{8.5 \left[\frac{21 \times 72 \times 5}{10000 \times 12} + \frac{24 \times 65 \times 5}{10000 \times 13} \right]}{8.5 \left[\frac{14 \times 80 \times 9}{10000 \times 16} + \frac{15 \times 70 \times 4}{10000 \times 7} \right]} = \frac{630 + 600}{630 + 600} = 1 : 1$

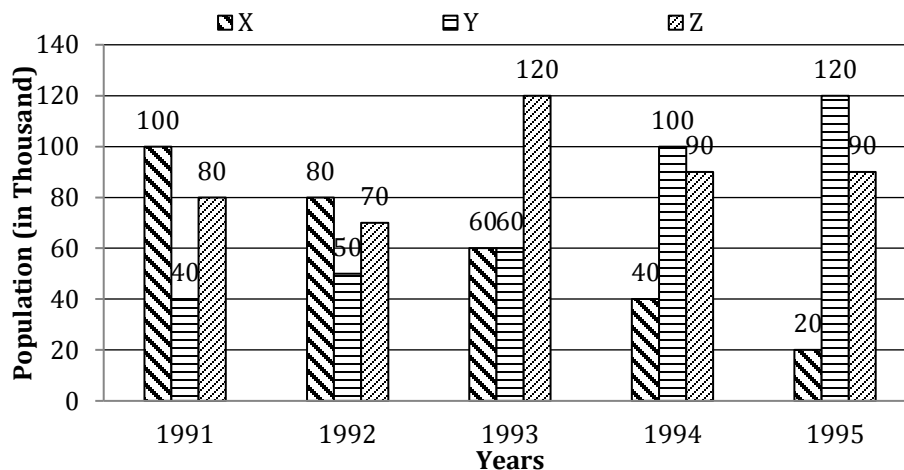
39. What is difference between total central angle of A, B and F together and C, E and F together ?
 (a) 49.4° (b) 45° (c) 50° (d) 50.4° (e) 50.8°

Sol. (d); $A + B + F = 21 + 24 + 10 = 55\%$
 $C + E + F = 16 + 15 + 10 = 41\%$
 Difference = $14\% = 14 \times \frac{18}{5} = 50.4^\circ$

- 40.** If 10% of adults from City A is graduate, then what is the ratio between graduate from City A and adult female population from city B ?
 (a) 25 : 63 (b) 63:25 (c) 63 : 29 (d) 29 : 57 (e) 63:250

Sol. (e); $\frac{1}{10} \left[\frac{850000 \times 21 \times 72}{100 \times 100} \right] = 12852$ graduates are 10% of Adults from city A
 Adult females from B = $\frac{24}{100} \times \frac{65}{100} \times \frac{5}{13} \times 850000$
 Ratio = $\frac{85 \times 21 \times 7.2}{24 \times 25 \times 85} = 63 : 250$

Directions (41-45): The following graph indicates the population of three different villages in five successive years, and bar graph shows ratio of male to female of three villages in five successive years.



Ratio of male to female population					
years \ Villages	1991	1992	1993	1994	1995
X	11 : 9	5 : 3	5 : 3	3 : 5	2 : 3
Y	3 : 5	2 : 3	8 : 7	3 : 7	1 : 1
Z	9 : 7	3 : 4	3 : 2	5 : 4	4 : 5

- 41.** What is the ratio between total number of males of village X & Y together in 1992 to the population of village Z in 1995 ?
 (a) 3 : 7 (b) 7 : 9 (c) 11 : 9 (d) 9 : 7 (e) 7:8

Sol. (b); Males in X in 1992 : $80 \times \frac{5}{8} = 50$ thousand
 Males in Y in 1992 : $50 \times \frac{2}{5} = 20$ thousand
 No. of population in Village Z in 1995 = 90 thousand
 Required Ratio = $\frac{70}{90} = 7 : 9$

- 42.** Which of the following villages shows continuous and uniform increase/decrease in their population over the years ?
 (a) Y (b) Z (c) X (d) X & Y (e) All of the above

Sol. (c); It is clearly visible from the graph that Population in village X decline continuously and uniformly

- 43.** Number of males in village Z over the years is how many times the number of females in village X over the years?(around two decimal places)
 (a) 1.76 (b) 1.56 (c) 2.76 (d) 2.54 (e) 1.45

Sol. (a); No of males in Z over the years = $80 \times \frac{9}{16} + 70 \times \frac{3}{7} + 120 \times \frac{3}{5} + \frac{5}{9} \times 90 + \frac{4}{9} \times 90$
 $= 45 + 30 + 72 + 50 + 40 = 237$ thousand
 No. of females in X over the years = $100 \times \frac{9}{20} + 80 \times \frac{3}{8} + 60 \times \frac{3}{8} + \frac{5}{8} \times 40 + \frac{3}{5} \times 20$
 $= 45 + 30 + 22.5 + 25 + 12 = 134.500$
 $\therefore$ Required Ratio = $\frac{237000}{134500} = 1.76$ times

44. Number of males from all villages in 1993 & 1994 together is how much percent more or less than the number of females of village Y in same years?(approx.)
 (a) 135 % less (b) 141.3 % less (c) 135 % more (d) 141.3 % more (e) 145.3% more

Sol. (d); No. of males in 1993 = $60 \times \frac{5}{8} + 60 \times \frac{8}{15} + 120 \times \frac{3}{5} = 141.5$ thousand

$$\text{No. of males in 1994} = 40 \times \frac{3}{8} + 100 \times \frac{3}{10} + 90 \times \frac{5}{9} = 95 \text{ thousand}$$

Total males = 236.5 thousand

$$\left. \begin{array}{l} \text{No. of females from Y in 1993} = 60 \times \frac{7}{15} = 28 \\ \text{No. of females from Y in 1994} = 100 \times \frac{7}{10} = 70 \end{array} \right\} \text{Total} = 98$$

Difference = 236.5 - 98 = 138.5 thousand.

$$\text{Required\%} = \frac{138.5}{98} \times 100 = 141.3\% \text{ more}$$

45. Find the difference between the average of females from village X and average population of village Z over the entire years?

(a) 63100 (b) 65200 (c) 63300 (d) 60000 (e) 63000

Sol. (a); Average No. of female in X over the years =

$$\frac{1}{5} (100 \times \frac{9}{20} + 80 \times \frac{3}{8} + 60 \times \frac{3}{8} + 40 \times \frac{5}{8} + 20 \times \frac{3}{5}) = \frac{134500}{5} = 26,900$$

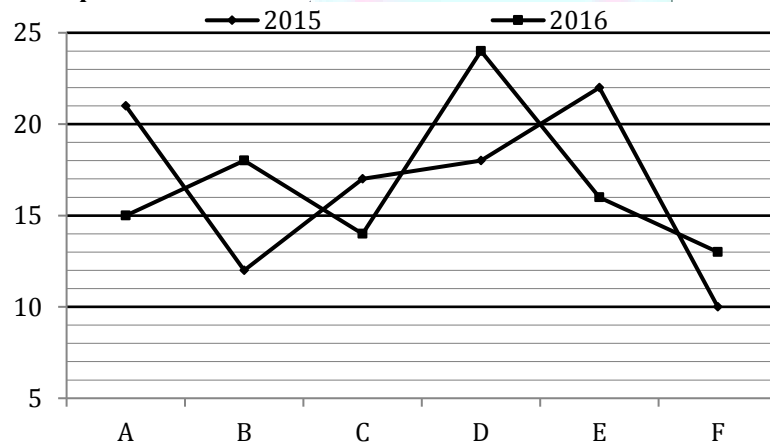
$$\text{Average population of village Z over the years} = \frac{1}{5} [80 + 70 + 120 + 90 + 90] = \frac{450000}{5} = 90000$$

$$\text{Required difference} = 90000 - 26900 = 63,100$$

Directions (46-50): Read the following graph carefully and answer the questions given below:

The line graph shows the percentage distribution of total number of students who got placed successfully in campus placement in two successive year after their graduation from six different universities while the table provides their female to male ratio.

Total placed students are 42500 and 44000 in 2015 and 2016 respectively.



Year → University ↓	2015 F : M	2016 F : M
A	11 : 4	1 : 2
B	10 : 7	7 : 9
C	5 : 12	9 : 11
D	13 : 12	2 : 1
E	1 : 4	1 : 4
F	3 : 2	5 : 3

46. Total number of females placed from University B in 2015 are what percent of total males placed from University C in same year ?

(a) $54\frac{14}{17}\%$ (b) $55\frac{14}{17}\%$ (c) $58\frac{14}{17}\%$ (d) $60\frac{14}{17}\%$ (e) $62\frac{14}{17}\%$

Sol. (c); Placed females from B in 2015 = $\frac{10}{17} \times \frac{12}{100} \times 42,500 = 3,000$

$$\text{Placed males from C in 2015} = \frac{12}{17} \times \frac{17}{100} \times 42,500 = 5,100$$

$$\therefore \text{Required percent} = \frac{3,000}{5,100} \times 100 = 58\frac{14}{17}\%$$

47. Find the difference between total females placed from university D in both years and that of from university E in both years.

(a) 7040 (b) 7470 (c) 7740 (d) 7570 (e) 7470

Sol. (c); Placed female from D = $\frac{13}{25} \times \frac{18}{100} \times 42,500 + \frac{2}{3} \times \frac{24}{100} \times 44,000 = 3978 + 7040 = 11,018$

$$\text{Placed females from E} = \frac{1}{5} \times \frac{22}{100} \times 42,500 + \frac{1}{5} \times \frac{16}{100} \times 44,000 = 1870 + 1408 = 3278$$

$$\therefore \text{Required difference} = 7,740$$

48. What is the ratio of male students from university F in 2015 to male students from university C in 2015 who got placement ?

(a) 3 : 4

(b) 3 : 1

(c) 4 : 3

(d) 1 : 3

(e) 1 : 4

Sol. (d); Required ratio = $\frac{\frac{2}{5} \times \frac{10}{100} \times 42500}{\frac{12}{17} \times \frac{17}{100} \times 42500} = \frac{1}{3}$

49. Find the average number of placed students from all universities except university D in 2015.

(a) 6790

(b) 6970

(c) 6975

(d) 7690

(e) None of these

Sol. (b); Required average = $\frac{42500}{5} \times \frac{1}{100} (21 + 12 + 17 + 22 + 10)$
 $\frac{425}{5} \times 82 = 6970$

50. Female students placed from university E in 2015 are what percent less than female students placed from university A in 2016 ?

(a) 25%

(b) 20%

(c) 15%

(d) 12%

(e) 16%

Sol. (c); Female students placed from university E in 2015 = $\frac{1}{5} \times \frac{22}{100} \times 42,500 = 1,870$
 Female students placed from university A in 2016 = $\frac{1}{3} \times \frac{15}{100} \times 44,000 = 2,200$
 $\therefore$ Required percentage = $\frac{2200 - 1870}{2200} \times 100 = 15\%$



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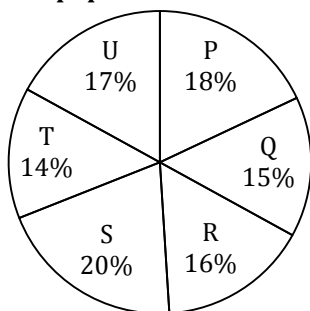
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PREVIOUS YEARS QUESTION

Directions (1-5): The following pie-chart shows the distribution of the total population of six villages and the table shows the percentage of illiterate population in these villages and the ratio of males to females among literate population.

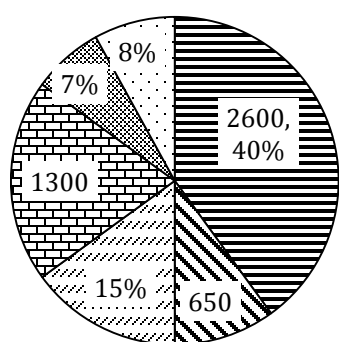
Total population = 8.5 lakhs



Village	% Illiterate	Males : Females
P	80	2:3
Q	75	4:1
R	60	5:3
S	72	4:3
T	65	2:5
U	70	8:7

- Find the average number of literate population of the given village except village R.
(a) 39010 (b) 30915 (c) 39015 (d) 42015 (e) 38015
- The number of literate males of village S is what percent of literate population of village T?
(a) $60\frac{15}{49}\%$ (b) $65\frac{15}{49}\%$ (c) $68\frac{15}{49}\%$ (d) $63\frac{15}{49}\%$ (e) $58\frac{15}{49}\%$
- Among the literate females of village R, 20% are graduate, 25% are postgraduate, 50% are having master degree. If the remaining females have PhD degree, then find the number females having PhD degree.
(a) 1025 (b) 2010 (c) 1030 (d) 1020 (e) 1012
- Find the total number of literate females in the village P, R and T together.
(a) 58510 (b) 68510 (c) 68150 (d) 65810 (e) 68015
- Literate males of village P are by what percent (approx.) more or less than literate females of Village U?
(a) 38% more (b) 35% more (c) 40% less (d) 45% less (e) 48% more

Directions (6 - 10): Eminent writer Bhule Tripathi has to publish a book of two different editions. Prior to publishing, the pages of books have been proofread by several scholars. Pie chart represents the no. of pages proofread by Scholars while the following table represents the errors found. Read the following questions and answer the questions.



- ☐ Mohit
- ☐ Kamal
- ☐ Prakash
- ☐ Raaj
- ☐ Logan
- ☐ Karolina Marine

Scholars	Number of mistakes found
Mohit	400
Kamal	130
Prakash	25
Raaj	130
Logan	76
Karolina Marine	85

- No of pages proofread by Karolina Marine and Mohit together is approximately how many times than pages proofread by Kamal and Prakash together?
(a) 2.5 (b) 2 (c) 3 (d) 1.5 (e) 1
- Pages per error is maximum for which person when considering minimum 600 pages being proofread?
(a) Prakash (b) Logan (c) Karolina (d) Mohit (e) Raaj
- Prakash refuses to read 40% of the pages allotted to him which is now distributed equally among Kamal and Logan. Now, the no of pages being proofread by Kamal is what % less than that by Mohit?
(a) 65 % (b) 66 % (c) 67.5 % (d) 68 % (e) 69%
- What will be the ratio of Errors found by Mohit and Kamal together to the average no of errors found by all of them?
(a) 530 : 141 (b) 543 : 200 (c) 427 : 143 (d) 734 : 353 (e) 147 : 141
- In order to complete the proceedings in time, One more scholar Siddharta Singh included in for proofreading. Everyone other than Logan and Karolina Marine was asked to allocate 10 percent of their pages allocated to them while Mohit was asked to allocate 30% of his pages to Siddharta Singh for proofreading. What will be the total no. of pages proofread by Siddharta Singh (approximately).
(a) 1030 pages (b) 1053 pages (c) 1073 pages (d) 1000 pages (e) 1100 pages

Direction (11-15): Study the given Chart carefully to answer the following questions.

Total number of people in the 5 villages comprising Men, women and children is shown in pie chart

Total number of people from all villages = 70,000

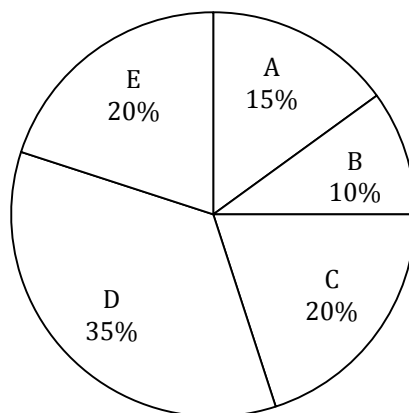


Table shows the Ratio of men to women and percentage of children in five villages.

(Person who have attained age equal to or above 18 years are men and women and person below 18 years are children)

	Men : Women (Above or equal to 18 years of age)	% of children (below 18 years)
A	7 : 8	28
B	9 : 5	15
C	3 : 4	18
D	13 : 12	20
E	2 : 3	15

11. What is the ratio of number of men from village C to the number of women from village A and B together ? (approximately)
(a) 0.8 (b) 0.5 (c) 1.1 (d) 0.4 (e) 1.8
12. What is the average of the number of children from all five villages ?
(a) 2530 (b) 2670 (c) 2850 (d) 2480 (e) 2702
13. Number of women from village C and E together are approximately what percent less or more than number of women from village B and D together ?
(a) 15% (b) 19% (c) 23% (d) 13% (e) 29%
14. If 25% children from village A attains 18 years of age after one year then what is the percentage increase in the number of adults (equal to or above 18 years) in village A (approximately)
(a) 7% (b) 12% (c) 6% (d) 10% (e) 18%
15. What is the number of women above or equal to 18 years from all villages together.
(a) 25472 (b) 29265 (c) 26583 (d) 14391 (e) 26568

Directions (16-20):The following pie-chart shows the percentage of passed candidate in SBI exam from cities X, Y, Z, K, L and M out of the total passed candidates from all six cities together in year 2010.

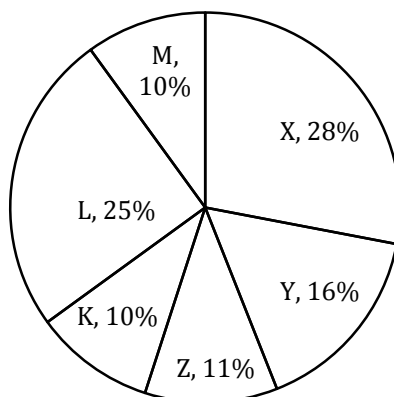


Table shows the percentage of fresher candidate who passed from each city out of the total candidates passed from that city in year 2010.

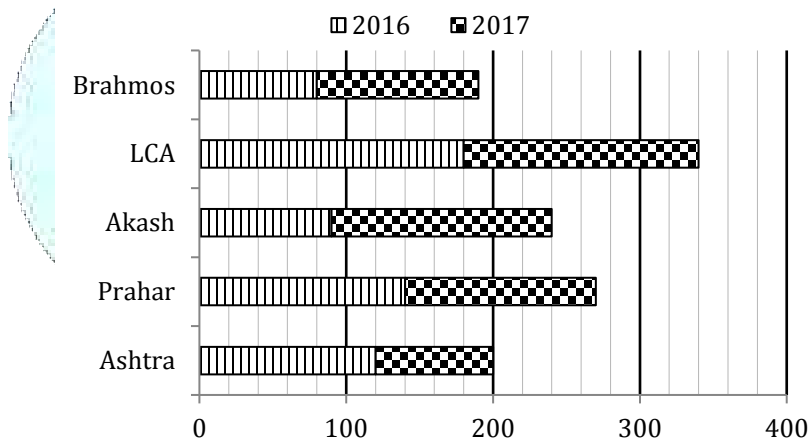
Cities	Percentage of fresher candidates passed in SBI exam
--------	---

X	20%
Y	25%
Z	15%
K	25%
L	12%
M	11%

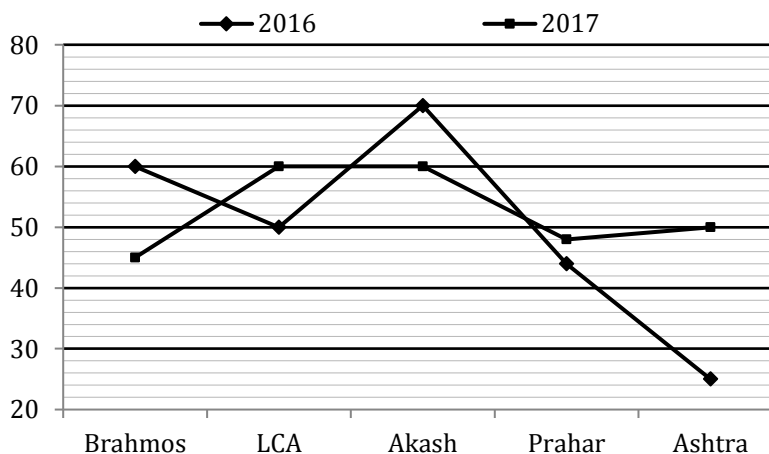
16. If in year 2010, total number of freshers passed from city K was 320, then how many freshers candidates passed the SBI exam from city L ?
 (a) 384 (b) 284 (c) 364 (d) 360 (e) 424
17. If in year 2010, total passed candidates from all cities was 1250, then what is the number of the non-fresher candidate from city X who passed the SBI exam in same year?
 (a) 140 (b) 210 (c) 420 (d) 280 (e) 320
18. If the non-fresher candidates passed from city Y in year 2010 were 180, then how many total candidates passed the SBI exam from all cities together?
 (a) 1450 (b) 1200 (c) 1500 (d) 1250 (e) 1650
19. If there is an increase of 10% and 20% in the number of passed candidates in city X and Y in year 2011 respectively from year 2010 and total passed candidate from city Z in 2010 was 770. Then what would be the difference in no. of passed candidates from city X and Y in year 2011?
 (a) 712 (b) 812 (c) 912 (d) 880 (e) 972
20. If total passed candidates from city Y in year 2010 was 320, then what is the ratio between the no. of freshers passed from city X and that of non-fresher passed from city Z?
 (a) 112 : 187 (b) 113 : 186 (c) 3 : 5 (d) 187 : 112 (e) None of these

Directions (21-25): Carefully study the following bar graph and line graph to answer the questions that follow.

Bar graph is showing the number of weapons of different types exported by DRDO in 2016 and 2017



Line graph is showing the percentage of weapons not exported

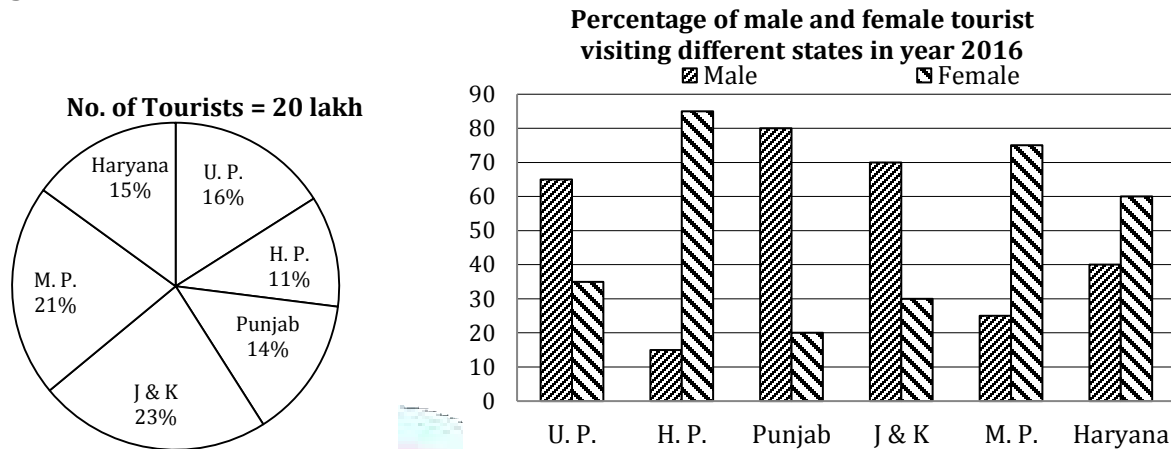


21. DRDO decided to import same number of Assault rifles as the number of Brahmos missiles manufactured by it in 2017, what is the number of Assault rifles imported ?
 (a) 320 (b) 210 (c) 190 (d) 200 (e) 220

22. Find the difference in the number of Prahar missile manufactured in 2016 and that Akash missiles not exported in 2017.
 (a) 22 (b) 25 (c) 35 (d) 20 (e) 28
23. What is the ratio of total number of Ashtra missiles manufactured in 2017 and that of LCA manufactured in 2016 ?
 (a) 5 : 9 (b) 3 : 10 (c) 4 : 9 (d) 2 : 5 (e) 9 : 4
24. Prahar missiles not exported in 2017 are by what percent more or less than Ashtra missiles that were not exported in 2016 ?
 (a) 200% (b) 120% (c) 300% (d) 150% (e) 210%
25. LCA exported in 2016 contributed what percent of total LCA produced in the given two years together ?
 (a) $21\frac{13}{19}\%$ (b) $27\frac{13}{19}\%$ (c) $26\frac{13}{19}\%$ (d) $19\frac{13}{19}\%$ (e) $23\frac{13}{19}\%$

Directions (26-30): Study the graph and answer the following questions

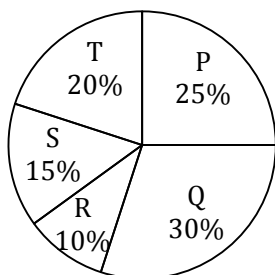
Percentage distribution of total tourists visiting different states in year 2016



26. If total tourist visiting in year 2017 in U.P. is increased by 25% from total tourist visiting U.P. in year 2016 and percentage of male and female tourist visiting U.P. in 2017 remains same as visiting in 2016 in U.P., then find the difference of male and female tourist visiting U.P. in 2017? (in lakh)
 (a) 1.6 (b) 2.1 (c) 1.2 (d) 1.4 (e) 2.2
27. Tourist visiting in state J&K in year 2016 speaks three language that is Hindi, English and Urdu are in ratio 2:3:5. Then find the total number of tourist who can speak both Urdu and Hindi? (in lakh)
 (a) 3.22 (b) 4.22 (c) 2.60 (d) 3.36 (e) 2.33
28. 20% of tourist who are visiting Haryana in year 2016 left Haryana and visited Punjab such that number of female tourist in Punjab remain same. Then find the ratio of male and female tourist in Punjab?
 (a) 14:71 (b) 71:14 (c) 2:5 (d) 5:2 (e) 81 : 14
29. If number of tourists visiting M.P. in year 2017 is increased by $7\frac{1}{7}\%$, then find the ratio of male and female tourists visiting M.P. in year 2017 ?
 (a) 5:13 (b) 5:12 (c) Cannot be determined
 (d) 3:5 (e) None of these
30. Female tourist visiting H.P in year 2016 is what percent of male tourists visiting M.P. in year 2016 ?
 (a) 128% (b) $278\frac{1}{21}\%$ (c) $158\frac{4}{21}\%$ (d) $178\frac{2}{21}\%$ (e) $168\frac{2}{21}\%$

Directions (31-35): Given below, pie graph show distribution of total number of male employee of HCL in five branches, P Q R S and T, and table shows ratio of male to female employee in each branch. Give the answer of the question according to given data:

Total male employee = 1500



Branch	Male : Female
P	15 : 8
Q	3 : 2
R	3 : 4
S	9 : 7
T	12 : 11

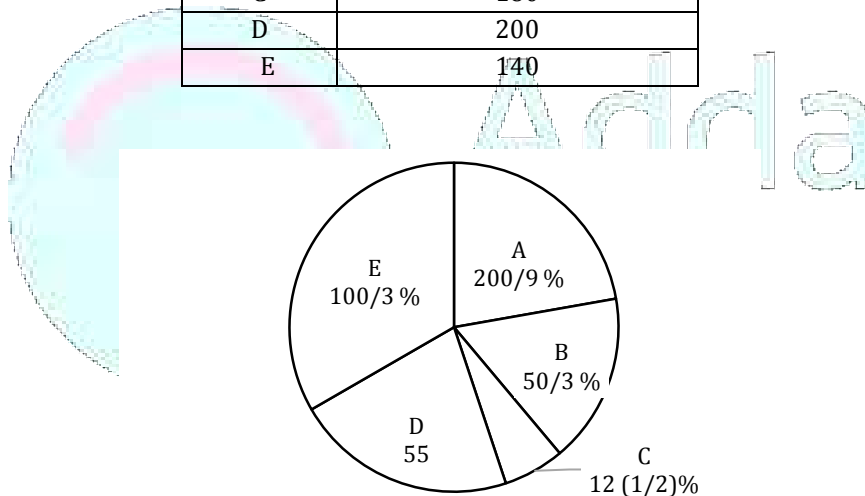
31. What is ratio between total male and female employee from branch T and R together to total number of female employee from branch S and T together ?

32. What is the difference between average of male employee from branch S, T and Q and average of female employee from branch S and R ?
 (a) $\frac{265}{2}$ (b) $\frac{275}{2}$ (c) $\frac{255}{2}$ (d) $\frac{285}{2}$ (e) None of these
33. 20% and 40% of male from branch T and Q respectively are married and ratio between married male to married female from company T and Q is 4 : 5 and 3 : 2 respectively. Then find total unmarried male and female employee from both branches ?
 (a) 900 (b) 860 (c) 870 (d) 890 (e) 990
34. 20% male employee from branch P are B. tech and 40% female employee from branch R are MBA. Then find total B.tech male employee from branch P, are how much percent more or less then total MBA female employee from branch T ?
 (a) 6.25% (b) 5% (c) 3% (d) 8.25% (e) 12.25%
35. Total male and female employee from branch T and S together is what percent of total male and female employee from branch R and Q together ?
 (a) $81\frac{7}{11}\%$ (b) $80\frac{7}{11}\%$ (c) $88\frac{7}{11}\%$ (d) 72% (e) $91\frac{7}{11}\%$

Directions (36-40): Table given below shows five types of article sold by a seller and selling price of each article. There is also a pie chart which shows the percentage distribution of these five articles (except article D), sold by the seller.

Distribution for article D is given in absolute value

Article	Selling Price of each article
A	150
B	120
C	180
D	200
E	140



36. Total amount got by selling article A is what percent more than the total amount got by selling article B?
 (a) $66\frac{2}{3}\%$ (b) $62\frac{2}{3}\%$ (c) 65% (d) $53\frac{1}{3}\%$ (e) 52%
37. What is the average amount got by selling article B, C and E together?
 (a) 11700 (b) 12560 (c) 10700 (d) 12500 (e) 10520
38. Suppose the seller wanted to sell another article named F whose total selling price is $11\frac{1}{9}\%$ more than the total selling price of article C and no. of article of F is equal to the no. of article of type E. Then find the difference between price of each article of F and that of each article B ?
 (a) 35 (b) 45 (c) 60 (d) 50 (e) 60
39. If the seller sells $\frac{2}{5}$ th of number of article of type C at 40% more than the price of each article of type B and sells the remaining article of type C at $\frac{3}{5}$ th of price of each article of type A. Then find the difference between total new price of article of type C and the average of total price of article of type B and D?
 (a) 2645 (b) 3246 (c) 3356 (d) 3646 (e) 3426
40. Find the ratio of average of total selling price of article A and E together to the average of total selling price of article B and C together?
 (a) 32 : 17 (b) 17 : 32 (c) 32 : 19 (d) 5 : 3 (e) 33 : 17

Directions (41- 45): Study the following pie chart and table carefully and answer the questions.

Pie chart shows the percentage distribution of cars sold by a seller in six different cities (except city F).

Distribution for city F is given in absolute value

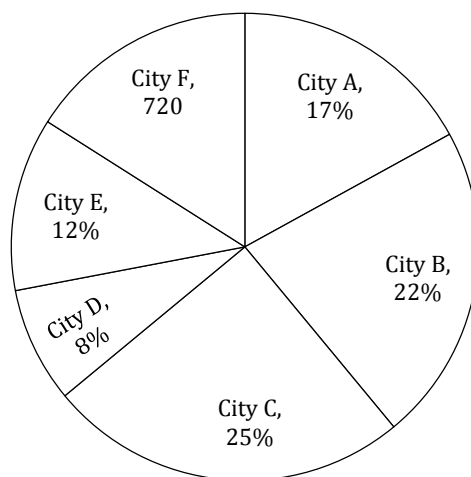


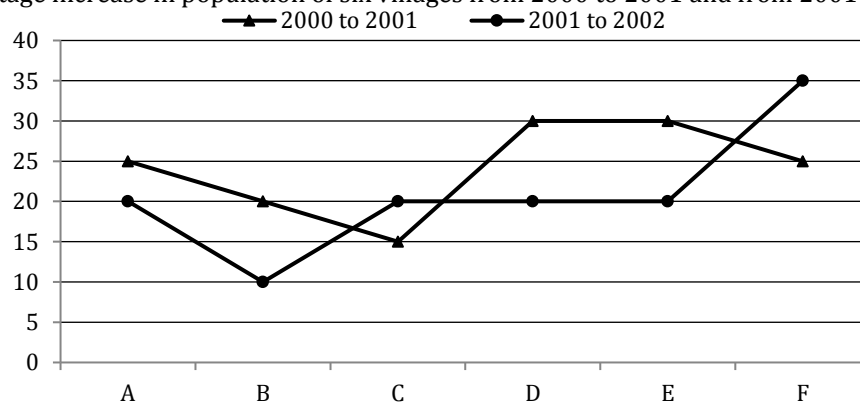
Table shows the ratio of no. of cars sold of company Tata to Company Suzuki sold in different cities.

City	Tata : Suzuki
A	7 : 8
B	5 : 4
C	2 : 3
D	5 : 7
E	8 : 7
F	9 : 7

41. What is the ratio of cars sold of company Suzuki to city A to the cars sold of same company in city E?
 (a) 21 : 34 (b) 34 : 21 (c) 31 : 21 (d) 34 : 19 (e) 34 : 31
42. What is the difference of average of cars sold of company Tata in city B and C together and average of Cars Sold of Company Suzuki in city D, F and E together?
 (a) 241 (b) 231 (c) 280 (d) 211 (e) 341
43. The number of cars sold of company Tata in city D is what percent more/less than the number of cars sold of company Suzuki in city C?
 (a) $83\frac{21}{27}\%$ (b) $72\frac{21}{27}\%$ (c) $77\frac{21}{27}\%$ (d) $68\frac{21}{27}\%$ (e) $62\frac{21}{27}\%$
44. If the seller stopped $9\frac{1}{11}\%$ of the supply of company Tata in city B and distributed the stopped supply of city B among the other five cities in equal number. Then find the new total Tata cars sold in city D and City A together?
 (a) 678 (b) 625 (c) 547 (d) 587 (e) 527
45. If cars sold in city C is increased by $11\frac{1}{9}\%$ and cars sold in city D is increased by $66\frac{2}{3}\%$. Find the difference of new total Tata cars sold in city C and D together to new total Suzuki cars sold in city C and D together? (ratio of car sold of Tata and Suzuki remain same)
 (a) 375 (b) 420 (c) 450 (d) 350 (e) 400

Directions (46 – 50): Read the following table carefully and answer the questions given below.

Line graph shows Percentage increase in population of six villages from 2000 to 2001 and from 2001 to 2002.



Actual population of these villages in 3 different years.

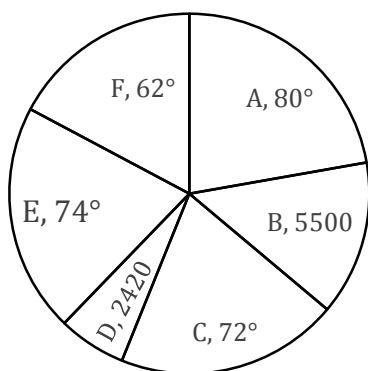
Village \ Years	2000	2001	2002
A	-	-	3750
B	-	1980	-
C	-	-	1518
D	-	-	-
E	1250	-	-
F	1200	-	-

46. What is the ratio of population of village E in 2002 to village A in 2000?
 (a) 41:50 (b) 37:45 (c) 48:31 (d) 44:53 (e) 39:50
47. Population of village A in 2000 is what percent more than population of village C in 2000?
 (a) $129\frac{3}{11}\%$ (b) $127\frac{3}{11}\%$ (c) $135\frac{3}{11}\%$ (d) $123\frac{3}{11}\%$ (e) $117\frac{3}{11}\%$
48. Ratio of Population of village C and D in 2000 is 22:27 respectively, what will be population of village D in 2002?
 (a) 1350 (b) 2108 (c) 1250 (d) 2106 (e) 2601
49. The population of village F in 2000 is what percent of the population of the same village in 2002?
 (a) $53\frac{7}{27}\%$ (b) $59\frac{7}{27}\%$ (c) $49\frac{7}{27}\%$ (d) $57\frac{7}{27}\%$ (e) $59\frac{7}{27}\%$
50. Total population in 2000 is approximate what percent less than the total population in 2002? (Ratio of Population of village C and D in 2000 is 22:27 respectively)
 (a) 33 (b) 39 (c) 37 (d) 38 (e) 40

Directions (51-55): The pie chart given below shows the distribution of number of literate persons in six villages in terms of absolute value or degree measures.

The table shows the value of percentage by which illiterate persons are more or less than literate persons in six villages.

Note: Difference between degree measure of village B and D is 28°



Villages	Percentage by which illiterates are more or less than literates
A	$13\frac{7}{11}\%$ more
B	$9\frac{1}{11}\%$ less
C	25% more
D	$18\frac{2}{11}\%$ more
E	25% less
F	$18\frac{2}{11}\%$ more

51. Find the total number of illiterate persons in village B and E together?
 (a) 11150 (b) 10105 (c) 11105 (d) 11050 (e) 10050
52. Illiterate persons of village A are what percent of total literate persons of all villages together?
 (a) $25\frac{25}{99}\%$ (b) $23\frac{25}{99}\%$ (c) $24\frac{25}{99}\%$ (d) $22\frac{25}{99}\%$ (e) $28\frac{25}{99}\%$
53. Find the ratio of total literate persons of villages A and D together and these of villages F and B together?
 (a) 65 : 51 (b) 51 : 76 (c) 56 : 51 (d) 51 : 56 (e) 51 : 57
54. If $15/26$ th of total illiterate persons in village F are males while $13/22$ th of total literate persons in same village are males, then find the total number of males in village F.
 (a) 5870 (b) 8680 (c) 6860 (d) 8750 (e) 8570
55. What is the average (in terms of degree measures) of the contribution of literate persons of villages B, C, D and E together?
 (a) 55.4° (b) 56.8° (c) 54.5° (d) 52.5° (e) None of these

Directions (56-60): Study the following graph and table carefully to answer the questions given below.

The annual income of Arunoday and Annual expenditure of Veer in tonnes from 2012-2016

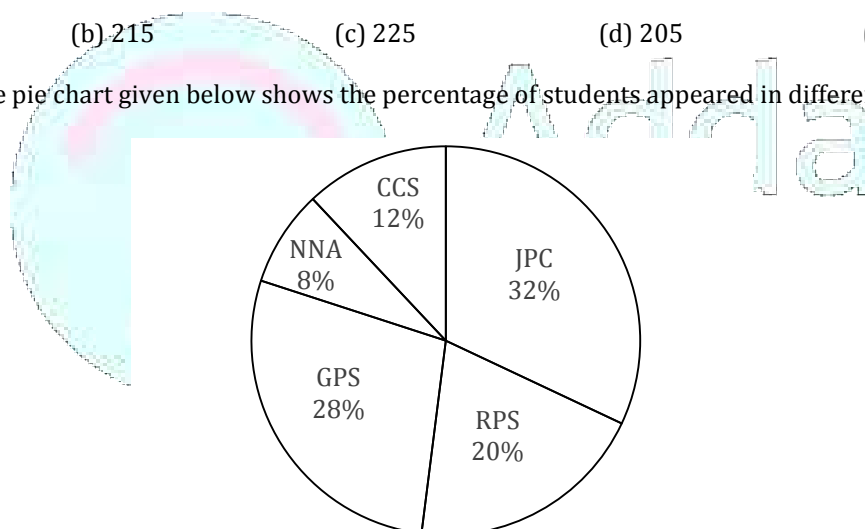
The table given below represents the respective ratio of the annual income of Veer and (Arunoday+Veer) and the respective ratio of the annual expenditure of Arunoday and (Arunoday+Veer)

Year	Annual Income	Annual Expenditure
2012	4 : 9	3 : 5
2013	4 : 11	2 : 5
2014	3 : 8	4 : 7
2015	2 : 9	5 : 8
2016	3 : 11	3 : 5

Note: Annual income = Annual expenditure + Annual saving

56. Find the difference (in thousands) in the average of annual savings of Arunoday in years 2012 and 2016 together and that of Veer in same two years together?
 (a) 160 (b) 165 (c) 155 (d) 175 (e) 150
57. Find the ratio of annual income of Veer in 2013 and 2015 together to annual expenditure of Arunoday in 2014 and 2016 together.
 (a) 6 : 11 (b) 5 : 7 (c) 3 : 7 (d) 6 : 7 (e) 7 : 6
58. Annual saving of Veer in 2015 and 2016 together is approximately what percent more or less than saving of Arunoday in 2013 and 2016 together ?
 (a) 82% (b) 88% (c) 85% (d) 90% (e) 86%
59. In 2017, Annual saving of Arunoday decreased by 12% while that of Veer increased by 15% as compared to that in 2016. Find their total expenditure in 2017 if their annual income in 2017 was same as in 2014 ?
 (a) 485 thousand (b) 495 thousand (c) 478 thousand (d) 458 thousand (e) 475 thousand
60. Find the average value (in thousand) of expenditure of Arunoday in 2015, saving of Veer in 2016 and income of Veer in 2013.
 (a) 210 (b) 215 (c) 225 (d) 205 (e) 250

Directions (61-65): The pie chart given below shows the percentage of students appeared in different schools of same state in year 2015.



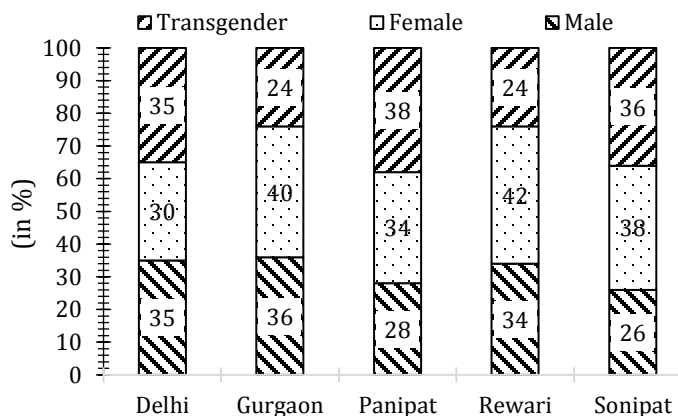
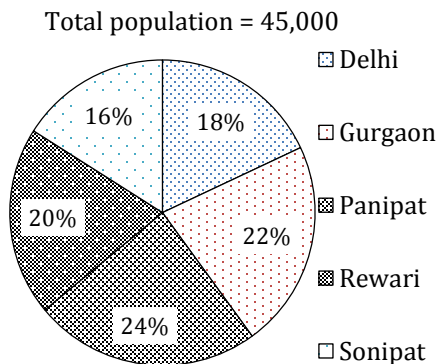
And the table given below shows the percentage of qualified students for different schools of same state in year 2015 and ratio of qualified boys to qualified girls.

School	Percentage qualified	Ratio of qualified boys to girls
JPC	32	1 : 3
RPS	28	5 : 2
GPS	25	3 : 2
NNA	30	1 : 2
CCS	24	5 : 3

61. If unqualified student in school JPC are 2176. Then find the difference between qualified boys and girls in school CCS.
 (a) 92 (b) 144 (c) 72 (d) 128 (e) 115
62. If qualified boys in school NNA is 120. Then qualified girls in school RPS is what percent more than the qualified boys in school NNA ?
 (a) 100% (b) 200% (c) 50% (d) 150% (e) 80%
63. If the difference of appeared student of school CCS and JPC is 1500. Then find the average number of qualified boys of school RPS, GPS and NNA ?
 (a) 275 (b) 250 (c) 325 (d) 225 (e) 260

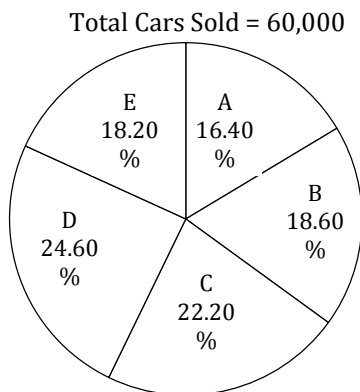
64. If the student appeared in 2016 of school RPS are 1200 which is $33\frac{1}{3}\%$ more than those in 2015, and qualified boys in school RPS in 2016 is same as qualified boys in GPS in 2015. Then find the number of qualified girls in RPS in year 2016 ?
[Given that percentage of qualified students in RPS are same in both years]
(a) 147 (b) 247 (c) 167 (d) 189 (e) 184
65. If there is another school formed in year 2016 with the name JSS which has number of qualified boys $33\frac{1}{3}\%$ more than the qualified boys in school NNA in year 2015, and number of qualified girls $66\frac{2}{3}\%$ less than qualified girls in school GPS in year 2015. Then find the ratio of qualified boys to qualified girls in school JSS ?
[Given that total appeared student in year 2015 is 9000].
(a) 7 : 8 (b) 8 : 7 (c) 3 : 8 (d) 8 : 5 (e) 6 : 7

Directions (66-70): Pie chart given below shows the percentage distribution of population in five cities. Graph given below show the percentage distribution of male, female and transgenders among them. Study the data carefully and answer the questions.



66. Maximum number of females in which city?
(a) Rewari (b) Delhi (c) Panipat (d) Gurgaon (e) Sonipat
67. Find the ratio of population of transgender in city Gurgaon & Panipat together to population of city Rewari except transgender?
(a) 17 : 18 (b) 19 : 18 (c) 18 : 19 (d) 17 : 19 (e) 18 : 13
68. Male population in city Gurgaon and Panipat together is how much more than female population of city Rewari and Sonipat together?
(a) 72 (b) 64 (c) 24 (d) 42 (e) 46
69. Transgender population of city Delhi and Sonipat together is what percent more than female population of city Gurgaon?
(a) $37\frac{2}{11}\%$ (b) $37\frac{3}{22}\%$ (c) $74\frac{1}{11}\%$ (d) $37\frac{1}{22}\%$ (e) $37\frac{5}{22}\%$
70. Female and transgender population of city Panipat is how much more than male & female population of city Gurgaon?
(a) 248 (b) 212 (c) 238 (d) 226 (e) 252

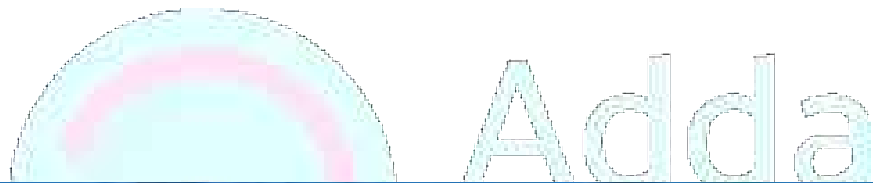
Directions (71 - 75): Pie chart given below shows total cars sold in five different cities. Table given below show the ratio of three models of cars sold in these cities. Study the data carefully and answer the follow questions.



Model \ City	X	Y	Z
A	3	4	5
B	4	3	2
C	6	5	4
D	7	8	9
E	4	5	5

71. 'Y' type car sold in city 'D' is what percent more than 'Z' type car sold in city 'A'.
(a) 18% (b) 14% (c) 28% (d) 20% (e) 10%

72. Find the average number of 'X' type car sold in city 'C', D and E together?
 (a) 4152 (b) 4251 (c) 4215 (d) 4125 (e) 4521
73. Find the ratio of 'X' and 'Y' type car sold in city 'A' together to 'X' and 'Y' type car sold in city 'B' together?
 (a) 37 : 41 (b) 62 : 41 (c) 41 : 62 (d) 41 : 37 (e) 42 : 37
74. Find the average no. of 'Y' car sold in city 'B', 'C' and 'D' together?
 (a) 4036 (b) 4630 (c) 4603 (d) 4306 (e) 4360
75. 'X' and 'Z' type car sold by in 'City 'D' together is what percent more than same type cars sold in city 'C' together?
 (a) $9\frac{31}{41}\%$ (b) $10\frac{30}{37}\%$ (c) $9\frac{21}{41}\%$ (d) $10\frac{20}{37}\%$ (e) 56%



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1. **(c);** Total literate population = $(0.2 \times 0.18) + (0.25 \times 0.15) + (0.28 \times 0.20) + (0.35 \times 0.14) + (0.3 \times 0.17) \times 850000$
 $= 195075$
 $\therefore$ Required average = $\frac{1}{5} \times 195075 = 39015$
2. **(b);** Required percentage = $\frac{\frac{4}{7} \times \frac{28}{100} \times \frac{20}{100} \times 850000}{\frac{35}{100} \times \frac{14}{100} \times 850000} \times 100$
 $= \frac{32}{49} \times 100 = 65\frac{15}{49}\%$
3. **(d);** Females having PhD degree = $\frac{5}{100} \times \frac{3}{8} \times \frac{40}{100} \times \frac{16}{100} \times 850000 = 1020$
4. **(b);** Number of literate females =
 $(850000 \times \frac{20}{100} \times \frac{18}{100} \times \frac{3}{5} + 850000 \times \frac{40}{100} \times \frac{16}{100} \times \frac{3}{8} + 850000 \times \frac{35}{100} \times \frac{14}{100} \times \frac{5}{7})$
 $= \frac{3}{5} \times 30600 + \frac{3}{8} \times 54400 + \frac{5}{7} \times 41650$
 $= 18360 + 20400 + 29750 = 68510$
5. **(c);** Literate males in village P = $\frac{2}{5} \times \frac{20}{100} \times \frac{18}{100} \times 850000 = 12240$
 Literate females in village U = $\frac{7}{15} \times \frac{30}{100} \times \frac{17}{100} \times 850000 = 20230$
 $\therefore$ Required percentage = $\frac{7990}{20230} \times 100 \approx 40\%$ less
6. **(b);** Total $\rightarrow 48\%$ by Mohit and Karolina Marine
 $40\% \rightarrow 2600$
 $\therefore 100\% \rightarrow \frac{2600}{40} \times 100$
 Total 6500 pages to be proofread
 $\therefore 650 \rightarrow 10\%$ by Kamal
 Total by Kamal and Prakash = 25%
 $\therefore$ Desired value = $\frac{48}{25} \approx 2$ times
7. **(a);** Logan and Karoline Marine are not applicable
 Mohit $\rightarrow \frac{2600}{400} = 6.5$ pages for one mistake
 Kamal $\rightarrow \frac{650}{130} = 5$ pages for one mistake
 Prakash $\rightarrow \frac{6500 \times \frac{15}{100}}{25} = 39$ pages for one mistake
 Raaj $\rightarrow \frac{1300}{130} = 10$ pages for one mistake
 Clearly, Prakash has taken 39 pages which is maximum for a mistake.
8. **(c);** Pages refused to be proofread by Prakash = $\frac{40}{100} \times \frac{15}{100} \times 6500 = 390$
 Now, Pages to be proofread by Kamal = $650 + \frac{390}{2} = 845$
 Desired value = $\frac{2600-845}{2600} \times 100 = 67.5\%$
9. **(a);** Error found by Mohit and Kamal = $400 + 130 = 530$
 Avg. = $\frac{530+25+130+76+85}{6} = 141$
 Ratio = $530 : 141$
10. **(c);** Mohit : $2600 \times \frac{30}{100} = 780$
 Kamal : $650 \times \frac{10}{100} = 65$
 Prakash : $\frac{15}{100} \times 6500 \times \frac{10}{100} = \frac{195}{2} \approx 98$
 Raj : $1300 \times 10\% = 130$

$\therefore$ Total pages proofread by Siddharth Singh = $780 + 65 + 98 + 130 = 1073$ pages

11. **(a);** Number of children from village C = $\frac{18}{100} \times \frac{20}{100} \times 70,000 = 2520$
 Number of men from village C = $\frac{3}{7} (14000 - 2520) = 4920$
 Number of women from village A and B together = $\frac{8}{15} (10500 - 2940) + \frac{5}{14} (7000 - 1050) = 4032 + 2125$
 Required ratio = $\frac{4920}{6157} \approx 0.8$
12. **(e);** Number of children from all five villages = $0.28 \times 10500 + 0.15 \times 7000 + 0.18 \times 14000 + 0.2 \times 24500 + 0.15 \times 14000$
 $= 2940 + 1050 + 2520 + 4900 + 2100 = 13510$
 Average = 2702
13. **(b);** Number of women from C and E together = $\frac{4}{7} (11480) + \frac{3}{5} (11900) = 6560 + 7140 = 13700$
 Number of women from village B and D together = $\frac{5}{14} (7000 - 1050) + \frac{12}{25} (24500 - 4900) = 2125 + 9408 = 11533$
 Required % = $\frac{2167}{11533} \times 100 \approx 19\%$
14. **(d);** Required % = $\frac{0.25 \times 0.28 \times 0.15 \times 70000}{(10500 - 2940)} = \frac{735}{7560} \times 100 = 9.72\% \approx 10\%$
15. **(b);** Number of women from all villages together = $\frac{8}{15} (10500 - 2940) + \frac{5}{14} (7000 - 1050) + \frac{4}{7} (11480) + \frac{12}{25} (24500 - 4900) + \frac{3}{5} (11900) = 4032 + 2125 + 6560 + 9408 + 7140 = 29265$
16. **(a);** Let total no. of candidate passed in exams for city K be x.
 $\therefore 25\%$ of x = 320
 $x = \frac{320 \times 100}{25} = 1280$
 $\therefore$ Total passed candidate from city K is 1280
 Let total no. of candidates passed from all cities be y.
 10% of y = 1280
 $y = 12800$
 total passed from city L = $\frac{25}{100} \times 12800 = 3200$
 $\therefore$ Total fresher candidates passed from city L = $\frac{12}{100} \times 3200 = 384$
17. **(d);** Non-fresher candidate who passed the exam from city X = $1250 \times \frac{28}{100} \times \frac{80}{100} = 280$
18. **(c);** Total passed candidates from city Y = $\frac{180 \times 100}{75} = 240$
 $\therefore$ total candidates passed from all cities = $\frac{240 \times 100}{16} = 1500$
19. **(b);** Total no. of candidates passed in 2010 = $\frac{770 \times 100}{11} = 7000$

No. of candidates passed from city X in 2010

$$= \frac{28}{100} \times 7000 = 1960$$

Candidates passed in 2011 from city X

$$= \frac{110}{100} \times 1960 = 2156$$

No. of candidates passed from city Y in 2010

$$= \frac{16}{100} \times 7000 = 1120$$

Candidates passed in 2011 from city Y

$$= \frac{120}{100} \times 1120 = 1344$$

$$\therefore \text{Required difference} = 2156 - 1344 = 812$$

20. (a); Total candidates passed = $\frac{320 \times 100}{16} = 2000$

Candidates passed from city Z = $\frac{11}{100} \times 2000 = 220$

Non-fresher candidates passed from Z

$$= \frac{85}{100} \times 220 = 187$$

Candidates passed from city X

$$= \frac{28}{100} \times 2000 = 560$$

Fresher candidates passed from X = $\frac{20}{100} \times 560 = 112$

$$\therefore \text{required ratio} = \frac{112}{187}$$

21. (d); Number of Brahmas missiles manufactured in 2017

$$= 110 \times \frac{100}{55} = 200$$

$$\therefore 200 \text{ Assault rifles were imported.}$$

22. (b); Prahar missiles manufactured in 2016

$$= 140 \times \frac{100}{56} = 250$$

Akash missiles not exported in 2017

$$= 150 \times \frac{100}{40} \times \frac{60}{100} = 225$$

$$\therefore \text{Required difference} = 25$$

23. (c); Required ratio = $\frac{80 \times \frac{100}{50}}{180 \times \frac{100}{50}} = \frac{4}{9}$

24. (a); Prahar missiles not exported in 2017

$$= 130 \times \frac{48}{52} = 120$$

Ashtra missiles not exported in 2016

$$= 120 \times \frac{25}{75} = 40$$

$$\therefore \text{Required percentage} = \frac{80}{40} \times 100 = 200\%$$

25. (e); Total LCA manufactured

$$= 180 \times \frac{100}{50} + 160 \times \frac{100}{40} = 760$$

$$\therefore \text{Required percentage} = \frac{180}{760} \times 100 = 23\frac{13}{19}\%$$

26. (c); Total tourist in year 2017 in U.P = $20 \times \frac{16}{100} \times \frac{5}{4} = 4$ Lakhs

And we know percentage of male and female tourist is same in 2017 as in 2016 for U.P.

$$\therefore \text{Required difference} = \frac{30}{100} \times 4,00,000 = 1,20,000$$

27. (a); Tourist visiting in year 2016 in J&K = $20 \times \frac{23}{100} = 4.6$ Lakh

Who can speak both Hindi and Urdu = $\frac{7}{10} \times 4.6 = 3.22$ Lakh

28. (b); Tourists who left Haryana

$$= 20 \times \frac{15}{100} \times \frac{20}{100} = 0.6 \text{ Lakh}$$

Female in Punjab in year 2016 = $20 \times \frac{14}{100} \times \frac{20}{100} = 0.56$ Lakh

Male in Punjab in year 2016 = $20 \times \frac{14}{100} \times \frac{80}{100} = 2.24$ Lakh

After increase male in Punjab = $2.24 + 0.6 = 2.84$ Lakh

$$\therefore \text{Required ratio} = \frac{2.84}{0.56} = \frac{284}{56} = 71 : 14$$

29. (c); Since percentage of male and female tourist of year 2017 is not given.

30. (d); Required percentage = $\frac{20 \times \frac{11}{100} \times \frac{85}{100}}{20 \times \frac{21}{100} \times \frac{25}{100}} \times 100 = \frac{187}{105}$

$$\times 100 = 178\frac{2}{21}\%$$

31. (c); Total employee from T and R

$$= \left(1500 \times \frac{20}{100} + 1500 \times \frac{20}{100} \times \frac{11}{12}\right) + \left(1500 \times \frac{10}{100} + 1500 \times \frac{10}{100} \times \frac{4}{3}\right)$$

$$= (300 + 275) + (150 + 200) = 925$$

Total number of female from S and T

$$= 1500 \times \frac{15}{100} \times \frac{7}{9} + 1500 \times \frac{20}{100} \times \frac{11}{12}$$

$$= 175 + 275 = 450$$

$$\text{Required ratio} = \frac{925}{450} = 37 : 18$$

32. (b); Average of male employee from S, T and Q

$$= \frac{1500 \times \frac{15}{100} + 1500 \times \frac{20}{100} + 1500 \times \frac{30}{100}}{3} = \frac{225 + 300 + 450}{3} = 325$$

Average of female employee from branch S and R

$$= \frac{1500 \times \frac{15}{100} \times \frac{7}{9} + 1500 \times \frac{10}{100} \times \frac{4}{3}}{2} = \frac{175 + 200}{2} = \frac{375}{2}$$

$$\text{Required difference} = 325 - \frac{375}{2} = \frac{275}{2}$$

33. (d); Total unmarried male from T and Q

$$= \left(1500 \times \frac{20}{100} - 1500 \times \frac{20}{100} \times \frac{20}{100}\right) + \left(1500 \times \frac{30}{100} - 1500 \times \frac{30}{100} \times \frac{40}{100}\right)$$

$$= (240 + 270) = 510$$

Total unmarried female from T and Q

$$= \left(1500 \times \frac{20}{100} \times \frac{11}{12} - 1500 \times \frac{20}{100} \times \frac{20}{100} \times \frac{5}{4}\right) + \left(1500 \times \frac{30}{100} \times \frac{2}{3} - 1500 \times \frac{30}{100} \times \frac{40}{100} \times \frac{2}{3}\right)$$

$$= (275 - 75) + (300 - 120)$$

$$= 200 + 180 = 380$$

$$\text{Required sum} = 510 + 380 = 890$$

34. (a); Total B. Tech male employee from branch P

$$= 1500 \times \frac{25}{100} \times \frac{20}{100} = 75$$

Total MBA female employee from branch R = $1500 \times \frac{10}{100} \times \frac{4}{3} \times \frac{40}{100} = 80$

$$\text{Required \%} = \frac{80 - 75}{80} \times 100 = 6.25\%$$

35. (c); Total (Male + female) employee from branch T and S

$$= \left(1500 \times \frac{20}{100} + 1500 \times \frac{20}{100} \times \frac{11}{12}\right) + \left(1500 \times \frac{15}{100} + 1500 \times \frac{15}{100} \times \frac{7}{9}\right)$$

$$= (300 + 275) + (225 + 175) = 975$$

Total male and Female employee from branch R and Q

$$= \left(1500 \times \frac{10}{100} + 1500 \times \frac{10}{100} \times \frac{4}{3}\right) + \left(1500 \times \frac{30}{100} + 1500 \times \frac{30}{100} \times \frac{2}{3}\right)$$

$$= (150 + 200) + (450 + 300) = 1100$$

$$\text{Required \%} = \frac{975}{1100} \times 100 = 88\frac{7}{11}\%$$

Solution (36-40)

Percentage distribution of article of type D

$$= 100 - \left(\frac{100}{3} + \frac{200}{9} + \frac{50}{3} + 12.5\right) = \frac{275}{18}\% = 55$$

$$\text{Total number of article} = 100\% = 55 \times \frac{1800}{275} = 360$$

$$\therefore \text{No. of article of type A}$$

$$= 360 \times \frac{200}{900} = 80$$

$$\text{No. of article of type B}$$

$$= 360 \times \frac{50}{300} = 60$$

$$\text{No. of article of type C}$$

$$= 360 \times \frac{12.5}{100} = 45$$

$$\text{No. of article of type E}$$

$$= 360 \times \frac{100}{300} = 120$$

Article	No. of article
A	80
B	60
C	45
D	55
E	120

$$36. (a); \text{Total selling price of article of type A} = 80 \times 150 = 12000$$

$$\text{Total selling price of article of type B} = 60 \times 120 = 7200$$

$$\therefore \text{Required percentage} = \frac{12000 - 7200}{7200} \times 100 = 66\frac{2}{3}\%$$

$$37. (c); \text{Required average}$$

$$= \frac{1}{3}(60 \times 120 + 45 \times 180 + 120 \times 140)$$

$$= \frac{32100}{3} = 10700$$

$$38. (b); \text{Total selling price of article C} = 45 \times 180 = 8100$$

Therefore, total selling price of article F

$$= 8100 \times \frac{10}{9} = 9000$$

$$\text{No. of article of type F}$$

$$= \text{No. of article of type E} = 120$$

$$\therefore \text{Price per article type F}$$

$$= \frac{9000}{120} = 75$$

$$\therefore \text{Required difference} = 120 - 75 = 45$$

$$39. (d); \frac{2}{5}\text{th of article of type C}$$

$$= \frac{2}{5} \times 45 = 18$$

Total selling price of $\frac{2}{5}\text{th}$ of article of type C

$$= 18 \times 120 \times \frac{140}{100} = 3024$$

$$\text{Price of } \frac{3}{5}\text{th of article of type C}$$

$$= 27 \times 150 \times \frac{3}{5} = 2430$$

Total new price of article C

$$= 3024 + 2430 = 5454$$

Required difference

$$= 5454 - \frac{(60 \times 120 + 200 \times 55)}{2}$$

$$= 5454 - 9100 = 3646$$

$$40. (a); \text{Required ratio} = \frac{80 \times 150 + 120 \times 140}{45 \times 180 + 60 \times 120} = \frac{28800}{15300} = 32 : 17$$

$$41. (b); \text{Total cars sold by manufacturer}$$

$$= \frac{720}{16} \times 100 = 4500$$

$$\therefore \text{Required ratio} = \frac{\frac{8 \times 17}{15 \times 100} \times 4500}{\frac{7 \times 12}{15 \times 100} \times 4500}$$

$$= \frac{8 \times 17}{7 \times 12} = 34 : 21$$

$$42. (a); \text{Required difference}$$

$$= \frac{1}{2} \left[\frac{5}{9} \times \frac{22}{100} \times 4500 + \frac{2}{5} \times \frac{25}{100} \times 4500 \right] - \frac{1}{3} \left[\frac{7}{12} \times \frac{8}{100} \times 4500 + \frac{7}{15} \times \frac{12}{100} \times 4500 + \frac{7}{16} \times 720 \right]$$

$$= \left[\frac{550 + 450}{2} \right] - \left[\frac{210 + 252 + 315}{3} \right]$$

$$= 500 - 259 = 241$$

$$43. (c); \text{Required percentage}$$

$$= \frac{\frac{25}{100} \times \frac{3}{5} \times 4500 - 4500 \times \frac{8}{100} \times \frac{5}{12}}{\frac{25}{100} \times \frac{3}{5} \times 4500} \times 100 = 77\frac{21}{27}\%$$

$$44. (e); \text{Stopped no. of cars which are not sold to city B}$$

$$= \frac{5}{9} \times \frac{22}{100} \times 4500 \times \frac{1}{11}$$

$$= 550 \times \frac{1}{11} = 50$$

$$\text{Since distributed equally} = \frac{50}{5} = 10$$

$\therefore$ new total Tata cars sold to city D and A.

$$= \frac{5}{12} \times \frac{8}{100} \times 4500 + \frac{7}{15} \times \frac{17}{100} \times 4500 + 20$$

$$= 150 + 357 + 20 = 527$$

$$45. (d); \text{New total car sold to city C}$$

$$= \frac{25}{100} \times 4500 \times \frac{10}{9} = 1250$$

$$\text{Tata cars sold to city C} = 1250 \times \frac{2}{5} = 500$$

$$\text{Suzuki cars sold to city D} = 1250 \times \frac{3}{5} = 750$$

New total cars sold to city D

$$= \frac{8}{100} \times 4500 \times \frac{5}{3} = 600$$

$$\text{Tata cars sold to city D} = 600 \times \frac{5}{12} = 250$$

Suzuki cars sold to city D

$$= 6000 \times \frac{7}{12} = 350$$

$$\therefore \text{Required difference} = (750 + 350) - (500 + 250) = 350$$

$$46. (e); \text{population of E in 2002}$$

$$1250 \times \frac{(100+30)}{100} + \frac{(100+20)}{100} = 1950$$

Population of A in 2000

$$= 3750 \times \frac{100}{125} \times \frac{100}{120} = 2500$$

$$\text{required ratio} = \frac{1950}{2500} = 39 : 50$$

$$47. (b); \text{Population of village A in 2000} = 2500$$

$$\text{Population of village C in 2000} = 1518 \times \frac{100}{120} \times \frac{100}{115} = 1100$$

$$\text{Required percentage} = \frac{2500 - 1100}{1100} \times 100 = 127\frac{3}{11}\%$$

$$48. (d); \text{population of D in 2000} = \frac{27}{22} \times 1100 = 1350$$

$$\text{Population of D in 2002} = 1350 \times \frac{130}{100} \times \frac{120}{100} = 2106$$

$$49. (e); \text{Population of F in 2002} = 1200 \times \frac{125}{100} \times \frac{135}{100} = 2025$$

$$\text{required percentage} = \frac{1200}{2025} \times 100 = 59\frac{7}{27}\%$$

$$50. (a); \text{total population in 2000} = 2500 + 1650 + 1100 + 1350 + 1250 + 1200 = 9050$$

total population in 2002 = 3750 + 2178 + 1518 + 2106 + 1950 + 2025 = 13527

required percentage = $\frac{13527-9050}{13527} \times 100 \approx 33\%$

Solution (51-55)

Degree measure of B + D = $360^\circ - (80^\circ - 72^\circ - 74^\circ + 62^\circ) = 72^\circ$

Also, difference b/w degree measures of B & D = 28°

$\therefore$ Degree measure of B and D is 50° and 22° respectively as B is more than D.

Now total literates in all villages together

$$= \frac{5500}{50} \times 360 = 39600$$

51. (c); Illiterate persons in village B

$$= \left(1 - \frac{1}{11}\right) \times 5500 = 5000$$

Illiterates in Village E

$$= \frac{75}{100} \times \frac{74}{360} \times 39600 = 6105$$

$$\therefore \text{Total sum} = 5000 + 6105 = 11105$$

52. (a); Illiterate persons of village A

$$= 39600 \times \frac{80}{360} \times \left(1 + \frac{3}{22}\right) = 10,000$$

$$\therefore \text{Required \%} = \frac{10,000}{39600} \times 100 = 25 \frac{25}{99}\%$$

53. (d); Required ratio = $\frac{A+D}{B+F} = \frac{80^\circ + 22^\circ}{50^\circ + 62^\circ} = \frac{51}{56}$ **54. (b);** Total illiterate male persons in village F

$$= 39600 \times \frac{62}{360} \times \left(1 + \frac{2}{11}\right) \times \frac{15}{26} = 4650$$

Total literate male persons in village F

$$= 39600 \times \frac{62}{360} \times \frac{13}{22} = 4030$$

$$\therefore \text{Total number of males in village F} = 4650 + 4030 = 8680$$

55. (c); Required average

$$= \frac{1}{4}(50 + 72 + 22 + 74) = 54.5^\circ$$

56. (b); For Arunoday:

In 2012 :

Annual income = Rs. 650 thousands

Annual expenditure = $400 \times \frac{3}{2}$

= Rs. 600 thousands

$\therefore$ saving (annual) = 50 thousand

Similarly, annual saving in 2016

$$= 800 - \left(200 \times \frac{3}{2}\right) = 500 \text{ thousand}$$

For Veer,

Annual saving in 2012

$$= \left(650 \times \frac{4}{5}\right) - 400 = 120 \text{ thousands}$$

Annual saving in 2016

$$= \left(800 \times \frac{3}{8}\right) - 200 = 100 \text{ thousands}$$

$\therefore$ Required difference of average

$$= \frac{1}{2}[(500 + 50) - (120 + 100)] = 165 \text{ thousands}$$

57. (d); income of Veer in 2013 and 2015

$$= 700 \times \frac{4}{7} + 700 + \frac{2}{7} = \text{Rs. 600 thousand}$$

expenditure of Arunodayin 2014 and 2016

$$= 300 \times \frac{4}{3} + 200 \times \frac{3}{2}$$

= Rs. 700 thousand

$$\therefore \text{Required ratio} = \frac{600}{700} = 6 : 7$$

58. (c); Saving of Veer in 2015 and 2016

$$\left(700 \times \frac{2}{7} - 150\right) + \left(800 \times \frac{3}{8} - 200\right)$$

= Rs. 150 thousands

Saving of Arunodayin 2013 and 2016

$$= 700 - \frac{2}{3} \times 300 + 800 - 200 \times \frac{3}{2} = 1000 \text{ thousand.}$$

$$\therefore \text{Required percentage} = \frac{850}{1000} \times 100 = 85\%$$

59. (a); Annual saving of Arunoday in 2017

$$= \frac{88}{100} \times 500 = 440 \text{ thousand}$$

Annual saving of Veer in 2017

$$= \frac{115}{100} \times 100 = 115 \text{ thousand}$$

$\therefore$ Their total expenditure

$$= (650 - 440) + \left(650 \times \frac{3}{5} - 115\right)$$

$$= 210 + 275$$

= 485 thousands

60. (e); Required average

$$= \frac{1}{3} \left[\frac{5}{3} \times 150 + \left(800 \times \frac{3}{8} - 200\right) + 700 \times \frac{4}{7} \right]$$

$$= \frac{1}{3} (250 + 100 + 400)$$

= Rs. 250 thousand

61. (c); Qualified student in school JPC = $\frac{2176}{68} \times 32 = 1024$

$$\text{Total appeared student in school JPC} = \frac{102400}{32} = 3200$$

$$\text{Total appeared student in school CCS} = \frac{3200}{32} \times 12 = 1200$$

$$\text{Total qualified student in school CCS} = 1200 \times \frac{24}{100} = 288$$

$$\text{Required difference} = 288 \times \frac{2}{8} = 72$$

62. (a); Qualified student in school NNA = $120 \times 3 = 360$.

$$\text{Total appeared student in school NNA} = \frac{360}{30} \times 100 = 1200$$

$$\text{Total appeared student in school RPS} = \frac{1200}{8} \times 20 = 3000$$

$$\text{Total qualified girls in school RPS} = 3000 \times \frac{28}{100} \times \frac{2}{7} = 240$$

Required percentage

$$= \frac{240-120}{120} \times 100 = 100\%$$

63. (d); Total appeared student = $1500 \times 5 = 7500$

$\therefore$ Required average

$$= \frac{7500}{3} \left[\frac{20}{100} \times \frac{28}{100} \times \frac{5}{7} + \frac{28}{100} \times \frac{25}{100} \times \frac{3}{5} + \frac{8}{100} \times \frac{30}{100} \times \frac{1}{3} \right] = \frac{675}{3} = 225$$

64. (a); Appeared student in year 2015 in school RPS = $\frac{1200}{4} \times 3 = 900$

Total appeared student in year 2015 = $900 \times 5 = 4500$

$\therefore$ qualified boys in school GPS in 2015 = $4500 \times$

$$\frac{28}{100} \times \frac{1}{4} \times \frac{3}{5} = 189 = \text{qualified boys in school RPS in 2016}$$

$$\therefore \text{qualified girls in school RPS in 2016} = 1200 \times \frac{28}{100} - 189 = 147$$

65. (b); Qualified boys in school NNA in 2015 = $9000 \times \frac{8}{100} \times$

$$\frac{30}{100} \times \frac{1}{3} = 72$$

Qualified boys in school JSS in year 2016

$$= 72 \times \frac{4}{3} = 96$$

Qualified girls in school GPS in 2015

$$= 9000 \times \frac{28}{100} \times \frac{25}{100} \times \frac{2}{5} = 252$$

Qualified girls in school JSS in 2016

$$= 252 \times \frac{1}{3} = 84$$

$$\therefore \text{Required ratio} = \frac{96}{84} = 8 : 7$$

66. (d); Number of females in Delhi

$$= 45000 \times \frac{18}{100} \times \frac{30}{100} = 2430$$

Number of females in Gurgaon

$$= 45000 \times \frac{22}{100} \times \frac{40}{100} = 3960$$

Number of females in Panipat

$$= 45000 \times \frac{24}{100} \times \frac{34}{100} = 3672$$

Number of females in Rewari

$$= 45000 \times \frac{20}{100} \times \frac{42}{100} = 3780$$

Number of females in Sonipat

$$= 45000 \times \frac{16}{100} \times \frac{38}{100} = 2736$$

Maximum number of female is in Gurgaon

67. (c); Transgender population in Gurgaon and Panipat together

$$= 45,000 \left[\frac{22}{100} \times \frac{24}{100} + \frac{24}{100} \times \frac{38}{100} \right]$$

$$= 45000 \times \frac{24}{100} \left[\frac{60}{100} \right] = 6480$$

Male and female together in Rewari

$$= 45000 \times \frac{20}{100} \times \frac{76}{100} = 6840$$

$$\text{Required ratio} = \frac{6480}{6840} = \frac{18}{19}$$

68. (a); Male population in city Gurgaon, Panipat together

$$= 45000 \left[\frac{22}{100} \times \frac{36}{100} + \frac{24}{100} \times \frac{28}{100} \right]$$

$$= 4.5 \times [792 + 672] = 6588$$

Female population in city Rewari and Sonipat together

$$= 45000 \left[\frac{20}{100} \times \frac{42}{100} + \frac{16}{100} \times \frac{38}{100} \right]$$

$$= 4.5 \times [840 + 608] = 6516$$

$$\text{Required difference} = 6588 - 6516 = 72$$

69. (d); Transgender population of city Delhi and Sonipat together

$$= 45,000 \left[\frac{18}{100} \times \frac{35}{100} + \frac{16}{100} \times \frac{36}{100} \right]$$

$$= 4.5 \times [630 + 576] = 5427$$

Female population of city Gurgaon

$$= 45,000 \times \frac{22}{100} \times \frac{40}{100} = 3960$$

$$\text{Required}\% = \frac{5427 - 3960}{3960} \times 100$$

$$= 37 \frac{1}{22}\%$$

70. (e); Female & transgender population of Panipat

$$= 45,000 \times \frac{24}{100} \times \frac{72}{100} = 7776$$

Male & Female population of city Gurgaon

$$= 45000 \times \frac{22}{100} \times \frac{76}{100} = 7524$$

$$\text{Required difference} = 7776 - 7524 = 252$$

71. (d); 'Y' type car sold in city 'D'

$$= 60000 \times \frac{24.6}{100} \times \frac{8}{24} = 4920$$

$$'Z' \text{ type car sold in city 'A'} = 60000 \times \frac{16.4}{100} \times \frac{5}{12} = 4100$$

$$\text{Required \%} = \frac{4920 - 4100}{4100} \times 100 = 20\%$$

72. (b); 'X' type car sold in city 'C' = $60000 \times \frac{22.2}{100} \times \frac{6}{15}$

$$= 5328$$

$$'X' \text{ type car sold in city 'D'} = 60000 \times \frac{24.6}{100} \times \frac{7}{24}$$

$$= 4305$$

$$'X' \text{ type car sold in city 'E'} = 60000 \times \frac{18.2}{100} \times \frac{4}{14} = 3120$$

$$\text{Required average} = \frac{5328 + 4305 + 3120}{3}$$

$$= \frac{12753}{3} = 4251$$

73. (c); 'X' and 'Y' type car sold in city 'A' together

$$= 60000 \times \frac{16.4}{100} \times \frac{7}{12} = 5740$$

'X' and 'Y' type car sold in city 'B' together

$$= 60000 \times \frac{18.6}{100} \times \frac{7}{9} = 8680$$

$$\text{Required \%} = \frac{5740}{8680} = \frac{41}{62}$$

74. (e); 'Y' type car sold in city 'B' = $60000 \times \frac{18.6}{100} \times \frac{3}{9}$

$$= 3720$$

$$'Y' \text{ type car sold in city 'C'} = 60000 \times \frac{22.2}{100} \times \frac{5}{15}$$

$$= 4440$$

$$'Y' \text{ type car sold in city 'D'} = 60000 \times \frac{24.6}{100} \times \frac{8}{24} =$$

$$4920$$

$$\text{Required average} = \frac{3720 + 4440 + 4920}{3}$$

$$= \frac{13080}{3} = 4360$$

75. (b); 'X' and 'Z' type car sold in city 'D' together

$$= 60000 \times \frac{24.6}{100} \times \frac{16}{24} = 9840$$

'X' and 'Z' type car sold in city 'C' together

$$= 60000 \times \frac{22.2}{100} \times \frac{10}{15} = 8880$$

$$\text{Required \%} = \frac{9840 - 8880}{8880} \times 100 = \frac{960}{8880} \times 100 =$$

$$10 \frac{30}{37}\%$$

PRACTICE SET (LEVEL-I)

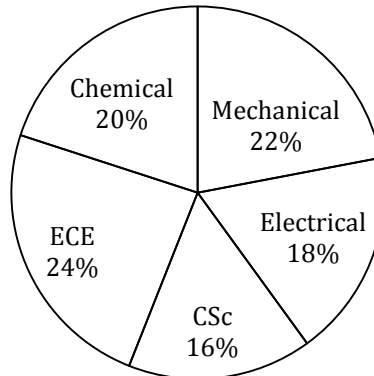
Direction (1-5): Study the following table carefully and answer the questions that follow.

The table shows the percentage of students of five branches who vote in favour of different lecturers participating for election of DEAN in a college. The pie chart shows the percentage distribution of students in different branches.

Lecturers→	K. L. Bali	VirajTyagi	Varun Gandhi	Yogesh Gupta	Deepak Mittal	Prince Saini
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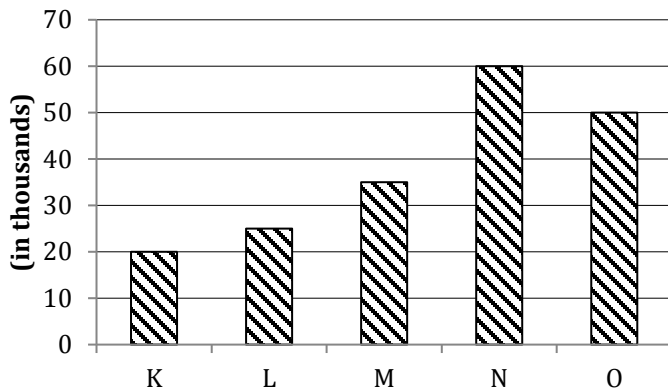
Branches↓						
Mechanical	14	15	16	22	20	13
Electrical	16	12	18	22	12	20
CSc	22	8	14	23	15	18
ECE	20	15	10	16	18	21
Chemical	12	18	22	20	16	12

Total number of students is 5000



- Find the difference between total number of votes casted for varun Gandhi and that of Deepak Mittal?
(a) 26 (b) 28 (c) 34 (d) 38 (e) 32
- Total number of students who favour Prince Saini are what percent of total number of students in ECE?
(a) $65\frac{11}{12}\%$ (b) $69\frac{11}{12}\%$ (c) $71\frac{11}{12}\%$ (d) $68\frac{11}{12}\%$ (e) $67\frac{11}{12}\%$
- Among the supporters of Varun Gandhi in mechanical branch, boys to girls are in ratio of 5 : 3. If total number of girls in all branch are 36 times of mechanical girls, who in supporting varun Gandhi. Then find the total number of boy in all branch?
(a) 2634 (b) 2264 (c) 2642 (d) 2624 (e) 2614
- Find the ratio of total number of students who support K.L. Bali to number of students in chemical branch?
(a) 437:500 (b) 407:500 (c) 427:500 (d) 471:500 (e) 417:500
- By what percent of total votes Yogesh Gupta beats Virajtyagi?
(a) $4\frac{19}{50}\%$ (b) $5\frac{19}{50}\%$ (c) $6\frac{19}{50}\%$ (d) $7\frac{19}{50}\%$ (e) $8\frac{19}{50}\%$

Directions (6-10): Study the bar-graph and table given below carefully and answer the question accordingly. Bar-graph shows the number of people in five different cities and table shows the percentage of male in five cities and the ratio of literate and illiterate people in five different cities.



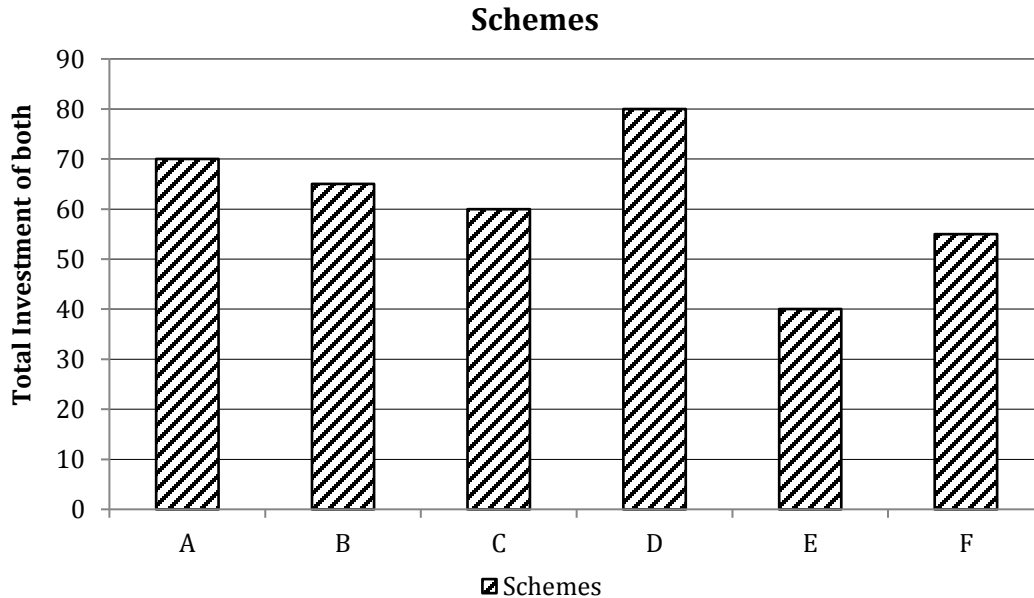
City	Percentage of Males	Literate : Illiterate
K	40%	1 : 3
L	60%	7 : 3
M	20%	1 : 4
N	75%	3 : 2
O	68%	9 : 11

- If 40 percent of male from city K went to city L then what is the total number of males in city L ?
(a) 17200 (b) 19200 (c) 16200 (d) 18200 (e) None of these
- What is the percentage of literate people from city L to the illiterate people from city M?
(a) 52.5% (b) 72.5% (c) 62.5% (d) 42.5% (e) None of these
- If 30 percent of male from city M are illiterate. Then find the ratio of illiterate male to the illiterate female from city M?
(a) 21 : 259 (b) 21 : 269 (c) 22 : 259 (d) 23 : 259 (e) None of these
- What is the average of illiterate people in five cities?

10. (a) 25000 (b) 20400 (c) 21000 (d) 23000 (e) None of these
 What is the ratio of total females from city M and O to the total illiterate males from city K and city L?
 (a) 2 : 5 (b) 7 : 5 (c) 5 : 7
 (d) Cannot be determined (e) None of these

Directions (11-15): Read the given data carefully and answer the given question

Bar graph shows Total Investment in thousand made by Abhimanyu and Arunoday in 6 different schemes (A, B, C, D, E, & F). Table shows the ratio of investment of Abhimanyu to Arunoday.

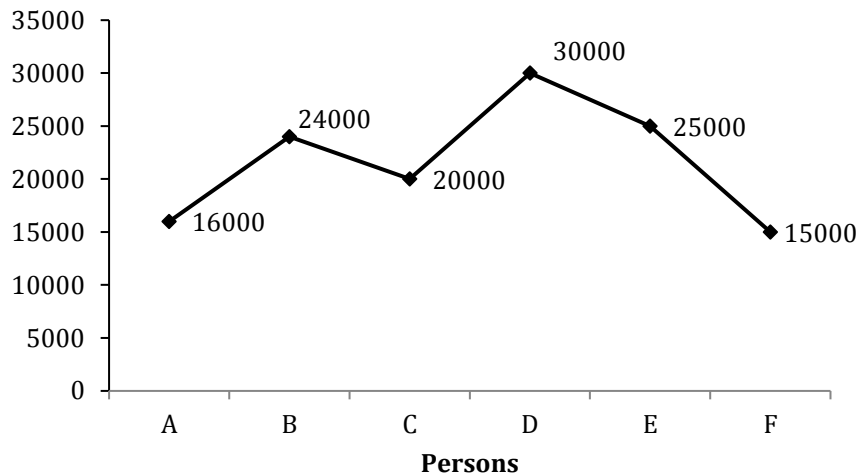


Scheme	A	B	C	D	E	F
Ratio of Investment	11 : 14	7 : 13	1 : 1	2 : 3	3 : 7	2 : 3

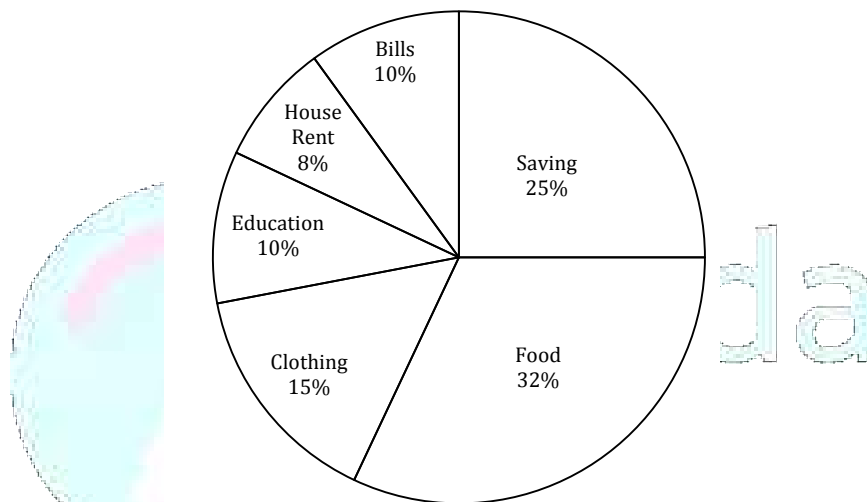
11. If scheme A offers simple Interest at R% per annum and total interest obtained from schemes A for 2 years is 11200 then find R% and share of interest of Abhimanyu
 (a) 8%, 4928 (b) 6%, 7312 (c) 5%, 5724 (d) 8%, 5321 (e) 8% 4982
12. What is ratio of total investment of abhimanyu in scheme A, B and D together to the total investment made in scheme C, E and F by Arunoday
 (a) 913 : 2222 (b) 1652 : 1325 (c) 1711 : 1820 (d) 1820:1711 (e) 1802:1711
13. Average of investment made by Abhimanyu in scheme A, C is what % of average of investment made by Arunoday in scheme E and F (Approximately)
 (a) 80% (b) 75% (c) 100% (d) 90% (e) 110%
14. If scheme B and C offers simple interest at the rate of 10% and $\frac{100}{3}\%$ respectively then find the total interest obtained from scheme B & C in 3 year given that Abhimanyu withdraw his total amount from scheme B in 3rd year whereas Arunoday withdraw his total amount in second year from scheme C.
 (a) 52520 Rs (b) 53373 Rs (c) 57225 Rs (d) 62250 Rs (e) 57252 Rs.
15. What is the ratio of total amount invested by Abhimanyu to total amount invested by Arunoday in all the schemes together except schemes E and F.
 (a) $\frac{2311}{3189}$ (b) $\frac{2225}{1333}$ (c) $\frac{1553}{1120}$ (d) $\frac{1852}{2021}$ (e) $\frac{2301}{3198}$

Directions (16-20): Study the following graphs carefully and answer the questions given below:

Graph given below line graph shows the monthly expenditure by 6 persons



Given below is the pie chart, which shows the percentage breakup of monthly Income of person B. Income = Expenditure (bill+house rent+education+clothing+food)+ Saving

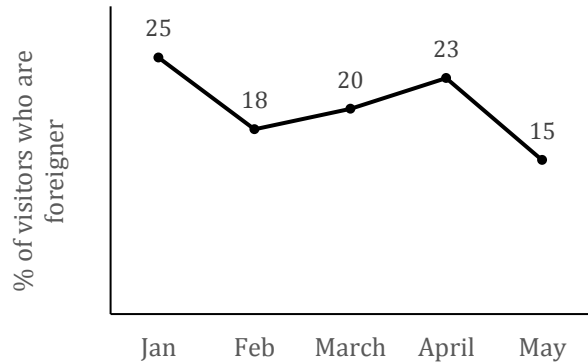
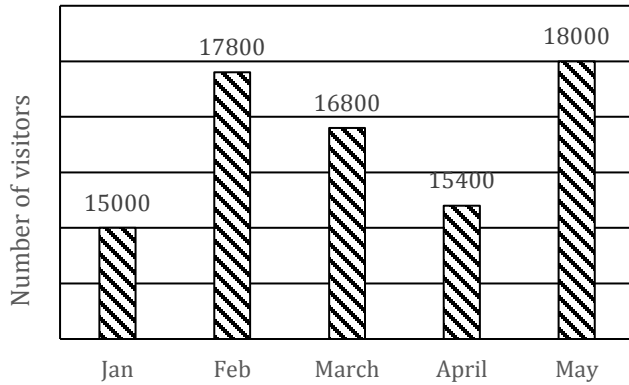


Note: Consider person B don't have any other expenditure.

16. If ratio of expenditure of person D on food and clothing is 8 : 7 and expenditure of D on clothing is $16\frac{2}{3}\%$ more than the expenditure of B on clothing then what is the sum of expenditure of person D on food and clothing together ?
 (a) 10500 (b) 8700 (c) 12000 (d) 14000 (e) 12500
17. If D and F spend 20% and $\frac{100}{3}\%$ of their monthly expenditure on house rent then expenditure of B on Education is what percent of expenditure of D and F together on house rent ?
 (a) $29\frac{1}{11}\%$ (b) $13\frac{1}{13}\%$ (c) $133\frac{1}{3}\%$ (d) 45% (e) $29\frac{7}{11}\%$
18. If income of C is $\frac{1100}{3}\%$ more than expenditure of B on clothing then saving of C is approximately what percent more or less than saving of B ?
 (a) 30% (b) 40% (c) 70% (d) 60% (e) 35%
19. If saving of A, B, C, D, E and F are in the ratio 1 : 2 : 1 : 3 : 2 : 1 then what is the average of their income ?
 (a) 30520 (b) 70252.33 (c) 28333.33 (d) 27089.33 (e) 28166.66
20. Expenditure of person E on bill is equal to the expenditure of B on bills. If Income of B is decreased by $17\frac{1}{2}\%$ then expenditure on bill for person E is changed by what percent ?
 (a) 20% (b) 165 (c) 12.5% (d) 17.5% (e) 8.5%

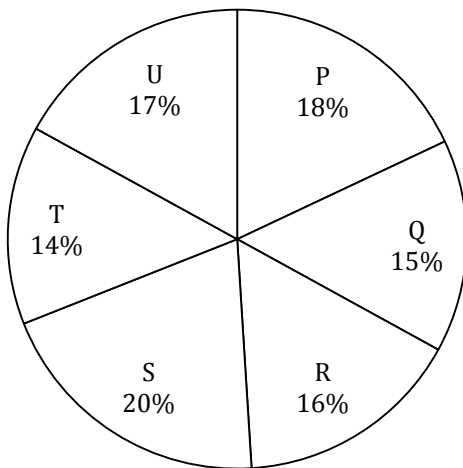
Directions (21-25): Given below is the bar graph which shows the total number of persons who are visiting Taj Mahal in 5 different months of years. Line graph shows the percentage of foreigners in number of visitors in different months.

Note → total visitors = Indian Visitors + Foreign visitors



21. If in June Month of same year total visitors increases by 20% over the January Month. And total foreigner increases by 30% then total Indian visitor in June month is what percent of total foreigners visiting Taj Mahal in Jan and Feb together. (approximately)
- (a) 188% (b) 198% (c) 178% (d) 208% (e) 125%
22. If 50% and 40% of total people who visited Taj Mahal in Feb and May respectively are females and ratio of male to female in foreign visitors in Feb and May is 5 : 4 and 3 : 1 respectively then, find total Indian females who visited Taj Mahal in Feb is what percent of total Indian males visiting Taj Mahal in May. (approximately)
- (a) 85% (b) 90% (c) 80% (d) 75% (e) 95%
23. Total foreign visitors in Feb and March month together are how much more/less than total Indian visitors in month of Jan and April together.
- (a) 18528 (b) 17268 (c) 14548 (d) 12500 (e) 16544
24. 50% of Indian visitors and $16\frac{2}{3}\%$ of foreign visitors in Feb month are married couples. If 50% of remaining Indian visitor are females and $66\frac{2}{3}\%$ of remaining foreign visitors are males then find the total number of males who visited Taj Mahal in February.
- (a) 7349 (b) 8258 (c) 9345 (d) 9870 (e) 8569
25. Foreign visitors in Feb month are what percent more or less than foreign visitors of May month.
- (a) $14\frac{2}{7}\%$ (b) $18\frac{2}{3}\%$ (c) $16\frac{2}{3}\%$ (d) $15\frac{2}{3}\%$ (e) $14\frac{2}{3}\%$

Directions (26-30): Given below is the pie chart which shows the percentage distribution of females in 6 villages out of total female population in these 6 villages and table shows the percentage of males out of total population in each village



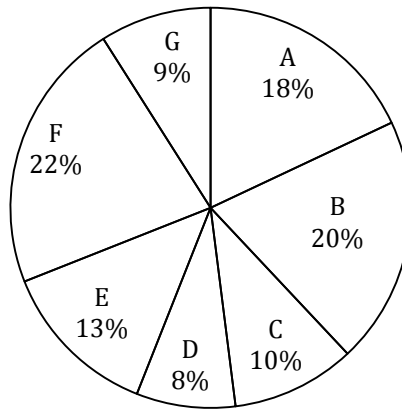
Total females = x

Villages	% of males in villages
P	$\frac{700}{13}\%$
Q	$\frac{400}{9}\%$
R	$\frac{1500}{31}\%$
S	$\frac{500}{9}\%$
T	$\frac{1100}{25}\%$
U	$\frac{1300}{30}\%$

26. Total population of village U is what percent of total population of village T?
- (a) 100% (b) 120% (c) 75% (d) 80% (e) 90%
27. What is the ratio of total females from village S and T together to the total of males from village R and U together?
- (a) 17 : 14 (b) 21 : 23 (c) 13 : 17 (d) 14 : 15 (e) 12 : 17
28. If $15\frac{5}{13}\%$ of total population of village P is illiterate and 20% of female in village P are illiterate then what percent of male are illiterate in village P?

- (a) $\frac{100}{7}\%$ (b) $\frac{90}{7}\%$ (c) $\frac{50}{3}\%$ (d) $\frac{80}{7}\%$ (e) $\frac{100}{3}\%$
29. If number of males in village P is equal to 17850 then what is the total number of males and females in village T and U together?
- (a) 36500 (b) 46750 (c) 48250 (d) 42300 (e) 54200
30. Total females in village R is 27200 and 30% of females in village R and 20% of males in village R are not registered in voter list. In an election, votes polled in village R is 60% of total population of village R. What is the difference in total registered voters who did not cast vote and total persons who are not registered in voter list?
- (a) 4120 (b) 5150 (c) 5530 (d) 4250 (e) 5440

Directions (31-35): Given below is the pie chart which shows the percentage distribution of books of publisher 'X' sold by 7 different books store in year 2016. Table shows the ratio of books sold of publisher X to publisher Y in these seven book stores. Some values are missing in the table. You have to calculate these values if required to answer the questions.



Total books sold of Publisher X = 25,700

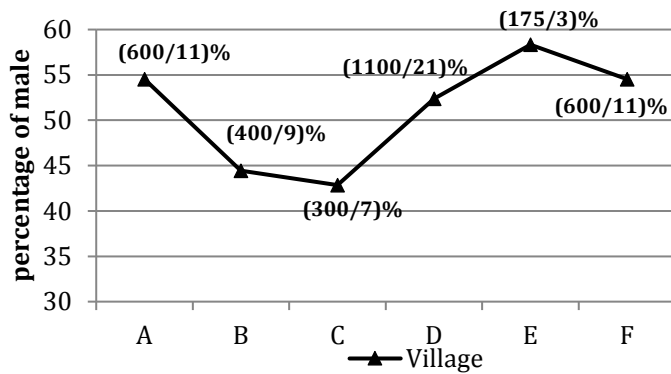
Book store	Ratio of books sold of publisher X to publisher Y
A	3 : -
B	- : 5
C	2 : 3
D	- : -
E	13 : 5
F	11 : -
G	3 : 4

31. What is the total number of books sold by store A and B together if books sold by store A of publisher Y is $33\frac{1}{3}\%$ more than that of publisher X and Books sold by store B of publisher X is 20% less than that of publisher Y.
- (a) 22359 (b) 21257 (c) 20256 (d) 23244 (e) 22556
32. What is the total number of books sold by store D if books sold of publisher Y in store D is 25% more than that of books sold by store D of publisher X
- (a) 2520 (b) 4020 (c) 4626 (d) 4422 (e) 4528
33. Books sold by store E, F and G together of publisher X is what percent more or less than books sold by these store of publisher Y if books sold by store F of publisher Y is $\frac{100}{11}\%$ more than that of books sold by F of publisher X.
- (a) $\frac{400}{31}\%$ (b) $\frac{300}{41}\%$ (c) $\frac{200}{9}\%$ (d) $\frac{100}{9}\%$ (e) $\frac{100}{11}\%$
34. If in year 2017 total books sold by store E is increased by $33\frac{1}{3}\%$ over previous year and ratio of books sold of publisher X and Y by store E in 2017 is 11 : 13 then books sold by store E of publisher X in 2016 is what percent more or less than that of books sold of publisher X by store E in 2017.
- (a) $\frac{200}{11}\%$ (b) $\frac{200}{9}\%$ (c) $\frac{100}{11}\%$ (d) $\frac{100}{9}\%$ (e) None of these
35. Average of books of publisher X sold by store B and C together is what percent more or less than that of average of books of publisher Y sold by store E and G together
- (a) $\frac{1100}{12}\%$ (b) $\frac{1100}{17}\%$ (c) $\frac{1300}{17}\%$ (d) $\frac{1400}{7}\%$ (e) $\frac{1700}{11}\%$

Directions (36-40): Given below is the line graph which shows the percentage of male out of total population in 6 villages in year 2016.

Table shows the total illiterate females and percentage of females who are literate in the given villages in year 2016

Total females = Total literate females + Total illiterate females



Village	Illiterate females	% of females who are literates
A	6000	40%
B	4500	55%
C	–	35%
D	3500	–
E	1000	80%
F	2000	60%

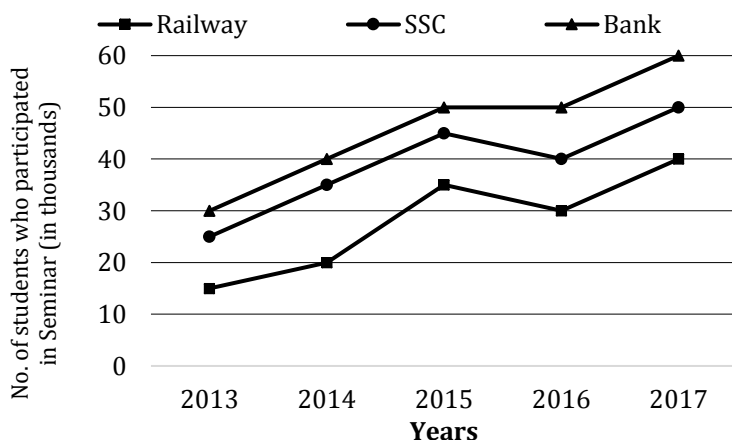
Note : In table some data are missing. If these data are required in any question then find them first and then proceed.

36. Total literate females from village E are what percent less than total males from village A.
 (a) $33\frac{1}{3}\%$ (b) $12\frac{2}{5}\%$ (c) $16\frac{2}{3}\%$ (d) $66\frac{2}{3}\%$ (e) $14\frac{2}{7}\%$
37. If total illiterate female in village C are $8\frac{1}{3}\%$ more than total males in village A, then total illiterate females in village C are how much more than total literate females in village F.
 (a) 6,000 (b) 10,000 (c) 5,000 (d) 4,500 (e) 6,500
38. If total literate female in village D is $44\frac{4}{9}\%$ more than total illiterate female in village B, then find the difference between total males in village D and total males in village F.
 (a) 6000 (b) 7000 (c) 8000 (d) 7000 (e) 5000
39. What is the ratio between total males in village F to the total males in village B.
 (a) 3 : 4 (b) 4 : 5 (c) 5 : 2 (d) 4 : 3 (e) 5 : 4
40. If the illiterate female from village C decreases by $\frac{200}{13}\%$ and literate female from same village increases by $\frac{100}{13}\%$ in year 2017 but total person (Male + female) remains same in village C in year 2017 then number of male in village C is decreased or increased by what percent.
 (a) $3\frac{23}{31}\%$ (b) $9\frac{29}{39}\%$ (c) $21\frac{6}{7}\%$ (d) $33\frac{3}{4}\%$ (e) None of these

Directions (41-45): The following line graph shows the total no. of students who are preparing for three different exam viz. SSC, Bank and Railway who participated in a seminar organized by career power in five different years.

The table shows the ratio of male to female who participated in seminar.

Study both the graph carefully and answer the questions that follows.

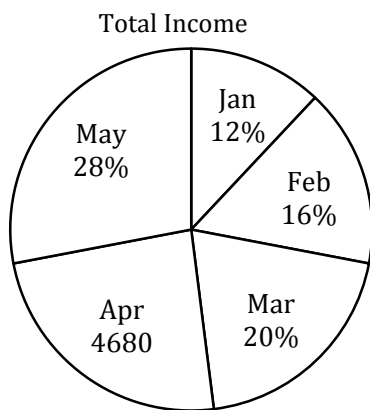


Year	Ratio of male to female who participated in seminar		
	Bank(M:F)	SSC(M:F)	Railway(M:F)
2013	4:1	4:1	4:1
2014	5:3	5:2	7:3
2015	7:3	7:2	6:1
2016	3:2	5:3	3:1
2017	2:1	7:3	3:2

41. The total no. of male students in 2014 from all exam who are participating in seminar is what percent of total no. of female students from all exam who are participating in seminar in 2013?
 (a) $547\frac{1}{7}\%$ (b) $457\frac{1}{7}\%$ (c) $455\frac{1}{7}\%$ (d) $452\frac{1}{7}\%$ (e) $745\frac{1}{7}\%$
42. If 10% of male students and 5% of female students preparing for Bank exam in year 2015 asked questions to the speaker in seminar and 10% of total students preparing for SSC asked the question to the speaker in the same year, then total no. of student from banking who asked question is what percent of the total no. of students from SSC who asked question in year 2015?

- (a) $90\frac{4}{9}\%$ (b) $92\frac{4}{9}\%$ (c) $94\frac{4}{9}\%$ (d) $97\frac{4}{9}\%$ (e) $98\frac{4}{9}\%$
43. What is the average no. of male students preparing for Railway exam who participated in seminar throughout all the five years ?
- (a) 22 thousands (b) 25 thousands (c) 20.5 thousands (d) 21.5 thousands (e) 19.5 thousands
44. Total no. of male students preparing for Bank exam in 2016 and 2017 together is approximately what percent more than the total no. of female students preparing for SSC exam who participated in the seminar together in the same years?
- (a) $133\frac{2}{3}\%$ (b) $133\frac{1}{3}\%$ (c) $138\frac{1}{3}\%$ (d) $131\frac{1}{3}\%$ (e) $135\frac{1}{3}\%$
45. In 2012 the career power had organized the seminar in which the total no. of student participating in seminar who are preparing for bank, SSC and Railway exam is 10%, 20% and 25% less than that in 2013 respectively and total no. of boys who participated in the seminar in year 2012 preparing for Bank, SSC and railway exam were 1000, 1500 and 2000 less than that in 2013 respectively. Find the total no. of girls participated in seminar in 2012 preparing for Bank, SSC and Railway exam together?
- (a) 6,850 (b) 5,670 (c) 7,650 (d) 6,750 (e) 7,550

Directions (46-50): pie chart given below shows the income of Satish on five different months Table shows his saving percentage and ratio of his total expense on three different items X, Y and Z. Study the data carefully and answer the following questions : Some value given in percentage and for April month absolute value is given in the Pie chart.



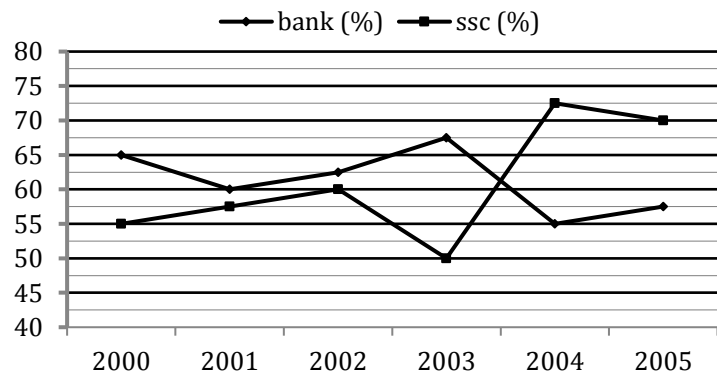
Month	Saving (%)	Rate of expense
Jan	$57\frac{9}{13}\%$	3 : 5 : 7
Feb	$38\frac{6}{13}\%$	7 : 8 : 9
Mar	52%	4 : 5 : 4
Apr	$38\frac{8}{9}\%$	4 : 3 : 4
May	$43\frac{1}{3}\%$	6 : 7 : 4

46. Expense in the month of Jan on 'Z' is what percent less than expense in the month of Feb on 'X'?
- (a) 25% (b) 20% (c) 22.5% (d) 17.5% (e) 15%
47. Find the difference between expense on 'Z' in the month of April to the expense on 'X' in the month of May?
- (a) 48 (b) 52 (c) 56 (d) 60 (e) 64
48. Total expense in the month of Jan is what percent of the total saving in the month of April?
- (a) $53\frac{36}{91}\%$ (b) $51\frac{55}{91}\%$ (c) $48\frac{36}{91}\%$ (d) $45\frac{55}{91}\%$ (e) $54\frac{36}{91}\%$
49. Find the average expense on 'X' in the month of Jan, Mar and May together?
- (a) 618 (b) 612 (c) 622 (d) 628 (e) 632
50. Find the ratio of total expense in Feb to total expense in April?
- (a) $\frac{96}{143}$ (b) $\frac{57}{125}$ (c) $\frac{87}{143}$ (d) $\frac{57}{143}$ (e) $\frac{87}{125}$

Directions (51-55): Study the data given below carefully and answer following questions based on these data.

Given below is the table which shows number of students participated (in thousand) in BANK exam and SSC exams from year 2000 to year 2005. There is also a line graph which shows percentage of qualified students in BANK exam and SSC exam

Year	No. of student (in thousand)	
	BANK	SSC
2000	85	90
2001	90	100
2002	95	105
2003	110	85
2004	80	85
2005	90	95

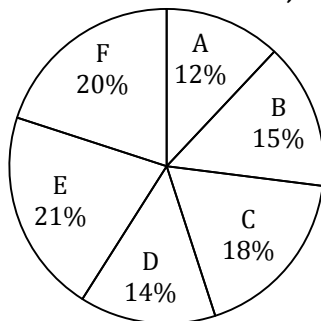


51. Number of qualified candidates in BANK exam in 2002 is what percent more or less than the failed candidates of SSC exam in 2001.
 (a) $41\frac{12}{17}\%$ (b) $39\frac{12}{17}\%$ (c) $36\frac{12}{17}\%$ (d) 42.5% (e) $35\frac{12}{17}\%$
52. Maximum growth of qualified candidates in BANK exam is recorded in which year, comparison to previous year?
 (a) 2001 (b) 2002 (c) 2003 (d) 2004 (e) 2005
53. Ratio between the total failed student in 2004 in both exam and qualified student of BANK exam in 2000 is:-
 (a) 351 : 442 (b) 451 : 342 (c) 442 : 453 (d) 229 : 189 (e) 475 : 442
54. Find the average number of students qualified in SSC exam in all year. (Consider nearest integer).
 (a) 55938 (b) 54620 (c) $56770\frac{5}{6}$ (d) 52940 (e) 58478
55. Find the difference between sum of qualified student of SSC exam in 2002, 2003, 2004 and sum of qualified students of BANK exam in 2001, 2003, 2005.
 (a) 11350 (b) 12455 (c) 13775 (d) 12875 (e) 14780

Directions (56-60): Pie-chart shown below shows percentages of cars sold by six Honda dealers.

Table shows the ratio of three type of cars out of total cars sold by different dealers. Study the data carefully and answer the following questions:

Total Cars sold = 12,000

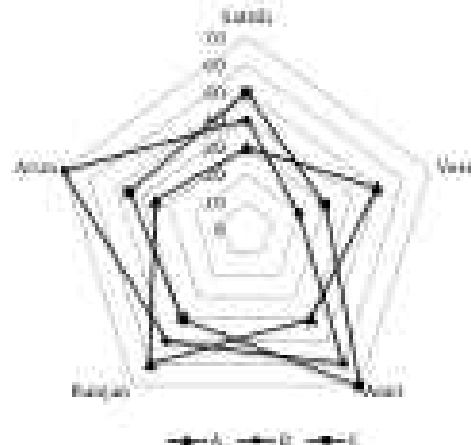


Type of Cars → Dealers ↓	Accord : Civic : City
A	4 : 2 : 3
B	3 : 4 : 3
C	7 : 4 : 4
D	6 : 8 : 7
E	3 : 6 : 5
F	5 : 4 : 6

56. What is the difference between the number of Accord cars sold by dealers D and E together and the number of City cars sold by dealers B and F together?
 (a) 360 (b) 420 (c) 540 (d) 480 (e) 460
57. The number of Accord and Civic cars sold by dealer A together is what percent of the number of Civic and City cars sold by dealer D together?
 (a) 90% (b) 80% (c) 75% (d) 60% (e) 50%
58. What is the average number of Civic cars sold by dealers A, B, D and E together?
 (a) 670 (b) 710 (c) 690 (d) 650 (e) 680
59. What is the ratio of the number of Civic to City cars sold together by dealer B to that by dealer E?
 (a) 11 : 7 (b) 7 : 11 (c) 5 : 8 (d) 8 : 5 (e) 7 : 12
60. Out of six dealers, which dealer sold the minimum number of City cars?
 (a) B (b) C (c) D (d) E (e) A

Direction (61- 65): Five traders bought 3 types of markers i.e. A, B and C on different price. The table given below shows the total number of markers bought and Radar graph shows per unit price of markers A, B and C to traders.

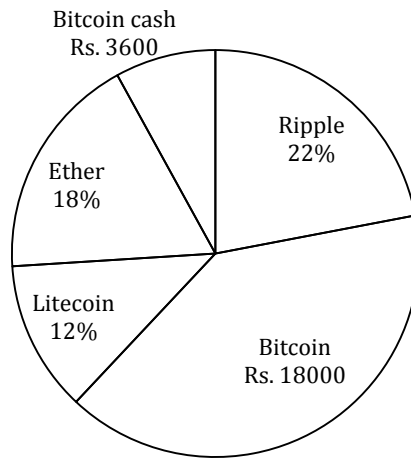
Traders	Total markers bought	A, B and C
Satish	165	3 : 4 : 4
Veer	195	4 : 5 : 4
Amit	175	2 : 2 : 1
Ranjan	210	4 : 6 : 4
Arun	198	1 : 3 : 5



61. Amount spend by Satish and Veer on Marker B is how much less than amount spend by Arun and Ranjan on marker C?
 (a) 3150 (b) 3250 (c) 3350 (d) 3450 (e) 3550
62. Total amount spent by Amit is what percent of the total amount spent by Veer?
 (a) 125% (b) 150% (c) 165% (d) 175% (e) 140%
63. Amount spend on marker A by Amit, Ranjan and Arun together is how much more or less than amount spend on marker C by Satish, Veer and Amit together?
 (a) 180 (b) 160 (c) 80 (d) 60 (e) 40
64. What is the ratio of the total amount spent by Satish to the total amount spent by Ranjan?
 (a) 7: 15 (b) 4: 5 (c) 7: 13 (d) 9: 14 (e) 7:16
65. Arun sold marker A at 10% profit, marker B at 20% profit and Marker C at 30% profit. Find the amount Arun got after selling all the markers?
 (a) 12,826 (b) 10,340 (c) 11,456 (d) 12,656 (e) 11,734

Directions (66-70): Study the following pie chart and table to answer the questions that follow.

The pie-chart shows the distribution (either in percentage or in absolute value) of investment made by Arunoday in five Cryptocurrencies in 2017 in month of November.



The table shows the profit or loss (by selling respective currencies at the end of November) in terms of percentage with respect to total amount invested on that currency.

Cryptocurrency	%Profit or % loss
Ripple	$90\frac{10}{11}\%$ Profit
Bitcoin	$22\frac{2}{9}\%$ loss
Litecoin	$27\frac{7}{9}\%$ loss
Ether	$55\frac{5}{9}\%$ profit
Bitcoin cash	$33\frac{1}{3}\%$ profit

66. Find the price per unit of Ripple bought by Arunoday if he sold 160 units at Rs. 75 each and remaining at Rs. 46 each. (take purchasing price rounded to nearest integer)
 (a) Rs. 38 (b) Rs. 32 (c) Rs. 22 (d) Rs. 28 (e) Rs. 16
67. Find the difference in the profit earned on Ripple and Bitcoin cash.
 (a) Rs. 7650 (b) Rs. 7850 (c) Rs. 7500 (d) Rs. 7800 (e) Rs. 5000
68. Profit earned on Litecoin and Ether together are what percent of the total investment made by Arunoday ?
 (a) $6\frac{2}{3}\%$ (b) $8\frac{2}{3}\%$ (c) $6\frac{1}{3}\%$ (d) $4\frac{4}{7}\%$ (e) $8\frac{1}{3}\%$
69. Find the overall profit or loss on Bitcoin, Litecoin and Ripple together.
 (a) Rs. 3550 (b) Rs. 3800 (c) Rs. 3500 (d) Rs. 3680 (e) Rs. 3050
70. When Arunoday bought Bitcoin, the rate was Rs. 12,00,000 per Bitcoin. Had he not sold the currency in the end of November but only at the end of December when the price of Bitcoin increased by 0.6% as compared to the price at which he bought it, then find the new profit or loss on Bitcoins.
 (a) Rs. 1080 (b) Rs. 2280 (c) Rs. 505 (d) Rs. 1108 (e) Rs. 108

Directions (71- 75): Given below is the bar-graph which shows the total number of persons who are visiting Hotel Shivoy in 5 different months of year.

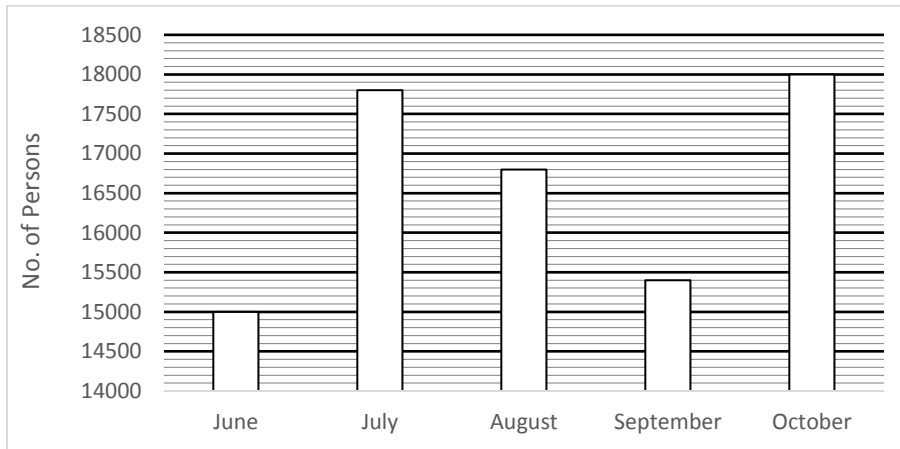


Table shows the percentage of foreigners in number of persons visiting in different months.

Months	% of persons who are foreigners
June	25%
July	18%
August	18%
September	23%
October	15%

Note- Total person = Indians + foreigners

71. If total Indian visitors in November are $2\frac{1}{17}\%$ of the total Indians visitors in October and total visitors are $\frac{4}{3}$ rd of the total visitors in June. Then find the difference of foreigner and Indian who visited in November?
 (a) 19820 (b) 18315 (c) 19370 (d) 17370 (e) 20210
72. If the ratio of male and female foreigners visiting Hotel Shivoy in month of September is 3 : 4, then foreigner males are what percent of foreigner females in September?
 (a) 75% (b) 85% (c) 92% (d) 60% (e) 70%
73. If $33\frac{1}{3}\%$ of foreigners visiting in August are married and 25% of Indians are also married, and $\frac{1}{4}$ th of the remaining foreigner are unmarried females and $33\frac{1}{3}\%$ of the remaining Indians are unmarried females. Then find the total females visiting Hotel Shivoy in month of August (there are no polygamy males or females)?
 (a) 6252 (b) 5468 (c) 6220 (d) 6174 (e) 6184
74. If the ratio of Indian male and female visitors in June is 2 : 3 and ratio of foreigner male & female visitors in August is 1 : 5, then find the ratio of Indian females visitors in June to foreigner males in August?
 (a) 365 : 28 (b) 28 : 375 (c) 375 : 28 (d) 355 : 28 (e) 375 : 23
75. Foreigner visitors in October are what percent of Indian visitors in June?
 (a) 28% (b) 27% (c) 36% (d) 32% (e) 24%

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PRACTICE SET (LEVEL-I) SOLUTIONS

1. (c); Total votes casted for varun Gandhi

$$= \frac{5000(22 \times 22 + 18 \times 22 + 14 \times 16 + 24 \times 10 + 20 \times 22)}{10000} = 790$$
 Total votes casted for Deepak mittal=

$$\frac{5000(22 \times 20 + 18 \times 12 + 16 \times 15 + 24 \times 18 + 20 \times 16)}{10000} = 824$$
 Required difference = $824 - 790 = 34$
2. (b); Required percentage

$$= \frac{\frac{5000}{10000}(22 \times 13 + 18 \times 20 + 16 \times 18 + 24 \times 21 + 20 \times 21)}{5000 \times \frac{24}{100}} \times 100$$

$$= \frac{\frac{1}{2}(286 + 360 + 288 + 504 + 240)}{1200} \times 100$$

$$= \frac{839}{1200} \times 100 = 69 \frac{11}{12} \%$$
3. (d); Girls in mechanical = $\frac{3}{8} \times \frac{16}{100} \times \frac{22}{100} \times 5000 = 66$
 $\therefore$ Total number of girls = $66 \times 36 = 2376$
 So, total number of boys = $5000 - 2376 = 2624$
4. (e); Number of students supporting K.L.
 Bali = $\frac{5000}{10000}(22 \times 14 + 18 \times 16 + 16 \times 22 + 24 \times 20 + 20 \times 12)$

$$= \frac{5000 \times \frac{20}{100}}{834} = \frac{417}{1000} = \frac{417}{500}$$
 $\therefore$ Required ratio = $\frac{834}{1000} = \frac{417}{500}$
5. (c); Total votes casted for Yogesh gupta

$$= \frac{5000(22 \times 22 + 18 \times 22 + 16 \times 23 + 24 \times 16 + 20 \times 20)}{10000}$$

$$= \frac{1(484 + 396 + 368 + 384 + 400)}{2} = 1016$$
 Total votes casted for vijay tyagi

$$= \frac{5000(22 \times 15 + 18 \times 12 + 16 \times 8 + 24 \times 15 + 20 \times 18)}{10000}$$

$$= \frac{1(330 + 216 + 128 + 360 + 360)}{2} = 697$$
 Difference in votes $1016 - 697 = 319$
 $\therefore$ Required percentage = $\frac{319}{5000} \times 100 = 6 \frac{19}{50} \%$
6. (d); Number of males in city K = $\frac{40}{100} \times 20,000 = 8000$
 Number of males who left city K = $\frac{40}{100} \times 8000 = 3200$
 Number of males in city L = $\frac{60}{100} \times 25,000 = 15000$
 Total number of males in city L after Males who joined city L
 $= 15000 + 3200 = 18200$
7. (c); Literate people from city L
 $= \frac{7}{10} \times 25,000 = 175,00$
 Illiterate people from city M
 $= \frac{8}{10} \times 35,000 = 28000$
 $\therefore$ Percentage = $\frac{17500}{28000} \times 100 = 62.5\%$
8. (a); 30 percent of male from city M
 $= \frac{20}{100} \times 35,000 \times \frac{30}{100} = 2100$
 $\therefore$ 2100 male from city M are illiterate
 Female from city M who are illiterate
 $= \frac{4}{5} \times 35,000 - 2100$
 $= 28000 - 2100 = 25900$
 $\therefore$ Ratio = $\frac{2100}{25900} = 21 : 259$
9. (b); Required average = $\frac{15000 + 7500 + 28000 + 24000 + 27500}{5} = 20400$
10. (d); Since the illiterate males from city K and city L cannot be determined.
11. (a); According to Question

$$11200 = \frac{70,000 \times R \times 2}{100}$$

$$R = 8\%$$

$$\text{Share of Abhimanyu} = \frac{11}{25} \times 11200 = 4928$$

$$12. (c); \text{Required ratio} = \frac{\frac{11}{25} \times 70 + \frac{7}{20} \times 65 + \frac{2}{5} \times 80}{\frac{1}{2} \times 60 + \frac{7}{10} \times 40 + \frac{3}{5} \times 55}$$

$$= \frac{30.8 + 22.75 + 32}{30 + 28 + 33} = \frac{85.55}{91} = 1711 : 1820$$

13. (c); Average of investment made by Abhimanyu in scheme A and C together is

$$= \frac{\frac{11}{25} \times 70 + \frac{1}{2} \times 60}{2} = \frac{60.8}{2} = 30.4$$

Average of investment made in scheme E & F by Arunoday

$$= \frac{\frac{7}{10} \times 40 + \frac{3}{5} \times 55}{2} = \frac{28 + 33}{2} = \frac{61}{2}$$

$$\Rightarrow 30.5$$

$$\text{Required } \% = \frac{30.4}{30.5} \times 100 \approx 100\%$$

14. (c); Interest obtained from scheme B

$$= \frac{7}{20} \times 65 \times \frac{20}{100} + \frac{13}{20} \times 65 \times \frac{30}{100}$$

$$= 4.55 + 12.675$$

$$= 17225 \text{ Rs}$$

Interest obtained from C

$$= \frac{1}{2} \times 60 \times 3 \times \frac{1}{3} + \frac{1}{2} \times 60 \times \frac{1}{3} = 30 + 10$$

$$= 40 \text{ thousands} = 40,000 \text{ Rs}$$

$$\text{Total Interest} = 57225 \text{ Rs}$$

15. (a); Required ratio

$$= \frac{\frac{11}{25} \times 70 + \frac{7}{20} \times 65 + \frac{1}{2} \times 60 + \frac{2}{5} \times 80}{\frac{14}{25} \times 70 + \frac{13}{20} \times 65 + \frac{1}{2} \times 60 + \frac{3}{5} \times 80}$$

$$= \frac{30.8 + 22.75 + 30 + 32}{39.2 + 42.25 + 30 + 48}$$

$$= \frac{115.55}{2311} = \frac{2311}{159.45} = \frac{2311}{3189}$$

16. (c); Expenditure of B on clothing = $\frac{24000}{75} \times 100 \times \frac{15}{100} = 4800$

$$\text{Expenditure of D on clothing} = \left(100\% + 16 \frac{2}{3}\%\right) 4800 = 5600$$

$$\text{Sum of expenditure of D on food and clothing} = \frac{5600}{7} \times 15$$

$$= 800 \times 15 = 12000$$

17. (a); Sum of Expenditure of D and F on house rent =

$$30000 \times \frac{20}{100} + \frac{1}{3} \times 15000$$

$$= 6000 + 5000 = 11000$$

$$\text{Expenditure of B on Education} = \frac{24000}{75} \times 100 \times \frac{10}{100} = 3200$$

$$\text{Required percentage} = \frac{3200}{11000} \times 100 = \frac{320}{11} = 29 \frac{1}{11} \%$$

18. (c); Expenditure of B on clothing = 4800

$$\text{Income of C} = \left(100\% + \frac{1100}{3}\%\right) 4800$$

$$= \frac{14}{3} \times 4800 = 22400$$

$$\text{Saving of C} = 22400 - 20000 = 2400$$

$$\text{Required } \% = \frac{5600}{8000} \times 100 = 70\%$$

19. (c); Saving of A, B, C, D, E and F are 4000, 8000, 4000, 12000, 8000 and 4000 respectively

$$\text{Income of A, B, C, D, E and F are 20000, 32000, 24000, 42000, 33000 and 19000 respectively}$$

$$\text{Required average} = \frac{170000}{6} = 28333.33$$

20. (d); Expenditure of E on bills = $32000 \times \frac{10}{100} = 3200$
 Income of B after decrement = $(100\% - \frac{35}{2}\%) 32000$
 $= 26400$
 Now expenditure of B on bills = $26400 \times \frac{10}{100} = 2640$
 Required percentage = $\frac{3200-2640}{3200} \times 100 = 17.5\%$

21. (a); Total visitors in June
 $= \frac{6}{5} \times 15000 = 18000$
 Foreigner in June month
 $= 1.3 \times \frac{1}{4} \times 15000 = 4875$
 Total foreign visitors in Jan and Feb together
 $= \frac{1}{4} \times 15000 + 17800 \times \frac{18}{100}$
 $= 3750 + 3204 = 6954$
 Required % = $\frac{(18000-4875)}{6954} \times 100 \approx 188\%$

22. (a); Total females visiting in Feb month
 $= 17,800 \times \frac{1}{2} = 8,900$
 Total males visiting in May month
 $= 18000 \times \frac{3}{5} = 10800$
 Foreign females visitors in Feb
 $= \frac{4}{9} \times 17800 \times \frac{18}{100} = 1424$
 Foreign male visitors in May
 $= \frac{3}{4} \times 18000 \times \frac{15}{100} = 2025$
 Required % = $\frac{8900-1424}{10800-2025} \times 100$
 $\approx \frac{8900-1400}{10800-2000} \times 100$
 $\approx \frac{7500}{8800} \times 100 \approx 85\%$

23. (e); Total foreign visitors in Feb and March month together
 $= 17800 \times \frac{18}{100} + 16800 \times \frac{20}{100}$
 $= 3204 + 3360 = 6564$
 Total India visitors in Jan and April
 $= 15000 \times \frac{75}{100} + 15400 \times \frac{77}{100}$
 $= 11250 + 11858 = 23108$
 Required difference = $23108 - 6564 = 16544$

24. (c); Total married Indian couples in Feb
 $= \frac{1}{2} \times 17800 \times \frac{82}{100} = 7298$
 Total married foreign couples in Feb
 $= \frac{1}{6} \times 17800 \times \frac{18}{100} = 534$
 Remaining Indian male visitors = 3649
 Remaining foreign males visitors = 1780
 Required total number of males
 $= \frac{7298}{2} + \frac{534}{2} + 3649 + 1780 = 9345$

25. (b); Required % = $\frac{3204-2700}{2700} \times 100 = \frac{504}{27} \%$
 $= \frac{56}{3} \% = 18\frac{2}{3} \%$

26. (b); Total population of female in village U = $\frac{17}{100}x$
 So,
 $\frac{17}{100}x = (100 - \frac{1300}{30})\% \rightarrow \frac{17}{100}x$
 $\frac{1700}{30} \% \rightarrow \frac{17}{100}x$

$$100\% \rightarrow \frac{17}{100}x \times \frac{30}{1700} \times 100$$

$$\text{Population of female in village T} = \frac{14}{100}x$$

$$(100 - \frac{1100}{25})\% \rightarrow \frac{14}{100}x$$

$$100\% \rightarrow \frac{14}{100}x \times \frac{25}{1400} \times 100$$

$$\text{Required \%} = \frac{\frac{17}{100}x \times \frac{30}{1700} \times 100}{\frac{14}{100}x \times \frac{25}{1400} \times 100} \times 100$$

$$= \frac{30x}{25x} \times 100 = 120\%$$

27. (a); Total females from village S and T together
 $= (20\% + 14\%)x = \frac{34}{100}x$

$$\text{Males from village R} = \frac{16x}{16} \times 15 = 15x$$

$$\text{Males from village U} = \frac{17x}{17} \times 13 = 13x$$

$$\text{Required ratio} = \frac{34x}{(15x+13x)} = \frac{34}{28} = 17 : 14$$

28. (d); Total females in village P = $0.18x$
 Total males in village P = $\frac{0.18x}{(100\% - \frac{700}{13}\%)} \times \frac{700}{13} \%$

$$= \frac{0.18x}{\frac{600}{13}} \times \frac{700}{13} = 0.21x$$

$$\text{Total population of village P} = 0.21x + 0.18x = 0.39x$$

$$\text{Total illiterate in P} = \frac{2}{13} \times 0.39x = 0.06x$$

$$\text{Total male illiterate in P} = 0.06x - \frac{20}{100} \times 0.18x$$

$$= 0.06x - 0.036x = 0.024x$$

$$\text{Required percentage} = \frac{0.024x}{0.21x} \times 100 = \frac{80}{7} \%$$

29. (b); $\frac{700}{13} \% \rightarrow 17850$
 $\frac{600}{13} \% \rightarrow 17850 \times \frac{13}{700} \times \frac{600}{13} \rightarrow 15300$
 $\frac{18x}{100} = 15300$
 $x = 85000$

$$\text{Total females in T and U together} = \frac{30}{100} \times 85000$$

$$= 26350$$

$$\text{Males in village T} = \frac{14}{100} \times 85000 \times \frac{25}{1400} \times \frac{1100}{25} = 9350$$

$$\text{Male in village U} = \frac{17}{100} \times 85000 \times \frac{30}{1700} \times \frac{1300}{30} = 11050$$

$$\text{Total male and female in village T and U together} = 26350 + 9350 + 11050 = 46750$$

30. (e); Total male and female who are not registered in voter list

$$= \frac{30}{100} \times 27200 + \frac{20}{100} \times \frac{27200}{(100\% - \frac{1500}{31}\%)} \times \frac{1500}{31} \% =$$

$$8160 + \frac{1}{5} \times \frac{27200}{\frac{1600}{31}} \times \frac{1500}{31} \%$$

$$= 8160 + 5100 = 13260$$

$$\text{Total population of village R}$$

$$= \frac{27200}{(100\% - \frac{1500}{31}\%)} \times 100\% = \frac{27200}{\frac{1600}{31}} \times 100 = 52700$$

$$\text{Total polled in R} = \frac{60}{100} \times 52700 = 31620$$

$$\text{Total registered voter who did not cast their vote} = 52700 - 13260 - 31620 = 7820$$

$$\text{Required difference} = 13260 - 7820 = 5440$$

31. (a); Total books sold by store A

$$= 18 \times 257 + \frac{18}{3} \% \times 4 \times 25700$$

$$= 18 \times 257 + 24 \times 257$$

$$= 257 \times 42$$

Total books sold by store B

$$= 20 \times 257 + \frac{20 \times 257 \times 5}{4} = 257 \times 45$$

Total books sold by both store

$$= 257(45 + 42) = 257 \times 87 = 22,359$$

32. (c); Total books sold by store D

$$= 8 \times 257 + 8 \times 257 \times \frac{5}{4} = 257(8 + 10)$$

$$= 257 \times 18 = 4626$$

33. (b); Total books of publisher X sold by store E, F and G together = 44×257

Total books of publisher Y sold by store E, F and G together

$$= 25700 \left(\frac{13\%}{13} \times 5 + 22\% \times \frac{12}{11} + 9\% \times \frac{4}{3} \right)$$

$$= 25700(5\% + 24\% + 12\%)$$

$$= 25700(41\%)$$

$$\text{Required percentage} = \frac{257(44-41)}{257 \times 41} \times 100 = \frac{3}{41} \times 100 = \frac{300}{41}\%$$

34. (a); Total books sold by store E in 2017

$$= \frac{4}{3}(13 \times 257 + 5 \times 257)$$

$$= 4 \times 257 \times 6$$

$$= 257 \times 24$$

$$\text{Required percentage} = \frac{257 \times 13 - 257 \times 24 \times \frac{11}{24}}{257 \times 24 \times \frac{11}{24}} \times 100$$

$$= \frac{257(13-11)}{257 \times 11} \times 100 = \frac{200}{11}\%$$

35. (c); Average of books of publisher X sold by store B and C

$$= \frac{30 \times 257}{2} = 15 \times 257$$

Average of books of publisher Y sold by store E and G

$$= \left(13 \times 257 \times \frac{5}{13} + 9 \times 257 \times \frac{4}{3} \right) \frac{1}{2}$$

$$= (5 \times 257 + 12 \times 257) \frac{1}{2} = 8.5 \times 257$$

$$\text{Required percentage} = \frac{15 \times 257 - 8.5 \times 257}{8.5 \times 257} \times 100 = \frac{6.5}{8.5} \times 100$$

$$= \frac{13}{17} \times 100 = \frac{1300}{17}\%$$

36. (d); Total literate female from E

$$= \frac{1000}{20} \times 80 = 4000$$

$$\text{Total female in village A} = 6000 + \frac{6000}{60} \times 40$$

$$= 10,000$$

$$\text{Total male in village A} = \frac{10000}{100\% - 54\frac{6}{11}\%} \times 54\frac{6}{11}\%$$

$$= \frac{10000}{\left(100\% - \frac{600}{11}\%\right)} \times \frac{600}{11}\% = 12000$$

Required percentage

$$= \frac{8000}{12000} \times 100 = 66\frac{2}{3}\%$$

37. (b); Total males in village A = 12000 (solved above)

$$\text{Illiterate female in village C} = 12000 \left(1 + \frac{1}{12} \right)$$

$$= 12000 \times \frac{13}{12} = 13000$$

$$\text{Total literate female in village F} = \frac{2000}{40} \times 60 = 300$$

$$\text{Required value} = 13000 - 3000 = 10000$$

38. (e); Total literate female in village D

$$= \left(1 + \frac{4}{9} \right) \times 4500 = \frac{13}{9} \times 4500 = 6500$$

$$\text{Total female in D} = 3500 + 6500 = 10000$$

$$\text{Total males} = \frac{10000}{\left(100\% - 52\frac{8}{21}\%\right)} \times 52\frac{8}{21}\%$$

$$= \frac{10000}{\frac{10}{21}} \times \frac{11}{21} = 11000$$

$$\text{Total females in F} = 2000 + \frac{2000}{40} \times 60 = 5000$$

$$\text{Total males in F} = \frac{5000}{\left(1 - \frac{6}{11}\right)} \times \frac{6}{11}$$

$$= \frac{5000}{\frac{5}{11}} \times \frac{6}{11} = 6000$$

$$\text{Required difference} = 11000 - 6000 = 5000$$

39. (a); Total males in F = 6000 (Solved above)

$$\text{Total females in B} = 4500 + \frac{4500}{45} \times 55 = 10000$$

$$\text{Total males in B} = \frac{10000}{\left(1 - \frac{4}{9}\right)} \times \frac{4}{9} = \frac{10000}{5} \times 4 = 8000$$

$$\text{Required ratio} = 3 : 4$$

40. (b); Let, total persons in 2016 in C = 9,100

So males in C in 2016

$$= 9100 \times \frac{3}{7} = 3900$$

Now illiterate females in 2017

$$= 5200 \times \frac{65}{100} \times \frac{11}{13} = 2860$$

And literate female in 2017

$$= 5200 \times \frac{35}{100} \times \frac{14}{13}$$

$$= 1960$$

Now total females in 2017 = 4820

Percentage increase in males in 2017 with respect to previous year

$$= \frac{(5200 - 4820)}{3900} \times 100 = 9\frac{29}{39}\%$$

41. (b); Total male students participating in seminar in 2014

$$= \left(\frac{5}{8} \times 40 + \frac{5}{7} \times 35 + \frac{7}{10} \times 20 \right) \text{ thousands}$$

$$= 64 \text{ thousands}$$

Total female students who participated in 2013

$$= \left(\frac{1}{5} \times 30 + \frac{1}{5} \times 25 + \frac{1}{5} \times 15 \right) \text{ thousands}$$

$$= 14 \text{ thousands}$$

$$\therefore \text{Required percentage} = \frac{64}{14} \times 100 = 457\frac{1}{7}\%$$

42. (c); Total students (both male and female) preparing for banking who asked questions in seminar in 2015

$$= \frac{10}{100} \times \frac{7}{10} \times 50 + \frac{5}{100} \times \frac{3}{10} \times 50$$

$$= (3.5 + 0.75) = 4.25 \text{ thousands}$$

Total students preparing for SSC who asked question in 2015

$$= \frac{10}{100} \times 45000 = 4.5 \text{ thousands}$$

$$\therefore \text{Required percentage} = \frac{4.25}{4.5} \times 100 = 94\frac{4}{9}\%$$

43. (c); Required average = $\frac{1}{5} \times \left(\frac{4}{5} \times 15 + \frac{7}{10} \times 20 + \frac{6}{7} \times 35 + \frac{3}{4} \times 30 + \frac{3}{5} \times 40 \right) = \frac{1}{5} \times 102.5 = 20.5 \text{ thousands}$

44. (b); Total no of male students preparing for bank exam in 2016 and 2017 together

$$= \frac{3}{5} \times 50 + \frac{2}{3} \times 60 = 70 \text{ thousands}$$

Total no of female students preparing for SSC exam in 2016 and 2017 together

$$= \frac{3}{8} \times 40 + \frac{3}{10} \times 50 = 30 \text{ thousands}$$

$$\therefore \text{Required percentage} = \frac{40}{30} \times 100 = 133\frac{1}{3}\%$$

45. (d); In 2012,

Total students who participated in seminar preparing for Banking

$$= 30,000 \times \frac{90}{100} = 27,000$$

$$\text{SSC} = \frac{80}{100} \times 25,000 = 20,000$$

$$\text{Railway} = \frac{75}{100} \times 15,000 = 11,250.$$

No. of boys who participated in seminar in 2012

$$\text{Banking} = \frac{4}{5} \times 30,000 - 1000 = 23,000$$

$$\text{SSC} = \frac{4}{5} \times 25,000 - 1500 = 18,500$$

$$\text{Railway} = \frac{4}{5} \times 15,000 - 2000 = 10,000$$

$$\therefore \text{Required no of girls} = (27,000 - 23,000) + (20,000 - 18,500) + (11,250 - 10,000) = 6,750$$

46. (d); Total Income = $\frac{4680}{100} \times 100 = 19500$

$$\text{Income in Jan} = \frac{24}{100} \times 19500 = 2340$$

$$\text{Expense in month of Jan} = 2340 \left[1 - \frac{750}{1300} \right] = 990$$

$$\text{Expense in month of Jan on 'Z'} = \frac{990}{15} \times 7 = 462$$

$$\text{Expense in month of Feb} = 19500 \times \frac{16}{100} \times \left[1 - \frac{500}{1300} \right] = 1920$$

$$\text{Expense in month of Feb on 'X'} = \frac{1920}{24} \times 3 = 560$$

$$\text{Required\%} = \frac{560 - 462}{560} \times 100 = 17.5\%$$

47. (b); Expense of 'Z' in Apr = $4680 \times \left[1 - \frac{350}{900} \right] \times \frac{4}{11} = 1040$

$$\text{Expense on 'X' in May} = 19500 \times \frac{28}{100} \times \left[1 - \frac{130}{300} \right] \times \frac{6}{17} = 1092$$

$$\text{Required difference} = 1092 - 1040 = 52$$

48. (e); Total expense in month of Jan

$$= 19500 \times \frac{12}{100} \left[1 - \frac{750}{1300} \right] = 990$$

$$\text{Total saving in month of April}$$

$$= 4680 \times \frac{350}{900} = 1820$$

$$\text{Required\%} = \frac{990}{1820} \times 100 = 54 \frac{36}{91}\%$$

49. (c); Expense on 'X' in Jan

$$= 19500 \times \frac{12}{100} \left[1 - \frac{750}{1300} \right] \times \frac{3}{15} = 198$$

$$\text{Expense on 'X' in Mar}$$

$$= 19500 \times \frac{20}{100} \times \left[\frac{100 - 52}{100} \right] \times \frac{4}{13} = 576$$

$$\text{Expense on 'X' in May}$$

$$= 19500 \times \frac{28}{100} \times \left[1 - \frac{130}{300} \right] \times \frac{6}{17} = 1092$$

$$\text{Required average} = \frac{1}{3} [198 + 576 + 1092]$$

$$= \frac{1}{3} \times 1866 = 622$$

50. (a); Required ratio = $\frac{19500 \times \frac{16}{100} \times \left[1 - \frac{500}{1300} \right]}{4680 \times \left[1 - \frac{350}{900} \right]} = \frac{1920}{2860} = \frac{96}{143}$

51. (b); Number of qualified candidates in BANK exam in

$$2002 = 95000 \times \frac{62.5}{100} = 59375$$

$$\text{Failed candidates in SSC exam in 2001} = 100000 \times \frac{42.5}{100} = 42500$$

$$\text{Required percentage} = \frac{59375 - 42500}{42500} \times 100$$

$$= \frac{16875}{425} \% = 39 \frac{12}{17}\%$$

52. (c); Qualified candidates of BANK exam in different year,

$$\text{In year, 2000} \Rightarrow 85000 \times \frac{65}{100} = 55250$$

$$2001 \Rightarrow 90000 \times \frac{60}{100} = 54000 \text{ decrease}$$

$$2002 \Rightarrow 95000 \times \frac{62.5}{100} = 59375 \text{ increase}$$

$$2003 \Rightarrow 110000 \times \frac{67.5}{100} = 74250 \text{ increase}$$

$$2004 \Rightarrow 80000 \times \frac{55}{100} = 44000 \text{ decrease}$$

$$2005 \Rightarrow 90000 \times \frac{57.5}{100} = 51750 \text{ increase}$$

Maximum growth is recorded in 2003 i.e; 74250 - 59375 = 14875

53. (e); Total failed student in 2004

$$= 80000 \times \frac{45}{100} + 85000 \times \frac{27.5}{100} = 59375$$

Qualified students of BANK exam in 2000 = 55250

$$\text{Required ratio} = 59375 : 55250 = 475 : 442$$

54. (c); Required average

$$= \frac{1}{6} \left[90000 \times \frac{55}{100} + 100000 \times \frac{57.5}{100} + 105000 \times \frac{60}{100} + 85000 \times \frac{50}{100} + 85000 \times \frac{72.5}{100} + 95000 \times \frac{70}{100} \right]$$

$$= \frac{1}{6} [340625] = 56770 \frac{5}{6}$$

55. (d); Sum of qualified student in SSC exam = 105000 ×

$$\frac{60}{100} + 85000 \times \frac{50}{100} + 85000 \times \frac{72.5}{100} = 167125$$

$$\text{Sum of qualified student in BANK exam} = 90000 \times$$

$$\frac{60}{100} + 110000 \times \frac{67.5}{100} + 90000 \times \frac{57.5}{100} = 180000$$

$$\text{Required difference} = 180000 - 167125 = 12875$$

56. (d); Number of Accord cars sold by dealers D and E together

$$= \left(\frac{6}{21} \times \frac{14}{100} + \frac{3}{14} \times \frac{21}{100} \right) \times 12000 = 480 + 540 = 1020$$

Number of City cars sold by dealers B and F together

$$= \left(\frac{3}{10} \times \frac{15}{100} + \frac{6}{15} \times \frac{20}{100} \right) \times 12000 = 540 + 960 = 1500$$

$$\text{Required Difference} = 1500 - 1020 = 480$$

57. (b); Number of Accord and Civic cars sold by dealer A together = $\frac{6}{9}$ of 12% = 8%

$$\text{Number of Civic and City cars sold by dealer D together} = \frac{15}{21} \text{ of } 14\% = 10\%$$

$$\text{Required Percentage} = \frac{8}{10} \times 100 = 80\%$$

58. (c); Total number of Civic cars sold by dealers A, B, D and E together

$$= \left(\frac{2}{9} \times \frac{12}{100} + \frac{4}{10} \times \frac{15}{100} + \frac{8}{21} \times \frac{14}{100} + \frac{6}{14} \times \frac{21}{100} \right) \times 12000$$

$$= 320 + 720 + 640 + 1080 = 2760$$

$$\text{Required Average} = \frac{2760}{4} = 690$$

59. (b); Civic and City cars sold together by dealer B

$$= \frac{7}{10} \text{ of } 15\% = \frac{21}{2}\%$$

Civic and City cars sold together by dealer E

$$= \frac{11}{14} \text{ of } 21\% = \frac{33}{2}\%$$

$$\text{Required Ratio} = \frac{21}{2}\% : \frac{33}{2}\% = 7 : 11$$

60. (e); Percentage of City cars sold by:

$$\text{Dealer A} = \frac{3}{9} \text{ of } 12\% = 4\%$$

$$\text{Dealer B} = \frac{3}{10} \text{ of } 15\% = 4.5\%$$

$$\text{Dealer C} = \frac{4}{15} \text{ of } 18\% = 4.8\%$$

$$\text{Dealer D} = \frac{7}{21} \text{ of } 14\% = 4.67\%$$

$$\text{Dealer E} = \frac{5}{14} \text{ of } 21\% = 7.5\%$$

$$\text{Dealer F} = \frac{6}{15} \text{ of } 20\% = 8\%$$

Hence, dealer A sold the minimum number of City cars.

$$61. (d); \text{Marker B bought by Satish} = \frac{165}{11} \times 4 = 60$$

$$\text{Marker B bought by Veer} = \frac{195}{13} \times 5 = 75$$

$$\text{Amount spend by Satish \& Veer on marker B} = 60 \times 40 + 75 \times 20 = 2400 + 1500 = 3900$$

$$\text{Marker C bought by Ranjan} = \frac{210}{14} \times 4 = 60$$

$$\text{Marker C bought by Arun} = \frac{198}{9} \times 5 = 110$$

$$\text{Amount spend by Ranjan \& Arun on Marker C} = 60 \times 40 + 110 \times 45 = 2400 + 4950 = 7350$$

$$\text{Required difference} = 7350 - 3900 = 3450$$

$$62. (b); \text{Amount spend by Veer} = \frac{195}{13} \times 4 \times 50 + \frac{195}{13} \times 5 \times 20 + \frac{195}{13} \times 4 \times 30 = 15[200 + 100 + 120] = 6300$$

$$\text{Amount spend by Amit} = \frac{175}{5} [2 \times 40 + 2 \times 60 + 70] = 35 [80 + 120 + 70] = 9450$$

$$\text{Required \%} = \frac{9450}{6300} \times 100 = 150\%$$

$$63. (c); \text{Amount spend on maker A by Amit, Ranjan and Arun} = \frac{175}{5} \times 2 \times 40 + \frac{210}{14} \times 4 \times 60 + \frac{198}{9} \times 1 \times 35 = 2800 + 3600 + 770 = 7170$$

$$\text{Amount spend on Maker C by Satish, Veer and Amit} = \frac{165}{11} \times 4 \times 50 + \frac{195}{13} \times 4 \times 30 + \frac{175}{5} \times 1 \times 70 = 3000 + 1800 + 2450 = 7250$$

$$\text{Required difference} = 7250 - 7170 = 80$$

$$64. (d); \text{Total amount spent by Satish} = \frac{165}{11} [3 \times 30 + 4 \times 40 + 4 \times 50] = 15 [90 + 160 + 200] = 6750$$

$$\text{Total Amount spent by Ranjan} = \frac{210}{14} [4 \times 60 + 6 \times 50 + 4 \times 40] = 15 [240 + 300 + 160] = 10500$$

$$\text{Required ratio} = \frac{6750}{10500} = \frac{9}{14}$$

$$65. (a); \text{Required value} = \frac{198}{9} \left[1 \times 35 \times \frac{11}{10} + 3 \times 70 \times \frac{12}{10} + 5 \times 45 \times \frac{13}{10} \right] = 22 [38.5 + 252 + 292.5] = 12826$$

Direction (66-70)

Currencies	Amount invested	Profit or loss
Ripple	9900	9000 profit
Bitcoin	18000	4000 loss
Litecoin	5400	1500 loss
Ether	8100	4500 profit
Bitcoin cash	3600	1200 profit
Total = 45000		

$$66. (b); \text{Total amount after selling Ripple} = 9900 + 9000 = \text{Rs. } 18900$$

$$\therefore 18900 = 160 \times 75 + x \times 46$$

$$\Rightarrow x = \frac{18900 - 12000}{46} = 150 \text{ units}$$

So, Arunoday bought a total of $160 + 150 = 310$ units of Ripple

$\therefore$ Price per unit of Ripple

$$= \frac{9900}{310} \approx 32 \text{ Rs.}$$

$$67. (d); \text{Required difference in profit} = 9000 - 1200 = 7800 \text{ Rs.}$$

$$68. (a); \text{Required percentage} = \frac{4500 - 1500}{45000} \times 100 = 6\frac{2}{3}\%$$

$$69. (c); \text{Required profit} = 9000 - 4000 - 1500 = \text{Rs. } 3500$$

$$70. (e); \text{Quantity of Bitcoin bought}$$

$$= \frac{18000}{12,00,000} = 0.015 \text{ units}$$

$$\text{New selling price} = \frac{100.6}{100} \times 12,00,000 = 12,07,200$$

$$\therefore \text{Amount collected after selling at new price} = 1207200 \times 0.015 = \text{Rs. } 18108$$

$$\text{So, required profit} = 18108 - 18000 = \text{Rs. } 108.$$

$$71. (c); \text{Total Indians in November}$$

$$= \frac{35}{1700} \times \frac{85}{100} \times 18000 = 315$$

$$\text{Total visitors in November}$$

$$= \frac{4}{3} \times 15000 = 20000$$

$$\therefore \text{Foreigners visiting in November}$$

$$= 20000 - 315 = 19685$$

$$\therefore \text{Required difference}$$

$$= 19685 - 315 = 19370$$

$$72. (a); \text{Males visiting who are foreigner in September}$$

$$= \frac{3}{7} \times \frac{23}{100} \times 15400 = 1518$$

$$\text{Females foreigner in September}$$

$$= \frac{4}{7} \times \frac{23}{100} \times 15400 = 2024$$

$$\therefore \text{Required percentage} = \frac{1518}{2024} \times 100 = 75\%$$

Alternately,

$$\text{Required ratio} = \frac{3}{4} \times 100 = 75\%$$

$$73. (d); \text{Foreigner who are married}$$

$$= \frac{18}{100} \times 16800 \times \frac{1}{3} = 1008$$

$$\text{Indians who are married}$$

$$= \frac{82}{100} \times 16800 \times \frac{1}{4} = 3444$$

$$\text{remaining foreign females visitors}$$

$$= \frac{2016}{4} = 504$$

$$\text{Remaining Indian females visitors} = \frac{10332}{3} = 3444$$

$$\therefore \text{Required total females}$$

$$= \frac{1008}{2} + \frac{3444}{2} + 504 + 3444 = 6174$$

$$74. (c); \text{Required ratio}$$

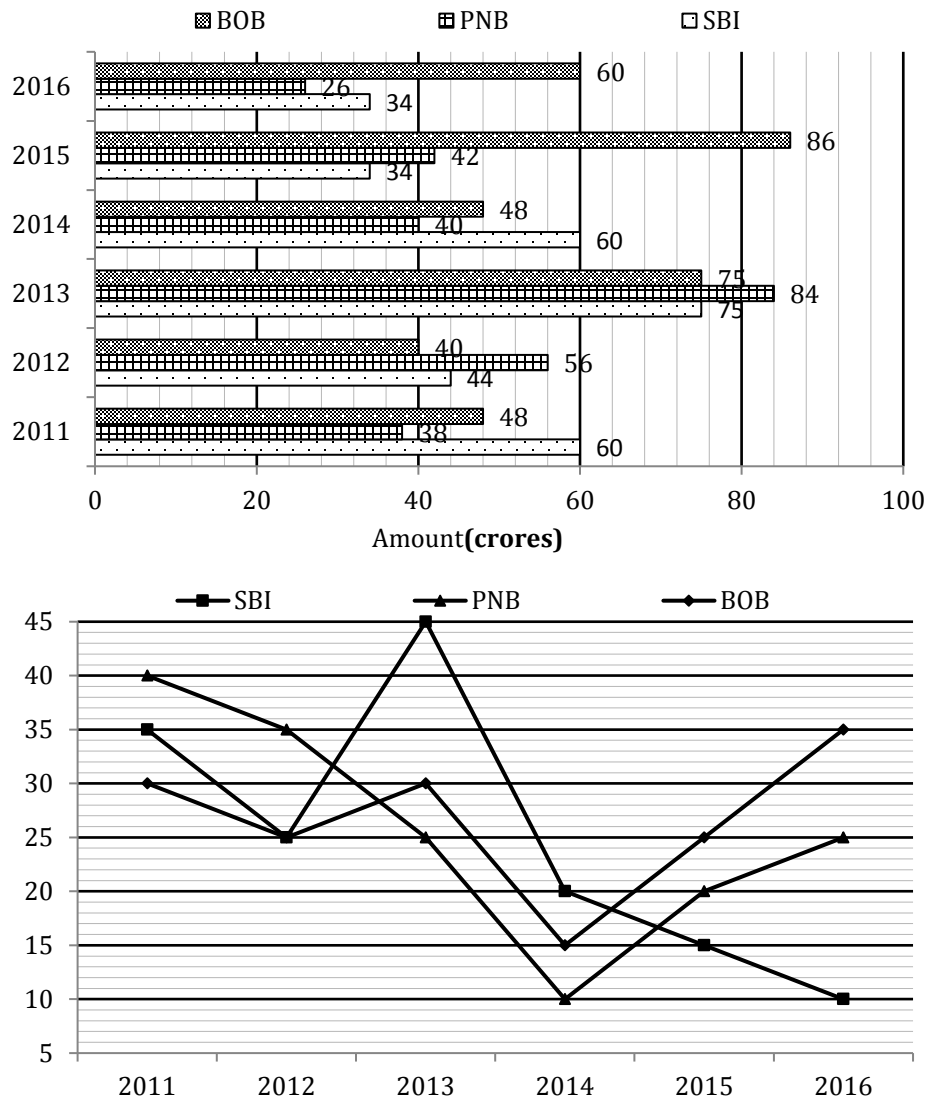
$$= \frac{\frac{3}{5} \times \frac{75}{100} \times 15000}{\frac{1}{6} \times \frac{18}{100} \times 16800} = \frac{6750}{504} = 375 : 28$$

$$75. (e); \text{Required percentage} = \frac{\frac{15}{100} \times 18000}{\frac{75}{100} \times 15000} \times 100 = 24\%$$



PRACTICE SET (LEVEL-II)

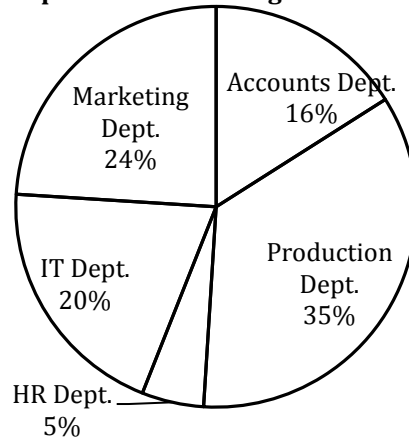
Directions (1-5): Given below is the bar graph showing the amount of loan (in crores) sanctioned by three banks namely SBI, PNB and BOB in six different years and The line graph shows the percentage of total loan that is sanctioned as Education Loan by the banks. Read the graphs carefully and answer the following questions.



- Find the difference between (in crores) the total educational loan sanctioned by PNB from year 2011 to 2013 and total educational loan sanctioned by BOB from year 2014 to 2016?
(a) 5.6 (b) 5.1 (c) 6.1 (d) 7.6 (e) 6.7
- Find the average of loan amount (in crores) sanctioned by SBI which is sanctioned not for education throughout the given years except year 2015?
(a) 34.87 (b) 35.67 (c) 33.87 (d) 38.37 (e) 38.67
- By approximately what percent the educational loan sanctioned by PNB in year 2012 and 2016 together is more or less than the educational loans sanctioned by SBI in year 2015 and 2016 together?
(a) 143% (b) 207% (c) 148% (d) 121% (e) 140%
- What is the ratio of total loan amount sanctioned by SBI to total educational loan sanctioned by PNB throughout the given years?
(a) 7030:747 (b) 3670:747 (c) 3007:477 (d) 3700:747 (e) 3070:747
- Educational loan sanctioned by BOB in year 2014 is what percent of total sanctioned loan amount by PNB in that year?
(a) 18% (b) 22% (c) 17% (d) 23% (e) 16%

Directions (6-10): Study the following pie chart and answer the following question given below : -
Number of male in the organization is two times of the number of females in the organization

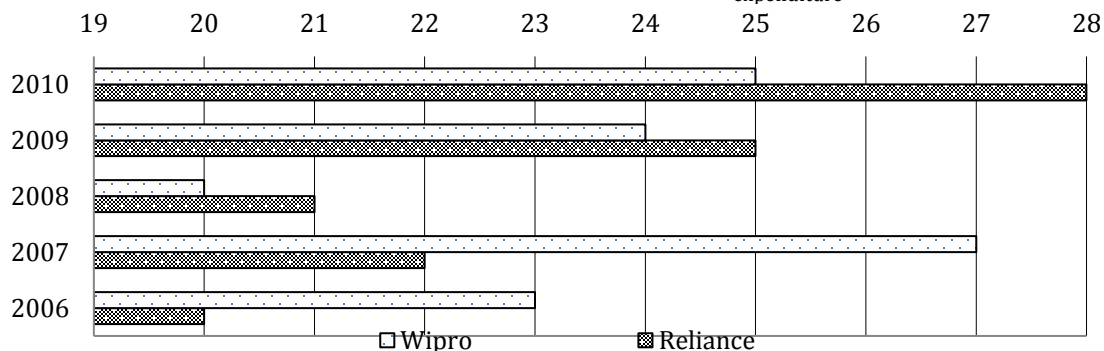
Percentage breakup of male employees working in various departments of an organization



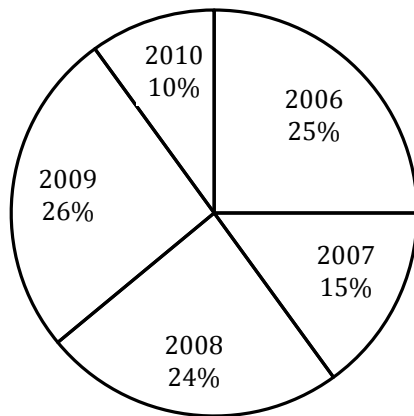
Number of females in Each Department	
Production Department	2400
Accounts Department	3200
HR Department	2800
IT Department	3800
Marketing Department	2100

6. Number of males in Accounts, IT and Marketing department together are approximately what percent more/less than the total number of females in the same departments?
 (a) 85% (b) 89% (c) 94% (d) 79% (e) 95%
7. Out of the total number of employees from HR department, 30% of the employees got promoted then find the ratio of the number of employees from HR department who get promoted to the total number of male employees in the organization?
 (a) 28600 : 1269 (b) 1170 : 28779 (c) 1269 : 14300 (d) 1269 : 28600 (e) 28779 : 1170
8. If one male from production department can do a work in 20020 days and the female from production department are 20% less efficient than that of male in production department. Find the total no. of days by employees from production department to do the same job?
 (a) $\frac{2002}{2009}$ days (b) $\frac{2002}{1096}$ days (c) $\frac{1008}{1001}$ days (d) $1\frac{809}{1193}$ days (e) $\frac{2097}{2002}$ days
9. Find the difference between the average number of males in all departments of the organization and the average of females in all departments of the organization?
 (a) 2860 (b) 2440 (c) 2630 (d) 2920 (e) None of these
10. Number of females in IT department is what percent of the total number of employees in Organization except HR department (approx.)
 (a) 12% (b) 8% (c) 6% (d) 15% (e) 10%

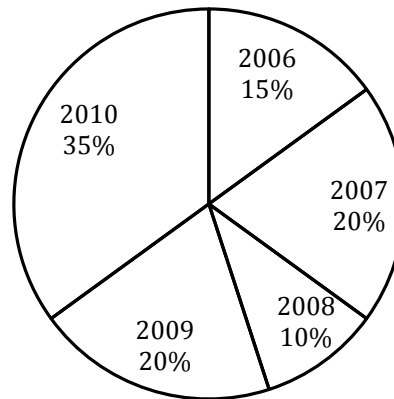
Directions (11-15): Given below is the line graph showing the profit percentage of 2 manufacturing firms, Reliance and Wipro from year 2006 to 2010. The pie charts show the percentage distribution of total income of each company in different years. Study the data carefully and answer the following questions {Note: $\% \text{profit} = \frac{\text{income} - \text{expenditure}}{\text{expenditure}} \times 100$ }



Reliance (total income=1500 Cr)

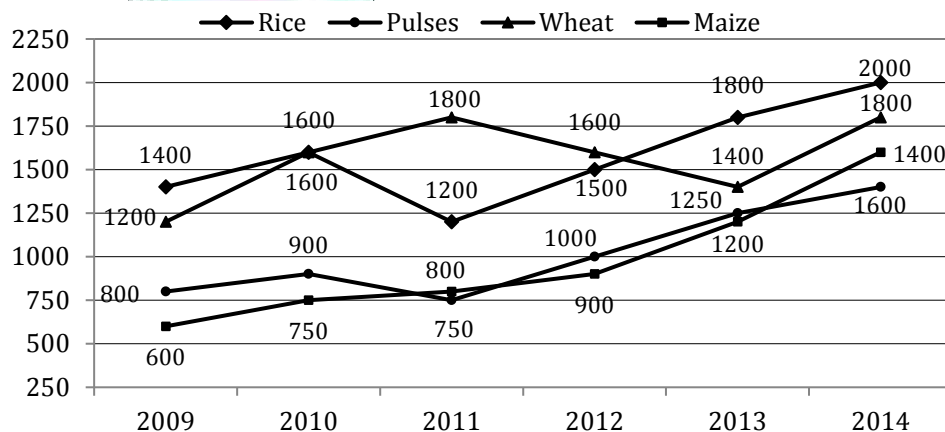


Wipro (Total income=1800 Cr)



11. The expenditure of Reliance in 2008 is approximately how much percent less/more than that of Wipro in 2009 ?
(a) 3% (b) 4% (c) 5% (d) 7% (e) 8%
12. The total income of Reliance in years 2006, 2007 and 2008 together is what percent more than the total income of Wipro for the same years ? (Calculate up to two decimal points)
(a) 16.66% (b) 18.51% (c) 27.4% (d) 20.4% (e) 22.22%
13. The income of Wipro in 2010 is approximately how much less/more than the average expenditure of Reliance for years 2009, 2010 and 2007 ?
(a) 107% (b) 407% (c) 307% (d) 207% (e) 507%
14. What is approximately the collective profit of Reliance and Wipro for years 2009 and 2010 ?
(a) 350 cr. (b) 270 cr. (c) 290 cr. (d) 307 cr. (e) 298 cr.
15. Calculate the difference of expenditures of Reliance and Wipro in 2011 if their expenditure in 2011 is 80% and 90% of their income in 2010 respectively ?
(a) 567 cr. (b) 257 cr. (c) 447 cr. (d) 120 cr. (e) None of these

Directions (16-20): Study the following line graph and table carefully to answer the given questions.
Total production under various heads(in tons)



Percentage of total production used only under these three heads-

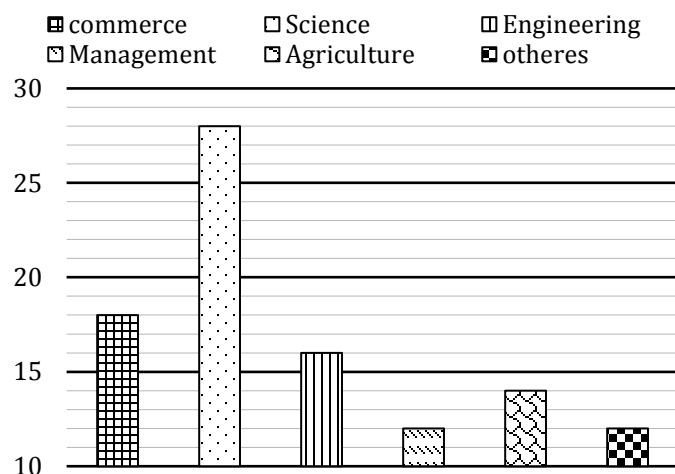
Year	Export	PDS Supply	Supply in open market
2009	42%	38%	-
2010	55%	-	25%
2011	48%	22%	-
2012	-	20%	18%
2013	-	33%	32%
2014	40%	-	35%

16. The average production of wheat and Mazes in 2011 is what percent of the supply in open market in year 2009 ?
 (a) 162.5% (b) 150.4% (c) 170% (d) 165.4% (e) None of these
17. What is the ratio of export in 2012 to PDS supply in 2014 ?
 (a) 31 : 19 (b) 31 : 17 (c) 41 : 31 (d) 19 : 13 (e) None of these
18. What is the difference between the overall production of maize in years 2009 to 2014 and production that exported in 2012 ?
 (a) 3000 tons (b) 2500 tons (c) 2550 tons (d) 2200 tons (e) None of these
19. The average production of all four crops in 2012 is what percent less than the average export in year 2009 and 2014 ? (Calculate up to two decimal points)
 (a) 52.8% (b) 40.46% (c) 53.71% (d) 43.18% (e) None of these
20. If 20% of the supply in open market in 2011 is retained and stored, then calculate the approximate percent decrease/increase in supply in open market in 2011 as compared to 2010 ?
 (a) 10% (b) 13% (c) 12% (d) 15% (e) 8%

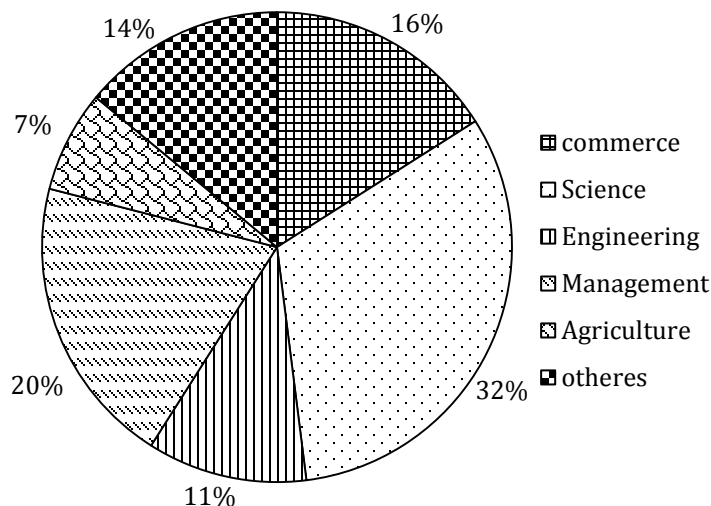
Directions (21-25): Study the following charts carefully and answer the questions given below:

Discipline wise breakup of number of candidates appeared in interview and Total no. of candidates selected after interview by the organization.

Total No. of candidates appeared in the interview=25750(in thousands)
Percentage of students in different discipline

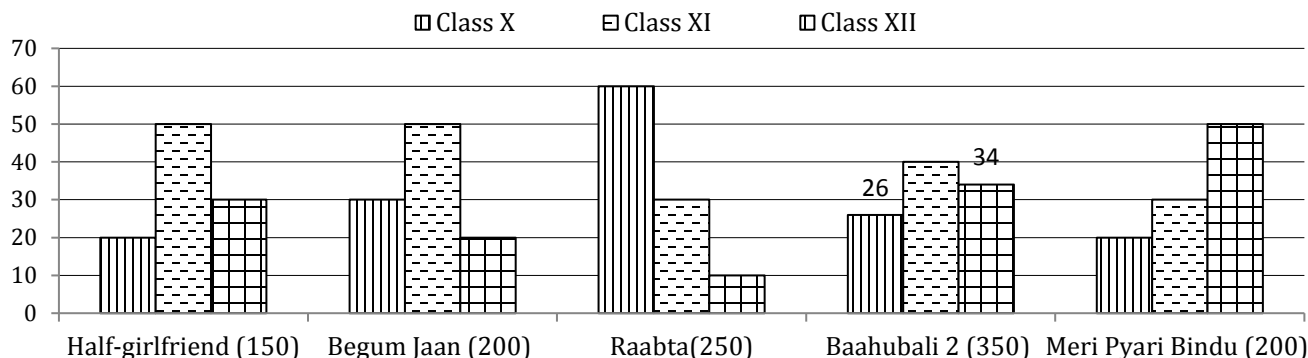


Total No. of candidates selected after interview=7300(in thousands)
Percentage of students in different discipline



21. The number of candidates who were selected after interview from commerce is what percent of number of candidates who were not selected after interview from Agriculture ? (Rounded off to two decimal points)
 (a) 37.75% (b) 46.43% (c) 42.5% (d) 52.5% (e) 55.55%
22. Calculate the average number of candidates who got selected after interview from Science, Engineering and Agriculture as a percentage of the number of candidates who did not get selected from same three disciplines ? (Rounded off to two decimal points)
 (a) 35.16% (b) 32.34% (c) 30.46% (d) 37.42% (e) 33.43%
23. The average number of candidates selected after interview from 'others' and Management is what percent of the number of candidates who appeared in interview from Engineering ? (Rounded off to two decimal points)
 (a) 24.26% (b) 28.18% (c) 30.12% (d) 34.17% (e) 29.14%
24. What percent of the total appeared candidates is the sum of the students who were selected after interview from Science and those who were not selected from others ? (Rounded off to two decimal points)
 (a) 25.45% (b) 21.26% (c) 12.74% (d) 17.10% (e) 17.89%
25. What is the ratio of students appeared in interviews from Management and those who got selected in interview from Commerce, others and Engineering together ?
 (a) 3017 : 2747 (b) 3015 : 2991 (c) 3090 : 2993 (d) 3417 : 2817 (e) None of these

Directions (26-30): Study the bar graph given below and answer the following questions:



Note: no Student went to watch more than one movie.

The bar graph shows the percentage break-up of number of students for different films in a theatre.

The table below shows total number of students in classes X, XI and XII.

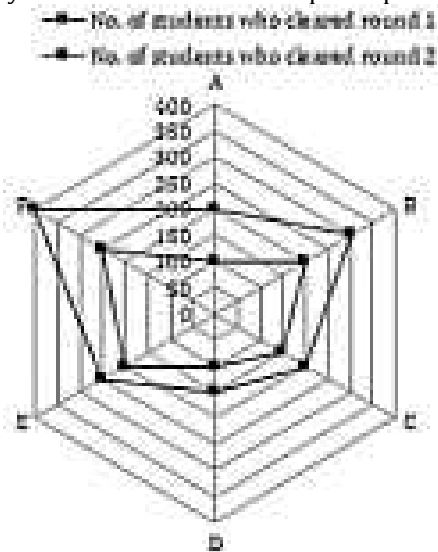
Class	No. of students
X	420
XI	480
XII	400

26. If the ratio of girls and boys, who went to watch 'Raabta' from class X, was 7 : 8, then what percent of the total number of students from class X who went to watch a movie is the number of girls who went to watch Raabta ? (approximate)
 (a) 19% (b) 23% (c) 25% (d) 15% (e) 21%
27. What is the difference between the number of students from class XI who did not go to watch any movie and that of the number of students from class XII ?
 (a) 19 (b) 41 (c) 27 (d) 21 (e) None of these
28. If all the students from class XII who did not go to watch any movie, later changed their mind and went to watch 'Meri Pyari Bindu', then calculate the % mark-up in the number of students who watched 'Meri Pyari Bindu' ?
 (a) 30% (b) 25.5% (c) 35.5% (d) 42.6% (e) None of these
29. The number of students from class XI and XII who watched Raabta is what percent of the number of students from class X who watched Begum Jaan, Half girlfriend and Meri Pyari Bindu ? (approximate)
 (a) 81% (b) 72% (c) 75% (d) 77% (e) 70%
30. Number of student who not go to watch movie from all the three classes is what percent of the total student.
 (a) $12\frac{1}{2}\%$ (b) $11\frac{7}{13}\%$ (c) $13\frac{2}{5}\%$ (d) $14\frac{7}{13}\%$ (e) $5\frac{1}{13}\%$

Directions (31-35): Study the graph given below and answer the questions that follow.

The table shows the no. of students from different schools in a contest. The graph shows the no. of students who cleared round 1 and round 2.

Note : Only those who clear round 1 participate in round 2.

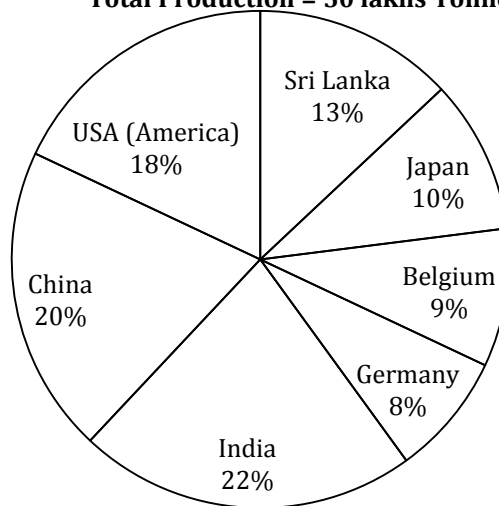


No. of students from different schools in a contest	
A	400
B	300
C	300
D	250
E	450
F	600

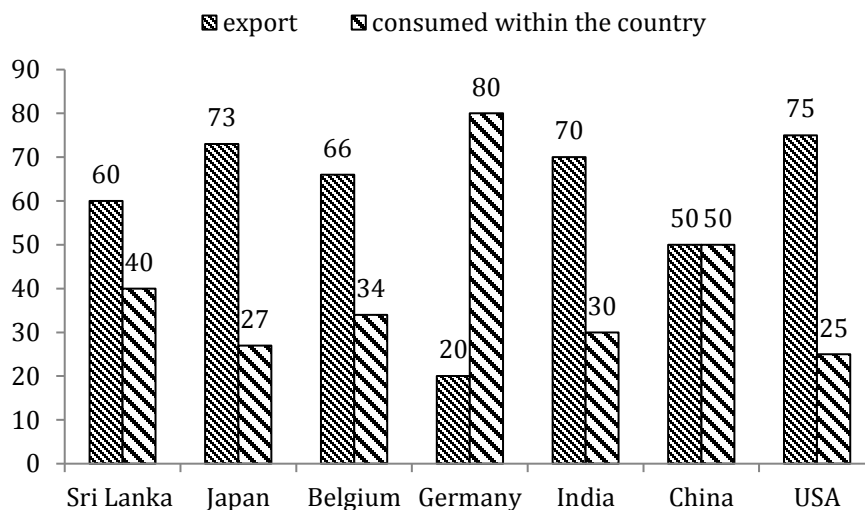
31. What is the ratio of number of students from school A and C together who were rejected in round 2 to the number of students from school B and F together who were rejected in round 1 ?
 (a) 3 : 8 (b) 2 : 5 (c) 7 : 8 (d) 3 : 7 (e) None of these
32. The number of students rejected in round 2 from all the schools together is what percent of the number of students rejected in round 1 from all the schools together ?
 (a) 75% (b) 50% (c) 20% (d) 25% (e) None of these
33. The ratio of girls to boys from school E who got rejected in round 2 is 2 : 3 and that from school C is 1 : 4. Then find the ratio of number of girls from both these schools who got rejected in round 2 to that of boys from the same school ?
 (a) 7 : 10 (b) 3 : 8 (c) 3 : 7 (d) 7 : 8 (e) None of these
34. If 20% of the students from school C, D and E together, who cleared round 1, failed to attend round 2, then find the number of students from these schools, who got rejected in round 2 ?
 (a) 40 (b) 60 (c) 50 (d) 30 (e) None of these
35. Number of student who cleared round 2 from all school are what percent of the total student from all school.
 (a) 40% (b) 50% (c) 30% (d) 39% (e) 25%

Directions(36-40) : Study the graphs given below & answer the questions that follow.

Production of wheat in different countries
Total Production = 50 lakhs Tonne



Percentage of export and consumed within the country out of total production is also given-

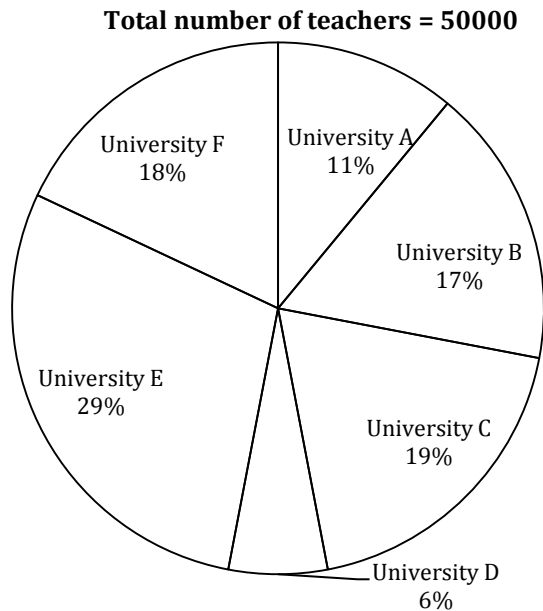


36. If out of the amount consumed in India, 91/11% was wasted due to bad storage, then find the difference between actual amount consumed in India and the amount exported by Srilanka ? (in tonnes)
 (a) 90000 (b) 100000 (c) 10000 (d) 80000 (e) None of these

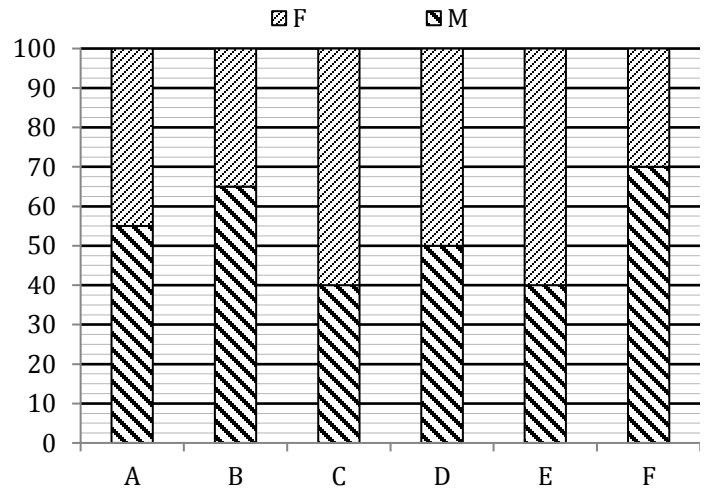
37. If the production in USA includes the production in Mexico and production in Mexico was $44\frac{4}{9}\%$ of the total given production under the head 'USA (America)', then production in Mexico is equal to the production in which % of the given countries ?
 (a) Belgium (b) Germany (c) Sri Lanka (d) Japan (e) None of these
38. Calculate the ratio between the average amount of production exported from China and Belgium to the average amount of production consumed in Japan and Germany ?
 (a) 797 : 555 (b) 797 : 565 (c) 797 : 455 (d) 797 : 465 (e) None of these
39. If Germany and USA donate 30% of their export to financially backward countries, then what percent of their total production do these two countries donate together ? (calculate up to two decimal points)
 (a) 19.41% (b) 18.64% (c) 21.46% (d) 17.42% (e) 17.17%
40. What is the difference between the quantity exported by Sri lanka, Japan and USA to the quantity consumed within these countries?
 (a) 3.3 lakh ton (b) 2.2 lakh ton (c) 8.3 lakh ton (d) 8.1 lakh ton (e) 10 lakh ton

Directions (41-45): Study the following pie chart and answer the following questions.

Percentage-wise distribution of teachers in six different universities



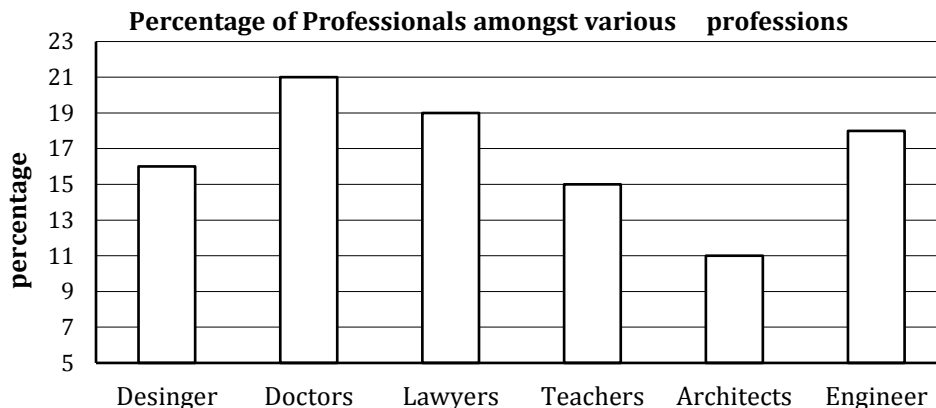
Percentage of male and female in different universities



41. If one-third sixth of the number of teachers from university F is professors and the salary of each professor is Rs. 96000, what will be the total salary of all the professors together from university F ?
 (a) Rs. 307.2 lakh (b) Rs. 32.64 lakh (c) Rs. 3.072 lakh (d) Rs. 3.264 lakh (e) None of these
42. Difference between the total number of teachers in university A, university B and university C together and the total number of teachers in university D, university E and university F together is exactly equal to the number of teachers of which university ?
 (a) University A (b) University B (c) University C (d) University D (e) University F
43. Number of female teachers in university D is approximately what per cent of the total number of male teachers in university B together ?
 (a) 55% (b) 59% (c) 27% (d) 45% (e) 35%
44. Total number of male teachers from university B and C together are what percent more/less than the total number of female teachers from university D and E together ? (calculate upto two decimal points)
 (a) 9.18% (b) 9.38% (c) 9.47% (d) 9.31% (e) 10.31%
45. What is the ratio between the number of male teacher of universities A to the number of female teachers of university C.
 (a) $\frac{220}{99}$ (b) $\frac{190}{99}$ (c) $\frac{121}{228}$ (d) $\frac{89}{190}$ (e) $\frac{250}{99}$

Directions (46-50): Study the following graphs carefully and answer the questions given below:

Survey Conducted on 10500 People to find out six various professional in the town and percentage of female among those six professional. Graph shows the percentage of various professional out of total professional (Consider every person has a profession from these 6 profession)



Percentage of female professionals

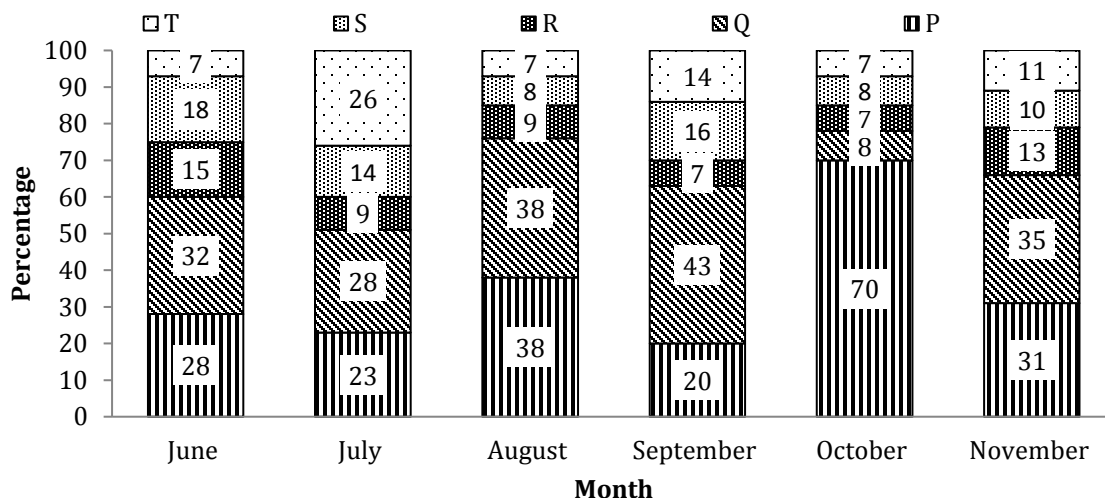
Doctors	20%
Engineers	60%
Architects	40%
Teachers	80%
Lawyers	40%
Designers	35%

46. What is the ratio of the male Engineers and male designers together to the same occupation female professionals together in the town?
 (a) 44: 41 (b) 55: 53 (c) 31: 35 (d) 44: 35 (e) None of these
47. The total number of lawyers in town is **approximately** what percent of the total number of doctors in town?
 (a) 95 (b) 98 (c) 90 (d) 85 (e) 81
48. What is the difference between the total number of male and female professionals in the town?
 (a) 1284 (b) 1134 (c) 1054 (d) 1164 (e) 1124
49. Female Doctors are what percent of the female teachers in the town?
 (a) 42 (b) 28 (c) 15 (d) 35 (e) 32
50. What is the ratio of the number of male architects to the number of male teachers in the town?
 (a) 11: 5 (b) 3 : 2 (c) 5: 11 (d) 2 : 3 (e) 6:11

Directions (51-55) : Read the following chart and answer the following questions given below it.

There are five companies which produces Diwali Fireworks in different months.

Bar graph shows Percentage wise distribution of production by these five companies in six different months



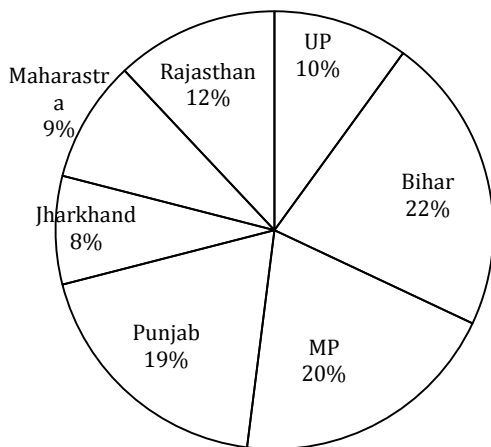
Following table shows the production of fireworks in units by these companies together in different months

June	P+Q	1920
July	R+S+T	1715
August	Q+R	1833
September	P+Q+T	3311
October	R	266
November	S+T	1218

Note- first row implies the production of P and Q together in June and as all rows.

51. If in December the no. of fireworks produced was 20% more than the firework produced in the month of November and the ratio between the fireworks produced in december by the company P, Q, R, S and T is 11 : 13 : 17 : 5 : 41, then find the no. of firework produced by company T in the month of December?
 (a) 3370 (b) 3140 (c) 3280 (d) Can't determined (e) 3480
52. No. of fireworks produced by company P and R together in July is Approximate what percent more/less then that of company S and T together in the month of November ?
 (a) 6% (b) 8% (c) 10% (d) 12% (e) 15%
53. What is the ratio of the total no. of fireworks produced in the month of July, August and September together to the total no. of fireworks produced in the month of October and November together?
 (a) 131 : 29 (b) 143 : 97 (c) 153 : 17 (d) 41 : 32 (e) 39 : 32
54. In October month, 70% of the fireworks produced by P was sold, 75% of the fireworks produced by S was sold and 50% of the total no. of fireworks produced by Q, R and T was together sold, then find the no. of fireworks sold by all of these companies in October?
 (a) 2508 (b) 2602 (c) 2498 (d) 2568 (e) 2580
55. In which month the difference between the no. of fireworks produced by all off the companies together to the previous month is maximum.
 (a) November (b) July (c) August (d) September (e) October

Directions (56-60): Given below is a pie chart which shows the percentage distribution of population in 7 states in year 2016 and table shows the ratio of male to female and ratio of married to unmarried person in the respective states.



Total population in 2016 = 500 million

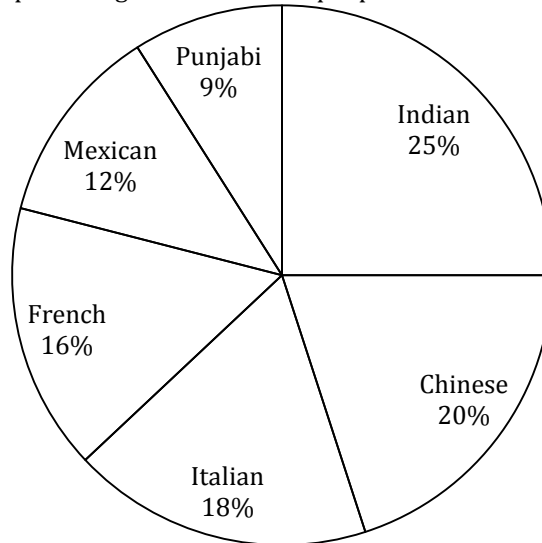
State	Ratio of married to unmarried persons		
	M : F	Married	Unmarried
Bihar	6 : 5	3	8
MP	3 : 2	9	11
Punjab	11 : 8	4	1
UP	3 : 2	3	2
Rajasthan	7 : 5	3	1
Maharashtra	5 : 4	7	2
Jharkhand	3 : 5	2	3

NOTE:- Number of married male/female = $\frac{\text{married persons}}{2}$

56. What is the ratio of unmarried female from state MP and Punjab together to the married female from UP and Bihar together?
 (a) 7 : 10 (b) 13 : 20 (c) 17 : 22 (d) 23 : 25 (e) 18 : 23
57. Unmarried males from state Jharkhand and Rajasthan together are what percent of total married persons from state M.P.?
 (a) 25% (b) 68% (c) $\frac{127}{4}\%$ (d) $\frac{130}{3}\%$ (e) $\frac{122}{3}\%$
58. What is the difference between average of married male from state Bihar and M.P. together to the average of married female from Maharashtra and Jharkhand together.
 (a) 6 million (b) 8 million (c) 9 million (d) 9.5 million (e) 10 million

59. Married females from state Punjab and UP together are what percent of Unmarried males from state Maharashtra and Jharkhand together.
 (a) $\frac{10600}{29}\%$ (b) $\frac{10580}{29}\%$ (c) $\frac{9823}{29}\%$ (d) $\frac{12385}{29}\%$ (e) $\frac{10540}{29}\%$
60. If $\frac{100}{3}\%$ of married persons get divorce from state Bihar, then unmarried females are increased by what percent in Bihar state.
 (a) $16\frac{1}{6}\%$ (b) $16\frac{2}{7}\%$ (c) $33\frac{1}{3}\%$ (d) $16\frac{2}{3}\%$ (e) $14\frac{2}{7}\%$

Directions (61-65): Pie chart shows the percentage distribution of people who likes different types of food.



The table shows the ratio of male to female above and below 25 years who likes different types of food
Note: Consider no person has exactly 25 years age.

	Below 25 years	Above 25 years
	Male : Female	Male : Female
Indian	8 : 5	9 : 7
Chinese	4 : 3	5 : 4
Italian	5 : 6	7 : 3
French	11 : 12	13 : 12
Mexican	3 : 4	5 : 6
Punjabi	5 : 4	3 : 5

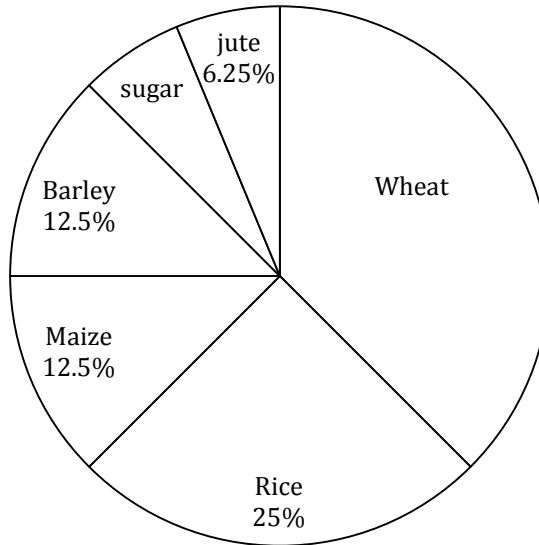
61. Total number of people who likes Italian is 48 lakh. Total number of people comprising male above 25 years who likes Punjabi and females below 25 years who like Punjabi food is 10 lakh, then find the total number of people who likes Punjabi for people below 25 yr.
 (a) 8 L (b) 10 L (c) 9.5 L (d) 15 L (e) 14.4 L
62. If the total number of female below 25 years who likes Chinese and male above 25 years whose likes Chinese is 9.6 lakh and 14 lakh respectively, then what is the total number of people who likes Indian. (in lakhs)
 (a) 55.5 (b) 64.6 (c) 72.8 (d) 59.5 (e) 57.2
63. If people below 25 years who likes French are 23,000 and difference between people below 25 years and above 25 years who likes French is 11,600 then, find the difference between total number of people who likes Mexican and Indian.
 (a) 43244 (b) 46254 (c) 27950 (d) 50130 (e) None of these
64. If total number of people who likes Indian is 1 lakh then what is the difference of people whose likes Mexican and French.
 (a) 1 L (b) 2.5 L (c) 0.5 L (d) 0.16 L (e) 0.2 L
65. Total number of people who likes Chinese is 80,000 then find the number of people above 25 years who likes Mexican if number of people above 25 years who likes Mexican and people below 25 year who likes Punjabi are same.
 (a) 30,000 (b) 20,000 (c) 25,000 (d) Can't be determined (e) None of these

Directions (66-70): Given below is the pie chart which shows the percentage distribution of production of 6 different types of crop. Table shows the ratio of amount exported to the amount consumed within India, total sale obtained on selling whole quantity produced and percentage of sales obtained from export.

Note → Some values are missing in the pie chart and table, you have to calculate the values if required to answer the question.

Total production = total amount consumed in India + total amount exported

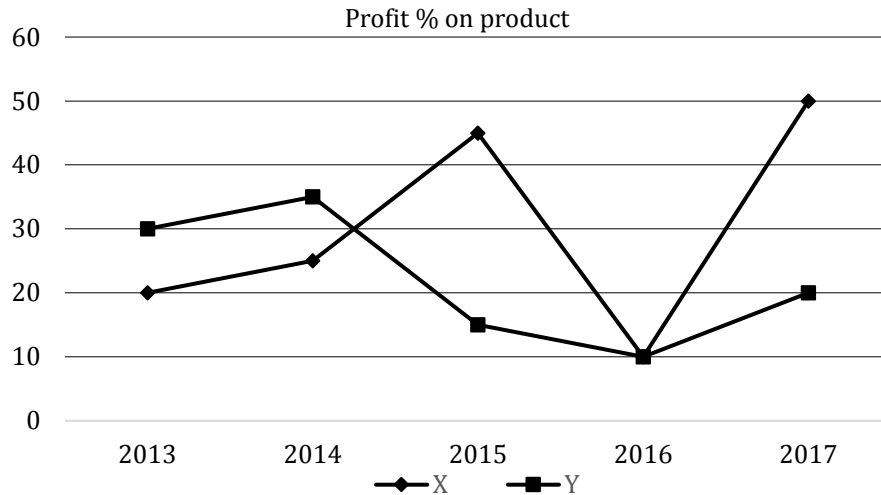
Total Production = 200000 tonnes



Crops	Ratio of amount exported to the consumed within India	Percentage of total sale obtained in export	Total sale (in Rs.)
Wheat	7 : 8	—	—
Rice	2 : 3	45%	6,00,000
Maize	—	35%	4,00,000
Barley	4 : 1	—	5,00,000
Sugar	—	$44\frac{4}{9}\%$	—
Jute	1 : 1	40%	3,12,500

66. If selling price per tonne of jute in India is equal to two times of selling price per tonne of Barley which is exported, then find the total sales of Barley in India?
 (a) 250000 (b) 200000 (c) 350000 (d) 400000 (e) 300000
67. Selling price of one tonne of Rice which is exported is what percent more or less than selling price per tonne of Jute which is sold in India?
 (a) 52% (b) 55% (c) 47% (d) 48% (e) 53%
68. If amount Maize exported is 25% of amount of Maize consumed in India then price per tonne of maize consumed in India is what percent of price per tonne of Jute consumed in India.
 (a) $66\frac{2}{3}\%$ (b) $33\frac{1}{3}\%$ (c) $43\frac{1}{3}\%$ (d) $53\frac{1}{3}\%$ (e) $45\frac{1}{3}\%$
69. If quantity of sugar consumed in India is 30% of the quantity of Barley exported and total production of Sugar is 250/3% less than total quantity of wheat produced from then find the total quantity of sugar exported
 (a) 4000 (b) 7000 (c) 6000 (d) 4500 (e) 6500
70. If the total sale of wheat was Rs. 5719000 and the ratio of selling price of one unit of wheat exported to selling price of one unit of wheat consumed within India is 2 : 3, then Find the selling price of one unit of wheat exported from India. It is also given that total production of wheat is 600% of total production of sugar
 (a) Rs. 60.2 (b) Rs. 40.2 (c) Rs. 30.1 (d) Rs. 50.2 (e) None of these

Directions(71-75): The line graph given below shows the profit percentage on a product P sold by two companies X and Y in five years and table shows the production (in quintals) of products P and Q by both companies in five years. Study the graph and table carefully to answer the questions based.



company \ year	X		Y	
	products		products	
	P (in quintals)	Q (in quintals)	P (in quintals)	Q (in quintals)
2013	40	60	50	100
2014	100	60	90	50
2015	70	30	110	30
2016	60	30	30	20
2017	50	100	50	70

71. In 2013 ratio between cost price per quintal of product P and cost price per quintal of product Q for both company is 4 : 7. What is the approximate profit percent on product Q in 2013 for both company, if ratio of profit of both product in 2013 is 2 : 3 (P : Q) by both company. (profit percent on product Q for both company is same)
- (a) 8% (b) 16% (c) 10% (d) 14% (e) 12%
72. In 2014 and 2016 cost price of both products per quintal are same for company X and profit percentage are also same for both product for company X then profit on product P is what percent of the profit on product Q in these two years for company X.(approx.)
- (a) 180% (b) 145% (c) 172% (d) 155% (e) 160%
73. What is the ratio of total selling price of product Q for company Y in year 2016 to 2017 if cost price of the products per quintal are in ratio 3 : 4 (P : Q) for both years 2016 and 2017 and profit for both product remain same in both year (2016, 2017) for company Y.
- (a) 89:310 (b) 34:41 (c) 91:300 (d) 95:341 (e) 65:344
74. What is the total profit earned by company Y in the year 2013,2014 and 2015 on product Q if cost price is 250 per quintal all over the year and profit percent on product Q is same as profit on product P of company Y.
- (a) 12370 (b) 10000 (c) 13000 (d) 15250 (e) 12000
75. Find the difference b/w the total production of P product and production of Q product by both companies in all over the years.
- (a) 150 quintals (b) 100 quintals (c) 129 quintals (d) 130 quintals (e) 120 quintals

PRACTICE SET (LEVEL-II) SOLUTIONS

1. (c); Ed. loan sanction by PNB
 $= (38 \times .40 + 56 \times .35 + 84 \times .25)$
 $= 15.2 + 19.6 + 21 = 55.8 \text{ cr}$
 Ed. Loan sanction by BOB
 $= (48 \times .15 + 86 \times .25 + 60 \times .35)$
 $= 7.2 + 21.5 + 21 = 49.7 \text{ cr}$
 $\therefore$ Required difference $= (55.8 - 49.7) = 6.1 \text{ crores}$

2. (d); Required average
 $= \frac{(60 \times .65 + 44 \times .75 + 75 \times .55 + 60 \times .80 + 34 \times .90)}{5}$
 $= \frac{(39 + 33 + 41.25 + 48 + 30.6)}{5}$
 $= \frac{191.85}{5} = 38.37$

3. (b); Ed. Loan by PNB in 2012 and 2016
 $= (56 \times .35 + 26 \times .25)$
 $= 19.6 + 6.5 = 26.1 \text{ crores}$
 Ed. Loan by SBI in 2015 and 2016
 $= (34 \times .15 + 34 \times .10) = 5.1 + 3.4 = 8.5 \text{ crores}$
 $\therefore$ Required percentage
 $= \frac{26.1 - 8.5}{8.5} \times 100 = 207.06 \approx 207\%$

4. (e); Required ratio
 $= \frac{(60 + 44 + 75 + 60 + 34 + 34)}{(38 \times .40 + 56 \times .35 + 84 \times .25 + 40 \times .10 + 42 \times .20 + 26 \times .25)}$
 $= \frac{3070}{747}$

5. (a); Required percentage $= \frac{48 \times \frac{15}{100}}{40} \times 100$
 $= \frac{7.2}{40} \times 100 = 18\%$

Solutions (6-10)

Departments	Male	Female
Production Dept.	10010	2400
Accounts Dept.	4576	3200
HR Dept.	1430	2800
IT Dept.	5720	3800
Marketing Dept.	6864	2100
Total	28600	14300

6. (b); Number of male in Accounts, IT and Marketing Dept.
 $= 4576 + 5720 + 6864 = 17160$
 Number of female in Account, IT and Marketing Dept.
 $= 3200 + 3800 + 2100 = 9100$
 Required % $= \frac{17160 - 9100}{9100} \times 100$
 $= \frac{8060}{9100} \times 100 = 88.57\% \approx 89\%$
7. (d); Required Ratio $= \frac{30}{100} (2800 + 1430) : 28,600$
 $= 1269 : 28600$
8. (d); M $\rightarrow$ 20020 days
 F $\rightarrow$ 25025 days
 Required No. of days

$$1 / \left(\frac{10010}{20020} + \frac{2400}{25025} \right) = 1 / \left(\frac{1}{2} + \frac{95}{1001} \right) = 1 \frac{809}{1193}$$

9. (a); Required difference $= \frac{1}{5} (28600 - 14300)$
 $= \frac{1}{5} \times 14300 = 2860$

10. (e); Employees except HR dept. $= 42900 - 2800 - 1430 = 38670$

Female in IT department = 3800

$$\text{Required \%} = \frac{3800 \times 100}{38670} \approx 10\%$$

11. (a); Expenditure of Reliance in 2008 $= \frac{24}{100} \times 1500 \times \frac{100}{121} \approx 298 \text{ cr.}$

$$\text{Expenditure of Wipro in 2009} = \frac{20}{100} \times 1800 \times \frac{100}{124} \approx 290 \text{ cr.}$$

$$\text{Required \%} = \frac{298 - 290}{290} \times 100 \approx 3\%$$

12. (b); Total income of Reliance in years 2006, 2004 and 2008

$$= \frac{(25 + 15 + 24)}{100} \times 1500 = 960 \text{ cr.}$$

Total income of Wipro in years 2006, 2004 and 2008

$$= \frac{(15 + 20 + 10)}{100} \times 1800 = 810 \text{ cr.}$$

$$\text{Req. percentage} = \frac{960 - 810}{810} \times 100 = \frac{1500}{81} = 18.51\%$$

13. (c); Average expenditure of Reliance in years 2007, 2009 and 2010

$$= \frac{\left(\frac{15}{100} \times 1500 \times \frac{100}{122} + \frac{26}{100} \times 1500 \times \frac{100}{125} + \frac{10}{100} \times 1500 \times \frac{100}{128} \right)}{3}$$

$$\approx \frac{610}{3} = 205$$

$$\text{Req. percentage} = \frac{\frac{35}{100} \times 1800}{205} \times 100 = 307\%$$

14. (d); Profit of Reliance in 2009 $= \frac{26}{100} \times 1500 \times \frac{25}{125} = 78$

$$\text{Profit of Reliance in 2010} = \frac{10}{100} \times 1500 \times \frac{28}{128} = 32.8125$$

$$\text{Profit of Wipro in 2009} = \frac{20}{100} \times 1800 \times \frac{24}{124} = 70$$

$$\text{Profit of Wipro in 2010} = \frac{35}{100} \times 1800 \times \frac{25}{125} = 126$$

$$\text{Total profit} = 78 + 32.8125 + 70 + 126 = 306.8125 \text{ cr.}$$

15. (c); Expenditure of Reliance in 2011 $= \frac{80}{100} \times \frac{10}{100} \times 1500 = 120 \text{ cr.}$

$$\text{Expenditure of Wipro in 2011} = \frac{90}{100} \times \frac{35}{100} \times 1800 = 567 \text{ cr.}$$

$$\text{Difference} = 567 - 120 = 447 \text{ cr.}$$

16. (a); Supply in open market in 2009 $= \frac{20}{100} \times 4000 = 800 \text{ tons}$

$$\text{Average production of wheat and mazes in 2011} = \frac{1800 + 800}{2} = 1300 \text{ tons}$$

$$\text{Req. Percentage} = \frac{1300}{800} \times 100 = 162.5\%$$

17. (b); Export in 2012 $= \frac{62}{100} \times (1600 + 1500 + 1000 + 900) = 3100 \text{ tons.}$

$$\text{PDS supply in 2014} = \frac{25}{100} \times (2000 + 1800 + 1400 + 1600) = 1700 \text{ tons}$$

$$\text{Ratio} = 3100 : 1700 = 31 : 17$$

18. (c); Export in 2012 = 3100 tons

Overall production of maize in the given years =
600 + 750 + 800 + 900 + 1200 + 1400

= 5650 tons

Difference = 5650 - 3100 = 2550 tons

19. (d); Average production of all four crops in 2012 =
 $\frac{1600+1500+1000+900}{4} = 1250$ tons

Average export in 2009 and 2014

$= \frac{\left(\frac{42}{100} \times 4000 + \frac{40}{100} \times 6800\right)}{2} = 2200$ tons

Required % = $\frac{2200-1250}{2200} \times 100$

$= \frac{950}{2200} \times 100 = 43.18\%$

20. (a); Actual amount of crop supplied in open market in 2011

$= \frac{80}{100} \times \frac{30}{100} \times 4550 = 1092$ tons.

Amount supplied in open market in 2010 = $\frac{25}{100} \times 4850 = 1212.5$

Decrease% = $\frac{1212.5-1092}{1212.5} \times 100 = 9.9\% \approx 10\%$

21. (a); No. of candidates selected after interview from Commerce = $\frac{16}{100} \times 7300 = 1168$

No. of unselected candidates after interview from agriculture = $\frac{14}{100} \times 25750 - \frac{7}{100} \times 7300 = 3094$

Required % = $\frac{1168}{3094} \times 100 = 37.75\%$

22. (b); No. of students who got selected = $\frac{(32+11+7)}{100} \times 7300 = 3650$

No. of students who did not selected = $\frac{(28+16+14)}{100} \times 25750 - 3650$
 $= 14935 - 3650 = 11285$

Required % = $\frac{3650}{11285} \times 100 = 32.34\%$

23. (c); Average no. of selected candidates from 'others' and management = $\frac{1}{2} \times \frac{(20+14)}{100} \times 7300 = 1241$

Required % = $\frac{1241}{\frac{16}{100} \times 25750} \times 100 = \frac{1241}{4120} \times 100$
 $= 30.12\%$

24. (d); Students selected from 'Science'

$= \frac{32}{100} \times 7300 = 2336$

Unselected students from 'others'

$= \frac{12}{100} \times 25750 - \frac{14}{100} \times 7300 = 3090 - 1022 = 2068$

Req. % = $\frac{(2336+2068)}{25750} \times 100 = \frac{4404}{25750} \times 100 = 17.10\%$

25. (c); No. of students appeared from 'management'

$= \frac{12}{100} \times 25750 = 3090$

No. of students who got selected from Commerce, others and Engineering together = $\frac{(16+11+14)}{100} \times 7300$

$= \frac{41}{100} \times 7300 = 2993$

Req. Ratio = $\frac{3090}{2993}$

26. (a); No. of students from class X who watched 'Raabta' =
 $\frac{60}{100} \times 250 = 150$

No. of girls from class X who watched 'Raabta' =
 $\frac{7}{15} \times 150 = 70$

Total no. of students from class X who went to watch a movie

$= \frac{20}{100} \times 150 + \frac{30}{100} \times 200 + \frac{60}{100} \times 250 + \frac{26}{100} \times 350 +$
 $\frac{20}{100} \times 200$

$= 30 + 60 + 150 + 91 + 40 = 371$

Required % = $\frac{70}{371} \times 100 \approx 19\%$

27. (b); No. of students from class XI who didn't watch any

movie = $450 - \left(\frac{50}{100} \times 150 + \frac{50}{100} \times 200 + \frac{30}{100} \times 250 + \frac{40}{100} \times 350 + \frac{30}{100} \times 200\right)$

$= 450 - (75 + 100 + 75 + 140 + 60)$
 $= 480 - 450 = 30$

No. of students from class XII who didn't watch any

movie = $400 - \left(\frac{30}{100} \times 150 + \frac{20}{100} \times 200 + \frac{10}{100} \times 250 + \frac{34}{100} \times 350 + \frac{50}{100} \times 200\right)$

$= 400 - (45 + 40 + 25 + 119 + 100)$
 $= 400 - (329) = 71$

Req. Difference = $71 - 30 = 41$

28. (c); Using solution of previous question,

Required % = $\frac{71}{200} \times 100 = 35.5\%$

29. (d); No. of students who watched Raabta from class XI and XII = $\frac{30+10}{100} \times 250 = 100$

No. of students from class X who watched Begum Jaan, Half Girlfriend and Meri Pyari Bindu = $30 + 60 + 40 = 130$

Req. % = $\frac{100}{130} \times 100 \approx 77\%$

30. (b); No. of student not go to watch movie

From class X = 49

From class XI = 30

From class XII = 71

Required % = $\frac{150}{1300} \times 100 = 11\frac{7}{13}\%$

31. (a); No. of students from school A and who were rejected in round 2 = $(200 - 100) + (200 - 150)$
 $= 100 + 50 = 150$

No. of students from school B and F who were rejected in round 1 = $(500 - 300) + (600 - 400)$
 $= 200 + 200 = 400$

Req. Ratio = $\frac{150}{400} = \frac{3}{8}$

32. (b); No. of students from all schools who got rejected in round 2 = $(200 - 100) + (300 - 200) + (200 - 150) + (150 - 100) + (250 - 200) + (400 - 250)$
 $= 100 + 100 + 50 + 50 + 50 + 150 = 500$

No. of students from all schools who got rejected in round 1 = $(400 - 200) + (500 - 300) + (300 - 200) + (250 - 150) + (450 - 250) + (600 - 400)$
 $= 200 + 200 + 100 + 100 + 200 + 200 = 1000$

Req. % = $\frac{500}{1000} \times 100 = 50\%$

33. (c); No. of students from school E who got rejected in round 2 = $250 - 200 = 50$

No. of students from school C who got rejected in round 2 = $200 - 150 = 50$

Required ratio = $\frac{\frac{2}{5} \times 50 + \frac{1}{5} \times 50}{\frac{3}{5} \times 50 + \frac{4}{5} \times 50} = \frac{3}{7}$

34. (d); No. of students from school C who attended round 2 =
 $= \frac{80}{100} \times 200 = 160$

No. of students from school D who attended round 2
 $= \frac{80}{100} \times 150 = 120$

No. of students from school E who attended round 2
 $= \frac{80}{100} \times 250 = 200$

No. of students rejected in round 2 = $(160 - 150) + (120 - 100) + (200 - 200)$
 $= 10 + 20 + 0 = 30$

35. (a); Total student cleared round 2 = $100 + 200 + 150 + 100 + 200 + 250 = 1000$

Total student = $400 + 500 + 300 + 250 + 450 + 600 = 2500$

Required % = $\frac{1000}{2500} \times 100 = 40\%$

36. (a); Actual amount consumed in India = $\frac{10}{11} \times \frac{30}{100} \times \frac{22}{100} \times 50,00,000 = 3$ lakh tonnes

Amount exported by Sri Lanka = $\frac{60}{100} \times \frac{13}{100} \times 50,00,000 = 3,90,000$

Difference = $390000 - 30000 = 90000$ tonnes

37. (b); Production in Mexico = $\frac{4}{9} \times 18/100 \times 50$ lakh tonnes
 $= \frac{8}{100} \times 50$ lakh tonnes = 8% of 50 lakh tonnes
 $=$ production in Germany

38. (c); Average amount exported from China and Belgium =
 $\frac{(\frac{50}{100} \times \frac{20}{100} + \frac{66}{100} \times \frac{9}{100}) \times 50 \text{ lakh tonnes}}{2} = 398500$ tonnes

Average amount consumed in Japan & Germany =
 $\frac{(\frac{27}{100} \times \frac{10}{100} + \frac{80}{100} \times \frac{8}{100}) \times 50}{2} = 227500$ tonnes

Ratio = $\frac{398500}{227500} = \frac{797}{455}$

39. (d); Required percentage
 $= \frac{(0.3 \times 0.75 \times 0.18 + 0.3 \times 0.2 \times 0.08) 50 \text{ lakh tonnes}}{(0.18 + 0.08) 50 \text{ lakh tonnes}} \times 100$
 $= 17.42\%$

40. (d); Srilanka

Exported = 3.9 lakh ton

Consumed = 2.6 lakh ton

Japan

Export = 3.65 lakh ton

Consumed = 1.35 lakh ton

U.S.A.

Export = 6.75 lakh ton

Consumed = 2.25 lakh ton

Required difference = $14.3 - 6.2 = 8.1$ lakh ton

41. (e); No. of teacher from F = 18% of 50000 = 9000

No. of professors from F = $\frac{1}{36} \times 9000 = 250$

$\therefore$ Required salary = $250 \times 96000 = 240$ lakhs

42. (d); Difference in both = $(53 - 47)\%$

$= (53 - 47)\% = 6\%$

Thus, the difference is exactly equal to the no. of teachers in university D.

43. (c); No. of male teachers in university B

$= \frac{65}{100} \times \frac{17}{100} \times 50000 = 5525$

No. of female teachers in university D = $\frac{50}{100} \times \frac{6}{100} \times 50000 = 1500$

Required percentage = $\frac{1500}{5525} \times 100 = 27.14\% \approx 27\%$

44. (b); Total no. of male teachers from B and C together

$= \frac{65}{100} \times \frac{17}{100} \times 50,000 + \frac{40}{100} \times \frac{19}{100} \times 50000$

$= 5525 + 3800 = 9325$

Total no. of female teachers from D and E together

$= \frac{50}{100} \times \frac{6}{100} \times 50000 + \frac{60}{100} \times \frac{29}{100} \times 50000$
 $= 1500 + 8700 = 10,200$

Required % = $\frac{10200 - 9325}{9325} \times 100 = 9.38\%$

45. (c); Required ratio = $\frac{11 \times 500 \times \frac{55}{100}}{19 \times 500 \times \frac{60}{100}} = \frac{121}{228}$

46. (a); Male Engineers + Male Designers = 40% of (18% of 10500) + 65% of (16% of 10500) = $756 + 1092 = 1848$
 Female engineers + female designers = 60% of (18% of 10500) + 35% of (16% of 10500) = $1134 + 588 = 1722$
 $\therefore$ required ratio = $\frac{1848}{1722} \Rightarrow 44 : 41$

47. (c); Required % = $\frac{19}{21} \times 100 = 90\%$ approx

48. (b); female Professionals = 10500 (20% of 21% + 60% of 18% + 40% of 11% + 80% of 15% + 40% of 19% + 35% of 16%) = 4683

Male = $10500 - 4683 = 5817$

Difference = $5817 - 4683 = 1134$

49. (d); Required %

$=$

$\frac{20\% \text{ of } 21}{80\% \text{ of } 15} \times 100\% \Rightarrow \frac{20 \times 20}{80 \times 15} \times 100\% \Rightarrow \frac{420}{12} \Rightarrow 35\%$

50. (a); Required ratio = $\frac{60 \times 11}{20 \times 15} \Rightarrow 11 : 5$

51. (c); December $\rightarrow \frac{120}{100} \times (\frac{1218}{21} \times 100) = 6960$

Required no. of firework produced by T

$= \frac{41}{87} \times 6960 = 3280$

52. (b); Fireworks produced by P and R in July

$= \frac{(23+9) \times 1715}{(9+14+26)} = 35 \times 32 = 1120$

Fireworks produced by S and T in November

$= \frac{(11+10) \times 1218}{21} = 1218$

Required % = $\frac{1218 - 1120}{1218} \times 100 = 8.04\% \approx 8\%$

53. (e); Required Ratio = $(3500 + 3900 + 4300) : (3800 + 5800)$
 $= 11700 : 9600 = 117 : 96 = 39 : 32$

54. (a); Required No. of fireworks sold
 $= (0.7 \times 2660) + (0.75 \times 304) + 0.5 \times (836)$
 $= 1862 + 228 + 418 = 2508$

55. (a); July $\rightarrow (3500 - 3200) = 300$

August $\rightarrow (3900 - 3500) = 400$

September $(4300 - 3900) = 400$

October $(3800 - 4300) = 500$

November $(5800 - 3800) = 2000$

56. (b); Married persons from MP = $20 \times 5 \times \frac{9}{20} = 45$ million

Unmarried female from state MP = $20 \times 5 \times \frac{2}{5} - \frac{45}{2}$

$= 17.5$ million

Similarly,

Unmarried female from Punjab = $19 \times 5 \times \frac{8}{19} - 19 \times$

$5 \times \frac{4}{5} \times \frac{1}{2} = 40 - 38 = 2$ million

Married female from Up and Bihar together

$= 10 \times 5 \times \frac{3}{5} \times \frac{1}{2} + 22 \times 5 \times \frac{3}{11} \times \frac{1}{2}$

$= 15 + 15 = 30$ million

Required ratio = 39 : 60 = 13 : 20

- 57. (d);** Unmarried males from state Jharkhand
 $= 8 \times 5 \times \frac{3}{8} - 8 \times 5 \times \frac{2}{5} \times \frac{1}{2} = 15 - 8 = 7$ million
 Unmarried males from State Rajasthan
 $= 12 \times 5 \times \frac{7}{12} - 12 \times 5 \times \frac{3}{4} \times \frac{1}{2}$
 $= 35 - 22.5 = 12.5$ million
 Required percentage = $\frac{7+12.5}{20 \times 5 \times \frac{9}{20}} \times 100$
 $= \frac{19.5}{45} \times 100 = \frac{130}{3} \%$

- 58. (a);** Average of married males from state Bihar & M.P. together
 $= \frac{1}{2} \left(22 \times 5 \times \frac{3}{11} \times \frac{1}{2} + 20 \times 5 \times \frac{9}{20} \times \frac{1}{2} \right)$
 $= \frac{1}{2} \left(15 + \frac{45}{2} \right) = \frac{75}{4}$ million
 Average of married female from Maharastra and Jharkhand together
 $= \frac{1}{2} \left(9 \times 5 \times \frac{7}{9} \times \frac{1}{2} + 8 \times 5 \times \frac{2}{5} \times \frac{1}{2} \right) = \frac{1}{2} \left(\frac{35}{2} + 8 \right)$
 $= \frac{51}{4}$ million
 Required difference = $\frac{75}{4} - \frac{51}{4} = 6$ million

- 59. (a);** Married females from state Punjab and UP together
 $= 19 \times 5 \times \frac{4}{5} \times \frac{1}{2} + 10 \times 5 \times \frac{3}{5} \times \frac{1}{2} = 38 + 15$
 $= 53$ million
 Unmarried males from state Maharastra
 $= 9 \times 5 \times \frac{5}{9} - 9 \times 5 \times \frac{7}{9} \times \frac{1}{2}$
 $= 25 - 17.5 = 7.5$ million
 Unmarried males from Jharkhand
 $= 8 \times 5 \times \frac{3}{8} - 8 \times 5 \times \frac{2}{5} \times \frac{1}{2}$
 $= 15 - 8 = 7$ million
 Required percentage
 $= \frac{53}{(7.5+7)} \times 100 = \frac{530}{145} \times 100$
 $= \frac{10600}{29} \%$

- 60. (e);** Unmarried females in Bihar
 $= 22 \times 5 \times \frac{5}{11} - 22 \times 5 \times \frac{3}{11} \times \frac{1}{2}$
 $= 50 - 15 = 35$ million
 Number of persons who get divorced
 $= \frac{1}{3} \times 22 \times 5 \times \frac{3}{11} = 10$ million
 So, Number of females who got divorce
 $= \frac{10}{2} = 5$ million
 Required percentage
 $= \frac{5}{35} \times 100 = \frac{100}{7} \%$
 $= 14\frac{2}{7} \%$

- 61. (e);** Total people who likes Punjabi = $\frac{48}{18} \times 9 = 24$ lakh
 Let total people below 25 yr who likes Punjabi = x
 Then, $(24 - x) \times \frac{3}{8} + \frac{4}{9}x = 10$
 $x = 14.4$ L

- 62. (d);** Total number of people below 25 years who likes Chinese = $\frac{9.6}{3} \times 7$
 $= 3.2 \times 7 = 22.4$ L
 Total number of people above 25 years who likes Chinese = $\frac{14}{5} \times 9$
 $= 2.8 \times 9 = 25.2$ L

Total people who likes Indian = $\frac{(22.4+25.2)}{20} \times 25 = 59.5$ Lakh

- 63. (c);** Let total people who like French food = x
 Then,
 $23000 - (x - 23000) = 11600$
 $46000 - x = 11600$
 $x = 46000 - 11600 = 34400$
 Required difference = $\frac{34400}{16} \times 13 = 27950$

- 64. (d);** Total people in the town = $\frac{1}{25} \times 100 = 4$ L
 Required difference = $\frac{4}{100} \times 4 = 0.16$ L

- 65. (d);** we can't determine these values individually.

- 66. (b);** Total sale of Jute in India = $\frac{3}{5} \times 312500 = 187500$ Rs
 Amount of Jute consumed in India
 $= \frac{1}{2} \times \frac{6.25}{100} \times 200000 = 6250$
 Price per unit of jute in India = $\frac{187500}{6250} = 30$ Rs/tonnes
 Price per tonne of Barey export = 15 rs/tonnes
 Total barley exported
 $= \frac{4}{5} \times \frac{12.5}{100} \times 200000 = 20000$ tonnes
 Total sale barley in India
 $= 500000 - 20000 \times 15 = 200000$ Rs

- 67. (b);** Rice exported
 $= \frac{25}{100} \times 200000 \times \frac{2}{5} = 20000$ tonne
 Total sale of rice exported
 $= \frac{45}{100} \times 600000 = 270000$
 Selling price of one tonne of exported rice
 $= \frac{270000}{20,000} = 13.5$ Rs/ tonnes

- Jute consumed in India
 $= \frac{1}{2} \times \frac{6.25}{100} \times 200000 = 6250$
 Total sale of Jute in India = $\frac{60}{100} \times 312500 = 187500$
 Per tonne price of jute consumed in India
 $= \frac{187500}{6250} = 30$ Rs /tonne
 Required % = $\frac{30-13.5}{30} \times 100 = 55\%$

- 68. (c);** Let amount of maize consumed in India = x
 $x + \frac{25}{100}x = \frac{12.5}{100} \times 200000 = 25000$
 $\frac{125x}{100} = 25000 \Rightarrow x = 20000$

- Total sale of maize in India = $\frac{65}{100} \times 400000$
 $= 260,000$
 Per tonne price of maize consumed in India
 $= \frac{2,60,000}{20,000} = 13$ Rs/ tonne
 Price per toone of jute consumed in India = 30 Rs/ tonne
 Required percentage
 $= \frac{13}{30} \times 100 = 43\frac{1}{3} \%$
69. (e); Quantity of barley exported
 $= \frac{4}{5} \times \frac{12.5}{100} \times 200000 = 20000$ tonne
 Quantity of sugar consumed in India = 30×200
 $= 6000$
 Let total wheat produced = x
 So,
 Total sugar produced = $\left(100\% - \frac{250}{3} \% \right)$ of x

$$= \frac{50}{3} \% \text{ of } x = \frac{x}{6}$$

So,

$$200000 = x + \frac{x}{6} + \frac{56.25}{100} \times 200000$$

$$x + \frac{x}{6} = 87500$$

$$x = 75000$$

$$\text{Total sugar produced in India} = \frac{75000}{6} = 12500$$

Total sugar exported

$$= 12500 - 6000 = 6500 \text{ tonne}$$

- 70. (a);** Let total sugar produced is x
 so total wheat produced will be $6x$
 percentage distribution of production of sugar and
 wheat = $100\% - 56.25\% = 43.75\%$
 percentage distribution of production of wheat
 $= \frac{43.75}{7} \times 6 = 37.5\%$

$$\text{Amount of wheat exported} = \frac{7}{15} \times \frac{3}{8} \times 200000$$

$$= 35000 \text{ t}$$

$$\text{Amount of wheat consumed} = \frac{8}{15} \times \frac{3}{8} \times 200000$$

$$= 40000 \text{ t}$$

Let, selling price of one tonne of wheat exported be
 Rs. $2x$ and that consumed be Rs. $3x$

$$\text{Then, } 35000 \times 2x + 40000 \times 3x = 5719000$$

$$\text{or, } 190000x = 5719000$$

$$\text{or, } x = 30.1$$

Selling price of one tonne of wheat

$$\text{Exported from India} = \text{Rs. } 2x = \text{Rs. } 2 \times 30.1 = \text{Rs. } 60.2$$

- 71. (e);** Let cost price of product P and Q per quintal for both
 company be $4x$ and $7x$ respectively

$$\text{Profit on product P by company X} = \frac{(4x \times 40) \times 20}{100} = 32x$$

$$\text{Profit on product P by company Y} = \frac{(4x \times 50) \times 30}{100} = 60x$$

$$\text{Total profit on P} = 92x$$

$$\text{Total profit on Q} = \frac{92x}{2} \times 3 = 138x$$

Cost price of product Q by both companies

$$= 7x (60 + 100) = 7x \times 160$$

$$\text{Required } \% = \frac{138x}{7x \times 160} \times 100 \approx 12\%$$

- 72. (c);** Let cost price per quintal of product P and Q be x
 In 2014

$$\text{Profit on P} \rightarrow \frac{100x \times 25}{100} = 25x$$

$$\text{Profit on Q} \rightarrow \frac{60x \times 25}{100} = 15x$$

In 2016

$$\text{Profit on product P} = \frac{60x \times 10}{100} = 6x$$

$$\text{Profit on product Q} = \frac{30x \times 10}{100} = 3x$$

$$\text{Required } \% = \frac{25x + 6x}{15x + 3x} \times 100 \approx 172\%$$

- 73. (a);** Let cost price of P and Q per quintal be $3x$ and $4x$
 respectively.

In 2016

$$\text{Profit} = \frac{(3x \times 30) \times 10}{100} = 9x$$

$$\text{Selling price of product Q} = (4x \times 20) + 9x = 89x$$

In 2017,

$$\text{Profit} = \frac{(3x \times 50) \times 20}{100} = 30x$$

$$\text{Selling price of product Q} = (4x \times 70) + 30x = 310x$$

$$\text{Required ratio} = \frac{89x}{310} = 89 : 310$$

- 74. (c);** Profit in 2013 $\frac{250 \times 100 \times 30}{100} = 7500$

$$\text{Profit in 2014} = \frac{250 \times 50 \times 35}{100} = 4375$$

$$\text{Profit in 2015} = \frac{250 \times 30 \times 15}{100} = 1125$$

$$\text{Total profit} = \text{Rs. } 13000$$

- 75. (b);** Production of product P by both
 Companies = 650 quintals
 Production of product Q by both companies = 550
 quintals
 Required difference = 100 quintals



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